

ngspice-46 — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

September 2026

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ngspice-46 — Full Change Report

Every modification applied to ngspice-46 in this project, with its reason. The baseline is the pristine ngspice-46 source tree (as vendored in the project's `original/` snapshot, which already contains OpenVAF-Reloaded's stock OSDI support); the current state is the tree committed in this repository under ngspice-46/. Each entry links the enhancement write-up in [enhancements_doc/](#) that carries the full engineering detail and the verifying example suite. The companion document [openvaf_changes_full-report.md](#) covers the compiler side.

Maintenance note: this report is updated whenever an enhancement touches ngspice sources. If you are reading it alongside a newer tree, the per-enhancement index at the end tells you the last change it covers.

Scope summary. Two new source files and 51 modified ones carry all the functional changes (~2,100 substantive diff lines). A further ~100 `Makefile.in` files and `config.h.in` differ only because the auto-tools were regenerated with a current libtool ([E-77](#)); they contain no hand-written changes and are not listed individually. Everything below is behavior, interface, or diagnostics.

1. The OSDI ABI, version 0.4 → 0.7

`src/osdi/osdi.h` is the authoritative C contract between ngspice and the `.osdi` objects `openvaf-r` produces. The project extended it six times — three times additively (new exported symbols old loaders simply never look up), three times with genuine version bumps (layout changes):

Change	Kind	Enhancement	Why
OSDI_ABSDELAY_COUNTS / OSDI_ABSDELAY_INFOS exports	additive symbols	E-1	<code>absdelay()</code> needs simulator-side waveform history and synthetic delay-equation rows; the descriptors tell ngspice which nodes/offsets each delay slot uses
OSDI_LAST_CROSSING_COUNTS / OSDI_LAST_CROSSING_INFOS exports	additive symbols	E-6	<code>last_crossing()</code> needs waveform history and crossing interpolation — same additive pattern as <code>absdelay</code>
OSDI_TERM_SHORT_COUNTS / OSDI_TERM_SHORT_INFOS exports	additive symbols	E-401	$V(a,b) <+ 0$ between two TERMINALS cannot be served by node collapsing (terminals are simulator-allocated), so the compiler emits a real 0 V source and lists the branch; the simulator drops its current when the netlist already ties those terminals to one node, which is what stops the redundant equation being a singular row
<code>OsdiNode.nodeset</code> field	ABI 0.5 (node stride change)	E-45	Verilog-A net initializers (<code>electrical a = 5.0;</code>) become solver initial-guess nodesets; every node record carries the hint
<code>num_ac_stim_src / ac_stim_sources / load_ac_stim</code> descriptor tail	ABI 0.6 (descriptor grows)	E-51	full <code>ac_stim()</code> support: a Verilog-A module can <i>be</i> the AC stimulus, so the descriptor must enumerate its AC right-hand-side sources

Change	Kind	Enhancement	Why
load_noise destination stride 1 → 2 (signed (flat, react) power pairs)	ABI 0.7	E-54	correct noise physics: operating-point-dependent factors and <code>ddt()</code> -shaped ($j\omega$) noise need a complex amplitude per source, $re + j\omega \cdot im$, so coherent same-named sources (LRM 4.6.4) sum with correct sign and frequency shape

Two additive flag conventions ride on the existing `eval()` interface (not version bumps):

- `EVAL_FLAG_IS_FINAL_STEP` (bit $1 \ll 21$) in the `eval input` flags — `@(final_step)` firing ([E-53](#));
- `eval return` flags for `$finish/$stop` requests and `$discontinuity`-driven step rejection ([E-55](#)).

Pairing consequence: this ngspice requires $OSDI \geq 0.7$ and rejects older objects with a clear version message; `.osdi` files from older compilers must be recompiled ([E-54](#)).

2. The OSDI runtime — `src/osdi/`

`osdi.h` — `PARAM_KIND_*` unsigned shifts

The parameter-kind flag macros were `PARAM_KIND_MASK` ($3 \ll 30$) and `PARAM_KIND_OPVAR` ($2 \ll 30$) — both shift into the sign bit of a signed int, which is undefined behavior (a UBSan build flags it on every OSDI load, at `osdiinit.c:41`). The shifts are now unsigned ($3u \ll 30$, $2u \ll 30$, and the two zero/one companions for consistency). The bit patterns are unchanged (`0xC0000000 / 0x80000000`), so the OSDI ABI is byte-identical, and they now match the u32 constants `openvaf-r` emits in `metadata/osdi_0_4.rs`. Found by the 2026-07 ASan/UBSan pass.

`osdiaccept.c` — new file ([E-55](#), [E-1](#), [E-6](#))

The accepted-timepoint hook (`DEVaccept`) OSDI previously lacked. It advances the `absdelay/last_crossing` waveform histories only at *accepted* points (so rejected timesteps don't corrupt the delay lines) and reads the per-attempt latched `$finish/$stop` request flags at the one boundary where honoring them is safe. Why: acting on `$stop` mid-Newton-iteration was treated as a step failure and ground the timestep into a rejection loop; `$finish` was ignored outright.

`osdidefs.h` (104 diff lines)

- `OsdiExtraInstData` grows the per-instance fields behind every runtime feature: `dt/temp` offsets with given-flags (instance temperature), `eval_flags`, `absdelay` waveform-history rows and pre-allocated KLU/SPARSE matrix-pointer sets for the delay-equation rows (re-pointed on every DC↔AC transition, mirroring the regular Jacobian handling), and the `last_crossing` history ([E-1](#), [E-6](#), [E-55](#)).
- `ALIGN` macro renamed `OSDI_ALIGN`: the macOS SDK's `<sys/param.h>` defines an unrelated `ALIGN`, producing a redefinition warning in every OSDI translation unit ([E-77](#)).

`osdiitf.h` / `osdiext.h`

The registry-entry structure gains the `absdelay/last_crossing` descriptor pointers, and `OSDIpendingRequests()` is exported so the analyses can ask "did any device request `$finish/$stop` at this accepted point?" ([E-1](#), [E-6](#), [E-55](#)). It also gains the terminal-short pointers and the `OsdiTermShortInfo` record ([E-401](#)).

osdiregistry.c (68 diff lines)

- Loads the `absdelay/last_crossing` descriptor arrays from each `.osdi` and threads them into the registry entries (E-1, E-6).
- Loads the terminal-short arrays the same way (E-401).
- Requires OSDI ≥ 0.7 (E-54).
- Descriptor-count hardening (E-228): `load_object_file` strided/indexed the descriptor arrays by the counts the `.osdi` itself declares (`OSDI_NUM_DESCRIPTOR`s, and per-descriptor `num_params/num_opvars/num_terminals/...`) with no bounds, so a corrupt, truncated, or ABI-mismatched object that still `dlopens` but declares an out-of-range count read past the array (the inner loop dereferences each `param_opvar` entry's `name[]` pointers) $\rightarrow$ SIGSEGV during `pre_osdi`. Two generous ceilings (`OSDI_MAX_DESCRIPTOR`s 4096, `OSDI_MAX_DESC_COUNT` $1 < < 20$, far above any real model) now reject such a file with a clean diagnostic — the outer count before it sizes the registry / strides the array, and each descriptor's own count fields in the loop before they index anything (a single load-time choke point protecting all downstream consumers). Targets the egregious/out-of-range count (the truncated / byte-corrupted / ABI-drift signature — an ABI mismatch reads a pointer's bits as a count $\rightarrow$ huge); a count that lies within the plausible range is an internally inconsistent binary indistinguishable from a valid one and is out of scope, as is a hostile object (a `.osdi` is native code whose loading executes it).

osdiinit.c (8 diff lines)

- Registers the `DEVaccept` hook (E-55).
- Publishes the synthetic instance-temperature parameter under both spellings, `dt` and the conventional `dtemp` every built-in device uses; `n1 a b mod dtemp=10` used to fail with *"unknown parameter"* while the underlying plumbing was exact (E-80).

osdiload.c (441 diff lines — the largest OSDI change)

- Stamps the `absdelay` synthetic rows (`y_synth`, `z`) for DC, AC and transient, driving the delay from the recorded history (E-1).
- Evaluates `last_crossing` from the recorded waveform with linear interpolation between the bracketing timepoints (E-6).
- Passes the final-step flag through to `eval()` (E-53).
- Latches `eval-return` flags per timepoint attempt (`point_eval_flags`, OR-ed across Newton iterations, reset per attempt) so `$finish/$stop` under solution-dependent conditions survive to the accepted-point boundary (E-55).
- Folds the stride-2 signed noise-power pairs (`fac · | fac | semantics`) when transferring `load_noise` results (E-54, E-42).

osdisetup.c (178 diff lines)

- Creates the synthetic delay-equation unknowns and matrix entries for `absdelay` (E-1).
- Applies the OSDI nodesets (`Osdinode.nodeset`) as solver initial guesses (E-45).
- Terminal shorts (E-401): `collapse_nodes` documents that terminals can never be collapsed (ngspice allocates them, not OSDI) and skips such a pair — which left $V(a,b) <+ 0$ between two terminals connecting NOTHING, a silent open circuit where the model wrote a short. The compiler now emits a real 0 V source there and lists the branch in `OSDI_TERM_SHORT_INFO`s; setup drops that branch's current unknown (`drop_node`, the same renumbering an ordinary collapse-to-ground does) whenever the two terminals do not resolve to two DISTINCT CONNECTED circuit nodes. Without that half the equation is redundant wherever the netlist already ties the terminals together — the normal wiring of a self-heating model — and a redundant 0 V source is a singular row.

- A `$fatal/$finish` raised during setup (models rejecting their configuration by design — HiSIM's `\$port_connected` guards) used to surface as ngspice's baffling *"impossible error — can't occur"*; it now reads *"a Verilog-A device rejected its configuration during setup"* (E-56).
- An internal OSDI node that appears in no Jacobian entry — a net structurally decoupled from the matrix, e.g. an explicit ground `gnd` reference whose branch contribution $V(p, gnd) \leftarrow \dots$ drops the `gnd` column — used to be allocated its own solver row. That all-zero row/column is benign under Sparse (it decouples to $V=0$) but makes the KLU matrix structurally singular, so KLU returned a wrong DC answer (groundcontrib: $v(p)=0$ not 1.5; hierbranch: branch currents 0). Such nodes are now tied to ground (node 0) at setup, matching how OpenVAF already treats them and fixing both solvers (E-116).

osdiacld.c (89 diff lines)

- AC loading gains the `ac_stim` right-hand-side injection: the partitioned AC-stimulus source array is loaded through `load_ac_stim` and added to the AC RHS with exact magnitude and phase, `$mfactor` applied linearly (deterministic signals scale with multiplicity) (E-51, E-26).
- Complex ($j\omega$ -shaped) noise coupling entries participate in the AC matrix (E-54).

osdinoise.c (93 diff lines)

- Per-instance grouping of same-named noise sources: coherent summation of complex amplitudes $(a + j\omega \cdot b) \cdot T$ before squaring, per LRM 4.6.4 — same-named sources correlate (including exact anti-phase cancellation), differently-named ones stay independent (E-42).
- Reads the stride-2 signed pairs of ABI 0.7 (E-54).

osditrunc.c (34 diff lines)

- Honors `$discontinuity(n)`'s next-step clamp: a negative `bound_step` sentinel from the model limits the step after the event (E-24).
- Honors the step-rejection return flag: when an event fired inside the step, `OSDItrunc` requests `delta/8` (with a $20 \cdot \text{CKTdelmin}$ termination floor) so the integrator bisection onto the discontinuity instead of extrapolating across it (E-55).

osdiparam.c, osdisetup.c, osdiregistry.c, osdiinit.c, osdidefs.h — the collapse-owner map

A terminal collapsed onto an internal node ($V(d, di) \leftarrow 0$, the `rd = 0` default path of essentially every compact model) reported a terminal current of exactly 0.0: the model writes its current into `di`'s residual and terminal `d`'s slot stays zero, while `OSDIask` read only `descr->nodes[t].resist_residual_off`. With `rd` and `rs` both zero every terminal read 0.0 and the reported currents summed to zero, so a Kirchhoff check passed on $0 = 0$. The solution was never affected — `osdiloadd.c` stamps the whole group into one matrix row, which is why nothing a user plots ever moved.

`OSDIask` now sums the terminal's whole collapse group, resistive residual plus (in a transient) the integrated reactive residual, striding the state cursor over non-members exactly as `osdiloadd.c` does. The grouping cannot be recovered after setup — `write_node_mapping` replaces the local mapping with global node numbers, and two terminals on one net share a global number while ground additionally collects grounded terminals, nodes collapsed to ground, E-116's decoupled internal nodes and E-401's dropped term-short flow nodes. It is therefore snapshotted the moment `collapse_nodes` returns, into one `uint32_t` per descriptor node trailing `OsdiExtraInstData(osdi_collapse_owner)`, holding `t + 1` for "owned by terminal `t`" so that the clobbered zero means "setup has not run" (E-416).

osdiparam.c, osdisetup.c, osdiinit.c, osdiregistry.c, osddefs.h, cktsens.c, breakp2.c — the collapse decided once, re-decided later

OSDItemp re-runs `setup_instance` on every temperature update, so a model can re-decide its node collapse — but nothing derived from the first decision can be rebuilt there (the node mapping, the matrix element pointers, the state layout, E-416's owner map). The file's own `// TODO` check that there are no changes in node collapse? is now answered: OSDItemp clears `collapsed[]`, re-decides, compares against a snapshot taken in `OSDIsetup`, and restores the snapshot unconditionally before the error check, because the matrix implements the snapshot. Clearing first is the whole fix — OpenVAF's collapse callback only ever STORES TRUE, so without it a collapse that stopped happening is invisible, which is the direction the defect takes. The snapshot is one `bool` per collapsible pair trailing E-416's owner map in the same instance block.

Three consumers acted on the mismatch. `cktsens.c` perturbed a collapse-selecting parameter and differenced two loads that shared one matrix element: the branch's `+g` and `-g` cancelled on the same diagonal and what survived was $\text{eps} * E / (Y * \text{delta}^2)$ — roundoff whose magnitude tracks $1/\text{rd}$ and whose sign moves with unrelated parameters ($2.5289\text{e-}02$ printed where the truth is $2.2676\text{e-}04$). It is now reported as 0 with a warning naming instance and parameter; it cannot be computed correctly, since `DEVsetup` runs only at the base value and re-running it would allocate nodes and trip `cktsens`'s own `CKTlastNode` guard. `.dc temp` ran a whole sweep on one topology — exactly 0 A sweeping into a collapse, a plausible 2x error sweeping out of one — and now warns once per instance; a hard error would mean propagating five bare `CKTtemp(ckt)`; returns in `dctrcurv.c` and changing error handling for every device model in the tree. `breakp2.c` stopped skipping the per-terminal expansion at exactly two terminals, where `@dev[i_p]` had been a length-1 scalar while `@dev[i_d]` was a waveform at three; the bare `@dev[i]` is kept beside it and synthesized names are de-duplicated against explicit `.save` entries (E-417).

osdisetup.c, outitf.c, com_measure2.c — three things nobody checked

`absdelay()` and `last_crossing()` are the only analog operators whose OUTPUT row the compiler deliberately leaves empty: `ngspice` fills it from its own history buffer, through matrix elements `OSDIsetup` allocates later. E-116's decoupled-node scan can only see coupling the COMPILER declared, so it found the output node in no Jacobian entry, tied it to ground, and `eval()` read `CKTrhs0ld[0]` -- exactly 0.0, forever. The value was right only when the model also CONTRIBUTED it, because that is what put the node into a Jacobian entry; `transition()` shares the same lowering and worked only because its output feeds the slew residual. Those nodes are now marked used before the scan decides. E-415 had looked for this in the compiler and reverted when the descriptor came out byte-identical -- correctly, since nothing was wrong there. No OpenVAF change and no ABI change: the descriptor already carries the infos and `osdiregistry.c` already parses them.

`.save` validated nothing. Nothing between `settrace` and the per-point read looked a name up, and `getSpecial`'s caller discards `INPaName`'s `E_NODEV/E_BADPARM`, so a misspelled device, a bogus parameter or an unexpanded wildcard produced a registered vector that stayed 0 long in silence. The check now calls `INPaName` itself -- the same routine the per-point read uses -- and a hierarchical spelling is REWRITTEN rather than rejected, so E-410's `@x1.r1[i]` resolves while the display name stays the user's. Unresolvable entries are warned about and still added, because dropping them would change what the plot contains.

`meas ...` when interpolated a crossing-free interval. The initialisation counted a crossing in the FIRST interval -- which runs from the operating point to the first timepoint and so straddles thresholds for reasons unrelated to the waveform -- and the evaluation, which deliberately starts one sample in, applied that leftover count to a later interval, dividing by a difference that was exactly zero or one ULP: $-inf$, $1.15292\text{e+}05$ s in a 3 us run, negative times. The count is no longer taken, every interpolation is gated on the interval actually bracketing the target (mirroring `measure_at`), and a `.dc` sweep restart resets to sample 0 so the previous sweep's last point is never a bracketing endpoint (E-418).

pz filled the simulator-stamped row nowhere. Because those rows are the simulator's to fill, every load path has to fill them -- `osdload.c` does for `dc/tran` and `osdiacld.c` for `ac`, but `osdipzld.c` did for neither, so the row was identically zero, the matrix was singular at every trial `s`, every trial looked like a root, and pz blamed the netlist: "the input signal is shorted on the way to the output". This predates the fix above for a model that CONTRIBUTES the value (its row was already live), and marking the nodes used would have widened it to every observed-only model. The AC stamp cannot be reused: `e^-s*t` overflows across pz's own search range and a transport delay has infinitely many roots, so there is no finite pole-zero set to find. `absdelay` is therefore stamped as the zero-delay wire -- the same linearization the operating point uses -- and warned about once per instance; `last_crossing` is stamped exactly, with no warning, its small-signal sensitivity being zero. All three cases now report the delay-free pole to the digit, including the two that aborted before E-418 (E-418).

osdi/Makefile.am

Adds `osdiaccept.c` to the build (E-55).

3. Analyses — `src/spicelib/analysis/`

`dctran.c` (46 diff lines)

- Advances OSDI delay/crossing histories via the accept path and skips history corruption on rejected steps (E-1, E-6).
- Issues the dedicated `@(final_step)` evaluation at the converged end of a successful transient (E-53).
- Honors accepted-point `$finish` (ends the analysis cleanly, firing `final_step` at the requesting point), `$stop` (pauses resumably), and aborts with a clear error on `$fatal`'s `E_PANIC` instead of retrying it as nonconvergence (E-55).

`dcop.c` (9) and `acan.c` (10)

- Fire `@(final_step)` at their successful end (an operating point fires both step events — a single point is first *and* last) (E-53).
- An AC job's operating-point phase carries the analysis name bit (only the name — adding the reactive `CALC_*` bits would wrongly enable integration at an op), so `analysis("ac")` holds through the whole AC run per LRM 4.6.1 and `@(initial_step("ac"))` fires in AC (E-53).

`dctrcurv.c` (150) + `include/ngspice/trcvdefs.h` (13)

- Generic `.dc @inst[param]` sweeps (a new sweep code): the DC sweep previously hardcoded `Vsource/Isource/Resistor/temperature`, so sweeping any other device parameter was a fatal error. The generic sweep resolves `@inst[param]` through the device's own parameter tables, refreshes per point via `DEVtemperature` (for OSDI exactly the `alter` path), nests with other sweep variables, and restores the original value afterwards (E-62).
- Fires `final_step` at the last sweep point; honors a `$finish` raised mid-sweep (E-53, E-55).

`noisean.c` (37)

- A singular AC matrix during noise analysis crashed ngspice with a SIGABRT: the code ignored `NIacIter`'s return and the adjoint solve asserted on the unfactored matrix. It now aborts cleanly ("AC solution failed at ... Hz") (E-56).
- Honors deferred `$finish`/`$stop` raised at the noise operating point and fires `final_step` on success (E-55, E-53).

span.c (31) + **cktspdum.c** (14)

- 1-port S-parameter analyses enabled: the "we need at least two ports" error was over-strict (a 1-port is a plain reflection measurement; the matrix machinery is N-general) — and it was hiding the cadjoint crash fixed in **dense.c** (E-64).
- **Rbase** published into the **sp** plot from port 1's **z0**, making the Touchstone writers work with no manual `let Rbase = 50` incantation (E-64).
- Stock NaN fixed in **donoise** noise-parameter extraction: the uncorrelated noise conductance G_u is analytically zero for fully-correlated single-source topologies, and floating-point rounding could land $\sqrt{Y_{cor}^2 + G_u/R_n}$'s argument at $-10^{-18} \rightarrow \text{NaN}$ for the noise figure. Clamped to the physical range ≥ 0 — found by parity testing against an OSDI resistor that produced the exact textbook NF where the built-in returned NaN (E-63).

cktdisto.c (16)

- **.disto** silently reported zero distortion for OSDI devices — the distortion kernel needs higher-order Taylor coefficients the OSDI ABI cannot provide (first derivatives only), and such devices were skipped without a word. A prominent warning now names each affected OSDI device type (E-62).

4. Frontend — **src/frontend/**

plotting/pyplot.c (new) + **com_pyplot.c** (new) + **plotit.c**, **commands.c** (E-94)

A new interactive command, **pyplot**, plots simulated vectors with matplotlib — a Python counterpart to gnuplot. **ft_pyplot()** (**plotting/pyplot.c**) mirrors **ft_gnuplot()**: it writes a `<file>.data` table and a `<file>.py` matplotlib script and **system()**s **python3** (synchronously for a PNG, backgrounded for an interactive window). A new "pyplot" device arm in **plotit()** routes to it, so it accepts the same plot expressions as **plot/gnuplot**; the **com_pyplot()** wrapper mirrors **com_gnuplot()**, and **pyplot** is registered in both command tables. Two set variables tune it: **pyplot_terminal=png** renders headless (Agg) to a PNG, and **pyplot_python** picks the interpreter (default **python3**). Purely additive (E-94). The output file base name is optional (E-95): **com_pyplot()** treats the first word as a file name only when it is not a plot expression (no `(`, and not a vector by **vec_get()**), otherwise it defaults to **pyplot** — so **pyplot v(out)** plots directly (the command's minimum argument count was lowered from 2 to 1). **ft_pyplot()** also grew stacked subplots (**set pyplot_subplots=N** $\rightarrow$ `plt.subplots(nrows, 1, sharex=True)`, **N** traces per panel; **vs** stays the x-axis, so it is not a panel separator) and matplotlib style sheets (**set pyplot_style=<name>**, **dark** $\rightarrow$ **dark_background**, guarded) (E-98). The headless **pyplot_terminal** was extended from PNG-only to the vector export formats **svg** and **pdf** (**set pyplot_terminal=svg/pdf** $\rightarrow$ `fig.savefig(<file>.<fmt>)`, matplotlib picking the writer from the extension), and a figure size control was added (**set pyplot_figsize="W,H"** $\rightarrow$ `plt.subplots(..., figsize=(W, H))`; quote the value so ngspice keeps the comma) (E-99). A **-eye** flag turns **pyplot** into a one-line eye-diagram renderer over the E-207 eye analysis: **com_pyplot()** detects the **-eye** marker, runs **com_eye()** on the trailing tokens (which folds the waveform and leaves **eye_wave/eye_t** + the scalar metrics in a fresh current eye plot), then a new **ft_pyplot_eye()** reads those back and writes a **hist2d**-based persistence eye script (**LogNorm**, **turbo**, empty cells masked; **threshold** + eye-centre lines, eye-height/width double-arrows, a metric title), launched by the same **system(python ...)** mechanism as **ft_pyplot()** and honouring the same **pyplot_terminal/pyplot_python/pyplot_backend/pyplot_figsize/pyplot_style** settings. **ft_pyplot()** and the **gnuplot/asciiplot** back-ends are untouched (E-208).

Four further groups of additions. Appearance controls — seven set variables (**pyplot_grid**, **pyplot_legend**, **pyplot_markers**, **pyplot_axhline**, **pyplot_axvline**, **pyplot_dpi**, **pyplot_transparent**) threaded into **ft_pyplot()**, all additive so a control-free plot is byte-identical to before (E-296). A **-fft** mode — **ft_pyplot()**'s **bool hist** became an **int mode** (**LINE/HIST/FFT**); **-fft** reuses the time-

domain data table and the emitted script resamples onto a uniform grid (`np.interp`, because transient data is adaptively sampled) before `np.fft.rfft`, scaled by `2/sum(window)` so a tone reads back its amplitude (`pyplot_fft_window/_db/_points/_logf`) (E-297). Complex-aware AC views — a new `ft_pyplot_ac()` renderer for `-bode` (mag dB / phase deg vs log-f, stacked, `np.unwrap`), `-nyquist` (imag vs real) and `-polar` (mag at phase), emitting a `<vec-index> <freq> <re> <im>` table that KEEPS the imaginary part an ordinary `pyplot` discards (E-298). And finishing touches — cross-run overlay (`pyplot_tran1.v(out) tran2.v(out)`) now sizes the data table by the longest scale so a finer second run is not truncated, plus a `pyplot_cursor` hover crosshair (matplotlib's built-in `Cursor`, window-only) (E-299). A `pyplot_mplcursors` variable then selects the `mplcursors` package (data cursors that snap to a trace) as the cursor backend, turning the cursor on by itself, with a try/except fallback to the built-in `Cursor` if the package is absent where the emitted script runs (E-300). The cursor gating was then consolidated so `pyplot_cursor` is the single master switch (off by default, the only enable); `pyplot_mplcursors` only selects the backend when the cursor is on (E-301).

postcoms.c (414 diff lines) + **postcoms.h** + **commands.c** (16)

The Touchstone I/O subsystem (E-64, E-72):

- **wrsnp** (with `wrs2p` dispatching to the same handler): Touchstone v1 for any port count — the classic 2-port S11 S21 S12 S22 column order preserved byte-identically, $N \geq 3$ in the spec's row-major layout with at most four complex pairs per line;
- writer options `wrsnp <file> [r|m|db] [s|y|z] [hz|khz|mhz|ghz]`, any combination in any order: MA/DB formats, Y/Z parameters (normalized to Rbase, $Y \cdot R / Z/R$, per the v1 spec), frequency units — the option line reflects every choice;
- **rdnsnp**: reads any Touchstone v1 file into a new plot with a Hz frequency scale and complex vectors matching the `.sp` plot's conventions (MA/DB converted back, Y/Z de-normalized, the 2-port column order handled, Rbase published so imports round-trip) — measured data diffs against simulation in one `let` expression.

Why: E-63 showed the S-parameter *analysis* was exact but its results could not leave ngspice in the industry-standard format (`wrs2p` demanded a never-created Rbase vector), and nothing could bring VNA data in.

outitf.c (16)

- Integer operating-point variables recorded per point: `getSpecial` masked with `IF_REAL` only, so an integer opvar saved with `.save @n1[flag]` produced garbage in transient vectors (E-32).
- The plot-memory guard's message printed `%Id` — a Windows-only `printf` length modifier that is undefined behavior elsewhere and rendered the message as the numberless "memory required (Id Bytes)". Now `%zu`: real byte counts on every platform (E-77).

resource.c (27)

`ft_ckspace()` — the "approaching max data size" warning (`RSS > 95%` of `RSS + free RAM`) — now honors `set no_mem_check` (the same opt-out the fatal output-size estimate in `outitf.c` respects; one variable governs both memory guards) and warns once per excursion above the threshold, re-arming below 90%, instead of repeating on every check through a long analysis (E-81).

display.c (1)

`#undef NODEV` before the local macro definition — the macOS SDK's `<sys/param.h>` defines an unrelated `NODEV` (E-77).

inpcom.c, parse.c, vectors.c, device.c + analysis/dctrcurv.c — bracketed names (E-408)

Three defects on the path from a name written in a deck to the object it denotes, none of which reported anything.

A leading zero split a node in two. `n[1]`, `n[01]` and `n[001]` were three distinct nodes — `print all` listed all three — while the vector lookup canonicalised the index before resolving it, so all three probes returned the `n[1]` value. With a single node the same mismatch is worse: a netlist that built `m[02]` was unreachable by every spelling, its own included, because the lookup only ever asked for `m[2]`. Enhancement-221's own header settles the question (*"the scalar names produced use the same bracket form, so a bus `a[0:1]` and an explicit `a[0]` denote the same node"*), so `inp_canon_bus_index_token()` canonicalises the index in the netlist, negative indices included; it is a byte-identical no-op when the spelling is already canonical.

@dev[param] could not name a bracketed parameter. `show` lists a bus device's terminal currents `i_a[0]..i_a[3]` (E-394) and its array parameter elements `ap[0]..ap[2]` with correct values, and the instance line can set them, but four separate splitters each cut the name at the inner `]` and reported *"no such parameter `i_a[0]`".* All four had to change because each serves a different command: `PPlex` (`parse.c`) lexes the token at all, `vec_get` (`vectors.c`) serves `print` and `let`, `com_alter` (`device.c`) serves `alter`, and `DCTfindInstParam` (`dctrcurv.c`) serves E-62's `.dc @inst[param] sweep`, which had been aborting with a fatal error. The E-269 wildcard alias `@*[param]` works *because* of the old first-`]` split, so a name beginning with `[` keeps it deliberately.

A bus range expanded everywhere except where the answer is read. `[lo:hi]` expanded on instance lines, R/C node lists, subcircuit calls and port lists, but not on `.save/.print/.plot` (the card produced nothing) or `.ic/.nodeset` (warned and ignored). Those cards were skipped by design: `inp_expand_buses` admits lines carrying bare node fields, and an output card writes `v(a[0:3])`. `inp_bus_rewrite_wrapped()` emits one copy of the whole token per index, so the wrapper and any `=value` ride along; a spaced `.ic v(a[0:1]) = 0.25` absorbs its value into the token first. A token naming two ranges has no unambiguous expansion and stays literal.

com_sweep.c + spiceif.c — a wildcard sweep knob is put back (E-409)

`sweep @*[wavelength_nm] ...` ran correctly and then printed

```
Error: no such device or model name
PError: syntax error in line segment
    @*[wavelength_nm]
near
    wavelength_nm]
```

The name looks mangled, which is what made the message alarming, but nothing was wrong with the parameter: the expression lexer's specials set contains `*`, so `@*[p]` cannot lex as one token and the parser reports the leftover text.

The message was the harmless part. `sw_read_knob()` is how E-385 captures the nominal it puts back at the end of a sweep, and it parses the knob name as an expression — the call that cannot succeed for a wildcard. The read reported failure, and E-385's deliberate all-or-nothing rule then skipped restoring everything: a wildcard sweep left every target at its last swept value, and with `-vs` it took a concrete co-knob down with it (`@dev1[wavelength]` left at 3.0 although it restores perfectly on its own).

One number could not fix it. A wildcard sets every matching model or instance to one value, but the values it overwrites can all differ, so undoing it needs one reading per target. `if_saveparam_wildcard()` / `if_restoreparam_wildcard()` walk exactly the same targets in exactly the same order as the existing `if_setparam_wildcard{,_instance}` — same device-type loop, same model and instance chains, same `paramlookup(..., inout=set)` predicate — so index *i* of the saved array names the same target on the way out as on the way in, reading through the parameter's askable twin and the same `doask/doset`

pair a plain alter/altermod uses. Saving stays all-or-nothing (a set-only or vector-valued parameter refuses the whole capture), and the replay is pinned to the circuit the readings came from, so a .param co-knob that re-sources the deck cannot have them replayed onto a different one.

spiceif.c, dctrcurv.c, com_sweep.c, gens.c — a subcircuit device without its type letter ([E-410](#))

A resistor inside a subcircuit is @r.x1.r1[resistance], while the node beside it is plain x1.m. The obvious spelling @x1.r1[resistance] named nothing.

The letter is not an accessor convention — r.x1.r1 is the device's real name. ngspice flattens a subcircuit by rewriting its cards back into the deck and re-parsing them as ordinary element lines, and the parser takes the device type from the FIRST CHARACTER of the card (inppas2.c), so the flattened refdes has to keep a type letter in front: the card x1.r1 a m 1k would otherwise be re-read as another subcircuit call. Two details confirm it: translate_inst_name() (subckt.c) exempts x devices, whose names already start with the right letter, so x2 inside x1 is plain x1.x2; and NODES, having no type, keep plain hierarchical paths.

The reconstruction needs no search. The letter prepended is literally the leaf name's own first character, and a device's name must already begin with its type letter, so x1.r1 can only mean r.x1.r1 — at any depth, and for OSDI devices alike (x1.x2.nd1 -> n.x1.x2.nd1). if_find_instance_hier() is consulted only after the exact lookups fail, so every name that resolves today is unaffected.

Following [E-408](#)'s lesson, every consumer is routed through it rather than just print: finddev() (print/let/alter/altermod), finddev_special(), DCTfindInstParam() for [E-62](#)'s .dc @inst[param] sweeps, and sw_fp_bind() for the sweep knob. For sweep the old behaviour was not even an error — the knob silently failed to bind and the three points came out identical.

show needed a different answer: its grammar takes the query's first character as the device-TYPE selector and uses :/#, not ., as the subcircuit delimiter, so x1.r1 parsed as type x and could never match a resistor. dgen_hier_match() is consulted alongside that grammar as a whole-word alternative and can only add a match. A first attempt widened the strcmp inside the decomposition instead; it was measured as DEAD CODE (the query bails at the type check long before) and reverted.

an option matched by substring, and three called unknown ([E-451](#))

Two findings in option handling, both from asking the question [E-450](#) had just answered for savecurrents: which OTHER options are decided by searching the line for a word?

inp.c -- **seed=** and **cshunt=** were found by substring. eval_opt() located them with a bare strstr over the option line, so any option whose NAME MERELY ENDED in the watched text was taken as that option:

.options seed=7	sunif -> 4.3854057230e-01	(correct)
.options myseed=7	4.3854057230e-01	identical
.options noseed=7	4.3854057230e-01	the spelling that reads "off"
.options xseed=7	4.3854057230e-01	
baseline, no seed set:	6.8709117128e-01	

and for cshunt the answer moves by SIX ORDERS OF MAGNITUDE -- a node reading 1.0000000000e+00 reads 6.9209970972e-07 under .options nocshunt=1e-6, and the same for mycshunt= and xcshunt=. Each of these ALSO prints Warning: unknown option 'nocshunt', which makes it worse rather than better: the user is told the option was not recognised, and it changes the answer anyway. For a Monte Carlo run the seed case means a mistyped option name silently changes the random sequence while being reported as ignored. Matched as whole tokens now, on [E-450](#)'s boundary rule; seedinfo too, so noseedinfo no longer switches it on and seed= no longer sees the seed inside it.

NOT a defect, though it reads like one: the loop is for (card = deck; ...) with no .options test, which looks as though any deck line carrying seed= would hijack the generator. It does not -- the function's own comment says *"Input is the option deck (already sorted for .option)"*, and measurement agrees: a .param seed=7, a model parameter named seed, and a comment containing seed= all leave the sequence at baseline.

spiceif.c -- three options took effect while being reported unknown. Asking which flagged names demonstrably change a run: scale=2 doubles @m1[w] (1e-6 -> 2e-6), rseries=100 moves a transient, and **autostop** truncates one from 567 rows to 2 while being reported as a name ngspice does not know. That is the case [E-447](#) fixed for savecurrents, seed and numdgt, with three names it did not cover, and its reasoning applies unchanged -- a warning that fires on a setting the run then honours teaches the user to ignore the check [E-438](#) added. scale and rseries are read out of the deck's own option cards, so like seed they never reach the .options parameter table the check consults. **sca1m** is flagged too and is deliberately NOT registered: it could not be shown to change anything, and this list is for options that demonstrably take effect.

savecurrents could be requested but never declined ([E-450](#))

inp.c -- a bare substring search decided the option. Whether .options savecurrents was in force came down to strstr(options->line, "savecurrents"), so EVERY option line merely CONTAINING the word switched it on, whatever the line actually said:

.options savecurrents	ON	(correct)
.options savecurrents=0	ON	<- says off, does on
.options savecurrents=false	ON	
.options nosavecurrents	ON	<- ngspice's own no<option> convention
.options mysavecurrentsxyz	ON	<- any identifier containing the word

nosavecurrents is the one that matters, because it is not an invented spelling: noacct, noinit, nomod and nopage are all real options, so a no prefix is exactly what a user reaches for. Once on there was no way back -- the feature could be requested but never declined.

Silent in every case, because nothing goes wrong when it fires: a deck that quietly saves every terminal current still simulates correctly, it just carries vectors the user asked not to have. On a large deck that is the difference between a small rawfile and a very large one, and on an OSDI device it is one vector per terminal -- which is how it surfaced, a user noticing i_<terminal> vectors they had not asked for.

The option is also reachable ONLY from an option CARD -- the deck, an .included file or a .lib section -- and not from spinit, .spiceinit, set or a .control option, because inp_savecurrents() is a netlist pre-pass run before the shell-variable machinery matters. That is the same "too late" ordering [E-411](#) recorded and the reason [E-449](#) reads its own option from the option lists. It narrows where the wrong answer can come from, and makes it stickier once there.

The line is now read as TOKENS: a token belongs to the family when it is exactly savecurrents or begins savecurrents_ (the declared variants being savecurrents_bsim3, savecurrents_bsim4, savecurrents_mos1), so an unrelated identifier no longer matches; a no prefix or a 0/false/no/off value turns it off; the later card wins, as a repeated .param or .options value already does. WHICH card is returned is deliberately unchanged -- the caller re-searches that one line for the _bsim3/_bsim4/_mos1 names to choose the MOS current set, so the FIRST enabling card is returned exactly as the old first-match did, and splitting the family across two cards behaves as it always has ([E-399](#): no wider than the evidence).

autobus across a subcircuit boundary ([E-449](#))

[E-444](#) lets one node name stand for a whole Verilog-A bus port, and does it in INP2N because that is where the model -- and so the bus width -- is known. Inside a .subckt that is too late.

subckt.c, inp.c -- the bus bits went to nodes that went nowhere. A definition declaring `a[0:4]` has the five formals `a[0]..a[4]`, so a device line writing the bare `a` matched none of them. Flattening turned it into the local node `x1.a` and `INP2N` expanded THAT, leaving `x1.a[0]..x1.a[4]` -- five floating nodes with the device wired to them instead of to the ports, the five bits all reading one identical voltage and the device contributing nothing.

Nothing was reported, because every terminal DID receive a node, so [E-402](#)'s under-connected warning had nothing to say. Turning the option ON therefore REMOVED the diagnostic the same deck gets with it off -- `autobus OFF` warns "4 of the 6 terminals ... are not connected", `autobus ON` says nothing at all and `rc=0`. That is exactly the shape [E-445](#) recorded when `autobus` indexed ground: switching a feature on removing an existing warning.

The fix is at the SUBSTITUTION, not at device parse. A token that is not a formal, but for which formals token[i] exist, expands to those formals' actuals; it needs no model, only the `.subckt` line, so it works at flattening time. The option is not reachable from there in the obvious way: `cp_getvar("autobus")` is false during flattening, because `inp_subcktexpend()` runs before `inp_dodeck()` publishes the options, AND the `.option` cards are no longer in the deck either, having been split into the option lists -- a deck scan finds nothing. `inp.c` reads them from those lists and passes the answer in, the same way it already reads `scale` a few lines above the expansion call. Restricted to OSDI instance lines, so a bare `a` on an R/C/L line inside the same subcircuit stays the ordinary local node it always was.

Bit order comes from the declaration, not the table. The bits are emitted by ASCENDING BIT INDEX, because a compiled model always orders a bus port's terminals that way whatever direction the Verilog-A declared -- `inout [4:0] a` still yields terminals `a[0]..a[4]`, measured rather than assumed -- and the instance line is positional. A DESCENDING `.subckt` declaration therefore binds in reverse, which is [E-411](#)'s rule (the written order decides) applied one level up; a model declared `[4:0]` still matches its own flat form.

The other subcircuit form was already correct and is unchanged. `.subckt mysub a b` declares two SCALAR formals, so both tokens on the device line ARE formals, flattening substitutes them normally, and `INP2N` expands the caller's own names: the subcircuit has two pins and the bits live in the caller's scope as `p[0]`, `p[1]`, `q[0]`... That was verified here against a fully-written-out reference for one instance and for two instances with different actuals. The two forms are different interfaces -- one passes a bus through as a unit, the other exposes one pin per bit -- and both are pinned.

Out of scope, deliberately: writing ONE BIT where the whole bus is expected (`N1 a[0] b` inside a subcircuit whose port is `a[0:4]`) still creates floating `n0[0]..n0[4]` silently. `a[0]` is a formal, so flattening resolves it to the actual `n0`, and by `INP2N` nothing distinguishes it from a legitimate top-level line whose bus base happens to be called `n0`. It is BYTE-IDENTICAL before and after this change; [E-445](#) guarded the two spellings it could recognise, this one is not recognisable at either end, and a heuristic for it belongs in its own investigation ([E-399](#): a fix must be no wider than its evidence). Bus shorthand on a subcircuit INSTANCE line (`Xi a b inner`) is likewise unsupported, but fails loudly with "too few nodes" rather than mis-simulating.

a node named after a constant ([E-448](#))

`ngspice` keeps a permanent `const` plot of twelve vectors — `c`, `e`, `i`, `pi`, `kelvin`, `boltz`, `echarge`, `planck`, `TRUE`, `FALSE`, `yes`, `no` — and `vec_get()` falls back to it when the current plot has no such vector. That fallback is how a bare `pi` resolves and it stays. It also answered a node of that name with the constant, in three distinct ways.

evaluate.c — `v(X)` answered with the speed of light. `v(X)` asks for a node voltage and a constant is never one, but the name was resolved before the context was considered, so a node that had been renamed or mistyped was answered rather than refused: `v(c)` gave `2.9979245800e+08` where `v(nosuch)` correctly gave "no such vector". The other eleven names are more dangerous than the speed of light because they look like results — `v(no)` is a flat `0.0`, and `v(i)`, `v(yes)` and `v(TRUE)` a flat `1.0`, values

indistinguishable from a grounded node or a real unity signal. It defeated an existing guard: [E-431](#) refuses a sweep -output expression that never resolves, and that refusal works — but the constant walked through it because the name *did* resolve, so -output `vo=v(c)` was drawn as a flat three-point curve while -output `vo=v(nosuch)` was refused. In a sweep the output IS a curve plotted against the knob, so a flat line reads as a legitimate finding. A node context no longer accepts a constant-plot binding. The related forms `v(c,0)`, `vm/vp/vr/vi/vdb(c)`, `i(c)` and `mag(v(c))` were checked and already refused — `v` is the only entry in the function table with a null handler, which is why it alone had the hole.

evaluate.c, parse.c — a bus bit named `c[0]` was unreachable. `.option autobus` ([E-444](#)) builds the node names `c[0]..c[4]` for a Verilog-A bus port called `c`. Those nodes were created correctly and the circuit solved correctly around them, yet none could be named: `display` listed `c[0]` and `print` all printed it as `5.0000000000e-01`, while `print c[0]` answered "indexing a scalar (c)" and `print v(c[0])` answered "bad v() syntax". [E-444](#)'s own five-bit ladder, with the bus renamed from `a` to `c`, printed nothing at all. [E-224](#) already preferred the literal name `a[0]` over "index 0 of a vector `a`", but only when the base was an UNRESOLVED zero-length placeholder — and for all twelve constant names a vector called `c` always exists, so the literal path was never tried. The gate is now the literal vector's own existence.

The silent one needs no constant. The same gate failed whenever the base resolved to a LONGER vector, which in a transient run is every node. A circuit holding both a scalar node `q` and a bus bit `q[0]` answered `print q[0]` with `7.5000000000e-01` — a scalar, node `q`'s value at `t=0` — instead of the bus bit's 59-row waveform, with no diagnostic. In an `op` run the same expression errors instead, because every vector is length-1, so the failure mode flipped with the analysis type. This is what argues the fix is about bracketed names generally rather than about the constant plot.

vectors.c — a bare name still resolves to the constant, but not in silence. With the two above fixed, `v(c)` and `c[0]` can no longer reach the fallback at all; what remains is a BARE `c`, genuinely ambiguous since the constant plot exists precisely so bare `pi` works. It still resolves to the constant and now says so, but only when the name also exists as a vector somewhere else — so every ordinary use of a constant stays silent.

One rule, four places. The bracketed-name rule was re-derived in `checkvalid()` and `PP_mknode()` in `parse.c` and in `op_ind()` and `apply_func()` in `evaluate.c`, each with the same too-narrow gate. [E-408](#) recorded what happens when copies of one rule drift apart. It is now a single `e448_literal_index()`, declared in `evaluate.h` so `parse.c` shares it. The three places that turn a double index into the `[%d]` text -- the helper, `op_ind()`'s ambiguity warning and `apply_func()`'s reconstruction for the `v()` form -- round through [E-274](#)'s `idx_floor()` rather than a bare `(int)` cast, which is undefined behaviour for `1e308/inf/NaN`. The helper and `apply_func()` build the SAME name, one to validate it and one to resolve it, so rounding them differently would mean the name checked is not the name looked up. Sixteen index expressions give identical values and exit codes before and after.

A deliberate change of meaning: where BOTH a vector `x` and a vector literally named `x[0]` exist, `x[0]` now means the literal one. That is the only case where a working expression changes meaning, and the reading it replaces is the silent-wrong-answer case above; indexing there needs a temporary and the warning says so. Ordinary indexing is untouched — nothing is named `myvec[3]` — and [E-433](#)'s quoting still works.

One change made and withdrawn. A `checkvalid()` hook was added so `v(c)` would report once rather than once per sweep point, on the evidence that a 21-point sweep printed 22 errors where -output `v(nosuch)` printed one. That evidence was an artifact of the probe: the two paths emit different wordings ("no such vector" vs "is not available or has zero length") and the filter matched only the first. Counted properly both emit 23 lines — pre-existing sweep behaviour for any unresolved output. Reverted.

the guard that covered one spelling of a bad value ([E-447](#))

Ten degenerate or invalid inputs accepted in silence. What makes them one enhancement is that in every case a working guard already sat a few lines away, covering a DIFFERENT spelling of the same mistake.

cktsopt.c — a negative **gmin** wrecked the operating point in silence. A diode's 0.63 V drop became -0.001 V at **gmin=-1** and -1e-9 at -1e6, **rc=0** with nothing printed, and the error grows smoothly so there is no threshold at which it announces itself. Every sibling was already guarded -- **reltol=0**, **abstol=-1**, **vntol=-1**, **chgtol=-1**, **trtol=-1**, **maxord=0/99** and **temp=-300** all warn, most from [E-426](#). **gmin** and **gshunt** were the two holes. Zero stays legal ("no **gmin**").

restemp.c — **scale=0** silently shorted a resistor. A resistance WRITTEN as 0 warned and was clamped, while the same zero reached through **scale=0** was silent AND unclamped (**v(nb)** read exactly 0.0), and **scale=-1** surfaced only as a downstream "singular matrix". **scale** multiplies the instance's own resistance, so a non-positive one is a dead short or an active element; it falls back to 1 with a warning now.

vsrpar.c, **isrpar.c** — **trrandom** with a TYPE outside 1..4 was a silently DEAD source. 0, 5, 9, -1 and 100 all gave rms 0.0 with **rc=0**; 1-4 give proper noise. A typo'd type number removed the stimulus from the circuit without a word.

diosetup.c — an invalid diode **level** was accepted. **level=2** -- a level the model knows about but does not implement -- was refused with a Fatal error, while **level=99** and **level=-1** were accepted silently and behaved as level 1. The check tested one specific value instead of a range.

cktsopt.c — **cshunt** set from **.control** was silently ignored. The option works by adding a capacitor to every voltage node at setup: **eval_opt()** scans the deck's own **.option** cards and publishes **cshunt_value** for **INPpas4()**. The **.control** path stores a value nothing reads. The card form is BIT-IDENTICAL to writing the capacitor by hand (6.9209970943e-07), and this was the only card-only option of nineteen tested. Behaviour kept, silence removed -- and the notice fires only when no card supplied the value, so correct decks stay quiet.

vsrask.c, **isrask.c** — **show** claimed a source had EVERY waveform. All eight share one coefficient array and all eight queries answered from it, so a source declared only **sin(0 2 3k)** reported 0/2/3000 EIGHT times -- under **sin** and under each of **pulse**, **exp**, **pwl**, **sffm**, **am**, **trnoise**, **trandom**. A dc-only source correctly showed -- for all eight, so the code separated "none" from "some" but not WHICH. Inactive waveforms now report as EMPTY rather than as an error, so **show** renders a bare -- instead of a column of **<<NAN, error>>**.

spiceif.c — three real **.options** keywords called unknown. **savecurrents**, **seed** and **numdgt**; **savecurrents** demonstrably works in the very run that calls it unknown (without it **@r1[i]** is a scalar, with it a transient waveform). E-446 downgraded this class because 176 of 179 **cp_getvar** names are shell variables -- that reasoning does not cover these three.

commands.c, **isrc.c**/**isrpar.c** — two diagnostics that named the wrong thing. **snload**'s help said one file while the command required two, so the documented form answered "too few args". And **pwl (... r=)** on a current source failed with the generic "unknown parameter (r)"; **r/td** are voltage-source-only (ISRC's **pwl** evaluator is a different, older implementation), so they are now declared purely to refuse them with the reason. Scope: a precise diagnostic, NOT feature parity -- porting repeat/delay means replacing ISRC's **pwl** evaluator and belongs in its own enhancement.

THREE reported defects were NOT. (a) **m=0** — E-426 examined exactly this and left it silent on purpose: it is the "disable this instance" idiom, its comment says so AT THE SITE and its suite asserts the silence. (b) **@r[resistance]** reporting the nominal value — E-426 settled that convention too and documents **1/@r1[conductance]** as the effective route (1200 ohm at 227 C in its own suite); **@c** and **@l** DO fold temperature in, so the three devices disagree, but that is a settled convention rather than a defect to flip

inside a set of guard fixes. Both were caught by E-426's suite failing in regression -- the mechanism working. (c) A misspelled option `method=bogus` looked silently ignored; ngspice prints "setAnalysisParm(options) ci_curOpt: unsupported integration method" at the end of the run, and the pre-E-447 binary prints it too. My probe's message filter missed it. Unchanged.

what the netlist wrote, quietly discarded (E-446)

Six places where something written explicitly in the deck was thrown away or reinterpreted with nothing printed. Three share one root cause.

ZERO USED AS THE "ARGUMENT NOT SUPPLIED" SENTINEL. `vsrclload.c` and `isrclload.c` decided whether an optional waveform argument had been given by testing `coeffs[n] != 0.0`, so a legitimately written 0 was indistinguishable from an omitted argument.

`vsrclload.c`, `isrclload.c` — an explicit **`TD1=0`** on an EXP source became the timestep. `V1 a 0 exp(0 1 0 1m 5m 1m)` starts one step late; every printed row carried the exact analytic value of the PREVIOUS timepoint to ten decimals. Because the substituted value IS the timestep, the answer depended on the timestep -- err -3.7e-03 at a 10 us step, -4.0e-02 (~4 %) at 100 us -- and writing `TD1` as `1e-30` instead of 0 made it exact. Both source types affected; SIN/PULSE/PWL/SFFM were exact throughout, which pinned it to EXP. Delays are now selected on whether the argument was SUPPLIED; the time constants keep a zero guard because they are divisors; omitted-argument defaults are untouched.

`isrclload.c`, `vsrclload.c` — PULSE **`PW=0`** and **`PER=0`** meant different things to a V and an I source. With `PW=0` the voltage source emitted no pulse and the current source one that never ended; with `PER=0` they disagreed the other way. A zero pulse WIDTH is now the degenerate pulse it asks for and a zero PERIOD means "do not repeat", on both. A zero rise/fall time still falls back to the timestep (documented SPICE, and what `vsrcl` already did).

`vsrclpar.c`, `isrclpar.c` — a PWL list with an odd token count. One point is left without a value, and a value was invented -- differently by each source: `pwl(0 0 1m 1 2m)` on a V source was BYTE-IDENTICAL to `pwl(0 0 1m 1 2m 0)`, ramping the stimulus to zero at a time never assigned one, while the I source ate the stray token AS a value and held it. Two different guesses at one ambiguous input, so it is refused. The neighbouring checks already warned about non-increasing and duplicate times; this was the gap between them.

`inp2dot.c` — a third `.dc` sweep source was neither run nor refused. `.dc V1 0 2 1 V2 0 2 1 V3 0 2 1` produced 9 rows, not 27, with `V3` pinned at its DC value. Proven by making `V3` dominant (`R3 = 1 ohm` against `R1=R2=1k`) and watching every value stay in the 1e-3 range. Fixed with it: surplus `.ac` arguments were dropped in silence -- any number, numeric or not, gave byte-identical output -- while `.tran` and `.dc` both refuse what they cannot use.

`ptfuncs.c` — the sign of a negative base. The default evaluated `pow(fabs(x), y)`: `(-2)**3` came out +8 and `(-2)**1` came out +2, so odd exponents were wrong and even ones coincided. Everything else disagreed -- `pwr(-2, 3)` is -8, a Verilog-A model's own `pow(-2, 3)` is -8, and both the LTspice and HSPICE branches preserve the sign; only PSPICE mode agreed, where `|x|^y` is that dialect's documented PWR convention. E-399's rule applies: an expression must not mean different things depending on whether the netlist or the model computed it. A negative base with an INTEGER exponent now carries its sign; with a non-integer exponent there is no real result and the historical magnitude is kept rather than a NaN, which would poison the Jacobian (E-256/E-440's reasoning). Positive bases untouched.

`capask.c` — **`@c[capacitance]`** folded in `m=` while **`@r[resistance]`** reported the written value, so one accessor idiom meant two things by device type. The capacitor now reports what its instance was given; `capload.c` applies `m` to the matrix stamps itself, so the simulation does not move.

One reported defect was NOT one. An AM source with a 5th argument emits a constant -- correct, because ngspice's AM is `AM(V0 VMO VMA FM FC TD PHASEM PHASEC)` and the 5th argument is the

CARRIER FREQUENCY, so `am(1 0 100 1k 0)` sets `FC=0` and leaves the offset. Against the right argument order AM matches analytic to $1e-12$. Unchanged.

nine silent failures made loud ([E-445](#))

Unrelated in mechanism, identical in shape: the simulator completed, returned 0 and printed nothing. Each fix is paired below with the sibling that was always handled correctly — that pairing is the argument, because it shows the guard already existed nearby and one path was missing it.

fourier.c — a bare `.four` card segfaulted. `dotcards.c` warns “no nodes given” and declines to save anything, but does not DROP the card, so the empty command still reached `fourier()`, which read the fundamental frequency straight off a NULL wordlist. `.four` was the only one of 25 zero-argument dot-cards that died; `.print` and `.plot` are handled a few lines above it, print the identical message, and were always safe. Needed transient results to exist, and survived subcircuit flattening.

fourier.c — an overflowing fundamental was accepted. `.four 1e400` overflows to +INF and produced a full report: THD 201.971 % with every harmonic frequency printed `inf`, against $1.91095e-11$ % for the valid case. The surrounding validation was already thorough — 0, -1000, abc, a literal `inf`, a literal `nan`, and even `1e-400` (which UNDERFLOWS to 0.0 and is caught by the `<= 0.0` test) were all refused. Overflow was the one hole.

inp2r.c, inp2c.c, inp2l.c — a comma silently changed a device value. `R1 in nb 1,5k` built a 5 kOhm resistor: `,` is a token separator, so the line parsed as the value 1 followed by a stray unlabeled 5k, and an unlabeled trailing number OVERWROTE the value already read. That overwrite is how `R1 a b rmod 2k` legitimately works — but there the value position held a NAME, so nothing had been read. Decisive evidence: every other separator obeys ngspice’s documented “ignore trailing text” rule (`1;5, 1:5, 1_000` → `1; 1.2.3` → `1.2`), and only the comma reached the overwrite.

inpcom.c — an over-wide array instance collapsed to ONE device. Past [E-441](#)’s 8192 cap the range failed to parse and the line was emitted unchanged, as a single device named `r[0:8192]: R[0:8191]` → 8192 devices (correct), `R[0:8192]` → 1 device (9x wrong), `R[0:19999]` → 21x wrong, all silent. The identical range in a NODE field already produced [E-443](#)’s warning; only the instance-NAME position was quiet, because [E-441](#) made `inp_expand_buses` skip an element line’s first token so the malformed check never saw it. A lone `R[2]` is still an ordinary instance name — a scalar bit is not a list.

inp2n.c — `.option autobus` indexed tokens that cannot carry an index. `N1 0 b bd` produced five ordinary FLOATING nodes `0[0]..0[4]` instead of ground, and `N1 a[0] b bd` produced `a[0][0]`. With `b` driven at 1 V the correct current is $-1.9375e-3$; the shorthand gave $5.4e-20$ — the device contributing nothing, silently. The same line with the option OFF is reported by [E-402](#)’s under-connected warning, so turning the feature on removed a diagnostic the feature-off path already had. Only BUS ports are checked: a scalar port is never indexed, so 0 or `a[0]` there stays legal.

com_sweep.c — a failed sweep point published the PREVIOUS solution. [E-438](#) counted these and said so, but the recorded VALUE was whatever the read-back returned — and a failed run leaves the earlier plot in place. Three different priors gave three different “results” (0.5 / 0.8 / 0.2), so this was the old solution being republished, not a constant. On a plot it is a flat shelf that looks physical, and `wr-data` wrote the rows unmarked. They are now NaN, which keeps the array shape — every output stays aligned against the sweep scale, exactly why [E-438](#) declined to drop the points — while plotting as a gap. `montecarlo` was the guarded sibling all along: same cause, same binary, already excluding failed samples.

subckt.c — a legal 20-deep hierarchy was called infinite recursion. MAXNEST bounded nesting at 21 and reported *infinite subckt recursion* on exhaustion, which a strictly non-recursive chain does not have; confirmed a DEPTH cap, not a name-length one (same boundary for short and long names). Now 256: a finite hierarchy costs exactly its own depth in passes, so the headroom is only paid by a pathological deck, and runaway recursion is still bounded by the existing million-instantiation cap. The message no

longer asserts a cause it cannot distinguish.

cktsopt.c — an iteration limit below the solver floor was discarded in silence. NIiter raises any limit below 100 to 100. E-426 found that floor, judged it deliberate and upstream, and left it alone; that judgement stands. What was missing is that nobody was told: the value is stored, option echoes it back, and a circuit needing 13 Newton iterations still took 13 under itl1=3 with both fallbacks disabled. The floor stays; setting a limit below it now says so, and only for a limit given explicitly — the shipped itl2=50 and itl4=10 are themselves below it.

spiceif.c — four working **.four** options were reported unknown. **.option fourgridsize=10** moves the reported grid from 200 to 10 AND printed “unknown option”. A warning that fires on a setting the run honours teaches the reader to ignore E-438’s check. nfreqs, nperiods, polydegree and fourgridsize are registered. Scope is deliberately those four: of the 179 cp_getvar names in the sources 176 are flagged, but most are shell variables whose documented home is set in **.control**.

One reported defect that was not one. An OSDI integer parameter declared from [1:inf) takes n=0.5 as 1 while refusing n=0.4, which looks like a range check running after rounding. E-399 established that a netlist value assigned to an integer parameter must round half away from zero, to match what Verilog-A does for the same conversion inside a model. 0.5 becomes 1, and 1 is inside the range. Rejecting it would contradict E-399 and refuse LRM-conformant decks — unchanged.

inp2n.c, spiceif.c — **.option autobus** (E-444)

A Verilog-A inout [0:4] a compiles to five OSDI terminals named a[0]..a[4], so a netlist had to spell all five out. With the opt-in option, one token per PORT is enough: N1 a b busdev expands to N1 a[0] a[1] a[2] a[3] a[4] b busdev, and a[2] elsewhere binds to the same node.

The information was already in hand. INP2N resolves the model BEFORE it binds nodes, and dev->termNames[] holds exactly those names — E-402 was already reading that table to REPORT which terminals a short line left unconnected. Terminals are grouped into ports (consecutive base[i] sharing a base = one bus port, a bracket-free name = one scalar port) and a line supplying one token per port is expanded. The bit indices are copied from the model’s own terminal name, so a port declared [4:1] expands to a[1]..a[4] rather than an assumed 0..n-1.

Opt-in because a short instance line already means something: it leaves trailing terminals unconnected, the \port_connected idiom BSIMSOI, BSIM-CMG/IMG/BULK, BSIM6 and PSP-HV rely on. Without the option nothing changes. Even with it the readings barely overlap — expansion needs the token count to equal the PORT count, and an all-scalar model has port count == terminal count, so a short line there can never be mistaken for the shorthand. The option name is registered with E-438’s .options checker so it is not reported as unknown.

Expansion needs the model, so it happens at device-parse time, not netlist-read time; the consequence is that it does not show up in listing e the way E-441’s array instances do.

inpcom.c — bracketed index lists (E-443)

base[lo:hi] (E-221 for nodes, E-441 for instance names) is joined by an explicit list and by the two written together: a[1,3,5], R[1,3,5], a[0:1,7]. Only the SOURCE of the indices is new, so every reading those enhancements established carries over — a list in a node field binds consecutive terminals, a list on an instance name makes one card per element, and a node list on an array instance is taken in step with it. Indices are used in WRITTEN order, neither sorted nor deduplicated, because the order is what binds nodes to terminals; a descending list therefore has no direction to get wrong, which is why E-411’s reversed-binding warning still fires for a[1:0] and stays quiet for a[3,1].

A lone **a[2]** is deliberately NOT a list. It is a scalar bus bit and stays a node name — E-221’s contract is that a[0:1] and an explicit a[0] denote the same node. That single condition is what makes the change additive: the only spellings that now mean something are ones that previously meant nothing.

The parse is split into `inp_bus_indices()` (the bracket contents) and `inp_bus_index_parse()` (the base around them), and four consumers share it: `inp_expand_bus_token`, `inp_expand_array_instances`, `inp_bus_rewrite_wrapped` (so `.save v(a[1,3,5])` expands, its old range-specific parse REMOVED rather than left beside the shared one) and the malformed-list diagnostic.

That diagnostic exists because a malformed list is not harmless: `R2 a[1,] 0 1k` re-tokenises the line and ngspice built a resistor with NO VALUE from it, warning only "Value of resistor r2 is too small, set to 1e-12" — a silent miswire in the new syntax. A bracket group that looks like an index list and is not one now says so, on a test tight enough that `mem[addr]` and `a[2]` stay silent.

inp.c, commands.c — listing tree (E-442)

A new listing form that draws the subcircuit hierarchy rather than the flat device list. `listing e` answers "what is actually simulated" — one line per device, structure only implicit in the dotted names — which is the wrong shape for "what does this design contain". Each `X` is labelled with the subcircuit it instantiates and drawn where it is USED, so the same subcircuit appears once per instantiation, and a summary reports instances, devices and depth. `listing t` is accepted alongside the existing `l/p/d/e/r`.

The walk is over `ci_origdeck`, the deck as read with `.subckt` blocks intact, because the expanded deck no longer records which subcircuit an instance came from. A useful consequence: array instances (E-441) are already expanded there, so `X[0:2]` shows as three instances.

Two details the walk has to get right. The subcircuit name is NOT "the last token before the parameters": `numparam` rewrites `X1 in 0 sub PARAMS: rv=2k` into `x1 in 0 sub 2k` before this runs, so the marker is gone and counting tokens picks `2k`. Resolving against the collected definitions — the last token that names a `.subckt` — is unambiguous where counting is not. And a `.control` block's commands begin with a letter exactly like element lines, so without skipping those blocks `option` and `listing` were drawn and counted as devices, and the last-child detection broke, leaving the final top-level entry with a tee where it needed a corner.

No "defined but never instantiated" report: ngspice comments such a definition out of the deck (`*subckt spare p n`) long before this runs, so it could never fire. (`undefined`) and (`recursive`) are defensive labels — both conditions abort the deck during expansion, so this command is not reached with either.

inpcom.c, parse.c, vectors.c, device.c — array instances (E-441)

A range on an instance NAME now means an array of devices: `R[0:3] a b r=1k` is four resistors in parallel, `N[0:3] a[0:3] b model` gives each element its own bit, and `N[0:3] a[0:3] a[1:4] model` builds a chain.

The design turns on one fact: the instance name selects the reading. Enhancement-221 already gave a range in a NODE field a different and equally useful meaning — one device with a wide port (`X1 bus [0:3] sub`) — and decks depend on it, so the two cannot both apply to a line. A scalar name keeps E-221's in-place expansion; a range on the name switches to one card per element with node ranges indexed *in step*. Nothing that parses today changes meaning, since an instance name carrying `[lo:hi]` was not a usable device name before.

`inp_expand_array_instances()` runs immediately before `inp_expand_buses()` and consumes the ranges belonging to the array reading, so whatever survives is E-221's by definition. Both passes read a range through one shared parser, `inp_bus_range_parse()`, so they cannot disagree about what a range is — and an XSPICE port group `[d1 d2]` has no base, is refused by that parser, and is therefore invisible to both.

Refusals: a width mismatch (`N[0:3] a[0:1] b`) is named and the deck is rejected — an earlier revision printed the error and let the line through, whereupon ngspice built one resistor literally named `r[0:3]` and completed the run with a circuit nobody described; and an XSPICE A-device array is refused,

because an A-device already uses brackets for port groups so its name is ambiguous by construction. Separately, E-221 no longer bus-expands the FIRST token of an element line: it is a name, never a node list, and expanding it buried the real complaint under a message about a device called `r[3]`.

Addressing the elements needed the accessor to cope with a bracketed NAME. The lexer in `parse.c` ended the token at the first `]`, and the splits in `vectors.c` and `device.c` took everything before the first `[` as the device name, so `@r[2][resistance]` looked for a device `r` with a parameter 2 — the element was visible to show but not addressable. The split turned out to live in FIVE places: `dctrcurv.c` (`.dc` failed fatally, `card` and `command` alike) and `outitf.c`'s `parseSpecial` (a `save` that matches nothing loses the WHOLE plot, so `save @r[1][i]` ended the run with "no data saved ... analysis not run") were both found by verifying the feature across subcircuit hierarchy and the `.control` surface after the first round. `ft_accessor_param_start()` is the shared rule and is deliberately narrow: a bracket group holding nothing but an integer AND immediately followed by another `[` belongs to the name, which leaves `@nd1[i_a[0]]` (E-408) and `@*[ [param] ]` (E-269) alone and keeps a lone `@r[2]` meaning device `r`, parameter 2.

Also fixed here: `inp_rem_levels()` reads `p->line->level`, i.e. into the cards, so the scope tree must be freed before the deck. Of `inp_readall`'s rejection sites the `inp_poly_2g6_compat` one had that order and the `inp_vdmos_model` one had it reversed; the new path was written by copying the latter and crashed with `EXC_BAD_ACCESS` at offset 8 (the `level` member) the first time it was taken. Both are now in the correct order — the `vdmos` site had been a use-after-free waiting for its first caller.

inpcom.c — a descending bus range binds terminals in reverse (E-411)

`nd1 d[3:0] 0 mm` against a model declaring `inout [3:0] a` compiles clean, simulates clean, and connects the terminals backwards. Both sides say `[3:0]`, which is exactly why it reads as correct.

Two conventions disagree. The COMPILER declares a bus port's terminals in ASCENDING bit order whatever direction the declaration uses -- `hir_def`'s `item_tree` lowering sorts the endpoints and loops upward, "declare from lsb to msb (ascending), matching natural bit order; direction of the original [msb:lsb] only affects range checks" -- so `inout [3:0] a` and `inout [0:3] a` give the same positional terminal list. The NETLIST does honour direction: E-221 expands `d[3:0]` to `d[3] d[2] d[1] d[0]`. Since the instance line's Nth node binds to the model's Nth terminal, a descending node list reverses the mapping.

Measured with one conductance per bit (`a[k] -> (k+1) mS`, so the current at a node identifies the bit it landed on): wired `d[0:3]` the nodes read 1, 2, 3, 4 mS; wired `d[3:0]` they read 4, 3, 2, 1 -- and the model's own declaration direction changes nothing, the `[3:0]` and `[0:3]` rows being identical.

`inp_expand_bus_token()` now reports whether the range ran msb-first, and `inp_expand_buses()` warns on element instance lines only. The scope came from scanning what the shipped suite actually does with descending buses rather than from taste: a `.subckt` descending PORT list is a deliberate interface choice that E-221 documents and `busnodes_examples` tests; output and IC cards bind nothing; and a range too wide to expand never binds at all (E-338's guard), so warning there would be false. The opt-out is set `nobusdirwarn` in `.spiceinit` -- a set inside `.control` is too late, because bus ranges expand while the netlist is read, and the message says so.

osdi/osdiloadd.c — an operating point that changed with frequency (E-412)

An OSDI operating-point variable read after `.ac` returned the SMALL-SIGNAL solution rather than the bias, and returned a different value at every frequency: an `opvar` defined as nothing but `vbias = V(a, c)` gave 0.4939733 after `op` (the true bias, from solving the circuit's KCL cubic by hand) but 0.4823410 / 0.4819028 / 0.0473587 after `.ac` at 1 k / 10 k / 1 meg.

An operating point cannot depend on frequency, which is what separates this from a tolerance argument -- the value is identical at `reltol` 1e-3, 1e-6 and 1e-10, and a multi-point sweep leaves the LAST point's.

CAUSE: [E-53](#) fires `@(final_step)` by issuing one dedicated evaluation per instance when an analysis completes, at `CKTrhsOld` -- which after a frequency sweep holds the small-signal solution. The model is genuinely re-evaluated, not mis-read: opvars chosen to tell those apart confirmed it (`3V+1` and `V^2` both matched the ac solution to nine digits, while a constant opvar stayed 42). Built-ins are unaffected, having no such post-analysis evaluation.

NOT WRONG, and established before fixing anything: the analyses themselves. A bias-dependent `white_noise(kf*V^2)` source integrated to `onoise_spectrum = 1.342573180366e-01`, matching the dc-bias prediction exactly and not the ac-bias one (`7.21e-02`). This is a READBACK defect; it is also not sticky, any following `op/dc/tran` restoring the correct value.

The evaluation cannot simply be skipped -- it is the only thing that fires `@(final_step)`, and `@(final_step("ac"))` and the noise variant are supported and tested by `finalstep_examples`. So `OSDIfinal-Step` snapshots the instance data around the call and restores it, for `MODEAC` and `MODEACNOISE` only: the event bodies still run and their `$strobe` output stands, while what the evaluation wrote is discarded -- correct, since those results are by design never loaded into the matrix or RHS. DC, DC-sweep and transient are deliberately not snapshotted, their final solution being a real operating point.

WHY IT SURVIVED: with a purely resistive device the dc and ac solutions coincide, so the natural small test case cannot see it at all.

breakp2.c + osdi/osdiparam.c — `savecurrents` for a multi-terminal OSDI device ([E-413](#))

`.options savecurrents` on a four-terminal compact model produced `@nd1[i] : current, real, 0` long -- a vector that exists and holds nothing, with no warning -- while a built-in BJT in the same deck produced four filled waveforms (`@q1[ic]`, `@q1[ie]`, `@q1[ib]`, `@q1[is]`).

CAUSE: `inp_savecurrents()` is a TEXTUAL pre-pass over the deck. It runs long before any `.osdi` is loaded, so it cannot know a compact model's terminal names and emits the same bare `@dev[i]` that R/C/L use -- its own comment ([E-394](#)) says exactly that. E-394 defines the bare `i` alias only for TWO-terminal devices, where it is unambiguous; for anything wider the save named a parameter that does not exist, so the vector was registered and never written.

The values were never wrong, only uncapturable: read as scalars after `op` the terminal currents were exact (`1.000e-3 / 4.000e-3 / 1.200e-2 / -1.700e-2`, summing to zero by KCL) and an explicit `.save @nd1[i_d]` always worked. What was missing was the waveform, which is what `savecurrents` exists to provide.

Since the names cannot be known during a textual pre-pass, the expansion moves to where they can: `ft_getSaves()` runs at analysis start with the circuit set up, so a bare `@dev[i]` belonging to an OSDI instance that does not define it is expanded there into one entry per terminal, via a new `OSDIterminal-Names()`. The two-terminal case is deliberately untouched -- `@dev[i]` is real there and is byte-identical to the pre-413 binary -- and built-ins never enter the path.

5. Parser and device registry — **src/spicelib/**

parser/inpgtok.c (10)

A `.model` card override silently dropped its first parameter when the type name and the opening parenthesis had no space between them (`modname devtype(p1=v1 ...)`): `INPgetNetTok`'s scan did not treat `(` as a token boundary, so the type token swallowed the paren and the first parameter name. Found while verifying event-function counters whose model-card overrides mysteriously kept defaults ([E-8](#)).

parser/inpgmod.c (5)

`find_model_parameter()` dereferenced `*(device->numModelParms)`, which is NULL for built-ins that take no model cards (VCVS, CCCS, ...) — any *referenced* `.model m vcv()` card segfaulted, with no OSDI involved at all (an ordinary MOS instance sufficed). This was the true root of the long-documented “module named like a built-in crashes” gotcha. One NULL guard; both shapes now produce clean, located errors ([E-76](#)).

parser/numparse.c — out-of-range int test

`ft_numparse` decides whether a parsed number is integer-valued with the round-trip `(double)(int)val == val`. Casting a double that lies outside `int` range to `int` is undefined behavior, reached by any control-language literal `>= 2^31` (e.g. `let x = 3e9`) — a UBSan build flags it at `numparse.c:166`. The value is now range-checked (`val >= -2147483648.0 && val < 2147483648.0`) before the cast; a value outside `int` range is, by definition, not `int`-representable, so the answer is unchanged for every in-range input (boundary-tested at `INT_MIN`, `INT_MAX`, and $\pm(2^{31})$). Found by the 2026-07 ASan/UBSan pass.

snp2va.c — lstsq_real heap overflow on a tiny Touchstone file

`lstsq_real` (the Householder-QR least-squares primitive behind `pre_snp`'s vector fit) documented itself as requiring an overdetermined system ($m \geq n$), but its back-substitution read `b[i]` and `A[i*n+i]` for every unknown `i` up to `n-1`, past the `m`-row buffers when $m < n$. `pre_snp` on a Touchstone file with very few frequency points (≤ 3) drives the fit underdetermined — the stacked system has fewer rows than poles, and the buffers shrink to a `malloc(0)` — so the back-substitution read off the end: a heap-buffer-overflow (ASan: READ of size 8 ... 7 bytes after a 1-byte region), reachable from an ordinary coarse `.snp` measurement. The back-substitution now leaves any unknown with no constraining row ($i \geq m$) at zero instead of reading past the matrix; for the normal $m \geq n$ path the guard never fires and the fit is bit-identical (verified: a 21-point file fits to 2 poles, rms 1.77e-05, before and after). Found by the 2026-07 ASan/UBSan pass (heavy-deck sweep, via a degenerate `pre_snp` input).

snp2va.c — unbounded filename-derived port count → heap corruption ([E-227](#))

`parse_touchstone` infers the port count `N` from the file extension `.snp` (`N = atoi(dot+2)`) with no upper bound. A filename such as `.s2147483647p` (`N = INT_MAX`) slips past the parse — the record stride `rec = 1 + 2*N*N` and the per-record inner-loop bound `N*N` share the same wrapped `int`, so `N*N` overflows to a small/zero value and the token layout stays self-consistent — but `N` is stored in `out->N` and then used to size the downstream vector-fit allocations (`malloc(nf*N*N*sizeof(cplx))`, `malloc(N*N*sizeof(cplx))`). The overflowing allocation and the writes that follow corrupt the heap (the program typically aborts later on an unrelated `malloc`, so it presents as flaky). The brute-force fallback already caps port inference at 512; the filename path now honors the same bound — `if (N > 512) N = 0`; falls back to inferring `N` from the data. Valid files (`.s2p`, ...) are far below the cap and unaffected. Found by the 2026-07 Touchstone-parser fuzzing pass (`rdsnp` was clean over 4,500 iters; `pre_snp` hit this on 103/3,600). Verify: `examples/snpfuzz_examples/verify_snpfuzz.py`.

parser/inp2n.c — an OSDI instance line with too FEW nodes ([E-402](#))

The node count was bounded on one side only: `numnodes > *dev->terms` errors with “*too many nodes connected to instance*”, while too FEW was never asked about. The binding loop then wrote `GENnode(fast)[i] = -1` for each unsupplied terminal and said nothing, so a typo silently produced a different circuit — a 4-terminal device given 3 nodes still simulates, with the omitted pin dangling and every branch touching it dead (`v(g)` 1.333 fully connected vs 2.000 mistyped, `rc=0`, empty diagnostic stream). Every built-in rejects the same mistake.

Not fixed by rejecting it. That -1 is the SAME sentinel `osdisetup.c` reads as "this terminal is deliberately unconnected" (`terminals[i] == -1 ? connected_terminals = i`) — the LRM optional-terminal feature that 32 corpus models query through `\$port_connected` (BSIMSOI, BSIM-CMG/IMG/BULK, BSIM6, PSP-HV). A dangling node is not a substitute either: electrically it matches an omitted pin, but it reports `\$port_connected == 1` where the omission reports 0.

So the lower bound WARNS, naming each absent terminal from `dev->termNames`, and states the consequence — including that an omitted terminal is not grounded, since that is the natural assumption and it is wrong (grounding the pin moves the measured node by 67%). Semantics unchanged; regression 322/322 with the warning firing zero times across the suite.

parser/inp2n.c, frontend/spiceif.c — .option silentports=ground (E-482)

E-481 turned the warning above off. What it could not do was make the deck run, and this is the other half.

Ten of the twelve corpus models that declare an optional pin tie it off with a POTENTIAL contribution (`Temp(t) <+ 0.0`), which puts nothing into a node `ngspice` allocated itself, so the private node `<inst>#<term>` has nothing driving it and the operating point dies on singular matrix. E-402's answer was *write 0 for the pin*, and a schematic front end that hides the pin cannot write anything — on the E-402 reproducer, silencing removed five warning lines and left all six singular-matrix reports, quiet and still broken.

`silentports_mode()` replaces E-481's `silentports_enabled()` and returns three states. It is read ONCE per instance line into `sp_mode`, which decides both the message and the binding — reading the variable twice would let a `.control` block that changes it mid-parse warn about a terminal it then grounds, or the reverse. `SP_OFF` warns and leaves the terminal dangling (unset — E-402, unchanged); `SP_QUIET` — the bare card, `=dangle`, `=quiet`, and the on-words `1/true/yes/on` — silences the report and leaves it dangling, which is E-481's behaviour preserved exactly; `SP_GROUND` — `=ground` alone — binds it. In `SP_GROUND` the binding loop no longer writes `GENnode(fast)[i] = -1` for an unsupplied terminal; it hands `INPtermInsert` a copy("`0`") and calls `bindNode`, exactly as for a token the line supplied. `INPpas2` inserts ground into the terminal symbol table with a hard-coded char `*groundname = "0"` before walking any device card, so "`0`" always resolves to the existing ground node and never creates one — checked with the OSDI card placed FIRST in the deck, and from inside a subcircuit, where it binds the GLOBAL ground.

Ground is asked for by name because it makes the model read `\$port_connected() == 1` and build the branches it would otherwise skip: a different circuit from the one the netlist describes. A word the user typed is the right gate for that; the bare card is not, which is why the bare card keeps E-481's meaning rather than acquiring this one.

The binding has to happen here and not in `osdisetup.c`. That file reads the -1 sentinel to decide how many terminals were connected and passes the count to `setup_instance` as `connected_terminals`, which is what the model reads back through `\$port_connected`. Bind in `INP2N` and everything follows on its own: the count is right, `\$port_connected` reports 1, no private node is created, and no collapse machinery is involved. Grounding later would have to undo a decision instead of never making it.

The query order is load-bearing. `silentports_mode()` asks `cp_getvar` for the CP_STRING FIRST, then CP_REAL for `=1/=0`, and CP_BOOL LAST — where it now means only what it is being asked, the bare card with no value. E-467 gave `cp_getvar` a CP_BOOL COERCION that answers TRUE for any value which is not an off-word: exactly right for a two-state option, which is what E-481 relied on, and fatal for a three-state one, because it swallows every value word and reports them all as a plain "on". Measured with the BOOL query still first: `silentports=quiet GROUNDED` the terminal, and so did `silentports=banana`. An unrecognised word is named once per distinct spelling and falls back to the DEFAULT rather than to either ON state — a typo must not be what silently drops a diagnostic or changes a circuit — following `ngspice`'s own handling of a bad enumerated value (`.options`

method=banana reports *unsupported integration method* and continues).

spiceif.c keeps the name registered in BOTH if_is_option()'s list and the type dispatch beside it; only the comment changes, since the dispatch answers what TYPE the name publishes and the VALUE is decided in INP2N.

analysis/dcpss.c, frontend/com_qpss.c — harmonic-balance convergence ([E-483](#))

QPSShb was called with its convergence bound and iteration cap compiled in — QPSShb(ckt, f1, f2, K1, K2, 0, 0, 60, 1e-10, ...). tol is an ABSOLUTE bound on the residual norm, so what is reachable depends on the circuit: the diode two-tone deck in distoexact_examples settles at 2.3e-15, while an FET amplifier carrying tens of milliamps floors around 1e-8 and cannot satisfy 1e-10 however many iterations it is given.

The way it failed was the defect. The Newton iteration reached a residual of 9.482e-09 at iteration 4 — nine orders down, quadratic — then sat on that number flat to seven digits for the remaining 55. Having “failed” the level, the E-135 source-stepping ladder halved its step and walked back to lambda = 0, 1022 Newton iterations in all, and returned a bare E_ITERLIM. A good answer found in five iterations was discarded after minutes of work — six to seven at hb 4 4, which is indistinguishable from a hang.

com_qpss.c gains qpss_knob(), which reads qpss_tol and qpss_maxiter the way qpss_verbose beside it is read. Both published types are asked, because E-454's lesson is that the SPELLING decides the type (=1e-8 a CP_REAL, = 1e-8 a CP_STRING), and the parse is strtod rather than atoi so qpss_maxiter=2e2 means 200 and not 2 — E-478's trap. A present-but-unusable value is reported and the default kept.

dcpss.c gains the stall test inside QPSShb's Newton loop: QP_STALL_RUN (4) consecutive iterates improving the residual by less than QP_STALL_RATIO (0.1%) is a stall, and a stalled residual is accepted as the answer only if it sits QP_STALL_ACCEPT (1e6) below the residual the level opened with. A stall at a residual that never came down is still a failure and falls through to the ladder — it reaches that verdict in a few iterations instead of maxiter. stall_accept is cleared whenever the ordinary tol test carries a level, so the closing message reports which test actually did the work.

Acceptance does not change the answer, which is the load-bearing claim: against the same circuit converging normally under a loosened bound, the fundamentals are BIT-IDENTICAL and the third-order products agree to 1.1e-05 relative, about 0.0001 dB on OIP3.

analysis/dcpss.c, frontend/com_qpss.c — believing an accepted residual ([E-484](#))

[E-483](#) made the bound reachable and taught the Newton loop to recognise a stall. It left a hole: the STALL path reported what it had settled for, the ORDINARY tol path reported nothing. A deck that loosened the bound far enough had its solution accepted with a clean converged in 3 iterations.

On a two-tone FET amplifier, set qpss_tol=1e-1 at K=4 accepts a residual that came down by only ~100x and the third-order products land 6 dB out — 27.818 dBm against the 33.976 dBm the trustworthy K=2 run gives. That is worse than the failure E-483 fixed: a refusal is honest, a wrong number is not, and E-483 had handed users a knob whose obvious misuse was silent corruption.

The reduction is now reported on the ordinary path too, naming the residual, the reduction, the bound that permitted the early stop and both remedies. The run is not aborted and the spectrum still prints, because the user may know exactly what they are doing.

QP_LOWRED_WARN is deliberately a DIFFERENT constant from QP_STALL_ACCEPT: one decides whether to accept a stalled residual at all, the other whether to believe an accepted one, and conflating them would tie a diagnostic to a control decision. Its value is calibrated on measurement — 101x puts OIP3 6.16 dB out and is warned, 55345x agrees with the K=2 answer to 0.033 dB and is not — with suite

checks holding the bar from both sides, since a warning that fires on a good answer is one people learn to ignore (E-445's note).

It does not fix **hb 4 4**. The Newton step degrades as harmonics are added (5.3e8x reduction at K=2, 7.8e4x at K=3, 1.5e2x at K=4), and drive amplitude, beta, is, alpha, cgs/cgd, the passive network, the topology and the DFT oversampling were each ruled out by measurement; pss_solve pivots and tmalloc is calloc. qp_build_matrix's harmonic-domain Jacobian remains the open question.

com_qpss.c also takes a const char *-> char * qualifier fix: E-483's qpss_knob() declared its name parameter const while cp_getvar takes a non-const pointer, two discards-qualifiers warnings in a tree kept at zero.

xspice/cm/cmutil.c, five **icm/analog/** models, **inp2dot.c**, **inpcom.c**, **inp.c**, **com_measure2.c** — guards that used the bad value (E-485)

Eight sites of one shape: the code establishes that its input is unusable, says so, and then answers from it.

cm_climit_fcn had its bail-out COMMENTED OUT. The five lines restoring a safe output sat inside / * ... */ , so the helper detected the crossed linear range, printed, and fell through. They could not be uncommented: they assign the LOCALS and return, while the out-parameters are written at the end of the function, so an early return would have left *out_final uninitialised — very likely why they were disabled rather than repaired. The repair fixes the INPUT: limit_range is a smoothing half-width, so clamping it to half the limit span leaves the thresholds coincident (hard limiting at the declared bounds) and every downstream branch valid. ilimit goes from 24.48 to 0.437 against rails of ± 1 ; the message from 26 copies per op to one (a report-once flag, since this is a raw printf with no instance context — E-480's INIT gate is not reachable here); and its text no longer names CLIMIT in a deck that has none.

limit, **int**, **d_dt** never computed **linear_range**. An oversized limit_range carried them to 24.5057, 95.04 and 24.25 against limits of ± 1 , and to 249999.75 at limit_range=1e6 — unbounded, linear in the parameter, silent, and all three declare those limits as MANDATORY parameters. Same clamp, reported at INIT. E-468's comment at limit/cfunc.mod:149 says it added its checks "as the CLIMIT sibling already does"; it ported two ideas and not CLIMIT's own guard.

pwl's monotonicity guard ended in **break** — leaving only the checking loop, after which the table was built from the rejected data and x=[0 2 1] answered 5.5 for an input of 0.5. It cannot return in place: x/y are STATIC_VAR-owned, so freeing double-frees and not freeing leaves a half-built table. The test moves beside the size_error check — before allocation, reading the parameter directly, refusing every evaluation — with both messages now INIT-gated.

hyst and **slew** had no checks. An inverted in_low/in_high pair and a hyst wider than the span left the block dead at 0.0; a negative rise_slope drove the output to -2.0 and a zero slope disabled limiting. Reported once at INIT and repaired to the nearest sane reading. slew's report is emitted from the INIT block, not beside the repair, which sits in the MIF_TRAN branch where INIT has passed — the shape E-480 had to move out of LIMIT.

dot_sens validated nothing while dot_ac in the same file rejects the same arguments by name; a reversed range swept a FABRICATED decade, 1e6 $\rightarrow$ 1e7 ascending. **dot_disto** refused them while reporting a DEVICE-parameter fault it does not have, identically for two faults. Both now share inp_sweep_args_ok(), which names the card and the argument.

.include and **source** accepted a DIRECTORY: fopen() on a directory succeeds on macOS, the BSDs and glibc, so the include contributed nothing and the deck solved a different circuit (v(out) 1.0 instead of 0.5) in silence. Both now test S_ISREG, the property their existing checks assume; .lib was already guarded.

com_measure2.c clamped a negative FROM and then printed the window the user asked for. The clamp moves to where `m_from` is what the report prints, and is announced.

NOT changed, deliberately: sweep's negative step is corrected on purpose (`com_sweep.c:2119`, "fix an obvious sign slip"); XSPICE Limits: are enforced; and `.nodeset` on an OSDI internal node really is ignored.

frontend/device.c, frontend/outitf.c, parser/inp2r.c — three names the simulator would not look up ([E-493](#))

Round 53 found the same shape three times: a name the user wrote was not looked up where it was written, and what came back described something else.

showmod <MODEL NAME> COULD NOT FIND THE MODEL. The device generator's grammar reads a bare word as an INSTANCE name, and only a #-prefixed one as a MODEL name. So with `.model dm` used by D1, `showmod d1` printed the model and `showmod #dm` printed the model, while `showmod dm` answered "No matching instances or models" -- of a model that plainly exists. The command's own help calls its argument "models", and its write sibling takes the model name directly (`altermod dm is=1e-12` works), so the one command dedicated to models was the one that could not be handed one. It affected OSDI models identically, the defect being in the name grammar rather than in any device.

RETRY RATHER THAN REINTERPRET. The bare-name-is-an-instance reading runs first and unchanged, so `showmod d1 --` and any name that is both a device and a model -- behaves exactly as before. Only when NOTHING matched is each bare word retried with the # the grammar wants, and a name that is neither ends at the same message it always did.

A SAVED NAME THAT MATCHED NOTHING WAS DROPPED IN SILENCE. Pass 1 of the output setup walks each `.save/.probe` item against the analysis's own vector names and marks the ones it places; anything it failed to match simply stayed unmarked and the run continued without it. So `.save v(n) v(nosuch)` recorded `v(n)`, dropped the typo and said nothing: the analysis succeeded and the vector the user asked for was merely absent. `.probe v(nosuch)` reaches the same path, which is why a mistyped NODE there was silent while a mistyped SOURCE in the same card is reported by the measure-source pass ("Could not find the instance line for ..."), and why Enhancement-418 already said it for the `@dev[param]` spelling ("no such device, so this vector will stay empty"). The plain node spelling was the one route left quiet.

It WARNS rather than refuses: an absent vector is not a wrong answer, and a deck that saves a node it does not always build is a real idiom. The check honours `savesused[]` first -- so the `all/allv` keywords and items belonging to another analysis are never reported -- and falls back to matching the name itself under `save all`, which `.probe` turns on and where Pass 1 never runs.

A RESISTOR MODEL NAMED **r** WAS UNREACHABLE. INP2R excluded the token `r` from being a model name outright. That exclusion is necessary for `R1 a b r=1k`, where `r` is the keyword that writes the resistance and must not be read as a model, but it also locked out a model actually CALLED `r`: `.model r r rsh=1k` with `R1 a 0 r l=1u w=1u` bound neither model nor value, so the device fell through to the default and came out as 1 mOhm with "resistance too low or not given" -- a message about the symptom, when the cause is that the model named on the line was never looked up. Every other name works, including every other resistor keyword (`rsh`, `l`, `w`, `tc1`, `tc2`, `temp`, `dtemp`, `m`, `scale`, `ac`, `dc`, `noisy`, `short`, `narrow`); only this one letter was unreachable.

The two spellings are distinguishable by what FOLLOWS the token: the keyword form is `r=`, a model name is not. Asking that instead keeps `r=1k` meaning what it always did and makes a model called `r` reachable. A deck with no such model is unaffected -- INPlookMod fails and the existing else branch restores the line and builds the default model, exactly as for any other unrecognised token.

Deliberately unchanged: `showmod` by device name, by `#model` and bare, and `show`, including a name that is both a device and a model; `altermod <model>`, which already worked; `.save all`, `.save allv`

and `.probe alli`, and any item belonging to another analysis; `@dev [param]` saves, which E-418 already speaks for; and `r=1k`, `r = 1k`, `R=2k`, a plain value, `r=1k tc1=0`, a value with `m=`, and `r=4k` winning when a model named `r` also exists.

`parser/inp2{e,g,s}.c`, `parser/inppas3.c`, `analysis/cktop.c`, `KLU/klusmp.c` — a node named only in a control position (E-492)

E, G and S take a controlling node PAIR, and those two names were bound exactly the way the output pair is -- `INPtermInsert`, which CREATES the node. So a typo simply invented a node and the run continued against it.

For E and G the invented node has no path to ground, the matrix goes singular, and the user is told "singular matrix: check node nosuch" -- a node they never wrote, reported as a fault in their circuit.

S IS WORSE. A switch only READS its control voltage to decide open or closed and stamps nothing for it, so the matrix stays non-singular, the solve succeeds, and the answer is silently wrong: `S1 a b ctl 0 sw` gives `v(b) = 0.999001` while `S1 a b nosuch 0 sw` gives `9.99999e-07` -- a factor of a million from one mistyped character, `rc = 0`, with no diagnostic at all. A phantom reference is dangerous exactly where it is READ but not STAMPED.

Every other route already answered this question. `.ic` and `.nodeset` report "IC on non-existent node - %s, ignored"; F, H, W and a B-source's `i()` report "unknown controlling source"; and all thirteen output constructs name a vector that does not exist. `warn_physics` even validates the switch's own `ron`, `roff` and `vh`. Only the controlling-node pair skipped it, under any option.

The mechanism is Enhancement-429's, unchanged. E-429 added `CKTnode::devRef` -- "was this node ever named by a device, or made by the simulator?" -- so that a `.tf` card may legitimately precede the devices defining its nodes while a typo is still caught. A control reference is the same kind of reference: it names a node without making it real, so it no longer sets `devRef`, and `CKTnodePhantom()` answers the question as it stands. Marking is left to `INPtermInsert` for every other position, so a control node that IS connected somewhere -- including one that is the source's own output, `E1 out 0 out 0 2` -- stays marked and is never reported. The check runs in PASS 3 for the same reason `.ic`'s does: only once every device card has been read is "did anything connect to this?" answerable.

It REFUSES rather than warns. E and G already fail on their own; a switch does not, and would otherwise carry on and answer from a node that is not in the circuit -- the detect-announce-then-use-it-anyway shape Enhancement-485 had to undo eight times in one round.

CKTop blamed Verilog-A for faults that were not Verilog-A's. E-378 added the abort message and E-399 narrowed its entry to the `E_PANIC` checks, arguing the test is exact because `E_PANIC (1)` and `ITERLIM (103)` are distinct values. That is true -- but the invariant the MESSAGE relies on is "E_PANIC means a Verilog-A `$fatal`", and `E_PANIC` has around ten producers: `cktsetup`'s missing model and device lists, `osdiparam`, `osdiload`, `osdisetup`, `dcps`, `cktpzstr`, `ifeval`'s bad node type and CIDER's `twosolve`. Measured: `.option klu` with a netlist whose matrix holds only current sources -- no Verilog-A device anywhere in the deck -- printed "a Verilog-A device raised `$fatal`" and pointed at an `OSDI(fatal)` message that does not exist, through `op`, `dc`, `ac` and `tran` alike, since each calls CKTop for its operating point. `CKTvaFatalRaised` is now set only where a `$fatal` is actually detected, and cleared on entry to CKTop so a previous analysis's fatal cannot be inherited.

KLU printed nine lines for one condition. `PreOrder`'s own comment says an empty matrix is legitimate -- "XSPICE pure digital circuits produce empty KLU matrix" -- and it returns success for one. But all three sites printed "Error (...): KLU Matrix is empty" on every call, and `Solve` then added "KLU numeric object is NULL" and "KLU symbolic object is NULL", which are CONSEQUENCES of the same empty matrix rather than separate faults. Re-entered per Newton iteration, one such circuit produced nine or more lines, none of which named what had actually happened. It now says once, in words, what the state is.

Deliberately unchanged: a control node connected anywhere else -- the source's own output, one defined later in the deck, ground, or one passed through a subcircuit port; F, H, W and B-source `i()`, which already named their missing controlling source; a real Verilog-A `$fatal`, still named as such and still pointing at its OSDI(fatal) line; and an ordinary circuit under either solver, or a mixed-signal KLU run, which get no empty-matrix note.

misc/printnum.c, evt/evtprint.c, parser/ptfuncs.c, and seven more — an unbounded number used as a length, and four wrong ones (E-491)

Round 51 found ten defects sharing one shape: a number the deck supplies, used without being measured against what it was about to control.

THE CRASHES. `set_numdgt` is the user's print precision and nothing bounded it. `printnum()` formatted with `sprintf(buf, "%. *e", cp_numdgt, num)` into caller buffers of `BSIZE_SP(512)`; `evtprint.c` formatted into a `char[100]`. `%. *e` needs `n+9` bytes, and both thresholds land exactly there: `set_numdgt=510` with print aborts, `set_numdgt=94` with `eprint` traps, both from a plain batch deck with no interactivity. `numdgt=94` is an ordinary "give me lots of digits" value.

`printnum()`'s OWN COMMENT had recorded the hazard -- "It can cause buffer overruns. The size of `buf` is unknown, so `cp_numdgt` can be large enough to cause `sprintf()` to write past the end of the array" -- without bounding it. The safe DSTRING sibling `printnum_ds()` sits directly below and cannot overflow, which is precisely why `fourier`, `wrdata`, `write`, `display` and `diff` were unaffected and only the `print` family crashed. `printnum()` now takes the buffer's size and CLAMPS the precision to what fits rather than truncating: `snprintf` alone would stop mid-number and hand the reader a value that is not the one computed, while beyond ~17 significant digits the extra places are zero padding, so a clamped column is the same number, just narrower -- and it says so once. `evtprint` uses the same clamp, so the two printing paths cannot disagree about how wide a number may be. Both sites trace by `git -S` to `4f29ffad`, the vanilla UPSTREAM import.

THE WRONG NUMBERS. `PTdivide` added `PTfudge_factor` to EVERY divisor rather than only a zero one, and that factor is `gmin * 1e-20` -- not even a fixed perturbation, but one that scaled with an unrelated convergence option. One over Boltzmann's constant came out 42% low under `.option gmin=1e-3` and 88% low under `gmin=1e-2`; `1/0` returned `1e26`, `1e32` or `1e50` depending on `gmin` alone. A deck's arithmetic must not move because the user reached for a convergence aid, so the nudge now applies only to an exact zero and uses a fixed epsilon -- a non-zero divisor is used as written, `1/0` is the same number in every deck, and the value keeps the default `gmin`'s `1e-32` so nothing already correct changes.

The B-source reduced trig arguments with `x - (int)(x/2pi)*2pi`. That cast is undefined above `2^31*2pi` (~1.35e10), and `sin(1e20)` returned `+0.9993` where the answer is `-0.6453` -- wrong sign, wrong magnitude, no diagnostic. Both OTHER evaluators in this simulator were already right: `numparam` and a Verilog-A model's own `sin()` each match `libm` exactly, making the B-source the SOLE OUTLIER. That is the divergence Enhancement-399 forbids, and `ptfuncs.c`'s own comment quotes the rule. `libm` reduces correctly for every finite double, so the macro is gone.

THE SILENT REFUSALS. A `sens` filter matching no parameter produced no plot and said nothing, returning OK and leaving the PREVIOUS analysis current for a following `print` -- and the trailing words on a `sens` line ARE filter patterns, so `sens v(b) r8` is a legitimate restriction that works, which made a mistyped filter indistinguishable from one that did its job. The interactive `meas` enforced its analysis keyword for INTERVAL measurements -- Enhancement-468 restored that -- and ignored it for POINT ones, so `meas tran m FIND v(n) AT=0.5` read a DC sweep as a transient; the check now runs once for every measurement type, in the command only, since the `.meas` card already gated correctly. `s_xfer` blamed the SOLVER for a model error: an all-zero denominator went NaN and was reported as "Dynamic `gmin` stepping failed", while a numerator longer than the denominator was detected, announced once per evaluation (1238 times over 300 steps), and the run then returned `rc=0` with the device contributing nothing -- both knowable from the parameters before any solve. `printf("oops ")` reached users on

STDOUT from two default: arms in inptree.c, one of them the very printer the "internal check of parse tree ... failed" diagnostic uses to SHOW the offending expression, so that message rendered as the word "oops" -- reachable from .param p={ln(0)}. show <dev> : <bogus> printed ?????????? where print, alter, altermod, wrdata and sweep all name the parameter. And a duplicate user .func was silently last-wins while shadowing a BUILTIN warned (E-467), so two includes each defining a helper displaced one another unannounced.

ONE COMMENT CORRECTED RATHER THAN OBEYED. Enhancement-440's comment claimed the singular-value routes "clamp to HUGE" and that clamping "keeps the Jacobian finite, which is what lets Nlitter reach for gmin or source stepping". Neither is what the code does: PTeval in ifeval.c treats a result equal to HUGE as an ERROR FLAG, returning E_PARMVAL and aborting -- the opposite of continuing -- and the routes were never uniform (sqrt(-1) and pwr(0,-1) abort, log(0) returns -1e99 and runs on, and PTdivide no longer reaches HUGE at all). Recorded rather than unified: making log(0) abort, or pow(0,-1) not abort, would each change the answer a working deck gets today. What was wrong was the description, and a reader trusting it would have drawn exactly the wrong conclusion about which of these keeps a simulation alive.

Deliberately unchanged: ordinary precisions (6, 12, 17 digits format exactly as before, and only a value that cannot fit its field is narrowed); 1/0 still returns a large finite number so the solve continues; interval measurements keep E-468's behaviour; a sens run with no filter or with one that matches; and .func still resolves to the last definition -- only the silence is gone.

osdi/osdiinit.c — an operating-point variable whose name the simulator also supplies (E-505)

ngspice writes its own instance parameters — m, temp, dtemp, dt — into the instance lookup table before anything the model declared, so a model with an operating-point variable of one of those names computes it on every evaluation and can never read it back. @n1[temp] returned the ambient temperature and @n1[m] the multiplier. The name is legal Verilog-A and nothing said a word, in the compiler or the loader.

It is named at load time rather than refused: the rest of the model works, and refusing would break a model that runs today over a name it never reads.

Operating-point variables only. A model may legitimately declare m or dtemp as its own instance PARAMETER — that is what E-394's has_m exists for, a CMC-style model scales by its own m, and the loader routes the deck's value into it, so nothing is shadowed. The first version of this warning fired for both and linguard_examples caught it — that suite asserts the parameter case is silent, and drop-guard_examples now pins it from the other side.

The three compiler-side halves of this enhancement are in the openvaf-r report.

osdi/osditrunc.c, osdi/osdidefs.h — a model may not demand an unbounded step count (E-504)

Two ways for a Verilog-A model to hang the transient indefinitely — over 150 s with no output, no error, and no "timestep too small".

\$bound_step(1e-18) is a perfectly legal positive request and was taken literally. The "timestep too small" test compares against CKTdelmin, which for a 12 ns analysis is ~5e-20 — far below the 1e-18 being asked for — so nothing ever fired. The step was not too small for the solver; it was too small to finish. No clamp value makes 1.2e10 steps work, so the rule is stated in the only terms that bound the run: a model may not force more than **E504_MAX_MODEL_STEPS** (10^6) steps across the analysis window. Beyond that the bound is clamped and the instance is named once. A model wanting genuinely fine resolution is unaffected, and ngspice's own adaptive stepping is untouched — this bounds only what a DEVICE may demand.

\$discontinuity(0) outside any conditional pinned the timestep to the last accepted delta, which could then never grow, so once the retry had cut it the transient crawled for the rest of the analysis. E-55 already

made the *retry* edge-triggered for exactly this reason; the cap needed the same floor. `$discontinuity(-1)` means “no discontinuity”, is fine, and stays untouched and silent.

The companion compiler-side fixes — `transition`'s negative rise time, a non-positive `$bound_step`, a negative noise power and `idtmod`'s zero modulus — are in the `openvaf-r` report. The noise sign is deliberately NOT fixed here: by the time the power reaches `osdnoise.c` the contribution factor has been folded into it as `fac*|fac|`, and E-42 uses that sign to sum same-named sources coherently.

Verified by `examples/rtdomain_examples/verify_rtdomain.py` — 16 checks under both linear solvers, 11 of which fail on the shipped binaries.

`spicelib/analysis/cktdojob.c`, `frontend/com_sweep.c` — setup reuse for decks containing a built-in semiconductor (E-503)

E-471 reuses the matrix between sweep points instead of tearing the circuit down and rebuilding it, and offered that only to circuits built entirely from types whose topology is fixed *unconditionally* — the linear elements and sources — plus OSDI, whose collapse is re-decided and compared on every `CKTtemp`. Every other device type refused reuse outright, so one transistor anywhere disabled it for the whole circuit: 3.46x at 300 sections, 5.28x at 600, 10.09x at 1200.

That is a per-type gate for a per-parameter hazard. A built-in semiconductor decides its node collapse in `DEVsetup`, from a knowable set of parameters and from nothing else — BJT `rc/rb/re/rco`, diode `rs/rsw/vp/tt`, JFET `rd/rs`, and the Mos1/2/3/6/9 family `rd/rs/rsh/nrd/nrs`. Each row was read off the device's own setup routine by auditing every `CKTmkVOLT` call and the condition guarding it, not from the parameter documentation, which does not say which parameters build nodes; `rco` gates a fourth BJT node through its `...Given` flag rather than its value and was missed on the first pass, which is what the audit caught.

The sweep now declares the parameters it varies, and reuse is allowed for these types when no declared name builds a node: 2.28x / 3.25x / 6.69x recovered at 300 / 600 / 1200 sections. The remaining gap to the all-linear time is the transistor's own evaluation, which no setup reuse can remove.

The absence of a declaration is what keeps it safe. `sw_request_reuse()` CLEARS the declaration rather than leaving whatever the last caller wrote, so a caller that cannot enumerate what it varies inherits nothing and keeps E-471's stricter gate: Monte Carlo binds its draws through the deck, and a `.param` knob reaches a model parameter through an expression this code cannot see — either could be feeding the very `rc` that decides a collapse. The match is whole-token, so a sweep of `rsh` is not read as a sweep of `rs`, and a type absent from both tables is still refused.

Where the declaration lives is part of the fix. It is a file-static in `cktdojob.c`, not a field on `CKTcircuit`. The first version put it in the struct, which grew it by 256 bytes and shifted every field after it — and made the `argguard` suite fail about two runs in three, its `warn_physics` count coming back 1, 2 or 3 at random for the same deck. The logic was not the cause: 256 bytes of padding alone, with none of this code, reproduces it. Something else in `ngspice` reads that struct by a layout it did not get from the current header. That is a real latent defect, it is recorded as an open finding, and this enhancement must not be what makes it reachable; held outside the struct the layout is byte-for-byte unchanged and `argguard` is stable ten runs out of ten.

Verified by `examples/reusedev_examples/verify_reusedev.py` — 16 checks under both linear solvers, 8 of which fail on the shipped binary. Every reuse check compares the swept curve against `.option reusesetup=0` and requires bit-identical results, having first asserted that the curve actually moves: a comparison of two flat lines proves nothing, and E-471's own notes record a naive version that silently froze a collapse and drew one.

frontend/parser/numparse.c, **frontend/com_{envelope,emir,eye,qpss,reduce,hbosc,compose,rfstab}.c** — the guard that refuses the wrong value and admits NaN ([E-502](#))

Round 60 swept the commands that produce a report — emir, eye, reduce, envelope, qpss, hbosc, stb, rfstab, compose — and found one guard written the same way in six of them:

```
if (x <= 0.0) { refuse; }
```

Every comparison with NaN is false. That test refuses 0, refuses a negative number, and lets NaN straight through — one ! away from correct.

A one-token SIGSEGV. envelope <node> nan <tstop> crashed five runs out of five with EXC_BAD_ACCESS address=0x10 in SMPmatSize. fc is NaN, so $T = 1/fc$ is NaN, so the settling transient is built as tran nan nan 0 nan — which ngspice refuses. The refusal was never checked, the matrix was never built, and EFanalysis dereferenced it one call later. Both halves are fixed: the argument is refused at the front, and envelope now checks that its own internal transient ran. That second check matters beyond this bug — any refused transient leaves CKTmatrix NULL, and a command must never use the result of an internal analysis without asking whether it happened. That is [E-438](#)'s shape, where a failed sample kept the previous plot and was counted as a pass.

A limit that is never exceeded. emir jmax nan reported "*0 segments over Jmax*" with every segment ok, on a grid with two genuine violations. reduce nan reported "*26 nodes -> 26 nodes (1.0x)*" and wrote a file called reduced.sp that reduced nothing. qpss v(o) nan 1.1e6 printed a 25-row spectrum in which every frequency, magnitude and phase was nan, and exited 0. eye -ui nan reported an eye height of 0 — a fully closed serial link. Each of those guards proves its own intent by working: jmax 0, reduce -1, -ui 0 and qpss with two equal tones were all correctly refused. This is [E-501](#)'s NaN spec bound one command over: a limit used only in a comparison is not a strict limit when it is NaN, it is *no limit at all*.

Two returned a different answer rather than a blank. eye -tstart nan was neither refused nor blank — the "skip samples before tstart" test is also a comparison, so NaN never skipped, the startup transient the flag exists to exclude was folded into the eye, and RMS jitter came back 9.86e-13 against 1.50e-15, 660x, with no diagnostic. qpss ... maxorder nan silently became order <= 1, dropping every intermodulation product — the entire reason that command exists — with only a quiet order <= 1 in a banner. Both reached their clamp through an undefined double->int conversion, as did emir top nan (landing on 1) and compose lin=1e12, which landed on INT_MAX and really allocated 17 GB: the vector was genuine, v[2147483000] read back correctly.

emir could not tell a given width from a default. It computes current density from @r[w], which answers 1e-5 for a resistor whose deck line never gave a width — so the segment most likely to be the oversight was analysed as a comfortable 10 um conductor and reported ok, while the header of com_emir.c documents that such segments are *skipped*. The inverse was also true: w=0 and w=-0.5u were skipped and reported as "*no width given*", the one thing they were not. emir now asks the instance (RESwidthGiven, reading resistor internals as spicelib/analysis/rcreduce.c already does for reduce) and reports the two reasons separately.

The idiom was already in the tree, and three commands that got it right were left alone: hbosc refuses K nan because its test is $K \geq 1$; compose refuses NaN on every parameter it takes; stb delegates its frequency spec to ac. The guards now share ft_argpos, ft_argfinite and ft_argcount, added next to ft_numparse itself. They take the raw token so the diagnostic names what the user typed, and they parse with ft_numparse rather than strtod, because these are SPICE numbers and thick 0.5u / -ui 0.5n / 1meg are how the documentation writes them — E-501 shipped a guard built on strtod, it refused dynamic 20u, and the aging suite caught it. ft_argcount checks the value before the cast, because converting a NaN or an out-of-range double to int is undefined.

Verified by examples/nanguard_examples/verify_nanguard.py — 34 checks under both linear solvers, 23 of which fail on the shipped binary.

frontend/com_aging.c, frontend/com_sweep.c, frontend/com_optimize.c, frontend/runcoms2.c — aged state across an internal reset, and what the loop commands claim ([E-501](#))

The aged device came back fresh. `aging` ([E-157](#)) writes an accumulated dose into each device's age parameter with `alter`, then leaves the circuit standing so the analyses that follow see the degraded part. That dose lives only in the running circuit — it has no deck representation, because it was never in the deck. So every command that re-sources the deck between evaluations threw it away on the first one and ran the rest of the loop on a fresh circuit, in silence.

That is `wcd`, `highsigma` and `optimize -center`: the three commands whose entire purpose is to characterise a part were the three that could not characterise an *aged* one. On the suite's deck a 95-year dose moves the worst-case distance from 8.66 sigma to 1.81 sigma — a part that sits comfortably inside its limit when new and barely inside it at end of life — and `wcd` printed 8.66 for both, with nothing in its output to say which circuit it had measured.

`montecarlo` escaped, but only by accident: [E-346](#)'s fast path arms whenever a random value binds — and prints *"no per-sample reset"* when it does — so the usual Monte Carlo never re-sourced at all. Its fallback path re-sources like the others.

`aging` already issues its writes as ordinary `alter` commands, so it now records them as it writes them, and the loop commands replay that record after their internal reset. The record is per-run — `aging` clears it before writing — which preserves [E-157](#)'s stated property that a dose is an *absolute* target and calling it twice is idempotent. A reset the user types must still mean "the deck as written", or there is no way back to a fresh device short of reloading the file; so the internal resets are marked as internal, `com_rset` drops the record when it is not marked, and only the marked ones replay.

aging accepted targets it could not age to. Its numbers went through a bare `atof()` and its guard was `t_target <= 0`. Every comparison with NaN is false, so `aging nan` walked through it and wrote NaN doses into every device while reporting them aged; `aging abc` aged to 0 s; and `dynamic`'s stop time was never checked to be a number at all, so `aging 1e8 dynamic verbose` consumed `verbose` as the stop time and then complained about `agerate`, a token the user had never typed. The guards are now written `!(t > 0.0) || !finite(t)`, and the stop time is checked before it is consumed so the message names what was actually typed. This is the shape [E-497](#) fixed in `setseed` and `disto`.

A spec bound that is never violated. A limit is only ever used in a comparison, so a NaN limit is not a strict limit — it is *no limit at all*, indistinguishable from omitting the flag. `montecarlo ... -max nan` reported `yield : 100.000%` with a Wilson interval. `montecarlo -spec -max/-min, wcd -max/-min/-tol/-step, highsigma -max/-min/-scale` and `optimize -spec -max/-min` are now parsed the way the rest of each command's numbers already were — with `ft_numparse`, not `strtod`: the first version of this guard used `strtod` end to end and refused `dynamic 20u`, the documented spelling in [E-157](#)'s own example, which the existing `aging` suite caught. A negative bound stays legal, which is why the spec readers are not [E-499](#)'s `-tol` reader; that one refuses negatives, correctly for a tolerance and wrongly for a limit.

A yield with nothing behind it. A yield is a statement about variation. A deck with no random parameter — or with one the metric does not depend on — produced 50 identical samples and reported `100.000%` (50/50) with a 95% CI of [92.865%, 100.000%]: an interval that looks tight precisely because every sample was the same sample. [E-495](#) warns when an individual `agauss(...)` call is degenerate; this is the whole-run version, and it catches what no per-call check can.

optimize published the score but not the answer. `optimize -center` published `dcenter_yield` and `dcenter_cpk` — what it *scored* — but never the centred knob value, the thing the command exists to find. A `-dparam` is a `numparam` symbol that never becomes a vector, so the shipped `dcenter_demo.cir` asked for it as one and printed `Warning from checkvalid: vector xc is not available` on every run since the example shipped. Knob values are now published as `optimize_<name>`, and the demo prints `optimize_xc = 5.0022` — the design centre its own comments predict.

aging param aimed somewhere odd. `aging 1e8 param w` wrote 4.095e8 into a MOSFET's *width* — a transistor 409,500 km across — reported “*1 device aged*”, and exited 0, because aging selects devices that *have* the named parameter and a width qualifies. The name stays the user's to choose, since models spell their aging state differently, so this warns and proceeds; but when the same device also exposes a plain age, choosing something else is far more likely a slip than an intent.

Verified by `examples/agestate_examples/verify_agestate.py` — 30 checks under both linear solvers, 19 of which fail on the shipped binary.

frontend/com_dl.c, frontend/inp.c, frontend/com_presnp.c, osdi/ — **pre_osdi -va** compiles Verilog-A and loads it in one step ([E-500](#))

A deck that ships its own models needed a separate build before it could run: compile each `.va` by hand, then point `pre_osdi` at the objects. `pre_osdi -va rmod.va gmod.va` now invokes `openvaf-r` on each source and loads what it produces, in one command.

Where the objects go. Into an `osdi/` directory beside the netlist, not beside the working directory and not next to each source. It keeps a model directory clean when a deck carries a dozen sources, and it makes an input/output collision structurally impossible: [E-452](#) recorded `openvaf-r m.va -o m.va` destroying the source and exiting 0, and an object written to `osdi/<stem>.osdi` cannot name the input whatever the input was called. The `.va` argument resolves against the netlist too, exactly as `load_osdi( )` already resolves the object it is handed. That was a bug in the first version of this change — the object went beside the netlist while the source was looked up beside the working directory, so the command worked from one directory and not from another. The two halves of one command must not disagree about what a relative path means.

Which compiler. `osdi_find_openvaf()` is `pre_snp`'s existing lookup, exported rather than reimplemented, so both generators share one policy: the `openvaf` `ngspice` variable, then `$OPENVAF`, then `$SPICE_LIB_DIR/openvaf-r`, then `PATH`. Searching `PATH` alone would have been the obvious implementation and the wrong one — for anyone running the shipped bundle the compiler sits in `$SPICE_LIB_DIR` and is frequently nowhere else.

Recompiling is the default; **.option osdicache** opts out. The instinct is to cache by default, and it is wrong here: a `.va` timestamp records whether the SOURCE changed and says nothing about the COMPILER having changed, so while `openvaf-r` is itself under development a cached object is one built by a version that may no longer exist. That is the shape of [E-453](#), whose cache key omitted its own codegen settings. The option is read from the deck's options cards, not through `cp_getvar` — `pre_` commands run before the circuit is set up and before any option is published, the trap [E-464](#) recorded for `autobus` — and is matched as a whole word rather than with `strstr`, because [E-451](#) shipped an option whose name was decided by a substring search. Every spelling that means off means off.

-f forces the rebuild, not just the reload. [E-229](#) added the flag for an edit -> recompile -> re-source loop; under `-va` the compile is part of that loop, so honouring `-f` for the load alone reloaded the very object the author was trying to replace — the one case the flag exists for. This was the second defect in the first version and the sharper one: with **.option osdicache** set, an edited `.va` silently kept answering with the old model. The suite edits a source and checks the answer moves from 0.5 to 0.3333 only when `-f` is given.

The staleness test is strictly newer, never `>=`. `st_mtime` is one-second granular on POSIX and two-second on FAT, so an edit and a re-run inside one tick would otherwise load the object built from the previous text. A tie costs one needless recompile; the other way costs a wrong answer — and that choice is also what makes a coarse filesystem clock harmless rather than dangerous.

Portability. `stat/st_mtime` come from `<sys/stat.h>`, which `inpcom.c` has used on Linux, macOS and Windows for years. `mkdir` is the one call that differs and is guarded on `_WIN32` — defined by MSVC, MinGW, MinGW-w64 and clang-on-Windows — where `_mkdir()` comes from `<direct.h>`.

Mixing. `-va` only changes how a `.va` ARGUMENT becomes a path; anything else still goes straight to `load_osdi`, so the two mechanisms converge on one loader. Every arrangement works: either order on separate lines, either order on one line, `-f -va` beside a pre-built object, and the same model arriving by both routes at once (reported as already loaded, E-229's existing behaviour, not an error). `-va` on a line with no `.va` argument is a no-op.

Refusals name the cause. A missing source is *no such Verilog-A source* rather than `exit 512`; one that will not compile shows the compiler's own diagnostics plus all four ways to supply a compiler; and a read-only directory is not fatal — the objects are written beside the netlist and the fallback is announced, so a deck in a shared PDK or a CI checkout still runs.

Unchanged: plain `pre_osdi file.osdi`.

maths/KLU/klusmp.c, frontend/com_optimize.c, frontend/com_sweep.c — loop-command arguments, verdicts, and a KLU refactor kind (E-499)

Round 58 re-probed the sweep / optimize fast path after E-498 fixed the transient side. The reuse arithmetic was clean this time. What was wrong was everything around it — the arguments the commands accept and the verdict they print — and, underneath both, one genuine memory bug that setup reuse made reachable.

A complex refactor handed a real factorisation. `klu_z_refactor` refills an existing factorisation *in place*, walking the L and U index arrays that `klu_z_factor` built. Handed a real Numeric object it walks half-sized arrays with complex strides: it reads and writes past their ends, and a later `klu_free_numeric` frees whatever it scribbled. The malloc abort names object `0x3ff0000000000000` — the IEEE-754 bit pattern of the double 1.0, a matrix *value* being freed as a pointer. The mismatch is reachable because a Numeric outlives the analysis that built it: every AC/SP/NOISE run is preceded by a real operating point, and E-471's setup reuse keeps the matrix standing between sweep points, so the second and later points arrived holding the operating point's real factorisation. Under `.option klu, sweep -analysis ac` and `-analysis noise` returned 0.0 at every reused point, `optimize -analysis ac` fitted a parameter 10x wrong while reporting *converged*, and `-analysis sp` crashed on 9 of 10 runs. SPARSE was never affected, and neither was any analysis that stays real — `op`, `dc`, `tran`, `tf`, `pz` and `disto` were correct throughout, which is what kept the scope narrow enough to miss. The matrix now records which kind of factorisation its Numeric holds and a kind change forces a full factorisation; both directions are guarded, because a real refactor of a complex Numeric is wrong the same way round.

Four arguments parsed by a weaker parser than the one in the same file. `com_optimize.c` has always had `optnum()` — built on `ft_numparse`, so it reads SPICE suffixes *and* scientific notation — and uses it for the bounds, `-target` and `-spec`. Its integer options did not: `-maxiter`, `-samples` and `-swarmsize` called `atoi()` and `-tol` called `atof()`, which stop at the first character they cannot use. So `1k` meant 1000 in a bound and 1 in `-maxiter` on the same command line: `-maxiter 1k` and `-maxiter 2e2` ran one or two iterations and returned 1495 where `-maxiter 1000` returns exactly 1500, and `-samples 2e2` ran 2 Monte-Carlo samples while design centering still printed a yield and a Wilson confidence interval computed from them. `abc` was 0 in all four, in silence. `sweep lin 2e2` (200 points) and `montecarlo 2e2` (200 samples) were already right — and sweep had grown a strict count parser in E-478 for exactly this reason. All four now parse strictly and refuse a value they cannot use, because a count or a tolerance *is* the request. `-seed` had the matching hole: E-497 taught the `setseed` command to refuse 3.7, but the option spelling truncated it without a word, and `-seed 0 / -seed -3` left the run unseeded and not reproducible while the author had asked in writing for a fixed seed. All four `-seed` sites now apply the command's rule.

"converged" when nothing was optimised. The word describes the search stopping, not the answer being the one the author wanted. A parameter the objective cannot depend on — or a name that does not resolve at all — ends after three evaluations with the starting value handed back as the fit (1 k, 5 k and 9 k returned 1000, 5000 and 9000, each labelled converged); `-target x 0.4 0` printed `sum-sq residual`

= 0, the most convincing number the command can produce, for a fit that never happened; and an answer pressed against a search bound was not distinguished from a minimum. The objective's spread across the run, and the answer's position relative to its bounds, were both available and unused. Said now, as E-438 already says it for failed evaluations and as QPSS-HB says *STALLED above tol — accepted*.

A knob that names nothing. `sw_kind()` falls through to `SW_ALTER` for any name it does not recognise, and the comment on that fallback already describes what follows: `alter` reports "no such device", the sweep runs anyway over a knob that never moves, and the user gets a full set of points, `rc = 0`, and a plottable flat curve whose x-axis is named after the typo. E-435 removed that shape for a subcircuit-local model name and E-488 for `temp`; this removes it for every remaining spelling, exactly as E-431 did for an unresolved `-output`. A malformed `list` value did the same on a smaller scale — `list 1k inf 2k 3k` ran one point and called `inf`, `2k` and `3k` unrecognised, though `list 1k 2k 3k` proves the last two are read without complaint. The list check is deliberately narrow, firing only for a token that reads as a number and is still unusable, because the token after a list is very often the next knob.

-overlay discarded the timepoints the solver chose. It resampled onto a uniform grid of the same point count as the longest run. A transient places its points where the waveform moves, so keeping the count and throwing away the placement is exactly backwards: the overlay of an RC driven by a 0.2 us pulse reported a peak of 0.244 where every one of its own runs, and a standalone run, said 0.402 — 39% low, from the same 121 points. It now uses the union of the runs' own timepoints, bounded to a small multiple of the old size, and says which grid it used. Its per-point vector names were formatted with %g, six significant digits, so `list 1000.0001 1000.0002 1000.0003` produced three vectors all called **vn_1000** of which only one was addressable; they now carry the shortest text that reads back as the exact value, as E-494 settled for the same class of defect.

Two warnings that contradicted each other. A point whose analysis did not converge is recorded as NaN (E-445) and *was also* counted as an unresolved output, so one run said both *recorded as NaN, not as results* (true) and *those entries are zero* (false) about the same points. An output is now counted as unresolved only where the analysis converged.

Unchanged: a real fit is not annotated; every legitimate knob spelling (bare device, `@dev [param]`, `temp`, `deck .param`), a good `list` and a two-knob sweep still parse; a transient sweep and the reuse tally are untouched, so the KLU guard cannot be mistaken for disabling refactoring; and a legal `-seed` still pins the run.

devices/vsrc/vsrc-temp.c, osdi/osdisetup.c — per-run state left stale by the setup-reuse fast path (E-498)

E-471 lets `sweep`, `optimize` and `montecarlo -warm` keep a circuit standing between points: `CKTdoJob` skips `CKTunsetup/CKTsetup` and re-runs only `CKTtemp`. That reasoning was about topology, and for topology it is right — node collapse is re-decided on every `CKTtemp`, and reuse is offered only to device types whose topology is fixed. What was never asked is a different question: *which state does DEVsetup initialise that is not topology at all?* For almost every device the answer is none. For two of them it is a value the transient run walks forward and expects to find reset at time zero.

A voltage source stopped scheduling breakpoints entirely. `VSRCbreak_time` is the source's own breakpoint cursor: `VSRCaccept` arms the next edge only when `CKTtime >= VSRCbreak_time`, and `vsrcset.c` seeds it to -1.0 so the first accepted point of a run arms the first edge. Under reuse that line never ran again and the instance carried the previous run's break time — a value at or past that run's TSTOP — so the test was false at time zero and stayed false for the whole run. The consequence is not a perturbation: the source armed no breakpoints at all, and the stepper walked straight over every edge. A reused point landed on 0 of 5 corners of a piecewise-linear pulse where a standalone run landed on all five (81 native timepoints against 103).

Measured against a standalone run of the identical circuit, `maximum(v(n))` — a grid-independent quantity, so this is a different answer and not a different sampling — was +16%, +44% and -10% across a five-

point sweep, and 106% at other spacings, in silence. A 0.3 ppm change of the swept value was enough, because the schedule is either armed or it is not. Only a sweep's *first* point still had a correct schedule, so the same resistance returned two different answers depending on the direction of travel: $R = 1000$ gave 0.40217511 sweeping up and 0.33963131 sweeping down.

optimize inherits it through E-472. Fitting R so that $\text{maximum}(v(n))$ reaches 0.20 gave 2501.777336 with reuse off — which puts the peak at 0.2000000003 — against 2156.690117 with it on, giving 0.2264, 13% out, in 106 evaluations rather than 67. Both printed *converged*.

This contradicts E-471's own escape-hatch comment, which promises that reuse *changes nothing a user can see except the time* and moves a value by a few ulp. The magnitude is fourteen orders of magnitude past that, and the cause it names is refuted as well: KLU and Sparse, two entirely different orderings and factorisations, produce bit-identical deviations.

OSDI's **last_crossing** cache. `crossing_time[]` is per-run state too, and `osdiaccept.c` states its contract outright — the cache is left unchanged when no qualifying crossing is found, so $V(z)$ keeps reporting the last crossing per the LRM, and it *starts at 0.0 before any crossing has been observed*. Only OSDIsetup seeded it, so a reused point began holding the previous point's crossing: a 100 kHz sine over 40 us made `last_crossing(V(in), 1)` read $3e-05$ at time zero of points 2 to 5 instead of 0. Its sibling analog operator was already right — `absdelay` re-seeds `delay_hist[k][0]` in `osdiload.c` on `is_init_tran`, the first transient call of each run rather than at setup — and that is the pattern this now follows.

The fix. Both are re-armed in the device's `DEVtemperature` method, `VSRCtemp` and `OSDItemp`. `CKTtemp` runs once per job on both paths, reuse and rebuild — the reuse branch calls it explicitly, to re-decide OSDI node collapse — and does not run on a resume, because `CKTdoJob`'s reset branch is skipped there, so a continued transient keeps the schedule it was paused with. Setup keeps its own seed, now as the allocation-time default rather than the only place the value is ever set.

Why the existing suites stayed green. `reusesetup_examples` and `reuseloops_examples` contain no PULSE or PWL source between them — they exercise `tran 10u 1m`, but only with sources that were provably unaffected. The two shipped examples that do combine sweep, `tran` and a PULSE both use `overlay`, whose resampling onto a common grid damps the error below their tolerances. Both moved when this was fixed: `sweepwave_demo`'s `vout_2000[10]` went from $2.441425e-02$ to the correct $2.440859e-02$.

Found here, deliberately not fixed here. Under KLU, `sweep -analysis ac` (and noise) with the reuse fast path returns 0.0 for the reused points and crashes outright in roughly one run in ten (SIGTRAP). It is non-deterministic — byte-identical decks give different answers run to run — which points at memory rather than arithmetic. It reproduces on the shipped binary, is absent with `reusesetup=0`, absent under Sparse, and absent from a manual `alter` plus `ac` loop, so it belongs to the reuse path, but it is a different defect in a different layer and needs its own investigation. It is reported rather than quietly tested around; the two controls that would touch it are asserted under Sparse.

analysis/dsetparm.c, analysis/nsetparm.c, maths/misc/randnumb.c, frontend/inpcom.c + five XSPICE models — constraints the documentation states and the code did not check (E-497)

Round 56 tried a technique worth keeping: mine the manual for stated numeric constraints, then ask of each whether anything enforces it. Three of the five findings came out of one `pdftotext ngspice-manual.pdf | grep -E "must be|should be"`.

THE **disto** TWO-TONE RATIO WAS UNVALIDATED. The manual is explicit `-- f2overf1 "should be a real number between (and not equal to) 0.0 and 1.0"` -- and the ratio is not decorative, because it sets the second tone. Measured on a reactive circuit, the $2F_1$ - F_2 product read 1.695 at ratio 0, 1.630 at 1, 1.477 at 1.5 and 1.580 at -0.5, against 1.711 for a legal 0.5. At ratio 1 F_2 equals F_1 , so the plot `ngspice` still labels "IM: f_1 - f_2 " holds a product at DC; at a negative ratio the second tone sits at a negative frequency. All

accepted in silence, and the numbers look ordinary either way.

Both neighbouring cases in the SAME switch already test their value and return E_PARMVAL; D_F20VRF1 was a bare job→Df2ovrF1 = value→rValue;. Refused rather than clamped: the ratio IS the experiment being asked for, and any substitute would answer a different question without saying so.

The guard is NARROWER than the manual, and deliberately so. The first version took the manual at its word and refused everything outside (0,1); E-255's suite caught it in the very next regression. That suite measures the two-tone IM3 at f1 = 1.0 GHz and f2 = 1.3 GHz and proves the result machine-exact against an independent QPSS harmonic-balance engine, so F2 above F1 leaves all three products at distinct non-zero frequencies and is well posed -- keeping F2 below F1 is a convention, and a working verified deck is better evidence than the sentence in the manual. Refused now: only <= 0 (the second tone at DC or a negative frequency) and == 1 (F1-F2 would be DC and 2F1-F2 would be F1, so the plots would not hold intermodulation products at all).

setseed SILENTLY TRUNCATED A FRACTIONAL SEED. %d stops at the first character it cannot use, so 2.5 scanned as 2. Every other bad spelling is named (*"Cannot use 0 / -3 / abc as seed!"*) and the sibling command repeat names a fractional count outright (*"bad repeat argument 3.7"*), so this was the one way to be wrong quietly.

s_xfer INDEXED **int_ic** BY THE WRONG ARRAY'S SIZE. The initialisation loop reads **PARAM(int_ic[den_size - 2 - i])** -- den_size being **PARAM_SIZE(den_coeff)** -- and nothing anywhere consults **PARAM_SIZE(int_ic)**, though the manual states int_ic "must be of size one less as the array of values specified for den_coeff". Too short and the initial conditions the array does not reach read as zero, taking v(out) at 10 us from 7.00005 to 4.99995e-05; too long and the surplus was never looked at.

THE OSCILLATOR FAMILY WENT SILENTLY DEAD. sine, square, triangle and oneshot each announced a control/frequency array mismatch and returned WITHOUT SETTING AN OUTPUT, on every evaluation: rc=0, the source held at zero all run, and 2025 message blocks over a 2 ms transient -- about a million for a 1 s run. This is precisely the shape E-491 fixed in s_xfer's cfunc.mod and explained there (*"say it once and stop, the way file_source does when its file will not open"*); four models were left doing the old thing. d_osc and d_pwm perform the same check inside their INIT block and so already say it once -- deliberately untouched.

A DUPLICATE .param WAS THE ONLY ONE OF ITS FAMILY TO PASS IN SILENCE. .func has been reported since E-491, .model and .subckt long before that, .param not at all -- and it resolves the OTHER WAY from two of them, taking the LAST definition where .model/.subckt keep the FIRST. So two included files that each set vdd agree or disagree purely by include ORDER, and a .param written in the deck is silently displaced by a library included after it; both measured. Which value wins is not changed -- decks depend on last-wins -- it is only made audible, scoped to TOP-LEVEL cards by the same .subckt nesting count that already governs the .func scan, because a subcircuit's own parameters are legitimately redefined once per instance.

Also corrected, with no demonstrated consequence: dsetparm.c's D_STOP and nsetparm.c's N_STOP each reset the start frequency field when it was the stop frequency that was refused. Both return E_PARMVAL before either field is read, so nothing observable followed -- but the line said the opposite of what it did.

Deliberately unchanged: every meaningful disto ratio -- 0.001 to 0.999 AND above 1, including E-255's 1.3 -- and single-tone disto, plus the neighbouring points/start/stop checks; setseed with an integer, with surrounding blanks, and its existing messages; s_xfer with a correct int_ic, with none at all, and E-491's num>den guard firing first; a matching cntl_array/freq_array on all three oscillators; a single .param, two names on one card, a subcircuit's own parameters (three shapes), and the .func/.model/.subckt messages.

frontend/dotcards.c, frontend/breakp2.c, frontend/outitf.c + three headers — a plot keyword is not a signal name (E-496)

Reported from use, not from a hunt. With `.option saveused` in the deck,

```
pyplot v(a[0]) xlabel 'something'
```

printed *"save 'xlabel': nothing of that name is in this analysis, so no such vector is produced"*. `xlabel` is the plot command's own grammar; the author never asked for a vector of that name, and the message tells them one is missing.

WHAT WAS WRONG. E-469's `.option saveused` reads the control block before it runs and saves every vector it believes is mentioned there. Its bare-word scan took EVERY argument of an output command that was not a number, a redirection or an expression, with no knowledge of those commands' own keywords:

written	collected as vector names
plot v(a) xlabel 'x' ylabel 'y' title 'T'	xlabel, ylabel, title
plot v(a) vs v(b)	vs
meas tran m1 FIND v(b) AT 50u	tran, m1, find, at

All 22 plot keywords, `hardcopy` likewise, and `meas` worst of all, since it names its analysis, its result and its function as bare words.

THE ANSWER WAS NEVER AFFECTED. E-469 over-collects deliberately -- its own comment says under-saving would cost the answer -- and a `save` matching nothing produces nothing: the vector the author asked for comes back bit-identical with zero or three stray keywords collected beside it. The names were gathered in SILENCE until E-493 added the unmatched-save warning, which exposed this flaw rather than causing it.

THE FIX IS IN TWO PARTS, AND THE ORDER OF IMPORTANCE IS THE REVERSE OF THE OBVIOUS ONE.

(1) An INFERRED save is never reported. E-493's warning exists to catch a name the AUTHOR wrote and got wrong -- `.save v(nosuch)`. A name `saveused` guessed on their behalf is not that. Registration now records which it is (`db_auto` on `struct dbcomm`, carried into `save_info.autosaved`), and the warning skips the inferred ones. The flag is set only around `ft_saveused()`'s own `com_save()` call, so `.save`, `save`, `.probe` and `savecurrents` are untouched and still report an unmatched name exactly as before. This single change covers every keyword of every command, including any that part 2 fails to enumerate.

(2) The grammar is skipped by the scan. The plot family (`plot`, `pyplot`, `gnuplot`, `hardcopy`) now knows its own keywords, and `meas`/`measure` contribute no bare words at all -- the vectors they read arrive as `v(...)/i(...)`, which the reference scan already takes from every line whatever command it belongs to. This stops the pointless work, but on its own it would be the E-487 trap: a hand-maintained list that silently rots as keywords are added, and whose failures are invisible. It mirrors `CT_PLOTKEYWORDS` in `cpitf.c`, which cannot be reused because that table is built for tab completion and populated only in an interactive session. A keyword missing from the copy costs nothing but the old noise for that one word -- which is exactly why part 1, not part 2, answers the report.

Deliberately unchanged: everything about WHAT IS SAVED, pinned by comparing every value against the same run without the option -- a plain `plot`, a `plot` carrying keywords, `plot ... vs ...`, `meas` then `plot`, and `let r = ...` then `plot r` are identical either way; `all` still means everything; `wrdata`'s leading file name is still not a vector; an explicit `.save` still makes `saveused` stand aside entirely; a node GENUINELY NAMED `title` or `vs` keeps the value it has without the option, so the keyword skip

cannot cost a real vector; and `.save v(nosuch)`, `.probe v(nosuch)` and `.save v(nosuch)` beside `.option saveused` all still warn.

parser/inpgval.c, parser/inpgmod.c — a value nothing checked had been read ([E-507](#))

INPgetValue's IF_REALVEC path tests INPevaluate's error and refuses the value. Its IF_REAL and IF_INTEGER paths took the same error and threw it away, so a token the parser cannot read became 0 and was applied as if the deck had asked for zero.

numparam makes that ordinary rather than exotic: it substitutes the text of a `{...}` expression, and `{1/0}` substitutes `inf`. `.model nm nmos level=1 vto=0.5 kp={1/0}` therefore built a transistor with `kp = 0`, conducting `1e-12` against `1.25e-4` — exit code 0, no diagnostic. The same value written `inf` on that card is refused, and so is `{1/0}` on an instance line, as a built-in device's value, in an analysis argument and through `altermod`: the model card was one path out of six, and the conversion to zero happened *before* the range check, so a `from` clause could not catch it either.

INPgetValue cannot simply return NULL for these types the way its vector path does — none of its call sites test for NULL, so that would trade a wrong number for a null dereference. The failure is recorded (INPlastValueError) and the model-card path asks for it, that being the one caller with no other way to find out. It reports and keeps the parameter's default, which is what the surrounding `.model` conventions already do for an unknown or duplicated parameter.

parser/inpptree.c — an ordinary leaf was not in the release switch ([E-507](#))

PT_PLACEHOLDER is node type 0 and a normal leaf (inpptree.h: *"for i(something)"*), but it was missing from the expression-release switch, so freeing one fell to the default arm and printed *"Internal error: unhandled parse-tree node type 0 while releasing an expression"*. That reached users on an error path, where a partially built tree is released — `B1 a 0 v={z}` with `.param z={1/0}` — announcing an internal fault ahead of the real, correct diagnostic. Like every other leaf it has no child to release.

frontend/outitf.c — `save` called an operating-point variable a missing parameter ([E-507](#))

INPName both finds a name and asks the device for its value, and any failure of the second half arrived here as the first half's message. An operating-point variable is registered IF_ASK like any other askable parameter, so the name resolves — but at `save` time no analysis has run, so the ask fails and the user was told *"device has no parameter 'gv'"* about a name the device does have and that `print` and `meas` both resolve. The netlist `.save` form reported the same case correctly, so the two spellings of one request disagreed. Only E_BADPARM produces that message now.

The compiler-side halves of this enhancement are in the `openvaf-r` report.

parser/inpgmod.c, analysis/dctrcurv.c, numparam/xpressn.c, frontend/rawfile.c — seven ways a decision made once was never revisited ([E-495](#))

Round 55 probed MOSFET model binning, which no earlier round had touched, and found it soft in three ways. The shape it exposed -- a decision taken once and never asked again -- then led into the DC sweep, which had the same defect twice more.

THE BINNING TOLERANCE WAS ABSOLUTE, AND THE VALUES ARE METRES. `is_equal()` tested `fabs(a - b) < 1e-9`, which on a channel length is a slop of one nanometre applied whatever the geometry. A device up to 1 nm outside EVERY declared bin was silently placed in one -- `l=31n` bound to a bin reaching 30n, while 31.1n was refused, so the edge really was the fixed constant. Because the magnitude does not scale, as a FRACTION of the device it grows without bound as processes shrink: 0.03% of a 3 um width, but 5% of a 20 nm channel. The comparison is now relative (1e-12).

ADJACENT BINS OVERLAPPED, AND **.model** ORDER DECIDED. `in_range()` states its rule in its own comment as `min <= value < max`, and then also accepted `is_equal(value, max)`, closing the interval so a device on a shared boundary matched both neighbours. Reversing two **.model** cards moved `l=1.999u` -- a length unambiguously inside the LOWER bin -- into the upper one and changed `i(V1)` by 2.95x on an otherwise identical deck. Selection now runs the strict half-open rule first and the closed reading only where the strict rule matched nothing; that second pass is what still admits a device sitting exactly on the TOP bin's `lmax`, and because it never runs when the strict rule found a bin, nothing that selects a bin today can move.

AN OSDI MODEL COULD NOT BE BINNED AT ALL. The binnable set was eleven hardcoded built-in names, so a Verilog-A model written exactly as a BSIM PDK writes one died with *"Unable to find definition of model nv"* -- for a model defined twice, the symptom rather than the cause, and the same shape as the resistor named `r` in E-493. OSDI types are now asked through the predicate E-323 already provides, and the four bin limits are consumed on the card the way `level` is.

A DEGENERATE DISTRIBUTION SPEC WAS SILENTLY FLATTENED. `agauss/gauss` are right to refuse a variation or sigma that is zero or negative, but did it without a word, and the result does not look like a failure: every draw returns the nominal, so a Monte Carlo over a parameter that never moves reports a yield of 100% where the real spec gave 12% -- `agauss(1000,100,3)` against `agauss(1000,100,0)`, one character apart, silent under every option including `warn_physics`. Now audible once per run per function; the behaviour is unchanged because a deck may legitimately zero a variation. `unif`, `aunif` and `limit` are deliberately untouched -- a negative variation there describes the same symmetric interval as its absolute value.

A **.dc** SWEEP NEVER REVISITED WHAT SETUP DECIDED. E-471's own comment says **.dc** "sets the circuit up once and walks its points inside the analysis", and that a reused setup freezes the topology so "the sweep quietly draws a flat line". It gave the **sweep** command the machinery to notice and rebuild; **.dc** never got it. So a swept `l/w` leaving the bin the device was PARSED into kept the old bin and every point past the boundary used the wrong model (2.9x out), and a swept parameter changing an OSDI node collapse kept the matrix built for the old topology and returned a FLAT LINE -- -1.000e-03 at `rs = 0, 500 and 1000`, where the answers are -1e-3, -6.67e-4 and -5e-4.

Both siblings are correct: `alter` re-selects the bin through `if_set_binned_model()`, and `sweep` runs a whole job per point. Rebuilding a matrix mid-analysis is far wider than this evidence supports and the correct command already exists, so **.dc** now REFUSES the point it cannot compute and names `sweep` -- E-485's rule that a wrong answer is worse than a refusal. The detection is exact and free when nothing moves: the collapse is read from the flag `OSDItemp ALREADY` sets on every parameter write (E-417) and that `CKTdoJob` already consumes -- `DCTsetInstParam` calls `DEVtemperature` and simply never asked -- and the bin is checked by reading the four limits off the model the device is bound to. The refusal deliberately does NOT borrow E-427's message about a value "the device refused ... also refused by `alter`", which would be false on both counts.

THE ASCII RAWFILE WAS ONE DIGIT SHORT OF A DOUBLE. `#define DEFPREC 15` is the precision of a `%.*e`, so it emitted SIXTEEN significant digits where seventeen are needed before an IEEE754 double reads back as itself; write then load changed values in the last ulp while the BINARY format, which stores the bits, was exact. Over 200000 random doubles `%.15e` fails to round-trip 51390 of them and `%.16e` none. The comment calling 15 "(max)" was wrong twice.

Deliberately unchanged: bin selection that works today, in either card order, including the top bin's `lmax`; `alter` and `sweep`; `unif/aunif/limit` and a healthy `agauss`; a value the device really refuses; an ordinary source sweep, an `m` multiplier sweep, and a **.dc** sweep of a single UNBINNED model; the binary rawfile and an explicit `rawfileprec`.

parser/ptfuncs.c, frontend/inpcom.c — two ways a B source expression disagreed with every other value path (E-494)

Round 54 compared the B source expression path against the paths that read the same number on the same deck. It answered differently twice: once about the sign of a value, once about the value itself.

x/0 LOST ITS SIGN. E-491 removed a gmin-derived fudge factor from PTdivide and replaced it with a fixed epsilon, writing the nudge as `arg2 = (arg1 >= 0.0) ? PTDIV_EPS : -PTDIV_EPS;`. The intent was to keep the sign of the numerator; the effect was the opposite, because a negative numerator was then divided by a NEGATIVE epsilon and the quotient came out positive either way. `B0 b0 0 v=v(p)/0` with `v(p) = -3` returned `+3e+32` where it should return `-3e+32`, and the constant spellings `-1/0`, `(0-2)/0`, `-1.0/0.0` and `-1/(1-1)` were all positive too, as was `E0 b0 0 vol='-1/0'`. Every case E-491 measured had a positive numerator, which is why its own suite did not see this.

A divisor of exactly zero carries no sign to recover, so the real choice is which side zero is approached from, and it has to be made once rather than per operand. The epsilon is now unconditionally positive -- zero from above -- so the quotient keeps the sign of the numerator, which is what $\lim(x/\epsilon)$ as ϵ tends to zero from above gives, and what the comment E-491 left in place already described. Everything else E-491 established is pinned unchanged: the magnitude of `1/0` is still `1e32`, an ordinary non-zero divisor is still used exactly as written, and `1/1.38064852e-23` still does not move when gmin does.

NUMERIC LITERALS WERE ROUNDED TO 11 SIGNIFICANT DIGITS. Every B source line is rewritten before parsing by `inp_modify_exp()`, which re-emitted each numeric literal it found with `"%18.10e"`. So `B0 b0 0 v=1.2345678901234567` reached the parser as `1.2345678901e+00`, a relative error of `1.9e-11`, while the *same literal* written on an R, C or V card, in a `.param`, or as an OSDI `.model` parameter kept all seventeen digits. The B source was the only path in the simulator that could not carry a double.

It was not merely a printout, because the rounding reached the arithmetic: `B0 b0 0 v=tan(1.5707963267948966)` returned `-1.96e11` against libm's `1.633123935319537e+16`. The tangent was right; its argument was not.

MORE DIGITS WOULD NOT HAVE FIXED IT. `INPevaluate()` reads this text back on the way in and accumulates the mantissa by hand as `mantis = 10 * mantis + digit`, which loses a bit once the accumulator passes 2^{53} . Emitting more digits than the value needs is therefore not free: a fixed `"%.17e"` turns the sixteen digits of `0.7853981633974483` into eighteen and brings it back 1 ulp out, when the user's own text would have survived intact. The new `inp_num_text()` emits the shortest decimal text that reads back as the same double, trying 15, 16 then 17 significant digits and keeping the first that round-trips. Seventeen digits always suffice for an IEEE754 double, so a finite value cannot fall out of the loop unmatched.

Deliberately not widened. `INPevaluate`'s own scaling still costs 1-2 ulp on a few values, but a V source shows it identically on the same literals, so the residue is shared by every value path and is a wider question than the one measured here; the suite pins parity and a 2 ulp bound rather than bit-exactness, against an 11-digit rounding that had put those same values `1.0e5-1.6e5` ulp out. And `mkb()` in `inpp-tree.c` folds a constant / with a raw division and no zero guard, bypassing PTdivide entirely, but probed from B, E and G sources in constant, parenthesised, nested and mixed forms it never produced an infinity -- every constant division still arrives through PTdivide with the corrected sign -- so the branch appears unreachable from user input and was left alone rather than fixed past the evidence.

Deliberately unchanged: the sign and magnitude of a positive `x/0` and its gmin independence; ordinary divisors (`2/4`, `-2/4`, `2/-4`, `-2/-4`, `0/5`, `1/1e-30`); the SPICE suffixes `100p`, `5MEG`, `1k`, a leading-dot `.5` and exponent forms; `pwl` B sources, which `inp_bsource_compat()` excludes from the rewrite; current B sources, node references mixed with literals, and the value surviving a transient; and E/G expressions, which never passed through `inp_modify_exp()` at all, since only b lines do, and were already exact.

parser/inp2n.c, frontend/subckt.c — mixing shorthand and written-out bits on a bus port (E-490)

`.option autobus` (Enhancement-444) lets one netlist token stand for a whole Verilog-A bus port, and it fires on a token count equal to the `PORT` count. Positional binding covers the other complete form, a count equal to the `TERMINAL` count. A line that leaves one port in shorthand while writing another port's bits out is `NEITHER`, so it fell through both and bound positionally against the flat terminal list. For `N1 a b[0:2] bmix against inout [0:4] a; inout [0:2] b`, the token `a` went to `a[0]`, `b[0]` to `a[1]`, `b[1]` to `a[2]` -- measured on a model whose eight bits each carry a different conductance, every node sat one or more terminals off and every current was wrong.

The only thing said was Enhancement-402's warning naming the terminals left over at the tail: the symptom, not the cause. A user who supplies the two nodes it asks for still has a circuit wired entirely wrong, and now with no warning at all. Enhancement-445's diagnostic -- "already carries an index, so it cannot be expanded as the bus port" -- exists for exactly this mistake, but sits `INSIDE` the port-count check. The guard and the failure were one `if` apart.

Nothing here is ambiguous, so the fix does not guess. It walks the ports left to right and lets each token declare which form it is in: a bare name on a bus port is shorthand for that port's bits; a token already carrying an index -- or ground, which E-445 established can never be indexed -- means that port was written out, so take one token per bit. `N1 a 0 0 0 bmix` reads correctly under the same rule. The rewrite is accepted only when the walk consumes **EXACTLY** the tokens the line has, and the walk is bounded by that count as well, because `gettok_instance` cannot tell where the node tokens end: without the bound, a port claiming more bits than the line has left would swallow the model name and report against it.

Where no reading exists, it refuses. If the counts do not reconcile then the bits written do not match the width the model declares, and nothing can repair that. The refusal happens where the port and both counts are still in hand to say so, naming the port, its width, the tokens written, and the two counts that would have worked. This replaces a wrong answer, not a working deck; warning and binding anyway is the detect-announce-then-use-it-regardless shape Enhancement-485 had to undo eight times in one round.

One rule, one reader. Deciding the mixed form needs to know whether a token already carries a bit index, and that is spelling-dependent: `a[0]` always, `a_0_` only while `.option autobus=kicad` is on. `subckt.c` already answered it for `.subckt` formals, and a second copy in `inp2n.c` would have been free to disagree about the KiCad spelling -- the two-readers-of-one-rule shape Enhancement-454 had to repair in this same option -- so the rule moved to `INPbusTokenIndexed`, beside `INPbusBitSuffix`, and `subckt.c` now delegates to it.

Two smaller fixes in the same option. `.option autobus=1` reported a style that does not exist: the on-word list held `true`, `yes` and `on` but not `1`, the mirror of the `0` in the off-word list. The comment on the early return explains how -- "bare flag, or =1: a `NUMBER`, not a string" -- which is true of `set autobus=1` but not of a deck `.option card`, which publishes a string. The feature worked; the message was simply wrong. And Enhancement-445's own message named a terminal where a port was meant, reporting a port declared `[0:2]` as "the bus port 'b[0]'", an index the user never wrote and could not act on.

Unchanged by design: with `.option autobus` off nothing moves, the whole path being gated on it; an all-shorthand line and an all-explicit line take the paths they always did; a short line with no shorthand token -- every bit written out, trailing terminals omitted -- keeps E-402's warning and the `$port_` connected idiom it protects; and E-445's two refusals still fire on a full-width line.

Measured and needing no fix, in the survey that preceded this change: ports of unequal width, descending `[4:0]` and non-zero-based `[1:5]` ranges, a 16-bit port, three buses of sizes 2/3/4, bus/scalar/bus interleaving, one token feeding two ports (which superposes exactly), uppercase tokens, `.option autobus` placed after the instance, two instances in parallel, `m=` multipliers, all four subcircuit shapes including reversed formals, `autobus=kicad` on unequal widths, and too-many-nodes as a hard error. A

[0:0] port connects correctly in both spellings; its node is named a rather than a [0], which is inherent -- at a full token count a width-1 bus is indistinguishable from a scalar.

frontend/com_sweep.c — sweep temp swept nothing (E-488)

sweep resolves a knob as a .model parameter, an instance/device parameter, or a symbolic deck .param. The GLOBAL circuit temperature is none of those, so a bare temp fell through to the instance/device branch, alter temp=... found no such device, and the sweep ran to completion over a knob that never moved -- a full set of points, rc = 0, and a perfectly plottable FLAT curve, which reads as "temperature has no effect on this circuit". Enhancement-431 removed that shape for an unresolved -output, and Enhancement-435's own comment in this file describes it for subcircuit-local model names; temp was a third instance, and the one users are most likely to meet, because every OTHER route already worked (.option temp=, set temp=, alter @dev[temp]=, sweep @#[temp]).

The obvious implementation does not work. Writing ckt->CKTtemp -- what .dc temp does -- left the curve flat, because CKTdoJob opens with ckt->CKTtemp = task->TSKtemp. .dc survives that only because its entire sweep runs INSIDE one CKTdoJob; sweep runs a fresh analysis command per point, so the write was discarded before the next point was solved. The TASK is what has to move, so the knob is applied by issuing option temp= -- how the frontend already moves it, and the same shape as the alter/altermod this file uses for every other knob. It also carries the value through the guarded OPT_TEMP funnel in cktsopt.c instead of around it, inheriting Enhancement-426's absolute-zero refusal and Enhancement-440's sanity check rather than copying them.

Deliberately NOT identical to **dc temp**. Over a node collapse that moves with temperature, dc temp returns 0.0 where a static op at the same temperature returns 0.5 -- it holds one setup for the whole sweep and never rebuilds (round-24, still open, NOT addressed here). sweep runs a fresh analysis per point and Enhancement-471's logic rebuilds where the collapse moves, so it is right. The suite pins all three series against the static op and asserts that dc temp DISAGREES, so a later pass that "fixes" sweep into agreement fails rather than quietly adopting the wrong answer.

Strictly additive: a deck .param temp is tested FIRST and still wins, so a deck defining and referencing its own temp sweeps exactly as before. Also matched to the oracle: an unphysical range is refused up front and sweeps nothing (left to the funnel alone, bad points are ignored one at a time and the axis still carries the value, so a row LABELLED -600 C held the answer for whatever temperature was in force), and the axis carries SV_TEMP so plot labels it as dc temp's does.

Not fixed: sweep's other unresolvable knobs -- a bogus name, a missing device, a missing parameter -- still warn and then sweep flat. Only temp names something that exists.

analysis/dcpss.c, com_hbosc.c, com_hb.c, typedef.c — RF results that were printed and then lost (E-487)

An RF result that is only PRINTED cannot be plotted, printed again, wrdata'd, diffed against a reference, or read by a script. Six of the eight RF entry points published a nutmeg plot; hbosc and phasenoise printed a table and stored nothing, leaving the session with hbosc's own startup transient as the current plot (hbosc runs a tran internally to seed the oscillation, so setplot after a successful run reported Current tran1).

One signature extended, its sibling not. Enhancement-209 added struct hbspectrum *out to HBanalyze so com_hb could publish its spectrum as vectors. HBOSCanalyze computes the same (2K+1)*N two-sided spectrum in the same layout and never got the parameter — the same shape E-486 spent its length on. It now takes it, on the same ownership contract, and PhaseNoiseAnalyze gains a matching struct pnspectrum *out that collects the curve beside the table it was already printing.

One publisher, not two. com_hb's hb_publish() was static; copying it into com_hbosc.c is how the two drifted apart to begin with. It is now hb_publish_spectrum(), shared and parameterised, so the

driven and autonomous cases cannot diverge again. `hbosc` additionally publishes `osc_freq`, because for an autonomous circuit the frequency is part of the ANSWER rather than an input.

The plot NAME is where this gets subtle. `ft_plotabbrev()` matches by SUBSTRING, so "hbosc" hit the hb pattern and the oscillator spectrum came out named hb1 — indistinguishable from a driven run in the same session — while "phasenoise" hit noise and collided with `.noise`. Both now have entries ahead of the pattern that shadows them, which is the rule E-383 wrote into that table. `plotorder_examples` asserts that rule for "every name in the tree", but its NAMES table is HAND-MAINTAINED and was passing over these two vacuously; it now knows about them, and mis-ordering the hbosc entry makes it fail with `hbosc->hb (want hbosc)`.

Types. `loadpull` created every vector `SV_NOTYPE` including two with a real type available; `pout_dbm` and `gain_db` are dB and now carry `SV_DB`. `gamma_re/gamma_im`, `pae`, `eff` and `stb's loopgain` are deliberately left untyped — they are dimensionless ratios and the enum has no member for that.

Not changed: `loadpull` leaves its intermediate transients in the session, 61 plots for `-n 9` and 333 for `-n 21`. That is a real problem, but discarding plots a user may want to inspect is a behaviour change on a shipped command, so it is recorded rather than made silently.

analysis/cktsens.c, xspice/enh/enhtrans.c, xspice/mif/mifmpara.c, eleven `icm/` code models — a guard one sibling had and the other did not ([E-486](#))

Twelve sites of one shape: the check exists somewhere in the tree, and is simply absent next door. Often the sibling is in the same directory; once it is ten lines away in the same function; once it is the same file's own history.

table2D segfaulted on a file that declares its own size. The data-row loop is driven by the FILE — while `(*cThisPtr)` — and wrote `table_data[lLineCount - 1]` with no upper bound, while every *other* dimension of that same file was checked: the x row, the y row, the width of each individual data row, even a premature EOF inside the comment block. A file declaring 3 y values and supplying 5 data rows indexed past the allocation and crashed — `EXC_BAD_ACCESS` at address 0 in `cm_table2D` — rc 139, no diagnostic, from a twelve-line table file. Too FEW rows was the mirror: the shortfall stayed as `calloc's` zeros, the table went to `sf_eno2_set` as though complete, and a probe in the missing region returned `0.0`, a physically plausible "no current". E-247 had already worked in this exact file ("fix OOB READ + interpolation UB on degenerate/too-small tables"), addressing the read side and the too-small case and leaving the WRITE and the too-many case. The sibling `table3D` refuses the truncated file outright, so the two disagreed about one input and only one of them crashed. Both directions are now bounded, and `table3D` — bounded by its `iz/iy` loops, hence never crashing — now warns about the surplus it had been dropping in silence.

count_steps() signalled an error with a value used as a count. It is declared to return a POINT COUNT, and E-362's overflow guard signalled failure with `return(E_PARMVAL)`. `E_PARMVAL` is 11, and the sole caller assigned the result straight to `nfreqs` with no error test, so an "impossible" sweep ran exactly eleven points — and because that return happened before `*stepsize = s`, the step was never written and every frequency after the first collapsed to zero. Beneath it sat two silent repairs of values the USER STATED: a DEC/OCT sweep starting at 0 Hz had `low` rewritten to `1e-3`, and a stop at or below the start had `high` rewritten to a decade above it. Both rewrote only the LOCAL copy, so the count came from the repaired bounds while the sweep still ran from `job->start_freq` — which is why `.sens ac dec 5 0 1meg` printed a full table of 0 Hz rows and `.sens ac dec 5 1k 1k` swept a full decade past the stop asked for (6 rows where `.ac` gives 1). The arithmetic predicts each observed count exactly: 6 for dec, 3 for oct, 5 for lin. `.ac`, `.noise`, `.disto` and `.sp` all get both cases right, so the rules are now `.ac's` own (`acan.c`); errors leave the function out of band in an `int *errp`, and `*stepsize` is written on every path.

Ten more of the same shape. `d_state's cm_set_indices()` presets its indices to 0 and returned FALSE — "no error" — when NOTHING matched, so a transition naming a state the file never defines ran ROW

0 silently (proved by construction: row 0 of 0s 0s holds the outputs at 0, row 0 of 1z 1z holds them at 1). `file_source`, the file-driven sibling of `pwl`, had no monotonicity check at all, so a time column that steps backwards, repeats, or starts below zero silently produced a different waveform. `xfer`'s `table` path refuses a negative or out-of-order frequency while its `file` path, ten lines below in the same function, validated nothing. `hyst`'s `input_domain` was unbounded where `pwl/pwlts` declare `[1e-12 0.5]`, letting an `input_domain` of `1e6` drive the output to 250000 against declared limits of 0 and

1. core called `PARAM_SIZE` exactly once, on `H_array`, and indexed `B` with it. `mlin`, `cpLine` and `cpmLin` had no limits on any geometry — a negative microstrip length returned 0.5011 against 0.4992 for the same line at `+1e-2`, a plausible answer from a geometry that cannot exist. `poly()` reported each of its two faults with the OTHER one's message. And `d_state`, `d_process`, `real_gain` and `oneshot` were the only models of twenty-plus with no declared limit on their delays.

Deliberately unchanged, with the built-in devices as the oracle. NEGATIVE capacitance and inductance are not defects: the built-in `C` accepts `-1u` and produces exactly the sign-inverted response the XSPICE model produces, and the built-in `L` diverges the same way (`7.5e+288` against `1.8e+285`, both ending in the same timestep abort). Only `c/l` of `ZERO` is fixed — a real division by zero that had surfaced as "Timestep too small; cause unrecorded". `mlin`'s `rho = 0` was first classified as an ordinary perfect-conductor idealisation; measuring it returned a bare NaN, so it is strictly positive after all, while `t`, `tand` and `d` at zero stay legal. `xfer`'s duplicate frequencies stay legal because the `table` path allows them — the fix gives the file path *that* rule, not a stricter one. `oneshot` is bounded at `[0 -]` rather than the digital `[1e-12 -]`: it is an analog model, only a negative delay was measured to do harm, and a zero rise delay works.

`git -S` puts both headline defects at `4f29ffad`, the vanilla upstream import — pre-existing upstream defects, not regressions from this tree.

device noise routines — a nominal temperature added to a difference ([E-403](#))

`res/resnoise.c`, `bjt/bjtnoise.c`, `dio/dionoise.c`, `vbic/vbicnoise.c`, `hicum2/hicum2noise.c` all computed the thermal-noise temperature delta as

```
dtemp = inst->REStemp - ckt->CKTtemp + (model->REStnom - CONSTCtoK);
```

`NevalSrcInstanceTemp` evaluates `4*k*(CKTtemp + param2)*G`, so `param2` is a DELTA. The third term adds the NOMINAL temperature expressed in CELSIUS (27 by default) to that difference, so an instance sitting at ambient got `+27 K`: `rd x 0 1k temp=27` read 4.4% more noise voltage (9% power) than the identical `rd x 0 1k`. Instance temperatures are stored in Kelvin, so the first two terms were already the whole answer; the third is dropped. `temp=100` and `dtemp=73` -- the same temperature written two ways -- now agree exactly, at the analytic ratio.

Surfaced as `sens` poisoning every later noise in a session: `sens` restores parameter VALUES but leaves the `...tempGiven` flag set, which flips the noise routine into the branch above. Correcting the arithmetic fixes both, since an instance at ambient then contributes a zero delta whether or not the flag is set.

devices/dev.c (51)

- Duplicate device-type registration warned and skipped: `osdi_add_device()` appended every descriptor unconditionally, and the model-card lookup scans front-to-back — so a module name duplicated across two loaded libraries silently resolved to the *first* registration (loading an updated model library gave the stale device with no hint), and a module named like a built-in corrupted model creation. Registration now checks the table case-insensitively and skips duplicates loudly ([E-76](#)).
- **pre_osdi** path dedup: loading an already-loaded file notes it and skips ([E-76](#), [E-81](#)).

- **pre_osdi -f** force reload (E-229): the three dedup gates (path, dlopen handle, device-name) plus dlopen's own path cache meant a recompiled `.osdi` never took effect without restarting ngspice. A `-f/-force` flag (parsed in `com_osdi`, `frontend/com_dl.c`) now force-reloads: `load_osdi(path, force)` stages a byte-for-byte copy of the recompiled file under a fresh unique path and loads *that* (a new path is always re-read, sidestepping the cache; no `dlclose`, which on macOS does not reliably unmap and where overwriting a mapped file faults the process), removes the staged copy once mapped, and `osdi_add_device(..., replace)` swaps the registered device to the new descriptor in place at its existing table index. The old SPICEdev + mapping are left resident so a circuit still built against the prior model stays valid. Without `-f` the re-load is still skipped (now with a hint pointing at `-f`).

osdi/osdiparam.c (E-93)

Setting a fixed (**localparam**) OSDI parameter from the netlist used to be swallowed silently: `openvaf` exports every `localparam` as a parameter, its "given" flag is forced false, and the written value is overwritten by the parameter-init default — so `.model ... N=8` on a frozen structural width parameter changed nothing, with no feedback. `OSDIparam/OSDImparam` now test the new `PARAM_FLAG_FIXED` descriptor flag (set by `openvaf`, a free bit of the parameter `flags` field) and emit *"parameter 'N' is a fixed (localparam) value and cannot be set from the netlist; ignored"* instead of silently dropping it (E-93).

6. Maths — **src/maths/**

dense/dense.c (11 substantive lines; the file also carries a whitespace/line-ending normalization from editing)

cadjoint() had no 1×1 base case: its cofactor loop allocated 0×0 and then negative-sized minors, killing ngspice with `malloc: can't allocate -8 bytes` — the crash that had been hiding behind the "at least two ports" guard in `span.c`. The adjugate of `[a]` is `[1]` (the empty minor's determinant is 1 by convention), making `cinverse` of a 1×1 equal $1/a$; a $100\ \Omega$ load on a $50\ \Omega$ port now writes a proper `.s1p` with $S_{11} = 1/3$ exactly (E-64).

sparse/spfactor.c (6)

The Markowitz-product overflow guards compare a double against `LARGEST_LONG_INTEGER (= LONG_MAX)`, which is not exactly representable as a double; the conversion is now explicit — identical semantics, intentional and warning-free (E-77).

KLU/klu_multiply.c (16)

`SMPmultiply`'s KLU path passed `NULL` internal↔external ordering maps to `KLU_matrix_vector_multiply`, which dereferenced them unconditionally (`&NULL[n] = 0x14` for `n = 5`) and segfaulted — so `.option linesearch` crashed under `.option klu`. The line search is the only caller of `SMPmultiply` under KLU, so this path had never been exercised. A `NULL` map now means the identity ordering (`ext = i + 1`, since the KLU CSR is 0-based and the RHS/Solution vectors are 1-based), making the KLU matrix-vector product — and hence the E-111 line-search residual merit — correct under KLU, with a merit sequence numerically identical to Sparse 1.3 (E-112).

KLU/klusmp.c (SMPcaSolve)

`SMPcaSolve` is the complex adjoint (transposed) solve used by noise and S-parameters. Its Sparse branch calls `spSolveTransposed`, but the KLU branch called the *non-transposed* `klu_z_solve` — silently wrong for any asymmetric matrix (every circuit with a transistor or controlled source), which is why `.noise` was disabled under KLU. It now calls `klu_z_tsolve` (transposed), so KLU noise matches

Sparse exactly (including OSDI device models). This is the core of [E-113](#), which also removes the KLU refusal guards in `noisean.c` / `pzan.c` (single-ended pole-zero runs under KLU). Balanced-output pole-zero was Sparse-only at first but [E-172](#) closed that too (a KLU `SMPcAddCol` branch plus a union-pattern reservation in `CKTpzSetup`), so no analysis is Sparse-only under KLU any more.

spicelib/analysis/cktsens.c

Sensitivity builds an auxiliary perturbation matrix `delta_Y` (holding $\partial Y / \partial p$) via `SMPnewMatrix`, which allocates a plain Sparse 1.3 matrix (`delta_Y->CKTkluMODE = 0`, no `SMPkluMatrix`). It is only *multiplied* against the solution (`SMPmultiply` $\rightarrow$ `spMultiply`), never factored, and `DEVbindCSC` always binds into `ckt->CKTmatrix`, not `delta_Y` — so it is correctly Sparse in every case. But two KLU-only setup blocks were gated on the main matrix's `CKTkluMODE`, so under `.option klu` they ran and dereferenced the NULL `delta_Y->SMPkluMatrix` — a segfault on every DC/AC `.sens` deck. The blocks now gate on `delta_Y->CKTkluMODE` (its own flag, `0`), keeping `delta_Y` Sparse under KLU exactly as it already is under the default solver, while the main `Y` matrix stays KLU. DC and AC sensitivity now match Sparse exactly ([E-114](#)).

A second, independent defect: the analysis altered the circuit it measured. Each derivative is taken by perturbing a parameter, reloading, and writing the original value back — which restores the *number* but not the model's given state, since every device setter marks its parameter as supplied and no device API offers an un-set. For a model whose behaviour is selected by whether a parameter was given rather than by its value, the model is left reinterpreted. The BJT is the sharp case: `ibe/ibc` default to 0 and ungiven, and `bjttemp.c` keys off `BJTBESatCurGiven` && `BJTBCsatCurGiven` — ungiven, the junction saturation currents fall back to `is`; given, they are taken literally, and `sens` had just made them given with the value 0. `BJTBESatCur` became 0, a transistor with no saturation current at all, and every later analysis in the session solved the dead device: 12.8% wrong on a single stage, 101% on a differential pair, with no diagnostic and cured only by reset. It was not a tolerance artifact (bit-identical at `reltol` 1e-3 and 1e-12), not a stale initial guess (`.nodeset` did not change it, and `sens` run first corrupted just the same), and of twenty commands driven against the same probe — `alter`, `altermod`, `sweep`, `montecarlo`, `optimize`, `tf`, `pz`, `disto`, `noise`, `hb`, `pss`, `sp` — `sens` was the only one that did it. `sens_save_models` / `sens_restore_models` now snapshot every model struct before the perturbation loop (device-agnostically, via `DEVmodSize` and `ckt->CKThead[]`), restore them after it and on the error paths, and re-run `CKTtemp()` so each instance's derived values are rebuilt from the restored models ([E-440](#)).

spicelib/analysis/distoan.c

Distortion is a complex analysis (Volterra series solved at the harmonic / intermodulation frequencies via `NIDIter` $\rightarrow$ `SMPcSolve`), but `distoan.c` had no KLU code at all: under `.option klu` the KLU matrix stayed in *real* mode, the complex solves ran against an unconverted matrix, and every harmonic came back zero — a silent wrong answer. The fix mirrors `acan.c`: before the solve loop it converts the matrix to complex (`DEVbindCSCComplex` per device + `KLUmatrixIsComplex`), and on exit converts back to real (`DEVbindCSCComplexToReal`) so a later real analysis in the same session is unaffected. KLU distortion now matches Sparse bit-for-bit (single-tone, two-tone intermodulation, and OSDI models) ([E-115](#)).

misc/equality.c (AlmostEqualULps)

`AlmostEqualULps` is ngspice's ULP-distance float comparison — the primitive behind breakpoint matching, `.measure` triggers, transient truncation, PSS, and source timing (18 files). It reinterprets each double as an `int64_t`, maps the sign, and compared them with `intDiff = llabs(aInt - bInt)`. That signed subtraction overflows `int64_t` when the two keys straddle zero, and `llabs(INT64_MIN)` is itself undefined — surfaced by a UBSan build. It is not merely theoretical UB: `AlmostEqualULps(-2.0, +2.0)` returned TRUE (declaring `-2` and `+2` "almost equal"), because those two keys differ by exactly `INT64_MIN`, whose `llabs` stays negative and slips under `maxULps`. The difference is now taken in `uint64_t` as larger-minus-smaller, where the wraparound is well defined and the true magnitude

of any two-`int64_t` difference always fits; the comparison against `maxUlp`s is unsigned. Behavior is identical wherever the old code did not overflow, and correct (`FALSE`) where it did. Found by the 2026-07 ASan/UBSan pass.

7. Build system

xspice/icm/makedefs.in (4)

The XSPICE codemodel (`.cm`) link rule hardcoded the pre-macOS-10.3 idiom `-bundle -flat_namespace -undefined suppress`, deprecated loudly by the modern linker on all 14 codemodel links (CI macOS builds included). Now `-bundle -undefined dynamic_lookup` — the modern spelling for plugins whose host symbols resolve at load time (E-77).

Autotools regeneration (the ~100 **Makefile.in** files + **config.h.in**)

Regenerated by `./autogen.sh` with `libtool 2.5.4` during the zero-warning work — a build tree whose configure predates `libtool 2.4.7` selects the deprecated `darwin -undefined suppress` flag whenever `MACOSX_DEPLOYMENT_TARGET` is unset. No hand-written content (E-77).

8. Per-enhancement index

Every enhancement that touched `ngspice`, oldest first:

E-1 *osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c, osdisetup.c, dctran.c (+accept path)*

`absdelay()`: synthetic delay rows + waveform history

E-6 *osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c*

`last_crossing()`: history + interpolated crossings

E-8 *parser/ingtok.c*

`.model` card first-parameter drop fix

E-24 *osditrunc.c*

`\$discontinuity` next-step clamp

E-25 *osdi callbacks (get_simparams)*

expose `analysis_name` + simulator `simparams`

E-32 *outitf.c*

integer `opvars` recorded per point

E-42 *osdinoise.c*

coherent same-named noise grouping (LRM 4.6.4)

E-45 *osdi.h (ABI 0.5), osdisetup.c*

Verilog-A `nodesets` applied to the solver

E-51 *osdi.h (ABI 0.6), osdiacld.c*

full `ac_stim` AC-RHS injection

- E-53 *dctran.c, dcop.c, dctrcurv.c, acan.c, noisean.c, osdiload.c*
 @(final_step) + analysis-name phase bits
- E-54 *osdi.h* (ABI 0.7), *osdinoise.c, osdiload.c, osdiacld.c, osdiregistry.c*
 node-free complex noise factors, stride-2 pairs
- E-55 *osdiaccept.c* (new), *osdiload.c, osditrunc.c, osdiinit.c, dctran.c, dctrcurv.c, osdiitf.h, Makefile.am*
 \ \$finish/\ \$stop/\ \$fatal honored; \ \$discontinuity step rejection
- E-56 *noisean.c, osdisetup.c*
 noise SIGABRT fix; setup-rejection diagnostics
- E-62 *dctrcurv.c, trcvdefs.h, cktdisto.c*
 generic .dc @inst[param] sweeps; .disto warning
- E-63 *span.c*
 donoise NaN clamp (stock defect, found by parity)
- E-64 *span.c, cktspdum.c, dense.c, postcoms.c/.h, commands.c*
 Touchstone export, auto-Rbase, 1-port .sp, cadjoint 1x1
- E-72 *postcoms.c/.h, commands.c*
 Touchstone round 2: MA/DB/Y/Z/units + rdsnp reader
- E-76 *dev.c, inpgmod.c*
 duplicate-registration warning, load dedup, stock .model segfault fix
- E-77 *osdidefs.h, display.c, outitf.c, spfactor.c, makedefs.in* (+autotools regen)
 zero-warning build, 33 → 0
- E-80 *osdiinit.c*
 dtemp instance-parameter alias
- E-81 *resource.c, dev.c*
 memory-warning opt-out + once-per-excursion; reload-note hint
- E-93 *osdi.h, osdiparam.c*
 warn (not silently ignore) when a netlist sets a PARA_FLAG_FIXED (localparam) OSDI parameter
- E-94 *plotting/pyplot.c* (new), *com_pyplot.c* (new), *plotit.c, commands.c*
 new pyplot command — plot vectors with matplotlib (a Python gnuplot)
- E-95 *com_pyplot.c, commands.c*
 make the pyplot output file name optional (defaults to pyplot)
- E-98 *plotting/pyplot.c*
 pyplot stacked subplots (pyplot_subplots=N) + matplotlib style sheets (pyplot_style)
- E-99 *plotting/pyplot.c*
 pyplot vector export formats (pyplot_terminal=svg/pdf) + figure size (pyplot_figsize)

E-110 *optdefs.h, tsdefs.h, cktsopt.c, cktntask.c*

`.option errpreset=conservative|moderate|liberal` — one knob for a coordinated tolerance/robustness set; explicit options override regardless of order (`moderate` = historical defaults)

E-111 *optdefs.h, tsdefs.h, cktdefs.h, cktsopt.c, cktdojob.c, cktntask.c, cktdest.c, niiter.c*

`.option linesearch` — globalized (damped) Newton via Armijo backtracking on a new KCL-residual merit $\|F\| = \|G \cdot x - b\|$; result-neutral, off by default

E-112 *maths/KLU/klu_multiply.c*

KLU support for `.option linesearch` — `SMPmultiply`'s KLU path passed NULL ordering maps that `klu_matrix_vector_multiply` dereferenced (SIGSEGV); NULL now means identity ordering. Line search verified merit-identical KLU vs Sparse

E-113 *maths/KLU/klusmp.c, spicelib/analysis/noisean.c, pzan.c*

KLU support for noise + single-ended pole-zero — `SMPcaSolve`'s adjoint KLU branch used the non-transposed solve (silently wrong noise on asymmetric matrices); now `klu_z_tsolve`. Guards removed; balanced-output pz keeps a targeted guard

E-114 *spicelib/analysis/cktsens.c*

KLU support for sensitivity — the auxiliary perturbation matrix `delta_Y` is Sparse, but two KLU setup blocks gated on the *main* matrix's flag dereferenced its NULL `SMPkluMatrix` (segfault on every DC/AC `.sens`); now gated on `delta_Y`'s own flag. DC/AC `.sens` match Sparse exactly

E-115 *spicelib/analysis/distoan.c*

KLU support for distortion — `.disto` is a complex analysis but `distoan.c` had no KLU code, so under KLU the matrix stayed real and every harmonic came back zero (silent wrong answer); now converts real↔complex around the solve loop like `acan.c`. Matches Sparse bit-for-bit

E-116 *osdi/osdisetup.c*

KLU wrong-DC fix — an OSDI internal node with no Jacobian entry (a decoupled ground reference) was given an all-zero solver row that made the KLU matrix structurally singular; now tied to ground. Fixes the `groundcontrib` and `hierbranch` KLU_XFAILs

E-117 *configure.ac (+configure), spicelib/analysis/dcpss.c*

Periodic steady state (PSS) productionized — built by default (was experimental `--enable-pss`, so `.pss` was unimplemented in shipped builds); ~230 lines of shooting-loop trace routed through `set ngdebug` (232 → 31 lines by default); fail-fast KLU guard (PSS hangs under KLU) directing `.pss` to `.option sparse`

E-118 *spicelib/analysis/dcpss.c*

PSS runs under KLU — the shooting loop hung under KLU (`klu_refactor`'s reused pivots inflated the truncation error into a ~20M-step timestep explosion); forcing a full re-factor (NISHOULDREORDER) each PSS step makes KLU converge to the same result as Sparse. E-117's Sparse-only guard removed

E-119 *pssdefs.h, spicelib/analysis/dcpss.c*

Retain the PSS periodic operating point — PSS sampled the node voltages per period for its DFT then freed them, and never captured device states; now the voltages and `CKTstate0` states are captured per sample and retained on the `PSSan` job (with frequency + dims) as the substrate for the periodic small-signal suite (PAC/pnoise/PXF). Self-checks that the retained samples reproduce the fundamental

E-120 *spicelib/analysis/dcpss.c*

Periodic small-signal Jacobian harmonics — walk the retained operating point, re-linearize each sample (CKTload MODEINITSMSIG) and stamp $G+jC$ (CKTAcLoad $\omega=1$), read the osc-node diagonal $\rightarrow g(t)$, $c(t)$, DFT to harmonics. The blocks the PAC conversion matrix is built from. (Fixed a complex-mode-not-set bug where $C(t)$ accumulated across samples.) Verified: RC gives $G=1/R1$, $C=C1$, no harmonics

E-121 *spicelib/analysis/dcpss.c*

PAC conversion-matrix engine — extend E-120 from the osc diagonal to every matrix nonzero, complex-DFT to harmonics G_k , C_k , assemble the $(2M+1)N$ harmonic conversion matrix $H_{\{nm\}}=G_{\{n-m\}}+j\omega_m \cdot C_{\{n-m\}}$, inject a unit current at the osc node in sideband 0 and solve by dense complex LU (pss_solve), report the sideband conversion gains. The numerical heart of PAC/pnoise/PXF. Verified: linear RC gives sideband-0 = AC driving-point $|Z|(f0/2) = 303.3 \Omega$ with the ± 1 sidebands at floating-point zero (no spurious conversion)

E-122 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

.pac command (periodic AC) — the user-facing analysis on the E-121 engine. .pac Fguess StabTime OscNode Points Harmonics SC_iter Steady_coeff <dec|oct|lin> Npts Fstart Fstop runs PSS then sweeps the input frequency, solving the conversion matrix at each point and emitting the 0-th-sideband node responses as a complex PAC Analysis plot(print/plot/wrdata). Reuses the PSS analysis via a PSSdoPAC flag; engine factored into pac_extract_harmonics (once) + pac_solve_at (per freq). Verified: swept sideband-0 $|b(f)| ==$ analytic AC driving-point $|Z(f)|$ to $1.6e-7$ across 10 kHz–1 MHz

E-123 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

finish .pac — (1) source-referenced stimulus: capture the netlist AC-source RHS from CKTAcLoad as the sideband-0 stimulus B_0 (a periodic-AC transfer/conversion gain), unit-current-at-osc-node fallback; (2) multi-sideband output: optional trailing maxsideband Ksb emits every conversion sideband $f_{in}+k \cdot f0$ ($k=-Ksb \dots Ksb$) as its own vector — sideband 0 keeps the node name, others <node>_usb<k>/<node>_lsb<k> via IFnewUId. Verified: RC with V1 AC 1 gives sideband-0 = low-pass transfer $1/\sqrt{1+(2\pi fRC)^2}$ ($0.998 \rightarrow 0.157$) with $b_{usb1}/b_{lsb1} \sim 0$ (no conversion)

E-124 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

.pnoise command (periodic noise) — fold each device's noise through the conversion matrix. pac_solve_adjoint solves $H^T \Psi = e_{\{out,0\}}$; pnoise_sweep loads the sideband-k adjoint into CKTrhs/CKTirhs and calls the existing device noise routines (NevalSrc, OSDI load_noise) once per sideband, accumulating $\Sigma_k S \cdot |\Delta \Psi_k|^2$ — reuses every device noise model via a minimal local NOISEAN context. .pnoise <pss> OutNode InSrc <dec|oct|lin> Np Fstart Fstop; emits onoise_spectrum/inoise_spectrum. Verified: linear RC reduces exactly to .noise ($4kTR/(1+(2\pi fRC)^2)$), matches the .noise reference to every digit)

E-125 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

.pxf command (periodic transfer function) — the adjoint of PAC, completing the PSS $\rightarrow$ PAC $\rightarrow$ Pnoise $\rightarrow$ PXF suite. pxf_sweep solves $H^T \Psi = e_{\{out,0\}}$ per frequency and dots each sideband block with the AC-source pattern $B0 \rightarrow x_{f_k} = \Sigma_j \Psi_k(j) \cdot B0(j)$, the input $\rightarrow$ output transfer at each sideband. .pxf <pss> OutNode <dec|oct|lin> Np Fstart Fstop [maxsb]; emits $x_f + x_{f_usb<k>}/x_{f_lsb<k>}$. Verified: by the identity $(H^{-1}B)_{out} = (H^{-T}e_{out})^T B$ the sideband-0 transfer is bit-identical to the PAC response ($0.998 \rightarrow 0.157$ low-pass), conversion sidebands $\sim 2e-16$

E-126 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

cyclostationary noise — `.pnoise ... cyclo` evaluates each device's noise PSD $S(f)$ at every PSS sample's bias (CKTload per sample) and folds it through the time-domain adjoint transfer $A_s(j\omega) = \sum_k \Psi_k(j\omega) e^{j2\pi k s/P}$, averaging: $\text{onoise}(f) = (1/P) \sum_s S(f, t_s) \cdot |\Delta A_s|^2$. Reuses the device noise routines. Verified: (1) reduces exactly to `.noise` on the linear RC (bias-independent thermal $\rightarrow$ Parseval); (2) a flicker resistor carrying the RC current gives $\text{onoise} \cdot f = R I^2 \cdot K_F \cdot \langle I^2 \rangle = 4.88e-10$ (uses the period-average $\langle I^2 \rangle$, matched to 5 digits)

E-127 *include/ngspice/{optdefs,tskdefs,cktdefs}.h, spicelib/analysis/{cktsopt,cktnntask,cktdojob,cktop,cktdest}.c, maths/ni/niiter.c*

pseudo-transient continuation — `.option ptcont` embeds $f(x)=0$ in a backward-Euler pseudo-transient $f(x) + G_{ps} \cdot (x - x_{\text{prev}}) = 0$ and marches $d\tau$ small $\rightarrow$ large ($G_{ps} \rightarrow 0$). G_{ps} diagonal via the gmin path; the $G_{ps} \cdot x_{\text{prev}}$ RHS coupling (added in NIiter) makes each step a stable-trajectory move, not a static gmin step. New `pseudo_transient` in the CKTop cascade, off by default. Verified under KLU+Sparse: result-neutral on normal circuits; on a stiff no-limiting exp, reaches the correct DC 0.837922 V where plain Newton lands on a spurious 70.5 V (21/21)

E-128 *include/ngspice/{optdefs,tskdefs,cktdefs}.h, spicelib/analysis/{cktsopt,cktnntask,cktdojob}.c, spicelib/analysis/dctran.c*

LTE-based dynamic integration-order control — `.option dynorder` selects the Gear order per step from the local-truncation-error limit instead of the stock $1 \leftrightarrow 2$ toggle, so orders 3–6 (already coded in NIcomCof but never used) are actually exercised. The selector compares the raw LTE-limited step (uncapped CKTtrunc) at the current order and its ± 1 neighbours and moves with hysteresis ($1.2\times$), a settling hold after each change (let the BDF divided differences rebuild), and an order-dependent step-growth cap ($2\times$ at order $\leq 3 \rightarrow 1.3\times$ at 6). Off by default; bounded by `maxord`; inert at the default `maxord=2`. Stiff-transient guards (post-breakpoint order hold + leaky-bucket rejection-rate order drop) keep it from collapsing the step on a violent slew. Verified under KLU+Sparse: RC decay $3\text{--}5\times$ fewer steps at matched accuracy with monotone error; smooth RLC ringdown $8.9\times$ fewer steps AND more accurate (0.13% vs 0.34%); nonlinear rectifier matches stock to 5 sig figs; the transistor-level $\mu A741 \pm 5 \text{ V}$ slew completes under `dynorder` and matches fixed Gear-2

E-129 *frontend/outitf.c*

sweep progress bar — the throttled Reference value status line (redrawn in place every 0.25 s during a sweep) now carries a live progress bar + percentage, e.g. Reference value : $5.91926e-04$ [=====] 74%. `outp_progress_frac()` computes the 0–1 fraction per analysis: transient from $(\text{CKTtime} - \text{TSTART}) / (\text{TSTOP} - \text{TSTART})$, AC/noise from the frequency's linear/log position in the band, DC from the accepted-point count over the nested step-count product; analyses with no span (op, ...) keep the plain line. Fixed-width (no stale chars), std-out status-line only (never in the rawfile/wrdata), same `!ft_norefprint` && `!cp_background` gating. Verified 22/22: printed % matches the analytic sweep fraction to 0.5 % across tran/AC/DC/noise; op shows no bar

E-130 *frontend/com_optimize.{c,h}, commands.c, options.c, include/ngspice/ftext.h, spicelib/analysis/cktdojob.c, frontend/outitf.c*

built-in Nelder-Mead optimizer — `optimize -param <name> <init> <lo> <hi> [-param ...] -analysis <cmd> -minimize <expr> [-maxiter N] [-tol T] [-verbose]` varies device/alter parameters, re-runs an analysis, and minimizes a scalar objective expression via a derivative-free downhill simplex in normalized [0,1] space (scale-invariant across orders-of-magnitude params). Sub-commands dispatched synchronously through `cp_coms` (not the deferring re-entrant `cp_evloop`); the hundreds of inner analyses are silenced via a new `ft_`

optimizing flag gating the banner/row-count/E-129 progress bar at source (`-verbose` to show). Verified 9/9 against analytic optima: DC divider $R1 \rightarrow 2333.3 \Omega$, AC low-pass $R1 \rightarrow 2756.6 \Omega$, 2-D compound objective $R1=3k/R2=2k$ exactly

E-131 *frontend/com_checkpoint.{c,h}, commands.c, com_commands.h, Makefile.am, spicelib/analysis/dctran.c, include/ngspice/cktdefs.h*

transient checkpoint / restart — new `savestate <file> / loadstate <file>` commands serialize the full transient integration state (solution vector `CKTrhsOld`, device state history `CKTstates[]`, time/step/order/mode, pending breakpoints) to a binary file and resume it, including in a fresh process — so a long run survives a crash, splits across sessions, or moves between machines (stock ngspice could only resume in memory). `Dctran()` gains a `CKTcheckpoint`-gated branch that opens a fresh output plot (no live plot to 666-relink across a reload), initializes the XSPICE breakpoint markers, and fixes up the breakpoint list before jumping into the stepping loop; the rhs length is keyed off `SMPmatSize+1` (not `CKTmaxEqNum`). A stored signature rejects a mismatched circuit; Sparse-solver only (KLU rejected with a clear message). Verified 19/19: resumed waveform bit-identical to an uninterrupted run for RC-step/pulse/diode built-ins and $\sim 2e-7$ for a compiled OSDI diode, across a separate process

E-132 *spicelib/analysis/dcpss.c, spicelib/parser/inp2dot.c, spicelib/analysis/psssetp.c, include/ngspice/pssdefs.h*

periodic S-parameters (`.psp`) — small-signal scattering parameters around a PSS operating point, including conversion between the input frequency and its sidebands $f_{in+k} \cdot f_0$ (mixers / switched circuits, where a static-DC `.sp` cannot see the conversion). Sits on the PSS $\rightarrow$ conversion-matrix suite (E-117–126): after PSS, `psp_sweep` excites each RF port (`portnum/z0`, the `.sp` framework) by driving its branch source ($V=1$, like `.sp`'s `VSRCspupdate`) through the shared $(2M+1)N$ conversion matrix, reads per-sideband port waves in the same Kurosawa power-wave convention, and forms $S^{(k)}=B^{(k)} \cdot A^{-1}$ (dense-complex `cinverse/cmultiply`). `pac_solve_at`'s matrix assembly factored into a reusable `pac_build_matrix`; `PSSdoPSP` flag + `psp` PSS param + `dot_psp` card. Because $S=B \cdot A^{-1}$ is excitation-basis-invariant, sideband 0 reduces exactly to `.sp` for a time-invariant network. Runs under both linear solvers (the conversion matrix is a standalone dense LU; PSS runs under both since E-118). Verified 8/8: sideband-0 matches `.sp` to $\sim 1e-16$ for 1/2/3-port resistive + reactive networks (magnitude and phase) incl. OSDI Verilog-A devices (G + reactive `ddt` stamps), conversion sidebands correctly ~ 0

E-133 *frontend/com_qpss.{c,h}, commands.c, com_commands.h, Makefile.am*

quasi-periodic (two-tone) steady state (`qpss`) — `qpss <expr> <f1> <f2> [periods] [max-order]` computes the two-tone steady-state spectrum: every mixing product $k_1 \cdot f_1 + k_2 \cdot f_2$ including third-order intermodulation (IM3 at $2f_1-f_2 / 2f_2-f_1$) that single-tone AC/PSS cannot show. For commensurate tones (common beat $fb=\gcd(f_1, f_2)$) it runs an ordinary transient over a few beat periods to reach steady state, then evaluates the Fourier coefficient directly at each exact intermod frequency (a direct DFT, exact — no FFT-bin rounding) and labels it by the 2-D index (k_1, k_2) . Deliberately not a slow beat-frequency shooting PSS; a front-end command driving a transient, so solver-independent and works with built-in and OSDI devices. Verified 11/11: analytic fundamentals/IM3/3f of a cubic, no even-order products, the 3:1 IP3 slope law (fund $\times 2$, IM3 $\times 8$ per $2\times$ drive), beat-frequency derivation, and an OSDI cubic matching the built-in

E-134 *spicelib/analysis/dcpss.c, frontend/com_hb.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h*

Harmonic Balance (`hb <f0> <K>`) — solves the periodic steady state in the frequency domain by Newton (each node a truncated Fourier series), instead of integrating in time; the real analysis behind ngspice's unimplemented `WITH_HB` stub. Residual $F_k = I_{R,k}(V) + [dq/dt]_k - I_{s,k} = 0$ with the E-121 $(2K+1)N$ conversion matrix as the exact Jacobian. Per iteration: inverse-DFT $V \rightarrow v(t_s)$; drive DC+AC device loads at those voltages for the resistive current $I_R = G \cdot v$

– b (companion b saved before `acLoad`, so it's the ACTUAL current not the tangent $G \cdot v$), and $G(t)/C(t)$; DFT; dense complex Newton (`pss_csolve`). Nonlinear reactive with NO charge extraction: $dq/dt = C(v) \cdot v'$, so the reactive current is the conversion matrix's $j\omega C$ term applied to V . Solver-independent (KLU + Sparse): the dense complex Newton is HB's own; the sparse solver only *reads* $G(t)/C(t)$ off the device matrix, so `hb_extract` carries the same `#ifdef KLU` complex-CSC binding (`DEVbindCSCComplex`) as the PAC extraction, and `com_hb` copies the task's KLU mode before building so a bare `hb` honours `.option klu` — verified bit-identical under both. Built-in + OSDI. Verified 8/8 vs `transient/fourier`, quadratic convergence: nonlinear R (analytic 3rd harmonic), R+C, a real diode rectifier (junction limiting), OSDI varactor $Q(v)$ 2nd harmonic, KLU==Sparse parity

E-135 *spicelib/analysis/dcpss.c, examples/hb_examples/verify_hb.py*

HB source-stepping continuation — makes E-134 Harmonic Balance robust on strongly-driven circuits where a cold full-strength Newton diverges ($|F| \rightarrow 1e69$). `HBanalyze`'s Newton loop is wrapped in an adaptive homotopy: every independent source scaled by λ : $0 \rightarrow 1$, each level warm-started from the last converged V . The residual becomes $F = I_R + I_C - \lambda \cdot I_s$; on level success V is checkpointed and $d\lambda$ grows ($\times 1.7$), on failure (Newton exhausted, residual non-finite, or singular Jacobian) V is restored and $d\lambda$ halves; $d\lambda < 1e-5 \rightarrow$ reported as no reachable steady state. First level is full strength ($d\lambda=1$) so easy circuits converge at $\lambda=1$ immediately, bit-identical to the plain solve. Automatic (no new syntax); set `hb_verbose` shows the λ ramp, summary reports iterations + continuation steps. Verified 9/9: a strongly-driven diode rectifier (5 V into 20 Ω , sharp $I_S=1e-14$) diverges cold but converges in 3 continuation steps, matching `transient fourier` to $<0.1\%$ for DC/ $f_0/2f_0$; the 8 existing HB checks stay bit-identical

E-136 *spicelib/analysis/dcpss.c, frontend/com_qpss.c, frontend/commands.c, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpss_hb.py*

two-tone Harmonic Balance (`qpss <expr> <f1> <f2> hb [K1] [K2]`) — the TRUE, incommensurate-capable quasi-periodic steady state, a frequency-domain HB engine alongside the E-133 transient `qpss` (which is unchanged, commensurate-only). Each node is a 2-D Fourier series $v(t) = \sum \sum V_{\{k1,k2\}} e^{j\{k1\omega_1+k2\omega_2\}t}$; `QPSShb` samples devices on a 2-D phase grid (θ_1, θ_2) — time never appears, so incommensurate tones (no beat period) just work — and 2-D DFTs to the conversion matrix $H_{\{n\},\{m\}} = G_{\{n-m\}} + j\omega_m \cdot C_{\{n-m\}}$ (`qp_build_matrix`), Newton-solved by `pss_csolve` with the E-135 source stepping. Sources captured by an oversampled least-squares APFT ($\Gamma^H \Gamma$) $I_s = \Gamma^H b$ (a square Vandermonde is catastrophically ill-conditioned past a few harmonics). Reactive needs NO charge extraction ($dq/dt = C(v) \cdot v'$); the converged operating point is retained (`qpss_hb_saved`) for `qpac` (E-137). Solver-independent (KLU + Sparse) — same `#ifdef KLU` complex-CSC binding as PAC/HB. Built-in + OSDI. Verified 7/7: analytic cubic $|IM3(2, -1)| / |3rd(3, 0)| = 3$, even products ~ 0 , 3:1 IP3 slope, incommensurate $\sqrt{2}$ tones (E-133 cannot), HB==transient (commensurate), KLU==Sparse; E-133 `verify_qpss.py` stays 11/11

E-137 *spicelib/analysis/dcpss.c, frontend/com_qpac.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpac.py*

two-tone small-signal QPAC (`qpac <f_in>`) — completes the quasi-periodic gap: the two-tone analogue of PAC. Run after `qpss ... hb`, `QPACanalyze` injects a small signal at f_{in} around the retained QPSS operating point (`qpss_hb_saved`) and reports the response at every sideband $f_{in} + k_1 \cdot f_1 + k_2 \cdot f_2$, mixing it through the SAME 2-D conversion matrix $H_{\{n\},\{m\}} = G_{\{n-m\}} + j\omega_m \cdot C_{\{n-m\}}$ (`qp_build_matrix` at f_{in}) that the QPSS Newton used as its Jacobian. Stimulus placed in the $(0, 0)$ sideband — a netlist AC-source `RHS B0` (captured bias-independent at the op-point) or a unit-current fallback — one dense `pss_csolve`. Adds no device evaluation $\rightarrow$ solver-independent (KLU + Sparse) by construction. Verified 7/7: reduce-to-AC (pump $\rightarrow 0$) $\sqrt{2}$ direct $(0, 0) =$ plain `.ac` response = R, sidebands vanish), v^2 -pump conversion ratio $| (1, 1) | / | (2, 0) | = 2$, equal-tone symmetry, clean no-op-point error, KLU==Sparse; E-136 (7/7) + E-133

(11/11) unaffected

E-138 *spicelib/analysis/dcpss.c, frontend/com_qpnoise.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpnoise.py*

two-tone small-signal QPnoise (qpnoise <output_node> <f_in>) — quasi-periodic noise, the two-tone analogue of pnoise (E-124), on the retained qpss ... hb operating point. Reports output + input-referred noise density at f_in, folding every device's noise over all sidebands f_in+k1·f1+k2·f2 (mixer/PA noise conversion invisible to a static .noise). QPnoiseAnalyze biases the devices at the QPSS operating point, then qp_solve_adjoint transposes the 2-D conversion matrix (qp_build_matrix) and solves $H^T \Psi = e_{\{out, (0,0)\}}$ — one adjoint gives the transimpedance from every (node, sideband) to the output; each device's DEVnoise computes $S \cdot |\Psi|^2$ (reading the transfer from CKTrhs/CKTirhs) and the sum over all Nh harmonics is the folded onoise; input-referred via the QPAC gain. Reads only retained data → solver-independent (KLU + Sparse) by construction. With no pump the conversion matrix is block-diagonal so only (0,0) survives ☐ reduces to .noise. Verified 6/6: reduce-to-noise (pump→0 ☐ onoise == plain .noise == 4kTR exactly), conversion active under pump, inoise=onoise/gain², clean no-op-point error, KLU==Sparse; E-133 (11/11) + E-136 (7/7) + E-137 (7/7) unaffected

E-139 *spicelib/analysis/dcpss.c, frontend/com_qpnoise.c, frontend/commands.c, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpnoise.py*

cyclostationary QPnoise (qpnoise <output_node> <f_in> cyclo) — upgrades E-138 to a time-varying device PSD. Under a two-tone pump a device's bias swings over the period (a diode's shot noise 2qI_D(t) spikes when it conducts), so QPnoiseAnalyze gains a cyclo branch using the identity $onoise = (1/P) \cdot \sum_s S(t_s) \cdot |A_s|^2$, where $A_s(j) = \sum_{\{(k1,k2)\}} \Psi_{\{(k1,k2)\}}(j) \cdot e^{j2\pi(k1 s1/P1+k2 s2/P2)}$ is the inverse 2-D DFT of the adjoint transfers (the time-domain transimpedance at phase sample s). Each sample re-biases the devices at the retained op-point (qp_synth, P1/P2 now stored in qp_harm) with per-sample junction settling (the E-134 MODEINITFLOAT walk — else a limited diode reports a stale bias and the "cyclostationary" result collapses onto the stationary one), evaluates S(t_s), folds through A_s, and averages. By Parseval reduces to the stationary sum (and .noise) when S is constant. Verified 10/10 (6 E-138 + 4 cyclo): cyclo reduce-to-noise, Parseval (bias-independent thermal PSD ☐ cyclo==stationary even under pump), a hard-pumped diode where cyclo differs from stationary by ~8× (switching-mixer cyclostationary enhancement, visible only with the settling), cyclo KLU==Sparse; E-133/136/137 unaffected

E-140 *spicelib/analysis/dcpss.c, frontend/com_hbosc.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/phasenoise_examples/verify_phasenoise.py*

oscillator phase noise (hbosc <oscnode> <K> [fguess] [tstab] + phasenoise <fstart> <fstop> [pts]) — closes the periodic/phase-noise gap with the phase-noise piece. HBOSC-analyze is an autonomous harmonic balance: an oscillator has no source, so $F(V)=I_R+[dq/dt]=0$ is solved for the harmonics AND the unknown oscillation frequency ω_0 . The conversion matrix $dF/dV=H$ is singular (right null space = phase mode $u_k=j\omega_k V_k$, since the oscillator's phase is free), so Newton runs on the bordered system $[H, \partial F/\partial \omega_0; u^*, 0]$ (nonsingular by bordering, $\partial F/\partial \omega_0=I_C/\omega_0$, gauge row u^*), transient-seeded (hbosc runs a short transient from the deck's .ic, reads amplitude + zero-crossing frequency), converging quadratically (LC osc: 4 iters to $|F|=3e-12$; inductors fit the $G+j\omega C$ matrix via branch-current MNA). PhaseNoiseAnalyze builds the adjoint of H at OFFSET df with the unit at the carrier sideband (m=1) and folds device noise (DEVnoise); as $df \rightarrow 0$ the limit-cycle matrix goes singular through the phase mode so the transimpedance diverges as $1/df \rightarrow$ folded noise $1/df^2$, normalized to carrier power $2|V1|^2 \rightarrow L(df)$ in dBc/Hz. Verified 8/8 on an LC oscillator: autonomous HB converges to f0+describing-function amplitude, $L(df)$ has the -20 dB/dec skirt near carrier flattening into the noise floor at a physical level (≈ -147 dBc/Hz @1kHz), thermal scaling $L \propto T$ ($2 \times T \rightarrow +3$ dB exactly, pinning the absolute), clean no-op-point error, KLU==Sparse; pnoise (9/9) + qpnoise (10/10) unaffected

E-141 *spicelib/analysis/dcpss.c, frontend/com_qpxf.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpxf.py*

two-tone small-signal QPXF (`qpxf <output_node> <f_in>`) — the quasi-periodic transfer function, the ADJOINT of QPAC (E-137), completing the two-tone small-signal suite (QPSS→QPAC→QPnoise→QPXF ≡ PSS→PAC→pnoise→PXF). QPXFanalyze runs one adjoint solve $H^T \Psi = e_{\{out, (0,0)\}}$ (`qp_solve_adjoint`, reused from QPnoise E-138) and dots each sideband block of Ψ with the netlist AC-source pattern $B0$ (the QPAC stimulus, captured at the op-point) → the transfer $H_{\{(k1,k2)\}} = \sum_j \Psi_{\{(k1,k2),j\}} \cdot B0_j$ from an input at every sideband $f_{in} + k1 \cdot f1 + k2 \cdot f2$ to the output, all from ONE adjoint. By the reciprocity identity ($H^{-1}B$)_{out}=($H^{-T}e_{out}$)^TB the sideband-(0,0) transfer is bit-identical to the QPAC response (the PXF↔PAC cross-check, E-125). Reads only retained data → solver-independent (KLU + Sparse). Verified 6/6: reciprocity (QPXF(0,0)==QPAC(0,0) bit-identical), conversion-sideband reciprocity, reduce-to-XF (pump→0 $\square$ (0,0)=plain transfer=R, sidebands vanish), clean no-op-point error, KLU==Sparse; QPSS 11/11 + QPSS-HB 7/7 + QPAC 7/7 + QPnoise 10/10 unaffected

E-142 *spicelib/analysis/dcpss.c, frontend/com_qpac.{c,h}, com_qpnoise.c, com_qpxf.c, commands.c, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpss_sweep.py*

input-frequency sweep for the two-tone small-signal analyses. `qpac/qpnoise/qpxf` gain a `<dec|oct|lin> N fstart fstop` sweep of `f_in` that emits a plottable ngspice plot (conversion gain / noise figure / image-rejection curves vs frequency), matching how `.ac/.pnoise/.pxf` sweep. QPACsweep/QPnoiseSweep/QPXFswep step `f_in` and reuse the exact single-frequency solve per point: `qpac` records per-node $|e_{(0,0)}|$ response, `qpnoise` records `onoise_spectrum/inoise_spectrum`, `qpxf` records `xf` (in-band (0,0)) + `xf_conv` (total conversion $\sqrt{\sum |H_{\{sb \neq 0\}}|^2}$). The analysis fills plain arrays; the front-end builds the plot with a frequency scale via the nutmeg vector API (`plot_alloc/dvec_alloc/vec_new`) — the correct layer for a front-end command (the analysis OUTp framework dereferences a live analysis job's JOBtype, which a front-end command lacks → was crashing). Shared helpers `qp_steptype/qp_sweep_maxpts/qp_emit_plot`. Verified 5/5: swept value at 0.3G == single-frequency `qpac/qpxf/qpnoise` (machine precision), reactive roll-off vs `f_in`, correct point count; single-frequency QPAC 7/7 + QPnoise 10/10 + QPXF 6/6 unaffected

E-143 *frontend/com_optimize.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optimize_ls_demo.cir*

gradient least-squares curve fitting for **optimize**. The built-in optimizer (E-130, derivative-free Nelder-Mead on a scalar `-minimize`) gains a least-squares mode: one or more weighted `-target <expr> <value> [<weight>]` measurements, optionally spread over several `-analysis` stages (each `-analysis` opens a stage; following `-targets` attach to it, so a single fit can combine e.g. a DC operating point AND an AC response), are fitted with Levenberg-Marquardt — a forward/backward finite-difference Jacobian J , normal equations $A=J^T J$, $g=J^T r$, damped solve $(A+\lambda \cdot \text{diag}(A)) \delta = -g$ via a small partial-pivot Gauss solver, standard λ up/down loop. Exploiting the sum-of-squares structure it converges in far fewer analysis runs than the simplex (27 vs 67 evals on a two-target RC fit). `-method nm|lm` overrides the auto choice (LS→lm, scalar→nm); the scalar `-minimize`/Nelder-Mead path is unchanged; `-minimize` and `-target` are mutually exclusive. Parameters are still applied in place with `alter` (no re-source) and the search runs in normalized [0,1] space. All changes in `com_optimize.c` (+ one help string); no new files, no ABI change. Verified 23/23 (E-130 checks [1]–[5] + E-143 [6]–[10]): 1-param/3-target/3-stage RC magnitude fit, 2-param DC+AC multi-analysis fit ($R1=3k, R2=2k$), LM-fewer-evals-than-NM, OSDI/Verilog-A diode extraction (recover both $i_s \rightarrow 1e-14$ and $n \rightarrow 1.2$ from two measured I-V points by weighted least squares), and input validation (lm-without-target, minimize+target, multi-analysis+scalar all rejected)

E-144 *frontend/com_optimize.c, frontend/inp.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optimize_dparam_demo.cir*

optimize tunes symbolic **.param** values via a new knob kind **-dparam**. **-param** remains an alter target (device/instance, changed in place); **-dparam** names a netlist **.param** symbol (e.g. **.param w=1u, R1={500*k}**) which — being expanded at parse time — is changed with **alterparam <name>=<value>** + a reset that re-sources the deck (re-evaluating **.param** expressions and re-stamping device values). Per parameter a kind (OPT_ALTER / OPT_DECKPARAM) is tracked; when any **-dparam** is present **opt_eval** applies ALL deck params first, re-sources ONCE, THEN applies the in-place alter params (reset rebuilds from the deck and would otherwise wipe an earlier alter — this ordering is what lets the two kinds mix; verified on a joint **.param**+device fit). No **-dparam** the E-130/-143 fast path is untouched (no reset, no perf hit). The two re-source banners (Reset re-loads circuit ... and Circuit: ... in inp.c) are gated by the existing **ft_optimizing** flag so hundreds of inner re-sources are silent (only the final leave-at-optimum run shows); **ft_optimizing** re-asserted after each reset. Also fixed 5 pre-existing **-wsign-conversion** warnings in inp.c's ***ng_script_with_params** handler (argc/size unsigned). Closes the last E-143 follow-up. No new files, no ABI change. Verified 31/31 (E-130/-143 [1]–[10] + E-144 [11]–[15]): scalar **.param** fit (rtop→2333.3), mixed **-dparam**+**-param** joint fit (rtop=3k, R2=2k), **.param** in expression (k→6), least-squares **-dparam** (LM), quiet re-source (≤1 banner); AC + OSDI-device-value **.param** verified manually

E-145 *frontend/com_optimize.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optres.m, examples/optimize_examples/optimize_mparam_demo.cir*

optimize tunes **.model**-card parameters via a third knob kind **-mparam**. **-param** remains an alter instance target and **-dparam** a symbolic **.param** (re-source); **-mparam <@model[param]>** names a **.model**-card parameter (e.g. **@dmod[is]**, **@rmod[r]**) — which is NOT alter-reachable and NOT **.dc**-sweepable (only sources/resistors/instance params are) — and is changed in place with **altermod <name>=<value>**, no re-source (as cheap per eval as **-param**, unlike **-dparam**). A new kind value OPT_MODELPARAM; in **opt_eval** the deck params are re-sourced first (E-144), then the in-place knobs: alter for instances, altermod for models. **@model[param]** is passed straight to **altermod** (which accepts that accessor form), mirroring how **-param @m1[w]** emits **alter @m1[w]=...**. Circuits with only **-param**/**-mparam** keep the E-130/-143 fast path (no reset). All three knob kinds mix in one run. Change confined to **com_optimize.c** (+1 help line); no new source files, no **inp.c** change, no ABI change. Verified 39/39 (E-130/-143/-144 [1]–[15] + E-145 [16]–[20]): OSDI model-param fit (**@rmod[r]**→3k), built-in diode model-param fit (**@dmod[is]**→1.22e-14), determined model+instance joint fit (r=3k, R2=2k), **-mparam** does 0 re-sources (in-place fast path), and all three kinds (**-dparam**+**-mparam**+**-param**) coexist and converge

E-146 *frontend/com_sweep.c (new), frontend/com_sweep.h (new), frontend/commands.c, frontend/com_commands.h, frontend/inp.c, frontend/Makefile.am, examples/sweep_examples/**

universal **sweep** command + **.sweep** card. Sweeps ANY circuit knob, generalizing **.dc** (which steps only sources/resistors/instance params). **sw_kind** auto-detects the knob: **@X[y]** with **ft_sim->findModel(ckt,X)** non-NULL model param (altermod), else instance (alter); a bare name with **nupa_get_param(name,&found)** found **.param** (alterparam+reset), else device (alter) — so model params and **.params**, impossible with **.dc**, are now sweepable. **sweep <knob> (<start> <stop> <step> | lin|dec|oct N a b | list ...)** [**-analysis <cmd>**] [**-output <name>=<expr> ...**]: sets the knob, runs the inner analysis (default op) at each point, evaluates each output (last value; default = all node voltages like **.dc**), and emits a summary plot named **sweep** with the knob values as the scale (nutmeg vector API, front-end layer per E-142). The per-point analysis plots are retained (**tran1, tran2, ...**). **.sweep** card: **inp.c** collects a top-level **.sweep** line into the post-parse control list as a sweep command; a static **sweep_active** re-entrancy guard stops a **.param** re-source (reset re-runs the **.sweep** card) from recursing. New source files registered in **commands.c/com_commands.h/Makefile.am**. Verified 11/11 (verify_sweep.py): sweep R1 reproduces built-in **.dc** R1 bit-for-bit, model-param (**@rmod[r]**) + **.param** (rtop) sweeps vs analytic divider, auto-detection routing, **.sweep** card == command form (no recursion), AC named-

output vs analytic |H(1kHz)|, transient settled value, lin/list/step point counts, multi-output

E-149 *maths/misc/randnumb.c, include/ngspice/randnumb.h, frontend/numparam/xpressn.c, frontend/numparam/spicenum.c, frontend/commands.c, examples/lhs_examples/**

Latin-Hypercube Monte Carlo sampling (mcsample), closing the low-discrepancy-sampling gap vs commercial tools. A stratified sampler in randnumb.c holds the mode/N/sample-index/dimension-counter/seed; each random dimension's stratum permutation (Fisher-Yates over $0..N-1$) + per-sample jitter are generated lazily from a self-contained splitmix64 seeded by (seed, dimension) — reproducible and order-independent. mc_sample_uniform() = (perm[d][il]+jitter[d][il])/N; mc_sample_gauss() maps it through inv_normal_cdf (Acklam probit, rel err < 1.15e-9) so Gaussian tails stratify too. Sample boundary: spicenum.c's nupa_signal(NUPADECKCOPY) edge (once per reset re-source, guarded by the existing firstsignalS latch) calls mc_sample_advance() (step sample, rewind dimension) before that pass's .param draws. Draw hook (xpressn.c): agauss/gauss take one mc_sample_gauss() and aunif/unif/limit one 2*mc_sample_uniform()-1 when LHS is active, else the unchanged gauss1()/drand() (LHS-off is byte-identical to before). Command com_mcsample (randnumb.c, registered in commands.c both tables) parses lhs <N> [seed <s>] / random / off. Front-end only, so solver-independent. Verified 5/5 under both solvers (verify_lhs.py): stratification (each of 48 strata hit once vs random's ~32/48), two-param multi-dimension, ~130x lower Var(sample-mean) at N=32 (both converge to the same mean), seed reproducibility, analytic mean/sigma correctness

E-150 *frontend/com_sweep.c, frontend/commands.c, maths/misc/randnumb.c, include/ngspice/randnumb.h, examples/_setup.py, examples/highsigma_examples/**

high-sigma rare-event estimation (highsigma), the last statistical ROW versus a commercial simulator -- 4-6 sigma failure probabilities (SRAM / standard-cell yield) that plain MC cannot reach (1e-7..1e-9 needs 1e7..1e9 runs). Scaled-sigma importance sampling: a new MC_MODE_SSS in the E-149 sampler draws each Gaussian .param from the lambda-inflated normal ($z = \text{lambda} * \text{gauss1}()$) and accumulates that draw's log likelihood-ratio $\log(\text{lambda}) - (z^2/2)(1 - 1/\text{lambda}^2)$ into sss_logw; mc_sample_weight() = exp(sss_logw) reweights the sample so the estimator is unbiased for the nominal probability. Uniform .params are bounded so SSS leaves them uninflated (weight 1). Direction-free -- no gradient / sensitivity / most-probable-failure-point. mc_sss_config(N, lambda, seed) seeds the global PRNG for reproducibility; mc_sss_off() reverts. Command com_highsigma lives in com_sweep.c (reuses its sw_run_cmd reset/analysis runner and sw_eval_expr), registered in commands.c: highsigma <N> [-scale <lambda>] [-seed <s>] [-analysis <cmd>] -metric <expr> [-max <hi>] [-min <lo>] loops N samples (reset -> analysis -> eval metric -> spec test -> read weight), reports P(fail), relative error, equivalent one-sided sigma-to-fail ($-\Phi^{-1}(P)$) and failure count, and leaves them in the highsigma_* vectors/vars. The spec is -max/-min values (not a >/< inside the expr, which a control command reads as an I/O redirect). Front-end only, solver-independent; the verify is heavy (thousands of re-sources) so examples/_setup.py marks highsigma SPARSE_ONLY. Verified vs analytic Phi(-beta) (verify_highsigma.py): beta=4 -> sigma 4.000, P 3.16e-5 (true 3.17e-5); deep tail beta=5 (P 2.87e-7) recovered from 6000 runs where plain MC expects ~0.0017 failures; two-sided spec ~doubles P; seed reproducibility exact; two independent Gaussians combine as $N(., \sqrt{s_1^2 + s_2^2})$

E-151 *maths/misc/randnumb.c, include/ngspice/randnumb.h, frontend/numparam/xpressn.c, frontend/com_sweep.c, frontend/commands.c, examples/_setup.py, examples/yield_examples/**

process/mismatch correlations + packaged Monte Carlo yield -- closes the last two partial (warn) statistical rows vs a commercial simulator. (1) CORRELATIONS: mc_corr_config(k, matrix) Cholesky-factors a k x k correlation matrix (row-major, unit diagonal; rejects a non-positive-definite one) into lower-triangular corr_L; mc_corr_component(i) lazily draws the k underlying z's once per sample THROUGH mc_sample_gauss() (so LHS stratification / SSS weighting apply), computes $y = L * z$, caches it, and returns y[i]. The cache is reset in mc_sample_advance()

) which now runs its reset in EVERY mode (so correlation works under plain MC, not just LHS/SSS). New `.param` function `mvnorm(i)` (frontend/numparam/xpressn.c: added to `fmathS`, `XFU_MVNORM` enum in `fmathS` order, and the dispatch) returns the *i*-th correlated standard normal; with no matrix registered it degrades to an independent draw. `com_mccorr(mccorr <k> <m11>. <mkk> | off)` in `randnumb.c`. A reusable `mc_lhs_config(N, seed)` was factored out of `com_mcsample`. (2) YIELD: `com_montecarlo` in `com_sweep.c` (reuses `sw_run_cmd / sw_eval_expr`), registered in `commands.c`: `montecarlo <N> [-lhs] [-seed <s>] [-analysis <cmd>] (-spec <metric> [-max <hi>] [-min <lo>])...` runs *N* samples (reset -> analysis -> eval each spec), counts a sample PASS only if every spec is within limits, and reports the yield with a Wilson 95% score interval and per-spec violation counts, leaving `montecarlo_yield/montecarlo_npass/montecarlo_n`. `-lhs` uses `mc_lhs_config` for a lower-variance estimate; corners are the ordinary `.lib` selection. Front-end only, solver-independent; heavy deck so `examples/_setup.py` marks yield SPARSE_ONLY. Verified (`verify_yield.py`, Sparse): `mccorr rho=+0.7/-0.6 -> empirical corr 0.711/-0.614` with correct means/sigmas; `mvnorm` without `mccorr -> corr ~0`; non-PD matrix rejected; single two-sided spec yield 0.8660 (true $P(|Z| < 1.5) = 0.8664$); two independent specs multiply (0.7508 vs 0.7506); `-lhs` unbiased and ~800x lower yield-estimate variance; positive correlation raises the joint yield (0.75 -> 0.82). Demo: a matched divider yields ~100% process-correlated ($\rho=0.9$) vs ~74% independent

E-152 *include/ngspice/optdefs.h, include/ngspice/tskdefs.h, include/ngspice/cktdefs.h, include/ngspice/smpdefs.h, spicelib/analysis/cktsopt.c, spicelib/analysis/ckntask.c, spicelib/analysis/cktdojob.c, maths/ni/niinit.c, maths/KLU/klusmp.c, examples/klu_tuning_examples/**

KLU matrix reordering + scaling controls. The KLU direct solver ran on the compiled-in `klu_defaults` (AMD ordering, max row scaling, BTF on) with no way to change them, and its one exposed knob `klu_memgrow_factor` was broken (`task->TSKkluMemGrowFactor = (val->rValue == 1.2)` stored a boolean, not the value). E-152 exposes three new `.options --klu_ordering=amd|colamd, klu_scale=none|sum|max, klu_btf=on|off` -- as IF_STRING options (friendly names, mapped to KLU's integer codes 0/1, 0/1/2, 1/0 in the `cktsopt.c` handlers) and fixes the memgrow assignment. Plumbing mirrors the existing memgrow path exactly: new `OPT_KLU_ORDERING/SCALE/BTF` enums (`optdefs.h`); `TSKklu*/CKTklu*` int fields on the task (`tskdefs.h`), circuit (`cktdefs.h`) and SMPmatrix (`smpdefs.h`) structs; defaults 0/2/1 that MATCH `klu_defaults` set in `ckntask.c` (so behaviour is unchanged unless the user overrides) plus the task copy; `task->ckt` in `cktdojob.c`; `ckt->CKTmatrix` in `niinit.c`; and in `klusmp.c` `SMPnewMatrix`, right after `klu_defaults()`, `Common->ordering/scale/btf = Matrix->CKTkluOrdering/Scale/BTF`. KLU-only; changes only HOW the matrix factors, never the physical solution; the numerics are untouched. Verified (`verify_klu_tuning.py`, KLU-only) on a resistor grid: every ordering/scaling/BTF setting gives the physically identical solution (max relative spread $2.8e-14$); AMD vs COLAMD and `scale=max` vs `scale=none` change the factorization arithmetic so the full-precision result differs in its last digits (rel diff $\sim 1.2e-14 / 2.8e-14$ -- a deterministic, nonzero proof the knobs reach KLU); invalid values warn; a plain `.option klu` equals `amd/max/btf-on` bit-for-bit; a wide-dynamic-range network solves correctly under `none/sum/max`

E-153 *maths/ni/niiter.c, maths/KLU/klusmp.c, include/ngspice/smpdefs.h, include/ngspice/optdefs.h, include/ngspice/tskdefs.h, include/ngspice/cktdefs.h, spicelib/analysis/cktsopt.c, spicelib/analysis/ckntask.c, spicelib/analysis/cktdojob.c, examples/trustregion_examples/**

Levenberg-Marquardt trust-region Newton (`.option trustregion`, off by default), completing the damped/trust-region-Newton row alongside the E-111 line search. In the Newton loop (`niiter.c`): the E-111 KCL-residual merit is reused (gated on `linesearch||trustregion`); when a dimensionless damping $\lambda > 0$ is in effect, $\mu = \lambda * ||\text{diag}(J)||$ is added to the effective diagonal `gmin` (`trGmin`, passed to `SMPreorder/SMPluFac`) AND `mux_k` to the RHS (the E-127 pseudo-transient coupling with $x_{\text{prev}} = x_k$), so the solve yields the exact damped step $x_{k+1} = x_k - (J + \mu I)^{-1} F(x_k)$; a post-solve accept/reject block re-loads at the trial point, computes $||F(x_{\text{new}})||$, and either accepts (relax $\lambda = 0.25$) or rejects (restore x_k , grow λ , force another iteration),

reusing the E-111 state-save/limiting-reset machinery; a convergence guard forbids converging while $\lambda > 0$. The $\|diag(J)\|$ scale (new SMPdiagNorm in klusmp.c, for both the KLU and Sparse matrices) makes λ dimensionless (Marquardt / scale-invariant). Unlike the line search (which only shortens a fixed Newton direction) large μ re-aims the step toward steepest descent, regularizing an ill-conditioned Jacobian. Result-neutral: the fixed point is $F=0$ for any μ and $\lambda > 0$ on success, so it converges to the same operating point -- verified BIT-IDENTICAL to plain Newton (diode, BJT, resistor divider, plus transient) under both solvers. Option plumbing mirrors linesearch (OPT_TRUSTREGION, IF_FLAG, TSK/CKT fields + CKTtrLambda). Solver-independent. HONEST FINDING: instrumented step-rejection counting showed ZERO rejections on every circuit tried (diode strings, behavioral exponentials, cubics, negative-resistance oscillators) -- ngspice globalizes Newton at the DEVICE level (per-device junction limiting: limexp/pnjlim/fetlim, 30 families) which damps the controlling voltages before the residual is computed, so a residual-increasing overshoot never reaches the solver-level merit; the trust-region is thus a correct, safe solver-level regularization that stays inert on typical circuits

E-154 *frontend/com_envelope.c (new), frontend/com_envelope.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, spicelib/analysis/envelope.c (new), spicelib/analysis/Makefile.am, include/ngspice/cktdefs.h*

Envelope Following (envelope <node> <fc> <tstop> [nppp] [m] [maxm] [reltol] [settle]), the last missing analysis in the RF/periodic-steady-state suite. For a carrier-driven circuit whose amplitude/phase envelope varies slowly over many carrier periods, it samples the state once per carrier period $T=1/f_c$ and integrates the slow drift, jumping M periods at a time. The exact per-period map is $X_{n+1}=\phi(X_n)$, ϕ = one-period DAE integration; the envelope obeys $dX/dn \sim \phi(X)-X$. The naive forward-Euler jump $X_{n+M}=X_n+M(\phi(X_n)-X_n)$ is UNSTABLE on high-Q circuits (the one-period monodromy has unit-circle eigenvalues, so $I+M(\phi-I)$ amplifies $\rightarrow$ blow-up; this is what shelved an earlier attempt). The new engine (spicelib/analysis/envelope.c, EFanalysis) uses the IMPLICIT backward-Euler jump $X_{n+M}=X_n+M(\phi(X_{n+M})-X_{n+M})$, solved by Newton $[(1+M)I - M*\phi] dY = -G$ with $\phi=d\phi/dY$ the one-period monodromy (finite-differenced, one extra period-integration per state, dense $N \times N$ solve capped for modest circuits). A-stable, so it tracks a resonator's envelope without diverging; its fixed point $\phi(X)=X$ is the true steady state. Step size M is chosen by step-doubling local-truncation-error control. The one-period map reuses the transient primitives (NIcomCof + NIiter per fixed substep, the dctrans state rotation) on a fixed grid of $nppp$ points in TRAPEZOIDAL mode (backward-Euler damps a high-Q resonance); ϕ is SELF-STARTING (a frozen-history variant plateaued at a false low fixed point) with a sub-divided backward-Euler first sub-step to bound the restart damping. The com_envelope.c front-end command runs a short settling transient, calls EFanalysis, and emits an envelope plot ($_amp = 2|V_1|$, $_dc$, $_re$, $_im$ vs time; nutmeg vector API). Solver-independent. Verified against a full .tran under both linear solvers: EF amplitude tracks the transient across a $Q \sim 3160$ tank ring-up to $<3\%$ and to the steady state ($<0.5\%$), stays bounded over 3000 carrier periods (26 envelope samples), and tracks a $Q \sim 316$ tank to $\sim 1.6\%$. Completes the RF suite

E-155 *frontend/com_reduce.c (new), frontend/com_reduce.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, spicelib/analysis/rcreduce.c (new), spicelib/analysis/Makefile.am, include/ngspice/cktdefs.h, examples/reduce_examples/**

RC network reduction (reduce <fmax> [factor f] [file fname] [name subckt] [keep node ...]), moving the post-layout "RC reduction / model-order reduction" gap to done. Post-layout extraction produces enormous linear parasitic R/C networks (thousands-millions of interior nodes); reduce collapses the R/C network to a small ELECTRICALLY EQUIVALENT one preserving the port behaviour over DC.fmax, written as an ordinary .subckt of R's and C's. Method (spicelib/analysis/rcreduce.c, CKTreduceRC): TICER (Time-Constant Equilibration Reduction) -- Schur-complement (Gaussian) elimination of interior nodes of $Y(s)=G+sC$, kept first order in s so the reduced network is REALIZABLE as R's and C's (no model-order-reduction black box, no passive-

synthesis step). Per elimination of node n , for each neighbour pair (a,b) : $G[a,b] := G[a,n]G[n,b]/G[n,n]$; $C[a,b] := (G[a,n]C[n,b] + C[a,n]G[n,b])/G[n,n] - G[a,n]G[n,b]C[n,n]/G[n,n]^2$. The conductance update is the exact Schur complement so DC is preserved EXACTLY; the capacitance is matched to first order. A node is eliminated only when its self time-constant frequency $f_n = G_n/(2\pi C_n) > \text{factor} * f_{\max}$ (quasi-static in the band of interest); *factor* trades reduction against in-band accuracy, monotonically. PORTS (kept nodes) are auto-detected as every node touched by a device that is NOT a resistor/capacitor (sources, transistors, OSDI Verilog-A devices), plus ground and user keep nodes, via the generic GENnode/terminal-count device interface -- so built-in and OSDI devices alike. The `com_reduce.c` front-end command enumerates the built-in resistor/capacitor instances (`CKTtypelook`), builds dense G/C over the RC nodes, runs TICER, and emits the reduced subckt (`CKTnodName` for node names). Solver-independent (reads devices + own dense solve). First cut is dense (node cap ~ 2500); sparse TICER is the follow-up. Verified under both linear solvers: a huge factor eliminates nothing and the emitted subckt reproduces the full AC response BIT-for-BIT; a moderate factor cuts nodes (e.g. 25->6, 4x) with <0.25 dB in-band error and DC exact; the accuracy/reduction tradeoff is monotone in factor; an OSDI device on a node auto-marks it a kept port

E-156 *spicelib/analysis/rcreduce.c, frontend/com_reduce.c, include/ngspice/cktdefs.h, examples/reduce_examples/**

Sparse RC reduction -- makes the E-155 reduce command scale to real post-layout size. The dense NN TICER engine (capped ~ 2500 nodes) is replaced by a SPARSE one: the network is stored as per-node adjacency lists (`RCadj`: a growable edge list of {neighbour, g, c}), and interior nodes are eliminated in a MINIMUM-DEGREE order (a lazy binary heap keyed on current degree, like sparse LU) so fill-in stays tiny -- a degree-2 chain node merges two series elements with ZERO fill. A FILL GUARD (`maxdeg`, default 12) declines to eliminate a node once its degree grows past the threshold, so a dense mesh core (whose boundary Schur complement is inherently dense) is left intact instead of densifying the whole network (measured: unguarded, one mesh elimination created 3308 fill edges; guarded, 9). The TICER Schur-complement update, DC-exact conductance, frequency criterion (eliminate only when $f_n = G_n/(2\pi C_n) > \text{factor} * f_{\max}$), and automatic port detection are unchanged from E-155; in the edge representation the diagonal is implicit so eliminating a node just adds Schur edges among its neighbours. Node cap raised 2500 -> 5,000,000; a 65,017-node network reduces in ~ 4 s. New `maxdeg` command option (`com_reduce.c`) and `CKTreduceRC` parameter (`cktdefs.h`). Also fixes a terminal-ordering pitfall: the reduced .subckt's terminals are emitted in internal-index order (not the order keep nodes were typed), so instantiating with the wrong order silently swaps ports -- the command now prints the correct `x1 <ports...> <name>` instantiation line. Verified under both linear solvers: identity reduction bit-exact (0.00 dB), a moderate factor gives $\sim 4x$ fewer nodes at <0.25 dB in-band with DC exact, monotone accuracy/reduction tradeoff, OSDI auto-port, and an 8001-node network (past the old dense cap) reduces successfully

E-157 *frontend/com_aging.c (new), frontend/com_aging.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, examples/aging_examples/**

Device aging (`aging <t_target> [rate <opvar>] [param <ageparam>] [dynamic <tstop> [tstep]] [verbose]`), adding the reliability "stress -> degrade -> re-simulate (fresh/aged)" flow (HCI/NBTI/TDDDB) and moving those reliability gap rows to done. The command finds every aging-capable device in the loaded circuit, computes how much it has degraded after `t_target` seconds of operation, writes that back into the device, and RE-STAMPS the circuit so any analysis run afterwards (`op/dc/tran/ac`) sees the aged devices; it runs the fresh stress simulation first, leaving it as the current plot (a "fresh" baseline). The design is MODEL-AGNOSTIC: a device opts in purely by exposing, in its Verilog-A/OSDI model, a degradation-RATE operating-point variable (default `agerate`, in dose-units/second) and a per-instance AGE parameter (default `age`, (`*type=\"instance\"*`)). The engine's only job is to integrate the rate into a dose and feed it back; the model owns the physics that maps age to a parameter shift (a sublinear power law, an Arrhenius factor, ...). Participants are detected by scanning each device TYPE's instance-parameter table

(DEVICES[t]->DEVpublic.instanceParms) for the two keywords, so ordinary resistors/sources are silently skipped and probing never errors. Two modes: STATIC (default) reads the rate at the DC operating point and scales by the lifetime, $\text{age} = \text{agerate}(\text{op})t_{\text{target}}$; DYNAMIC (*dynamic* <tstop>) runs a transient over one representative window, trapezoidally (time-weighted) integrates the rate, and extrapolates, $\text{age} = (\text{INT } \text{agerate } dt / t_{\text{stop}})t_{\text{target}}$ -- capturing duty cycle. Because age is per-instance, devices sharing a .model but at different bias age independently. Implementation (frontend/com_aging.c) reuses the com_optimize synchronous command-dispatch pattern to run op/tran, the nutmeg expression engine to read @inst[agerate], and alter @inst[age]=... to write back; console chatter suppressed via ft_optimizing unless verbose. Solver-independent. Verified under both linear solvers with a demo square-law NMOS carrying an NBTI hook (agemos.va): enumeration picks exactly the ageable devices, the dose is exactly ratet_target , the threshold shift matches the analytic $\Delta V_{th} = dv_{th_ref}(\text{age}/\text{age_ref})^{0.25}$ law to 5 sig figs, aged drain current drops, degradation is monotone in lifetime, a near-threshold device loses a larger fraction of its current than a hard-driven one for the same shift, a 30%-duty pulsed gate ages at 0.30x the DC rate (dynamic), and a sub-threshold device accrues zero dose

E-158 frontend/com_emir.c (new), frontend/com_emir.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, examples/emir_examples/*

Power-grid EMIR (emir [rail V] [thresh frac] [thick m] [jmax A/m2] [n exp] [tref s] [top k] [verbose]), adding electromigration + IR-drop reliability sign-off and moving the last reliability gap row to done. After a DC solve of the power-distribution network the command reports two metrics. IR-DROP: for every node, the drop below the ideal rail ($\text{rail} - V(\text{node})$); the rail defaults to the highest node voltage (the supply pad) unless given; reports the worst node and every node past a threshold (default 10% of the rail). ELECTROMIGRATION: for every wire-segment resistor, the current DENSITY $J = |I|/(w\text{thickness})$, ranked; the worst-density segment, every segment past the current-density limit J_{max} , and a Black's-equation relative lifetime $\text{MTTF}/\text{ref} = (J_{\text{max}}/J)^n$ (Black: $\text{MTTF} \sim J^{-n}$; a segment exactly at J_{max} has $\text{MTTF} = \text{the reference lifetime } t_{\text{ref}}$, needing no process-specific prefactor). The physics the analysis exists to expose: EM is set by current DENSITY, not current -- a fat trunk carrying huge current can be safe while a thin wire at a fraction of that current voids first, which is why real grids taper wire width with carried current to hold J roughly constant. Implementation (frontend/com_emir.c) runs a fresh op (com_optimize synchronous-dispatch pattern), walks the current plot's node-voltage vectors (plot_cur->pl_dvecs filtered to SV_VOLTAGE) for IR-drop, and enumerates the resistor instances (the device type named "Resistor") reading each segment's current and width via the nutmeg expression engine (@Rk[i], @Rk[w]) -- so it needs no device-struct access and works for any resistor; segments without a width are skipped and counted. Both tables are qsort-ranked (IR by drop, EM by density) and truncated to top. Solver-independent (reads a DC solution + per-resistor currents). Verified under both linear solvers on a 3-segment tapered ladder off a 1 V rail: worst IR drop 0.30 V (30%) at the far tap matching the exact ladder solve and linear in load current; $J = I/(w\text{thick})$ exact; the worst-density segment is the narrow low-current wire, not the high-current wide trunk; the Black MTTF ratio between two segments equals $(J_2/J_1)^n$; with J_{max} between the trunk and leaf density exactly the under-sized segments fail; rail auto-detect equals an explicit rail; and an OSDI (Verilog-A) current load at a tap is handled identically to a current source

E-159 (no ngspice source change) examples/compactmodels_examples/*

Real production compact-model bring-up -- a validation/example enhancement that exercises the EXISTING toolchain on actual CMC (Compact Model Coalition) Verilog-A models, so no ngspice or openvaf-r source change. BSIM4 (the ~12,600-line industry-standard bulk MOSFET) is compiled through openvaf-r to OSDI (~6 s) and validated against ngspice's BUILT-IN BSIM4 -- the same model, so a rigorous self-check: the OSDI BSIM4.8 and native BSIM4.8.3 agree to ~2% on the Id-Vds output family and ~4.5% on the Id-Vgs transfer curve (the residual is the point-release version gap). EKV (EPFL compact MOSFET, which ngspice has NO built-in for) is brought up purely through OSDI, giving textbook saturation. A general finding surfaced: OSDI keeps ev-

ery internal node STATIC (no dynamic node collapsing), so BSIM4's internal drain/source nodes (di/si) -- which the default rdsmod=0 expects the simulator to collapse onto the external d/s -- are left floating and the device conducts ZERO current; enabling the external series-resistance nodes with rdsmod=1 (the model near-shorts them, 1e3 mho, when no S/D geometry is given) connects them and the device works. Models are compiled in place from the bundled OpenVAF integration-test sources (licenses stay with them). Verified under both linear solvers (6 checks)

E-160 (no ngspice source change) *examples/cmcsweep_examples/*, examples/_setup.py*

CMC compact-model coverage sweep -- the comprehensive follow-up to E-159, again a validation/example enhancement with no ngspice/openvaf-r source change (only _setup.py gains cmcsweep in its SPARSE_ONLY and REGRESSION_EXCLUDE sets so the slow 19-model compile runs single-solver and off the fast sweep). It drives EVERY real CMC (Compact Model Coalition) Verilog-A model bundled with OpenVAF through openvaf-r -> OSDI -> ngspice and reports a coverage matrix, the same idea as the E-84 LRM sweep but for production compact models. Result: 19 of 19 models COMPILE and 19 of 19 LOAD, spanning every device class -- MOSFET (BSIM3, BSIM4, BSIM6, BSIMBULK, EKV, HiSIM2, HiSIMSOTB, MVSG_CMC), FinFET (BSIMCMG, BSIMIMG), SOI (BSIMSOI, HiSIMHV), GaN HEMT (ASMHEMT), surface-potential (PSP102, PSP103), bipolar/SiGe-HBT (HICUML2, MEXTRAM), and diode (DIODE, DIODE_CMC). Deeper conduction+physics validation for a representative model per class: OSDI BSIM4 and BSIM3 match ngspice's BUILT-IN BSIM4/BSIM3 to ~2%/~7%; EKV gives monotonic saturating I-V; HICUML2 shows bipolar current gain beta ~ 100 on a Gummel plot; DIODE_CMC turns on exponentially (x16000 over 0.2-1.0 V). Two findings: (1) the E-159 internal-node/rdsmod=1 issue is BSIM4-SPECIFIC, not universal -- BSIM3 handles the zero-resistance case fine and conducts out of the box (its apparent zero at Vgs=1 V is just a high ~1.7 V default threshold); (2) HiSIMHV (6 terminals d,g,s,b,sub,temp) needs cosubnode=1 to enable the substrate node, and with the default cosubnode=0 its \$fatal correctly guards the node-count mismatch (confirming \$fatal works through OSDI). Models compiled in place from the bundled integration-test sources (licenses stay put). Verified (7 checks); the compile/load tiers are solver-independent

E-161 (no ngspice source change) *examples/dynmodels_examples/**

Dynamic (AC/RF) compact-model validation -- extends the E-159/160 DC validation into the dynamic domain, again a validation/example enhancement with no ngspice/openvaf-r source change. Where E-159/160 exercised DC I-V, this exercises the DYNAMIC behavior, which flows through a different code path: OSDI's REACTIVE (charge) Jacobian stamping and ngspice's .ac analysis, on the real production models (compiled in place from the bundled CMC sources). BSIM4 C-V: the gate capacitance Cgg(Vgs) is extracted from .ac as Im(I_gate)/omega and swept over gate bias; it rises from ~40 fF in subthreshold (overlap+fringe) to ~129 fF in inversion (approaching CoxWL) -- the textbook MOSFET C-V -- and the OSDI-compiled BSIM4 matches ngspice's BUILT-IN BSIM4 to under 1% at every bias (a tighter check than the ~2% DC I-V match, since Cgg is dominated by the version-independent oxide term). Cutoff frequency fT (the frequency where AC current gain |h21|=|I_out/I_in| falls to 1): OSDI BSIM4 fT ~ 3.5 GHz matches the built-in to ~1%; HICUML2 (SiGe HBT, which ngspice has no built-in for) has zero default transit time (t0=0) giving infinite fT, so a realistic dynamic parameter set (t0=10 ps, 1 fF junction caps) is supplied and the resulting fT lands right at the transit-time limit 1/(2pit0) ~ 15.9 GHz and rises with collector current -- textbook charge-control behavior. Verified under BOTH the Sparse and KLU solvers (.ac is supported by both), 4 checks

E-162 *frontend/inp.c, examples/hb_examples/verify_hb.py*

.hb dot-card for harmonic balance -- gives the E-134 hb command netlist dot-card parity with the PSS family (.pss/.pac/.pnoise/.pxf/.psp/.sp are dot-cards, while the HB family hb/qpss/hbosc were control-block commands only). A top-level .hb <f0> <K> [points] [maxiter] card now runs single-tone harmonic balance straight from the deck (no .control block needed). Rather than duplicating the HB machinery as a separate analysis JOB, .hb reuses

the deck->control mechanism Enhancement-146 introduced for `.sweep`: during deck loading (frontend/inp.c) a top-level `.hb ...` line is stripped of its leading `.` and appended to the post-parse control list, so it executes as `hb ...` (the `com_hb` command) once the circuit is built -- inheriting all of the E-134 engine (argument parsing, the E-121 conversion-matrix Jacobian, E-135 source-stepping continuation, solver independence). A boundary check (next char after `.hb` is whitespace or end-of-line) keeps it from swallowing any future `.hb*` card. Before this, `.hb` produced unimplemented dot command `'.hb'` -- the only `.hb` handler in ngspice was a disabled `#ifdef WITH_HB` upstream stub (`dot_hb`) that merely redirected to the PSS analysis, not the E-134 engine. Verified (examples/hb_examples/verify_hb.py, +2 checks): the `.hb` dot-card run in plain batch mode (no `.control`) converges and produces a harmonic-balance spectrum bit-for-bit identical to the `hb` command form on a junction-limited diode rectifier; it threads its optional `[points]/[maxiter]` arguments and coexists with a following `.control` block (deck order preserved); both Sparse and KLU give identical results. (Like `.sweep`, a bare command-style dot-card with no other analysis card prints a benign "no simulations run" batch notice even though HB ran.)

E-163 *frontend/inp.c, examples/qpass_examples/verify_qpass.py, examples/phasenoise_examples/verify_phasenoise.py*

.qpass / .hbosc / .phasenoise dot-cards -- completes the harmonic-balance family's netlist dot-card parity begun by E-162's `.hb`. Three more top-level command-style cards now run their analyses straight from the deck: `.qpass <expr> <f1> <f2> [periods] [maxorder]` (two-tone quasi-periodic steady state, E-133/136; add the `hb` keyword for the frequency-domain form), `.hbosc <oscnode> <K> [fguess] [tstab]` (autonomous harmonic balance for an oscillator, E-140; the deck needs a `.ic` to start the oscillation), and `.phasenoise <fstart> <fstop> [points]` (oscillator phase noise, E-140; run after `.hbosc`). Each reuses the same one-branch deck->control mechanism as `.hb` (E-162) / `.sweep` (E-146): in frontend/inp.c a top-level `.qpass/.hbosc/.phasenoise` line is stripped of its leading `.` and appended to the post-parse control list, executing as the corresponding command once the circuit is built -- inheriting the full E-133/136/140 engines and their solver independence. Because the bridge preserves deck order, `.hbosc` (which finds the oscillator PSS + frequency) and `.phasenoise` (which reads it) compose, so a complete oscillator phase-noise run needs no `.control` block. A per-card boundary check (the char after the name is whitespace or end-of-line) keeps the cards distinct -- in particular `.hbosc` is NOT matched by the `.hb` branch, whose own boundary check rejects the trailing `osc`. Verified: the `.qpass` dot-card produces a two-tone spectrum bit-for-bit identical to the `qpass` command (examples/qpass_examples/verify_qpass.py); `.hbosc + .phasenoise` in a deck reproduce the command form's oscillation frequency and phase-noise spectrum exactly with order preserved (examples/phasenoise_examples/verify_phasenoise.py); both under Sparse and KLU. No regression to `.hb/.sweep`.

E-164 *(no ngspice source change) examples/rfpa_examples/**

Large-signal RF characterization of a real transistor -- a validation/example enhancement (no ngspice/openvaf-r source change) that closes the production-model validation loop (DC E-159, coverage E-160, small-signal E-161, now large-signal). It drives a common-emitter amplifier built from the bundled HICUM/L2 SiGe HBT (compiled in place, with the E-161 dynamic params `t0=10ps + 1fF caps`) hard and extracts the large-signal RF figures of merit via TRANSIENT + Fourier/FFT: AM-AM gain compression (the power gain expands 17.8->20.7 dB then compresses -- the exponential-transconductance signature of a bipolar -- defining P1dB), harmonic distortion (the third harmonic follows the textbook 3:1 slope, $HD3 \sim A^3$), and two-tone third-order intermodulation (the IM3 products $2f_1-f_2 / 2f_2-f_1$ at ~ -30 dBc). IIP3, extracted from the single tone as $A/\sqrt{3*HD3}$ (the two-tone IM3 is exactly 3x the single-tone HD3 for a third-order nonlinearity), is constant at ~ 0.134 V across drive level, confirming the third-order model. A real finding: the frequency-domain harmonic-balance engines (`hb/qpass`, E-134/136, which just gained netlist dot-cards in E-162/163) do NOT converge on this stiff, many-internal-node production model in an amplifier configuration -- `hb` returns error 103 at any drive level (even 1 mV) and with extra iterations/

source-stepping, and the two-tone qpss is prohibitively slow (>2 min per point). Transient+FFT integrates reliably and is the robust route for large-signal RF on a full production compact model; HB/QPSS remain the tool of choice for lighter, well-conditioned circuits. Verified under both linear solvers (4 checks): transient small-signal gain matches .ac (7.744 vs 7.742), HD3 3:1 slope (62.9 vs 64), IIP3 constant (0.134 V, spread <2%), and gain compresses at high drive (7.74->5.97).

E-165 *(no ngspice source change) examples/modelnoise_examples/**

Production compact-model noise validation -- a validation/example enhancement (no ngspice/openvaf-r source change) exercising the last untested small-signal path, OSDI's .noise stamping of the models' own white_noise / flicker_noise sources, on production models compiled in place from the OpenVAF integration-test sources. BSIM4 (MOSFET): the output-noise spectral density $S_v(f)$ of a common-source amplifier is compared to ngspice's BUILT-IN BSIM4 over the full band and matches to ~1.5% everywhere, covering both the low-frequency 1/f FLICKER region (amplitude density $\sqrt{S_v} \sim 1/\sqrt{f}$) and the flat high-frequency THERMAL floor -- a stringent check since the flicker coefficients, thermal-noise model, and their bias dependence all have to line up. HICUM/L2 (SiGe HBT): no ngspice built-in, so validated against physics -- its default noise is WHITE (shot noise on the junction currents + thermal noise on the parasitic resistances, no flicker by default), a flat spectrum in contrast to the MOSFET's 1/f rise; with a small source resistance so the intrinsic device noise dominates, the output-noise floor tracks the collector SHOT-noise line $\sqrt{2qI_c R C^2}$ across two decades of bias current (within 6% at high bias where shot dominates) and scales as $\sqrt{I_c}$. Verified under BOTH the Sparse and KLU solvers (.noise works under both since E-113 fixed the KLU adjoint solve), 5 checks: BSIM4 spectrum matches built-in <4%, BSIM4 1/f flicker slope ($S_v(1\text{Hz})/S_v(10\text{Hz}) \sim \sqrt{10}$), BSIM4 flat thermal floor, HICUM white noise, HICUM shot-noise tracking + $\sqrt{I_c}$ scaling. Completes the production-model validation loop (DC E-159, coverage E-160, small-signal E-161, large-signal E-164, noise E-165).

E-169 *frontend/syntaxhl.c (new), include/ngspice/syntaxhl.h (new), frontend/commands.c, frontend/Makefile.am, main.c, examples/syntaxhl_examples/**

Interactive command-line syntax highlighting -- the interactive prompt now colors the command line AS IT IS TYPED. The command word is shown GREEN the moment it is a recognized command, RED when it can no longer become one, and left the terminal default while it is still a valid PREFIX being typed (so `pl` is neutral, `plot` green, `plt` red); numbers are yellow, "quoted strings" magenta, and -option flags cyan. The command word is classified exactly as the interpreter resolves it -- the same case-insensitive walk of the live command table `cp_coms` that `do_command()` uses, plus the control keywords (`if/while/foreach/begin/end/...`) -- so the color always agrees with whether the word will actually run. Implementation: a pure engine `cp_highlight_line()` (`frontend/syntaxhl.c`) tokenizes a line and wraps each token in ANSI SGR color escapes; live coloring overrides GNU readline's `rl_redisplay_function` (installed by `cp_synhl_init()` from `main.c` right after `rl_outstream` is set). The replacement `redisplay` redraws the colorized line only when the prompt plus line fit on one terminal row and otherwise defers to readline's own `rl_redisplay()`, so a wrapped line is never corrupted; the cursor is repositioned with a plain CR + CUF (cursor-forward) so mid-line editing is unaffected. Because that manual cursor positioning bypasses readline's internal position tracking, readline's own `accept_line` stops emitting the closing newline, so the Enter key is bound to a wrapper (`cp_synhl_accept`) that emits the newline itself when coloring is active before running the normal `accept_line` -- otherwise a command's output would begin on the input line (when coloring is off the wrapper is a no-op and readline handles the newline as usual). Gated three ways: it is on by default only at an interactive TTY (`isatty(rl_outstream)`), is disabled by `set no_syntax_highlight` and by the NO_COLOR environment convention, and never fires in batch (`ngspice -b`) or any piped/redirected session -- so simulation output is byte-for-byte unchanged (confirmed by the full example regression). Because as-you-type coloring needs the terminal in raw mode it requires a readline-enabled build, which the shipped binaries are (CI builds with `--with-readline=yes` and the committed binaries link `libreadline`). A new `synhl <command line>` command prints the colorized form of a line non-

interactively -- useful for previewing and, since it needs no terminal, what makes the coloring engine regression-testable in batch. Verified (examples/syntaxhl_examples/verify_syntaxhl.py, 11 checks): the engine via synhl in batch (valid command green, unknown red, valid prefix neutral, numbers/strings/options colored, case-insensitive) and the live as-you-type behavior driven through a pseudo-terminal (green/red/neutral on the fly, NO_COLOR and a non-tty suppress all color); the live layer auto-skips on a build without readline. Solver-independent (front-end only).

E-170 *frontend/syntaxhl.c, include/ngspice/syntaxhl.h, examples/syntaxhl_examples/**

Semantic syntax highlighting -- extends E-169 from lexical to SEMANTIC coloring of the interactive line: it now checks whether the SIGNALS and EXPRESSIONS typed actually exist and parse. (1) A signal reference `v(node)/i(source)/@device[param]` is looked up read-only in the current plot (`vec_fromplot`, no evaluation, no vector creation); it keeps the default color if it exists and is drawn RED if it does not. (2) Inside an otherwise-valid expression only the invalid signal reddens -- `print v(a)+v(zzz)` colors just `v(zzz)`, including signals nested in functions (`sqrt(v(a))-v(bad)`). (3) A genuinely malformed expression argument of an expression command (`plot/print/gnuplot/asciiplot`) reddens as a whole, tested with the real parser `ft_getpnames_from_string(FALSE)` with its tree freed. (4) Error output is drawn RED at an interactive terminal: `cp_err` is wrapped in a coloring stream (`funopen` on BSD/macOS, `fopencookie` on Linux) that prepends the red SGR to each write; `cp_curerr` is pointed at the wrapper too so ngspice's per-command stream reset `cp_ioreset()` keeps it in place instead of restoring and closing it. A half-typed expression is NOT treated as an error and stays neutral -- a signal whose parenthesis has not closed (`v(bP)`), an expression with unbalanced parens (`v(a)+v(b)`) or one ending in an operator (`v(a)+`). KEY FIX: the expression parser's error handler `PError` writes straight to `stderr`, bypassing the `cp_err` muting, so an early version spewed `syntax error in line segment ...` onto the prompt while typing; `expr_pares()` now also mutes `fd 2 (dup2 /dev/null)` across the parse and restores it. Signal-validity is scoped to the unambiguous `v()/i()/@` forms (bare words could be plot keywords like `vs`, functions like `sqrt`, or options `->` false reds). Same three-way gating as E-169 (interactive TTY, set `no_syntax_highlight`, `NO_COLOR`); the parser + vector lookups run per keystroke but are read-only and do NOT corrupt command execution (a subsequent `let z=v(a)+v(b)` still computes correctly). Verified (`verify_syntaxhl.py`, 19 checks total): the E-169 lexical checks plus a semantic layer driven through a pseudo-terminal after simulating a small circuit -- valid signal not red, invalid signal red, invalid-in-expression reddens only the signal, malformed expression red as a whole, half-typed stays neutral with no parser-error spam, error output red, `NO_COLOR` suppresses it. Solver-independent (front-end only).

E-171 *maths/KLU/klusmp.c, spicelib/analysis/pzan.c, examples/klupz_examples/**

KLU/Sparse solver audit -- KLU pole-zero fixed (complex determinant + pivot tolerance) -- a deep audit of the KLU<->SMP bridge found that pole-zero analysis under 'option klu' SILENTLY PRODUCED GARBAGE for any circuit with complex poles or zeros: a series RLC reported four bogus real poles (-100, -10, -0.89, 0) instead of its conjugate pair -5000 +/- j999987.5. Real-axis roots came out right (the mixed formula is accidentally correct for real pivots), which is why the existing single-real-pole RC regression never caught it. DEFECT 1 (`spDeterminant_KLU`): the complex determinant built each pivot as the mixed quantity $1/(U_x R_s), U_z R_s$ and took its complex reciprocal -- KLU's `Udiag` stores the ACTUAL pivots (its solve divides by them) while Sparse's `Diag` stores RECIPROCAL pivots (its solve multiplies), and the adaptation kept Sparse's reciprocal dance for the real part only. The real branch was worse: its product loop NEVER RAN (the loop index was left at N by the preceding permutation scan, so `det` was always +/-1.0), it divided instead of multiplying, and it never wrote `*piDeterminant` (`SMPcDProd` consumed an uninitialized stack value). Both branches computed the permutation sign as `#non-fixed-points/2`, wrong for any cycle longer than 2 (a 3-cycle is EVEN but counted odd). Rewritten: $\det(A) = \text{sign}(P) \cdot \text{sign}(Q) \cdot \text{prod}(U_{\text{diag}}[k] R_s[k])$ with parity computed exactly by cycle decomposition (new `PermutationParity` helper); the KLU determinant now matches Sparse to ~14 digits at every Muller trial point. DEFECT 2 (`SMPcReorder/SMPpreorder`): pole-zero calls `SMPcReorder` with `PivRel = 0.0`; Sparse's `spOrderAnd`

Factor SANITIZES a non-positive threshold to its stored default, but the KLU branch assigned it straight to Common->tol, making KLU's partial pivoting accept an EXACTLY-ZERO diagonal as pivot ($|diag| \geq 0max$ always holds). At PZ's s=0 trial an inductor branch has a 0.0 diagonal (-sL), so the factorization returned KLU_SINGULAR and PZ recorded a spurious root at the origin -- and the poisoned deflated Muller search never expanded past $|s| \sim 10$, yielding the garbage above. Both reorder entry points now mirror Sparse's sanitization (in-range PivRel still takes effect; DC/AC/tran pass the 1e-3 default and are byte-identical). With both root causes fixed, the E-113-era guard in pzan.c ('pole-zero finite-zero computation is not supported with option KLU' -- added when the zero search was seen to go spuriously singular, i.e. defect 2) is REMOVED; the finite-zero search now works under KLU and a genuine E_SHORT reports the same 'input shorted' diagnostic as Sparse. The balanced/differential-output PZ guard remains (SMPcAddCol genuinely has no KLU branch). Verified (examples/klupz_examples/verify_klupz.py, 6 checks): full pole/zero root-set parity between the two solvers plus analytic anchors on a series RLC (conjugate pair), RC lowpass (old-good case, no regression), lead network (finite real zero), RC highpass (origin zero), RLC bandstop (PURELY IMAGINARY zeros $+j1e6$, the hardest case), and RLC bandpass (the finite-zero search formerly guarded off) -- all six identical across solvers. Full 134-example regression clean. Residual scope (solver-independent, ~1990 Muller-on-determinant fragility): a twin-T's deflated conjugate zero pair can still stall under KLU (4 of 6 roots found) while on the bandpass it is SPARSE that hits its iteration limit (KLU converges cleanly).

E-172 *maths/KLU/klusmp.c, spicelib/analysis/cktpzset.c, spicelib/analysis/pzan.c, examples/_setup.py, examples/klupz_examples/**

KLU balanced pole-zero + full partial pivoting -- closes E-171's two scope items; no analysis card remains Sparse-only under KLU. (1) BALANCED / DIFFERENTIAL-OUTPUT pole-zero ('pz in 0 a b vol', non-ground output reference) was guarded off as unsupported: the algorithm's change of variables needs the per-trial column fold $col(b) += col(a)$ (SMPcAddCol), which had no KLU branch -- and KLU's CSC sparsity pattern is FIXED at COO->CSC conversion time, so the fold's fill-in cannot be created on the fly the way Sparse's spcCreateElement does. Fixed in two halves: CKTpzSetup reserves the UNION pattern before the conversion (every row of the solution column is also registered in the balance column via SMPmakeElt; duplicate (row,col) COO entries fold into a single CSC slot by the conversion's group labeling, so the reservation is safe), and SMPcAddCol gains a KLU branch that merge-walks the two columns' sorted row lists adding the complex values in place (with a defensive error if an addend row is missing, impossible after the reservation). The two pzan.c balanced-output guards are removed. (2) FULL PARTIAL PIVOTING for PZ factorizations: validating a fully-differential configuration exposed spurious far-field roots at $|s| \sim 1e19$. $.1e21$ (including a positive-real pole). An exact-rational (fraction-arithmetic) determinant of the very matrix KLU factored proved the factorization itself was numerically rotten at extreme $|s|$: at $s=4.52e19$ the exact determinant was $+4.36e11$ while the pivot product gave $-2.05e12$ -- wrong sign AND magnitude. Root cause is architectural: KLU's ordering is fixed at klu_analyze time (pattern-only AMD/BTF) while Sparse re-runs value-aware Markowitz ordering every PZ trial; PZ sweeps $|s|$ across ~20 decades, and with the relaxed default $tol=0.001$ the fixed ordering accepted catastrophically-cancelling pivots. KLU's only value-adaptive lever is the within-block partial pivoting, so the E-171 sanitization fallback is upgraded from 'keep current tolerance' to FULL partial pivoting ($tol=1.0$) whenever the caller passes an out-of-range PivRel -- only pole-zero does (0.0); explicit in-range tolerances (DC/AC/tran 1e-3) are honored unchanged. This single change eliminated the far-field junk AND cured the twin-T conjugate-zero-pair stall recorded in E-171's scope (what looked like vintage-Muller noise-floor fragility under KLU was largely this factorization inaccuracy; KLU now finds all 6 twin-T roots with notch-zero real parts $\sim 1e-12$, an order CLOSER to the exact 0 than Sparse's $\sim 1e-11$). Verified (verify_klupz.py, 6 -> 9 checks): the E-171 six plus the differential RC bridge (balanced output, formerly 'not supported'), a balanced output with a complex pole pair, and the twin-T notch -- all root sets identical across solvers. The dual-solver harness's balanced-pz KLU-skip detection is removed from examples/_setup.py (nothing remains Sparse-only). Full example regression 134/134.

E-173 *spicelib/analysis/cktpzeig.c (new), maths/dense/eig.c (new), maths/KLU/klusmp.c, spicelib/analysis/pzan.c, cktsopt.c, cktntask.c, cktdojob.c, include/ngspice/{smpdefs,optdefs,tskdefs,cktdefs}.h, two Makefile.am, examples/pzeig_examples/**

Eigenvalue-based pole-zero ('.options pzeig') -- a modern alternative root finder for .pz. The vintage driver hunts roots of $\det(G+sC)$ with a Muller iteration on determinant values and is famously fragile (iteration limits -- 'giving up after 231 trials' on a plain RLC bandpass -- noise-floor stalls, chaotic search paths; E-171/172 fixed the KLU-side defects but the ~1990 algorithm itself stays delicate under BOTH solvers). The new opt-in method computes every pole and zero at once as a dense eigenvalue problem, reusing the existing CKTpzSetup/CKTpzLoad machinery so it applies unchanged to the poles and zeros configurations, to balanced/differential outputs, and to both linear solvers. Steps: (1) the PZ-configured MNA matrix is AFFINE in s , $A(s) = G + sC$, so two loads ($s=0, s=1$) extract the dense pencil via a new solver-agnostic *SMPdenseExtractReal* (KLU: walk $A_p/A_i/A_x$ Complex; Sparse: walk *FirstInCol* through the $Int \rightarrow Ext$ translation maps); a third load at an arbitrary point VERIFIES affinity so a non-polynomial device falls back with a clear message instead of wrong roots. (2) C is structurally singular, so the pencil is solved by SHIFT-INVERT linearization: factor $(G + \sigma C)$ ONCE at a non-root shift (tried from a small ladder of shifts, with the circuit's own sparse solver), form the dense $M = (G + \sigma C)^{-1}C$ by n sparse solves; from $(G+sC)v=0 \Leftrightarrow Mv = \mu v$ with $\mu = 1/(\sigma - s)$, every finite root is $s = \sigma - 1/\mu$ and the pencil's infinite eigenvalues land harmlessly at $\mu = 0$ (dropped by a noise-floor threshold $64neps * ||M||$). (3) eigenvalues of the real dense M from a new self-contained eigensolver *maths/dense/eig.c* -- the classical balance / Hessenberg / Francis double-shift QR chain (EISPACK balance/elmhes/hqr lineage, written fresh, no LAPACK dependency), unit-tested standalone against companion matrices, complex conjugate pairs, a 50x50 tridiagonal with known spectrum, and a badly-scaled balance case. Roots are emitted as the same PZtrial list the Muller driver produces, so PZpost and all downstream output are unchanged. Option plumbing follows the standard task-option path (OPT_PZEIG $\rightarrow$ TSKpzEig $\rightarrow$ CKTpzEig flag $\rightarrow$ dispatch in pzan.c). RESULTS: RLC bandpass -- same roots as Muller with NO iteration-limit warning; 10-section RC ladder -- all ten poles matching the analytic tridiagonal-eigenvalue formula $s_k = -(2-2\cos((2k-1)\pi/21))/RC$ and Muller root-for-root; twin-T -- all 6 roots under both solvers; RLC bandstop -- purely imaginary zeros $+j1e6$ EXACT; balanced/differential output works; purely resistive circuits yield no roots and no crash. DEFAULT REMAINS MULLER (pzeig is opt-in). Dense $O(n^2)$ memory / $O(n^3)$ QR, capped at 2000 unknowns with a clear error above. Accuracy is dense-QR class (absolute error $\sim \epsilon * \text{dominant root magnitude}$): an exact origin-zero can print as $\sim 1e-7$ when poles sit at $1e6$ rad/s; Muller refines locally and can resolve wider per-root dynamic range, while pzeig never misses or invents a root -- the two methods agree to ≥ 6 digits on every battery circuit. Verified (verify_pzeig.py, 13 checks: both solvers x both methods + analytic anchors + default-method check). Full example regression 136/136.

E-174 *src/frontend/commands.c, examples/helpcmd_examples/**

help all crash fix (help string used as printf format) -- running `help all` (or `help montecarlo`) in the interactive shell crashed the shipped ngspice with 'Error: tvprintf failed / ERROR: fatal error in ngspice, exit(-1)'. Root cause: ngspice's help printer uses each command's help TEXT as a printf FORMAT string -- `out_printf(ccc[i]->co_help, cp_program)` in `com_help.c` (and the parallel `com_ahelp.c`) -- passing only one argument (`cp_program`, the program name, intended for a single `%s` as in the legitimate 'Report a %s bug.'). Any other '%' in a help string is an invalid/incomplete conversion specifier, so `tvprintf` fails and ngspice calls a fatal `exit(-1)`. The montecarlo command's help (added in E-151) read '... reports the yield with a Wilson 95% CI ...'; the '%' (percent-space) is the invalid specifier. `help all` walks the command table alphabetically and died at `montecarlo`; `help montecarlo` crashed directly too. Present in BOTH command tables (`spcp_coms` for ngspice, `nutcp_coms` for nutmeg). Not X11/terminal/solver specific -- plain format-string handling, so it reproduced on every build; it was invisible in headless CI only because `help all` is an interactive command not exercised by the batch example decks. Fix: escape the literal percent as `%%` (printf

convention) in both tables, so the line renders 'Wilson 95% CI' and help all completes. New regression guard examples/helpcmd_examples/verify_helpcmd.py with two layers: RUNTIME -- drives an interactive ngspice on a pseudo-terminal and asserts help, help all, and help montecarlo each run to completion (no crash, no 'tvprintf failed') and that the montecarlo line renders a literal '95% CI'; and a STATIC class-guard that scans commands.c and fails if any command help string carries a format hazard (a '%' that is not '%s' or '%%', or more than one '%s' since only one argument is passed), catching a future unescaped '%' even when that command is never exercised at runtime. Full example regression 137/137.

E-175 *spicelib/analysis/dcpss.c, examples/rfconv_examples/**

RF-suite audit -- the dropped parametric term in every periodic small-signal analysis -- a full audit of the RF suite (.sp/Touchstone, PSS/PAC/pnoise/pxf, PSP, HB, QPSS + the two-tone family, hbosc/phasenoise) driven by analytic anchors and cross-analysis invariants. CLEAN: .sp + Touchstone (17 checks: series-R/shunt-C/3-port star S-parameters exact to 10 digits, the asymmetric-2-port S21/S12 column-order trap, reciprocity, RI/MA/DB + Y/Z (v1 normalization) + GHz round-trips through wrsnp/rdsnp, N-port 4-pairs-per-line layout); HB (analytic cubic: fundamental $a1A+(3/4)a3A^3$ and $H3=(1/4)a3A^3$ exact to 7 digits, even harmonics at machine zero, Sparse/KLU identical); QPSS-HB/QPSS-tran (two-tone fundamentals, IM3 $(3/4)a3A^3$ exact, commensurate aliasing guard fires correctly). THE BUG: every LPTV small-signal conversion matrix scaled its reactive coupling by the COLUMN (input-sideband) frequency, $H_{nm} = G_{\{n-m\}} + j\omega_m C_{\{n-m\}}$. For a small signal riding a FROZEN periodic capacitance $C(t)=dQ/dv|_{\text{orbit}}$ the exact linearization is $di = d/dt[C(t)dv] = C\dot{d}v + Cdv_{\dot{}}$, and the product rule collapses to the ROW frequency: $H_{nm} = G_{\{n-m\}} + j\omega_m nC_{\{n-m\}}$. Using ω_m silently DROPS THE PARAMETRIC-PUMPING TERM $C\dot{d}v$ -- the very term that makes a pumped varactor convert. Every conversion sideband through a time-varying capacitance came out scaled by exactly ω_{in}/ω_{out} : on the audit circuit (1k -> OSDI varactor $C(v)=c0(1+\alpha v)$ pumped 1V@1MHz, 1mV probe at 250kHz) the +1/+2 sidebands were 3x/5x/7x/9x too small, with measured pre-fix/truth ratios matching the omega ratios to FIVE DIGITS (0.3333/0.2000/0.14286/0.11111) -- diagnosis confirmed beyond doubt. AFFECTED: pac, psp, pnoise, pxf, qpac, qpnoise, qpxf, and the phasenoise adjoint, for any circuit with pumped/nonlinear capacitance (varactors, junction and MOS capacitances -- every real mixer and oscillator). UNAFFECTED: LTI circuits (no off-diagonal C harmonics; $\omega_n == \omega_m$ on the diagonal) -- exactly what every prior regression check used; the E-171 accidental-correctness pattern again. THE SUBTLETY: the harmonic-balance residual uses the SAME builders and must KEEP the column frequency -- its reactive current is the exact chain rule $d/dt Q(v(t)) = C(v(t))\dot{v}(t)$ whose n-th harmonic is $\sum_m C_{\{n-m\}}(j\omega_m V_m)$. That is why HB and QPSS-HB always matched transient ground truth while the small-signal analyses did not, and why a blanket fix would have broken HB. FIX: a smallsig mode flag on pac_build_matrix and qp_build_matrix plus the two inline copies (the PAC adjoint and the phasenoise adjoint): smallsig=1 -> row frequency (restores the parametric term); smallsig=0 (HB/HBOSC/QPSS-HB residual + Jacobian) -> chain-rule column frequency, byte-for-byte unchanged. Regression safety proven directly: the linear-circuit PAC control produces IDENTICAL output pre/post fix, and HB/QPSS-HB outputs are byte-identical. Ground truth = plain transient + one-beat Fourier projection (independent of all conversion-matrix machinery); the fixed qpac matches all five sidebands within 2% (integrator error class). Also noted: cktspnoise.c is a dead fossil (#ifdef RFSPICE_ never compiled). Verified (examples/rfconv_examples/verify_rfconv.py, 5 checks x both solvers: truth vs QPSS-HB unchanged, truth vs fixed qpac, pre-fix signature absent, HB + QPSS-HB analytic anchors). Full example regression 138/138.

E-176 *spicelib/analysis/dcpss.c, examples/_setup.py, examples/pssdriven_examples/**

Driven-mode PSS shooting -- no frequency hunt, no breakpoint flood -- diagnosis of the E-175 audit's flagged residual ('PSS shooting robustness: the varcap PSS ran 17+ minutes without converging'). Instrumentation showed 9,627,161 accepted timepoints by $t=2\mu s$ (~0.2 ps per step, zero

LTE rejections): ngspice's PSS (the Lannutti shooting) is OSCILLATOR-oriented -- it hunts the fundamental frequency through mid-period forced breakpoints whose spacing is proportional to steady_coeff. On driven circuits (everything the E-117..126 periodic small-signal stack runs on) this is doubly pathological: (1) at the standard decks' steady_coeff=5e-6 the grid spacing is ~0.5 ps -- millions of steps per shooting cycle, and the convergence-tolerance knob doubles as grid density so tighter tolerance quadratically explodes runtime (even the default 1e-3 forces thousands of points/period -- why a linear RC .pss took ~4 minutes); (2) the frequency estimate can never settle EXACTLY on the source frequency, the cycle endpoint phase-drifts against the source, and the period residual FLOORS (~1e-4) -- shooting never converges and burns the full sc_iter budget at flood speed. FIX: a DRIVEN mode, auto-detected when the circuit contains a time-varying independent source (SIN/PULSE/... V or I source, funcTGiven): the period is pinned to the exact source period (no estimation, no estimator grid; each cycle is one plain-transient period; varactor residual 2e-1 -> 8e-9 in 17 cycles / 0.3 s); the shooting-phase max step is clamped to T/pspoints so the orbit is integrated on the SAME discretization as the retained samples (validation caught this: without the clamp LTE grows steps toward T/2 and shooting converges -- to 7e-9! -- onto the fixed point of that coarse discretization, several percent off the true orbit: retained swing 0.1651 vs analytic 0.15718; the old flood had been masking the effect by brute force); the post-FFT 'relaunch at the strongest spectral line' is disabled when driven (a rectifier-like circuit with a dominant 2nd harmonic must not retain the wrong period); the AUTONOMOUS oscillator path is untouched (byte-identical behavior verified on oscillator decks; DC-only-supply circuits never trigger detection). IMPACT: rc_pss ~4 min -> 0.05 s with the retained fundamental EXACTLY 1000000 Hz (the old hunt returned 999999.8976); psp example 406 s -> 0.9 s; the rfpss battery (5 verifies, 61 checks) minutes-each -> <0.5 s each; rfanalyses 'several minutes even Sparse-only' -> 0.56 s; the KLU PSS pass 'prohibitively slow' -> 0.14 s. HARNESS: rfpss and rfanalyses removed from both REGRESSION_EXCLUDE and SPARSE_ONLY -- the whole RF periodic small-signal suite now runs in EVERY regression sweep under BOTH solvers; the direct .pac-on-pumped-varactor check became affordable, closing the E-175 loop on the direct PAC path (sidebands match transient ground truth within 1%). Verified (examples/pssdriven_examples/verify_pssdriven.py, 6 checks x both solvers: driven detection + exact retained frequency, retained swing == |H| within 0.1%, retained-period self-consistency, direct .pac vs transient truth, autonomous non-trigger). Full example regression 145/145.

E-177 *spicelib/analysis/dcpss.c, examples/pnoise/pnoise_examples/**

pnoise noise-folding referee + the folded-flicker frequency fix -- closes the E-175 audit's last honesty-ledger gap: periodic noise analysis (pnoise/qnoise/phasenoise) was 'correct by construction, not correct by measurement' -- the folding of device noise through LPTV conversion had never been checked against anything independent. THE REFEREE (three independent layers, on a 1-node LPTV-conductance circuit $G(t) = g_0 + g_1 \sin(\omega_0 t)$ chosen so conversion transfers are $O(1)$): (1) a from-scratch Python conversion matrix, $\text{onoise}(f) = \sum_k |Z_k(f)|^2 * S(|f + kf_0|)$ -- same physics, zero shared code; (2) a TRNOISE transient Monte-Carlo (no shared code OR theory; 198-segment Welch PSD, with the pumped/unpumped RATIO cancelling the TRNOISE amplitude convention); (3) the LTI (pnoise == .noise) and white limits. VERDICT ON THE WHITE PATH: measured-correct -- pnoise matched the referee to 6 digits at every frequency and the MC arbiter confirmed the ~1.86x folding ratio within statistical error; the adjoint, sideband sum and thermal densities are right. THE BUG THE REFEREE CAUGHT: the stationary sideband loops set the noise-evaluation frequency ONCE, to the output frequency (data.freq = freq outside the k loop). But the sideband-k adjoint carries noise ORIGINATING at $|f + kf_0|$ -- a frequency-dependent source PSD (flicker $1/f$, noise_table E-109) must be evaluated at the source-side frequency. The proof was digit-exact in both directions: pre-fix pnoise reproduced the referee's deliberately-wrong evaluate-at-f model DIGIT-FOR-DIGIT (1.114323e-14 vs 1.114323e-14 at 1 kHz -- a 21% overestimate on this circuit, unbounded as $f \ll f_0$); post-fix it reproduces the correct model digit-for-digit (9.175472e-15). Why three generations of checks missed it: white noise is frequency-flat and LTI circuits have no $k \neq 0$ transfer -- the THIRD occurrence of the accidental-correctness pattern (E-171 determinant,

E-175 parametric term). *FIXED* in all three stationary loops: *pnoise* ($\text{data.freq} = |\text{freq} + k f_0|$ per sideband), *qpnoise* ($|f_{\text{in}} + k_1 f_1 + k_2 f_2|$ per harmonic), and *phasenoise* -- where it matters most: folded far-sideband blocks near a carrier were evaluated at the tiny OFFSET frequency, inflating the flicker ($1/f^3$) region. CYCLO SCOPE: the cyclo mode's time-domain identity $\text{onoise} = (1/P) \sum_s S(t_s) |A_s|^2$ treats the source PSD as frequency-flat across the folding span -- exact for modulated-white noise and for flicker on conversion-free circuits (the shipped E-125/126 use cases), not for flicker folded through $k \neq 0$ conversion (cannot be collapsed into that product with the aggregate device-noise API); documented at both cyclo sites, with the (now exact) stationary mode as the right tool when folded flicker matters. Verified (examples/pnoise_examples/verify_pnoise_examples.py, 5 checks x both solvers, ~2 s thanks to E-176): white folding vs referee ($\leq 0.1\%$), flicker folding vs referee ($\leq 0.5\%$, sample-0 bias read from the retained orbit), pre-fix signature absent, pump/no-pump ratio == referee ratio, LTI limit == .noise. The rfpss/qpss/rcyclo suites pass unchanged (conversion-free or white -- provably unaffected). Full example regression 146/146.

E-178 *spicelib/analysis/dcpss.c*, *spicelib/devices/dio/dionoise.c*, *examples/cyclofold_examples/*, *examples/rfpss_examples/*, *examples/qpss_examples/**

exact separable cyclostationary folding + the HB DC-source fix -- removes the cyclo mode's frequency-flat limitation E-177 documented. For the separable model $S(t, f) = m(t)^2 g(f)$, $\text{onoise}(f) = \sum_g \sum_q |B_q(g)|^2 * g_g(|f + q f_0|)$ with $B_q(g) = (1/P) \sum_s m_g(t_s) dA_s(g) e^{j q \theta_s}$ and $A_s = \sum_k \Psi_k e^{-j k \theta_s}$. Per-generator amplitudes recovered with NO device-API change: the noise-summary machinery (*prtSummary/outpVector*, one density per generator; family totals excluded by the name-prefix rule) plus load POLARIZATION against a fixed reference load $R = \Psi_0$ -- five DEVnoise sweeps per orbit sample ($A+-R$, $A+-jR$, R) give $c_{g,s} = \sqrt{S_g(t_s) dA_s(g)}$ up to a constant phase that cancels in $|B_q|^2$. Spectral shapes are MEASURED pointwise at the folded frequencies (no $1/f^{\text{EF}}$ assumption -- noise_table works) at several probe biases; flat generators keep the original time-domain identity (exact for ANY quadratic form incl. NevalSrc2-correlated pairs); stationary limit == the E-177 sum, flat limit == the old identity (Parseval). Ported 2-D to qpnoise ... cyclo ($B_{\{q_1, q_2\}}$ at $|f + q_1 f_1 + q_2 f_2|$). THE PHYSICS: 'flicker sees $\wedge^2$, white sees $\langle m^2 \rangle$ ' -- only the envelope's DC feeds the $1/f$ band; the old identity was 23% high ($8/\pi^2$) on $|\sin|$ -modulated flicker and 34% high on the conversion circuit; the new sweep lands on a from-scratch referee to $\leq 3e-4$ (the old binary reproduced the deliberately-flat model to $6e-5$ -- the error was the model, not the code). BONUS BUG (4th accidental-correctness occurrence): the cross-machinery check (same circuit through the QPSS-HB orbit) came out exactly 4x -- BOTH HB Newtons (hb E-134, qpss ... hb E-136) double-subtracted the DC sources (the settle-mode rhs folded into I_R already carries them; -lambdaIs subtracts them again): every DC bias voltage converged to exactly 2x, silently corrupting all bias-dependent noise and conversion (flicker $\sim I^{\text{AF}}$ off by 2^{AF} , shot 2x) while AC content and every AC-driven validation stayed exact. Fixed by adding the unscaled DC source block back (net DC drive exactly -lambdaIs_DC); qpnoise cyclo now agrees with pnoise cyclo DIGIT-FOR-DIGIT across the two orbit machineries. rcyclo's flicker law updated to the exact $\langle |I| \rangle^2 = (8/\pi^2) \langle I^2 \rangle$ (pinned to $1e-4$). 6 checks x both solvers; regression 147/147. FOLLOW-UP (same enhancement): the E-139 hard-pumped-diode deck exploded the cyclo path ($\sim 6.5e1 \text{ V}^2/\text{Hz}$) -- dionoise.c computes its three sidewall generators only when DIOresistSWGIVEN but the prtSummary block copies ALL DIONSRCS slots, so without a sidewall model the summary vector carried uninitialized stack garbage (latent in stock ngspice for decades; E-178 is the first consumer of per-generator densities). Fixed at the device (unwritten slots zeroed, dio/dionoise.c) and hardened in both cyclo consumers (finite non-negative PSD guards; garbage probe ratios fall back to the flat path). verify_qpnoise's 'cyclo > 2x stationary' diode expectation was an artifact of the garbage slots + doubled HB DC bias; replaced with a closed-form torus-average referee (resistive network => memoryless LPTV, instantaneous adjoints) matched to ~1%.

E-179 *spicelib/analysis/tfanal.c*, *spicelib/analysis/cktsens.c*, *frontend/com_measure2.c*, *examples/*

*stdaudit_examples/**

standard-analyses audit: referee battery + three fixes. The 'Standard analyses (analog)' gap-table rows are 1990s SPICE3 code whose VALUES had never been physics-checked. FIX 1 (tfanal.c, inherited from Berkeley SPICE3 1988): .tf i(vm) vin output impedance was ALWAYS 1e20 -- the output-impedance solve forces 1V on the sense branch and inverts the branch current as 1/MAX(1e-20, rhs), but that current is NEGATIVE for every passive network (the input-impedance path right above divides by -rhs); fixed with the same sign + fabs guard; feedback-amp referee exact (1000.990 == RL + node Thevenin). FIX 2 (cktsens.c): KLU AC sensitivity silently truncated to ONE frequency point -- the E-114 KLU complex-conversion block inside the frequency loop reused i (the OUTER loop variable) for its DEVmaxnum walk, so after the first point i=DEVmaxnum ended the sweep; the surviving point was correct, which is why E-62's single-point check passed. Fixed with a local index (dead second block hardened too); full sweep == analytic $dV/dC = -j\omega R/(1+j\omega RC)^2$ to 6 digits, both solvers. FIX 3 (com_measure2.c): .meas DERIV{ATIVE} was parsed but evaluation fell into an explicit 'currently not supported' stub (empty #if 0 measure_deriv() placeholder) -- implemented: 3-point Lagrange-quadratic derivative on the nonuniform time grid, AT= and WHEN forms sharing FIND's parse/locate flow, verified vs the analytic sine slope to 5 digits; DERIVATIVE/INTEGRAL long spellings accepted (only DERIV/INTEG matched); pre-existing -Wmissing-prototypes warning silenced (str_has_arith_char2 static). MEASURED-CORRECT: .disto HD2/HD3 == analytic diode Volterra kernels $\leq 0.06\%$ incl. a frequency-dependent load (harmonics at $Z(2\omega)/Z(3\omega)$ -- no E-177-style frequency bug), SIM2 intermods (f1+f2, f1-f2, 2f1-f2) incl. cascades, exact DISTOF1/DISTOF2 scaling, junction-cap harmonics == Harmonic Balance to ~6 digits (two independent engines); .noise onoise_total^2 == band analytic (kT/C) 0.06% + flicker log-integral 6 digits; .sens DC == central finite differences at a nonlinear OP (5-6 digits, model params incl.); nonlinear-OP .pz works via the driving-point form (the E-62 'quirk' was a bias source shorting the injection node -- ngspice's refusal is correct); tran/op/meas battery exact. 8 checks x both solvers; regression 148/148.

E-180 *frontend/com_checkpoint.c, examples/checkpoint_examples/**

checkpoint/restart under KLU (the last Sparse-only feature falls). E-131 guarded savestate/loadstate off under KLU, recording the reason as 'KLU factorization objects absent on the restore path' -- a mis-diagnosis. REAL cause: .option klu lives in the task (TSKkluMODE) and reaches the circuit only in CKTdoJob at analysis dispatch, but loadstate calls CKTsetup DIRECTLY first, so the matrix was built Sparse (SMPkluMatrix NULL); the resume dispatch then set ckt->CKTkluMODE=1 without re-setup (by design -- it must preserve the restored state), and the first NIiter dereferenced the NULL SMPkluMatrix (KLUloadDiagGmin store; reproduced EXC_BAD_ACCESS +0x90). savestate was never broken (the checkpoint file holds only solver-agnostic vectors). FIX: com_loadstate copies the task's solver selection and the E-152 KLU knobs (ordering/scale/BTF/memgrow) into the circuit BEFORE CKTsetup, exactly CKTdoJob's block; both guards removed. BONUS: cross-solver restore -- all four save x load combinations (fresh processes) continue exactly on the uninterrupted trajectory ($\leq 1e-9$ at $t=3.5ms$ of a 4ms diode+RC run). verify_checkpoint's 'KLU rejected' check replaced by the 4-combo matrix (22/22); solver-notes and gap-analysis tables updated. Nothing in the simulator is Sparse-only any more (E-172 closed the last analysis, E-180 the last feature). Regression 148/148.

E-181 *spicelib/analysis/dctran.c, spicelib/analysis/cktsort.c, spicelib/analysis/cktask.c, spicelib/analysis/cktdojob.c, include/ngspice/cktdefs.h, include/ngspice/tskdefs.h, include/ngspice/optdefs.h, examples/corenum_examples/**

core-numeric audit: the integrator certified exactly + **.options ordfix**. Nobody had ever verified that the Gear (BDF) corrector coefficients deliver their nominal order (orders 3-6 were dead code for ~30 years until E-128 selected them; E-128 verified order SELECTION, not coefficient CORRECTNESS). NEW OPTION ordfix=K (off by default): a fixed-order verification mode -- ev

ery converged step is accepted (the step is ruled by tmax, not the LTE), the first step is shrunk 1000x (an order-1 starter proportional to the pinned step imposes an $O(h^2)$ floor), the order ramps to K while the step is tiny, then the step grows gently ($\times 1.15$; large ratios destabilize high-order variable-step BDF). Each guard defeats a measurement trap that initially presented as a bug but is real numerics (incl. the LTE step preference $h_k = (c_k \cdot \text{tol})^{1/(k+1)}$) INVERTING at loose tolerance -- the E-128 controller's refusal to raise there is correct). CERTIFIED: accepted trajectories satisfy the exact variable-step BDF-k formula (Lagrange differentiation on the actual nodes) to $\leq 1.3e-13$ at every order 1-6 (~90 uniform stencils each); measured global slopes confirm orders 1-3; trap-vs-Gear dissipation on a lossless LC matches theory ($|R(iy)|=1$ vs order-dependent damping). Also referee'd with NO defects: solver precision == conditioning theory under both solvers (1e18-conductance-spread asymmetric mesh == exact-rational MNA to 1e-15); all four DC convergence-aid paths (default/noopiter/gminsteps=0/srcsteps=0) agree on a 40-diode hard-DC chain (spread 2.2e-6, KLU==Sparse 7e-13); xmu=0.5 == trap bit-identical, xmu=0.45 damps; lvltim=1 works. Drive-by: DCtran_step_quit prototype added to cktdefs.h (pre-existing -Wmissing-prototypes). 8 checks x both solvers; regression 149/149.

E-182 *frontend/plotting/plotit.c, frontend/plotting/pyplot.c, examples/pyplot_examples/**

pyplot autoscale by default -- the matplotlib bridge (E-94/95/98/99) pinned every generated axis with set_xlim/set_ylim taken from ngspice's grid-rounded internal plot ranges. User request: rely on matplotlib's autoscaling + fig.tight_layout() instead. plotit() always fills xlims[]/ylim[] (user args OR data-derived), and the statics that distinguish the user case (xlim/ylim from getlims) are freed BEFORE the device dispatch -- plotit now captures user_xlim/user_ylim flags at the free point and passes NULL limits to ft_pyplot unless the user gave xlimit/ylim explicitly (ft_pyplot already handled NULL; gnuplot/asciiplot back-ends untouched). verify_pyplot +2 checks (no pins + tight_layout on auto plots; exact pins for explicit limits). Regression 149/149.

E-183 *frontend/com_pyplot.c, frontend/plotting/pyplot.c, examples/pyplot_examples/, features/*

pyplot usability: distinct default names, deck-folder output, linewidth & backend. (1) REAL BUG: in window (interactive) mode pyplot launches the Python viewer in the BACKGROUND (python3 pyplot.py &); two plots that both omit the file name shared pyplot.py/pyplot.data, and the 2nd call overwrote them before the 1st viewer read them -- both windows showed the 2nd plot (its title and data). Invisible in PNG mode (synchronous). Fix (com_pyplot.c): a per-session counter names successive no-name plots pyplot, pyplot-2, ... (first unchanged). (2) The .py/.data/.png are written next to the CIRCUIT FILE via ngdirname(ci_filename) rather than ngspice's cwd; the script's loadtxt/savefig carry the deck-dir path so Python finds the data from any cwd; a bare in-folder deck name still lands in the cwd (unchanged). (3) set pyplot_linewidth=<w> -> linewidth in the plot()/step() calls. (4) set pyplot_backend=<name> -> matplotlib.use() ahead of import pyplot, overriding the auto backend (incl. the Agg forced by the png/svg/pdf terminals). All four opt-in; default output byte-unchanged. filename buffers in pyplot.c grown 128->1024 for full paths. verify_pyplot grows to 13 checks x both solvers; packaged as a standalone feature bundle under features/. Regression 149/149.

E-184 *frontend/outitf.c, examples/progressbar_examples/**

sweep progress bar reaches 100%. Fix to the E-129 progress bar: it sometimes froze below 100% at end of run (e.g. '[====] 82%' then 'No. of Data Rows'). CAUSE: the 'Reference value' status line is throttled -- redrawn only when >0.25 s has elapsed since the last update -- so the FINAL sweep point's print is almost always skipped (it lands within 0.25 s of the previous tick), freezing the bar at the last throttled value; reaching 100% was never reliable. FIX: outp_print_reference records the latest refval + a bar-shown flag every call; a new outp_finish_reference, called from fileEnd/plotEnd just before the 'No. of Data Rows' line, reprints the bar full at 100% in place using the sweep's TRUE endpoint (TRAN CKTfinalTime, AC ACstopFreq, NOISE NstopFreq, DC last source value). One-shot (reset at analysis start and after firing), silent for non-swept analyses (op/tf), same ft_optimizing/ft_noreprint/cp_background guards -- quiet/background/optimizer

runs and the rawfile unaffected. `verify_progressbar`'s end-of-run check tightened from $\geq 90\%$ to $=100\%$ (all four swept analyses); 22/22. Regression 149/149.

E-188 *spicelib/analysis/dcop.c, include/ngspice/cktdefs.h, frontend/com_sweep.c, examples/warmstart_examples/**

Monte Carlo warm-start (**montecarlo ... -warm**). Each MC sample (E-151) re-sources the deck and COLD-solves its DC bias point, running the full gmin/source-stepping homotopy from scratch even though consecutive samples move the operating point only slightly. Profiling: the solve, not the deck re-source, dominates for non-trivial circuits, and every op cold-starts -- two IDENTICAL ops back-to-back each burn the full homotopy (~52 Newton iters on a diode ladder; a warm .nodeset at the solution cuts it to ~5). CKTop tries a direct Newton from the caller's firstmode before any homotopy, but DCop always passes MODEINITJCT (cold junction voltages), discarding the previous solution. FIX: `montecarlo -warm` wraps the loop in `CKTsetWarmStart(1/0)`; DCop keeps a buffer (outside the CKTcircuit that reset recreates; indexed by equation number, stable across resets of an identical-topology deck) of the last converged CKTrhsOld, preloads it and calls CKTop with MODEINITFLOAT (use existing node voltages) when a valid same-size guess exists, and snapshots each converged solution. A poor guess just fails the first NIiter and falls through to the normal cold homotopy, so the converged point is identical -- only a cheap failed attempt is wasted. First sample cold, rest warm; when -warm is absent `dcop_warm_enable=0` and the path is byte-identical (one extra SMPmatSize read). CORRECTNESS: warm changes only Newton's starting point, so it converges to the same OP to within convergence tolerance -- `warm==cold` yield EXACTLY at `reltol=1e-6`, agreeing to within a couple boundary samples at default `reltol` ($\sqrt{3} \sim 3.8V$, where `reltol*|v|` ~ the narrow spec band; a tolerance effect, removed by tightening `reltol`). Iteration count $\sim 52 \rightarrow \sim 4$ (~13x); wall-clock benefit scales with per-iteration cost (large/hard-converging designs win most; small circuits are dominated by deck re-source + command dispatch). Shared DC-op code, so Sparse+KLU; composes with -lhs. 5 checks (exact at tight `reltol`, close at default, iteration cut, +lhs, +klu). Regression 152/152.

E-189 *frontend/com_sweep.c, examples/sweepwave_examples/**

Sweep waveform overlay (**sweep ... -overlay**). A usability follow-up to the E-146 sweep command: sweep steps any knob, runs an inner analysis per point, and records each output's LAST value into a summary sweep transfer curve -- but overlaying the full per-point WAVEFORMS to compare shapes meant `setplot`-ing each retained run by hand. `-overlay` (alias `-ov`) collects every point's whole output waveform, resamples them onto a common independent-variable grid, and builds one sweepwave plot with a `<output>_<value>` vector per (output, knob value), left current so `plot <out>_<val> ...` works at once -- the HSPICE .step overlay in one step. Three static helpers in `com_sweep.c`: `sw_eval_vec` (evaluate the output expr and copy its FULL waveform + scale into caller buffers; complex/AC stored as magnitude), `sw_interp` (binary-search piecewise-linear resample, flat outside range -- each run lands on its own adaptive time/freq grid, so a shared grid is required before a common scale vector), `sw_wavename` (clean nutmeg name, non-alnum $\rightarrow$ _). The sweep loop captures each waveform alongside the unchanged last-value recording; after the loop a common grid over [min,max] of all runs at the finest resolution is built, every waveform resampled onto it. GRACEFUL: a scalar inner analysis (op, no waveform) prints "overlay ignored (analysis '...' has no waveform to overlay)" and yields the ordinary summary -- never an error. Without the flag the path is byte-identical to E-146. Front-end only, solver-independent. Verify: overlaid RC step responses match $1 - \exp(-t/RC)$ ($RC=1/2/4\mu s$) to worst $< 1e-4$, distinct per-value vectors, graceful op, summary curve intact. 5 checks. Regression 153/153.

E-190 *frontend/com_sweep.c, examples/nested_sweep_examples/**

Nested multi-knob sweep (**sweep ... -vs ...**). Second usability follow-up to the E-146 sweep command (after E-189 `-overlay`). sweep stepped ONE knob; `-vs <knob> <spec>` (alias `-family`) adds OUTER knobs whose CARTESIAN PRODUCT forms a curve family: the inner (positional)

knob stays the x-axis of the summary sweep plot, and each combination of the outer knobs yields one curve per output, named `<output>_<outerknob>_<value>...` -- the parametric family of .dc ... SWEEP ... / HSPICE nested .dc. Up to SW_MAXKNOB=4 knobs, cartesian points capped at SW_MAXPTS. The three-form spec parser (lin/dec/oct, list, start/stop/step) was factored into `sw_parse_spec`, now shared by the inner knob and each `-vs` knob so all knobs accept the same forms. SET ORDERING (the subtle part): each knob is applied with its E-146 mechanism -- a .param needs `alterparam+reset` (re-source), an instance/model knob is in-place `alter/altermod` -- and a reset drops every in-place alter, so the loop iterates inner-fastest and re-resources ONCE only when a .param knob's value actually changed (outer-loop boundaries; an inner .param still resets every point, a pure in-place sweep never resets), staging all .params via `alterparam` before the single reset, then RE-APPLYING every in-place knob after the reset (so an inner alter survives an outer .param re-source). By construction a single knob reduces EXACTLY to E-146: one outer combination -> curve named `<output>`, E-146 messages, identical data indexing (existing sweep/sweepwave examples pass untouched). `-overlay` composes: each cartesian point's full waveform becomes a sweepwave vector named with EVERY knob value (inner first), so a single knob is `<output>_<value>` exactly as E-189. New static helpers: `sw_parse_spec`, `sw_set_deck/sw_set_inplace` (split from the old `sw_set`), `sw_famillyname` (summary curve name over outer knobs), `sw_pointname` (overlay name over all knobs; replaces `sw_wavename`). Verify: RC low-pass step, R inner x C outer, each family curve == $1 - \exp(-T/(R \cdot C))$ vs R to worst <5e-6, incl. a .param-OUTER knob (reset/alter ordering); cartesian run/curve counts; `-overlay` family names carry both knob values; single knob still names its curve vo. 5 checks. Regression 154/154.

E-191 *spicelib/analysis/acan.c, spicelib/analysis/span.c, examples/aclin2_examples/**

.ac lin 2 / .sp lin 2 off-by-one fix. Found while auditing the data-output path (`wrdata/wrsnp/save/rawfile` write): a ragged-length `wrdata` test wrote an AC vector one point short, which traced not to the writer but to the AC LINEAR frequency sweep. The AC (`acan.c`) and S-parameter (`span.c`) analyses computed the linear step behind `if (numberSteps - 1 > 1)` -- true only for $N \geq 3$ -- so exactly `lin 2` fell through to the single-point patch (`freqDelta = 0`). With a zero step the sweep loop (`freq += freqDelta; if (freqDelta == 0) goto endsweep;`) stops after ONE point, silently dropping the `fstop` point: a two-point linear AC/SP sweep produced one row. `dec/oct` (different step formula) and `lin 1/3/4/...` were all correct; only $N=2$. The single-point patch (Richard McRoberts) was meant for `lin 1; -1 > 1` over-reached by one. FIX: guard on `numberSteps > 1` in both files (the form `noisean.c` already used correctly), so $N=1$ -> one point at `fstart` (patch preserved), $N=2$ -> `freqDelta = fstop - fstart` -> two endpoints, $N \geq 3$ unchanged. The distortion analysis uses a different $N+1$ -point convention (unaffected); noise was already correct. Verify: RC low-pass, both `lin 2` endpoints are GENUINE solved points matching $1/(1+j\omega RC)$ to <1e-6 (not a duplicate); `lin 1` stays one point; `lin N` gives N linearly-spaced points ($N=3,5,10$); `.sp lin 2` gives two points with analytic series-R S-params ($S_{11}=1/3$, $S_{21}=2/3$). Both solvers. The wider output-path audit found the writers themselves CLEAN: Touchstone `wrs2p/wrsnp/rdsnp` round-trip across RI/MA/DB, S/Y/Z (Rbase normalization $Y \cdot R / Z / R$), Hz-GHz incl. complex angles; 2-port special S_{11} S_{21} S_{12} S_{22} order + N-port row-major both correct for round-trip AND spec interop; `wrdata` complex (`re,im`)/singlescale/ragged padding; `save` filtering restricts the vector set; `rawfile write/load` bit-exact at 12 digits. 5 checks. Regression 155/155.

E-192 *frontend/com_checkpoint.c, frontend/com_checkpoint.h, frontend/runcoms.c, examples/autosave_examples/**

Auto-checkpoint on interrupt (`set autosave=<file>`). Follow-up to E-131 checkpoint/restart, prompted by "does savestate fire on Ctrl-C?" -- it did not. E-131 gave `savestate/loadstate` but a checkpoint had to be taken by hand, so an interrupted long transient lost its progress. E-192: `set autosave=<file>` makes an interrupted transient auto-write an E-131 checkpoint. SAFE HOOK POINT (the crux): writing a file from a signal handler is undefined behavior, so E-192 does NOT checkpoint in `ft_sigintr` -- that handler only sets `ft_intrpt` and returns (during a run `ft_`

setflag is TRUE, so no longjmp). At the next ACCEPTED timepoint, dctran's IFpauseTest (which is OUTstopnow) sees the flag and returns 1, so dctran returns E_PAUSE; if_run maps E_PAUSE to return code 1; dosim (runcoms.c) reports "interrupted" (err equals 1). E-192 hooks exactly that interrupted branch -- on the MAIN thread, at a consistent timestep boundary, with ordinary buffered I/O. CHANGE: com_savestate's body factored into ckt_write_checkpoint(ckt, fname) (the command is now a thin wrapper); the hook calls the SAME function, so an autosave file is byte-for-byte what savestate writes. In the interrupted branch: if cp_getvar("autosave", ...) yields a filename AND the run was a transient (CKTmode and MODETRAN set, CKTtime above 0), call ckt_write_checkpoint. OPT-IN (no var means byte-identical old behavior, nothing written), TRANSIENT-ONLY (the MODETRAN / CKTtime guard blocks stale state after, e.g., an interrupted AC). Interactive-mode only: the SIGINT handler is installed only under non-batch mode, so a batch-mode Ctrl-C still terminates (unchanged). A real Ctrl-C (SIGINT), a stop when ... breakpoint, and a Verilog-A stop task all funnel through the same interrupted branch (they differ only in how the flag is set, upstream of E-192). CORRECTNESS: autosave file byte-identical to a manual savestate at the same pause; loadstate resumes it exactly (E-131 round-trip). Verify (examples/autosave_examples): stop-paused transient writes a checkpoint; equals a manual savestate (byte-identical); loads/resumes; opt-in gate (no var means no file); and a REAL Ctrl-C (SIGINT) via a PTY (interactive) fires it (lenient: skip if the run finishes before the signal, the hook being covered deterministically). E-131 checkpoint example still passes (savestate refactor). 5 checks. Regression 156/156.

E-193 *spicelib/analysis/dcpss.c, examples/pnoiseunits_examples/*, examples/rfpss_examples/verify_rcpnoise.py, examples/rfpss_examples/verify_rcyclo.py, examples/cyclofold_examples/verify_cyclofold.py, examples/pnoiseexamples/verify_pnoiseexamples.py, examples/rfpss_examples/rc_pac.cir*

.pnoise honors **sqrnoise** (noise-density units fix). Found auditing the RF/PSS suite. The ngspice manual (sec. .noise, pp. 326-327) defines onoise_spectrum/inoise_spectrum as a noise spectral DENSITY in V/sqrt(Hz) (or A/sqrt(Hz)) by default, switchable to the squared V^2/Hz form by the sqrnoise control variable. .noise obeys this (noisean.c reads cp_getvar("sqrnoise") and sqrt's onoise/inoise when unset). But .pnoise (periodic noise) ALWAYS emitted the squared V^2/Hz density and IGNORED sqrnoise -- so the same-named vectors carried DIFFERENT units across the two analyses (differing by a square), and the documented sqrnoise control silently did nothing for pnoise. Confirmed by the toggle: .noise flips 4.063e-9 (V/sqrt Hz) <-> 1.651e-17 (V^2/Hz) with set sqrnoise; .pnoise returned 1.651e-17 both ways. FIX: pnoise_sweep (dcpss.c) now reads sqrnoise via cp_getvar (added #include cpextern.h) and, when unset (the default), applies sqrt to the emitted onoise/inoise densities -> V/sqrt(Hz); set sqrnoise -> V^2/Hz. BOTH emit paths get the conditional sqrt: the E-178 cyclostationary folding branch and the standard adjoint-folding branch. Now .pnoise and .noise agree by default (match to 0.00). SCOPE: .pnoise only. The two-tone qpnoise command's single-point diagnostic already prints BOTH forms explicitly (V^2/Hz and V/sqrt(Hz)) so there is no ambiguity, and its swept + single-point paths share the V^2/Hz convention; an initial QPnoiseSweep change was REVERTED to avoid an internal swept-vs-single-point split (verify_qpss_sweep [3] compares them). EXAMPLES: rc_pnoise now checks the default V/sqrt(Hz) form (sqrt of the analytic) and cross-checks vs .noise without forcing sqrnoise (match to 0.00); rc_cyclo/cyclofold/pnoiseexamples are power-density relations (onoise == ..., referee sums |Z|^2S) so they set sqrnoise to keep V^2/Hz, and two LTI-limit checks that compared pnoise(V^2) to .noise(V/sqrt Hz)^2 became direct equalities. WIDER AUDIT (all correct): PAC driving-point + the +-1 CONVERSION SIDEBANDS (via the .pac card's trailing maxsideband arg) match a clean transient DFT to ~0.1%; Harmonic Balance matches a transient DFT to 4-5 digits (an apparent 12x .disto HD3 and 2.4x PAC-sideband discrepancy both dissolved into fourier-command interpolation error at high harmonics, resolved by a proper DFT). Also documented the previously-undocumented .pac trailing maxsideband arg (dcpss.c comment + rc_pac.cir). Verify (examples/pnoiseunits): default pnoise == sqrt(4kTR/(1+(wRC)^2)) (V/sqrt Hz); set sqrnoise == the density (V^2/Hz);

default^2 == squared (exact sqrt relation); default pnoise == default .noise (0.00). 4 checks. Regression 157/157.

E-194 *frontend/com_optimize.c, examples/psoopt_examples/**

Particle swarm optimization (**optimize -method pso**). A new search method for the built-in optimizer. E-130/143 gave two LOCAL methods -- Nelder-Mead simplex (**-method nm**, scalar -minimize) and Levenberg-Marquardt (**-method lm**, gradient least-squares over **-targets**) -- both of which descend into whichever basin the start point sits in, so on a MULTIMODAL objective they return the local minimum nearest the start. E-194 adds a GLOBAL, population-based, derivative-free method: PSO. N particles fly through the normalized $[0,1]^{np}$ box; each keeps its own best-seen point (pbest) and the swarm shares a global best (gbest); each iteration updates every velocity $v = \text{chi} * (v + \text{phi} * r1 * (\text{pbest} - x) + \text{phi} * r2 * (\text{gbest} - x))$ and moves $x = \text{clamp}_{01}(x + v)$ with the standard Clerc-Kennedy constriction ($\text{chi}=0.72984$, $\text{phi}=2.05$ so the swarm converges without exploding), velocities clamped to half the box, positions to $[0,1]$. Particle 0 starts at the user init point, the rest uniform-random, so the whole box is explored. Reuses the existing `opt_eval` (returns the scalar cost for both objective kinds), so PSO works for scalar -minimize AND -target least-squares -- unlike LM, which needs targets. New options: **-swarmsize <N>** (aliases **-swarm/-npart**; default auto = $10+4np$ capped at 60), **-seed <s>** (PSO draws from a small self-contained splitmix64 PRNG independent of ngspice's global RNG, so a seed gives byte-identical results), **shared -maxiter/-tol** (tol is the gbest-stagnation tolerance held over several iterations). The nm/lm dispatch is unchanged. Verify (examples/psoopt): on the classic multimodal $f(p) = \sin(p) + \sin(10 * p / 3)$ over $[2.7, 7.5]$ (global $p=5.1457$ $f^*=-1.8996$, several higher local minima) started at the $p=2.7$ corner, PSO finds the global (-1.8996 , $p=5.145$) while Nelder-Mead from the same start is trapped at a local minimum (-1.19992 , $p=3.387$); a fixed seed is exactly reproducible; independent seeds all reach the global basin; PSO also solves a -target least-squares objective (recovers p to a $1e-12$ residual) and a 2-D separable multimodal min ($f^*=-3.799$). Existing optimize example (nm+lm, 20 checks) unchanged. 6 checks. Regression 158/158.

E-195 *frontend/com_optimize.c, examples/deopt_examples/**

Differential evolution (**optimize -method de**). A second GLOBAL method for the built-in optimizer, after PSO (E-194), from the same population-based derivative-free family. DE keeps a population of NP vectors in the normalized $[0,1]^{np}$ box; where PSO pulls trials toward remembered bests, DE (classic DE/rand/1/bin) builds each trial from a scaled DIFFERENCE of random members and crosses it with the target: $v = a + F * (b - c)$ ($F=0.8$; a,b,c distinct and $!= i$), $u[j] = v[j]$ if $\text{rand} < \text{CR}$ or $j == j_{\text{rand}}$ else $x[i][j]$ ($\text{CR}=0.9$ binomial crossover, one dimension always taken), then greedy selection (keep u in place of $x[i]$ iff $\text{cost}(u) \leq \text{cost}(x[i])$). The difference vector b-c self-scales to the population's own spread -- large while members are far apart (broad exploration), shrinking as they converge (fine refinement) -- which makes DE robust on rugged / discontinuous / poorly-scaled landscapes with no per-problem step tuning. Reuses the existing `opt_eval` (scalar cost for both objective kinds), so it works for scalar -minimize AND -target least-squares like PSO. Shares PSO's infrastructure: the same self-contained splitmix64 PRNG (so -seed is byte-reproducible), the -swarmsize population option (default auto $10+4np$ capped at 60; clamped to ≥ 5 because DE needs 4 distinct members to form a mutant), and **shared -maxiter/-tol** (gbest-stagnation). The nm/lm/pso dispatch is unchanged; DE is only selected by **-method de** (aliases **diffevol**, **differenialevolution**). Verify (examples/deopt): on the multimodal $f(p) = \sin(p) + \sin(10 * p / 3)$ over $[2.7, 7.5]$ (global $p=5.1457$ $f^*=-1.8996$) started at the trapping $p=2.7$ corner, DE finds the global (-1.8996 , $p=5.14574$ -- matches p^* to 5 digits) while Nelder-Mead from the same start is trapped at a local minimum (-1.19992 , $p=3.387$); a fixed seed is exactly reproducible; independent seeds all reach the global basin; DE also solves a -target least-squares objective (recovers p to a $4e-13$ residual) and a 2-D separable multimodal min ($f^*=-3.799$); and DE and PSO both reach the global while the local NM does not. Existing optimize (nm+lm, 20 checks) and psoopt (6 checks) examples unchanged. 7 checks. Regression 159/159.

E-196 *frontend/com_optimize.c, examples/saopt_examples/**

Simulated annealing (**optimize -method sa**). A third GLOBAL method for the built-in optimizer, completing the set: after PSO (E-194) and DE (E-195), both population-based, SA is the classic SINGLE-WALKER global optimizer. It holds one current point in the normalized $[0,1]^{np}$ box; each step it proposes a random neighbour $x_n = \text{clamp}_{01}(x + \text{step} * U(-1,1))$ and applies the Metropolis rule -- accept if $\text{cost}(x_n) \leq \text{cost}(x)$ (downhill), else accept anyway with probability $\exp(-(\text{cost}(x_n) - \text{cost}(x))/T)$ (uphill). While the temperature T is high uphill moves are readily accepted so the walker climbs OUT of local minima; T is then cooled geometrically ($T *= \alpha$, α set so T drops ~ 4 decades over the -maxiter cooling levels, with $L=8+4np$ moves per level), so uphill moves become rare and it settles into a minimum. The best point ever visited is returned. ONE candidate per step (no population) -> the lightest-weight global method, the natural choice when each analysis is expensive. Everything auto-scales to the problem: $T_0 = \text{mean } |D\text{cost}|$ of a dozen random probes (so an uphill move of that size is accepted about half the time when hot), and the step size $= 0.30 * \sqrt{T/T_0} + 0.02$ shrinks as T cools (broad exploration hot, fine refinement cold) -- nothing to tune. Reuses `opt_eval` (scalar cost both kinds) -> works for scalar -minimize AND -target least-squares. Shares the self-contained `splitmix64` PRNG (-seed byte-reproducible); -maxiter is the number of cooling levels. The `nm/lm/ps/de` dispatch is unchanged; SA is only -method sa (aliases `anneal`, `simulatedannealing`). A single walker refines more loosely than a population, so SA reliably reaches the global BASIN rather than machine precision (a short `nm/lm` polish from its result sharpens it when that matters). Verify (examples/saopt): on the multimodal $f(p) = \sin(p) + \sin(10 * p/3)$ over $[2.7, 7.5]$ (global $p=5.1457$ $f^*=-1.8996$) from the trapping $p=2.7$ corner, SA finds the global basin ($f < -1.85$, $|p-p^*| < 0.05$) while Nelder-Mead is trapped at a local minimum (-1.19992 , $p=3.387$); a fixed seed is reproducible; independent seeds all reach the global basin; SA solves -target least-squares and a 2-D multimodal min; and all THREE global methods (sa/ps/de) reach the global while the local nm does not. Existing optimize (20) + pssopt (6) + deopt (7) examples unchanged. 7 checks. Regression 160/160.

E-197 *frontend/com_optimize.c, examples/opt100_examples/**

A 100-parameter circuit optimization (raised **optimize** limits). The built-in optimizer sized its per-run arrays from two compile-time macros; E-197 raises them from `OPT_MAXP` 16 to 128 (parameters) and `OPT_MAXT` 64 to 128 (least-squares targets), and lifts the auto-population cap for the PSO/DE population methods from 60 to 256, so a genuinely large problem fits in a single call. Exceeding a cap is reported cleanly (optimize: too many -param (max 128)), not crashed. Testbed: 100 independent 1 mA sources, each into an unknown resistor R_i to ground (so $v(n_i) = 1e-3 * R_i$), fitted to 100 targets $t_i = 1e-3 * (100 + 9800 * i/99)$ ohm -- a linear voltage ramp 0.1 .. 9.9 V, a well-posed separable 100-D least-squares. Levenberg-Marquardt is the right tool for a well-posed high-D fit and solves it to machine precision (sum-sq residual $2.5e-29$, rms $5e-16$, 619 evaluations, ~ 2 s) with all 100 parameters and 100 targets honored. Two fixes make the derivative-free GLOBAL methods FUNCTION at that scale (they are intrinsically far slower than LM in high dimension, but were failing outright): (1) adaptive DE crossover -- classic DE/rand/1/bin uses $CR=0.9$, mutating almost every coordinate of a trial; in high dimension a trial that perturbs $\sim n$ coordinates at once is essentially always worse than its parent and rejected by greedy selection, so DE freezes at its starting guess. E-197 sets $CR = (n \leq 16) ? 0.9 : 15.0/n$, capping the expected mutated-coordinate count at ~ 15 for large n (exactly the classic 0.9 for $n \leq 16$, so low-D DE is unchanged), and DE descends again -- from ~ 1240 to ~ 882 over 35 generations at 100-D. (2) Dimension-scaled convergence patience -- high-D population runs plateau for several generations between improvements, so the gbest-stagnation counter that stops PSO/DE would cut a still-descending run short; the threshold now grows as $8 + n/4$ (exactly 8 for $n \leq 3$, so small- n tests are unchanged), giving 33 generations of patience at $n=100$. DE at 100-D remains a genuinely hard global search that does not fully converge in a short budget -- intrinsic to global optimization in high dimension, not a defect; the message is the division of labor between LM (well-posed high-D fits) and the global methods (multimodal, now usable as dimension grows). Verify (examples/

opt100): LM fits all 100 params to 100 targets at machine precision (both raised caps proven at once); DE descends substantially at 100-D (self-calibrated against its own starting cost, $\geq 15\%$ reduction); a fixed -seed is byte-reproducible even at 100 dimensions; and a 129-parameter over-cap run is reported cleanly. Existing optimize (39) + psnoop (6) + deopt (7) + saopt (7) examples unchanged. 4 checks. Regression 161/161.

E-198 *frontend/com_stb.c, frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, examples/stb_examples/**

stb stability / loop-gain analysis. A new front-end command that measures a feedback loop's small-signal loop gain $T(f)$ and reports its phase and gain margin -- the everyday stability check for op-amps, regulators, PLLs -- which ngspice lacked (you had to hand-roll a .control script). It is the one-command equivalent of Spectre's stb (Middlebrook-Tian double injection). Usage `stb <Vprobe> <Iprobe> <ac-sweep>`: the user marks the loop break with a bias-transparent probe pair between the driving node A and the loaded node B -- a series `Vstb A B dc 0 ac 0` (+node = driver A, -node = load B) and a shunt `Istb 0 B dc 0 ac 0` (ground to load B). Breaking a loop and injecting a single test signal mis-reads the loop gain because of the LOADING at the break (a series voltage injection is exact only from a zero-impedance source into an infinite-impedance load); the double injection cancels that error. `stb` runs two AC sweeps by altering the probe AC magnitudes: voltage injection (`Vstb ac=1`) gives $T_v = -v(A)/v(B)$; current injection (`Istb ac=1`) gives the current loop gain from the probe branch current $a = i(Vstb)$ as $T_i = -a/(a+1)$; combined $T = (T_v T_i - 1)/(T_v + T_i + 2)$. *The combined T is INDEPENDENT of where the probe sits in the loop -- the property a single injection lacks -- verified directly (a loaded break gives the same margins as a clean break in the same loop). Numerically, the common clean-break case has a high load impedance so $a \rightarrow -1$ (all injected current returns through the shorted voltage probe) and $T_i \rightarrow \text{inf}$, which makes $T_i = -a/(a+1)$ divide by zero; `stb` evaluates the algebraically identical singularity-free form $T = -(T_v a + a + 1)/(T_v a + T_v + a + 2)$, exact at $a = -1$ where it reduces to $T = T_v$.* The complex loop gain is stored as vector `loopgain` vs frequency in a new `stb` plot (`plot db(loopgain) / plot 180/PI*cph(loopgain)`); phase margin = $180 + \text{angle}(T)$ at the first $\text{abs}(T)=1$ crossing, gain margin = $-\text{abs}(T)_{\text{dB}}$ at the first $\text{angle}(T)=-180$ deg crossing (both log-interpolated on an unwrapped phase). Implementation (`com_stb.c`, registered in `commands.c`, declared in `com_commands.h`): recovers the probe's two node names via `CKTinst2Node`, dispatches `alter/ac` through the command table (as `com_optimize/com_sweep` do), and reads the complex vectors $v(A)/v(B)/Vstb\#branch/frequency$ with `ft_evaluate`. Reuses the AC analysis, so both the Sparse and KLU solvers. Verify (`examples/stb`): PM/GM match the closed-form loop gain of an ideal-output op-amp loop; a loaded break (finite op-amp output impedance) gives the SAME margins as a clean high-impedance break (proving the current injection corrects the loading -- a single voltage injection is $\sim 10\%$ off and diverges past crossover); a 10x-gain design's reduced phase margin is tracked and matches analytics; the complex `loopgain` is stored; and an unknown probe is reported cleanly. 5 checks. Regression 162/162.

E-199 *examples/nport_examples/snp2va.py, examples/nport_examples/**

N-port Touchstone device (`snp2va.py`). Instantiate a measured or simulated S-parameter block (a filter, cable, connector, package, or amplifier stored in a Touchstone .sNp file) as a circuit element that works in AC AND transient. ngspice had no native n-port element: its .sp "ports" are tagged voltage sources used to MEASURE a circuit's S-parameters, and `rdsnp/wrsnp` (E-64/E-72) are reader/writer commands, not a device. Rather than a native convolution device (with its own history buffers and stability headaches), `snp2va.py` converts the file into a Verilog-A n-port realized with `laplace_nd`, so OpenVAF's OSDI laplace machinery (E-31 complex poles) supplies BOTH the AC response and the transient (recursive-state) response with no convolution code. Pipeline: parse Touchstone (any port count; S/Y/Z data; MA/DB/RI formats; Hz..GHz; arbitrary reference impedance) $\rightarrow S(f) \rightarrow Y(f) = (1/z_0)(I-S)(I+S)^{-1} \rightarrow$ common-pole vector fit (Gustavsen) $\rightarrow$ emit `I(p_i) <+ sum_j [ laplace_nd(V(p_j), num_ij, den) + e_ijddt(V(p_j)) ]`. *Two numerical details are essential. (1) The vector fit MUST be done in NORMALIZED frequency -- without it the se column*

is $\sim 1e9$ while the $1/(s-p)$ columns are $\sim 1e-7$, so the least-squares is hopelessly ill-conditioned and the poles never relocate (the pole relocation is done as polynomial roots of the sigma numerator via Durand-Kerner, not a matrix eigensolve). (2) Y is IMPROPER whenever a network has shunt capacitance ($Y \sim sC$ at high frequency), which `laplace_nd` -- a ratio of polynomials -- cannot represent (it gave a 63 percent AC error at the band edge); the fix is to give `laplace_nd` only the strictly-proper (pole) part and emit the s^*e term as an explicit `ddt` (a capacitance), dropping the AC error to $\sim 1e-6$. HARDENING: automatic order selection CLIMBS the pole count to capture high-order and delay-dominated responses (a pure delay is uniformly poor at low orders, so the selection must not quit early on a small between-order improvement), but stops at the KNEE once the error has already come down near a floor -- past that knee extra poles just fit measurement noise into an over-fitted, ill-conditioned, unstable model, so stopping there fits clean data to machine precision and noisy data at its noise floor. It breaks when a fit turns unstable (polynomial-root finding degrades at high degree) and returns the best STABLE fit. Stability is enforced (right-half-plane poles reflected into the left half plane) so the emitted model is always BIBO-stable even for a noisy non-passive fit; passivity is checked and reported (enforcement by residue perturbation is a documented non-goal for now). Pure Python standard library, NO numpy: least-squares via Householder QR, complex matrix inverse via Gauss-Jordan, and polynomial roots via Durand-Kerner are all reimplemented (each validated against numpy to machine precision during development). Verify (examples/nport): each check generates a Touchstone file from a network whose response is known exactly, runs `snp2va.py` + OpenVAF, and confirms the device matches the ORIGINAL network in ngspice -- an R-L-C resonator to $5e-6$ in AC (including the transmission peak) and $5e-3$ in transient (one model, both analyses); a 5-pole LC ladder (order selection scales past 2 poles); a 3-port star network to $4e-9$ on both coupled outputs; and a noisy measurement fit at its noise floor, reported non-passive, staying bounded in transient. Stress-tested to near machine precision on resonators, notches, high-Q blocks, and multi-pole ladders; a delay-dominated transmission line degrades gracefully (AC accurate over the fitted band, transient pulse edges ring -- inherent to rational fitting of a pure delay, for which a T/LTRA line is the right model). 6 checks. Regression 163/163.

E-200 `src/frontend/snp2va.c, src/frontend/snp2va.h, src/frontend/com_presnp.c, src/frontend/commands.c, src/frontend/inp.c, src/frontend/inpcom.c, src/frontend/Makefile.am, examples/presnp_examples/*`

Built-in `pre_snp` command. Folds the E-199 Touchstone-to-Verilog-A converter into ngspice itself as a C command, so the workflow no longer needs an external Python script: inside a `.control` block, `pre_snp <file.sNp> [module]` parses the `.sNp`, common-pole vector-fits it, writes the `.va`, and invokes `openvaf-r` to compile the `.osdi` -- a one-line analog of `pre_osdi`. `snp2va.c` is a faithful C port of `snp2va.py` with identical numerics: it uses C99 double *Complex* (`typedef'd cplx`, so it does not collide with ngspice's own `complex.h` macros) and reimplements the three vector-fitting primitives -- least-squares via Householder QR, complex matrix inverse via Gauss-Jordan, polynomial roots via Durand-Kerner -- plus the same parse $\rightarrow S(f) \rightarrow Y(f) = (1/z_0)(I-S)(I+S)^{-1} \rightarrow$ vector fit $\rightarrow$ emit $I(p_i) \leftarrow + \sum_j [\text{laplace_nd}(V(p_j), \text{num}, \text{den}) + e_{ij\text{ddt}}(V(p_j))]$ pipeline, the same normalized-frequency fit, the improper *es* (shunt-C) term split out as an explicit `ddt`, and the same automatic order selection returning the best STABLE fit (RHP poles reflected $\rightarrow$ BIBO-stable). It was checked numerically identical to the Python reference (resonator RMS $4e-6$, ladder $7e-7$, 3-port $3e-8$). The public entry point `snp2va_convert(snpfile, vafile, module, msg, msglen)` is called by the command wrapper `com_pre_snp` (`com_presnp.c`), which then shells out to `openvaf-r`. ORDERING: the command is registered under the name `snp`, so writing `pre_snp` in a deck triggers ngspice's existing `pre` mechanism (`inp.c` strips the `pre_` prefix and runs the command before the circuit is parsed). That alone is not enough -- `pre_snp` must run before the `pre_osdi` that loads its output, and control lines otherwise execute in deck order -- so the collected pre-commands are now executed in TWO PASSES: pass 0 runs every `snp` command (in deck order), pass 1 runs the rest. The `.osdi` a `pre_snp` generates is therefore guaranteed to exist by the time any `pre_osdi` loads it, EVEN IF the deck lists the `pre_osdi` line first; the ordering is a guarantee, not a convention the user must remember. `snp/pre_snp` are

also added to `inpcom.c`'s case-preservation list (beside `osdi/pre_osdi`) so the file-path argument keeps its original case. The compiler is located by `find_openvaf()`: the `openvaf` ngspice variable (set `openvaf=...` in `spinit` or on the command line -- a set inside `.control` runs too late, after the pre-commands), then the `OPENVAF` environment variable, then `$SPICE_LIB_DIR/openvaf-r`, then `PATH`; if none resolve, `pre_snp` reports the failing `openvaf-r` invocation and lists all four options. Verify (examples/presnp): each check builds a Touchstone file from a network whose response is known exactly, lets `pre_snp` do the whole convert+compile+load inside ngspice, and confirms the device matches the ORIGINAL network -- `pre_snp` writes the `.va` and compiles the `.osdi`; the device matches an R-L-C resonator to $5e-6$ in AC (including the transmission peak) and to $5e-3$ in transient (one compiled block, both analyses); a 3-port star network compiles and matches on both coupled outputs to $4e-9$; and with `pre_osdi` written BEFORE `pre_snp` the model still loads, proving the ordering guarantee. 5 checks. Regression 164/164. HARDENING (from an N-port stress sweep -- generate an N-port RLC network, extract S-params with `.sp`, round-trip through `pre_snp`, compare DC/AC/transient vs the original subcircuit): two defects the 2-pole checks never reached, both fixed in `snp2va.c`. (1) Order-selection double-free: the climb that keeps the best STABLE fit let its best and prev buffers alias and then freed the same allocation twice, so `pre_snp` aborted (SIGABRT) on any network whose fit needed ≥ 3 pole orders (the 2-pole resonator/star escaped by breaking at tolerance first); fixed by not freeing a buffer the other pointer still holds. (2) Transient divergence from a non-passive realization: each Y_{ij} is fit independently, so the model could blow up in transient ($v \rightarrow \sim 1e284$) even though every pole is in the LHP and AC is exact -- a single degree-Np rational per element has coefficients spanning $\sim \text{abs}(p)^{Np}$ ($\sim 1e79$ for 8 poles at $1e10$ rad/s, an ill-conditioned companion form), and the improper e^s terms form a capacitance matrix with no passivity constraint (an indefinite "negative-capacitance" matrix $\rightarrow$ unstable DAE); fixed by realizing each element as a PARALLEL bank of first/second-order laplace_nd sections (every coefficient $\leq O(\text{abs}(p)^2)$) and projecting the e-matrix onto the symmetric PSD cone (symmetrize $\rightarrow$ Jacobi eigendecompose $\rightarrow$ clamp negative eigenvalues to 0). The presnp example gained a 4-port coupled-ladder check (fit climbs past two orders; transient must stay bounded and match) that fails on the pre-prefix converter. Regression 164/164.

E-201 `src/frontend/snp2va.c`, `examples/presnp_examples/*`

`pre_snp` scalability (fast vector fitting + shared-pole realization). The E-200 converter struggled from about 8 ports up (an 8-port took ~ 190 s and larger port counts were infeasible). Two things scaled badly, both in `snp2va.c`: the pole identification stacked all NN elements into one dense least-squares of size $(2NsNN) \times (NN(Np+2)+Np) \rightarrow O(N^4)$ memory, $O(N^6)$ compute, re-solved dozens of times over the order climb (~ 8 TB of RAM at $N=100$); and the emitted model gave every one of the NN elements its own laplace_nd bank -- $O(N^2Np)$ filter sections and OSDI state. Four fixes rebuild the scalability. (1) FAST VECTOR FITTING (Deschrijver/Gustavsen): the stacked pole-ID matrix has block-arrow structure -- each element's residue columns are independent and couple only through the shared common-pole (sigma) columns -- so instead of forming that matrix, each element block is Householder-reduced on its own (a small $2Ns \times (Np+2)$ QR) and only its residual rows (orthogonal to that element's own columns) are stacked into one tall $((2Ns-Np-2)Ne) \times Np$ system solved for sigma; memory $O(N^4) \rightarrow O(N^2)$, compute $O(N^6) \rightarrow O(N^2NsNp^2)$. (2) RECIPROCITY: a passive network has symmetric Y, so when detected only the upper triangle $(N(N+1)/2)$ elements is fit and residues are mirrored ($\sim 2x$ fewer solves). (3) ITERATION CAP: pole relocation stops once the poles settle (relative move $< 1e-4$) instead of a fixed 12 sweeps (typically 3-5). (4) SHARED-POLE REALIZATION: all NN elements share the same poles, so the pole-filters are computed ONCE per input port -- a real pole gives one filter $V(pj)/(s-p)$, a conjugate pair two real basis filters $(s-\sigma)/D$ and ω/D , with the residue entering each output as a real scalar weight -- and each output current is a cheap weighted sum; this is $O(NNp)$ laplace_nd and OSDI state instead of $O(N^2Np)$, the difference between a model that compiles/simulates at large N and one that does not. Three latent bugs the stress test also surfaced, all fixed: (a) an order-selection double-free -- at higher pole counts a relocated pole set could lose its adjacent-conjugate-pair structure and build_basis/ctil_to_cres (which walk $i+=2$ on a complex pole) wrote one slot past the residue

array; fixed by a canon_poles step that snaps near-real poles to real and rebuilds exact adjacent conjugate pairs. (b) a 256-entry stack array -- the Touchstone parser assembled each frequency's matrix in a fixed cplx pv[1616], *hard-capping the port count at 16 (N=32 smashed the stack); moved to the heap, auto-detect limit lifted to 512.* (c) an OpenVAF crash -- a laplace_nd coefficient array assigned to a variable (as the shared realization does) crashes OpenVAF when a coefficient is an integer literal like '{1}' (which %.12g prints for 1.0); the emitter now forces a real literal '{1.0}'. Results: fit at N=8 220s -> 3.6s; N=64 fits in 47s (was ~860GB of RAM); N=100 fits in 2.75min at 9.6e-4 error; model laplace_nd count now NNp (65 at N=8, 449 at N=32) not $\sim N^2 \cdot Np/2$ (256, 7168). Correctness preserved: DC/AC/transient still overlay the original subcircuit (N=8 unchanged; N=16, previously infeasible, matches to 1.3%). The remaining bottleneck is OpenVAF's compile time for the (now far smaller) shared model (~26s at N=16, ~43s at N=20), so the practical end-to-end ceiling is $\sim N=20-24$ while the fit itself scales to N=100. Verify (examples/presnp): an 8-port coupled ladder converts+compiles in a few seconds with the compact shared model (65 laplace_nd for 8 ports) and matches the original network in AC; the E-200 order/realization guards and resonator/3-port checks still pass. 9 checks. Regression 164/164.

E-202 *src/maths/dense/dense.c, examples/spscale_examples/**

.sp S-parameter matrix inverse is $O(N^3)$, not $O(N!)$. The pre_snp stress test (E-201) noted a separate wall: producing the input Touchstone file with ngspice's own .sp analysis was itself pathologically slow -- an 8-port took ~13s and a 10-port ~18 minutes, growing by roughly a factor of N per added port. That is in the RFSPICE analysis, not the converter. Profiling put ~99 percent of an 8-port .sp run inside CKTspCalcSMatrix (cktspdum.c), which builds the port S-matrix at every frequency by inverting N x N complex matrices -- several per frequency, for the S, Y and Z matrices. The complex inverse cinverse (maths/dense/dense.c) was computed by the adjugate/determinant method (Cramer's rule): $\text{cinverse} = \text{cadjoint}(A) * (1/\text{cdet}(A))$. cdet is a RECURSIVE COFACTOR EXPANSION -- $O(N!)$ -- and cadjoint computes N^2 cofactors (each an $(N-1) \times (N-1)$ determinant), so cinverse was $O(N^3 N!)$; with several inverses per frequency across a whole sweep the cost explodes by a factor of N per port (6! -> 7! -> 8! is exactly the measured x7, x8, x9 per added port). FIX: cinverse and cinversedest now invert by GAUSS-JORDAN ELIMINATION WITH PARTIAL PIVOTING on the augmented [A I], in $O(N^3)$ -- a drop-in replacement with the same signature and the same result (the true inverse). Behavior preserved: the old cinverse never returned NULL (the adjugate always produced a matrix, garbage for a singular input), so the new one returns NULL only on a true allocation failure and zero-fills a singular matrix; the singular case does not arise for a well-posed .sp network, and the two cktspdum.c call sites that do not null-check are safe. cdet/cadjoint remain for any other callers. RESULTS: .sp at N=8 13s -> 0.3s, N=10 ~18min -> 0.02s (~50000x), N=16 0.03s, N=32 0.09s; the extracted S-parameters are unchanged, verified against the closed-form network to $\sim 2e-7$. The same inverse is used by the periodic S-parameter path (.psp/PSS in dcps.c), which benefits too; full example regression (touchstone, rfanalyses, psp, ...) unchanged. Together with E-201 the whole extract-or-import -> convert -> simulate workflow now scales to many ports. Verify (examples/spscale): a 12-port RLC ladder -- a port count that took minutes before -- runs through .sp + wrsnp, completes in a fraction of a second, and every entry of the extracted 12x12 S-matrix matches the closed-form network across the sweep to $\sim 2e-7$. 2 checks. Regression 165/165.

E-203 *src/frontend/com_measure2.c, src/misc/string.c, examples/acmargin_examples/**

.meas ac gain/phase margin (+ batch vdb/vp auto-save fix). An audit against a 'richer TRIG/TARG, FIND...WHEN, integral/RMS/derivative, AC margins' wish-list found that ngspice's .measure already covers TRIG/TARG (with VAL, RISE, FALL, CROSS, LAST, TD, AT), FIND...WHEN, AVG/RMS/INTEG/DERIV, MIN/MAX/MIN_AT/MAX_AT/PP, ERR, and param='...', across tran/ac/dc/sp -- all present and working. The one genuine gap was AC margins. The textbook way to get them from the existing primitives, .meas ac gm FIND vdb(out) WHEN vp(out)=-180, CANNOT work: vp() returns the phase WRAPPED to $(-180, 180]$, so on any loop whose true phase sweeps past -180 the wrapped phase jumps from about -179 straight to about +179 and never equals -180,

and the crossover is never found. FIX: two first-class .meas ac functions computed on the UNWRAPPED phase -- phase_margin (PM = 180 + the unwrapped phase at the 0 dB gain crossover) and gain_margin (GM = -gain(dB) at the -180 deg phase crossover). The phase is unwrapped by undoing the +360 deg atan2 jumps, the 0 dB and -180 deg crossings are found by linear interpolation, and the crossover frequency is interpolated in log space. A loop with no finite -180 crossover (a one- or two-pole loop, infinite gain margin) is reported as such rather than as a bogus number. Implemented as measure_margin() dispatched from get_measure2() in com_measure2.c. One subtlety: ngspice's phase (vp, and the measure 'p' vector type) honors a runtime cx_degrees flag that defaults to RADIANS, so radtodeg() is a no-op by default; the margin code computes phase in degrees explicitly. TWO BUGS the work surfaced, both fixed: (1) gettok_iv (misc/string.c), used only by the batch measure auto-save pass, copied the leading v/i then broke after the FIRST following character on n_paren==0 -- so vdb(out) became vd (and vm/vp/... were all truncated); the stray vd then failed to save ('can't parse vd') and left the node unsaved, so a batch .meas ac ... vdb(out) died with 'no data saved for AC'. Fixed to stop only once the parenthesised part has actually closed. (2) a companion strip_ac_vec reduces vdb(out) to v(out) so the base node is what gets .save'd. Verify (examples/acmargin): 5 checks against closed-form loop gains (buffered one-pole sections, so poles sit exactly where placed) -- a stable 2-pole loop's phase margin matches analytic to under 1 deg (84.89 vs 84.89) and its gain margin is correctly reported infinite; a 3-pole loop's phase margin (-42.75 deg) and gain margin (-15.92 dB) both match the closed form exactly; and a batch .meas ac ... vdb(out) dot-card auto-saves the node and returns the right value. 5 checks. Regression 166/166.

E-204 *src/include/ngspice/optdefs.h, src/include/ngspice/tskdefs.h, src/include/ngspice/cktdefs.h, src/spicelib/analysis/cktsopt.c, src/spicelib/analysis/cktntask.c, src/spicelib/analysis/cktdojob.c, src/spicelib/analysis/cktop.c, examples/convhelp_examples/**

auto-triggering DC convergence aids (.option convhelp). ngspice's CKTop operating-point cascade already auto-escalated through gmin stepping and source stepping, but the two strongest aids -- the globalized damped-Newton line search (E-111) and pseudo-transient continuation (E-127) -- only fired when the user hand-set .option linesearch or .option ptcont, and nothing reported which aid rescued a hard operating point. The ngspice-vs-Spectre gap analysis flagged exactly this: what remained was auto-triggering heuristics (reaching for these aids without the user asking) and folding them into the robustness presets. .option convhelp makes the whole cascade one automatic ladder -- Newton (with line search), then gmin step, then source step, then pseudo-transient, then optran -- and prints a one-line note naming the aid that produced the point. Under convhelp the op-point Newton, and its fallback homotopies (which call NIiter internally), run with line search enabled; E-111's line search reduces to a plain full Newton step whenever the full step already reduces the weighted residual norm, so points that converge easily take the same iterates and are unaffected. The E-127 pseudo-transient gate was widened so it also fires when convhelp is set, with no explicit .option ptcont needed. CKTop was refactored to a single done: exit that saves the caller's line-search flag on entry and restores it on every return (so the setting never leaks into the transient analysis that follows) and emits the aid note there. OFF BY DEFAULT: the default (convhelp unset) path is byte-for-byte the historical behavior with no new output. errpreset=conservative enables it through the E-110 TSKtolGiven given-flag mechanism (a new ERRP_CONVHELP bit), so an explicit .option convhelp=0 still overrides the preset regardless of .options line order. Plumbing mirrors E-127: OPT_CONVHELP enum and ERRP_CONVHELP bit (optdefs.h), TSKconvhelp bitfield (tskdefs.h), CKTconvhelp bitfield (cktdefs.h), the OPT_CONVHELP setter plus convhelp in the errpreset value table plus the convhelp OPTtbl keyword (cktsopt.c), copy and default-off (cktntask.c), CKTconvhelp = TSKconvhelp (cktdojob.c), and the auto-escalation plus reporting in CKTop (cktop.c). Verify (examples/convhelp): the option is accepted; on a battery of normal circuits (diode, BJT, two-diode, resistor network) convhelp is result-neutral (node voltages identical to a plain run) and silent (no aid note, since plain Newton already converges); on a deliberately stiff behavioral-exponential deck with no junction limiting -- root of $1e-14 * (\exp(V/0.026) - 1) = (100 - V)/100$, i.e. $V(1) = 0.837922$ V -- where plain Newton overshoots to a spurious 70.5 V root

(and even the transient-op fallback lands there), convhelp reaches the correct 0.837922 V via auto-fired pseudo-transient continuation, reports the aid, and changes the outcome versus plain Newton; errpreset=conservative enables the same ladder; and an explicit convhelp=0 overrides the preset. Both linear solvers (KLU and Sparse 1.3). 35 checks. Regression 167/167.

E-205 *src/frontend/snp2va.c, examples/lowrank_examples/**

pre_snp low-rank residue factorization. The E-200/201 shared-pole realization emits an $O(N^2)$ -term, $O(NNp)$ -filter Verilog-A model, so OpenVAF's compile time grows super-linearly with the port count and caps the practical N at ~ 20 -24 (an $N=32$ block compiles in ~ 80 s -- the vector fit itself scales to $N=100$, but the emitted model's compile does not). When an N -port block's ports couple through only a few shared modes -- the common case for multi-port filters, cavities, and packages with a shared plane -- each pole's $N \times N$ residue matrix has rank r much less than N . E-205 exploits that: per channel (the conductance d , the capacitance e , and each pole section) the emitter builds the $N \times N$ real weight matrix and, via a new in-C one-sided Jacobi SVD, picks the cheaper of a dense emit (a term per significant entry, one laplace_nd per input port) or a low-rank emit $W = UV^T$ (rank kept above $\text{tol_rank} = 1e-7$, so a clean singular-value gap compresses fully while a slowly-decaying channel stays dense). The key is INPUT COMBINING: because laplace is linear, $\sum_j V[j][m] \text{laplace}(V(p_j)) = \text{laplace}(\sum_j V[j][m]V(p_j))$, so the low-rank form filters the r COMBINED inputs $u_m = \sum_j V[j][m]V(p_j)$ ONCE (r filters, not N) and distributes them to the outputs via U -- collapsing filters from $O(NNp)$ to $O(rNp)$ (the piece that dominates the compile) and coupling terms from $O(N^2)$ to $O(Nr)$. A full-rank block has no such structure, so every channel keeps the dense form and is emitted bit-identically (the auto-detect leaves it alone); the E-201 PSD projection of the capacitance e is preserved, so the low-rank transient stays stable. The env var PRE_SNP_DENSE forces the old dense realization (escape hatch / A-B test). Measured on a 3-shared-mode block (openvaf-r compile of the emitted .va): $N=16$ 16.9s to 1.5s (96 to 8 filters), $N=24$ 42s to 5.7s (144 to 10), $N=32$ 81s to 11.6s (192 to 8) -- super-linear to roughly linear, about 7-14x, with the response exact (AC $\sim 4e-7$, transient $\sim 3e-5$ versus a forced-dense build of the same fit). New code: a jacobi_svd and an emit_filter helper plus the per-channel chooser and input-combined emit, all in snp2va.c. Verify (examples/lowrank): a 12-port/3-mode block emits far fewer filters low-rank than dense (7 vs 73); its AC ($3.9e-8$) and transient ($1.2e-4$, bounded) equal the forced-dense build of the same fit; a full-rank 8-port ladder gets no compression with bit-identical AC. 5 checks. Regression 168/168.

E-206 *src/frontend/com_optimize.c, examples/dcenter_examples/**

design centering / yield optimization (optimize -center). The capstone that fuses two whole sub-systems: the optimizer (Nelder-Mead / LM / PSO / DE / SA, E-130-197) and the Monte-Carlo yield suite (agauss/mccorr sampling + montecarlo specs, E-149-151). Design centering optimizes the NOMINAL design point to maximize parametric yield / Cpk under process variation. The outer optimizer searches the design knobs (-dparam/-param/-mparam, which feed the deck's agauss centres); the objective at each candidate is an INNER Monte Carlo of -samples N draws -- each reset re-samples the deck's process variation (agauss/.param, and any mccorr correlations) around the current centre -- scored against -spec [-max hi] [-min lo] limits (the same spec syntax as montecarlo) and reduced to the worst-case Cpk and the pass-fraction yield. The fusion is small and clean in opt_eval: the in-place-knob application was factored into opt_apply_inplace so the MC loop can re-apply the centre after each reset re-sources the deck, and a new opt_eval_center runs the inner MC and returns -min(Cpk); the method drivers (nm/ps/de/sa) are unchanged -- they call opt_eval, which now returns the centering cost. Cpk is the objective (yield is reported alongside) because raw yield is a granular step function that stalls the simplex, whereas $Cpk = \min(\text{gap to each active limit}) / (3 \sigma)$, worst-cased across specs, is continuous and monotone in yield -- so maximizing it centres the distribution in its spec window. A fixed inner seed gives every candidate the SAME process draws (common random numbers), so the objective is deterministic and smooth; with -lhs the stratified sample-mean is about zero, so the optimum lands on the analytic centre. Reuses the E-149/151 sampling (mc_lhs_config, mc_sss_off, setseed); lm is rejected for -center (Cpk is not a least-squares residual) and the default method is Nelder-Mead; results

are published as `dcenter_yield` and `dcenter_cpk` vectors. `com_optimize.c` only. Verify (examples/`dcenter`): a synthetic problem output $\sim N(xc, 0.5)$ with spec [4,6] (`agauss(m,v,3)` makes the actual $\sigma = v/3 = 0.5$, so the analytic centre is the midpoint 5 and the analytic Cpk is $1/(3*0.5) = 0.667$) is centred from an off-centre start $xc=4.0$ to $xc = 5.00$ with the expected Cpk 0.667 and a high yield; a montecarlo baseline at $xc=4.0$ sits near 50% while the centred design reaches ~96%; and a two-knob problem with a lower and an upper spec on two outputs centres both params independently ($xa \rightarrow 5$ on [4,6], $xb \rightarrow 10$ on [9,11]). 5 checks. Regression 169/169.

E-207 `src/frontend/com_eye.c` (new), `src/frontend/com_eye.h` (new), `src/frontend/commands.c`, `src/frontend/com_commands.h`, `src/frontend/Makefile.am`, `examples/eye_examples/*`

eye diagram / jitter analysis (eye command). A new application domain: the core signal-integrity measurement for serial links / SerDes / high-speed I/O, which ngspice lacked. `eye -ui [-tstart t0] [-threshold vth] [-window frac]` is a self-contained front-end post-processing command (in the `fft / linearize / meas` family, registered in `commands.c/com_commands.h/Makefile.am`), solver-independent. It reads the transient waveform and its time scale via `ft_evaluate`; auto-detects the two logic rails from the 20th/80th percentiles of the samples (`eye_level0`, `eye_level1`, `eye_amplitude`) and the decision threshold (default the midpoint, or `-threshold`); finds every threshold crossing by linear interpolation; estimates the UI phase as the circular mean of the crossings modulo UI; computes each crossing's time-interval error (TIE, relative to the ideal UI grid) and reduces it to `eye_jitter_rms` (standard deviation of TIE) and `eye_jitter_pp` (peak-to-peak); measures `eye_height` (the vertical opening at the sampling instant, one half-UI from the transitions -- min of the samples above threshold minus max of the samples below, taken in a $\pm$ window `UI slice`) and `eye_width` (UI minus the peak-to-peak jitter, the horizontal opening at the threshold), plus `eye_width_ber12 = UI - 14.069eye_jitter_rms` (the eye width at BER 1e-12 from the ± 7.035 sigma Gaussian random-jitter tail); and folds the waveform modulo $2*UI$ into the `eye_wave` vs `eye_t` vectors (built with `dvec_alloc/vec_new`, as `stb` and `sweep` build their result plots), whose scatter plot IS the eye diagram (plot `eye_wave` vs `eye_t`). No numerical-core changes. Publishes 11 permanent result vectors. Verify (examples/`eye`): an ideal 0/1 clock (perfectly periodic edges) recovers rails [0,1] and amplitude 1 with jitter about 1e-24 s (machine zero) and a full-UI, fully open eye; a clock whose edges carry a KNOWN injected Gaussian timing jitter ($\sigma = 20$ ps) is recovered to within a few percent (measured about 18.6 ps rms, 99 ps pp) and `eye_width` equals UI minus the measured peak-to-peak jitter; and the folded eye vectors are published and `wrdata eye_wave` writes the folded eye without crashing (the `eye_wave/eye_t` vectors live in their own fresh eye plot -- as `stb` does -- so `wrdata/plot` pair equal-length vectors; dumping the length-m eye vectors into the transient plot, whose scale is the full much-longer time vector, `segfaults`). 7 checks. Regression 170/170.

E-208 `src/frontend/plotting/pyplot.c`, `src/frontend/plotting/pyplot.h`, `src/frontend/com_pyplot.c`, `examples/pyplot_examples/*`

`pyplot -eye`: one-line matplotlib eye diagrams. A convenience bridge between the E-207 eye analysis and the E-94 matplotlib back-end: `pyplot [name] -eye -ui [-tstart t0] [-threshold vth] [-window frac]` runs the eye analysis AND renders the folded eye in one command. `plot eye_wave` vs `eye_t` draws a single connected line through the folded samples in file order (a scribble, not an eye); `pyplot -eye` instead draws the samples as a 2-D histogram (`hist2d`, log-scaled `LogNorm`, `turbo colormap`, empty cells masked with `cmin=1`) so overlapping traces accumulate into a persistence eye -- exactly how a sampling scope paints one -- annotated with the decision threshold (dashed), the eye-centre sampling instant at 1 UI (dotted), a vertical eye-height double-arrow, a horizontal eye-width double-arrow along the threshold, and a title carrying UI / eye height / eye width (% UI) / RMS jitter. Implementation: `com_pyplot()` detects the `-eye` marker (a single bare token before it is taken as the output base name; default `eye`), runs `com_eye()` on the trailing tokens -- which folds the waveform and leaves `eye_wave/eye_t` plus the scalar metrics in a fresh current eye plot -- then calls the new `ft_pyplot_eye()` in `plotting/pyplot.c`, which reads those back from the current plot, writes `.data` (the folded sample pairs) and a `.py` matplotlib script, and launches Python

with the SAME system(python ...) mechanism as `ft_pyplot()`, honouring every existing `pyplot_*` setting: `pyplot_terminal=png/svg/pdf` renders headless (Agg) and runs synchronously, otherwise an interactive window is launched in the background; `pyplot_python` picks the interpreter, `pyplot_backend` forces a matplotlib backend, `pyplot_figsize` sets the figure size, and `pyplot_style` applies a style sheet (a dark sheet flips the annotation colours to stay legible on a dark ground). No numerical-core changes; the eye math is entirely E-207's, and `ft_pyplot()` and the `gnuplot/asciiplot` back-ends are untouched. Verify (examples/pyplot): a self-contained data eye -- a pseudo-random NRZ bit stream (PWL) through a bandwidth-limiting RC channel ($\tau \sim 0.5 UI$) -- is rendered with `pyplot eyefig -eye v(rx) -ui 0.5n`: the eye analysis runs and reports its metrics, `eyefig.png` is a valid PNG, the generated script is a hist2d persistence eye referencing the metrics, and the no-name form defaults the base to `eye.png`. 4 checks (17 total in the pyplot suite). Regression 170/170 (extends the pyplot suite, no new folder).

E-209 *src/spicelib/analysis/dcpss.c, src/frontend/com_hb.c, src/include/ngspice/cktdefs.h, examples/hb_examples/verify_hb.py*

hb publishes its spectrum as nutmeg vectors. The E-134 harmonic-balance command (`hb`) used to only PRINT a spectrum table: `HBAnalyze()` freed the converged two-sided Fourier solution V_r/V_i as soon as the table was written, so the result could not be `plot/print/wrdata`-ed without scraping `stdout` (the `amp2n2222` HB study had to parse the table in Python for every figure). Now, after a converged `hb`, a fresh current '`hb`' plot holds a real scale `hbfrequency` (0, f_0 , $2f_0$, ..., Kf_0 ; $K+1$ points) and one COMPLEX vector per node (named by the node, e.g. `out`, `vcc`, `q1#branch`) carrying the single-sided amplitude, so `mag(out)/vp(out)` reproduce the printed

E-210 *src/frontend/inp.c, src/spicelib/analysis/dcpss.c, examples/rfpss_examples/rc_pss.cir, examples/rfpss_examples/verify_rcpss.py*

PSS polish: `.pss` dot-card runs in batch + complex spectrum vectors. Two usability fixes to the E-117 shooting-method periodic steady state (`pss`), in the spirit of E-209 (`hb`). (1) The `pss COMMAND` (inside `.control`) worked, but a top-level `.pss` card by itself silently did nothing in batch mode -- `ngspice -b deck.cir` with just `.pss ...` reported "no simulations run", unlike `.tran/.ac/.hb` which auto-run; a deck had to wrap it in `.control ... run ... .endc`. `.pss` is now dispatched to the `pss` command the same way E-146 (`.sweep`) and E-162 (`.hb`) handle their cards: during deck loading (`frontend/inp.c`) a top-level `.pss ...` line is stripped of its leading '`.`' and appended to the post-parse control list, so it executes as `pss ...` once the circuit is built. A boundary check (next char is whitespace or end-of-line) keeps it from matching a longer name, and it never matches `.psp` (periodic S-parameters, a distinct card). Decks that already wrap `.pss` in `.control ... run` keep working (one PSS run, unchanged output). (2) The `pss` analysis already published two plots -- `pss1` (time-domain periodic waveform) and `pss2` (frequency-domain spectrum) -- but `pss2` stored MAGNITUDE ONLY (`IF_REAL`): the DFT computed each harmonic's phase (`pssphases`) and then discarded it, so `vp(out)` was unavailable and print out gave a bare magnitude, inconsistent with AC analysis (whose node vectors are complex) and with the `hb` command's E-209 vectors. `pss2` now publishes COMPLEX node vectors (`IF_COMPLEX`): each harmonic is stored as `mag * e^{j*phase}` (the DFT returns the phase in degrees; the DC term is kept real), built into `IFcomplex` rows and emitted with `OUTpData` (as `PAC/PXF/PSP` already do) instead of the real-only `CKTdumpp`. `mag(out)` reproduces the old magnitude spectrum exactly and `vp(out)` now recovers the phase; the max-frequency-verification logic still reads the retained magnitude array (`pssResults`), so it is unchanged, and the time-domain `pss1` plot is unchanged (a real waveform). BACKWARD COMPATIBILITY: because `pss2` node vectors are now complex, print out / `wrdata` out emit `re,im` columns (exactly as AC analysis does) rather than a bare magnitude -- decks that parsed the magnitude directly should use `mag(out)`; the bundled `rc_pss.cir` was migrated from `print b` to `print mag(b)`. This is the same magnitude-vs-complex convention the rest of `ngspice`'s frequency-domain analyses use. The shooting-PSS numerical core, the retained periodic operating point, and every analysis built on it (`PAC/Pnoise/PXF/PSP`) are untouched. Verify (examples/rfpss): a top-level `.pss` card (no `.control`) auto-runs in batch and reaches convergence; the frequency-domain spec-

trum resolves as complex -- both `vm(out)` (magnitude) and `vp(out)` (phase) return, with a driven diode rectifier's harmonics showing the expected non-trivial phase; the existing PSS/PAC/PXF/solver-parity checks pass unchanged. 2 checks (27 total in the `rcpss` suite). Regression 170/170 (extends the `rpfss` suite, no new folder).

E-211 *src/spicelib/analysis/dcop.c, src/frontend/inpcom.c, examples/codeanalysis_examples/**

Code-analysis bug fixes: XSPICE DC-op else + LTSPICE table free. A static-analysis pass over the ngspice tree (clang's analyzer run on every project-modified file, each finding then confirmed by reading the code, false positives -- e.g. symbolic-size array indexing in static math helpers -- discarded) surfaced two real defects. BUG 1 (`dcop.c`): `DCop()` chooses between the event-driven XSPICE solver (`EVTop`) and the analog solver (`CKTop`) with a `BRACELESS` else whose body was originally `converged = CKTop(...)`, so `CKTop` ran only when there were no event-driven instances. Enhancement-188 (DC warm-start) inserted its preload code between the else and `CKTop`, so the else then covered only its first statement (`wsize = SMPmatSize(...)`) and `CKTop` ran UNCONDITIONALLY: for a deck with event-driven code-model instances (A-devices) BOTH `EVTop` and `CKTop` ran, `CKTop` re-solving the analog part and overwriting the event-driven DC result, and `wsize` was read UNINITIALISED at the warm-start guard (line ~110) and snapshot (line ~129) on the `EVTop` path (undefined behaviour, masked today only because DC warm-start is off by default). FIX: hoist `wsize = SMPmatSize(ckt->CKTmatrix)` above the branch (always initialised; the warm-start snapshot reads it on both paths) and add braces so the else covers the warm-start preload + `CKTop`, restoring the original exactly-one-solve semantics. The `#ifdef XSPICE` braces vanish together when XSPICE is not compiled, so the non-XSPICE build is byte-unchanged. BUG 2 (`inpcom.c`): `inp_compat()` converts an LTSPICE `E ... table=(...)` controlled source using an UNINITIALISED `char *ckt_array[100]` stack array; the LTSPICE branch sets only `ckt_array[1]`, and a single-pair table (`table=(x0,y0)`, `ipairs==1`) then `tfree(ckt_array[2])` -- a free of a garbage stack pointer (`ckt_array[2]` is written only on the other, non-LTSPICE, branch). FIX: zero-initialise the array (`char *ckt_array[100] = { NULL };`); ngspice's `tfree(NULL)` is a safe no-op (`txfree` guards `if (ptr) free(...)`), so a `tfree` of any slot not written on a given branch does nothing instead of freeing garbage. No functional change to correct analog decks -- the fixes remove undefined behaviour and a redundant analog solve on the XSPICE and LTSPICE-import paths respectively. Verify (examples/codeanalysis, 5 checks x both solvers): a mixed-signal DC op (an analog divider + an `adc_bridge`, an event-driven instance) solves the analog node to `v(a)=0.5` without error; a single-pair LTSPICE `E ... table=(0.5, 3)` imports and yields the constant 3 V; a plain analog DC op is unchanged (a would-fail before/after test is impractical -- the redundant `CKTop` leaves a simple circuit's DC point unchanged, and freeing a garbage pointer is UB that may or may not crash). 5 checks. Regression 171/171 (new example folder).

E-212 *src/frontend/breakp.c, src/frontend/device.c, src/frontend/vectors.c, src/frontend/dotcards.c, src/frontend/inpcom.c, src/spicelib/analysis/cktpzset.c, examples/crashfix_examples/**

Crash hardening: seven user-triggerable crashes. A static-analysis pass (clang analyzer) plus argument fuzzing of every frontend command and of malformed/degenerate netlists surfaced seven reproducible, user-triggerable crashes (SIGSEGV/SIGABRT) in STOCK ngspice, all in user-input handling (command parsers, the `.op` output path, netlist recursion) and none in the numerical core (device models, OSDI loader, Sparse 1.3, KLU, and the analysis drivers were audited and found clean). Each is a missing edge-guard. (1) `iplot` (`breakp.c`, `com_iplot`): `iplot -w/-d` with the flag as the trailing token consumes its value word (`wl=wl->wl_next`), guards it with `if(wl)` (proving `wl` can be NULL), then the loop-bottom `wl=wl->wl_next` dereferences NULL. FIX: guard the advance. (2) `altermod` (`device.c`, `com_altermod->com_alter_common`): `com_alter` guards `if(!wl)` but `com_altermod` does not, so a bare `altermod` reaches `com_alter_common` with an empty list `-> wlin=wl_head=NULL -> wl_nthelem(100,NULL)=NULL -> eq(wlin->wl_word, ")]")` derefs NULL. FIX: guard the empty `wl_head` with a clean error. (3) `measure` (`vectors.c`, `vec_get`): a malformed measure statement (`meas tran x FIND, ... WHEN, FIND v(1)` with no `WHEN/AT`, across `tran/dc/ac/sp`) leaves `meas->m_vec` NULL, which flows into `com_measure_when ->`

vec_get(NULL) -> copy(NULL)=NULL -> strchr(NULL, '.'). FIX: make vec_get NULL-safe (if !vec_name) return NULL) -- the common sink; every caller already handles a NULL return, so each trigger reports "no such vector" instead of crashing. (4) empty .op (dotcards.c, ft_cktcoms): a circuit with no non-ground nodes -- e.g. a deck whose every component is commented out - produces an op plot with no data vectors, so pl_dvecs is NULL; the former assert(pl_dvecs != NULL) aborted (SIGABRT; ngspice's autotools build defines no NDEBUG so asserts are live) and with -DNDEBUG the next line dereferenced NULL. FIX: guard if(plot_cur && plot_cur->pl_dvecs) and skip the empty op printout (the solver had handled the empty matrix correctly; this was purely output formatting). (5) KLU pole-zero (cktpzset.c): a bsearch miss for the drive pointer was reported with a printf and then dereferenced anyway (job->PZdrive_pptr = matched->CSC_Complex); latent (SMPmakeElt creates the element just above, so it is always in the bind table and the miss cannot occur today) but the guard was objectively broken. FIX: add the missing else/NULL path, matching the correct idiom in cktsetup.c. (6) .include recursion (inpcom.c, inp_read): inp_read tracked call_depth but never limited it, so a file that .includes itself -- directly, via an A->B->A cycle, or via a .lib section that .includes its own file -- recursed until the C stack overflowed (SIGSEGV); .lib files are de-duplicated by find_lib and read once, so the crash is always via .include. FIX: cap at INP_MAX_INCLUDE_DEPTH=50; a cycle now reports an error (a 49-level chain still reads fine). (7) .func recursion (inpcom.c, inp_expand_macro_in_str): the expander re-expands a substituted function body with no depth guard, so .func f(x)={f(x)} (or a mutual f<->g) expanded forever and overflowed the stack (each frame also holds a params[FCN_PARAMS] array ~8KB, so it fails near ~1000 deep). FIX: a macro_depth counter capped at INP_MAX_MACRO_DEPTH=100, reset on every error path. The netlist-recursion caps mirror the include/expression-depth hardening OpenVAF-r received in Enhancement-148. No functional change to correct decks -- the fixes turn crashes on malformed or degenerate input into graceful errors. Verify (examples/crashfix, 17 checks x both solvers): every repro now exits gracefully (no signal -- crashes are detected from the child's SIGSEGV/SIGABRT exit code, which the pre-fix binary returned) while every valid form still works (iplot -w 3 v(1); altermod nm vto=0.7; meas tran vmax MAX v(1) -> 1.0; a normal .op; KLU .pz; a legitimate nested .include; .func sq(x)={x*x} -> sq(4)=16). 17 checks. Regression 172/172 (new example folder).

E-215 *src/main.c, src/osdi/osdiload.c, src/include/ngspice/osdiitf.h, examples/plusargs_examples/**

command-line plusargs (+name[=value]) served to Verilog-A models. Lets ngspice -b deck.cir +corner=ff pick a corner or toggle a feature flag at run time without editing the deck -- the simulator half of E-215 (the compiler half is in the openvaf report). main.c treats a +-prefixed command-line argument as a plusarg rather than an input file (both the batch and interactive file loops) and calls the new ngspice_register_plusarg (osdiitf.h). get_simparams (osdiload.c) exposes each plusarg on the OsdiSimParas channel a compiled model already reads for \$simparam\$str (E-25), as namespaced simparams: numeric \$test\$plusargs\$name=1 (present), \$valset\$plusargs\$name=1 (given as name=value), \$valnum\$plusargs\$name (value via strtod), and string \$value\$plusargs\$name (value text). The base sim_params arrays are spliced onto the extended arrays only the FIRST time get_simparams sees any plusarg present, so a run with NO plusargs allocates nothing and is byte-for-byte unchanged; the extended arrays are built once (plusargs are constant for the run) and only the base numeric values + analysis_name are refreshed per call. No OSDI ABI change (the string/number channel already existed). Verify (examples/plusargs, 12 checks both solvers): a conductance set from +boost/+gain=25/+scale=2.5/+corner=ff|ss plus unmatched/unrelated/no-value edge cases. Regression 175/175 (new example folder).

E-216 *src/frontend/com_optimize.c, examples/pareto_examples/**

NSGA-II multi-objective / Pareto optimization. Adds a -method nsga2 to the optimize command that trades TWO OR MORE competing objectives and returns a Pareto FRONT of non-dominated designs, rather than the single optimum the scalar methods return (E-130 Nelder-Mead, E-194 PSO, E-195 DE, E-196 SA) -- the multi-objective generalization of E-206 design cen-

tering. Objectives are given as repeated -minimize / -maximize ; the existing optimize infrastructure is reused wholesale (the -param/-dparam/-mparam knobs, the -analysis-driven expression objectives via opt_eval_expr, the normalized [0,1] search space and the opt_rand RNG). New: a vector-valued objective evaluator opt_eval_objs (maximized objectives negated to a common minimization convention) and nsga2() implementing standard NSGA-II (Deb 2002) -- fast non-dominated sorting (Pareto rank via domination counts), crowding distance (per-front normalized neighbour gaps, boundaries at infinity), binary-tournament selection by the crowded-comparison operator, real-coded SBX crossover (eta=15) + polynomial mutation (eta=20, rate 1/n), and elitist parent+offspring (2N->N) survivor selection so a non-dominated design is never lost. The command prints the final front (each design's objectives + parameters, sorted by the first objective) and publishes each objective column as a permanent vector pareto1/pareto2/... so the front plots directly (plot pareto2 vs pareto1). nsga2 needs >=2 objectives and a single -analysis stage; population defaults to 20+4*np (override -swarmsize), generations -maxiter, -seed for reproducibility. The scalar methods and their single-objective/least-squares/centering paths are unchanged (a lone -minimize still seeds the scalar objective for nm/pso/de/sa). Verify (examples/pareto): the Schaffer-1 benchmark -- minimize $f_1 = px^2$ and $f_2 = (px-2)^2$ over px in $[-1,3]$, whose Pareto set is exactly px in $[0,2]$ tracing $f_2 = (\sqrt{f_1}-2)^2$, with the knob range wider than the front so NSGA-II must discover $[0,2]$. The returned front is a reasonable size, every point is genuinely non-dominated, it spans the trade-off (px reaches ~ 0 and ~ 2), and the published vectors lie on the analytic curve to machine precision ($8.9e-16$). 6 checks. Regression 176/176 (new example folder).

E-217 *src/frontend/com_pyplot.c, src/frontend/plotting/pyplot.c, src/frontend/plotting/pyplot.h, src/frontend/plotting/plotit.c, examples/pyplohist_examples/**

pyplot-hist: histogram plots. `pyplot -hist <sig> ...` renders each listed signal's VALUE distribution as a matplotlib histogram (plt.hist) instead of a trace vs time/frequency -- for the amplitude spread of a transient, a Monte-Carlo measurement, or a noise sample. Unlike -eye (E-208, a distinct analysis) -hist is a render MODE, so it is small and reuses the whole pyplot pipeline: com_pyplot detects the -hist marker anywhere in the arg list, unlinks it, and dispatches the remaining [name] signal-list to plotit with a distinct device string pyplohist; plotit resolves the signal expressions exactly as for a normal plot (v(out), db(...), vector arithmetic, node names) and passes the resolved vectors to ft_pyplot with a hist flag; ft_pyplot writes the same .data table and emits `ax.hist(value_column, bins, alpha, label)` per signal instead of `ax.plot(x,y)`. All existing pyplot facilities carry over unchanged (set `pyplot_terminal=png/svg/pdf` headless render, `pyplot_subplots` panels, style sheets, `figsize`, `backend`); two histogram-specific options are added -- `pyplot_hist_bins` (bin count, default matplotlib 'auto') and `pyplot_hist_density` (normalize to a density). Correctness details: the x-axis is the signal value and the y-axis the count (labels set accordingly); histogram panels do NOT share an x-axis (`sharex=False` -- different signals have unrelated value ranges, vs a line plot's shared time/freq axis); the FULL value length is histogrammed -- the data-table loop normally walks the shared scale vector, but a histogram of a raw let vector (e.g. a Monte-Carlo result) can have a scale whose length differs from the values', so for hist it walks the longest VALUE vector and pads shorter ones with NaN (the generated script filters via `~np.isnan`), fixing a silent truncation to the scale length; overlaid histograms get `alpha=0.6` transparency and a legend while a single one is drawn opaque. No change to the ordinary pyplot trace path or to pyplot -eye. Verify (examples/pyplohist): two signals with closed-form distributions -- `ramp=i/(N-1)` uniform on $[0,1]$ (flat histogram), `sine=sin(2pi*i/100)` arcsine on $[-1,1]$ (U-shaped, a sinusoid dwells near its peaks so the edges tower) -- checked analytically from the generated .data (uniform bins within 15% of the mean; sine edge bins » middle), plus the -hist path taken (plt.hist not plt.plot, non-shared x), the full 20000-sample length not truncated, and a valid non-trivial PNG. 6 checks. Regression 177/177 (new example folder).

E-218 *src/frontend/com_pyplot.c, src/frontend/plotting/pyplot.c, src/frontend/plotting/pyplot.h, src/frontend/plotting/plotit.c, examples/pyplotcontour_examples/**

pyplot -contour: 2-D contour maps for 2-D parameter sweeps. `pyplot -contour <z> <x> <y>` renders a filled matplotlib contour map of a quantity *z* over the (*x*,*y*) plane -- the natural view of a 2-D parameter sweep (a device current over a (*V*_{gs},*V*_{ds}) grid, a gain over an (*R*,*C*) grid, a noise figure over a (*freq*,*bias*) grid). Like `-hist` (E-217) and unlike `-eye` (E-208, a distinct analysis), `-contour` is a render MODE dispatched over the ordinary signal list: `com_pyplot` detects the `-contour` marker anywhere in the arg list, unlinks it, and dispatches the remaining `[name] z x y` to `plotit` with a distinct device string `pyplotcontour`; `plotit` resolves the three expressions exactly as for a normal plot (*v*(out), *db*(...), vector arithmetic, node names) into a three-vector list (*z* first, then *x*, then *y*); the new `ft_pyplot_contour` writes an (*x*,*y*,*z*) data table and a matplotlib script that TRIANGULATES the (*x*,*y*) points (`ax.tricontourf`). Because the plane is triangulated rather than reshaped into a rectangular mesh, gridded OR scattered sweep data plots identically -- no grid-dimension metadata (*nx*,*ny*) is needed, and a sweep with a ragged tail still works; *x*/*y*/*z* come from any 2-D sweep leaving three equal-length vectors (nested .dc, the sweep command family E-146/E-190, a .step grid, or .control vectors). All existing pyplot facilities carry over (set `pyplot_terminal=png/svg/pdf` headless render, style sheets, `figsize`, `backend`); three contour-specific options are added -- `pyplot_contour_levels` (level count, default matplotlib-auto), `pyplot_contour_lines` (overlay labelled black contour lines on the filled map), `pyplot_contour_cmap` (colormap, default viridis). The colorbar is labelled with *z*'s name and the axes with *x*'s and *y*'s. Robustness: `-contour` needs exactly three vectors, else it is rejected with a clear message (`pyplot -contour` needs exactly three vectors) rather than a confused plot; a degenerate 1-D (collinear) sweep cannot be triangulated, so the generated script catches the `qhull/matplotlib` failure and exits with a helpful line (a contour needs a genuine 2-D sweep -- the points must not be collinear) instead of a bare traceback. No change to the ordinary pyplot trace path, `pyplot -hist`, or `pyplot -eye`. Also removed a stale unused-variable warning (`ft_pyplot`'s `scale`) left by E-217. Verify (`examples/pyplotcontour`): a closed-form surface $z=x^2+y^2$ over *x*,*y* in $[-2,2]$ (a paraboloid whose contours are concentric circles, *z*=0 at the centre and *z*=8 at the corners) -- checked analytically from the generated .data: the `-contour` path is taken (`tricontourf` not `plot/hist`, a colorbar labelled *z*, axes *x*/*y*), the data table has three columns (*x*,*y*,*z*) for all 1681 rows, the column mapping is correct (*z* reconstructs x^2+y^2 to 9e-16), the sweep is genuinely 2-D (*x* and *y* each span the full range, *z* runs from ~0 at the centre to ~8 at a corner), and a valid PNG is rendered. A second demo (`bridge_dc_demo.cir`) exercises the feature on genuine simulation output: a diode-OR bridge whose output *V*(*c*) follows whichever of its two inputs is higher, swept with a nested .dc (*v*₁ drives node *a*, *v*₂ drives node *b*, so *V*(*a*)/*V*(*b*) are the two swept values at each point); `pyplot -contour v(c) v(a) v(b)` maps the output over the (*V*(*a*),*V*(*b*)) plane -- a max-like corner surface. It checks the nested .dc yields the flattened 51x51 = 2601-row 3-column grid, the axes are the two real swept sources (each spanning $[-1,1]$), and the output is ~0 when both inputs are low and rises when either input is high (the diode-OR signature, which confirms the columns map onto a real circuit result). 19 checks. Regression 178/178 (new example folder).

E-221 *src/frontend/inpcom.c, examples/busnodes_examples/**

array/bus node ranges in the netlist. ngspice has no bus/array node syntax: a node is an opaque string, and a range like `a[0:1]` is read as a single literal node name (so `R1 a[0:1] r=2k` gives `R1` only one node token and misparses `r=2k`). E-221 adds a netlist preprocessing pass that expands a range token `base[lo:hi]` into the scalar node sequence `base[lo] base[lo±1] ... base[hi]`, so buses can be written compactly: `R1 a[0:1] r=2k -> R1 a[0] a[1] r=2k`; `X1 n[0:3] mysub -> X1 n[0] n[1] n[2] n[3] mysub`; `.subckt mysub p[3:0] -> .subckt mysub p[3] p[2] p[1] p[0]`. The range supplies node tokens POSITIONALLY -- the 'two terminals' reading: `R1 a[0:1]` is a single resistor from `a[0]` to `a[1]`, not an array of resistors; multi-terminal instances, subcircuit calls and `.subckt` port lists expand the same way. Semantics: `base[lo:hi] -> base[lo] base[lo±1] ... base[hi]`, DESCENDING when `lo>hi` (Verilog convention `d[3:0] -> d[3] d[2] d[1] d[0]`); `a[0:0]` is a single element; the scalar names use the same bracket form so a bus `a[0:1]` and an explicit `a[0]` denote the SAME node (the two forms interoperate); only a WHOLE whitespace-delimited token that is exactly `base[int:int]` is expanded, so already-scalar names (`a[0]`), non-integer ranges (`a[x:y]`), malformed ones (`a[1:2:3]`),

device values, model names and XSPICE %vd[...] port groups are left untouched; .controlendc blocks are skipped and only element lines and .subckt lines are processed (.model/.param/other dot cards ignored); a range wider than 8192 elements is left literal (the device parser then reports it) so a typo like a[0:1000000000] cannot exhaust memory. Implementation: one pass inp_expand_buses in frontend/inpcom.c, run in inp_readall right after whitespace normalization and BEFORE subcircuit expansion and device parsing -- so every downstream consumer (numparam, .subckt expansion, the device parser, OSDI) sees the already-scalar node list and needs no changes; continuation lines are stitched earlier in inp_read, so each logical line is processed whole; the token test parses base (a plain node name) then [:] filling the rest of the token and emits the element list with DSTRING, a non-match copying the token verbatim. Verify (examples/busnodes): 6 checks, both solvers, all from node voltages/branch currents (solver-independent) -- R1 a[0:1] r=2k connects a[0]-a[1] so $I(R1)=(1-3)/2k=-1mA$; the scalar R2 a[0] addresses the same node as the bus element ($I=+0.25mA$, only correct if a[0] is one shared node); a subcircuit with a descending port bus p[1:0], called with X1 n[0:1], connects positionally ($I(Rs)=-5mA$); and non-integer (b[x:y]), malformed (c[1:2:3]) and scalar tokens are left literal while a .control let w=a[0:1] is not rewritten. One file; no device/solver/OSDI change; a deck with no base[lo:hi] token is untouched (a fast path skips any line without '['). Regression 181/181 (new example folder).

E-222 *src/frontend/inpcom.c, src/frontend/subckt.c, src/misc/string.c, examples/parserfuzz_examples/**

ngspice netlist-parser hardening (fuzzing): seven crashes/hangs -> clean errors. Fuzzing the netlist parser -- mutating real decks (byte flips, truncation, line duplication, delimiter/bracket/keyword/directive/number injection, and the new E-221 bus-range tokens) and running each under ngspice -b -- found seven ways to CRASH (SIGSEGV/SIGABRT, real memory-safety bugs since ngspice is C) or HANG on malformed input; a 12000-iteration campaign dropped the crash+hang rate from 26 to 1 (96%, all hangs eliminated). ROOT CAUSES, all fixed: (1) misc/string.c gettok_instance() returns an empty token WITHOUT advancing *s when it sits on '(' or ')', and frontend/inpcom.c get_number_terminals()'s OSDI ('n') case was the only multi-token case with no iteration cap -- a line starting with 'n' containing '(' (e.g. 'nan.func f(x)=...', nan lexing as a number) spun forever tmalloc-ing each pass; gettok_instance() now always advances (consumes the bracket as a one-char token) and the 'n' loop gained the same cap as its siblings. (2) inp_expand_macro_in_str(): a truncated '.model m' (no model type) ran nexttok off the end, leaving the scan pointer NULL, then strchr(NULL); NULL guards added on the .model skip and the scan loop. (3) frontend/subckt.c doit(): MAXNEST bounds subcircuit DEPTH, but a subckt that instantiates itself with branching factor ≥ 2 blows up to $2^{MAXNEST}$ instances (an effective hang) before the depth cap trips -- a total-instantiation cap now catches recursive subcircuits. (4) doit(): a bare 'X' invocation (nothing after the refdes) makes the find-last-token walk run off the FRONT of the line buffer -- an out-of-bounds read ending in strcmp(NULL); such a line is now skipped. (5) inp_modify_exp(): two copies into a fixed buf[512] with no bound -- an unterminated 'v' (no closing ')') and a long identifier token (a run of '[', which is an accepted char) overran the STACK buffer (stack smashing); both loops are now bounded by the buffer size. (6) doit(): a malformed '.subckt' can register a NULL name; the invoked-name match eq(su_name,...)=strcmp() then deref NULL; guarded. (7) translate(): gettok_noparens() returns NULL at the end of a malformed controlled-source line, then strcmp(NULL,'POLY'); guarded. Method: the fuzzer classifies OK/clean-error/CRASH/HANG over self-contained seed decks exercising the parser surface (devices, .subckt, .param, .model, .control, expressions, continuations, buses); each crash was triaged to its faulting instruction with lldb (conditional strchr/strcmp-with-NULL breakpoints), sample for the hangs, and Guard Malloc (libgmalloc) to pin the stack overrun, then read at the source; fixes applied one at a time, verified against the recorded crash corpus, and a fresh 12000-iteration run confirmed convergence. No valid deck changes behaviour. Verify (examples/parserfuzz): 10 checks -- a minimal deck per root cause (nan.func, .model in .control, a self-recursive subckt, a bare X line, an unterminated v, a 4000-'[' identifier, 4000 '{' in a subckt E-source) now yields a clean/bounded outcome (no signal/abort, no hang), and three valid decks (subckt hierarchy with an E-source and a parameter, a v(x)*v(x) B-source expression, a POLY source in a subckt) compute V to the analytic value. The

recorded 27-input crash corpus is all clean. RESIDUAL: one fuzz input in 12000 still crashes in XSPICE (xspice/mif/mifgetmod.c MIFgetMod) -- an 'a'-device whose model name matches a non-code-model (e.g. a diode .model) is processed as a code model, reading device->modelParms as the wrong type; a code-model type-confusion in the XSPICE subsystem, out of scope for this netlist-parser pass and flagged as a follow-up. Three files (frontend/inpcom.c, frontend/subckt.c, misc/string.c); no device/solver/OSDI change. Regression 181/181 (new example folder).

E-223 *src/xspice/mif/mifgetmod.c, examples/xspicemodel_examples/**

XSPICE a-device model-type validation (MIFgetMod): fixes the one residual crash left by the E-222 netlist-parser fuzzing pass. An a device is an XSPICE code-model instance; the last token on its card is a model name, which MIF_INP2A (xspice/mif/mif_inp2.c) hands to MIFgetMod (xspice/mif/mifgetmod.c). MIFgetMod looks the model up in the pass-1 table and, if it has not been instantiated yet, processes its .model card AS a code model -- it casts INPmodfast to MIFmodel, reads the device's XSPICE DEVpublic fields (param, num_param), and matches the .model parameters through the MIF parameter path -- with NOTHING checking that the resolved model is actually a code model. If an a-device's model name happens to match a NON-code-model .model (a diode, a BJT, a resistor model), MIFgetMod walked all of that machinery over a structure that is not a MIFmodel and over DEVpublic fields that a built-in SPICE device leaves unset, reading unrelated memory as the wrong type -> SIGSEGV. The crash bit hardest when the non-code-model's parameter keywords COLLIDE with the a-device's: a diode .model has is/n, so the parameter loop found a keyword match and dereferenced the type-confused setModelParm path. The fuzz-found trigger (the E-222 residual) was exactly this, reached through an E-221 bus-expanded a-device inside a subcircuit (a[0:0] D1 a b dm with .model dm d(is=1e-14 n=1)). FIX: a code model is exactly a device that carries a code-model evaluation function -- cmpp emits .cm_func = <fn> for every code model (xspice/cmpp/writ_ifs.c) and it is the pointer XSPICE calls to evaluate the model (mifload.c/evtload.c), while every built-in SPICE device sets .cm_func = NULL. MIFgetMod now validates DEVICES[INPmodType]->DEVpublic.cm_func != NULL (and INPmodType < DEVmaxnum) BEFORE the MIFmodel cast and the parameter loop, returning a clean 'model X is not a code model; an a device requires an XSPICE code-model .model' error otherwise. The guard sits before the cast, so it catches the model whether or not an ordinary device of the same name already instantiated it. A NULL guard on the inner strcmp was rejected as whack-a-mole (the crash merely shifts to the next type-confused access); the model TYPE is the thing to validate. Legitimate code models are unaffected (cm_func != NULL). Scope: XSPICE only, one file (xspice/mif/mifgetmod.c); no netlist-parser, device, solver, or OSDI change. Verify (examples/xspicemodel): 8 checks, both solvers -- six pathological decks (a-device -> diode with and without colliding params, BJT, resistor model, the bus-expanded-in-a-subckt fuzz repro, and an undefined model) now yield a clean bounded error (no signal/abort, no hang), and a legitimate adc_bridge code-model a-device still binds and simulates (analog divider node v(a)=0.5). Regression 182/182 (new example folder).

E-224 *src/frontend/parse.c, src/frontend/evaluate.c, examples/arraynodeprint_examples/**

Reference array/bus node voltages in **print/plot/let**. Enhancement-221 added array/bus node ranges to the netlist (R1 a[0:1] -> R1 a[0] a[1]), naming the nodes with LITERAL brackets so the node voltage vector is stored as a[0]. But the frontend expression parser uses [...] as its vector-INDEX operator, so in print/plot/let the token a[0] was parsed as 'element 0 of a vector named a'. There is no vector a, so print v(a[0]) and print a[0] silently failed (with a 'vector a is not available' warning) even though the node existed and solved correctly (it shows up in the .op node table and display). FIX: a literal-node fallback -- when the base name of an index expression is an UNRESOLVED name (a zero-length placeholder dvec) and the index is a constant and a node/vector named base[index] exists, use that node instead of indexing. It only fires when the base does not resolve, so ordinary vector indexing (realvec[3], where the base IS a real vector with non-zero length) is unaffected. Three frontend spots, because a[0] reaches the machinery three ways: (1) frontend/parse.c checkvalid() -- the pre-evaluation validity check rejected the whole

term because base *a* is a zero-length placeholder; for an INDX node over an unresolved base it now accepts the term when the literal node `base[index]` exists. (2) `frontend/evaluate.c op_ind()` -- evaluates the bare `a[0]` form; it now resolves the literal node name BEFORE evaluating the bare base, so no spurious 'no such vector *a*' is emitted. (3) `frontend/evaluate.c apply_func()` -- the `v()` node-access special case needed `arg->pn_value`, which an INDX op node lacks (it errored 'bad `v()` syntax'); it now reconstructs the literal node name from the INDX argument and resolves it. RESULT: `print a[0]`, `print v(a[0])`, array nodes inside expressions `v(a[0])-v(a[1])`, and `plot/let` all work; the recovered vector is a normal waveform (indexable, aggregatable) in transient. Branch current through the bus resistor stays the device parameter `@r1[i]` (ngspice has no current-between-two-nodes syntax; `i()` takes a source name). Scope: frontend only, two files (`parse.c`, `evaluate.c`); no device/solver/OSDI change. Verify (examples/arraynodeprint): 8 checks, both solvers -- `print a[0]/v(a[0])/v(a[1])`, the expression `v(a[0])-v(a[1])`, and `@r1[i]` read the correct DC values; ordinary indexing `unitvec(4)[2]` is unchanged; a genuinely-missing node `v(a[9])` stays a clean miss (no value, no crash). Regression 183/183 (new example folder).

E-225 *src/frontend/evaluate.c, src/frontend/com_measure2.c, src/frontend/fourier.c, src/math/cmaths/cmath4.c, examples/cmdfuzz_examples/**

Command/expression-evaluator crash hardening (fuzzing). With the netlist parser hardened (E-212/E-222/E-223), the fuzzer was turned on a previously-unfuzzed surface: the interactive `.control` command interpreter and the vector-expression evaluator (`print/let/plot/meas/fft/fourier` and the operators `?:`, indexing `[ ]`, ranges `[ : ]`, arithmetic). Mutating command lines/expressions over a running circuit and running each under ngspice `-b` found memory-safety crashes (SIGSEGV / SIGABRT, real since ngspice is C); 62 crashes in the first 2500 iterations, driven to 0 by FIVE fixes in three groups. GROUP A -- degenerate-input heap overflows in the math transforms (`fft/deriv/fourier` each crashed on a too-short (< 2-point) or synthetic vector like `fft(1)`, `deriv(vecmin(v(1)))` (a plot-derived scalar), `fourier 1k deriv(vecmin(v(1)))`; the malloc abort fired on a LATER allocation, so they looked non-deterministic): (1) `maths/cmaths/cmath4.c cx_fft()` -- the Green's FFT (rffts) derefs bit-reversal tables `BRLowArray[(M-1)/2]` that `fftInit()` only allocates for `M>2`, so an input of `<=4` points read unallocated memory; and `time/xscale` were sized by the data length while the scale loops fill `pl_scale->v_length` entries and the scale vector uses `fpts` points, so a vector shorter than the plot scale overran both -- pad to `N>=8`, size buffers for the largest fill, reject `length<2/scale-too-short`. (2) `maths/cmaths/cmath4.c cx_deriv()` -- the polynomial-fit derivative reads `degree+1` points per fit; a `length-1` input ran the fit/edge loops over too few points and overran the heap -- for `length<2` return a zero vector (derivative of a single point is undefined). (3) `frontend/fourier.c` -- a degenerate input gives a zero time span and overran the interpolation grid -- require `>=2` data points. GROUP B -- `frontend/evaluate.c ft_ternary()`: the `?:` operator evaluated its condition and selected branch with no NULL check, so when either failed to evaluate (`1?0[3]:9`, where `0[3]` indexing-a-scalar returns NULL) it hit `vec_copy(NULL)` / `cond->v_link2` and crashed -- the dominant crash; fixed with NULL guards (an audit confirmed `op_ind`, `op_range`, `binary-op` and `apply_func` already guard NULL). GROUP C -- `frontend/com_measure2.c`: measure error messages were formatted into a shared `char errbuf[100]` (passed by pointer to the `measure_parse_*` helpers); a failed measure writes its ENTIRE expression string as the vector name (`sprintf(errbuf, "no such vector as '%s'", meas->m_vec)`), so a measure expression longer than ~80 chars (trivially `meas tran m MAX (v(1)+v(1)+...)`) overran it -> SIGABRT (content irrelevant, only length, which is why it resisted reduction) -- fixed with a file-scope `MEAS_ERRBUF_SIZE` and a bounded `snprintf` at every write (28 sites). Note: `dowhile 1 ... end` (true constant condition, no break) is an intentional infinite loop, flagged as a HANG but correct. Scope: frontend/math only, four files (`evaluate.c`, `com_measure2.c`, `fourier.c`, `cmath4.c`); no device/solver/OSDI change. Verify (examples/cmdfuzz): 13 checks -- a minimal repro per root cause now yields a clean bounded outcome, and regressions confirm `fft(v(1))/deriv(v(1))`, a valid `?:`, and ordinary indexing still work. Regression 184/184 (new example folder).

E-226 *src/frontend/rawfile.c, examples/rawfuzz_examples/**

Rawfile-load crash hardening (fuzzing). Continuing the fuzzing campaign onto another untrusted-input surface: the nutmeg rawfile reader (`raw_read` in `frontend/rawfile.c`, reached by the `load` command). A `.raw` file is external data (often written by another tool), so its parser must survive malformed input. Mutating a valid `.raw` (produced by `write`) -- tweaking the header counts, corrupting the `Flags:/Variables:` lines, adding/removing value rows, truncating, flipping bytes -- and load-ing each found a NULL-dereference crash (SIGSEGV, memmove into 0x0). ROOT CAUSE: `raw_read` resets `flags = VF_PERMANENT` at the start of each plot and sets `VF_REAL / VF_COMPLEX` only when it parses a `Flags:` line; each vector is then allocated with `dvec_alloc(NULL, SV_NOTYPE, flags, npoints, NULL)`, which allocates `v_realdata` OR `v_compdata` according to those bits. A rawfile with NO valid `Flags:` line -- the line missing entirely, or carrying only unknown flags (`Flags: xyz`) -- leaves flags with neither `VF_REAL` nor `VF_COMPLEX`, so `dvec_alloc` allocates NO data array at all, and the value-reading loop does `fread(&v->v_realdata[i], ..) / writes v->v_compdata[i]` into a NULL buffer -> memmove into 0x0 -> SIGSEGV. (A bad No. Points: / No. Variables: count -- huge, negative, non-numeric -- was already handled cleanly; only the missing TYPE was fatal.) FIX: when the header is complete and vector allocation begins (at the `Variables:` line), if flags carries neither `VF_REAL` nor `VF_COMPLEX`, default to real (the common case) with a warning instead of allocating a typeless vector; the `VF_PERMANENT` reset is per-plot so the guard runs for every plot in a multi-plot file, and a valid file is unaffected. Also fuzzed in the same pass: the interactive `.control` command interpreter and shell layer (variable substitution `\$var`, `alter/altermod/alterparam`, analysis reruns, structured control `if/while/foreach/repeat/dowhile`) -- clean over 7500 iterations (already hardened by E-212/E-222/E-225). Scope: frontend only, one file (`rawfile.c`); no device/solver/OSDI change. Verify (`examples/rawfuzz`): 4 checks -- a valid `.raw` is written, then loaded with its `Flags:` line removed and with an unknown `Flags: xyz`, each now a clean bounded outcome (was SIGSEGV); a regression confirms a valid `.raw` still round-trips write -> load with the right length. Regression 185/185 (new example folder).

E-227 *src/frontend/snp2va.c, examples/snpfuzz_examples/**

Touchstone `.snp` parser crash hardening (fuzzing). Continuing the fuzzing campaign onto the Touchstone S-parameter parsers -- external network-data files (VNAs, EM solvers, other tools), so their parsers must survive malformed input. Two exist: `rdsnp` (the E-72 Touchstone-v1 reader) and `pre_snp` (the E-200/E-201 convert-to-Verilog-A path in `frontend/snp2va.c` that vector-fits the data and compiles it through `openvaf-r`). Mutating valid `.s1p/.s2p/.s3p` files -- corrupting the # option line, data values (NaN/Inf/overflow tokens, added/removed columns), duplicating/deleting rows, truncating, flipping bytes -- AND mismatching the port count encoded in the `.snp` filename extension, then running each through both readers: `rdsnp` was clean (4,500 iters); `pre_snp` hit a heap-corruption crash (SIGSEGV) 103 times, ALL on a filename with an absurd port count (e.g. `.s2147483647p`). ROOT CAUSE: `snp2va.c` infers the port count from the extension (`N = atoi(dot+2)`) with no upper bound; a huge `N` slips past the parse itself because the record stride `rec = 1 + 2*N*N` and the per-record inner-loop bound `N*N` use the same wrapped int (`N*N` overflows to a small/zero value, so the token layout stays self-consistent and no error is raised), but `N` is stored in `out->N` and used DOWNSTREAM to size the vector-fit allocations (`malloc(nf*N*N*sizeof(cplx))`, `malloc(N*N*sizeof(cplx))`) -- the overflowing/mismatched allocation and the writes that follow corrupt the heap (the program aborts later on an unrelated `malloc`, hence flaky). The brute-force fallback (used when the filename gives no usable count) already caps inference at 512 ports, so 512 is the parser's own sanity bound -- the filename path just wasn't held to it. FIX: hold the filename-derived count to the same limit -- `if (N > 512) N = 0`; falls back to inferring `N` from the data. A valid file is unaffected (its extension is far below the cap) and a genuinely mis-extended file now recovers `N` from the data layout instead of trusting an absurd filename. Scope: ngspice frontend only, one file (`frontend/snp2va.c`); no device/solver/OSDI change. Verify (`examples/snpfuzz`): 2 checks -- a valid `.s2p` copied to a `.s2147483647p` name loads via `pre_snp`

without crashing (run repeatedly, was SIGSEGV), and a valid .s2p still converts through the full pipeline (pre_snp -> .va -> openvaf-r -> .osdi). Regression 186/186 (new example folder).

E-228 *src/osdi/osdiregistry.c, examples/osdifuzz_examples/**

OSDI **.osdi** loader descriptor-count hardening (fuzzing). A .osdi model is an external, compiled shared library that ngspice dlopens and reads descriptor metadata from (load_object_file, reached by the pre_osdi command). The loader strided/indexed the descriptor arrays by the counts the file declares -- OSDI_NUM_DESCRIPTORs (outer loop stride by *OSDI_DESCRIPTOR_SIZE) and each descriptor's num_params (inner loop indexing param_opvar[param_id] and dereferencing its name[] pointers), num_opvars/num_terminals/num_noise_src/... (offset arithmetic) -- with NO bounds. A .osdi that still dlopens but declares a count larger than its real array (a truncated file, a byte-corrupted one, or one built by a compiler whose OSDI descriptor ABI differs -- the drift the loader's own comments warn about) read past the array -> SIGSEGV during pre_osdi (same unbounded-count class as E-227, here on a binary rather than a text surface). FIX: two generous ceilings in osdiregistry.c (#define OSDI_MAX_DESCRIPTORs 4096u, #define OSDI_MAX_DESC_COUNT (1u<<20)) far above any real compact model (the biggest ship a few dozen modules and ~1k params); after reading OSDI_NUM_DESCRIPTORs, reject the file (clean diagnostic, INVALID_OBJECT) if it exceeds the first, before it sizes the registry or strides the array; inside the descriptor loop, reject if any of a descriptor's count fields (num_nodes/num_terminals/num_params/num_instance_params/num_opvars/num_noise_src/num_jacobian_entries/num_collapsible/num_states) exceeds the second, before those fields index/size anything -- a single load-time choke point protecting every downstream consumer (setup, eval), since a rejected file never registers its devices. HONEST SCOPE: this targets the egregious/out-of-range count (the signature of a truncated/byte-corrupted/ABI-drifted object, where corruption reliably lands -- an ABI mismatch reads a pointer's bits as a count -> huge value); it is NOT a claim that the loader is proof against every malformed .osdi -- a count that lies within the plausible range is an internally inconsistent binary indistinguishable from a valid one, and a .osdi is native code whose mere loading executes it, so a hostile object is out of scope by construction. On macOS byte-corrupting a signed .osdi also breaks its code signature so dlopen rejects it before the loader runs; the residual exposure is chiefly Linux (no signing gate) and toolchain/ABI drift. Scope: ngspice only, one file (osdi/osdiregistry.c); no device/solver/compiler change. Verify (examples/osdifuzz): 3 checks -- two malformed .osdi built from tiny C stubs (one declaring a huge OSDI_NUM_DESCRIPTORs, one with a valid count but a huge per-descriptor num_params built from the real OsdiDescriptor struct in osdi.h) now load-reject cleanly (both a SIGSEGV rc=139 on the pre-fix binary), and a positive control confirms a real openvaf-r-built .osdi still loads and simulates (v(a)=1). Regression 187/187 (new example folder).

E-229 *src/frontend/com_dl.c, src/frontend/commands.c, src/spicelib/devices/dev.c, src/spicelib/devices/dev.h, examples/osdireload_examples/**

pre_osdi -f: reload a recompiled OSDI model without restarting. A Verilog-A developer's inner loop is edit .va -> recompile .osdi -> re-run, but interactively ngspice loaded each .osdi once: load_osdi (dev.c) is guarded by a path dedup, a dlopen-handle dedup (osdiregistry.c), and a device-name dedup (a type already in the global DEVices[] is ignored because the model-card lookup scans front-to-back and the first/stale entry wins -- see E-223); underneath, dlopen caches by path (a re-open returns the cached handle, never re-reading the recompiled file). So a recompiled model never took effect without restarting. FIX -- an opt-in -f/-force flag: com_osdi (com_dl.c) parses -f anywhere in the argument list and passes a force flag to load_osdi(path, force); on a forced reload of an already-loaded path it STAGES a byte-for-byte copy of the (recompiled) file under a fresh unique path and loads that (a new path is always re-read, sidestepping dlopen's cache; this avoids dlclose, which on macOS does not reliably unmap and where overwriting a still-mapped file faults the process), then removes the staged copy once it is mapped (the mapping keeps it alive on POSIX), and osdi_add_device(..., replace) swaps the registered device to the freshly loaded descriptor IN PLACE at its existing table index so no other device type's index

shifts. The previous SPICEdev and the library mapping it points into are intentionally left resident: a circuit still built against the old model keeps a valid device, and freeing descriptor-owned memory would risk a double free (documented caveat: don't re-run a pre-existing circuit built against the prior model version after reloading it). Behaviour is unchanged without `-f` (plain re-load still skipped, now with a hint pointing at `-f`); the `osdi` command help gains the flag. Scope: frontend + device registry, four files (frontend/com_dl.c, frontend/commands.c help text, spicelib/devices/dev.c, spicelib/devices/dev.h); no solver/analysis/OSDI-ABI/compiler change. Verify (examples/osdireload): compiles a resistor model at 1k then 2k, drives ngspice in pipe mode to load the 1k version (`i(v1)=-1mA`), recompiles the same file to 2k mid-session, confirms a plain re-load is skipped (still 1k), then `osdi -f` reloads it and the operating point changes to the 2k result (`i(v1)=-0.5mA`) -- one process, no restart. Regression 188/188 (new example folder).

E-231 *src/frontend/com_gnuplot.c, src/frontend/plotting/gnuplot.c, examples/csv_examples/**

CSV output for **wrdata** (**set wr_csv + wrdata -csv**). wrdata only ever wrote whitespace-separated columns (each vector prefixed by its own scale column); the knobs `wr_singlescale/wr_vecnames/wr_onespace/numdgt` tune spacing and precision but never the separator, so a genuine comma-separated export needed a post-processing pass. The only `-csv` in the tree was `show -csv` (device-parameter tables, not data). ADDED a first-class CSV mode. (1) `set wr_csv --` a boolean read in `ft_writesimple` (plotting/gnuplot.c) alongside the existing `wr_*` vars; when set it takes a dedicated, self-contained branch that writes a single shared scale column once (implies `wr_singlescale`, so the existing equal-length sanity check applies), a header row of vector names (implies `wr_vecnames`), and comma-separated values with `"%.*e"` (no sign-padding, no trailing separator) honouring `numdgt`; complex `.ac` vectors become two columns (real, imag). The branch writes and returns before the space-formatted writer, so DEFAULT wrdata output is byte-for-byte unchanged. (2) `wrdata -csv <file> <vec...> --` a per-call alias accepted in ANY argument position: `com_write_simple` (com_gnuplot.c) scans the wordlist for a `-csv` token, enables `wr_csv` for just that write, and restores the prior global state afterward (no leak). Because `plotit` COPIES its wordlist (never mutates/frees the caller's), the `-csv` node is spliced out for the call and relinked afterward, leaving the caller's list intact to free; the head case is detected as `csvnode == wl` (the incoming head's `wl_prev` is not necessarily null), and no-vector forms fall through the `if (wl)` guard writing nothing, node still relinked. Scope: ngspice frontend only, two files (frontend/com_gnuplot.c, frontend/plotting/gnuplot.c); no solver/analysis/device/OSDI/compiler change; default output preserved exactly. Verify (examples/csv_examples, 9 checks): default output unchanged (no commas); `set wr_csv` writes a header + comma-separated values with the right numbers; `wrdata -csv` in first/mid/last position is byte-identical to the option form; CSV numbers equal the default-format numbers bit-for-bit; the flag does not leak into the next plain wrdata; `.ac` emits real/imag columns under a frequency scale; `.tran` emits a time scale with uniform csv-parseable rows. Regression 190/190 (new example folder).

E-232 *src/maths/KLU/klusmp.c, examples/solverfix_examples/**

KLU solver-glue correctness hardening (source audit). A deep read of the active SMP dispatch layer -- `klusmp.c`, the ONLY sparse-matrix-package layer compiled when KLU is enabled (`sparse/Makefile.am` builds `spsmp.c` only if `!KLU_WANTED`; `klusmp.c` routes BOTH solvers via its `CKTkluMODE` branches) -- found four latent defects, all fixed, all behaviour-preserving for any circuit that actually solves (prior functional KLU bugs E-113/114/115/171/172 were already fixed and are confirmed still in place). (A) `SMPluFac` and `SMPsolve` dereferenced `KLUmatrixCommon->status` BEFORE their `== NULL` guard; the complex twin `SMPclufac` checks NULL first with an early return -- reordered both to match (`SMPsolve`, being void, now reads status only inside the non-NULL else). Common lives for the matrix lifetime so it is never actually NULL here; the fix aligns the paths and satisfies a static analyzer. (B, the substantive one) the complex solves `SMPcSolve/SMPcaSolve` copied the RHS with a plain identity map (`RHS[i+1]`), while the real `SMPsolve` routes RHS through the node-collapse permutation `KLUmatrixNodeCollapsingNewToOld`; if a structural-zero column ever collapsed a node (`map != identity`) the AC/noise/pz RHS

would be silently mis-ordered relative to the DC/tran solve. Dormant -- the map is the identity for any circuit that factors (an empty MNA column is structurally singular and does not solve), which is why KLU AC/noise/pz have always matched Sparse -- but the two paths were inconsistent; both complex solves now apply the identical gather/solve/scatter so a future change to collapse cannot quietly break complex analyses. (C) SMPcZeroCol indexed KLUmatrixAp[Col-1] with no ground guard (Ap[-1] if Col==0); added the Col >= 1 guard its sibling SMPfindElt already has (pole-zero passes indices >= 1, so it never fired). (D) SMPmultiply's real branch ended with a dead iSolution = iRHS; (reassigns a by-value local; a real matrix has no imaginary product) -- removed. Scope: ngspice only, one file (maths/KLU/klusmp.c); no analysis/device/OSDI/compiler change and no behavioural change for any solvable circuit. Verify (examples/solverfix, 4 checks): on an ASYMMETRIC net (a VCVS makes the MNA matrix non-symmetric, so the noise ADJOINT solve genuinely differs from the forward solve) KLU still agrees with SPARSE 1.3 to the bit on AC, on the noise spectrum (exercising the most-changed SMPcaSolve), and on pole-zero roots (SMPcaSolve + SMPcZeroCol + SMPcAddCol), plus a clean DC-op + transient run under KLU. Regression 191/191 (new example folder).

E-233 *src/maths/KLU/klusmp.c*

KLU glue deeper audit: finish the null-check reorder + correct the collapse map. Follow-up to E-232 from a deeper read of the same file; two more latent defects, both behaviour-preserving (one path never triggers, the other is confirmed-dead code). FIRST two scarier candidates were investigated and CLEARED empirically, not just by inspection: (i) a suspected per-analysis memory leak -- .pz/.sens reuse ckt->CKTmatrix and re-call SMPconvertCOOtoCSC + SMPpreOrder without freeing the prior arrays/Symbolic, which LOOKS like a leak -- was REFUTED by instrumenting the lifecycle over op->pz->sens: every convert/preorder sees a FRESH matrix (Ap==NULL, Symbolic==NULL) and SMPnewMatrix/SMPdestroy are balanced (5 new / 4 destroy / 0 overwrite-without-destroy); ngspice tears down and rebuilds the matrix per analysis, so nothing leaks (no fix made -- no bug). (ii) node-collapse is DEAD code: instrumenting the collapse branch and sweeping the whole 191-example regression (every OSDI model, pz, sens, noise) plus pathological topologies fired it 0 times, because OSDI collapses nodes at allocation time (osdisetup.c: a collapsed node gets no matrix column) so the COO never has a gap and Gmin fills every diagonal. FIXES: (3) E-232 fixed the dereference-before-NULl-check in SMPluFac/SMPsolve but the SAME pattern survived in SMPcReorder and SMPpreorder (they read KLUmatrixCommon->status before the ==NULL guard) -- reordered both to check NULL first with an early return, matching SMPcLUfac, and corrected two mislabeled diagnostics (the Common-NULl and Symbolic-NULl branches each printed a stray 'KLUnumeric object is NULl'). Common is never actually NULl, so behaviour is unchanged. (2) in SMPconvertCOOtoCSC the structural-zero-column compaction recorded NewToOld[reduced] = MatrixC00[i].col, but with TWO OR MORE gaps the RHS is a partially-reduced index from an earlier pass, not the true original, so the map would resolve to the wrong node and the real SMPsolve gather/scatter would mis-address the RHS -- fixed by chaining through the existing map (NewToOld[reduced] = NewToOld[MatrixC00[i].col], identity on the first pass so single-gap behaviour is unchanged); defensive since the path is confirmed dead. Scope: ngspice only, one file (maths/KLU/klusmp.c); no analysis/device/OSDI/compiler change and no behavioural change for any circuit. Verify: behaviour-preserving, so no new example -- the existing examples/solverfix_examples (KLU vs SPARSE bit-agreement on AC/noise/pz over an asymmetric net) plus the full 191/191 regression remain unchanged; fix 2's dead path cannot be exercised by construction (correctness is by the chaining argument). Regression 191/191.

E-234 *src/frontend/com_loadpull.c (new), src/frontend/com_loadpull.h (new), src/frontend/commands.c, src/frontend/com_commands.h, src/frontend/Makefile.am, examples/loadpull_examples/**

loadpull: power-amplifier load-/source-pull analysis. Load-pull (sweep the load impedance a PA output sees over the Smith chart and contour Pout/gain/PAE/efficiency to find the optimum load) is a PA-design staple ngspice lacked, and open-source load-pull is scarce. NEW loadpull command rides the existing .tran engine + alter (as optimize/sweep do) + pyplot -contour (

E-218). `loadpull -load <R> <L> <C> -out <node> -drive <Vsrc> -f <f0> [-supply <Vsrc>]` sweeps Gamma over ‘

E-235 *src/frontend/com_stb.c, examples/stbfix_examples/**

stb: fix a latent use-after-free in the probe lookup. The defensive follow-up flagged in E-234. `com_stb` resolved its voltage probe with `INPretrieve(&name, symtab)`, which replaces the pointer with the interned symbol-table string -- the same memory the voltage source's own name field points at -- and does NOT free the old copy; the following `tfree(name)` therefore double-freed the source's live name (and the symbol-table entry). It never bit in practice because `stb` runs once with no re-setup, but it is real memory corruption and the identical defect that bit `loadpull` in E-234 (there the per-point re-setups re-created the drive's branch node from that freed, allocator-reused memory). Fixed by dropping `INPretrieve` (top-level device names need no subckt translation -- `findInstance` does its own name match) and lowercasing a private copy instead (ngspice stores instance names lowercased), so only our own copy is freed; as a bonus the probe lookup becomes CASE-INSENSITIVE -- `stb Vprobe Iprobe` used to fail "no such probe source 'Vprobe'" and now resolves. The now-unused complex helper `stbsub` was removed to keep the build warning-free. Scope: ngspice frontend only, one file (`com_stb.c`); no solver/analysis/device/OSDI/compiler change; loop-gain results unchanged. Verify (`examples/stbfix`, 3 checks): a mixed-case probe name resolves and reports a loop gain (failed before); the loop gain is still correct (a gain-1e5 buffer loop -> ~100 dB DC loop gain, measured 99.91 dB); 60 repeated `stb` runs stay stable. Regression 193/193 (new example folder).

E-236 *src/frontend/com_measure2.c, src/frontend/com_measure2.h, src/frontend/measure.c, src/tclspice.c, examples/measovf_examples/**

.meas: fix a stack-buffer overflow on long measurement names. Found in a memory-safety deep dive; reproduces on the stock binary. `get_measure2()` formats each measurement result line with `sprintf(out_line, "%-20s= %.e ...", mName, ...)` into the caller's fixed stack buffer `char out_line[1000]` (`measure.c`). `mName` is the measurement NAME token, taken verbatim from the `.meas <analysis> <name> <func> ...` card via `cp_unquote` -- an unbounded user string with no length check. A `.meas` name longer than ~1000 chars overruns `out_line`; on macOS the stack canary fires and the process aborts (SIGABRT, exit 134), elsewhere it is plain stack corruption. All ten result-formatting sites share the flaw. E-225 had already hardened the sibling `errbuf[100]` in this same file to `snprintf` (after a measure-syntax fuzz campaign), but the fuzzing never exercised a long measurement NAME, so `out_line` was missed. Fix: thread the destination buffer size through `get_measure2(..., size_t max_out_line, ...)` and convert all ten `sprintf(out_line, ...)` to `snprintf(out_line, max_out_line, ...)`. The three callers pass the real size -- `measure.c` batch `.meas` passes `sizeof out_line`, the autostop check and the `tclspice.c` interactive path pass `NULL/0` (all ten sites are already guarded by `if (out_line)`, so they never reach `snprintf`). An over-long name is safely truncated to 999 chars; the measurement value itself is computed and stored exactly as before. Scope: ngspice frontend only; no solver/analysis/device/compiler change; measurement results unchanged. Verify (`examples/measovf`, 2 checks): a `.meas` with a ~4000-char name no longer crashes (exit 0, was 134); normal measurements still correct (MAX 1.0, AVG 0.6, 0.1->0.9 rise time 0.8 ns). Regression 194/194 (new example folder).

E-237 *src/frontend/dotcards.c, src/frontend/vectors.c, examples/nameovf_examples/**

.print/.plot/.four: fix stack-buffer overflows on long vector/node names. Sibling of E-236, same deep dive. The SPICE2-compat rewrites for output tokens, and the vector-name helper they feed, each copied an unbounded user name into a fixed `BSIZE_SP[512]` stack buffer: `fixem()` (`dotcards.c`) rewrites `v(a,b)->v(a)-v(b)` (+ `vm/vp/vi/vr/vdb` variants) via `sprintf(buf, "v(%s)-v(%s)", a, b)` (19 writes); `gettoks()` (`dotcards.c`) rewrites `i(x)->x#branch` via `sprintf(buf[513], "%s#branch", x)`; `vec_basename()` (`vectors.c`) does `strcpy(buf, v->v_name)`, reached by `.print/fft/spec/linearize`. A `.print/.plot/.four` output token whose `node/`

branch name(s) exceed the buffer overran the stack -- macOS aborts (SIGABRT/SIGTRAP, shell exit 133), elsewhere plain corruption. Fixing fixem alone was NOT enough: with valid long node names the rewrite succeeds and the long $v(a)-v(b)$ name then flows into `vec_basename` and crashes there, so all three needed hardening. Fix: size each scratch buffer to its input -- `fixem` allocates `strlen(string)+32` and every write is a bounded `snprintf(buf, bufsz, ...)`; `gettoks` builds the string with `fprintf`; `vec_basename` replaces the fixed buffer + `strcpy` with `copy()` of the source (allocates to fit), preserving branch semantics. No truncation, so a valid long differential still computes the correct value. The `vectors.c` recompile also surfaced 3 pre-existing `-Wconversion` warnings (`v->v_type`, enum `simulation_types`, passed to `dvec_alloc`'s int param), silenced with `explicit (int)` casts (build stays warning-free). Scope: ngspice frontend only; no solver/analysis/device/compiler change; output values unchanged. Verify (examples/nameovf, 4 checks): long $v(a,b)$ on nonexistent nodes no longer crashes; a VALID long differential $v(A,B)$ with $v(A)=2, v(B)=1$ runs and prints $v(A)-v(B)=1.0$ exactly (no truncation); long $i(x)$ via `.four` (`gettoks`) no longer crashes; short $v(1,2)$ still rewrites to $v(1)-v(2)=2.0$. Regression 195/195 (new example folder).

E-238 *src/frontend/dotcards.c, examples/malftoken_examples/**

gettoks: fix a NULL-deref crash on a malformed differential token. Third find of the same deep dive (after E-236/E-237), from fuzzing degenerate output tokens; reproduces on the stock binary. `gettoks()` parses the output tokens of `.save/.print/.plot/.four` (via `ft_savedotargs`) and `.measure vars` (via `com_measure2`). For a differential $v(a,b)$ it finds the comma `c` and close paren `r` (`r = strchr(t, ')'`), `c = strchr(t, ',')` and splits the second operand at `r`: `if (c != r) { *r = '\0'; ... }`. A MALFORMED token like $v(1,$ has a comma but NO `)`, so `r` is NULL while `c` is not -- `c != r` is then true and `*r = '\0'` dereferences NULL -> SIGSEGV (EXC_BAD_ACCESS at 0x0; lldb: `gettoks <- ft_savedotargs`). A one-character typo (missing paren) in a `.print/.save/.plot/.four/.measure` token crashes the simulator. Fix: require `r` non-NULL before splitting -- `if (r && c != r)`. When there is a comma but no `)`, the split is skipped and the token degrades to a harmless parse; every well-formed case is unchanged ($v(a,b)$ both non-NULL `c!=r` -> split; $v(a)$ `c==r` -> skip; $i(x)$ sets `c=r=NULL` -> skip). Scope: ngspice frontend only, one guard in `dotcards.c`; no solver/analysis/device/compiler change; well-formed parsing unchanged. Verify (examples/malftoken, 3 checks): `.print tran v(1,` no longer crashes; the malformed shape across `.save/.plot/.four/.meas + i()/vdb()/vm()` no longer crashes; well-formed $v(1,2)$ still rewrites to $v(1)-v(2)=1.5$. A broad malformed-token sweep (prefixes**x**bodies**x**cards, hundreds) is crash-free after the fix. Regression 196/196 (new example folder).

E-239 *src/spicelib/parser/inpptree.c, examples/funcarity_examples/**

Expression parser: fix a NULL-deref on a one-argument **min/max/pow/pwr**. Fifth find of the same deep dive (after E-235–E-238), from fuzzing the behavioral-source expression parser; reproduces on the stock binary. `PTeval()` (`ifeval.c`) evaluates the two-argument functions `pow/pwr/min/max` by reading their operands as `tree->left->left` and `tree->left->right` -- it assumes the argument is a `PT_COMMA` pair. The function-node builder `PT_mkfnnode()` (`inpptree.c`) looks the name up but NEVER validates argument count, so a one-argument call like `min(1)` builds a node whose `left` is a scalar `PT_CONSTANT` (its `->left` is NULL); at load time `PTeval` does `tree->left->left` -> NULL deref -> SIGSEGV (`PTeval <- IFeval <- ASRCload <- CKTload`). Reachable from any B-source, E/G source, or `.param` expression -- a one-line typo crashes the simulator during setup. Crash is specific to these 4 functions and the scalar (non-comma) one-arg form: `min()` is caught by the existing `if (!arg)` guard, `min(1,2,3)` yields a `PT_COMMA` (no deref), and single-arg forms of other functions (`hypot(1)`, `atan2(1)`) fall through the single-arg path and error cleanly. Fix: reject the wrong arity at parse time in `PT_mkfnnode` -- `if ((POW`

E-240 *src/xspice/icm/analog/s_xfer/cfunc.mod, bin//codemodels/analog.cm* (rebuilt), *examples/sxfer_examples/*

XSPICE **s_xfer**: fix an out-of-bounds crash on a static-gain transfer function. Sixth find of the

same deep dive (after E-235–E-239), from fuzzing XSPICE a-device instantiation; reproduces on the stock binary. First find in the XSPICE code-model library rather than the core simulator. The `s_xfer` analog code model (a Laplace transfer function $H(s)=\text{num}/\text{den}$ in controller-canonical form) allocates its integrator state array to `den_size` elements, then in the DC/transient branch unconditionally read `pout_pin = *(integrator[1])` (the $d(\text{out})/d(\text{in})$ partial) -- assuming a first- or higher-order denominator. A 0-order denominator (`den_size == 1`, i.e. a STATIC-GAIN transfer function like `s_xfer(num_coeff=[k] den_coeff=[1]) = constant k`) allocates integrator with a single element, so `integrator[1]` is one past the end -> read garbage -> SIGSEGV at load (`cm_s_xfer <- MIFload <- CKTload`). Depends only on `den_size==1`, NOT the coefficient values: `den_coeff=[1]` (valid unity denominator), `[5]`, `[0]` all crash; `[1 1]/[1 2 1]` fine -- so it bit ordinary static-gain input, not just degenerate decks. AC branch already safe (loops bounded by `den_size/num_size`). Fix: guard the read -- `if (den_size > 1) pout_pin = *(integrator[1]); else pout_pin = (num_size>0) ? *gain * *(num_coefficient[0]) : 0.0` (for `den_size==1`, `out = gainnum[0]/in`, so the partial is `gainnum[0]`); *the output value was already computed correctly. Zero denominator ($H=\text{inf}$) now gives clean non-convergence instead of a crash. Fix is in a .cm code model, so analog.cm was regenerated via `cmpp` and redeployed under `bin//codemodels/`; the ngspice binary is unchanged. Scope: XSPICE analog code-model library only (`cfunc.mod` + rebuilt `analog.cm`); no core simulator/solver/analysis/device/compiler change; dynamic-filter results unchanged. Verify (examples/sxfer, 4 checks): static-gain `s_xfer(num=[3] den=[1])` no crash & exact ($H=3 \rightarrow 12\text{ V}$); another static gain $H=0.4 \rightarrow 1.6\text{ V}$; zero denominator no crash; dynamic $H=1/(s+1)$ still correct (*

E-241 `src/frontend/com_fft.c, examples/fftnorm_examples/*`

fft/spec: fix the amplitude normalization for non-power-of-2 records. First numerical-correctness (wrong-number, not crash) find of a correctness-audit campaign that checked ngspice analyses against analytic + numpy oracles. `com_fft.c` has two paths: FFTW3 (exact length-point transform, `scale = length/2`, correct) and a Green's radix-2 FFT (no FFTW3) that zero-pads the length input samples up to the next power of two `N`. The non-FFTW path normalized the single-sided amplitude by the PADDED size (`scale = N/2`) instead of the true sample count (`length/2`), so every FFT whose sample count is not a power of two reported amplitudes too small by `length/N` -- up to 2x. A `.tran` essentially never yields exactly 2^k points, so this bit the common case silently. The DC bin is the clean discriminator (always exactly bin 0, no window/scalloping ambiguity): a signal with DC offset `D` read `D*length/N`, e.g. a 1025-sample record (padded to 2048) reported a DC of 2.0 as 1.0009. `spec` carried the same error in its power normalization (`intres = N*N` instead of `length*length`; the FFTW path already used `length*length`). Fix: `scale = length/2` and `intres = length*length` in the non-FFTW path, matching FFTW. The frequency axis was already padding-aware (`freq[i] = i/span * length/N`), so only the magnitude scale needed the change; power-of-2 records (`length == N`) and FFTW-linked builds are unchanged. Validated against numpy as an independent oracle: after the fix ngspice's `fft` matches `numpy.fft.rfft` of the same zero-padded record (normalized by `length`) bin-by-bin to $\sim 1\text{e-}9$. Scope: ngspice frontend only, two normalization constants in `com_fft.c`; `fft/spec` amplitudes now change for non-power-of-2 records (they become correct). Verify (examples/fftnorm, numpy-free, 4 checks): DC offset 2.0 reads 2.0 on a non-power-of-2 record; same on a maximally-padded record (1025->2048, where the bug halved it to ~ 1.0); DC reading independent of sample count across four record lengths; `spec` PSD tone peak length-independent. Regression 199/199 (new example folder).

E-242 `src/spicelib/devices/nport/*` (new device dir), `src/frontend/snp2va.c, src/frontend/snp2va.h, src/frontend/com_presnp.c, src/spicelib/parser/inp2n.c, src/spicelib/devices/dev.c, configure.ac, src/spicelib/devices/Makefile.am, src/Makefile.am, examples/nport_native_examples/, bin//ngspice (rebuilt)`

Native C n-port device + **pre_snp -native:** a pole/residue Y-parameter block stamped straight into the matrix, no OpenVAF compile. `pre_snp` (E-200/201/205) converts Touchstone -> Verilog-

A -> `openvaf-r -> .osdi`; the fit is cheap but the compile is the bottleneck and walls at ~24-32 ports ($0(N*Np)$ `laplace_nd` sections, $0(N^2)$ terms for full-rank coupling). This adds a built-in `nport` device (`new src/spicelib/devices/nport/`) that evaluates the same model $Y_{ij}(s) = d_{ij} + s * e_{ij} + \sum_k res_{ijk} / (s - p_k)$ directly. It stamps as a multi-terminal admittance (four-corner conductance per (i,j); no per-port branch unknowns -> N equations not $2N$): DC stamps $Y(0)$, AC the complex $Y(jw)$ into the (real,imag) slots, and transient a trapezoidal companion -- d a conductance, s*e a capacitor, each $res/(s-p)$ a first-order state $dx/dt = p*x + u$ whose shared pole states x_{jk} are advanced once per load and parked in `CKTstate0` (the stamp phase recovers the history $B = x_{n+1} - a * u_j$, so no per-instance scratch). Complex poles come as conjugate pairs so the summed current is real. It rides the generic N dispatcher (`inp2n.c` broadened from OSDI-only to also accept the `nport` model): `N1 p1..pN ref mymodel / .model mymodel nport(file="m.nport")`, ref an explicit reference terminal (tie to 0 for a ground-referenced Touchstone block; the extra terminal makes floating/differential blocks expressible). KLU supported via `nportbindCSC.c` (real/complex/complex->real bindings; a ground row/col keeps its Sparse trash pointer, like the RLC devices). `snp2va.c` was refactored so the shared parse+vector-fit lives in `snp_fit()`, feeding two emitters: `pre_snp -osdi` (default, unchanged VA->OSDI route) and new `pre_snp -native`, which writes the compact `.nport` for this device -- same fit, no compiler, scaling to hundreds of ports. Verify (examples/`nport_native`, 6 checks x 2 solvers, all vs closed-form oracles, no `openvaf-r` needed): RC one-port e-term (AC $1/Y$ + transient exponential); RLC one-port complex conjugate pole pair through resonance; Pi two-port off-diagonal d/e cross-coupling vs analytic MNA; `pre_snp -native` fits a known 2-port `.s2p` and reproduces its Y; a 20-port 6-pole full-rank model vs an analytic linear solve. Cross-checks (dev notes): native AC == OSDI backend to 1e-11 on `137mhz_bpf.s2p`; a 100-port block solves an AC point in ~90 ms (4.85e-10 vs analytic); KLU == Sparse bit-for-bit. Scope: ngspice only, additive; default `-osdi` path unchanged so existing decks are unaffected; no solver/analysis/compiler change. Regression 200/200 (new device dir + example folder).

E-243 `src/frontend/snp2va.c, examples/presnp_examples/verify_presnp.py, bin/*/ngspice (rebuilt)`

pre_snp -osdi: emit an explicit **ref** terminal so the instance line is identical for both **-osdi** and **-native**. E-242's native `nport` device carries an explicit reference terminal (`N1 p1..pN ref model`), but the original `pre_snp -osdi` emitted a Verilog-A module with only the N signal ports referenced to implicit global ground (`N1 p1..pN model`) -- so the two backends were not drop-in interchangeable. `snp2va.c` now emits module `m(p1..pN, ref)` and takes every branch relative to `ref`: `I(pi,ref) <+ ... V(pj,ref) ...` (and `laplace_nd(V(pj,ref), ...)`, `ddt(V(pj,ref))`). `ref` is a PLAIN electrical port -- NOT a ground-declared node: that keeps it connectable (tie to 0 for a ground-referenced block, or any node for floating/differential), and it appears in every `I(pi,ref)` stamp so it is well-posed (declaring ground `ref` would make it an internal 0V node -- no longer a terminal -- and risks the E-116 all-zero-row/structurally-singular-KLU failure a ground node in no Jacobian entry caused). The instance line is now identical for both backends. Compatibility: the `-osdi` module goes from N to N+1 terminals, so existing `-osdi` decks must add the `ref` node (tie to 0 to reproduce the old behavior); the `presnp` example decks were updated. Verify: the identical line `N1 p1 p2 0 model` through both backends matches to 1e-11 on `137mhz_bpf.s2p` (`ref=0` reproduces the prior result exactly); examples/`presnp` (9 checks: 2-/3-/4-/8-port `pre_snp` cases, AC+tran+ordering) passes with the `ref` terminal. Scope: ngspice only, the VA emitter in `snp2va.c`; `-native / .nport device / vector fit / .osdi compile path` all unchanged. Regression 200/200.

E-244 `src/spicelib/devices/nport/nportsetup.c, src/frontend/com_pyplot.c, examples/crashfix2_examples/, bin/ngspice (rebuilt)`

Crash-hardening round 2: two user-triggerable crashes in recently-added code (both reproduce on the shipped binary). Found by argument-fuzzing the newer devices/commands. (1) `nport` device (E-242): `NPORTsetup` never checked that the instance line connected all N+1 nodes the `.model`'s port count claims. A `.nport` with more ports than the instance wires (`N1 a 0 mod` for a 2-port), or

nports beyond NPORT_MAXTERMS, made setup stamp an UNBOUND (-1) terminal -> the generic N dispatcher leaves unconnected terminals as -1, and passing a negative row/col trips the sparse builder assert `Row>=0 && Col>=0` (spbuild.c:267 -> SIGABRT) / out-of-bounds. Fix (nportsetup.c): reject a .nport whose port count leaves no room for the reference ($N+1 > \text{NPORT_MAXTERMS}$) and verify every node `[0..N] >= 0` before stamping -> clean E_BADPARAM with a diagnostic naming the shortfall. (2) pyplot (E-217/E-218): the -hist/-contour render-mode markers were stripped by UNLINKING AND FREEING a node of the command's OWN argument wordlist -- but that list is owned/freed by the command loop. When the marker was the FIRST word, com_pyplot freed the list HEAD, which the command loop then freed again -> use-after-free/double-free (SIGSEGV) on `pyplot -hist v(out) / pyplot -contour ... (pyplot v(out) -hist, marker not first, was fine -- so the E-217/E-218 examples, which never put it first, always passed.)` Fix (com_pyplot.c): detect the marker by scanning (no mutation) and build a filtered COPY without it, freed by us at the end; the caller's list is untouched. Bonus: -hist as first arg now renders the histogram it silently degraded to a line plot before. Verify (examples/crashfix2, 9 checks x2 solvers): under-bound + over-max nport decks exit gracefully with the diagnostic; a correctly-bound 2-port still simulates; pyplot -hist/-contour first-arg and -hist-alone no longer crash; -hist-first now emits `plt.hist()`; trailing-marker form still works. Scope: ngspice only, device input validation + command-arg ownership; no numerical change, valid usage unaffected. Regression 201/201.

E-245 *src/frontend/com_measure2.c, src/frontend/spiceif.c, examples/crashfix3_examples/, bin//ngspice (rebuilt)*

Crash-hardening round 3: two NULL-deref crashes in CORE ngspice command parsers (stock code, both reproduce on the shipped binary). Found by argument-fuzzing meas and altermod. (1) meas: `measure_parse_stdParams` (com_measure2.c) splits each trailing param token on '=' with `strtok`; a LONE '=' token (a stray one, e.g. `meas tran m find v(x) when v(x)=0.5 = y`) makes `strtok` return NULL for the name, which was then passed to `strcascmp()` -> NULL deref SIGSEGV. The `pValue==NULL` branch was guarded but `pName` was dereferenced first. Only the interactive meas command hits it (the .meas dot-card pre-tokenizes differently); `key=val/=val/val=` are all fine, only a standalone '=' produces the NULL. Fix: treat a NULL `pName` as a clean syntax error before `strcascmp`. (2) altermod: `altermod nm c` where the 2nd token is a bare device-type letter (c/e/i/...) or a digit makes `com_altermod` treat it as another MODEL to alter with no `param=value`, so `paramlookup` (spiceif.c) is called with `param==NULL`; its model-parameter loop passed that straight to `eq()/strcmp()` (the instance-parameter loop just above already guarded `!param`) -> NULL deref SIGSEGV. Input-dependent: only tokens resolving to a real device type crash (c/e/i/0/1/2), others are dropped earlier. Fix: guard the model loop against a NULL `param` (and defensively a NULL keyword), matching the instance loop. Verify (examples/crashfix3, 12 checks x2 solvers): the stray '=' meas forms and `altermod nm <c`

E-246 *src/xspice/icm/analog/pwl/cfunc.mod, src/xspice/icm/analog/pwlts/cfunc.mod, bin//codemodels/analog.cm (rebuilt), examples/pwlfix_examples/*

XSPICE **pwl/pwlts**: fix an out-of-bounds heap read on `x_array/y_array` length mismatch. Second find in the XSPICE code-model library (after E-240's `s_xfer`), from fuzzing code-model parameters; confirmed with AddressSanitizer. `pwl` and `pwlts` take two INDEPENDENT vector parameters `x_array` (breakpoints) and `y_array` (ordinates); the XSPICE framework allocates each to its OWN length. Both models size their working table from `x_array` only (`size = PARAM_SIZE(x_array)+2`) then copy BOTH arrays with that single index (`y[i]=PARAM(y_array[i-1])`), so a `y_array` SHORTER than `x_array` reads past the end of the `y_array` parameter -> heap-buffer-overflow READ (ASan: `cm_pwl cfunc.c:321 / cm_pwlts cfunc.c:202`), pulling adjacent/uninitialised heap into the interpolation table (undefined behaviour; can crash outright once the mismatch runs past the mapped heap). Reachable from any deck with such an a-device (`.model m pwl(x_array=[0 .25 .5 .75 1 1.5 2 2.5] y_array=[0 1])`). The sibling models already guard this -- `onshot` checks `cntl_size != pw_size`, `multi_input_pwl` checks `PARAM_SIZE(x) != PARAM_SIZE(y)` -- but `pwl/pwlts`, the oldest PWL models, predate the check. A sweep of

the other analog models with paired arrays found no further cases (sine/square/triangle copy into their own cntl_size-sized buffers, so a short array zero-pads rather than overrunning; summer/mult are validated by the framework against the connection size). FIX: add the same equal-length guard `-- if (PARAM_SIZE(x_array)!=PARAM_SIZE(y_array)){cm_message_send(size_error);return;} --` before the alloc/copy in both models, aborting setup cleanly with `x_array` and `y_array` must have the same length instead of reading out of bounds. Fix is in a .cm code model, so `analog.cm` was regenerated via `cmpp` and redeployed under `bin/*/codemodels/`; the `ngspice` binary is unchanged. Scope: XSPICE analog code-model library only; no core simulator/solver/analysis/device change; valid equal-length pwl/pwlts usage unaffected. Verify (examples/pwlfix, 5 checks x2 solvers): a valid pwl interpolates exactly (in 0.5 -> 1.0 V); pwl with x longer than y, and the other way, both report the size error without crashing; a valid pwlts still runs; a mismatched pwlts reports the size error; self-skips if codemodels unavailable. The OOB was reproduced and shown fixed under an AddressSanitizer build of the code models. Regression 203/203 (new example folder).

E-247 *src/xspice/icm/table/table2D/cfunc.mod, src/xspice/icm/table/table3D/cfunc.mod, bin/codemodels/table.cm* (rebuilt), *examples/tablefix_examples/*

XSPICE **table2d/table3d**: fix out-of-bounds read + interpolation UB on degenerate or too-small tables. Third area of the XSPICE code-model deep dive (after E-240 s_xfer, E-246 pwl/pwlts), found by fuzzing table dimensions; confirmed with ASan/UBSan. Both models read a lookup table from a file whose first lines declare the axis point counts (ix, iy [, iz]); the loader (`init_local_data`) never validated them, so a small table broke two ways. (1) OUT-OF-RANGE RAMP (heap OOB read): for an input past the table edge the model ramps the derivative to zero using `xcol[1]-xcol[0]` (low side) or `xcol[ix-1]-xcol[ix-2]` (high side); a single-point axis (`ix==1`) has no `xcol[1]/xcol[-1]` -> heap-buffer-overflow READ (ASan: `cm_table2D cfunc.c:298`); `table3d` has the same for all three axes. (2) ENO INTERPOLATION (OOB/UB): the Madagascar ENO of order `p` reads a $2*(p-1)$ -point stencil per axis (order 2 needs 2 pts, 3 needs 4, 4 needs 6, 5 needs 8); the order param DEFAULTS to 3 (needs ≥ 4 pts/axis) but the loader only clamped it UP to a minimum of 2, never DOWN to what the table supports -> the default order on any table with a < 4 -point axis (a 3x3 table, or a 2-plane z axis like the shipped `test-3d-1.table` with `iz=2`) ran off the stencil -> UB/OOB in `eno2.c/eno3.c`. FIX (both models, right after the `order>=2` floor): compute `mindim = min(ix,iy[,iz])`; reject `mindim<2` with a clean error (a 1-point axis can be neither ramped nor interpolated); and clamp `interporder` to `mindim/2+1` (the inverse of the $2*(p-1)$ stencil rule) so a small table interpolates at a reduced but VALID order instead of reading out of bounds. Tables large enough for the requested order are byte-identical (every shipped example has `mindim/2+1 >= order`; the smallest are 7x7). Fix is in .cm code models, so `table.cm` was regenerated via `cmpp` and redeployed under `bin/*/codemodels/`; the `ngspice` binary is unchanged. No regression coverage impact: the regression `table_model` example uses OSDI Verilog-A table models, not these XSPICE code models. Scope: XSPICE analog code-model library only; no core simulator/solver/analysis change; tables large enough for the requested order unaffected. Verify (examples/tablefix, 6 checks x2 solvers): a valid 8x8 order-3 `table2d` interpolates exactly (`out=x+10y` at (2.5,3.0) -> 32.5); a single-x-point `table2d` and a degenerate `table3d` (`iz=1`) are rejected with a clean error; small 3x3 / 3x3x3 order-3 tables run (order clamped) without crashing. The OOB read and the ENO UB were reproduced and shown fixed across a full (dimension x order) grid under ASan/UBSan builds. Regression 204/204 (new example folder).

E-248 *src/spicelib/devices/cpl/cplsetup.c, bin/ngspice* (rebuilt), *examples/cplfix_examples/*

CPL coupled-line device: fix two out-of-bounds accesses (conductor count + R/L/C/G matrix sizes). First find in a CORE `ngspice` device (the XSPICE dive E-240/246/247 was the code-model library); found by fuzzing the p-device conductor count and matrix sizes, confirmed with ASan/UBSan; reproduces on the shipped binary. A CPL line has `noL` conductors (the instance dimension, set from the node count on the p card); each symmetric R/L/C/G matrix is its upper triangle, $noL*(noL+1)/2$ values. `CPLsetup/ReadCpL` (`spicelib/devices/cpl/cplsetup.c`) validated

NEITHER count. (1) UNDER-SPECIFIED MATRIX (heap OOB read): ReadCpL fills the matrices with $f = \text{CPLmodPtr}(\text{here}) \rightarrow \text{Rm}[\text{counter}]$ (and $\text{Lm}/\text{Cm}/\text{Gm}$) where counter runs $0..\text{noL}*(\text{noL}+1)/2-1$, but $\text{Rm}/\text{Lm}/\text{Cm}/\text{Gm}$ are allocated to the count the user actually gave (Rm_counter , ...); fewer values than the triangle needs $\rightarrow$ read past the end (ASan: heap-buffer-overflow READ at `cplsetup.c:474`). (2) TOO MANY CONDUCTORS (fixed-array overflow): ReadCpL uses `RLINE*lines[MAX_CP_TX_LINES]` and `CPLine.in_node[MAX_CP_TX_LINES]` with `MAX_CP_TX_LINES==8` (`include/ngspice/swec.h`), indexed by `noL`; a p card with >8 conductors writes past them (UBSan: index 8 out of bounds for 'NODE *[8]' at `cplsetup.c:430`). Both reachable from a valid-syntax netlist; on the release binary the reads pull adjacent heap $\rightarrow$ heap-layout-dependent garbage (here a downstream fatal-error exit, but could equally be wrong results or a crash). FIX: validate both up front in `CPLsetup`, right after `noL = here $\rightarrow$ dimension` and before any allocation -- reject `noL<1`

E-249 *src/spicelib/devices/urc/urcsetup.c, src/spicelib/devices/ltra/ltraset.c, bin//ngspice (rebuilt), examples/tlinefix_examples/*

URC + LTRA transmission-line input validation: reject a degenerate/unbounded lump count and negative line parameters. Two CORE transmission-line devices accepted out-of-range parameters that gave a silently wrong result or a resource-exhaustion hang instead of a clean error (same class as E-248 CPL). (1) URC (`urcsetup.c`): the uniform-RC line is expanded at setup into a ladder of n (the n = lumps) resistor+capacitor(/diode) sections -- one node group per lump -- and n is also an exponent of `pow(k,n)`; the user's n was NEVER bounds-checked. $n \leq 0$ ran the loop zero times $\rightarrow$ no lumps $\rightarrow$ output node left silently unconnected (`v(out)=0`, a wrong answer, no diagnostic); a large n (typo $n=100000000$, or $2e9$ near `INT_MAX`) exhausted memory / hung building the ladder (even ~ 20000 lumps takes tens of seconds; the source even carried a may want to limit lumps to `<= 100` comment). Fix: require `1 <= n <= URC_MAX_LUMPS (1000; auto-computed count is  $\sim 3..30$ )`, clean `E_BADPARAM` otherwise, validated after the auto-lump default so it covers both given and computed counts. (2) LTRA (`ltraset.c`): the lossy line picks its RLGC special-case with `!= 0` tests, so a NEGATIVE L or C passed and fell through to `sqrt(L/C)/sqrt(L*C)` in `LTRAtemp` $\rightarrow$ NaN characteristic impedance/delay and a degenerate all-zero run (a zero/too-few-param line was already the clean "at least two ... must be ... nonzero" error). Fix: reject any negative R/L/G/C up front (per-unit-length params must be physical). Scope: core ngspice, two device setups; binary rebuilt; no solver/analysis/numerical change; a validly-specified URC/LTRA line is unaffected. Verify (`examples/tlinefix`, 8 checks x2 solvers): a valid 5-lump URC divider and a valid LTRA lossless line still simulate; URC $n=0/-3/100000000$ each rejected cleanly (the huge case INSTANTLY, no hang); URC $n=1000$ (max) still runs; LTRA negative C rejected cleanly while `C=0` keeps the pre-existing nonzero error. Regression 206/206 (new example folder).

E-250 *src/xspice/icm/digital/d_lut/cfunc.mod, src/xspice/icm/digital/d_genlut/cfunc.mod, bin//codemodels/digital.cm (rebuilt), examples/dlutfix_examples/*

XSPICE `d_lut/d_genlut`: fix an undefined-behaviour `1<n` shift on the input-port count. Found with UBSan while sweeping the XSPICE code-model library (E-240/246/247 dive). Both digital lookup-table models size their table as $2^{(\#inputs)}$ via a left shift: `d_lut size=PORT_SIZE(in); tablelen=1<<size (cfunc.c:143-144)`, `d_genlut isize=PORT_SIZE(in); entrylen=(1<<isize) (cfunc.c:154/177)`. The input-port count was NEVER bounded: a left shift of a 32-bit int by ≥ 32 is UNDEFINED BEHAVIOUR (UBSan: "shift exponent 32 is too large for 32-bit type 'int'" at `cfunc.c:146/:177`), and a count in the high twenties asks for a multi-gigabyte table alloc. Reachable from a valid-syntax netlist -- an a-device with $\sim 30+$ digital inputs fanned into the LUT (e.g. all tied to one pulled-up net). On arm64 the release UB is benign (shift amount taken mod 32 $\rightarrow 1<<32==1$, no crash) so the table size is silently WRONG rather than loud -- which is exactly why it's worth fixing. Fix: cap the input-port count at `D_LUT_MAX_INPUTS / D_GENLUT_MAX_INPUTS (24; a real LUT has only a handful of inputs, and the table_values string is  $2^n$  chars)` BEFORE the shift, `cm_message_send` clean error + return otherwise. Fix is in .cm code models, so `digital.cm` was regenerated via `cmpp` and redeployed under `bin//`

codemodels/; the ngspice binary is unchanged. Scope: XSPICE digital code-model library only; no core simulator/solver/analysis change; a normal LUT is unaffected. Verify (examples/dlutfix, 4 checks x2 solvers): valid 2-input d_lut and d_genlut still simulate; a 32-input d_lut and d_genlut are each rejected with a clean "out of range" error, no crash. The 1<32 UB was reproduced under a UBSan build and shown fixed. Regression 207/207 (new example folder).

E-251 *examples/hb_examples/verify_hb.py (verification only -- no ngspice source change)*

Harmonic-balance tight nonlinear correctness verification. A rigorous tightening of the HB (E-134/135) correctness cross-check. The existing checks compare the HB spectrum against ngspice's own transient+fourier steady state, but at K=8 harmonics and a 1.5-3% tolerance. This proves -- and enforces in the regression -- that HB converges to the EXACT periodic steady state, not merely within a few percent. AUDIT: HB and tran+fourier are independent code paths, so their agreement is a real oracle once the transient reference is accurate. Using a purely RESISTIVE rectifier (cjo=0/tt=0 -> steady state reached instantly, so the transient's only error is its $O(dt^2)$ timestep, which is Richardson-extrapolated to $dt \rightarrow 0$) and sweeping K=8..48, the HB harmonics converge MONOTONICALLY to the $dt \rightarrow 0$ transient; at K=48 the match is $\sim 1e-7$ (h1/h2 exact, h3 -1.9e-7, h4 -3.1e-6, h5 +8.9e-7). The 13% apparent 5th-harmonic 'error' at K=8 is entirely HB truncation (aliasing of the discarded harmonics), NOT a modelling error -- it vanishes as K grows. TIGHTENED CHECK (verify_hb.py [10]): on the resistive rectifier, HB at K=24 matches a fine-dt (period/2000) transient fourier to < 0.5% for DC..4th (6x tighter than the loose checks, observed worst-case $\sim 2e-5$), and raising K from 8 to 24 strictly shrinks the 5th-harmonic mismatch (encoding 'converges as K grows' so a future regression that broke HB's harmonic content would be caught). Scope: verification only (verify_hb.py); no ngspice binary / HB-engine / analysis change; documents that HB is EXACT, not merely within tolerance. Regression 207/207.

E-252 *src/xspice/icm/analog/xfer/cfunc.mod, src/xspice/icm/analog/file_source/cfunc.mod, bin/codemodels/analog.cm (rebuilt), examples/filefix_examples/*

XSPICE xfer/file_source: fix two heap OUT-OF-BOUNDS WRITES in the file parsers. Found while sweeping the file-parser code models (d_state/d_source/xfer/file_source); confirmed with ASan. Unlike the earlier code-model finds (E-246/247/250, reads or bounded errors) these are OOB WRITES -- silent heap corruption. (1) xfer (read_file): reads a transfer-function file (Touchstone-style # option line + data), sscanf's up to 9 values/line and a state machine stores EVERY value (freq/real/imag triples), so a line with more than one record stores >3; but the alloc check reserved only 3 (if(i+3>size), ALLOC=1024) -> a multi-record line writes past the buffer at the 1024-double boundary (ASan: heap-buffer-overflow WRITE, cm_xfer cfunc.c: 232). Fix: reserve the sscanf max of 9 (if(i+9>size)); the loop appends at most count<=9/line since offset>=1 && span>=offset+2 are enforced). (2) file_source: stores one record/line = a timepoint + size channels = stepsize(=size+1) doubles, but reserved only size (vecallocated - size), one short; at the realloc boundary (count lands exactly at vecallocated-size once vecallocated grows out of stepsize alignment) the last channel writes one double past the end (ASan: heap-buffer-overflow WRITE, cm_filesources cfunc.c:326). Fix: reserve a full record (-stepsize). Both are heap OOB WRITES reachable from a valid-syntax netlist + crafted data file; release build corrupts adjacent heap silently rather than always crashing (UB either way). Siblings checked: d_source validates per-line token count vs declared width; d_state's fixed line buffer is fgets-bounded -- no analogous overrun. Fix is in .cm code models, so analog.cm was regenerated via cmpp and redeployed under bin/*/codemodels/; the ngspice binary is unchanged. Scope: XSPICE analog code-model library only; no core/solver/analysis change; a well-formed data file is unaffected. Verify (examples/filefix, 4 checks x2 solvers): valid transfer-function + file_source files simulate; an xfer multi-record file and a boundary-crossing file_source file run without overrunning. Both reproduced under ASan (cm_xfer/cm_filesources) and shown fixed. Regression 208/208 (new example folder).

E-253 *src/frontend/com_rfstab.c, src/frontend/com_rfstab.h, src/frontend/commands.c, src/frontend/*

com_commands.h, src/frontend/Makefile.am, bin//ngspice (rebuilt), examples/rfstab_examples/

rfstab: two-port stability & gain report -- the first RF design-aid on top of the S-parameter machinery. ngspice can compute S-parameters (.sp / n-port / Touchstone) but not turn them into the stability/gain figures of merit an RF designer works with. After a .sp analysis publishes S_1_1.. S_2_2 vs frequency, the new **rfstab** command post-processes them into the standard linear-two-port metrics per frequency: determinant $D=S_{11}S_{22}-S_{12}S_{21}$; Rollett $K=(1-$

E-254 *src/frontend/plotting/pyplot.c, src/frontend/plotting/pyplot.h, src/frontend/plotting/plotit.c, src/frontend/com_pyplot.c, bin//ngspice (rebuilt), examples/pyplotsmith_examples/*

pyplot -smith: Smith-chart plotting mode -- the second RF design-aid (after **rfstab**, E-253). A Smith chart is the working surface for two-port RF design (reflection coefficients, matching trajectories, stability/gain circles), but ngspice's modern matplotlib output (**pyplot**) had no Smith mode -- only the legacy in-terminal plotter did. **pyplot [name] -smith <complex vectors>** draws each listed complex vector (S_1_1, S_2_2, a load reflection Gamma, a stability/gain-circle trace) as a curve in the reflection-coefficient plane (Re->x, Im->y) over the standard Smith grid: the unit circle

E-255 *src/spicelib/analysis/cktdisto.c, bin//ngspice (rebuilt), examples/distoexact_examples/*

.disto proven machine-exact vs Harmonic Balance / QPSS-HB + behavioral-source (B) distortion warning. RF/distortion correctness frontier, in the E-251 HB-proof mold. (1) PROOF: .disto (the 1990 Volterra small-signal distortion analysis) reports the pure 2nd/3rd-order kernels scaled by the DISTOF1/DISTOF2 magnitudes; HB (hb) and two-tone QPSS-HB (qpss) are INDEPENDENT large-signal engines including all orders, whose k-th harmonic / mixing amplitudes converge onto the Volterra result with $\sim A^2$ leakage. So HB(A)->.disto as $A \rightarrow 0$ -- the HB-proof structure applied to distortion; because HB and .disto share ngspice's identical device model, model-constant ambiguity (VT, DC-op) cancels, so the agreement measures the .disto engine itself. Measured (diode + series R, both solvers): single-tone HD2 vs HB 2f tightens rel err $4.0e-5 \rightarrow 2.6e-7$ as $A:1e-3 \rightarrow 1e-4$ (HB->.disto), HD3 $\sim 4e-6$; two-tone IM3 (2f1-f2) vs QPSS-HB (2,-1) $\sim 1e-5$; .disto output scales EXACTLY as A^2/A^3 (DISTOF1/2 applied exactly, kernels amplitude-independent). The residual floors are the reference engine's own readout/

E-256 *src/math/ni/niiter.c, src/spicelib/analysis/cktop.c, src/include/ngspice/cktdefs.h, src/spicelib/parser/ptfuncs.c, bin//ngspice (rebuilt), examples/bsrcconv_examples/*

Fix SILENT spurious DC operating point for singular-derivative behavioral sources. A behavioral (B) source whose small-signal conductance is INFINITE at the $v=0$ initial guess -- $B I = \sqrt{v(n)}$ ($dI/dv = 0.5/\sqrt{v} \rightarrow \infty$), $I = 0.1/v(n)$, $I = \ln(v(n))$, $v^{0.5}$ -- pins the node at $v \sim 0$ so Newton takes a vanishing step; the convergence test (iterate-to-iterate VOLTAGE change only) sees no change and declares "converged" at a point that grossly VIOLATES KCL (0.42A imbalance on a 0.6V/1ohm divider). The false convergence also PRE-EMPTED the gmin/source-stepping fallbacks (they never ran). Silent + data-dependent ($i = 1m\sqrt{v}$) converges fine; a .nodeset finds the true root). The B-source AC derivative itself is correct (matches analytic $dI/dv \sim 1e-10$ across the function library) -- the defect is purely the DC convergence ACCEPTING a KCL-violating point. FIX (3 coordinated parts): (1) *niiter.c* false-convergence guard -- after the voltage test declares convergence, also check the KCL residual $F = Gx - b$ (the node-current imbalance, already computed for the E-111 line search); if the worst node merit $> 100x$ the current tolerance the point is spurious -> decline convergence so CKTop falls through to gmin/source stepping, which regularizes the singular node and finds the TRUE point. Separation is decisive: false convergence sits at merit $\sim 1/\text{reltol} \sim 1000$, every legitimate circuit < 1 (measured $1e-7..0.35$). (2) First-attempt isolation -- a new one-bit CKTdcFirstTry flag (*cktdefs.h*) set by CKTop (*cktop.c*) ONLY around the initial plain-Newton attempt, so the guard never fires inside a convergence-aid sub-solve (gmin/source stepping, pseudo-transient/optran, convhelp); an earlier unconditioned version broke convhelp/ptcont/corenum/tempphys by firing inside their homotopies. (3) *ptfuncs.c* **pwr(0,negative)** guard -> HUGE instead of raw

+inf (matches the existing /0 and sqrt(neg) guards) so a singular derivative stays finite rather than NaN-poisoning the Jacobian. Result-neutral for every circuit that already converged. Verify (examples/bsrcconv, both solvers): [1] $I=\sqrt{v}$ reaches analytic $v_0=0.178045$ with KCL satisfied ($i_{B1}=i_{R1}$), not the spurious $v\sim 0$; [2] $I=0.1/v$ reaches upper-branch $v_0=0.887298$; [3] result-neutral: $v^2/v^3/\exp/\tanh$ unchanged. Full regression 212/212 incl. the convergence-aid suites convhelp/ptcont/corenum/tempphys. Scope: DC Newton loop + a one-bit CKT flag + a parser math guard; binary rebuilt; confined to the first operating-point attempt.

E-257 *src/math/ni/niiter.c, bin//ngspice (rebuilt), examples/bsrcconv_examples/ (verify [4])*

Extend the E-256 false-convergence guard to the TRANSIENT operating point. E-256 fixed the silent spurious DC operating point for singular-derivative behavioral sources but gated the guard to MODEDCOP only; the transient operating point (MODETRANOP -- the bias a .t tran starts from without uic) was left uncovered and still false-converged. So a .t tran of a biased $B\ I=\sqrt{v(n)}$ started from the spurious $v\sim 0$ and showed a FAKE startup transient ($t=0\ v(n)=9.3e-17$ ramping $0\rightarrow 0.178$ over $\sim 20ms$) when the biased circuit should sit at 0.178045 from $t=0$. E-256 gated to MODEDCOP conservatively because the pseudo-transient/optran aids run in transient modes -- but the guard already fires only when CKTdcFirstTry is set (the initial plain-Newton attempt), and optran runs as a FALLBACK with CKTdcFirstTry=0, so it is safe to broaden the mode condition to MODEDCOP

E-258 *src/math/ni/niiter.c, src/spicelib/analysis/dctrcurv.c, bin//ngspice (rebuilt), examples/bsrcconv_examples/ (verify [5])*

Extend the false-convergence guard to the .dc sweep cold-start point. E-256/257 fixed the silent spurious operating point for singular-derivative behavioral sources for .op (MODEDCOP) and the transient op (MODETRANOP), but a .dc sweep still false-converged on its FIRST point: .dc V1.. of a $\sqrt{v}$ source gave $v(n)=9.3e-17$ at the first sweep value (true root 0.178045) while every later point (warm-started from the previous solution, away from the $v=0$ singularity) was correct. Why the guard missed it: it fires only when CKTdcFirstTry is set, which happens only inside CKTop -- but in the default (non-HS) path the .dc sweep (dctrcurv.c) solves each point with a DIRECT NIiter call, bypassing CKTop, and only falls back to CKTop if that solve FAILS; the false convergence "succeeds", so CKTop (and the guard) never ran. FIX (2 parts): (1) niiter.c -- the E-256/257 guard, which enumerated the op modes (MODEDCOP

E-259 *examples/integaccuracy_examples/* (verification only -- no ngspice source change)*

Transient integration accuracy PROOF (oracle-based, both solvers). A permanent regression guard for the transient integrator, in the E-251 (HB) / E-255 (.disto) mold: rather than checking one waveform, it proves the integrator MATH PROPERTIES vs closed-form analytics + Richardson self-convergence. The audit that produced it swept the whole transient path and found it CORRECT. [1] ORDER of accuracy on RC decay $\exp(-t/RC)$: global error $\sim dt^p$ with the theoretical p -- TRAP $p=1.94$, Gear2 $p=1.93$, Backward-Euler $p=0.94$ (order test freezes the LTE step controller $trtol=1e11$ so steps are uniform and the convergence is purely truncation order). [2] ENERGY signature on an LC $\cos(wt)$ over 30 periods: TRAP amplitude ratio 0.9999 (marginally stable/energy-conserving, AND the trapezoidal-ringing BE-switch is NOT firing spuriously), Backward-Euler 0.057 (dissipative) -- the fingerprint of a correct trapezoidal vs BDF. [3] BREAKPOINT: a PULSE edge into RC matches the piecewise-analytic charge (max err $6e-7$, pre-edge pinned 0). [4] RLC damped sinusoid vs $\exp(-at)(\cos wd\ t + (a/wd)\sin wd\ t)$ (max err $4.3e-6$, correct wd /envelope). [5] NONLINEAR charge: a diode+CJO rectifier converges under Richardson (TRAP $dt\rightarrow dt/2$) at order $p=2.00$, so the nonlinear device charge integration is 2nd order too. Both solvers give identical results (integrator is solver-independent). Separately confirmed (needs .options dynorder + set ngdebug, not asserted): higher-order Gear engages -- maxord=6 reaches order 6 and rejects $\sim 10x$ fewer steps than order 2 (14 vs 143 on the RLC), so the high-order BDF coefficients are functional not a no-op. Scope: verification only; no source/binary change. Regression 213/213 (new example folder).

E-260 *examples/integaccuracy_examples/* (verify [6]; verification only -- no source change)*

LTE step-controller accuracy PROOF on a stiff circuit. Extends E-259 (which verified the integration METHODS with a FIXED step, freezing the LTE controller to isolate truncation order) with the complementary property: the ADAPTIVE local-truncation-error step controller actually DELIVERS the accuracy requested via reltol -- on a stiff circuit, the hard case. Two independent decays $v_f = \exp(-t/1\mu s)$, $v_s = \exp(-t/1ms)$ (stiffness ratio 1000) integrated with one adaptive step (the controller must take tiny steps to resolve the fast decay then coarsen for the slow tail). Swept over reltol the delivered error vs the closed form shrinks MONOTONICALLY with no plateau: reltol $1e-3 \rightarrow 1e-9$ gives err(fast) $9.0e-3 \rightarrow 2.0e-6$ while internal steps grow $1032 \rightarrow 1884$. The error scales $\sim \text{reltol}^{0.6}$ -- the theoretical global-error rate for a 2nd-order method under LTE control (reltol is a LOCAL per-step tolerance; $\text{step} \sim \text{reltol}^{1/3} \Rightarrow \text{global err} \sim \text{reltol}^{2/3}$). Separately confirmed (not asserted): extreme stiffness ratio $1e6$ ($\tau = 1ns/1ms$) still tracks reltol, and a re-excited fast mode (1MHz square wave) is handled by refining $\sim 6x$ more steps at each edge. Verify (examples/integaccuracy [6], both solvers): on the stiff circuit err at reltol $1e-3/1e-5/1e-7$ is strictly decreasing and improves $\geq 10x$ (measured $\sim 247x$). Completes the transient-engine proof (E-259 methods + E-260 step controller). Scope: verification only; no source/binary change. Regression 213/213.

E-266 *src/spicelib/analysis/cktsetup.c, src/spicelib/analysis/cktpzset.c, src/include/ngspice/cktdefs.h, examples/solverannounce_examples/* (binaries rebuilt by CI)*

Announce the direct linear solver once, not on every analysis. CKTsetup() (and CKTpzSetup()) printed Using SPARSE 1.3 as Direct Linear Solver (or the KLU line) unconditionally at the top of setup, and CKTsetup runs once per analysis -- so the sweep command, which sets the knob and re-runs the -analysis command for every point (alter/altermod in place, or alterparam+reset which re-sources the deck), reprinted the line on EVERY point (a 5-point sweep, 5 times; a 100-point sweep, 100). Monte Carlo, optimize, and the pso/de/sa optimizers -- all of which re-run the analysis per point -- were equally noisy. FIX: an announce-on-change helper CKTannounceSolver(int klu) (defined in cktsetup.c, prototyped in cktdefs.h, called by both cktsetup.c and cktpzset.c) prints the line only when the active solver differs from the last announcement. The last-announced solver is tracked PROCESS-WIDE (a function-static), not per-circuit, because a .param sweep re-sources the deck and rebuilds the circuit each point -- a per-circuit flag would still repeat. Consequences: a multi-point sweep/Monte/optimize run announces the solver ONCE; a single analysis still announces once (unchanged); a genuine solver switch (.option klu/.option sparse) has a different mode and re-announces; a fresh ngspice process starts un-announced, so batch runs and the dual-solver test harness (which re-execs once per solver) still print it once each, and the HB (verify_hb) and benchmark suites that detect KLU by grepping for the line still see it. Result-neutral: the line is informational and part of no numerical comparison; full dual-solver regression unchanged. Verify (examples/solverannounce, 4 checks): a 5-point sweep announces SPARSE once (was 5x); a single op still announces once; a sparse->klu switch re-announces both; a KLU analysis still prints the KLU line. Scope: 3 source files (one small helper + two call-site swaps + one prototype); no analysis/result change.

E-267 *src/frontend/com_sweep.c, examples/sweepbus_examples/* (binaries rebuilt by CI)*

sweep records array/bus nodes under their natural names (**ph[0]**, not **ph_0_**). Sweeping a circuit whose outputs are array/bus nodes -- a node named ph[0] (E-221) -- stored the result vectors under MANGLED names: ph[0] became ph_0_, ph[1] ph_1_, so the sweep plot listed ph_0_, ph_1_, ... The sweep command (E-146/-189/-190) builds each result-vector name from the output plus, for a -vs family or overlay waveform, an appended _<knob>_<value> segment; that segment carries a float (_rt_1.5k, with ./-/e illegal in a nutmeg vector name), so the name builders sw_familynamename/sw_pointname mapped every non-alphanumeric char (except _) to _ -- but over the WHOLE string, INCLUDING the user's base output name, destroying a bus node's brackets. FIX (com_sweep.c): (1) a helper sw_append_sanitized(base, suffix) sanitizes ONLY the appended _<knob>_<value> segment, leaving the base name byte-for-byte intact, so ph[0] stays

ph[0] (and a -vs family curve is ph[0]_rt_2k); ordinary names are unchanged (a plain node has no special chars; an explicit -output name=expr still uses the clean given name). (2) A bare -output token that is a bus range ph[0:3] now expands to one output per index (ph[0] ph[1] ph[2] ph[3]), matching the netlist bus expansion (E-221), so sweep rt .. -output ph[0:3] records the four taps directly. Values are unaffected -- only the NAME the result is stored under changes (and range-token expansion); each output is still evaluated exactly as before. Verify (examples/sweepbus, 4 checks, a 4-tap bus divider): -output ph[0:3] records four vectors named ph[0]..ph[3] (range expanded, natural names); the mangled ph_0_.. names are gone; the recorded values are the correct divider ratios; a plain -output vo=v(out) is unaffected. Full dual-solver regression unchanged (the existing sweep suites use the -output name=expr form, whose clean names are untouched). Scope: one source file; no analysis/result change.

E-268 *src/frontend/spiceif.c, src/frontend/device.c, src/frontend/com_sweep.c, src/include/ngspice/ftext.h, examples/sweepwild_examples/* (binaries rebuilt by CI)*

Wildcard model-parameter knob **@*[param]**. Multiple .model cards can share a model-parameter NAME (two Verilog-A models both with a wavelength parameter). Nothing let you sweep that shared parameter across all such models in ONE knob: sweep @dev1[wavelength] (altermod) targets one model, and the .param idiom (.param wl, wavelength={wl} in each card, sweep wl) works but its .param knob commits with reset -- RE-SOURCING THE ENTIRE DECK at every point (slow) and needing the cards to reference the param. NEW: **@*[param]** sets param on every loaded model that has it, IN PLACE (altermod, no reset), so sweep **@*[wavelength]** ... co-varies all of them fast; it also works standalone (altermod **@*[wavelength]=1.55u**). IMPL: (1) spiceif.c if_setparam_wildcard(ckt, param, val) -- model params are a device-TYPE property, so one parmlookup per device type decides whether that type's models carry param; every model of a matching type (walking ckt->CKThead[typecode]->GENnextModel) is set with doset (the same path a concrete altermod @model[param]= uses); returns the count set and runs CKTtemp to push into live instances when called mid-run; models/types without the param are skipped. (2) device.c com_alter_common -- a wildcard device name **@*** routes to if_setparam_wildcard; a concrete @model[param] or device is unchanged; if no loaded model has the param, one clear warning. (3) com_sweep.c sw_kind -- **@*[param]** classified as an in-place altermod knob (like @model[param]), so the sweep applies it without reset; prototype in ftext.h. Additive -- **@*** was not previously a valid model name, so no existing alter/altermod/sweep behaviour changes. Verify (examples/sweepwild, 4 checks, both solvers): two wlmodel cards (R=wavelength1k) + one unrelated model without wavelength; sweep **@*[wavelength]** co-varies BOTH wlmodel devices (equal currents tracking 1/(wavelength1k)); the unrelated model is untouched; a concrete @dev1[wavelength] still targets only dev1; **@*[<absent>]** warns and changes nothing. Full dual-solver regression passes. Four source files.

E-269 *src/frontend/spiceif.c, src/frontend/device.c, src/frontend/com_sweep.c, src/include/ngspice/ftext.h, examples/sweepwild_examples/* (binaries rebuilt by CI)*

Instance wildcard **@#[param]** (alias **@*[param]**). E-268's **@*[param]** sets a MODEL parameter on every model that has it; this adds the INSTANCE counterpart -- set an instance parameter on EVERY device instance that has it, in place (alter/altermod, no re-source), so sweep **@#[scale]** ... co-varies an instance parameter across all instances. Two spellings: **@#[param]** (canonical; # marks instance) and **@*[param]** (alias -- the outer @dev[param] parse leaves param as [param, whose leading [is stripped). IMPL: (1) spiceif.c if_setparam_wildcard_instance(ckt, param, val) -- instance params are a device-TYPE property, so one parmlookup(..., do_model=0, ...) per type gates membership; every instance (walking each model's GENinstances->GENnextInstance) is set with doset(dev=inst, mod=NULL) -> setInstanceParm (the path a plain alter @dev[param]= uses). (2) device.c alter_set -- the device-name token picks the wildcard: **@#*** (token **#***) or **@*[param]** (token *** + param** starting [) -> instance wildcard; **@*** -> model wildcard (E-268); else ordinary if_setparam; no-match -> one warning. (3) com_sweep.c sw_kind -- **@#*/@*[...]** classified as in-place

altermod-style knobs (no reset); prototype in `fteext.h`. `@*=models`, `@#*/@*[...]`=instances; concrete `@dev[param]` unchanged. Additive -- these were not previously valid names. Verify (examples/sweepwild, 7 checks, both solvers) on a `wlmodel` with a `MODEL` param `wavelength` and an `INSTANCE` param `scale` (`R=wavelengthscale1k`): model wildcard co-varies both model cards AND reaches .model cards inside `SUBCIRCUITS` (they flatten to per-instantiation model copies); instance wildcard `@#*[scale]` and alias `@*[scale]` co-vary every instance; concrete `@dev1[wavelength]/@NX2[scale]` target just one; absent-param wildcards warn. Full dual-solver regression passes. Four source files.

E-270 *src/frontend/com_sweep.c, examples/sweepbounds_examples/* (binaries rebuilt by CI)*

sweep validates its numeric bounds (ASan/UBSan fuzz find). An ASan/UBSan instrumented build fuzzing the newest command parsers (`sweep/alter/altermod/loadpull/rfstab/stb/pyplot`) surfaced a bug in sweep's spec parser `sw_parse_spec`. sweep reads each bound of its `<start>` `<stop>` `<step>` (and `'lin`

E-271 *src/frontend/com_let.c, examples/letoob_examples/* (binaries rebuilt by CI)*

let no longer reads one byte before its buffer on an empty LHS (ASan/UBSan fuzz find). The same command-parser fuzz that produced E-270 flagged a one-byte out-of-bounds READ in `com_let`. `com_let` flattens everything after `let` into a heap string `p`, splits the RHS off at the first `=`, then NUL-terminates the destination vector name at the first `[`. It next trims trailing whitespace from the name: `for (q=p+strlen(p)-1; *q<=' ' && p<=q; q--); *++q='\0';`. When the LHS is a bare bracket -- `let [[ = ...`, where the leading `[` has just been turned into `'\0'`, leaving `p` an EMPTY string -- or is entirely whitespace, `strlen(p)==0` so `q` starts at `p-1`; the loop test evaluates `*q` BEFORE the `p<=q` guard can short-circuit, dereferencing `p[-1]` -- a read one byte before the allocation (ASan: `heap-buffer-overflow READ of size 1 at com_let.c, from let [[ = @#[r] v1 -o i(v1)`). The stray byte does not fault deterministically on the shipped build (almost always valid heap metadata), but it is a genuine OOB read on malformed input and could fault if `p` sat at the start of a page. FIX (`src/frontend/com_let.c`): test the bound FIRST -- `for (q=p+strlen(p)-1; p<=q && *q<=' '; q--)` -- so `&&` short-circuits and `q` is never read while it points below `p`. An empty or all-whitespace name then falls through unchanged to the existing "bad variable name" check, which rejects it cleanly. No valid `let` is affected (a one-line reordering of a loop condition). Verify (examples/letoob, 5 checks): the fuzz-found `let [[ = ..`, a single-bracket `let [ = 5`, and an all-whitespace LHS each error quickly with `bad variable name` and no crash/hang; a plain `let a = 5` and an indexed `let a[1] = 7` still assign correctly; the original input under an ASan build no longer reports the overflow. Full dual-solver regression passes. One source file.

E-272 *src/frontend/device.c, examples/alternull_examples/* (binaries rebuilt by CI)*

alter/sweep no longer deref a NULL parameter for an m-named device (ASan/UBSan fuzz find). The same command-parser fuzz crashed on `sweep mag(v(b)) 0 1k 5k 1k`. `com_alter_common` (`src/frontend/device.c`) supports altering a device's `PRINCIPAL` value with no named parameter -- `alter r1 = 2k` -- in which case the parsed param is NULL. Just before applying the value it runs a binned-MOSFET guard: `'if ((dev[0]=='m') && (eq(param,"w"))`

E-273 *src/math/cmaths/cmath2.c, examples/mathcast_examples/* (binaries rebuilt by CI)*

`cmaths` integer operators guard their (**int**) casts against out-of-range doubles (ASan/UBSan fuzz find). Continuing the expression-layer fuzz flagged `cmath2.c:765` on `1e30 % 5`; it is one of a cluster where a routine casts a double operand to `int` with a plain cast (UB when the value is non-finite or outside int range). (1) `cx_mod` (the `%` operator) computed `(int) floor(fabs(op))` per operand -- `1e30 % 5` is UB (UBSan) and on the SHIPPED build returned a garbage 2 rather than an error (a silently wrong result). (2) `cx_vector/cx_cvector/cx_unitvec` set the result length with `(int) fabs(arg)` -- `vector(1e30)/unitvec(1e30)` produced a saturated (`INT_MAX`-ish) length and then allocated and filled it: a multi-gigabyte `calloc` (macOS over-commit lets it succeed)

plus a billion-iteration fill loop, i.e. an apparent HANG on the shipped build (rc=142 under a watchdog), not merely a sanitizer report. FIX (cmath2.c): range-check before every (int) cast. cx_mod floors each operand into a double, then rchecks it lies in [0-or-1, INT_MAX] before the cast; a non-finite/too-large operand takes the existing argument out of range for mod path (rcheck->EXITPOINT->NULL). A shared helper cx_vec_len() validates the length for the three vector builders (fabs/cmag are >= 0, so one <= INT_MAX check also rejects NaN/inf); an out-of-range length prints vector length ... is out of range and returns NULL, which apply_func already handles (it checks if (!data) return NULL). Valid expressions unchanged -- 17 % 5 == 2, vector(4) == [0,1,2,3], unitvec(3) == [1,1,1], and any length within int range behaves exactly as before. Pre-existing (cmaths, not touched by the recent command work). Verify (examples/mathcast, 5 checks): 1e30 % 5, vector(1e30), unitvec(1e30) each error cleanly and fast (no UB, no hang, no garbage); valid 17 % 5 and vector(4) still evaluate correctly. Full dual-solver regression passes. One source file.

E-274 *src/frontend/evaluate.c, examples/idxcast_examples/* (binaries rebuilt by CI)*

An out-of-range vector index no longer invokes undefined behaviour (ASan/UBSan fuzz find). A fuzz of the expression layer flagged evaluate.c:779 on v(a) [1e308]. op_ind (vector indexing v[expr]) rounds the index with (int)floor(value+0.5); casting a double that is non-finite or outside int range to int is UB. The code ALREADY clamps the resulting index to [0, majsize-1] with a warning -- the only defect was the cast, which runs before that clamp. FIX (evaluate.c): a helper idx_floor() clamps the value to int range (NaN->0) before the cast, applied to all three casts (the real index and the real/imag parts of a complex [lo,hi] range index); v(a) [1e308] then resolves to the last element with the pre-existing warning, exactly as a large in-range index already did. Verify (examples/idxcast, 5 checks): vx[1e308], vx[1e30], a NaN index each resolve cleanly (no UB); valid vx[0] and range vx[0:2] still return the right elements. One source file; no change to any valid index.

E-275 *src/maths/cmaths/cmath4.c, examples/iffreal_examples/* (binaries rebuilt by CI)*

ifft() on a real vector no longer reads past its buffer (ASan/UBSan fuzz find). cx_ifft unconditionally did ngcomplex_t *indata = (ngcomplex_t *) data;. For a VF_REAL input, data is a plain double[length] (length8 bytes), so reading indata[i] for i<length walks length COMPLEX elements (length16 bytes) -- twice past the buffer (ASan heap-buffer-overflow READ: a 528-byte/33-element region read as 66 complex). cx_fft (Enhancement-225) already distinguishes real and complex input; cx_ifft did not. FIX (cmath4.c): for a VF_REAL input build a proper complex array (imag=0) sized length, use it, and free it before returning; a VF_COMPLEX input still points straight at data; a length>=2 guard (matching cx_fft) rejects a degenerate input. Verify (examples/iffreal, 3 checks): ifft(vx) on a real vector runs with no overflow; the fft->ifft round-trip keeps the length; complex ifft still works. One source file; no change to a valid complex ifft.

E-276 *src/maths/cmaths/cmath2.c, examples/rndcast_examples/* (binaries rebuilt by CI)*

rnd() guards its (int) cast against an out-of-range operand (ASan/UBSan fuzz find). cx_rnd (the rnd() random-integer function) turned each operand into a modulus with j = (int)floor(dd[i]); rand() % j. Casting a double outside int range to int is UB, so rnd(1e30) (or inf/NaN) tripped UBSan (cmath2.c:162). This is the same class as Enhancement-273, which hardened cx_mod and the cx_vector builders but not cx_rnd (at the time rnd(1e30) did not reproduce; the expression fuzz found the path). FIX (cmath2.c): a helper cx_rnd_i() clamps the value to int range (mapping NaN to 0) before the cast, applied to all three casts (the real operand and the real/imag parts of a complex operand); the result is unchanged for any in-range operand. Verify (examples/rndcast, 3 checks): rnd(1e30) and rnd(inf) resolve cleanly (no UB); a valid rnd(5) still yields an integer in [0,5). One source file; completes the E-273 (int)-cast hardening for cx_rnd.

E-277 *src/maths/cmaths/cmath4.c, examples/derivcx_examples/* (binaries rebuilt by CI)*

deriv() of a complex vector: fix a heap overflow AND a wrong result (ASan/UBSan fuzz find).

The fuzz flagged `cmath4.c:314` on `deriv((0.5, time))` (complex input). `cx_deriv` fits a degree-*d* polynomial over a sliding window; its COMPLEX branch diverged from its (correct) REAL branch in three places, each an out-of-bounds access on the last block AND a numerical error: (1) it read the data window `c_indata[j+i+base]` while fitting against the scale window `scale + i - degree + base` -- offset by degree, so it fit mismatched (x,y) pairs and read degree points past the end (heap-buffer-overflow READ); the real branch reads `indata + i - degree + base`. (2) The real-part output loop used `j <= i + degree/2` where the imag-part loop and the whole real branch use `j <= i - degree/2`, overrunning `scale[]/c_outdata[]`. (3) The tail indexed `scale[j+base]/c_outdata[j+base]`; the real branch's tail uses `j` (its FIXME notes `j+base` crashed), so a grouped (`base>0`) complex derivative overran both arrays. FIX (`cmath4.c`, complex branch only): align to the real branch -- read `c_indata[j + i - degree + base]`, bound the real-part output loop with `i - degree/2`, index the tail with `j`. Every overflow is gone AND the complex derivative is now numerically correct. Verify (examples/derivcx, 3 checks): `deriv((0.5, time))` runs with no overflow; `deriv(t + 2t*i)` now returns (1, 2i) -- the correct derivative, mid-vector; a real `deriv(2t)` still returns 2; a focused deriv stress over complex/real/grouped inputs is clean under ASan. One source file; real derivatives unchanged, complex derivatives now memory-safe and correct.

E-278 *src/maths/cmaths/cmath4.c, examples/scaleguard_examples/* (binaries rebuilt by CI)*

Transform functions `integ/deriv/fft` guard a length/scale mismatch (ASan/UBSan fuzz find). Continuing the expression-layer fuzz, `integ/deriv/fft` overran the heap on a vector whose length differs from its plot scale (`cmath4.c:500` on `integ(unitvec(200))`, :1065 on `fft(vector(5))`, :407 on `deriv(unitvec(200))`). `cx_integ/cx_deriv/cx_fft` index both the data (by length) and its plot scale; a SYNTHETIC vector (`vector(n)/unitvec(n)/an expression result`) carries the current plot's scale, whose length need not equal *n*. Two directions: (1) data LONGER than the scale (`integ/deriv` of `unitvec(200)` on a 66-pt plot) -- the loops read `pl_scale->v_realddata[i]` for `i<length`, past the end of the scale; (2) data much SHORTER than the scale (`fft(vector(5))` on a 66-pt plot) -- the Green's inverse-FFT sizes `datax` from `N=pow2(length)` but the output loop writes `tpts(=scale length)` points from `datax[2*i]`, so `N<tpts` overran `datax`. Only `cx_fft` had been guarded (Enhancement-225); the siblings had not. FIX (`cmath4.c`): `cx_integ/cx_deriv` reject a data vector LONGER than its scale (the direction that reads OOB) with a clean error -- a SHORTER data vector, as produced by `fft` (freq-domain, where group-delay `deriv(vp(3))` is legitimate), stays valid; `cx_fft` grows its transform size *N* (and *M*) to cover the output length `tpts` before allocating `datax`, a no-op for a well-formed transform where `pow2(length)>=tpts` already (the `fft->fft` roundtrip is unchanged). Verify (examples/scaleguard, 5 checks): `integ(unitvec(200))`, `deriv(unitvec(200))`, `fft(vector(5))` resolve cleanly (no overflow); the `fft->fft` roundtrip still returns a full-length vector; valid `integ(vx)/deriv(vx)` still work; a transform stress over mixed lengths/types is clean under ASan. One source file; completes for `integ/deriv/fft` the length/scale guarding E-225 gave `fft`.

E-279 *src/frontend/com_let.c, src/frontend/options.c, src/frontend/com_measure2.c, src/maths/cmaths/cmath4.c, examples/castguard_examples/* (binaries rebuilt by CI)*

The remaining unguarded (**int**) casts and the last unguarded scale-dependent transform (systematic audit, not fuzz). After the fuzz rounds a systematic AUDIT -- grep every (`int`) `floor/fabs(...)` of a runtime double, plus every `cmaths` routine taking a `struct plot *` and whether it guards its length against the scale -- found the tail of two already-fixed classes at sites the fuzzer never reached. (1) Unguarded (`int`)`floor(x+0.5)` of a user-supplied double (UB outside `int` range, the E-273/-274/-276 class): `com_let.c` (an index expression in an indexed `let`), `options.c` (`set numdgt / rawfileprec / measureprec` -- `set numdgt=1e30` tripped UBSan at `options.c:359`, `set rawfileprec=1e30` at :340), `com_measure2.c` (`meas ... rise/fall/cross=<n>` at :1580/:1584/:1588). (2) An unguarded scale-dependent transform (the E-278 class): `cx_mtimeavg` walks the scale data `dsc[j]` for `j < length-1`, so a vector longer than its plot scale -- `mtimeavg(unitvec(200))` on a shorter plot -- read past the scale (ASan heap-buffer-overflow @ `cmath4.c:1188`). FIX: a small clamping helper per file (`let_idx_int`, `opt_int`,

meas_int) converts to int only after clamping to int range (NaN -> 0), so an out-of-range option/count saturates instead of invoking UB; values already in range are unchanged. cx_mtimeavg gets the same length-vs-scale guard cx_integ/cx_deriv received in E-278. Verify (examples/castguard, 7 checks): set numdgt=1e30, set rawfileprec=1e30, set numdgt=-1e30, meas .. rise=1e308, mtimeavg(unitvec(200)) all clean where they previously tripped UBSan/ASan; a valid set numdgt=6 and mtimeavg(vx) still work. Four source files, guard-only.

E-280 *src/frontend/com_let.c, examples/letidxoob_examples/* (binaries rebuilt by CI)*

An out-of-range **let v[i] = x** wrote past the end of the vector (heap-buffer-overflow WRITE). Hardening the index cast (E-279) exposed a far more serious defect underneath. get_index_values(com_let.c) parses either one index (v[i]) or a range (v[lo:hi]). It validates low > high and high >= n_elem_this_dim -- but those checks lived INSIDE THE RANGE BRANCH ONLY. The single-index path set high = low and returned completely unchecked, so let vx[100] = 1 on a 66-element vector walked straight into the byte-offset arithmetic and performed a heap-buffer-overflow WRITE (ASan: WRITE of size 8) -- memory corruption from an ordinary typo, no exotic input required; a very large index additionally overflowed the index * n_byte_elem product in int (com_let.c:771). The range form vx[0:999] was correctly rejected all along. FIX: move both checks out of the range branch so they validate a single index too; an out-of-range assignment now reports index/high range (N) exceeds the maximum value (M) and changes nothing. Reads are unaffected -- op_ind clamps an out-of-range READ index with a warning (E-274) -- and every valid assignment, including the last element and the range form, behaves exactly as before. Verify (examples/letidxoob, 6 checks): let w[10]=99 on a 10-element vector, let w[999]=99, let w[1e308]=99 each rejected cleanly where the first two corrupted the heap; the last valid index (let w[9]=42) and a mid-vector assignment still assign; the range form still rejected. One source file, moving two existing checks.

E-281 *src/maths/cmats/cmath4.c, examples/derivgroup_examples/* (binaries rebuilt by CI)*

deriv() over-read on a partial block (grouping != length) (ASan fuzz find). cx_deriv walks its input in blocks: for (base = 0; base < length; base += grouping) for (i = degree; i < grouping; i += 1) ... fitting a window around i + base. grouping is the vector's v_dims[0]; for an ordinary vector that equals v_length, so there is a single block and the window always fits. But a vector whose declared dimension differs from its length -- as produced by a binary op on operands of UNEQUAL length, e.g. min(v(b), ac.v(b)) where a 66-point real and a 5-point complex vector yield length 66 with dims[0] = 5 -- leaves a PARTIAL LAST BLOCK: base climbs to length - 1 while the window still spans base + grouping - 1, reading past the end of the input (ASan heap-buffer-overflow READ @ cmath4.c:326, the same line E-277 had already realigned). FIX (cmath4.c): bound the inner loop with i + base < length in BOTH the real and complex branches, so the fit window can never reach past the input; a no-op whenever grouping == length (every ordinary vector), and a partial trailing block too short to hold a full window is simply skipped. Verify (examples/derivgroup, 4 checks): deriv(min(v(b), ac.v(b))) and deriv(max(v(b), ac.v(b))) clean where they previously over-read; an ordinary real deriv(2t) still returns 2; the complex derivative from E-277 is still correct (deriv(t + 2t*i) = 1 + 2i). One source file, a loop bound in each branch.

E-282 *src/frontend/plotting/agraf.c, examples/plotlabel_examples/* (binaries rebuilt by CI)*

asciipLOT read past its axis-label buffer on a 3-digit exponent (audit lead). Following up the plotting (int)floor(...) casts noted during the E-279 audit turned up a different, real defect one layer over. ft_agraf allocates its two axis-label lines as maxy + margin + FUDGE + 1 bytes (FUDGE=7) and NUL-terminates them at the last byte, budgeting the exponent width with sprintf(buf, "%1.1e", 0.0); shift = strlen(buf) - 7;. Formatting 0.0 ALWAYS yields a 2-digit exponent, so shift is 0 and the whole layout silently assumes two digits -- but real data can need three: denormal or very large values produce labels like 1.00e-320 (9 chars vs 8). The last label's memcpy(&line2[i + margin - ((j<0)?2:1) - shift], buf, strlen(buf))

then runs one byte too far and overwrites line2's terminating '\0', so the following `out_printf("%s\n%s\n", line2, line1)` walks off the end of the heap allocation -- ASan reports `heap-buffer-overflow READ` of size 82 on an 81-byte region inside `vsnprintf`. On the shipped build it does not necessarily fault; whether it prints heap garbage or crashes depends on what follows the buffer. FIX (`agraf.c`): remember the allocation bound in `line_end` (`maxy` is REASSIGNED further down as `maxy = spacing*nsp`, so the bound cannot be recomputed at the label loop), clamp the label memcopy so it can never reach the terminator slot (handling a negative offset too), and re-assert `line2[line_end] = '\0'` after the loop. Verify (examples/plotlabel, 5 checks): `asciiplot` of denormal 1e-320, 1e300, 1e-300 data and of both extremes together are clean where they previously over-read; an ordinary plot still renders its legend, axis rule and points; rendering confirmed BYTE-IDENTICAL to the pre-fix binary. One source file; no change to any rendered plot.

E-283 *src/frontend/plotting/agraf.c, src/frontend/points.c, src/frontend/display.c, examples/plotcoord_examples/* (binaries rebuilt by CI)*

Plot coordinate math no longer casts a non-finite double to `int` (extreme-data fuzz). The same output/formatting fuzz that produced E-282 found three more sites where a plot coordinate goes through `mylog10()` -- which returns +/-inf for zero, denormal or overflowed values -- or through a division by a range that can be zero, and is then cast to `int` (undefined behaviour). All are reached by ORDINARY plot commands on pathological data. (1) `agraf.c`: the decade `mag = (int)floor(mylog10(...))` -- data that overflows ($-1e308 * 6 \rightarrow -\text{inf}$, so `ylims[0]` is `+inf`) or is zero/denormal makes the argument non-finite; the limits `lmt/hmt = (int)floor(ylims[0] / tenpowmag)` are equally exposed since `tenpowmag = pow(10, mag)` can itself be 0 or inf. (2) `points.c` `ft_findpoint(): (int)((mylog10(pt) - tl) / (th - tl)) * (maxp - minp) + minp` is 0/0 for a degenerate range (`lims[0]==lims[1]`, or log endpoints sharing a decade). (3) `display.c`: the four screen-coordinate casts (log and linear, x and y). FIX: per-site clamping chosen so the result stays MEANINGFUL, not merely defined -- `agraf_decade()` bounds the exponent by `DBL_MAX_10_EXP` (the widest decade a double can represent) and maps NaN to 0, `agraf_int()` clamps the `lmt/hmt` ratios into `int` range; `ft_findpoint` computes the fraction in double, maps NaN (degenerate range) to 0 and clamps to [0,1] so the result always lands in `[minp, maxp]`; `disp_clamp_coord()` clamps each screen coordinate into the viewport. Verify (examples/plotcoord, 5 checks): `asciiplot` of overflowing ($-1e308$), denormal and all-zero data, and a min/max over denormal data, are clean where they previously invoked UB; an ordinary `asciiplot v(b)` still renders. Rendering additionally diffed against the pre-fix binary and confirmed BYTE-IDENTICAL for ordinary data. Three source files, guard-only.

E-284 *src/frontend/spiceif.c, src/frontend/device.c, src/frontend/com_sweep.c, src/include/ngspice/ftext.h, examples/wildparam_examples/* (binaries rebuilt by CI)*

Wildcard parameter diagnostics, and the **sweep** knob label (reported from real use). A Verilog-A parameter `L_um` could not be swept as `sweep @*[L_um]`, and ngspice's message named '`L_um`' -- which reads like the mixed-case name had been mangled by ngspice's case-insensitive netlist tokens. The case theory is a RED HERRING: the OSDI parameter lookup IS case-insensitive (`@*[L_UM]` resolves `L_um` correctly, verified). The real cause is a LEVEL mismatch that the diagnostics `hid: @*[param]` (and `@#*[param]`) is the INSTANCE wildcard, while a plain parameter `real L_um` in Verilog-A is a MODEL parameter -- an instance parameter needs (`* type = "instance"`). So the instance wildcard correctly matched nothing and reported no loaded instance has parameter '`L_um`': lower-cased, with no hint that the parameter exists at the MODEL level, and nothing pointing at the fix. Separately, sweep labelled every knob it applies in place as (model param) -- including the instance wildcards -- because `sw_kind()` returns `SW_MODEL` as a DISPATCH flag ("apply via altermod, no re-source") and the banner reused that flag as a description. FIX: (a) `spiceif.c` gains a probe `if_hasparam_wildcard(ckt, param, do_model)` reporting whether any loaded model (or instance) carries a settable parameter of that name -- it only probes, nothing is changed; (b) `device.c` uses it so a wildcard that matched nothing checks the OTHER

level and names the form that would work, in both directions (... but a loaded model does -- use the model wildcard '@*[l_um]' (an instance parameter needs (* type = "instance" *) in the Verilog-A), and the mirror suggesting @#[...]), while a parameter that genuinely does not exist still gets the plain message with no misleading suggestion; (c) com_sweep.c gains sw_knobdesc(), which classifies the knob from the NAME TOKEN, so the banner reads (instance param, wildcard) for @#[...]/@*[...] and (model param, wildcard) for @*[...]. Verify (examples/wildparam, 8 checks) on a model with a mixed-case MODEL parameter Wavelength and a mixed-case INSTANCE parameter L_um: both cross-level hints appear and name the right form; a truly absent parameter gets no bogus hint; sweep labels instance and model wildcards correctly; the matching wildcard still sweeps to the right values; and ALL-CAPS @*[[L_UM]] still resolves the mixed-case L_um, confirming the lookup was never case-sensitive. Four files; diagnostics and one banner string, no change to which parameters any wildcard actually sets.

E-285 *src/frontend/plotting/plotit.c, src/frontend/plotting/agraf.c, src/frontend/plotting/gnuplot.c, src/frontend/com_measure2.c, examples/veclenmix_examples/* (binaries rebuilt by CI)*

Output paths indexed one vector by another's length (extreme-data fuzz). The remaining findings from the extreme-data output fuzz all proved to be ONE class in four places: a vector's OWN length need not equal its plot SCALE's length -- any synthetic vector carries the current plot's scale, so let `y = vector(8)` on a 66-point transient plot has 8 points and a 66-point scale -- and a COMPLEX vector holds its data in `v_compdata`, leaving `v_realdata` NULL. (1) `plotit.c`: the transient resampling passed `v->v_realdata` together with `v->v_scale->v_length` to `ft_interpolate()`, which indexes the DATA by that length, reading ~58 elements past a shorter vector; for a complex vector it passed NULL outright -- a HARD SEGV on the shipped build (`asciiplot sqrt(-1*vector(10))` returns `rc=139`). (2) `agraf.c`: the bracketing indices lower/upper are computed against the X scale and bounded by `xscale->v_length`, but were then used to index each plotted VECTOR. (3) `gnuplot.c` (`wrdata`): the "no more data" guard tested only `i >= scale->v_length` while the branch it protects indexes `v->v_realdata[i]`. (4) `com_measure2.c`: the plain tran/dc branch read `d->v_realdata[i] / dScale->v_realdata[i]` with no NULL check, though the ac/sp branches beside it already had one -- measuring a complex vector dereferenced NULL (present twice, in `measure_at()` and `measure_deriv_at()`). FIX: clamp each index to the vector it actually addresses -- `plotit.c` passes the length the data and scale SHARE and skips the transient resampling entirely when any vector is not real (`all_vecs_real()`), since that block replaces the shared scale and must convert all vectors or none (complex vectors keep plotting through the normal path); `agraf.c` clamps lower/upper into `[0, v->v_length-1]` per vector; `gnuplot.c`'s guard also requires `i < v->v_length`; `com_measure2.c` takes the real part when `v_realdata` is NULL, exactly as its ac/sp branches do. Verify (examples/veclenmix, 7 checks): a vector shorter than its scale, a vector longer than its scale, a complex vector (previously `rc=139`), `wrdata` of a short vector, and a measure over a complex vector are all clean and still render/measure; an ordinary `asciiplot v(b)` and `meas tran ... max v(b)` are unaffected. Ordinary plot output AND `wrdata` output diffed against the pre-fix binary and confirmed BYTE-IDENTICAL; all eight decks from the output fuzz are clean. Four source files, guard/clamp only.

(E-25's simparam exposure lives in the OSDI callback table populated at load time; its diff rides inside the `osdiload.c/callbacks` changes.)

Build-warning cleanup (post-E-127): a full rebuild under the repo's -Wconversion flags surfaced latent warnings. `configure.ac` had -s (a strip flag) in the compile CFLAGS, so clang warned "argument unused during compilation: '-s'" on every source file (~1526 warnings) — removed (binaries are stripped separately by the fold and CI). -Wconversion also flagged a real latent bug in `maths/ni/nipzmeth.c` (`b = 0.0` was an assignment where `b == 0.0` was intended) — fixed — and benign `double→bool` conversions in `maths/cmaths/cmath4.c` — made explicit. Full-build warnings dropped 1903 → 394; the remainder are -Wconversion warnings in third-party SuiteSparse (KLU) and CMC device models (HiSIM), which are left to their upstream sources.

Every entry above is guarded by a verify script under [examples/](#) and was regression-checked against the full example arsenal, the compiler's integration suite, and the VA_TEST industry-model corpus at the time it landed.

E-296 *src/frontend/plotting/pyplot.c*

pyplot appearance controls. Seven additive set variables: `pyplot_grid` (on/off/x/y, overriding the grid-follows-axis default), `pyplot_legend` (off, or a matplotlib location -- underscore form since set keeps only the first word), `pyplot_markers` (a cycling marker `o s ^ D v * P X` ON the line, distinct from `pointstyle=markers` which drops the line), `pyplot_axhline/pyplot_axvline` (SI-aware reference lines via `ft_numparse`), `pyplot_dpi`, `pyplot_transparent`. With none set the emitted script is byte-identical to before. Verify ([examples/pyplotmore](#)): each control appears in the script AND the script executes; the default path is unchanged. One file.

E-297 *src/frontend/plotting/pyplot.c, plotit.c, com_pyplot.c*

pyplot **-fft** magnitude spectrum. `ft_pyplot's bool hist -> int mode (LINE/HIST/FFT)`. `-fft <sig>` reuses the time-domain (t,y) table; the emitted Python RESAMPLES onto a uniform grid (`np.interp`) before `np.fft.rfft`, because transient data is adaptively sampled and a raw FFT over non-uniform samples is wrong. One-sided, scaled by `2/sum(window)` so a pure tone reads back its amplitude. Options `pyplot_fft_window` (hann/hamming/blackman/rect), `pyplot_fft_db`, `pyplot_fft_points`, `pyplot_fft_logf` (log-f in Python, dropping the DC bin -- NOT the command's `xlog`, which would validate the `t=0` TIME scale and abort). Verify ([examples/pyplotmore](#)): a `2.0@1kHz + 0.5@3kHz` tone reads 2.0 and 0.5 (amplitude oracle, `rel<2%`). Three files.

E-298 *src/frontend/plotting/pyplot.c, plotit.c, com_pyplot.c*

pyplot complex-aware AC views **-bode/-nyquist/-polar**. An ordinary `pyplot v(out)` on AC data silently keeps only the REAL part (0.5 not 0.7071 at -3 dB). New `ft_pyplot_ac()` emits a `<vec-index> <freq> <re> <im>` table -- the imaginary part is CARRIED -- grouped by the first column so multiple/ragged vectors stay separate; `-bode = 20log10`

E-299 *src/frontend/plotting/pyplot.c*

pyplot overlay robustness + hover cursor + export note. Cross-run overlay via `ngspice's plot-name.vector refs (pyplot tran1.v(out) tran2.v(out))` already worked, but the data table was sized by the FIRST vector's scale, so a longer second run was truncated; it now walks the LONGEST scale across the plotted vectors (each still uses its own scale for x, shorter ones pad with NaN which matplotlib skips; a single run is unchanged since all share one scale). `set pyplot_cursor` adds a hover crosshair via matplotlib's built-in Cursor widget (no extra package), interactive-only (not emitted for a hardcopy). Pan/zoom/save (matplotlib toolbar) and the `<name>.data` export were already provided. Verify ([examples/pyplotmore](#)): a finer second run keeps every sample; cursor present in a window script, absent from a hardcopy. One file.

E-300 *src/frontend/plotting/pyplot.c*

pyplot **pyplot_mplcursors** cursor backend. On top of E-299's `pyplot_cursor` (built-in `matplotlib.widgets.Cursor` crosshair), `set pyplot_mplcursors` selects the `mplcursors` package -- data cursors that snap to a trace and show its (x,y) on hover -- AND turns the cursor on by itself. Unset, the built-in crosshair remains the default. The emitted script imports `mplcursors` inside a try and FALLS BACK to the built-in Cursor on ImportError, so a deck written where `mplcursors` exists still runs where it does not. Window-only (not emitted for a hardcopy), like `pyplot_cursor`; `mplcursors` must be importable by the interpreter named by `pyplot_python`. Verify ([examples/pyplotmore](#)): the `mplcursors` branch + fallback are emitted on `pyplot_mplcursors` alone, `pyplot_cursor` alone still emits only the built-in Cursor, neither appears in a hardcopy; the `mplcursors` branch was executed under an interpreter that has the package (calls `mplcursors.cursor(hover=True)`) and the fallback under one that does not. One file.

E-301 *src/frontend/plotting/pyplot.c*

pyplot **pyplot_cursor** is the single cursor master switch. E-299 (built-in Cursor) and E-300 (mplcursors) had drifted: `pyplot_mplcursors` turned the cursor ON by itself, so there was no one "off" knob. The enable condition went from `pyplot_cursor OR pyplot_mplcursors` to `pyplot_cursor` alone -- off by default, the ONLY thing that enables any cursor; `pyplot_mplcursors` now only SELECTS the backend when the cursor is on and does nothing on its own. Predictable, and matching every other optional pyplot behaviour. Verify (examples/pyplotmore): the full gating truth table -- default off, cursor-alone -> built-in, mplcursors-ALONE -> off, both -> mplcursors (with fallback), off in a hardcopy. One condition, one file.

E-302 *src/frontend/com_measure2.c*

.meas avg did not clip its window to [from,to]. Found by a closed-form oracle (a same-binary comparison is structurally blind to a uniformly present error). `measure_minMaxAvg()` trapezoid-summed only the samples INSIDE the window and divided by their span, without interpolating either boundary -- so the first and last partial intervals were dropped. `rms/integ (measure_rms_integral)` already clip exactly, so ngspice's own two answers for one window disagreed: `avg=1.27114` vs `integ/(to-from)=1.27324`, closed form 1.2732395; and the echoed `to=` was the first sample OUTSIDE the window (2.50280e-04 for a requested 2.5e-4). `O(dt/window)` error: 1.6e-3 -> 4e-7 here. Fix interpolates both ends and reports `sprev` as the window end, all guarded to `AT_AVG` so `min/max/min_at/max_at` keep whole-sample semantics. The `dc` branch is deliberately EXCLUDED -- a `dc` sweep may descend and enters its window at the high end, where a clip extrapolates (an early version of the fix turned a descending-sweep `avg` of 0.270250 into -3.79e+11); every `dc` measurement is byte-identical. Known remaining: **.meas dc avg** still truncates (0.270250 vs 0.2708333), and on a descending sweep `integ` returns 0.0 with `from= nan`. Verify (examples/measwindow): 28 checks vs closed-form integrals, windows landing BETWEEN samples; fails 14/28 on the pre-fix binary.

E-303 *src/frontend/com_measure2.c*

.meas dc avg window truncation, both sweep directions. Completes E-302, which fixed `tran/ac/sp` but excluded `dc` because a sweep may DESCEND (`dc v1 2 0 -0.001`) and enters its window at the HIGH end, where an unguarded clip extrapolates outside the bracketing samples. **.meas dc avg** averaged over [first sweep point inside, last inside] instead of [from,to]: 0.270250 against a closed-form mean-of- x^2 of 0.2708333, echoing `to=7.49000e-01` for a requested 0.75, while **.meas dc integ** over the same window was already correct. Fixed direction-agnostically in `measure_minMaxAvg()`: the clip works from the ACTUAL crossing between the previous raw sample and the current one -- open the window at the boundary crossed on the way in, close it on the way out -- so it fires at `from->to` ascending, `to->from` descending, and never on a sweep that does not cross. Guarded to `AT_AVG`, so `min/max/min_at/max_at` keep whole-sample semantics. The echoed window reports the upper end of the range actually covered, so a descending sweep no longer prints the nonsensical `from= 0.25 to= 0.25`. STILL OPEN (different function, `measure_rms_integral`): on a descending sweep `integ/rms` return 0.0 with `from= nan` -- the loop meets the first sample already above `to` and interpolates with index `i-1 == -1`, an OUT-OF-BOUNDS read, then breaks with an empty array. Verify (examples/measwindow): 38 checks, the same window run in BOTH sweep directions vs closed form; 32/38 on the pre-303 binary.

E-304 *src/frontend/com_measure2.c*

.meas dc integ/rms heap-buffer-overflow on a descending sweep. Completes E-303. On `dc v1 2 0 -0.001` these returned 0.0 with `from= nan` (rms printed nothing). `measure_rms_integral()` builds its clipped window assuming the scale ASCENDS -- it clips at `from` going in and `to` going out, and the widths `x[i+1]-x[i]` must be positive for the Simpson/trapezoid sums. A descending sweep starts at the top, so the FIRST sample is already above `to`: the end-clip fired at `i==0` and interpolated against index `i-1 == -1`, reading `v_realdata[-1]`. AddressSanitizer: heap-buffer-

overflow ... measure_interpolate com_measure2.c:195 / measure_rms_integral com_measure2.c:1415. It then broke with a one-element array, so while (i < xy_size-1) ran ZERO times (sums 0) and first was never set (m_measured_at unwritten = the nan). Fix: walk the samples in order of INCREASING SCALE rather than index order -- the identity for ascending data, so tran/ac/ascending-dc are byte-identical -- and make the has-a-previous-sample guard the traversal position $n > 0$ instead of $i > 0$, correct in both directions. integ 0.0 -> 0.135416 (closed form 0.135416667); rms -> 0.307458 (0.307459347). Also removed two dead reads in the toVal < 0.0 tail that indexed d->v_realddata[] with the INTEGRATION loop counter (past the end of the data) into variables nothing reads again, and took the window end from xScale[v_length-1] -- the smallest scale value when the sweep descends, not the last traversed. Verify (examples/measwindow): 44 checks, both sweep directions for avg/integ/rms vs closed form; 40/44 on the pre-304 binary; and the OOB is PROVEN gone -- ASan reports it before the fix and is clean after.

E-305 *src/maths/misc/randnumb.c, src/include/ngspice/randnumb.h, src/frontend/com_sweep.c, src/frontend/commands.c*

wcd -- worst-case distance / MPFP high-sigma. The named remainder of the statistical suite, and the complement E-150's own write-up says it does not implement. highsigma uses direction-free scaled-sigma IS; wcd works in STANDARDISED NORMAL space and asks a geometric question -- which point of the failure region is most probable? With the margin as $g(u) > 0$ for pass, that is the boundary point closest to the origin; its distance beta is the worst-case distance and, the density being spherical, $P_{fail} = \Phi(-\beta)$ -- the sigma number a designer quotes. Hasofer-Lind/Rackwitz-Fiessler iteration with forward-difference gradients: cost BOUNDED at a few iterations of $(1+ndim)$ sims, versus the $1e6-1e9$ samples plain MC needs merely to SEE a 4.5-6 sigma event. -is N refines FORM (exact for a linear margin, approximate when it curves) with mean-shift IS centred on the MPFP, carrying the exact ratio $\phi(z)/\phi(z-u^*)$: 986 failures seen in 2000 shifted samples for a $3.4e-6$ event. Implemented as two modes behind the EXISTING single Gaussian-draw funnel mc_sample_gauss(), so it composes with LHS/SSS rather than duplicating them: MC_MODE_WCD makes draws deterministic (dimension d returns a chosen $u[d]$, so the deck is a plain function $g(u)$) and MC_MODE_SHIFT does the shifted sampling. Dimensionality is DISCOVERED (a deck's Gaussian-draw count is unknown until it has been evaluated once) via a running maximum cleared per evaluation -- a single margin evaluation triggers several deck-copy passes and a later pass that draws nothing would otherwise wipe the count. Verify (examples/wcd): 19 checks vs the ANALYTIC GAUSSIAN TAIL, never a previous build -- FORM is exact for a linear margin so beta is exact to machine precision at 3/4/5/6 sigma and P matches $\Phi(-\beta)$ to $\sim 1e-8$; a 2-D case puts the MPFP on the symmetric point $(4/\sqrt{2}, 4/\sqrt{2})$, which is what shows the search finds the right DIRECTION and not a 1-D degenerate answer; a nominal-already-failing case gives signed $\beta = -3$. The four existing statistical suites pass unchanged.

E-306 *src/maths/cmats/cmath4.c*

The E-241 twin: **fft()** expression function normalised by the padded N. E-241 fixed an amplitude normalisation that divided by the ZERO-PADDED transform size instead of the input length, in the fft COMMAND (frontend/com_fft.c). The identical mistake survived in maths/cmats/cmath4.c `cx_fft` -- the vector-expression function reached by `let F = fft(v)`, a separate implementation of the same computation. Found by continuing the oracle campaign that produced E-241, and located with E-241's own discriminator: bin 0 is unambiguous (no window/scalloping) and $X[0] = Dlength$ for a DC offset D, so dividing by N reads back $Dlength/N$. On one signal (4001 samples padded to 4096, DC 2.0) the COMMAND read 2.000000 and the EXPRESSION read 1.953613 = $2.0 * 4001 / 4096$. NOT a convention question: `cx_fft` holds TWO complete implementations (complex-input and real-input) and in BOTH the FFTW branch already used the input length $((double)length)/2.0$, fpts) while Green's radix-2 used the padded size $((double)N)/2$, (double) N) -- the correct version sitting a few lines from the wrong one, an internal contradiction like avg vs integ in E-302. HAVE_LIBFFTW3 is undefined in this build so Green's is live. Fixed both Green branches to the input length. INDEPENDENT confirmation: nothing here touches

ifft, yet `ifft(fft(x))` went $2.3e-02 \rightarrow 1.1e-16$ (the pair only inverts when the forward normalisation is right), so `cx_ifft` was audited and deliberately left alone. Full audit of every Green-kernel caller: `com_fft.c` `fft x2` and `spec/PSD x2` correct (E-241); `cx_ifft` correct; `trannoise/1-f-code.c` cannot pad (`n_pts` grown to 2^n_{exp} by construction); `fft/ifft` are the only transform functions in the expression table so there is no spec twin. Verify (examples/fftexpr): 6 checks vs closed form (DC bin, round trip padded and unpadded, complex-input bin 0 =

E-311 *src/frontend/measure.c*

param/expr measurements now work in a **.control** block. Found oracle-checking the less-common **.meas** modes. The **param/expr** measure types are handled only by `do_measure()`'s second pass (via `nupa_eval`); the interactive **.control** **meas** command (`com_meas`) skips `do_measure` and calls `get_measure2()` directly, which has no **param/expr** case -- so **.meas tran pp param='a1-a2'** (dot-card) worked but **meas tran pp param='a1-a2'** (**.control**) failed with 'no such function as ...', down to a bare **param=a1**. Fix: in `com_meas` the prior results (`a1,a2`) are already single-valued ngspice VECTORS, so a **param/expr** measurement is `let <name> = (<expr>)` via the ordinary vector evaluator -- detected BEFORE `com_meas`'s single-valued-vector substitution loop (which would mangle the expression), reassembled from the tokens after `param=/expr=`, delimiter single quotes stripped, and re-lexed with `cp_lexer` so a spaced expression tokenises as the shell would. Works: `param=a1-a2` (unquoted), `param='a1-a2'` (quoted no-space), `param={sqrt(a1*a1 + a2*a2)}` (braces, spaces OK), `expr=`. Normal **meas** types (`max/min/avg/when/...`) route through `get_measure2` unchanged (byte-identical vs shipped). KNOWN LIMITATION left in place: `param='sqrt(a1*a1 + a2*a2)'` (single quotes + internal spaces) is mangled by the **.control** shell's own quote pre-expansion UPSTREAM of **meas** -- use braces; the dot-card handles it (no shell tokenizer). Verify (examples/measparam): 8 checks vs closed-form triangle values; 7/8 fail on the pre-fix binary.

E-312 *src/xspice/cm/cm.c, src/xspice/icm/analog/s_xfer/cfunc.mod*

XSPICE integrating code models are second-order accurate. Found oracle-checking the `s_xfer` Laplace transfer function against a closed-form transient: its error fell only as $O(h)$, not the $O(h^2)$ every native ngspice storage element delivers. Two independent causes. (a) SHARED across all integrating code models (`s_xfer`, `int`, `d_dt`, ...): `cm_static_integrate()`'s TRAPEZOIDAL order-2 arm was a self-flagged stand-in -- `cur = -0.5*ag[0]*intgr[1]` with a compensating `geq *= 0.5`, carrying its own `/* WARNING - This code needs to be redone */` -- which works out algebraically to backward Euler ($y(n)=y(n-1)+h*u(n)$), a first-order method. Fixed to the true rule $y(n)=y(n-1)+(h/2)(u(n)+u(n-1))$; the "one previous value" the WARNING said was missing is $u(n-1)$, stored in the SPARE state double `cm_analog_alloc` already reserves per integrator (`doubles_needed = bytes/sizeof(double)+1`), rotated through the `CKTstates[]` history for free -- the standard SPICE companion-model idiom of keeping a state and its rate in adjacent slots. `geq` is now `ag[0]` like the order-1 and Gear arms; a guarded legacy fallback preserves old numerics only where no spare exists (no real code model). (b) SPECIFIC to `s_xfer`: its controller-canonical feedback read the fed-back integrator states from the PREVIOUS timestep (`cm_analog_get_ptr(i,1)`), making the loop explicit/lagged one step, which caps $H(s)$ at $O(h)$ on its own even after (a). Now reads current-iteration states (offset 0) so the loop is solved implicitly within each Newton step (the DC/init pass already used offset 0). Both TRAP and Gear now converge $O(h^2)$. Convergence verified vs an exact $1/(1+\tau s)$ sine oracle (error ratio ~ 2 per h -halving before, ~ 4 after) and, for stability of the now-implicit feedback, vs SciPy `signal.lsim` across five transfer functions -- 1st-order, a $\zeta=0.05$ near-oscillatory resonator, a 3rd-order Butterworth, a stiff pair with poles three decades apart, and one with a numerator zero -- all converge and match to $<2.2e-4$. Verify (examples/sxferorder): 6 checks under both solvers, including a self-contained convergence-ORDER test (ratios ~ 4 , fitted $p \sim 2.1$) that fails on the pre-fix binary with no reference build required. 246-example regression green under both solvers.

E-315 *src/spicelib/analysis/tfanal.c, src/spicelib/analysis/cttic.c, src/spicelib/analysis/distoan.c*

Three analysis crashes on adversarial-but-valid input -> clean errors (command/netlist fuzz, E-222..228 / E-270..285 family). [6] `.tf` on a SINGULAR circuit -- a dangling inductor (l1 2 3 1, nodes 2,3 floating) makes the operating point singular; `tfanal.c` IGNORED CKTop's return, so the matrix was never factored and the following `SMPsolve()` asserted `IS_VALID(Matrix) && IS_FACTORED(Matrix)` (`spsolve.c:137, SIGABRT`). Fix: propagate the CKTop error before solving. [7] a second `.pz` over a URC device -- CKTic zeroes the RHS vectors with a loop the compiler vectorises into `memset(CKTrhs,...)`, but here `ckt->CKTrhs` is NULL, so it wrote through a NULL pointer (SIGSEGV in `_platform_memset`). Fix: CKTic returns cleanly when the RHS vectors/matrix are unallocated. [8] `.disto` with NO distortion sources (a plain resistor) -- `distoan.c`'s output section left `OUTpBeginPlot`'s result unchecked, so on failure `acPlot` stayed NULL and `OUTattributes(acPlot,...)` dereferenced it (SIGSEGV at `OUTattributes+268, addr 0x28`); the first plot checked its result but the five output-section plots did not. Fix: capture and check `OUTpBeginPlot` at each output plot, bail on NULL `acPlot`. Legitimate `.tf/.pz/.disto` unaffected -- guards fire only on the failure paths (valid `.tf` still gives 0.5 on a 1k:1k divider). Verify (examples/ngcrashanalysis): 4 checks.

E-316 `src/frontend/com_measure2.c`

`.meas AVG` ended one timestep short of `to`. Found continuing to oracle-check `.meas` against its own contract (E-302/303/304 family): AVG is the same quantity as `INTEG/(to-from)` over the identical window. `measure_minMaxAvg()`'s final window-clip (added by E-302, which made AVG integrate a trapezoid over `[from,to]`) guarded the WHOLE accumulation with `!AlmostEqualUlp(svalue, meas->m_to, 100)`, so when the first out-of-window sample fell within 100 ULPs of `to` the entire final trapezoid `[sprev,to]` -- a full timestep -- was skipped. AVG's window then ended one timestep short of `to`: for a $2.5e-7$ timestep the AVG window echoed `to=6.45990e-04` while `INTEG` (correct) echoed `to=6.46240e-04`, exactly $2.5e-7$ apart, so `avg != integ/(to-from)` by ~1.6%. `measure_rms_integral()` (`INTEG/RMS`) does NOT have this bug -- it always adds the final point, interpolating to `to` only when the sample overshoots, else using the `~to` sample as-is. Fix: move the `!AlmostEqualUlp` guard so it gates only the INTERPOLATION, not the accumulation, matching `INTEG` -- the final trapezoid is now always added, and AVG covers the full `[from,to]` window (`AVG==INTEG/(to-from)`, rel $1.6e-6$). MIN/MAX keep whole-sample semantics (branch is `AT_AVG-only`). Verify (examples/measavgwin): `AVG==INTEG/(to-from) + AVG window reaches to`, both fail on the pre-fix binary; the E-302/303/304 measwindow suite (44 checks) and E-311 measparam still pass.

E-318 `src/spicelib/devices/vsrc/vsrcload.c`

SFFM/AM voltage sources dropped the DC offset before the delay. Found in the correctness campaign, oracle-checking transient waveform sources. `vsrcload.c` evaluated SFFM and AM with `if (time <= 0) value = 0;` (after subtracting TD), returning 0 instead of the quiescent value at `time<=0` -- so an SFFM/AM source's OPERATING POINT (common `TD=0`) was 0 not `VO`, and over any pre-delay window it read 0, injecting a spurious startup transient (an SFFM(2,0.001,...) `~= 2V` source into an RC started at 0V and charged to 0.785V, vs a DC-2 source flat at 2V). Three oracles show `VO` is correct: the SIN case in the SAME function holds `VO+VAsin(phase) at time<=0`; ngspice's OWN current-source SFFM (`isrcload.c`) has NO such zeroing and evaluates the formula directly (the two SFFM implementations in one binary disagreed); SIN/PULSE/EXP/PWL all preserve their offset. Fix: hold the waveform's `time=0` value -- SFFM value = `VO + VAsin(phase + MDIsin(phasesem))`; AM value = `VO + (VMO + VMAasin(phasesem))*sin(phase)`; (both reduce to `VO` for zero phases). No example/regression circuit used SFFM/AM, so nothing else changes; the current-source path was already correct and is untouched. Verify (examples/sffmoffset): SFFM->1.5, AM->2.0 over a delayed window (SIN control->1.5), all failing on the pre-fix binary.

E-319 `src/frontend/com_qpss.c`

Transient-form QPSS leaked the fundamental into every mixing bin. Found in the correctness campaign via the RF steady-state oracle (a QPSS steady state must equal the HB-form and a long

transient + DFT). The transient-form qpss <expr> <f1> <f2> [periods] [maxorder] (as opposed to ... hb K1 K2) runs a two-tone transient then computes each 2-D harmonic $k_1f_1+k_2f_2$ by a TRAPEZOIDAL integral of $v(t)*\exp(-j2\pi f t)$ over the "last period" [tt[i0], tt[end]] of the RAW transient grid. That window is not exactly the beat period $T=1/\gcd(f_1,f_2)$ ($tt[i0] \geq wstart$ but not $==$, short by up to one tstep), is non-uniform, and has non-periodic endpoints, so the large DC/fundamental leaks $\sim tstep/T$ into EVERY mixing bin. On a LINEAR two-tone RC (every product with

E-320 *src/frontend/com_sweep.c, src/frontend/numparam/{spicenum.c,xpressn.c,numparam.h,numpaif.h}*

Sweeping a **.param** no longer forces a full circuit reset at every point. A **.param** has no live binding into the built circuit -- numparam folds each {expr} on a device line to a fixed numeric literal at parse time -- so the only way to change one is alterparam (rewrite deck text) + reset (re-parse + subckt-expand + CKTsetup + matrix REORDER), per point, even though topology never changes. E-320 adds a fast path to the sweep command: when every swept **.param** feeds ONLY addressable top-level device/model VALUES, com_sweep per point overrides the swept symbol(s) in the retained numparam dico (nupa_add_param), refreshes the derived-param closure (new nupa_recompute_params replays the dependent .param lines), re-evaluates each captured device value against the dico (new nupa_eval_expr -> the numparam formula() evaluator), and pushes it into the live instance with a resolved direct set (ft_sim->setInstanceParm = CKTparam -> DEVparam), refreshing each touched device type once via DEVtemperature -- NO reset, so the matrix ordering/factorization pattern is preserved. The template is the temper machinery (a temperature-dependent device value re-evaluated + alter'd in place every .dc step); the difference is that a parse tree cannot re-read an arbitrary .param cell (only time/temper/hertz bind to live ckt storage), so re-eval goes through numparam not IFeval. Two optimizations: each @dev [param] target is resolved to (instance, param-id) ONCE at arm time (no per-point lex or device-list walk), and identical value expressions are grouped so each unique expression is evaluated once per point. Conservative classifier: arms only when a scan of the ORIGINAL (pre-expansion) deck finds every swept-param occurrence to be a top-level device/model value slot (positional principal, or key={expr} named); DISARMS to the exact reset path on any subckt-body use (expression not retained past flattening), structural use (param in a node position, an instance/model/subckt name, a subckt call arg, .if/.elseif, .temp, an analysis card, .option, .ic, .nodeset, .global), or a derived .param (a non-swept param defined from a swept one). Because a disarm reproduces today's behaviour exactly, a sweep can only get faster, never change result. Measured on a 2500-stage ladder, 120-point sweep: param feeding ONE device in a large circuit 2.91s->0.26s (11.1x), param feeding ALL 5000 devices 3.42s->1.33s (2.6x); fast matches reset to $\sim 7e-9$ (numparam value-string formatting only), subckt fallback bit-identical. Scope: the sweep command (instance values fast, model values via textual altermod fallback); subckt-internal params and the optimize/Monte-Carlo .param reset sites stay on the reset path. Verify (examples/paramfastsweep): top-level param arms + matches closed-form $R2/(rval+R2)$; subckt param falls back yet identical.

E-321 *src/frontend/com_sweep.c, src/frontend/numparam/{spicenum.c,numpaif.h}*

.param fast-sweep, subcircuit tier -- subckt-INTERNAL device values now take the fast path. E-320 was limited to top-level values: a swept param feeding a device inside a subckt disarmed to reset, because numparam freezes each {expr} to a literal during flattening. The expression is recoverable: a flattened instance card (r.x1.r1 in out 1e3) carries card->linenum back to its subckt-DEFINITION body line (R1 a b {rval}), a top-level template whose original text numparam retains in dicoS->dynrefptr[linenum] (two instances of a subckt share the linenum). So capture now walks the FLATTENED deck (ci_deck) and, per device card, recovers the value expression via the new nupa_get_dynref(card->linenum) while the card's own first token gives the full hierarchical instance name -- top-level and subckt-internal captured by one mechanism, no name reconstruction. (Empirically the key is card->linenum, NOT card->linenum_orig, which uses a different counter -- inpcom.c line_number vs line_number_inc -- and is off by one.) SAFE across

subckt scope, since a subckt can shadow/derive-from/pass-through a swept global: (1) a Pass-1 scan of the original deck INCLUDING subckt bodies disarms on a swept param in a subckt header default, a subckt-call (X) argument, a subckt-body .param that shadows (swept LHS inside a subckt) or derives from (swept RHS) the param, or any structural slot (node/name, .if, .temp, analysis card, .option, .ic, .nodeset); (2) an arm-time SELF-CHECK re-evaluates each captured expression against the GLOBAL dico at nominal params and requires it to reproduce the value numparam already baked into the flattened card (read straight off the card, no ask-API), so any shadow/pass-through that slips past (1) disarms the whole path; (3) a disarm falls back to the exact reset path. Validated by a 10-deck battery -- 5 that must arm (top-level, direct subckt use, multi-instance, nested subckt, scaled expr) and 5 that must disarm (local .param shadow, formal-param shadow, formal pass-through, subckt-local derived, top-level derived) -- where EVERY deck's fast output equals reset bit-for-bit and every arm/disarm decision is as expected; the subckt fast path matches closed-form $R2/(rval+R2)$ to $9e-10$ and reset to $1.9e-9$. Speedup follows fan-out: a param feeding one device inside a subckt in a large fixed circuit $2.92s \rightarrow 0.24s$ (12.0x); a param feeding every device (400 subckt instances x 8 R) $2.50s \rightarrow 1.64s$ (1.5x). Scope: the sweep command (subckt-internal instance values fast, model values via altermod fallback); optimize/Monte-Carlo .param reset sites stay on the reset path. Verify (examples/paramfastsweep): top-level arm + subckt-internal arm + local-shadow fallback, all vs closed form (4 checks).

E-322 *src/frontend/com_optimize.c, src/frontend/com_sweep.{c,h}*

.param fast-sweep, optimizer tier -- the **optimize** command shares the fast-path engine. Every **-dparam** (symbolic .param) objective evaluation used to do **alterparam** + **reset** (re-parse + rebuild the whole circuit), and an optimization runs tens to hundreds of evaluations. E-322 exports the E-320/321 capture-classify-selfcheck-push engine (**sw_fp_build** / **sw_fp_apply** / **sw_fp_free**, de-static'd + declared in **com_sweep.h**) and wires it into the optimizer: **opt_fp_arm()** collects the **OPT_DECKPARAM** knob names and calls **sw_fp_build** ONCE before the search; **opt_eval** (Nelder-Mead / Levenberg-Marquardt / PSO / DE / SA) and **opt_eval_objs** (NSGA-II) push the candidate's deck-param values in place with **sw_fp_apply** instead of **alterparam+reset**; **sw_fp_free** at cleanup. The conservative classifier and arm-time self-check are unchanged, so a deck-param feeding a subckt shadow / structural slot / derived param disarms and the optimizer runs exactly as before. Two opt-outs: (1) **-center** design-centering (E-206) keeps reset -- its inner Monte-Carlo RE-SAMPLES process variation on each reset, a re-draw not a deterministic value push; (2) a device-count guard keeps the reset path on SMALL circuits -- the in-place apply has a fixed per-eval cost (numparam re-eval + dico ops) roughly independent of circuit size while a reset's cost grows with the deck, crossing at ~80 device instances, and because the in-place values differ from reset in the last few digits (numparam value-string formatting) an extremely tight -tol on a tiny circuit could otherwise send the two paths to different iteration counts; so **opt_fp_arm** counts flattened device instances and engages only at ≥ 80 . Correctness: across NM/PSO/DE the fast-path optimum is IDENTICAL to reset (122-device ladder fit: $r_{top}=361.914$ under NM and DE, 382.872 under PSO), and NSGA-II arms and returns the same Pareto front; all 41 existing optimize verify checks still pass. Speedup: a 4000-device fixed ladder, single **-dparam** knob, Nelder-Mead to convergence $0.94s \rightarrow 0.17s$ (5.6x); grows with circuit size, guarded to the sub-30ms reset on tiny circuits. Scope: **optimize -dparam** knobs, all methods except **-center** (**-param/-mparam** were already in-place). Verify (examples/optimize): a large-circuit **-dparam** optimization arms the fast path and still converges (2 new checks).

E-323 *src/frontend/com_optimize.c, src/osdi/osdiinit.c, src/include/ngspice/osdiitf.h*

.param fast-sweep, OSDI-aware optimizer guard. E-322 gated the optimizer's fast path behind a device-count guard (arm at ≥ 80 device instances) because on a small circuit a reset re-parses faster than the fast path's fixed per-eval overhead. That threshold was calibrated on RESISTORS and is wrong for OSDI (compiled Verilog-A) devices: a resistor reset just re-parses an element line, but an OSDI reset RE-INITIALIZES each compiled model instance (re-runs its setup/temperature callbacks), ~30x costlier per device. Measured (fast path on vs off, same binary, OSDI-resistor

ladder): the fast path wins at 3 OSDI instances (2.1x), scaling to 3.2x at 161 -- where resistors needed ~80 to break even. So the E-322 count guard would keep a SMALL OSDI -dparam optimization (common in device model extraction/fitting) on the slower reset path, forgoing 1.5-3x (a 300-instance OSDI fit measured 3.0x fast vs reset, same optimum). Fix: the crossover is about reset COST not device COUNT, so opt_fp_arm now walks the built circuit's instances and WEIGHTS each by device kind -- primitive=1, OSDI=OPT_FP_OSDI_WEIGHT=30 (measured ~3 OSDI ~= the reset cost of ~80 primitives) -- and compares the weighted total to the same 80 primitive-equivalent threshold. A handful of OSDI devices arms the fast path; a small all-resistor circuit still keeps the cheap reset. OSDI types are identified by a stable marker: every OSDI SPICEdev is built by osdi_create_spicedev and shares the OSDIparam instance-parameter setter (no built-in device uses it), exposed via a one-line osdi_devtype_is_osdi(type) helper in osdiinit.c; the optimizer call is #ifdef OSDI-guarded so a build without OSDI is unaffected (every instance weighs 1). Correctness is unchanged from E-322 -- the guard only decides WHETHER to arm, not what the fast path computes -- so a small OSDI -dparam optimization now arms and converges to the SAME optimum as the reset path. Verify (examples/optimize): a small OSDI -dparam optimization (compiled optres.va) arms the fast path and converges (2 new checks); all 43 optimize checks pass.

E-338 *ngspice-46/src/frontend/inpcom.c*

A full 64-bit bus range HUNG ngspice and grew without bound -- 3.6 GB after 4 s, 7.6 GB after 9 s, from ONE netlist line, in the released binary. R1 a[-9223372036854775808:9223372036854775807] r=2k. Found by fuzzing the array/bus node feature. ROOT CAUSE -- THE GUARD COULD NOT SEE THE REAL WIDTH: E-221 expands base[lo:hi] into scalar node names behind a deliberate BUS_MAX_WIDTH (8192) ceiling, computed as long width = (hi >= lo) ? (hi - lo + 1) : (lo - hi + 1). strtol saturates at LONG_MIN/LONG_MAX, so the full span makes hi - lo + 1 OVERFLOW a signed long -- undefined behaviour, which in practice wrapped to a small value, so the guard saw a tiny width and let it through. The loop then stepped from LONG_MIN toward LONG_MAX, about 1.8e19 iterations, appending to a string each time. The guard was not missing: it was computing its input with the very overflow it existed to prevent. ONLY THE FULL SPAN TRIGGERS IT -- the one-sided cases a[-9223372036854775808:0] and a[0:9223372036854775807] have width ~9.2e18, which fits and is correctly rejected. That is exactly why a corpus using only 32-BIT extremes (a[-2147483648:2147483647], width 4.29e9) missed it: that width is representable, so the guard worked. FIX: compute the span in UNSIGNED arithmetic (exact for every pair of longs) and compare the SPAN rather than the width, so the + 1 cannot overflow either; strtol setting ERANGE now also rejects an endpoint too large to represent rather than letting it be silently clamped to LONG_MIN/LONG_MAX, values that then look like an ordinary acceptable range. A netlist is untrusted input, so this is a resource-exhaustion defect reachable from a single line with no diagnostic and no bound. FUZZ: 432 decks, 0 crashes/hangs/runaway -- 54 range tokens (32- and 64-bit numeric extremes, malformed brackets, missing parts, nesting, control characters, XSPICE %vd[. ..] groups, a 4000-character base name, 40-deep bracket nesting) crossed with 8 placements (device nodes, subckt call, .subckt port list, value position, model name, print v(), bare print, let). CORRECTNESS: 13/13 numeric checks -- ascending AND DESCENDING ranges map ports in the right order, verified through an ASYMMETRIC subcircuit (1k on one port, 10k on the other) so a reversed mapping changes the measured current instead of being invisible; .subckt bus4 d[3:0] expands descending and maps positionally across four distinct currents; R1 a[0:1] 2k is ONE resistor (series 2k+4k = 0.5 mA); a bus element and an explicit scalar are the SAME node; v(a[0]) and bare a[0] resolve correctly (E-224) while ordinary vector indexing still works; non-integer/malformed/already-scalar tokens are untouched; the guard boundary expands at 8192 and stays literal at 8193. Verify (examples/busoverflow): 5 checks.

E-339 *ngspice-46/src/frontend/parse.c*

v() with THREE OR MORE node names SIGSEGV'd ngspice -- found by fuzzing the pyplot com-

mand family, but the defect is in the SHARED EXPRESSION PARSER. `print v(in,out,0)`, `let q = v(in,out,0)` and `pyplot v(in,out,0)` all died; `plot` SURVIVED, which is why an obvious spot-check passes and this was easy to miss. ROOT CAUSE -- TWO PATHS WITH DIFFERENT OWNERSHIP: `v()` takes at most two node names (`v(a)` or the differential `v(a,b)`), and a third was never rejected. `PP_mkfnnode`'s comma branch recurses, `PP_mkbnnode(MINUS, PP_mkfnnode(func,left), PP_mkfnnode(func,right))` then `free_pnode(arg)` -- that branch CONSUMES its argument, while the normal path at the end of the function BORROWS it (`p->pn_left = arg; pn_use++`). `free_pnode` recurses into children guarded by `pn_use`, so: with TWO names the children are plain nodes, each recursive call takes the BORROW path and bumps `pn_use` to 2, and the outer free merely decrements -- safe; with THREE OR MORE one child is ITSELF a comma node, so the recursive call takes the CONSUME path and frees it, and the outer `free_pnode(arg)` then walks into that already-freed child -- a DOUBLE FREE. Node existence is irrelevant: all-present `v(in,out,0)`, all-missing `v(a,b,c)` and mixed `v(in,out,zz)` crash alike -- it is purely the arity. FIX: reject the invalid arity where it is detected (Error: `v()` takes at most two node names.) rather than relying on the inconsistent ownership. Repairing the mismatch instead would mean changing `PP_mkfnnode`'s contract for every caller in order to make a syntactically invalid expression work; the arity is not legal in the first place, so rejecting is both smaller and more honest. The ownership mismatch is recorded as the underlying hazard. FUZZ: 122 `pyplot` invocations, 0 crashes/hangs/unexpected writes after the fix -- the eight mode flags (`-eye -hist -contour -smith -fft -bode -nyquist -polar`) crossed against missing/extra/repeated/conflicting arguments, hostile signal lists (unclosed `v()`, empty `v()`, 8000-character names, control characters, 200 signals) and hostile output base names (3000 characters, spaces, quotes, shell metacharacters, unicode, leading `-`). NOT A DEFECT, flagged and dismissed: `pyplot ./name` and `pyplot /abs/path` write outside the working directory -- that is the documented `pyplot [file] plotargs` contract and matches `wrdata/write`, so the fuzz policy was wrong rather than the command. Verify (examples/vfuncarity): 5 checks.

E-341 *ngspice-46/src/frontend/com_sweep.c*

sweep -analysis reset and **-analysis remcirc** SIGSEGV'd -- found by fuzzing `sweep/` optimize for RECURSION and STATE. `sweep` runs its `-analysis` argument as a command at every point, dispatched by name through `sw_run_cmd`, and nothing stopped a user naming `reset` or `remcirc` -- which free and rebuild (or remove) the very circuit the sweep is iterating over. The loop carried on with its resolved knob bindings, the old `CKTcircuit*` and the old plot. WHY THE FIX REJECTS UP FRONT RATHER THAN RECOVERING: the obvious repair -- notice the circuit changed and break out -- was tried FIRST AND MADE THINGS WORSE. The sweep's post-loop plot finalisation touches the same freed state, so breaking out merely moved the crash, AND it turned a previously WORKING case (`-analysis 'optimize ...'`, which resets internally but re-establishes its own state) into a NEW crash. That attempt was reverted. Rejecting a destructive analysis before the loop starts cannot regress anything, and nothing legitimate is lost: a sweep re-resources the deck itself when a knob needs it, so a user `reset` in the per-point analysis has no purpose. `optimize` is deliberately NOT rejected -- it resets internally, recovers, and works today. FUZZ (E-270 already covered numeric bounds, so this EXTENDED THE STRATEGY rather than repeating seeds): ROUND 1, argument surface -- 684 invocations, CLEAN -- 20 knob kinds (instance `param`, `@model[param]`, `.param`, the E-268/269 wildcards `@[p]/@#[p]/@*[[p]]`, malformed `@-forms`) x 29 sweep specifications (start/stop/step, lin

E-342 *ngspice-46/src/frontend/options.c, variable.c, variable.h*

Two ownership bugs in the synthetic user-variable list: a heap-use-after-free (SIGSEGV) and a double free (SIGABRT). `cp_usrvars()` synthesizes `$plots`, `$curplot`, `$curplottitle`, `$curplotname` and `$curplotdate` and returns a list its CALLERS free. That only works if the list owns every node. Two places assumed it did. [A] `cp_enqvar()` does NOT always allocate -- it clears its `tbfreed` out-parameter and returns a BORROWED pointer into `plot_cur->pl_env` when that live list already defines the requested name. `cp_usrvars()` declared `tbfreed`, passed it to all five calls and NEVER

READ IT, so it relinked the borrowed node (`tv->va_next = v`, orphaning the live list's tail) and the caller then freed a node somebody else still owned. Not hypothetical: a rawfile Option: line writes `pl_env`, so a loaded rawfile containing Option: `plots = 1` was enough -- SIGSEGV for all five names, ASan confirming allocate/free/use across three different functions (`raw_read -> cp_setparse`, `cp_getvar -> free_struct_variable`, `cp_usrvars -> cp_enqvar`). Fixed by honouring the flag and copying a borrowed variable; the five near-identical blocks became one loop so the rule cannot be applied to four of five sites. New `var_copy()` is written as the deliberate DUAL of `free_struct_variable()` -- deep, because that function recurses into `va_vlist` and a shallow copy would only move the bug. [B] found while fuzzing the fix for [A] and INDEPENDENT of it, no file involved: `cp_remvar()` freed the variable it looked up even when nothing had unlinked it. Only the `US_OK` arm unlinks; `plots` returns `US_READONLY` and the `curplot*` names return `US_DONTRECORD`, leaving the node inside `uv1` to be freed a second time by the trailing `free_struct_variable(uv1)`. `unset plots` aborted in `malloc` on a plain deck. Fixed by tracking whether the node is genuinely ours to free -- which also closes the same latent corruption for variables found in `variables`, `pl_env` and `ci_vars`. The `ci_vars` borrowed path is NOT reachable for these names today (`plot_cur` is always set, so all five return earlier) -- measured, not assumed -- but the fix keys on the ownership flag rather than the source list, so it covers that path anyway. Fuzz: 640 combinations (5 names x 8 Option: value shapes x 16 follow-up sequences), each run twice, plain and under ASan, all clean. No leak from the copy: 20000 iterations through `cp_usrvars()`, peak RSS 7.1 MB vs 7.2 MB control (LeakSanitizer unavailable on this host, so measured rather than asserted). Behaviour preserved and DIFFED against the old binary, not eyeballed. Found by tracing the E-341 performance finding through `cp_enqvar()` and asking what happens when `tbfreed` comes back 0. Verify (examples/`usrvarown`): 6 checks.

E-343 *ngspice-46/src/frontend/options.c, variable.c, include/ngspice/cpextern.h*

The $O(N^2)$ sweep cost E-341 documented but did not fix: FIXED, 26.6x at 16000 points (39.55 s -> 1.49 s). `cp_getvar(name,...)` looks up ONE name, but its first statement built ALL FIVE synthetic user variables (`$plots`, `$curplot`, `$curplottitle`, `$curplotname`, `$curplotdate`), searched the result and freed it again. Synthesizing `$plots` walks the live plot list and copies a string per plot, so every call cost $O(\text{number of plots in the session})$ regardless of what was asked for; `beginPlot()` does two such lookups per analysis ('`printinfo`', '`interp`') and a sweep creates a plot per point, so point `k` paid $O(k)$. WHY THE FIRST ATTEMPT FAILED (the whole lesson): E-341 tried making the construction LAZY -- build the list only after the `variables` search misses -- and measured 1.0x. `printinfo` and `interp` are normally UNSET, so that search always misses and falls through to needing the list anyway; laziness only helps the lookups that were never the problem. The gate has to be on WHICH NAME IS REQUESTED, not on when the list is built. Neither hot lookup is one of the five, so the fix builds NOTHING at all. `cp_usrvars()` is kept for `cp_vprint()`, which genuinely prints all five. Safe because both gated callers already search `pl_env` and `ci_vars` separately, which is the only other thing `cp_enqvar()` could return; ownership still follows E-342. Measured: 7.3x / 13.0x / 18.9x / 24.1x / 26.6x at 1k / 2k / 4k / 8k / 16k points -- the speedup GROWS with `N`, which is what removing a quadratic looks like. The regression suite itself, same 274 examples, went 748 s -> 479 s, because every analysis in every example was paying two of these lookups. NOT LINEAR YET, and the remainder is characterised rather than hidden: profiling the NEW binary at 64000 points puts 89% of what is left in `plot_alloc()`, a different function, which scans the whole plot list with a case-insensitive compare to pick a unique name. Two fixes were considered and REJECTED HERE: an exact-membership index (correct, but `plot_list` is mutated in ~9 places and a missed insertion hands out a DUPLICATE plot name), and a cheaper superset index (contained, but CHANGES BEHAVIOUR -- today plot names are reused after `destroy all`, and a superset would skip past them). Both are changes to plot lifetime, not to this lookup path. Semantics diffed byte-for-byte against the old binary, including a deck with `printinfo` and `interp` explicitly SET across `op/tran/ac`. E-342's two vectors re-checked. ASan clean. Verify (examples/`sweepyscale`): 5 checks.

E-344 *ngspice-46/src/frontend/com_sweep.c*

.model params now take the fast sweep's DIRECT SET -- the one tier E-320 left on the textual path. Model params now cost exactly what instance params cost (they were 2.1x-2.8x slower). The E-320..323 fast **.param** sweep resolves each swept device value to a slot ONCE and pushes it with `ft_sim->setInstanceParm`, skipping the per-point reset -- but `sw_fp_resolve()` bailed on 'if (!ckt

E-345 *ngspice-46/src/frontend/vectors.c, com_fft.c, linear.c, postcoms.c, spec.c, mw_coms.c*

The residual O(N) E-343 profiled but rejected fixing: FIXED. The sweep is now LINEAR -- 87x at 64000 points (54.97 s -> 0.632 s). `plot_alloc()` and `plot_add()` pick a plot name by counting a shared, monotone `plot_num` up until `<abbrev><plot_num>` is not the typename of any plot in `plot_list`. Plots are PREPENDED, so the colliding probe hits at the head in O(1); it is the SUCCESSFUL probe -- proving a name ABSENT -- that walks the whole list, with a tolower per character. One full walk per plot created, and a sweep creates a plot per point. Only the MEMBERSHIP TEST changed: a hash index of the typenames currently in `plot_list` answers it in O(1). The search still starts at the same shared `plot_num` and still counts up by one, which matters more than it looks -- `plot_num` is shared ACROSS abbreviations and only advances on a collision, so the sequence is `op1 tran1 op2 ac2 op3` not `op1 op2 op3`; and a number freed by `destroy all` is REUSED, which is exactly why E-343 rejected a 'remember every name ever issued' superset cache. Both copies of the loop (`plot_alloc` and `plot_add` carried the same one) now share `plot_unique_typename()`. E-343's OTHER objection -- an exact index is unsafe because `plot_list` is mutated in ~9 places and a missed insertion hands out a DUPLICATE plot name -- was answered by REMOVING the nine places rather than tracking them: `plot_new()` is now the SINGLE insertion point (six callers open-coded the same two lines: `com_fft x2`, `linear x2`, `postcoms`, `spec`), `killplot()` calls the new `plot_forget()`, `com_removecirc()` (which rewrites `plot_list` wholesale) calls `plot_forget_all()`, and the index builds LAZILY from `plot_list` so the static `const plot` needs no registration. PROVEN, not asserted: `plot_name_taken()` carries a PLOTNAME_SELFHECK block that answers the same question the OLD way and abort(s) on disagreement; the full suite was run with that build -- 276/276 OK, ZERO disagreements. The block stays in the source, unset, so it can be re-run against future changes. A REAL BUG was hit on the way and is worth recording: converting the open-coded prepends with a REGEX produced a DOUBLE INSERTION, because three sites ALREADY called `plot_new()` with the redundant two lines wrapped AROUND it, so the edit turned the trailing line into a second `plot_new()`. The second call ran `new->pl_next = plot_list` when `plot_list` was already new -- a SELF-LOOP -- and `vec_gc` spun on the circular list forever. Found by a stress case (fft) that hung, located by SAMPLING the hung process, not by guessing. A mechanical bulk edit needs each site read: the pattern being replaced was not the pattern actually there. Names verified IDENTICAL to the previous binary across 15 stress cases (mixed analyses, `destroy all`, `destroy first/middle/last`, `setplot new`, `fft`, `linearize x2`, `spec`, `write/load x2`, `load-then-destroy`, `twelve-then-destroy`, `remcirc`, `sweep-then-destroy`, `nested destroy`). Measured: per-point cost pinned at 10 us and the doubling ratio converged on exactly 2.0x -- the first time this sweep has been linear. With E-343, a 16000-point sweep went 39.55 s -> 0.163 s, about 240x. Verify (examples/plotname): 7 checks.

E-346 *ngspice-46/src/frontend/com_sweep.c*

A BUG FIRST, then the last tier: the fast **.param** path was silently FREEZING random draws that the reset path re-drew -- and fixing that IS the Monte Carlo tier. The fast sweep only re-evaluates brace expressions that MENTION the swept name. But `numparam` INLINES a `.param`'s expression into the device line during preprocessing, so `.param rv = agauss(...) + R2 a b {rv}` arrives as `r2 a b {(agauss(...))}` -- carrying the random call but NOT the swept name, so it was skipped and never touched again. (Same discovery: `nupa_recompute_params`, added in E-320 to refresh the derived-param closure, replays ZERO lines on such a deck -- there is no `.param` line left.) Measured over 5 points: reset gave `rv = 1097.7/1015.2/952.5/988.7/969.2`, fast gave 1161.1 FIVE TIMES. E-320's guarantee that results never change, only speed, had not

held for any deck with a random .param since it shipped. Fix: capture brace expressions that draw from the RNG (agauss/gauss/unif/aunif/limit/mvnorm) even with no swept name present. Once that exists MONTE CARLO NEEDS NOTHING MORE -- a sample is a re-draw plus an in-place push, and montecarlo arms the same machinery with NO swept knob (sw_fp_build(NULL,0)). Bit-identical to reset required THREE things, each found by a test that failed first: (1) DECK ORDER -- random binds sort ahead of the expression-grouped ones by capture index, since only they consume the RNG, so the stream matches however the deterministic binds are grouped; (2) NO CACHING -- a random bind is never served from the by-expression-text cache, because two devices with identical text must draw INDEPENDENTLY (verified: two {agauss(1000,300,3)} resistors give 1113.1 and 1099.5, not equal); (3) the SAMPLE BOUNDARY -- a deck re-copy signals one MC sample via nupa_signal(NUPADECKCOPY) -> mc_sample_advance(), which rewinds the per-sample dimension counter and steps the Latin-Hypercube sampler; skipping the re-source skipped that signal, which is exactly why plain draws matched reset (240/400 both) while -lhs did NOT (239 vs 258) and -lhs -warm did not (252 vs 258). Raising the boundary in the apply path made all four agree. The E-321 arm-time self-check is SKIPPED for random binds -- re-evaluating one draws a NEW value so it could never reproduce the baked one, and the draw would perturb the very stream that must match; their safety rests on the structural disarms, which were extended so a random gets the same scrutiny a swept name gets (a random reaching .temp, a subckt call or any structural slot disarms). Verified against an INDEPENDENT ORACLE, not a golden number: the same experiment hand-written as a reset loop in the control language -- montecarlo 124/200, oracle 124. Battery of 12 cases run on BOTH paths with discriminating yields (60.3/57.7/27.7/84.0/89.7/90.8%): plain, -lhs, -warm, gauss, aunif, limit, two randoms, same-expression-twice, random model param, two specs, derived random (disarms), structural random (disarms) -- all identical. Speed 1.8x/2.3x/2.4x at 20/200/1000 devices -- smaller than the sweep tiers because an MC sample re-draws and re-pushes EVERY random value; the saving is the re-source, not the pushes. CHANGES EXISTING RESULTS: a sweep over a deck with a random .param now re-draws it per point instead of freezing it -- which is what makes the fast path agree with reset again. Regression 278/278. Verify (examples/mcfastpath): 7 checks.

E-348 *ngspice-46/src/spicelib/parser/inp2dot.c, src/spicelib/analysis/dcpss.c*

.pss segfaulted on a short argument list -- and, separately, on a COMPLETE one whose **harmon-ics** was 0. Found by a new sweep shape: take a KNOWN-GOOD invocation of each command and run every PROPER PREFIX of its argument list. Existing campaigns could not see this -- cmdfuzz mutates argument VALUES (every value it supplies is present) and the parser fuzzers feed malformed FILES; a command that reads its fifth argument after checking only that a first exists is invisible to both. Across 38 commands pss was the ONLY offender: 0 args hung, 1-4 args SIGSEGV; qpss (8 args) and optimize (12) truncate cleanly. The stripped binary gave only the shape -- ldr d0, [x19] with x19 == 0, then fdiv, then a store -- and the UBSan build named it: dcpss.c:4990, load of null pointer of type 'double', i.e. Mag[0] = Phase[0] / (double)ndata. THREE failures stacked. (1) dot_pss() reads seven required values with no end-of-line check, and INPgetValue() cannot report 'nothing left to read' -- it returns 0 -- so a short card reached the analysis with points/harmonics silently 0. (2) DCpss() never validated: CKTharms is the LENGTH of every array the DFT writes into, and TALLOC(double, 0) is tmalloc(0), which returns NULL. (3) DFT() writes index 0 OUTSIDE any loop -- Mag[0], Phase[0], nMag[0], nPhase[0], Freq[0] - so its numFreq bound never protected them. That third point is why a parser guard ALONE would have been insufficient: pss 1meg 1u out 1024 0 50 5u is well-formed at full arity and crashed by the same route, a second entrance that never passes the truncation check. Fixed at all three layers: a PSS_NEED_ARG macro naming the missing argument (the idiom dot_dc() already uses a few hundred lines earlier in the same file, which dot_pss simply never adopted); DCpss() rejecting fguess <= 0, points < 1 and harms < 1 before anything is sized from them; and DFT() guarding its own unconditional index-0 stores. WHAT WAS DELIBERATELY NOT TIGHTENED, because it was measured rather than assumed: every parameter was tested at 0 and negative, and sc_iter and steady_coeff are harmless at both (rc=0), so they are still accepted -- narrowing

the accepted input on no evidence could break a working deck. NO VALID ANSWER MOVED: 22 correct invocations (op/dc/ac/tran/tf/pz/sens/noise/disto/four/pss plain and uic/pac/pnoise/hb/sp) were run on the shipped and rebuilt binaries and every printed number compared EXACTLY -- all identical, the one apparent difference being Total analysis time, i.e. the clock. Truncation sweep 0 crashes from 5. Verify (examples/pssargs): 7 checks.

E-349 *ngspice-46/src/spicelib/parser/inp2dot.c*

One transposed letter in a node name killed the process -- **op** then **tf v(ouy) v1** and the session is gone, with every loaded circuit, plot and let-defined vector. Seven commands did it: tf, pz, noise, sens, pss, pxf, psp. Found by a STATEFUL CROSS-COMMAND fuzz -- random sequences of state-mutating commands followed by a canary circuit whose answer is known exactly (1 V across a 1k/3k divider, v(b) must be 0.75). The harness was first aimed at the pre-E-342 binary, where it found 38 aborts in 200 sequences, so a clean run means the detector works rather than that it is asleep. ROOT CAUSE: INPtermInsert() is CREATE-OR-FIND. That is right for a device card, where the device DEFINES its nodes (R1 in out 1k is what brings out into existence), and wrong for an analysis card, which only REFERS to one. A mistyped name was quietly created as a new unconnected node. The milder consequence is arguably the worse bug -- before setup nothing crashes and the invented node reads back as a perfectly plausible v(bogus) = 0.000000e+00, no error, no warning. The fatal one: from the .control section CKTsetup() has already snapshotted the node list into prev_CKTlastNode and sized the matrix, so the extra node makes the tail check ending CKTunsetup() fail -- and that check calls controlled_exit(EXIT_FAILURE). The check is CORRECT and was firing on a real inconsistency; the defect is upstream of it. This never bit in a plain netlist purely because of ORDERING: a .tf card in the deck is parsed BEFORE CKTsetup() runs, so the invented node is inside the snapshot and the tail still matches -- it is reachable only interactively, which is exactly where a typo is most likely and losing the session hurts most. Fix: analysis cards resolve through INPtermSearch(), the lookup-only half of the pair that already existed alongside INPtermInsert() and was simply never used here; an unknown name is reported (no such node: ouy) instead of created. DECK ORDER STILL WORKS, and this is the constraint that shaped the fix rather than an afterthought -- a .tf card may legally sit ahead of the devices that define its nodes, so creation remains permitted while the circuit is NOT yet set up, and is refused only once CKTisSetup is true, i.e. deck parsing is over. Applied at 14 call sites across dot_noise, dot_tf, dot_sens, dot_pss, dot_pac, dot_psp, dot_pnoise, dot_pxf and dot_hb; dot_pz reached the same insert INDIRECTLY through INPgetValue(IF_NODE) and its four node arguments are now read explicitly so they take the same path. Verified: 0 of 14 commands kill the process, from 7; the same reference as a deck card still parses; a .tf card placed BEFORE its devices still returns transfer_function 0.75, output_impedance 750, input_impedance 4000; 22 valid invocations compared digit-for-digit against the shipped binary, all identical; and the stateful fuzz that found it went 58/60 to 60/60. Verify (examples/nodetypo): 6 checks.

E-350 *ngspice-46/src/frontend/com_sweep.c*

A sweep never put its .param back, and after two sweeps **reset** could not either. .param rload = 3k, sweep it 1k..5k, sweep it again 2k..10k, then reset + op gave 0.909 -- rload still 10k while the deck said 3k. One sweep was fine; the second broke it. ROOT CAUSE: the two implementations disagreed about the state they LEAVE BEHIND, which is exactly what E-320 guarantees they never do ("a sweep can only get faster, never change its result") -- it held for the numbers, not for the aftermath. The reset path drives each point with alterparam, which edits the parameter PERMANENTLY, so a later reset re-sources a deck that now carries the last swept value; the fast path never touches the deck, so reset did restore. Same command, same netlist, different circuit afterwards, decided only by which path happened to arm. WHY THE SECOND SWEEP: arming self-checks each captured expression against the value numparam baked into the flattened card at nominal (a guard against a subckt-local shadow), but the first sweep left the dico holding its last swept value while the flattened card still carried the nominal one -- so the check could not pass for an honest reason and DISARMED. The second sweep of a parameter therefore fell silently back

to the reset path: no error, no banner, the fast .param path armed line simply stopped appearing -- and the reset path is the one that edits the deck permanently, which is why the damage needed exactly two sweeps to surface. Both symptoms, one root cause: the sweep never restored the parameter. FIX: capture each swept .param before the first point and restore it at cleanup -- deck text via sw_set_deck (what reset re-resources) AND the numparam dico via nupa_add_param + nupa_recompute_params (what the NEXT sweep reads and what arming self-checks against). BOTH HALVES ARE NEEDED and the first attempt shipped only the deck half: with the dico still dirty the second sweep read the first sweep's last value as its own nominal and dutifully restored THAT, giving 0.833 instead of 0.909 -- better, and still wrong. The capture is all-or-nothing (if any swept name cannot be read back nothing is restored) rather than putting some parameters back and not others. Restoring is the direction that makes the two paths agree; making the fast path permanent instead would have satisfied the letter of the E-320 invariant while leaving reset unable to undo a sweep and every repeat sweep disarmed, and it matches how ngspice already treats a swept source in .dc, which does not leave it parked at the last step. DELIBERATELY UNCHANGED: the live circuit still holds the last swept point, so a bare op afterwards answers with that value -- true of both paths before this change and unchanged by it; only the deck and dico return to nominal, so reset means something again. Verified: reset gives 0.75 after 1/2/3/ mixed-spec sweeps (was 0.909 from the second on); 3 of 3 sweeps arm the fast path (was 1 of 3); derived params and subckt-internal values restore exactly; and 12 sweep batteries compared digit-for-digit against the shipped binary -- plain/list/dec, repeated sweeps, derived, subckt-internal, model params, Monte Carlo draws twice over to catch RNG-stream drift, and a two-knob family, 5-11 numbers per case, none vacuous -- ALL IDENTICAL, so nothing a sweep computes moved. Regression 282/282. The example is a proven trigger, not decoration: on the pre-fix binary 4 of its 5 checks fail, and the one that passes is precisely the invariant that must hold on both. Verify (examples/sweeprestore): 5 checks.

E-351 *ngspice-46/src/osdi/osdisetup.c, ngspice-46/src/osdi/osdidefs.h*

sens killed the process on any OSDI model carrying an INTERNAL NODE -- which is every production compact model. BSIM, HICUM, PSP and EKV all allocate internal nodes for their terminal resistances, so .sens was effectively unusable with real Verilog-A models, and it failed by taking the session down: "Internal Error: node allocation in DEVsetup() during sensitivity analysis, this will cause serious troubles!, please report this issue!" then exit(1). FOUND BY a campaign built on the observation that the previous OSDI sweep (83/83 clean) was LOPSIDED -- 40 dc and 22 ac checks against only 2 sens, 2 disto and 4 pz. Those thin three are exactly where an OSDI device is most likely to differ from a built-in, because they do not merely solve the circuit, they consume the DERIVATIVES the model returns. Three axes were run (OSDI vs equivalent built-in on both solvers; Sparse vs KLU on the same OSDI deck; repeat-invariance): 181/184, and all three failures were this one bug, on the single model in the set with an internal node. ROOT CAUSE: cktsens.c re-invokes every model's DEVsetup() to stamp the perturbation matrix and requires that doing so allocate NO nodes -- it snapshots CKTlastNode around the call and calls controlled_exit(EXIT_FAILURE) if it moved. The requirement is reasonable and every built-in device meets it by guarding allocation on "not already allocated": the inductor does `if (here->INDbrEq == 0) { CKTmkCur(...); here->INDbrEq = tmp->number; }` and INDunsetup() zeroes INDbrEq again. OSDI had NO such guard -- its setup called CKTmkVolt/CKTmkCur for each internal node unconditionally, so the second call, the one sens makes on a circuit that is still set up, allocated a fresh set and tripped the check. The check was CORRECT and firing on a real inconsistency; the defect is upstream of it. FIX: the same idiom the built-ins use. Each instance records in OsdiExtraInstData the node numbers its internal nodes were given, and a later setup reuses them instead of allocating. OSDIunsetup() deletes those nodes so it also clears the record -- the OSDI counterpart of INDunsetup() zeroing INDbrEq -- because otherwise a genuine re-setup after teardown would hand out numbers for nodes that no longer exist. Reuse is conditional on the node COUNT matching, since collapsing depends on parameters that a lter can move; if it changes, the record is dropped and nodes are allocated afresh. NOT MERELY NON-FATAL -- CORRECT, checked two

independent ways because "it stopped crashing" proves nothing: an internal-node model ($r=3000 + rs=10$) and a plain one ($r=3010$) are the SAME circuit (operating points agree to 14 digits), so their sensitivity to the shared built-in R1 must be identical -- and is, at $-1.8718770344518e-04$ from both decks, every printed digit, against a closed form $-RL/(R1+RL)^2 = -1.87187890622571e-04$ (the $1e-6$ residue is the sensitivity method's own perturbation step, not the fix). The model's own parameter agrees too: $n1_r$ $6.21886087563079e-05$ vs closed form $6.21886679809205e-05$. Verified: campaign 184/184 (from 181/184); sens completes on both solvers; sens twice in one session gives 41 bit-identical values; a reset $\rightarrow$ sens $\rightarrow$ reset cycle returns the operating point to $7.50623441396507e-01$, guarding that the node record never outlives the nodes; all 7 other analyses on the same model unaffected. Regression 283/283. The example is a proven trigger: on the pre-fix binary 5 of its 7 checks fail, and the two that pass are precisely the ones that should -- the model WITHOUT an internal node, and the analyses other than sens. Verify (examples/osdisens): 7 checks.

E-352 *ngspice-46/src/osdi/osdidisto.c (new), ngspice-46/src/osdi/osdi.h, osdiinit.c, osdiregistry.c, osdiext.h, ngspice-46/src/include/ngspice/osdiitf.h, ngspice-46/src/spicelib/analysis/cktdisto.c*

SUPERSEDED BY E-359 (the capability stands and is still verified by the same oracles; the compiler-emitted-tensor IMPLEMENTATION below was replaced by numerical differentiation of the analytic jacobian in ngspice, removing the compile-time regression, the 30MB objects, the runtime penalty and the ABI bump). **.disto** now includes a Verilog-A model's nonlinearities; previously it printed a warning and left them out. Distortion is a VOLTERRA analysis: it needs each device's Taylor expansion of $I(v)$ to THIRD order, not the operating-point linearisation the Jacobian provides. Built-in devices hand-code those coefficients (diodset.c computes id_x2/id_x3), which is why only FOUR of ~58 built-ins implement DEVdisto at all -- diode, BJT, BSIM1, MES -- and the OSDI ABI stopped at first derivatives, so cktdisto had nothing to ask an OSDI device for. KEY ENABLER, verified before anything was built: OpenVAF's autodiff is ALREADY arbitrary-order (mir_autodiff's DerivativeIntern chains through previous_order, with a comment anticipating 8th order), and nested $ddx(ddx(ddx(...)))$ reproduces closed form to every printed digit. So the compiler work was not implementing higher-order AD but ASKING for it: build_taylor_tensors re-runs auto_diff over its own first-order results for the second, and again for the third. NO 3-VARIABLE CEILING: ngspice's framework caps at three controlling variables (Dderivs holds derivatives "w.r.t 3 variables"; the DFx helpers take up to 27 scalar arguments with no fourth-variable form, DF_nF₁₂ taking a struct because the list outgrew the calling convention). That cap belongs to those hard-coded helpers, not to the mathematics -- the contribution is a symmetric tensor contraction -- so osdidisto.c contracts directly over however many inputs a device has and never calls $D1x/DFx$. Proven with an ideal mixer $I(out)=kV(a,ref)V(b,ref)$ whose ONLY nonlinearity is the mixed partial (both diagonal 2nd derivatives identically zero, so a dropped cross term gives exactly zero): matches closed form $0.5kA^2R$ EXACTLY. TWO SILENT-ZERO BUGS found on the way, both indistinguishable from a linear model: (1) the tensors were OPTIMISED AWAY -- ensure_optbarriers protects residuals/noise/jacobian but the Taylor values feed nothing else in the MIR, so the optimiser deleted them and every coefficient arrived as zero (they also take the mfactor scaling, since m parallel devices inject m times the distortion current); (2) NOTHING STORED THEM -- eval's jacobian/residual stores are gated on CALC_ flags that .disto never sets. Plus one wrong-by-a-constant bug only a reference implementation could catch: $D1nF12$ returns $0.5S2vF12(...)$ and $S2vF12$ ALREADY sums both index orderings, so without that 0.5 the mixed-tone products came out exactly 2x -- and because IM3 consumes the $f1-f2$ kernel its error was a non-obvious 2.246x. KNOWN LIMITATION AT THE TIME, SINCE LIFTED BY E-353: a model whose contribution goes through \$limit emitted NO tensors (the residual depends on the limited value, not the raw voltage read; build_jacobian handled that via intern.lim_state and the tensor pass did not). Limiting is standard in production diode/BJT/MOS models, so this was a real restriction; E-353 folds the limited values into the derivative chain and removes it. What remains unreachable is a nonlinearity in a GROUND-REFERENCED probe (the tensors are indexed by model input and a bare $V(a)$ is not one, having no hi/lo pair). Registering DEVdisto

removed `cktdisto.c`'s blanket warning, which would have turned such a model into a SILENT ZERO -- worse than before -- so `OSDIdisto` warns for itself. VERIFIED against four oracles, two involving no simulator at all: HD2/HD3 vs a closed-form polynomial (0.0 and $1.2e-16$); the A^2/A^3 scaling laws (4.0000, 8.0000); f_1+f_2 , f_1-f_2 and IM3 vs the built-in diode ($1.87e-06$, the bound the 0.24 ppm $\$vt$ difference justifies); IM3 vs transient+FFT in a DIFFERENT DOMAIN, converging 5.7% -> 1.4% -> 0.21% as the drive halves, exactly the A^2 signature of the 5th-order content a 3rd-order truncation excludes; and the multi-variable cross term (0.0). TWO HARNESS TRAPS recorded because both would have produced a confident wrong answer: the first IM3 transient was measuring SPECTRAL LEAKAGE from the fundamentals rather than IM3 (exposed by the A^3 law giving 2.93 instead of 8, not by the value looking wrong); and a first parameter choice made HD3 IDENTICALLY ZERO (the two contributions cancel exactly when $c_3 = 2c_2^2/Y_{tot}$, and $2(1e-4)^2/2e-3$ is precisely the $1e-5$ chosen) -- a zero that would have passed a naive check for the wrong reason, the same trap as the old campaign's "HD2 ~ 0 on a LINEAR network". The suite now refuses to score a both-zero comparison. Regression 284/284. The example is a proven trigger: on the pre-fix binaries 5 of its 6 checks fail, and the one that passes is the one that should. Verify (examples/osdidisto): 6 checks.

E-353 `ngspice-46/src/osdi/osdidisto.c`, `examples/limitdisto_examples/`, `examples/osdidisto_examples/`

SUPERSEDED BY E-359 (`$limit` models still work -- the suite below passes unchanged -- but the chain-fold no longer exists; differencing the real jacobian makes the problem unable to arise). **.disto** now works for Verilog-A models that use `$limit` -- i.e. every production compact model. E-352 gave OSDI devices distortion analysis, but only for models reading their controlling voltages DIRECTLY; a model passing one through `$limit` contributed NOTHING, and limiting is how every production diode/BJT/MOS model converges, so the feature did not reach the models people actually run. ROOT CAUSE: `$limit` hands the model a LIMITED copy of a branch voltage and the residual is written in terms of that copy, so differentiating the residual by the RAW voltage read yields zero -- nothing in the expression depends on it. The model is not linear, but every derivative the tensor pass asked for was. This was never a problem for AC or transient: `build_jacobian` has always folded the limited values back in via `intern.lim_state`; E-352's `build_taylor_tensors` simply did not perform the same fold. FIX: each model input now maps to a CHAIN of autodiff unknowns rather than a single one -- new `taylor_unknown_chain` returns the limited values from `intern.lim_state.raw` (each with the sign it contributes) followed by the input's own unknown. The 2nd-order pass sums over every PAIR drawn from the two chains and the 3rd-order pass over every TRIPLE, each term carrying the XOR of its entries' signs; an entry is negated when the model limits a REVERSED branch, `$limit(V(ref,a),...)`. VERIFICATION uses an exact expectation with no appeal to another simulator: `$limit` is a convergence aid, so at convergence the limited value equals the actual one and a model written with it is THE SAME DEVICE as the same model written without it -- every distortion product must agree. Both spellings are generated from ONE body string, which is what guarantees they differ only by `$limit` and cannot drift apart. Five shapes cover the chain combinatorics (where this can go wrong -- a model limiting a single branch never puts more than one entry on either side of a pair): two independently limited diagonals; a cross term with BOTH inputs limited (2x2 pairs); a cross term with ONE input limited (asymmetric chains, 2x1); a THIRD-order cross term whose chain has 3 entries, so the triple loop genuinely runs; and a reversed-branch limit, the only spelling producing a NEGATED chain entry. All 13 live comparisons agree, worst $7.1e-10$ -- convergence noise, since the two spellings settle $8.3e-11$ apart (limiting changes the Newton path, not the solution) and d^2/dv^2 of $\exp(v/vt)$ amplifies that by $1/vt^2$. THREE WAYS THIS COULD HAVE PASSED WHILE WRONG, each caught and closed because each produces a confident green: (1) BOTH-ZERO comparisons -- 'no distortion' is precisely the pre-fix behaviour, so scoring 0 == 0 as agreement would certify the bug; the suite scores a both-zero product as a FAILURE except where the mathematics requires zero (a bilinear term `kv1v2` has no third derivative, so its IM3 is correctly zero and is asserted as such). (2) COMPARING MAGNITUDES -- a dropped sign on a negated chain entry flips the result and leaves

E-359 *ngspice-46/src/osdi/osdidistonum.c (new), osdidistonum.h (new), osdidisto.c, osdi.h, osdiregistry.c, Makefile.am, src/include/ngspice/osdiitf.h, examples/osdidisto_examples/**

.disto for Verilog-A rebuilt on NUMERICAL differentiation of the analytic jacobian; the compiler-side tensor machinery is DELETED. E-352 obtained the 2nd/3rd-order Taylor coefficients by having the compiler emit a symbolic closed form. The capability was right but the cost was out of all proportion to what .disto does: on ASMHEMT 707k intermediate derivatives -> 1.4M MIR instructions -> a 30.2MB .osdi and a 126s compile against a 3.6s baseline (35x; hisimhv 20x), plus -- because the tensors were computed inside eval() -- a 3.3-3.9x RUNTIME penalty on every DC sweep, transient, AC and noise run, and an OSDI 0.8->0.9 ABI bump that made every already-compiled model unusable without a rebuild. ROOT MISTAKE: .disto needs the coefficients at ONE point, once per instance, once per analysis; E-352 computed a closed form valid at EVERY point in order to evaluate it once. WHAT REPLACES IT: the model already publishes its FIRST derivatives analytically -- that is the jacobian every analysis loads -- so the higher ones follow by differencing THAT at the operating point ($d^2 R_r/dv_p dv_q = d/dv_q[J_{rp}]$; $d^3 = d^2/dv_q dv_s[J_{rp}]$). Differencing an already-analytic first derivative rather than the residual is what keeps it accurate. Two properties make it cheap: the tensors are evaluated at the DC OPERATING POINT so they are frequency-INDEPENDENT (built once at D_SETUP, reused by every mode and every frequency), and they are built in NODE coordinates where an instance has 5-20 nodes rather than the 52 'model inputs' ASMHEMT reports. Entries keep the exact (row, col...) shape the analytic path produced, so the whole verified Volterra contraction is reused UNCHANGED -- only the tensor source and the kernel differ. NODE COORDINATES ALSO LIFT A REAL LIMITATION: E-352 indexed by model input (a branch voltage with a hi/lo pair), so a nonlinearity in a GROUND-REFERENCED probe V(a) had no pair, was never recorded, and the device reported 'contributes no distortion' and returned zero; it now works and is pinned to a closed form (-4.16666665e-02 vs exact -4.16666667e-02). E-353's \$limit chain-fold disappears for the same reason -- differencing the real jacobian sees what the simulator sees, so there is no chain to fold, and E-353's seven-shape suite (written specifically to break that logic) passes unchanged against an implementation containing none of it. ACCURACY, measured against EXACT CLOSED FORMS rather than against the old implementation: cubic 4.0e-09/5.4e-09, exponential diode 2e-10/~1e-10, internal-node 3.6e-12/1.2e-08, ground-referenced 4e-09, MEXTRAM vs the analytic implementation 5e-09/1e-07 -- all below the 1.9e-06 floor the built-in diode oracle already sits at (a 0.24 ppm \$vt difference). THE STEP SIZE IS THE WHOLE GAME: truncation is $O(h^2)$ and roundoff $O(\epsilon/h)$ at 2nd order but $O(\epsilon/h^2)$ at 3rd, so one step cannot serve both (a single $1e-3 \cdot V$ step put the result 8.5e-5 out); and the right yardstick is not the operating voltage but the CURVATURE scale ($v_t = 26mV$ for a diode, volts for a polynomial), so a voltage-relative step left a cubic biased at 0V 11% wrong. The 2nd-order pass measures

E-360 *ngspice-46/src/osdi/osdidisto.c, examples/osdidisto_examples/**

A circuit with TWO OR MORE different Verilog-A models reported distortion for only the LAST one; every other model contributed ZERO, silently. Introduced by E-359 and shipped with it. CAUSE: DEVdisto is dispatched once per DEVICE TYPE -- cktdisto.c walks DEVICES[] and calls each type's handler in turn -- and every distinct .osdi registers as its own device type, so a circuit using two Verilog-A models calls OSDidisto TWICE for every mode, D_SETUP included. E-359 cached the numerical tensors in ONE GLOBAL array and cleared it at D_SETUP, so model B's setup destroyed model A's tensors; in the mode passes A's instance pointers then failed the consistency check and every instance was skipped. That check is the only reason this produced NO numbers rather than WRONG ones -- without it model A would have contracted model B's tensors against A's kernels and produced confident garbage. FIX: the cache is keyed by descriptor (OsdNumModelCache{descr, ent, n, cap}; numcache_for(descr,create) finds-or-adds, numcache_reset drops only that model's entries), so each model only ever clears its own; lookup is linear in the number of MODEL TYPES, a handful, not in instances. WHY THE TESTS MISSED IT: every distortion test -- the six E-352 checks, the seven E-353 \$limit shapes, every model in the corpus

campaign -- used EXACTLY ONE .osdi. The bug needs two. Single-model coverage was thorough enough to hide a whole-feature failure. The new check [7] compares the cubic branch's value WITH a diode present against the same branch ALONE, so it fails on the silent zero rather than merely asserting non-zero; against the pre-fix binary it reports '0.00000000e+00 alongside a diode vs 6.24999998e-03 alone'. Also stress-tested: 50 instances (cache growth past its initial capacity of 32; first and last match the single-instance reference) and three model types with .disto run TWICE in one session (every branch correct on both runs, so the per-model reset is idempotent). Verify (examples/osdidisto): 7 checks. Regression 285/285.

E-361 *ngspice-46/src/osdi/osdidistonum.c, ngspice-46/src/spicelib/analysis/distoan.c, examples/osdidisto_examples/**

Two sanitizer findings in .disto that every ordinary build tolerates silently. (1) HEAP-BUFFER-OVERFLOW, mine from E-359: numdisto_jac_at copied the solution vector with for (i = 0; i <= n; i++) where n = CKTmaxEqNum, but CKTrhsOld holds CKTmaxEqNum entries -- ngspice's own code copies it as CKTmaxEqNum * sizeof(double) -- so the last valid index is n-1 and this read one element past the end on EVERY .disto run with a Verilog-A device. ASan: 'READ of size 8 ... 0 bytes after 32-byte region', a 4-double region faulting at index 4, matching exactly. It is a READ, so nothing was corrupted and the computed values are unchanged by the fix, but it is undefined behaviour on the main path; the same off-by-one was in the node-range guard (gp > n where the valid range is gp < n). (2) (int)NaN IN THE SWEEP POINT COUNT, pre-existing ngspice in distoan.c and reproducible with NO OSDI device in the circuit at all: a linear sweep with start == stop leaves DfreqDelta at zero so the tolerance term is 0/0, and a decade/octave sweep with ZERO steps does $\exp(\log(10)/0) = \text{inf}$ and then evaluates $0 * \text{inf}$ -- converting NaN to int is undefined, so a nonsense request quietly ran with whatever point count the hardware produced instead of being refused. FIX: zero-step decade/octave sweeps are rejected the way an unknown step type already is, and the linear division is guarded (where DfreqDelta is non-zero the term is $\text{floor}(\text{DfreqDelta} * \text{reftol} / \text{DfreqDelta}) = \text{floor}(\text{reftol}) = 0$ for any sane reftol, so zero is the value consistent with every other case and normal sweeps are bit-identical). NOTE: this second one had been firing under the project's OWN test deck all along -- osdidisto check [4]'s mixer sweeps `lin 1 1e6 1e6`, precisely the start == stop trigger -- and nothing ever failed because an un-sanitized build tolerates it. Neither defect produced a wrong number: one is an out-of-bounds READ, the other an undefined conversion that happened to yield a usable value on this hardware. That is the class only a sanitizer finds, and the argument for running the suites under one after any change involving raw index arithmetic. Verify (examples/osdidisto): 8 checks, the new one asserting a zero-step sweep produces no output rather than inventing some (an ordinary build cannot observe the UB itself). Both suites 8/8 and 7/7 under ASan+UBSan with values unchanged. Regression 285/285.

E-362 *ngspice-46/src/spicelib/analysis/{distoan,dctran,cktsens,dctrcurv}.c, examples/sweepguard_examples/* (new)*

Fuzzing analysis-card sweep parameters against an ASan+UBSan build: SEVEN pre-existing defects, all reachable from ordinary input, three of the counts involved being used as ALLOCATION SIZES and one reaching the allocator NEGATIVE. E-361 found one (int)NaN conversion in .disto's point count; that looked like a family rather than a one-off, so this drives every sweep-taking analysis (.disto/.ac/.noise/.sens/.sp/.dc/.tran/.four/.fft) with degenerate and extreme parameters. FINDINGS: distoan.c DECADE and OCTAVE -- the count goes +-inf (a zero or denormal start frequency makes log() diverge) or exceeds INT_MAX, and the (int) cast is undefined, feeding DstorAlloc(..., NoOfPoints+2); distoan.c LINEAR -- DnumSteps+1 is plain INT arithmetic and overflows at INT_MAX BEFORE any double-valued guard can run; distoan.c LINEAR -- the final count could be negative, reaching the allocator as ASan's 'requested allocation size 0xfffffffffff0'; dctran.c -- CKTtimeListSize overflows and is used both by TMALLOC and by osdiaccept.c to size a buffer; cktsens.c x2 -- the same cast pattern driving the sensitivity sweep; dctrcurv.c -- .dc has NO bound at all, advancing by a step and comparing against the stop value, so `dc V1 0 1 1e-`

30 (a typo for 1e-3) runs forever with no diagnostic, and a step below the ULP of the start never advances the value at all. .tran already declines the equivalent request and a zero step is already refused, so .dc was the outlier. ONE FIX MADE THINGS WORSE FIRST: clamping CKTtimeList-Size to a representable value regressed tran 1e6 1e30 1e6 from erroring to HANGING, because ngspice's existing rejection of absurd transients WAS this very overflow producing an allocation size that failed -- clamping made the allocation succeed and the run proceed forever. The accidental rejection had to become a deliberate one (E_PARMVAL); absurd transients now return rc=1 exactly as before and ordinary ones are bit-identical. TWO FINDINGS WERE NOT BUGS and were deliberately not 'fixed': an enormous but finite sweep is SLOW, not hung, and a timeout cannot tell them apart -- sens ac dec 1000000 over 300 decades measured 0.05/0.20/1.78/17.4s at 1/10/100/1000 steps (linear, so simply large), and ac oct 1000000 1e300 1e6 takes 12.8s identically on the shipped and fixed builds, exceeding the fuzzer's timeout only under ASan's ~5x slowdown. The scaling check that distinguishes them is documented in the harness. Final fuzz run 400 iterations: SANITIZER=0, CRASH=0, no real hangs. Verify (examples/sweepguard): 3 checks pinning eleven repros plus six ordinary sweeps that must be unaffected; a proven trigger -- against the pre-fix build it reports three HANGs. Its 'produces no output' check passes even pre-fix, because an ordinary build cannot observe an undefined conversion, only its consequences, which is why the sanitizer harness (fuzz_analysis.py) ships alongside it rather than being folded into the regression. Regression 285/285.

E-364 *src/osdi/osditrnoise.c (new), src/osdi/{osdidefs.h,osdiload.c,osdisetup.c,osdi.h,Makefile.am}*

TRANSIENT NOISE for OSDI devices: Verilog-A noise sources are now injected into .tran. .noise linearises about an operating point, so it cannot show jitter, noise-induced switching or oscillator phase noise; ngspice has had trnoise for independent sources for years, but OSDI devices were the only silent components in an otherwise noisy circuit. NO ABI OR COMPILER CHANGE was needed -- the compiler already emits a full parametric description that NOTHING read: noise_source_type[], load_noise_params() (per-source power and exponent, i.e. $S(f)=\text{power}/f^{\text{exp}}$ at the CURRENT bias), and each source's node pair and correlation name. load_noise_params and noise_source_type were dead ABI, partly because the NOISE_TYPE_* names lived only in the compiler-side header and had never been mirrored into ngspice's osdi.h (now fixed). ACTIVATION IS AUTOMATIC with no option to set: it turns on when the circuit already has a trnoise source and adopts that source's noise timestep, so all generators share one grid; it is deliberately NOT keyed on 'the device declares noise sources', which every real compact model does and would mean always-on. AMPLITUDE LAW, derived not fitted: f_alpha is Kasdin's fractional-noise method, $H(z)=(1-z^{-1})^{-(\alpha/2)}$ so

E-365 *src/include/ngspice/cktdefs.h, src/spicelib/analysis/{cktpzset.c,cktsetup.c}, src/frontend/com_hb.c*

pz followed by hb returned a SILENTLY WRONG harmonic-balance result (DC term 0.2500 instead of the correct 0.2439, 2.5% out) and, underneath it, read freed memory. CKTpzSetup opens with NIdestroy(ckt); NIinit(ckt); -- freeing ckt->CKTmatrix and building a DIFFERENT one -- while leaving CKTisSetup asserted, so every device's cached matrix-element pointer (bound by CKTsetup into the old matrix) dangles. com_hb then guarded its setup with 'if(ckt->CKTmatrix == NULL

E-366 *src/frontend/com_qpss.c, src/include/ngspice/klu-binding.h*

Two more sites of the E-365 stale-binding class, found by continuing the SAME sequence-fuzzing campaign against the FIXED build. Widening the pool (RF/steady-state commands, sweep/wcd, and SOLVER SWITCHING option klu/option sparse between analyses -- added deliberately, since KLU and SPARSE keep separate per-instance pointer sets) took the yield from 500 iterations/1 signature to 700/3. (1) com_qpss.c carried the identical guard com_hb.c had -- 'if(ckt->CKTmatrix == NULL

E-367 *src/frontend/typesdef.c, src/frontend/com_sweep.c*

sweep produced plots nobody could name. (1) `plot_alloc()` names a plot from the `plotabs[]` table in `typesdef.c` and falls back to the literal "unknown" when there is no entry -- and EVERY plot type this project added was missing: `sweep`, `sweepwave`, `hb`, `envelope`, `eye`, `loadpull`, `rfstab`, `stb`. `sweep` therefore created `unknown4/unknown7/unknown10`. All eight registered. ORDER IS LOAD-BEARING: `ft_plotabbrev` returns the FIRST entry whose pattern is a substring of the plot name, so "sweepwave" must precede "sweep" or a waveform plot would be abbreviated "sweep" and collide with the point plot (the same rule already explains why a plot named "spectrum" matches the earlier "sp" entry, never "spect"). The struct is `{p_name, p_pattern}` -- the SECOND field is matched and the FIRST is returned, which reads backwards from the table and is now documented in-source, because getting it wrong yields a table that compiles and matches nothing. (2) The summary printed ' sweep ' as a LITERAL, so the one hint the command gave for returning to an earlier sweep -- `setplot sweep` -- always failed with "no such plot named sweep"; it now prints the real name (`sweep3`), in BOTH summary branches (single-knob and the -vs/-family curve family), and the stale header comment in `com_sweep.c` documenting the plot as named `sweep` is corrected. The numbers are not 1,2,3 because `plot_unique_ttypename` draws from a counter SHARED by every plot type and advances it only far enough to avoid a collision -- a 3-point sweep burns `op1/op2/op3` and lands on `sweep3`. That is pre-existing behaviour and is NOT changed; what changes is that you no longer have to guess. FOUND ALONGSIDE, LEFT OPEN: `print alle` is dead in every build configuration -- `findvec_alle()` sits behind `#if defined(XSPICE) && defined(SIMULATOR)` with a `#ifndef SIMULATOR` stub returning NULL beside it, and `-DSIMULATOR` is set only on `ngspice_CPPFLAGS/libngspice_la_CPPFLAGS` (`main.c`, `ngspice.c`, `sharedspice.c`) while every `src/frontend` source is compiled ONCE into `libfte.la` with `AM_CPPFLAGS` alone, which both the executable and the shared library link. Confirmed by inspection: the string "DigitalData" is absent from the binary. Adding `-DSIMULATOR` to `src/frontend/Makefile.am` was tried and REVERTED -- `ngproc2mod/ngmultidec/ngmakeidx` also link `libfte.la` but not the XSPICE libraries, so enabling the block breaks them at link time under `--enable-oldapps`. The guard is unsatisfiable BY DESIGN, and a real fix means moving the event-node lookup into a simulator-only object. Two latent items behind that dead code are recorded so they are not re-discovered: `findvec_alle` overwrites the `pl_ttypename` `plot_alloc` just assigned with a hardcoded copy("dig1") (leaking it, leaving the registered completion keyword pointing at a name no plot has, and giving every digital plot the same name), and `get_all_type` has a misplaced paren -- `tolower(word[0] != 'a')` instead of `tolower(word[0]) != 'a'` -- defeating its own case-insensitivity, currently harmless only because callers lowercase first (verified: `print ALLV` and `print allv` behave identically). Verified by examples/`sweepname_examples`: 6/6 fixed vs 1/5 on the pre-fix binary (got ['unknown3'], no 'into plot' message). Regression 291/291.

E-368 *src/frontend/typesdef.c*

The periodic small-signal analyses named their plots wrong; E-367's claim that the naming was fixed was too broad. E-367 found its eight missing plot types by grepping for STRING LITERALS passed to `plot_alloc()`, which is blind to the path every standard and RF analysis uses: `outitf.c` calls `plot_alloc(run->type)` with a RUNTIME string handed to `beginPlot(analName)` by each analysis (mostly `ckt->CKTcurJob->JOBname`). Checking properly found seven more, four of them COLLIDING with an unrelated analysis -- worse than "unknown", because the name looks right. `ft_plotabbrev` returns the FIRST entry whose pattern is a substring of the plot name, and each of these contains a more general pattern: `PXF Analysis` matched nothing at all -> unknown; `PAC Analysis` contains "ac" -> collided with ordinary AC; `PSP Analysis` contains "sp" -> collided with S-parameters; `PNoise Analysis`, `qnoise` and `phasenoise` all contain "noise" -> collided with ordinary noise AND with each other; Frequency Domain Periodic Steady State Analysis (QPSS) contains "periodic" -> collided with PSS. Each now has its own pattern placed BEFORE the general one its name contains. TWO ORDERING TRAPS: "qnoise" contains "pnoise", so the `qnoise` entry must precede the `pnoise` entry; and PSS/QPSS differ only by their leading words (Time Domain vs Frequency Domain Periodic), so the QPSS pattern must carry "frequency domain periodic" rather than the "periodic" they share. Verified by examples/

periodicnames_examples: 9/9 fixed vs 4/9 pre-fix (pac->ac2, pxf->unknown2, pnoise->noise2, PSS/QPSS both pss). The FOUR REGRESSION CONTROLS -- ordinary noise, ac, tran, sp -- pass on BOTH binaries, which is what shows the change is targeted: those names are relied on elsewhere in the repo (setplot noise1, print noise1.noise_spectrum in noisecorr/noisejw/stdaudit/tempphys/physcheck) and none moved; no deck that runs pnoise selects a plot by name either. HARNESS TRAP RECORDED: the setplot listing has a blank line immediately after its header, so a non-greedy regex reaching to the first blank line captures NOTHING -- an early harness reported "no plot" for all 25 analyses and called the run clean. SCOPE STATED HONESTLY: envelope/eye/loadpull use literals and were registered by E-367, so they are correct by construction but no deck here reaches those paths; sens produced no plot under every invocation tried, consistent with the already-open sens breakage and not a naming problem. Regression 292/292.

E-369 *src/spicelib/devices/vsrc/vsrcbindCSC.c*

Closes the E-365/366 stale-binding class. E-366 fixed two sites and left the third OPEN on purpose, because it could not be closed with another guard: the binding was NOT NULL but STALE, and no NULL test distinguishes those. THE ASYMMETRY IS THE BUG: VSRcbindCSC assigns the pole-zero binding only INSIDE `if (here->VSRcibrIbrPtr)`, and that pointer is allocated ONLY by a pole-zero analysis -- so on any later analysis the gate is false, the assignment never happens, and the binding keeps its previous value, pointing into the BindStruct that SMPdestroy() freed when the pz matrix went away. Both consumers (VSRcbindCSCComplex, VSRcbindCSCComplexToReal) then dereference it, because THEIR guard is `VSRcbranch != 0` -- a property of the device -- rather than "was this binding re-established for THIS matrix". ASan on `option klu ; pz ; ac:` heap-use-after-free READ of size 8 in VSRcbindCSCComplex, freed by SMPdestroy, reallocated by SMPconvertCOOtoCSC. FIX: clear the binding BEFORE the gate, so a stale value can never survive a matrix rebuild -- one line, fixing both consumers at once since both NULL-test it; the duplicated `(x != 0) && (x != 0)` copy-paste slip is collapsed in the same pass. NO OTHER DEVICE NEEDS THIS: vsrc is the only one that hand-writes a pz binding block (grep "Pole-Zero Analysis" across every *bindCSC.c returns one file); every other binding goes through the klu-binding.h macros, which are self-consistent -- CREATE and CONVERT are gated on the SAME (`a>0 && b>0`) condition, so a skipped create implies a skipped convert. HONEST LIMIT, stated in the doc and the example: the PRE-FIX binary PASSES every behavioural check -- the freed BindStruct entry still held the right CSC_Complex pointer, so the UAF read plausible values and the numbers came out correct, verified deliberately on a 40-node ladder with a tran between the pz and the ac to churn the heap. This is a MEMORY-SAFETY fix, not a wrong-answer fix; the behavioural checks are a regression guard, and the check that DISCRIMINATES runs under a sanitizer (the example runs it when NGSPICE_ASAN is set and SKIPS LOUDLY otherwise, so nobody mistakes a pass for proof). UB that happens to give the right answer today is still UB -- the same read against recycled memory is how E-365 presented, 2.5% wrong and silent. examples/klubind_examples 8/8. Regression 294/294.

E-370 *src/spicelib/devices/urc/urcinit.c*

Every `.pz` re-expanded the URC subcircuit, creating nodes past the allocated RHS. FOUND BY RUNNING AN EXISTING FIXTURE UNDER A SANITIZER: a sweep of all 185 shipped decks with each solver forced produced exactly one real finding, in `ngcrashanalysis_examples/pz_urc.cir` -- the E-315 fixture, passing ever since E-315 stopped it SIGSEGVing. Nothing had run it under a sanitizer; the crash was gone, the corruption underneath was not. ASan: heap-buffer-overflow READ of size 8 in RESload (CKTload->NIiter->CKTop->PZan), buffer allocated by Nireinit. RESload indexes `ckt->CKTrhsOld` by node number, so the read past the end means a resistor's node number exceeded the allocated RHS. CAUSE: urc declared `.DEVsetup = URCsetup` AND `.DEVpzSetup = URCsetup`, but URCsetup is not a matrix-entry allocator like its neighbours -- it is a SUBCIRCUIT EXPANDER (CKTmkVolt per lump, CKTtrtElt per element) with NO idempotency guard, and CKTpzSetup calls DEVpzSetup for every device on EVERY pz job. Each `.pz` therefore expanded the URC again, creating fresh internal nodes AFTER NIinit had sized

the RHS. That is exactly why RESsetup/CAPsetup are safe in the same slot and URCsetup is not: they only allocate matrix entries, which is idempotent. URC was the only expander wired up this way. Even a SINGLE .pz triggered it (CKTsetup expands once, CKTpzSetup expanded again); the overflow took a second pass to cross the allocation boundary. Visible symptom on an ordinary build: doAnalyses: device already exists, existing one being used + aborted run. FIX: .DEVpzSetup = NULL. The URC stamps nothing itself -- DEVload, DEVacLoad and DEVpzLoad are all NULL -- and the RES/CAP instances it creates are ordinary elements under their own device types, for which CKTpzSetup already calls RESsetup/CAPsetup; that is what actually re-binds the matrix. NOT A SOLVER BUG despite being found in a KLU/Sparse hunt: it reproduces identically under both. BEHAVIOURAL CHANGE, not hidden: on the E-315 fixture the fixed build gets FURTHER -- instead of aborting on a spurious duplicate device it runs the analysis and reports singular matrix: check node 3, which is CORRECT (that fixture's node 3 is genuinely floating); the duplicate expansion had been masking a real topology diagnostic behind a spurious one. verify_ngrashanalysis.py still 4/4. examples/urcpz_examples 10/10 fixed vs 5/10 pre-fix. Regression 294/294.

E-371 *src/frontend/vectors.c, src/frontend/outitf.c, src/frontend/spec.c, src/frontend/postcoms.c, src/frontend/com_fft.c, src/frontend/rawfile.c*

Per-type plot numbering, and a date on every plot. Reported from user observation: a 500-point sweep produced sweep500, not sweep1, and the plot's Date field was empty. (1) NUMBERING. plot_unique_typename() counted up a SINGLE counter shared by every plot type. A sweep runs one analysis per point and KEEPS its plot (deliberate -- overlay and waveform inspection rely on it), so a 500-point sweep created op1..op500, each collision pushing the shared counter, and by the time the sweep's own plot was named the counter stood at 500. The number looked like the point count; it was really "how many plots exist". Each abbreviation now carries its own counter, so the first sweep is sweep1 whatever else ran. A tran/ac/tran/ac/noise/sweep/sweep deck goes from sweep12 op12..op3 noise3 to sweep2 op10..op1 noise2; the two agree except in the pathological case where one type's plots number an unrelated type. WHAT HAD TO BE PRESERVED: E-345 exists because this path was QUADRATIC over a sweep (89% of a 64000-point sweep in plot_alloc->cieq->tolower), fixed by starting the search from the shared monotone counter plus a hash membership index. A per-type counter restarting at 1 each time would have reintroduced exactly that quadratic, so the counter is REMEMBERED per type (nghash on the lowercased abbreviation) and the search resumes from it -- E-345's timing check still passes at 22->21 us/point, ratio 0.93 over a 4x longer sweep. destroy HAS TO WALK IT BACK DOWN: destroy all must restart numbering at tran1, which examples/lifecycle_examples (E-81) pins -- and that is what CAUGHT the first version of this change. destroy unlinks plots one at a time through plot_forget() rather than plot_forget_all(), so plot_forget() now lowers that type's counter to the freed number. That also makes a single destroy op2 recycle the number (the shared counter went on to op4), which is the behaviour destroy all always had, so it makes the two consistent rather than adding a rule. (2) DATES. Only the analysis path set one: outitf.c did pl->pl_date = copy(datestring()) after plot_alloc, so every plot created DIRECTLY by a command -- sweep, hb, envelope, eye, loadpull, stb, rfstab, qpac -- left pl_date NULL and print rendered it as (null). The stamp now happens in plot_alloc(), the single point where every plot is created, covering all eight command paths and any future caller. The five sites that set their own date became redundant and were removed, except rawfile.c, where a loaded rawfile's OWN date must win -- that one frees the stamp before replacing it, so centralising does not leak. VERIFIED by updating and extending examples/plotname_examples -- the E-345 example whose expectations this deliberately alters, and the file that documents this behaviour: 9/9 fixed vs 4/9 on a true pre-fix build (all six files reverted). The tran-has-a-date row PASSES on both, as the control showing the analysis path always had one and only the command paths changed. Regression 294/294.

E-372 *src/frontend/variable.c*

unset plots reported a bogus "Internal Error". Found by a PLOT-LIFECYCLE sequence fuzzer

-- random analyses interleaved with plot-management commands (destroy, setplot, unset plots, write/load, fft, linearize, remcirc) under ASan. That area was chosen deliberately: it is where E-342 (borrowed-pointer UAF via unset plots) and E-345 came from, and E-371 had just added a new allocation and new pointer arithmetic to it. Fired in 18/120 cases and minimised to ONE command on the SHIPPED binary: unset plots -> "Error: plots is read-only." followed by "cp_remvar: Internal Error: var 112". TWO DEFECTS in that one line. (1) THE MESSAGE PRINTED AN ASCII CODE, NOT A NAME: the signature is cp_remvar(char *varname) and the format was "...var %d", *varname, which dereferences the string to its FIRST CHARACTER -- 112 is 'p' for plots, and unset curplot reported 99, which is 'c'. The diagnostic never named the variable it complained about (verified against the ordinals). (2) IT WAS NOT AN INTERNAL ERROR: the branch is guarded by if (*p), and *p non-NULL only means the variable WAS FOUND in one of the lists walked just above -- the normal state for plots and curplot -- so it fired on 100% of ordinary unsets of them. An internal error valid input reaches every time carries no signal. FIX: drop both spurious prints. For US_READONLY the "%s is read-only" message is already the complete answer and nothing is unlinked or freed there (correct for a read-only variable), so removing the noise changes no behaviour; for US_DONTRECORD the case's own comment says "Do nothing..." and its siblings curplotname/curplottitle/curplotdate were always silent, so curplot now matches them. The other Internal Error in the same function ("US val %d", i) is CORRECT and left alone -- i really is an int and that default branch is unreachable from valid input. CHECKED AS A CLASS: a grep for a %d conversion fed a dereferenced char* across the frontend returns no other occurrence. VERIFIED by examples/unsetvar_examples (no sanitizer needed): 9/9 fixed vs 5/9 pre-fix. The FOUR CONTROLS -- curplotname, curplottitle, curplotdate and an ordinary set/unset round-trip -- pass on BOTH binaries, which shows the change is confined to the two affected cases and did not break unset itself. Re-running the fuzzer against the fixed build: 400 iterations, 0 findings, 382/400 cases holding live plots at the end (a real clean result, not a harness that stopped working); harness committed under examples/unsetvar_examples/fuzz/ and excluded from the regression sweep. PROCESS NOTES RECORDED: the investigation began by checking whether E-371 had leaked (it added a datestring() allocation per plot) -- it had not, killplot() frees pl_date and peak RSS stayed flat 7152->7168 kB across 202->6060 plots created and destroyed. That leak check was FIRST attempted with ASAN_OPTIONS=detect_leaks=1 and came back clean MEANINGLESSLY, because macOS/arm64 answers "detect_leaks is not supported on this platform" -- a sanitizer that silently does not run is indistinguishable from a clean result unless you check. Regression 295/295.

E-373 *src/frontend/rawfile.c*

A rawfile write/load round trip lost the x-axis column and renamed the sweep axis. FRESH AXIS: E-226 fuzzed rawfile LOADING with malformed input looking for crashes; this asks instead whether a rawfile written by ngspice and then loaded by ngspice still holds the same data. Round-trip identity is a perfect oracle (no reference implementation needed) and catches silent fidelity loss a crash fuzzer cannot see; it also lands on freshly-changed code, since E-371 had just touched rawfile.c. Two independent defects, NEITHER of which corrupted a value. (1) print LOST THE X-AXIS COLUMN for every loaded plot -- before write: Index v-sweep v(mid) became after load: Index v(mid), so nothing indicated which x-value each row belonged to. print prepends the scale column only when pl_ndims is non-zero (postcoms.c); outtif.c initialises it to 0 and sets it to 1 when analysis data arrives, but pl_ndims appeared NOWHERE in rawfile.c, so every loaded plot carried 0 and failed the gate. PROVEN NOT INFERRED: before writing the fix, a one-line probe setting pl_ndims=1 on load was inserted, shown to restore the column exactly, and reverted. THE op CASE IS THE CONTROL THAT MATTERS: pl_ndims=1 could have made print INVENT a column for a plot with no real scale (an op plot's variable 0 is just data, and the reader assigns pl_scale to variable 0 unconditionally) -- it does not, because print's inner pl_scale && !vec_eq(bv, pl_scale) test still suppresses it, and op prints the same header before and after; checked explicitly, since that is why the fix is safe. (2) THE .dc SWEEP AXIS WAS RENAMED v-sweep -> v(v-sweep) in the written file. The v(...) form means "the voltage

AT node X” -- right for a node probe, wrong for a sweep AXIS. A .dc plot’s scale is the synthetic, voltage-typed vector v–sweep, which has no v(prefix, so the writer wrapped it. It did NOT compound (verified over three successive write/load passes: the prefix test leaves v(v–sweep) alone), so the damage was a one-time rename. Fix: add ‘

E-374 *src/frontend/trannoise/wallace.c, src/math/misc/randnumb.c, src/main.c, src/sharedspice.c*

setseed did not seed TRANSIENT NOISE. Found by a correctness campaign over the command set: two identical decks with `setseed 42` produced DIFFERENT noise waveforms, while `setseed 42` then `rnd(100)` returned exactly 50.0 both times -- so the command worked, just not for the generator transient noise draws from. CAUSE: `#define WaGauss` selects the Wallace normal generator, and `initw()` opened with `srand((unsigned int) getpid()); // srand(17); TausSeed();`. `initw()` runs at STARTUP and fills its two pools from that `getpid()`-derived stream, and `GaussWa` draws from those pools; a later `setseed` reset `srand` and the Tausworthe state but nothing refilled pools that already existed. FIX, in three parts (moving the seeding alone is not enough): (1) `initw()` no longer calls `srand()`, so it seeds from whatever state the caller established and can never clobber a user seed; (2) the two startup call sites do the `srand(getpid())` themselves, so an UNSEEDED run stays random per process; (3) `com_sseed()` calls `destroy_wallace(); initw();` so a new seed actually rebuilds the pools. TRAP RECORDED: the first version of the `com_sseed` change was guarded `#if defined(WaGauss) && defined(SIMULATOR)`, and `SIMULATOR` is NOT defined for library sources like `randnumb.c` -- exactly the situation E-367 documented after finding `print alle` dead for the same reason. The whole fix compiled out silently and `setseed` still did not reproduce. Guard is now on `WaGauss` alone. That is twice this macro has disabled working code; treat `#ifdef SIMULATOR` outside `main.c/ngspice.c/sharedspice.c` as a bug until proven otherwise. VERIFIED by `examples/setseed_examples`: 5/5 fixed vs 4/5 pre-fix. Three properties, the third mattering as much as the first (a fix that pinned one constant sequence would satisfy the first and be useless): same seed -> byte-identical waveform; different seed -> different waveform; NO `setseed` -> still varies run to run. Two controls pass on BOTH binaries: `rnd()` was never broken, and the seeded stream is still real noise (3.5 mV peak-to-peak over 160 points), not a frozen constant. CONSEQUENCE: this falsified a claim in `docs/internals/ngspice_internals/ngspice_transient_noise_analysis.md` that `setseed` fixes the stream -- untrue when written, true now; the note was corrected and its seed-averaged statistics stand, having been means over genuinely independent runs. Regression 297/297.

E-377 *src/osdi/osdi.h, examples/simparamdiag_examples/**

OSDI diagnostics were unreadable and carried no severity. Found by the correctness campaign over all 94 \\${}-prefixed system functions, when a wrong `\$simparam\${str("analysis")}` reported `OSDI(debug) n1: unknown \${simparam_stranalysisOSDI(debug) n1: unknown \${simparam_str....` FOUR defects in one line. (1) NO SEPARATOR: `concat("unknown \${simparam_str", name)` joins with nothing, so the function name and its argument run together and the reader cannot see where one ends -- the numeric `\${simparam}` path had the same bug; now `unknown \${simparam\${str "analysis", and reported with the \${simparam\${str` spelling the user actually writes rather than the internal `\${simparam_str`. (2) NO NEWLINE: `osdi_log` writes `fprintf(dst, "%s", msg)` and no message carried a `\n`, so consecutive reports concatenated into one line -- which also HID the repetition, the old output looking like two lines where the fixed one shows 373 (the same 373 reports were always there). (3) WRONG SEVERITY FOR EVERY MESSAGE IN THE OSDI LAYER -- the one with reach beyond `\${simparam: ngspice's osdi.h` had `#define LOG_LVL_MASK 8`, but the level occupies the low THREE bits (DEBUG 0 .. FATAL 5) so the mask must be 7; 8 selects bit 3, which no level ever sets, so `lvl & LOG_LVL_MASK` was 0 = `LOG_LVL_DEBUG` for EVERY level. `\$display, \$info, \$warning, \$error` and every fatal diagnostic alike were labelled "OSDI(debug)" and written to `STDOUT` -- severity invisible to the user and to any log scraping, and `\$error/\$warning` never reaching `stderr`. OpenVAF's own copy of the header (`osdi/header/osdi_0_4.h`) has always said 7, so the two sides of the ABI disagreed on how to decode the level. Measured: before, all four severities print "OSDI(debug)"

on stdout; after, OSDI n1:/OSDI(info) n1: on stdout and OSDI(warn) n1:/OSDI(err) n1: on stderr. (4) A LEAK: the message was malloc'd, passed to osdi_log and never freed -- free was not even declared in the runtime's NO_STD extern block -- so a failing operating point leaked 373 allocations, not one; both call sites now go through one helper that frees. THE REPETITION IS FIXED BY E-378: making the messages visible exposed that there were 373 of them, left open here as "a convergence-path change" -- an understatement. CKTop read the fatal's E_PANIC return as "did not converge" and walked the gmin/source-stepping ladder, re-evaluating the device each pass, then blamed timestep too small. E-378 aborts the operating point on a fatal, so the report now appears once. Regression 302/302. Verify (examples/simparamdiag): 12 checks, a proven trigger -- pre-fix 1/12, and that single pass is a negative check that succeeds for the wrong reason on the old binary.

E-378 *src/spicelib/analysis/cktop.c, examples/opfatal_examples/**

A Verilog-A \$fatal now ABORTS the operating point instead of driving the convergence ladder. CKTop reads any non-zero return from NIiter as "did not converge" and answers it by working through gmin stepping, source stepping, pseudo-transient continuation and optran. A Verilog-A fatal arrives through that same channel as E_PANIC, so the operating-point solver could not tell "this model refuses to evaluate" from "this circuit is hard to solve" -- and responded to a fatal by trying harder. Every aid re-evaluates every device, so the model re-raised the same fatal on each pass: measured at 373 messages for one device, 746 for two, 1119 for three -- exactly 373*N, one per device evaluation, 373 being how many evaluations the whole ladder performs before giving up. THE DAMAGING PART IS NOT THE VOLUME: the run then ended with Error: Transient op failed, timestep too small, naming CONVERGENCE, so a model with a typo'd \$simparam name told the user their circuit would not converge while the real cause sat several hundred lines above. THE SAME GUARD ALREADY EXISTED IN THE OTHER PATH: E-55 added it to the transient time-stepping loop in dctran.c, with a comment that the error was otherwise "swallowed by the retry logic"; the operating-point path never got it (guards before this change: dctran.c 2, cktop.c 0). A .tran is affected too, since a transient computes its operating point first -- which is why the transient case still emitted 44 messages despite E-55's guard: it never reached the loop where that guard lives. THE FIX tests for E_PANIC after the plain Newton solve AND after each aid (a model may only fatal at a bias some later aid reaches), then reports the real cause. The test is EXACT rather than heuristic: E_PANIC is 1 and the non-convergence code E_ITERLIM is E_PRIVATE+3 = 103, distinct values, so it cannot mistake a stalled Newton solve for a fatal. Result: \$fatal in an .op 380 messages/"timestep too small" -> 1/"aborting"; unknown \$simparam 373 -> 1; \$fatal through .tran 44 -> 1. Regression 303/303. Verify (examples/opfatal): 8 checks, a proven trigger -- pre-fix 2/8, and the two that pass pre-fix are the ACCEPT checks (an ordinary circuit still solves; a circuit forced onto the aids with .options noopiter still converges through gmin stepping), which are there because a guard that aborted too eagerly would break every circuit that legitimately needs an aid.

E-380 *src/spicelib/analysis/dctrcurv.c, examples/dcstate_examples/**

A .dc sweep inherited integration coefficients from a preceding analysis -- a 45% SILENT wrong answer. dio load.c gates its charge branch on 'MODEDCTRANCURVE

E-381 *src/frontend/com_stb.c, examples/stbrestore_examples/**

stb handed its probe sources back ZEROED, so a following .ac returned all zeros. stb measures loop gain by injecting through two EXISTING sources -- it drives one with ac = 1 while holding the other at ac = 0, then swaps them -- and when finished "restored the probes to quiescent" by setting BOTH to zero. Zero is only quiescent if that is where the probe started: a source carrying a real AC drive had that value destroyed, and any following .ac came back with EVERY node exactly 0.00000000e+00, with no warning and no error (@v1[acmag] 1.0 -> 0.0; vm(mid) 0.3333333333 -> 0.0000000000). FIX: read each probe's acmag before the first injection and write those exact values back, instead of forcing zero. Only the magnitude is saved, because the injection

writes `ac = N` which sets `acmag` and leaves `acphase` untouched -- verified rather than assumed, and the example asserts it so the assumption cannot rot. SCOPE is narrower than it first looks: in the documented usage the probes are dedicated injection sources sitting at `ac 0`, where restoring zero happens to be right. It becomes wrong when an already-driven source is named as the probe, which is legal and is what the fuzzer did (`stb V1 V2` with `V1` the deck's own `ac 1` source), and nothing indicates the drive has been destroyed. FOUND BY cross-analysis STATE fuzzing with a NUMERIC oracle -- `result(B after A) == result(B alone)` -- the same instrument as E-380; re-running the 196-pair campaign against the E-380 binary left four mismatches and this was the only genuine one. The other three were --> hb deviations of $1e-17..1e-23$ on near-zero harmonics whose RELATIVE deviation reached 30%, which looks alarming and means nothing -- the classic no-absolute-floor trap, the same one that produced a false $2.12e-07$ "mismatch" in the metamorphic campaign; with `atol=1e-12` all three are within tolerance. WHAT THIS CLOSES OUT: the chase began from an asymmetry in `dcps.c` -- five `spSetComplex` calls and zero `spSetReal` -- which turned out NOT to be a defect. Matrix mode is managed per factorization by the SMP layer (`SMPcLUfac/SMPcReorder` set complex, `SMPluFac/SMPreorder` set real), so there is nothing to restore; `dcps.c` sets it manually because it assembles $G + jC$ by hand (`spSetComplex` -> `CKTload` for the real halves -> `CKTacLoad` for the imaginary halves -> `factor`), which is the intended pattern and the reason it is the only file in the tree calling `spSetComplex` directly. Regression 305/305. Verify (examples/stbrestore): 7 checks, a proven trigger -- pre-fix 4/7. The four that pass on BOTH binaries are the accept half (a probe that started at `ac=0` still ends at `ac=0`; `acphase` survives; `stb` still reports a loop gain; `stb` run twice reports the same one), and `stb`'s own answer is BIT-IDENTICAL across the fix (

E-382 *src/frontend/com_loadpull.c, examples/lprestore_examples/**

loadpull left the user's tuner at its last swept point. It sweeps the R, L and C of the user's own matching network across the Smith chart, and when finished left them wherever the last grid point put them -- the old code said so itself, `/* restore something sane on the load (last set values are fine) */`, a comment with no code under it. The last set values are not a result, merely where the loop stopped: `RL 50 -> 84.83 ohm, LL 1e-15 -> 1.34e-8 H`. Any following analysis then ran against the wrong network -- on the test deck an `.ac` moved from 0.4789 to 0.6765, a 41% error, silently. THE SAME CLASS AS E-381 (`stb` handing its probe sources back ZEROED): both commands borrow parts of the user's circuit, drive them, then guess at what "putting them back" means -- one guessed zero, the other guessed "wherever we stopped". `sweep` already had it right (E-350 captures each swept parameter's nominal value and restores it at cleanup) and this follows that precedent. Two neighbours were checked and are NOT defects: `optimize` leaves the best point because that is its deliverable, and `aging` writes back the accumulated dose because that is the whole command. A TRAP WORTH RECORDING, and the reason this row is precise: a `@name[param]` string built in C must be LOWER-CASED and fails SILENTLY if it is not. Two wrong explanations preceded this one, so what follows is what was MEASURED. (1) THE PARSER REJECTS UPPERCASE -- `A/B` at a single call site, same context, same instant: `raw=<@RL[resistance]> len=0` versus `lower=<@rl[resistance]> len=1; ft_getpnames_from_string("@RL[...]", TRUE)` returns NULL. (2) IT FAILS INSIDE `PPparse`, BEFORE `checkvalid` -- instrumenting both branches showed "PPparse FAILED", and that path is `return NULL` with NO diagnostic, so `if_getparam` (which would have printed "no such device or model name") is never reached; `checkvalid` does warn on a zero-length vector and was confirmed not to fire. (3) TYPING THE SAME TEXT WORKS because the command never sees uppercase -- instrumenting `com_print`'s wordlist showed it already holding `@rl[resistance]` when the user typed `@RL[resistance]`, so the fold happens BEFORE command dispatch. That is why `print/alter/show/sweep` all accept uppercase (none passes it on) while a string a command builds internally from its own argument words bypasses the fold. (4) NOT ESTABLISHED: where that pre-dispatch fold lives -- not `com_print`, not `ft_getpnames_quotes`, not a command-table flag, and not a blanket fold of `.control` lines, since unquoted echo UNQUOTED MixedCaseWORD preserves case. Left stated rather than guessed at. The first attempt blamed placement instead (reading before `loadpull`'s priming transient), a red

herring settled only by instrumenting the actual code path. Regression 306/306. Verify (examples/lprestore): 7 checks, a proven trigger -- pre-fix 3/7. The four failures are the defect (R and L left moved, the 41% .ac shift, and source-pull mode leaving its own tuner at 70.69 ohm); the three passing on BOTH binaries are the accept half -- loadpull still reports a peak Pout (3.9576 dBm, identical across the fix), running it twice still reports the same optimum, and CL was already unchanged because the last grid point happened not to move it. examples/loadpull_examples passes 4/4 unchanged.

E-383 *src/frontend/typesdef.c, examples/plotorder_examples/*, examples/sweepname_examples/verify_sweepname.py, examples/plotname_examples/verify_plotname.py*

Four plot-type entries in **plotabs[]** that could never be reached, so four analyses' plots were named after a DIFFERENT analysis. `ft_plotabbrev()` returns the FIRST entry whose pattern is a SUBSTRING of the plot name, and E-367/E-368 both wrote that ordering rule into the source -- then E-367 broke it with its own entry: `envelope` went in at the BOTTOM of the table, below { "op", "op" }, and "envel-OP-e" contains "op", so the entry was dead and every envelope plot came out `op1`. `setplot envelope1` found nothing, and the plot took a number out of the operating-point sequence. Auditing every plot name in the tree that reaches `plot_alloc()` -- the literals, the names `qp_emit_plot()` passes, and the descriptive `anaName` strings arriving via `plot_alloc(run->type)` -- found three more of the same shape: `qpac->pac1` (collides with `.pac`), `qpxf->pfx1` (collides with `.pfx`), and `spectrum->sp1` (collides with `.sp`; BOTH `spec` and `fft` set `pl_name` to that literal, so both were affected). A COLLIDING name is worse than an unnamed one because it looks right: `.pac` plus `qpac` in one session gave `pac1/pac2` with nothing to tell them apart. `qpnoise` was already correct, since E-368 had placed it ahead of `pnoise/noise`. The spectrum fix is a deliberate change to two stock ngspice commands' plot names (`spec` and `fft`) -- { "spect", "spect" } had been sitting in the table unreachable for exactly that purpose, and E-367's comment had recorded the result as an unfixable quirk. WHERE it goes is the constraint: the noise analysis names its plot "Noise SPECTral Density Curves ...", so one row too high would have renamed every ordinary noise plot and broken `setplot noise1` across this repo; it sits below { "noise", "noise" }. E-345's example asserts the `fft` name directly and its expectation was updated WITH this change rather than around it -- the full regression is what surfaced that, which is why it is run before the fix is called contained. `plot_alloc("digi")` in `findvec_all()` looks like the same defect and is deliberately NOT given an entry: the next line overwrites `pl_typename` with a hardcoded "digi", so an entry would be dead on arrival, and the path proved unreachable from every invocation tried (`print alle`, `plot alle`, event nodes confirmed via `eprint`) -- adding an untestable entry is the defect being removed. Since two of the four defects were introduced BY the enhancements fixing this table, the ordering invariant is now asserted against the table itself rather than left to deck coverage. `loadpull` leaving 17 `tran` plots behind was checked and is NOT a defect -- sweep keeps 334 `op` plots for the same documented reason. Verified by 25 checks (25/25 fixed, 12/25 pre-fix; the 11 accept checks pin `op/tran/ac/noise/sp/.pac/.pfx/qpnoise` and E-367's `sweep/eye/hb` on BOTH binaries).

E-384 *src/spicelib/devices/vsrc/vsrcpar.c, src/spicelib/devices/vsrc/vsrcset.c, src/spicelib/devices/vsrc/vsrctemp.c, src/spicelib/analysis/span.c, src/osdi/osdiregistry.c, src/include/ngspice/devdefs.h, src/spicelib/devices/dio/dio.c, src/spicelib/devices/mes/mes.c, examples/sensstate_examples/**

A transient after **sens** came back with EVERY NODE EXACTLY ZERO -- including the node the source drives -- plus five more defects around **sens** and the RF ports. THE BIG ONE IS NOT IN `sens`: `VSRCparam`'s `pwr` and `freq` cases set here-->`VSRCfunctionType = PORT UNCONDITIONALLY`, and EVERY voltage source carries those parameters, not just ports. `sens` perturbs every settable real parameter of every device, so it wrote `pwr/freq` on ordinary sources, flipped them to `PORT` (power 0), restored the `VALUE` afterwards and never the function type -- the deck's `SIN` was gone for the rest of the session. `alter @v1[pwr]=0` did the same damage in one line with no `sens` involved. Blast radius measured before the fix: `tran` and `envelope` ZEROED, `hb/pss` failing with ‘

E-385 *src/spicelib/devices/vccs/vccsset.c, src/spicelib/devices/cccs/cccsset.c, src/spicelib/devices/vsrc/vsrcpar.c, src/spicelib/devices/vsrc/vsrcrtemp.c, src/frontend/com_sweep.c, examples/staterestore_examples/**

Three more commands that left the user's circuit changed behind them -- found by a STATE-RESTORATION AUDIT that goes after the CLASS rather than instances (E-380 .dc coefficients, E-381 stb probes, E-382 loadpull tuner, E-384 sens/PORT were the first four). Oracle: for every (instance, parameter) pair, the value after a command must equal the value before. (A) sens .. . ac KILLED VCCS AND CCCS SOURCES -- @g1[*gain*] 1e-3 -> 0 and a following .ac returned 0.0 where the answer is 1.0. Not a bug in sens: VCCSparam folds the multiplier into the coefficient when *gain* is written (if (VCCSmGiven) VCCScoeff *= VCCSmValue) and VCCSmValue was never defaulted to 1 -- res does exactly that in ressetup.c. sens perturbs every settable real parameter, so it wrote *m* (setting *mGiven*), read it back as 0, wrote that 0 back as the restore, and the next *gain* write multiplied by zero. Explains the scope exactly: VCCS/CCCS have a settable *m*, VCVS/CCVS do not, and only the first two were hit; and why one frequency point was harmless while three were fatal. (B) sweep NEVER PUT AN alter/altermod KNOB BACK -- @r1[*resistance*] 2000 -> 2199, following op 3.8% wrong. E-350 covered only the .param path. Fix follows E-350's all-or-nothing rule; needs its own reader because sw_eval_expr returns 0.0 on failure, indistinguishable from a knob that is legitimately zero. Later follow-up: that reader evaluated the knob name as a VECTOR EXPRESSION, which only the @dev[*param*] spelling satisfies -- so a BARE device knob (sweep V1 0 5 1, sweep R1 1k 5k 1k) could be SET but never read back, and the sweep left the device at its LAST swept value: V1 stuck at 5 V and every later analysis in the session using it, R1 at 5k, I1 at 1 mA, with only a could not be read warning that names neither the knob nor the cost. The APPLY side never had the trouble -- alter <dev>=<val> resolves a NULL parameter to the device's IF_PRINCIPAL one -- so the reader now resolves that same principal keyword from the device tables and reads @<dev>[<keyword>] through the proven expression path, capturing by construction the value the sweep is about to overwrite. The E-437 shape once more: the guard existed and one spelling never reached it. A first attempt using if_getparam(.. ., NULL, ...) was WRONG and made it worse -- it returned V1's branch current (-5.01e-04) and R1 as 1e-12, and the restore then wrote those back; the before/after differential caught it. sweep-restore 13/13 (up from 5), the six new checks failing on the unfixed binary and the two controls passing on both. (C) AND E-350's OWN PATH RESTORED ONLY HALF OF WHAT IT MOVED -- it put the numparam table back, but on the E-320 FAST path the point loop pushes values straight into the live circuit, so devices kept the last point's values: .param rl=3k + sweep rl lin 3 1k 5k left @r2[*resistance*]=5000 and the next op 19% wrong. Nominals are now pushed back the same way each point was, before sw_fp_free() drops the binds. E-384's FIX COVERED ONLY HALF ITS CLASS: if (!VSRCfuncTGiven) left the DC-ONLY source unprotected (no waveform -> guard passes), so sens still turned it into a PORT and a following tran read 0.0 where the answer was 1.0. The decision moved to VSRCtemp, inside the block that already knows whether the instance is really a port; VSRCparam's pwr/freq cases no longer touch the waveform selector at all. E-384's own example passes either way because its deck carries a SIN -- the audit is what caught it. HARNESS TRAPS, both load-bearing: the BEFORE snapshot must be BOUNDED by the AFTER marker or it swallows the AFTER block and (building a dict) later values overwrite earlier ones -- before == after, EVERY command clean; caught only by a POSITIVE CONTROL against a known-defective binary. And the control must be a UNION of benign analyses subtracted PER PAIR, not op vs op (same analysis reproduces the same operating point) and never by NAME (*p* is settable on some devices, an output on others). LIMITATION stated: 114/292 pairs are operating-point dependent and subtracted, so a corruption of one of those is masked. ALSO OBSERVED, NOT FIXED: sens_cplx reads uninitialised memory (denormal garbage varying run to run). The campaign harness ships as examples/staterestore_examples/audit/audit_state.py (31 commands x 292 pairs), labelling each pair [INPUT]/[output] from the device tables. Verified by 20 checks (20/20 fixed, 9/20 pre-fix; the 8 accept checks pin an explicit *m*=2 still doubling the gain, sens's own analytic numbers, VCVS/CCVS, plain tran/ac, E-350's .param restore, and bit-identical S-

parameters, on BOTH binaries; check 14 is an audit canary -- a deliberate alter the embedded audit must detect).

E-386 *src/spicelib/devices/{res,cap,ind,dio,bjt,vccs,vcvs,cccs,ccvs}/ask.c, src/frontend/spiceif.c, examples/senscplx_examples/*

The sensitivity queries returned the PREVIOUS query's value. `print @r1[resistance] -> 1000`, then `print @r1[sens_cplx] -> 1000`; `print @r1[i] -> 5e-4`, then `print @c1[sens_cplx] -> 4.99999616e-04`. Every `*_QUEST_SENS*` case had the shape `if (ckt->CKTsenInfo) { ... write... } return(OK);` -- and `CKTsenInfo` is set only by the SENS2 analysis, which is not compiled in, so on any ordinary run the handler wrote NOTHING and still returned OK. The caller then read whatever was already in its `IFvalue`. FIRST SEEN AS UNINITIALISED MEMORY (denormal garbage 2.12736e-314 varying between runs) -- same defect in disguise: the frontend's `IFvalue` is a static reused by every query (`spiceif.c`, `doask`, which sets only `pv.iValue`), and `sens_cplx` is `IF_COMPLEX`, so a preceding REAL query leaves the imaginary half stale -- a double read of bytes that were never a double is exactly what a denormal like 2.1e-314 is. Interleaving queries makes it plain: the answer is the last thing you asked for. FIXED IN THE HANDLERS, not the caller, because two other callers pass an uninitialised STACK `IFvalue`: `dctrcurv.c` (which saves the result as a parameter's NOMINAL to restore after a .dc sweep) and `cktsens.c`'s `sens_getp` (which feeds `sgen` the value it will write back). Both latent today -- they only ask for parameters whose handlers do write -- but a contract of "returns OK, may not have written" cannot be made safe downstream. All 60 cases initialise their output before the `if`: 10 device types (`res`, `cap`, `ind`, `mutual`, `dio`, `bjt`, `vccs`, `vcvs`, `cccs`, `ccvs`) x six queries (`sens_dc/sens_real/sens_imag/sens_mag/sens_ph/sens_cplx`). Zero is what the surrounding code already chose -- the MAG and PH cases explicitly set 0 when the response magnitude is zero. `doask` hardened as well (`memset` before use; `pv.iValue` set AFTER, since it is an INPUT). SCOPE MEASURED by querying every parameter of seven device types immediately after a sentinel: 44 returned the sentinel pre-fix, 2 after -- and those two are `@r1[r]/@r1[ac]`, genuine aliases of resistance that really are 1000. Verified by 8 checks (8/8 fixed, 6/8 pre-fix). One check passes on BOTH binaries deliberately: querying the sensitivity parameters FIRST, with no preceding query, already returned 0 because the static starts zeroed -- that is what made the defect easy to miss, since a probe asking only the suspect parameter sees nothing wrong. The 4 accept checks pin ordinary parameters (keyed by INSTANCE -- `gain` exists on both a VCCS and a VCVS and a name-keyed check silently kept only the last), operating-point readbacks asserted SELF-CONSISTENTLY as $p = i^2 R$ rather than against a literal from a simpler deck, `sens`'s own analytic numbers, and show `all : all`.

E-394 *src/frontend/inpcom.c, src/frontend/inp.c, src/osdi/osdiparam.c, src/osdi/osdisetup.c, src/osdi/osdiload.c, src/osdi/osdiinit.c, src/spicelib/devices/nport/{nport.c,nportparam.c,nportdefs.h}, examples/osdiplumb_examples/**

Six ngspice+OSDI plumbing defects; FIVE are the same shape -- it works for a BUILT-IN device and silently does not for an OSDI one. (1) THE SUBCIRCUIT MULTIPLIER NEVER REACHED AN OSDI DEVICE: `X1 a 0 sub m=3` scales a built-in R/diode inside the subckt exactly while an OSDI device contributed 1x -- in DC, AC, transient, thermal noise, flicker noise, charge AND S-parameters -- and `\$mfactor` read inside the model was always the device's OWN `m`, never multiplied by any enclosing `X` at any nesting depth. The cause is ONE CHARACTER: `inp_fix_subckt_multiplier` appends `m={m}` to each device line inside a multiplied subckt but skips lines whose first letter names a device with no multiplier, and 'n' was in that skip list -- **N** is the OSDI dispatcher. PDKs wrap compact models in multiplied subckts, so this under-counted device area with NO diagnostic. **N** also hosts the native n-port, which has no multiplier and ERRORED on an unexpected `m=`, so `nport` now ACCEPTS `m` and reports a non-unity value rather than the multiplier being dropped in silence there too (applying a real multiplier would mean scaling a stateful convolution and its history terms -- no finding here concerns that device). (2) NESTED SUBCIRCUIT MULTIPLIERS DID NOT COMPOUND, for built-in devices as well -- only the OUTERMOST `m=` survived (2 around 3 gave 2x; 2x3x5 gave 2x, not 30x) because an existing

`m=` was multiplied ONLY in HSPICE compatibility mode and otherwise a second `m={m}` was appended and won. A subckt that DECLARES `m` as a parameter is a separate case and was always right (X's `m` binds it and must not also multiply) -- pinned in the accept half. (3) INSTANCE `temp=` WAS NOT CONVERTED CELSIUS->KELVIN: every built-in adds `CONSTCtoK` at parameter-set time (`dioparam.c`) and ngspice's own OSDI code acknowledges the convention where it hands `tnom` over as `CKTnomTemp - CONSTCtoK`, but the OSDI path stored the raw number and used it as the Kelvin device temperature -- `temp=75` reached the model as `$temperature=75` and `$vt=6.5 mV` instead of 30 mV, `-2.5e+16 A` on a VA diode where the correct answer is `-4.85e-07 A`; `temp=0` made `$vt` EXACTLY ZERO so `limexp(V/\$vt)` divided by zero and the op failed outright; `temp=-40` gave a NEGATIVE absolute temperature and a negative thermal voltage; and nothing warned. `temp` also STACKED with `dtemp` (`75+10` meaning 85) instead of overriding -- it now overrides and prints `restemp.c`'s own "Instance temperature specified, `dtemp` ignored", with the same sensitivity-analysis silence. `.temp/.option temp/dtemp/temp` sweeps were all correct throughout; only this path was raw. (4) `.option scale` NEVER REACHED AN OSDI MODEL: each built-in applies it in its own parameter setter (`cp_getvar("scale")`) and nothing scales an OSDI instance parameter -- nor can it, the OSDI ABI carries no units -- so the Verilog-A channel `\$simparam("scale")` is the answer, and real models ask for it (the EKV model in this project's own `VA_TEST` corpus has `SIMPARSCALE \$simparam("scale",1.0)`) while ngspice's ten-entry `simparam` table omitted it, leaving the model on 1.0 while a built-in MOSFET in the same netlist scaled `l=2 -> 2e-6`. `shrink/imax/rthresh` deliberately NOT added: ngspice has no such option, so the honest outcome is the model's own default. (5) `.options savecurrents` PRODUCED NOTHING FOR OSDI DEVICES -- `@r1[i]` appeared for a built-in resistor and the compact model beside it produced no current vector at all, `@n1[i]` did not exist, so the only way to see a terminal current was to EDIT THE MODEL. Every OSDI instance now answers to `i_<terminal>` and a two-terminal one also to the bare `i` that R/C/L use, taken from the device's own stamp into that node's KCL row (resistive residual + the integrated charge derivative in a transient) so it is EXACT, not a finite difference -- verified equal-and-opposite, KCL to 3e-16 on a 3-terminal device, and `CdV/dt` in a transient. SCOPE BOUNDARY STATED: `savecurrents` is a TEXTUAL pre-pass and a model's terminal NAMES are not knowable there, so >2-terminal devices are read per terminal by name (`.save @n1[i_d]`) rather than auto-saved. (6) `\$simparam\$str("analysis_name")` CONTRADICTED `analysis()`: E-53 taught the `ANALYSIS_` flags to consult the running job so an AC job's op phase reports `ac`, but the string channel stayed on the CKTmode-only derivation and `OSDIfinalStep` carried a THIRD derivation testing `MODEAC` without `MODEINITSMSIG` -- so inside ONE evaluation a plain op reported `name=ac` while `analysis("ac")` was false, and an AC job's op phase reported `name=dc` while `analysis("ac")` was true. The in-source comment claiming "the same convention as `analysis()`" had become FALSE. THE MISTAKE WORTH RECORDING: making the other two match `OSDIload` is the obvious repair and is WRONG -- `examples/finalstep_examples` pins that `ac`-qualified events stay SILENT in a plain op, and propagating `OSDIload`'s derivation made `final_ac` fire there. `OSDIload` was the odd one out: it set `CALC_REACT_JACOBIAN` and `ANALYSIS_AC` together from 'MODEAC

E-395 `src/spicelib/parser/inpdp.c, src/spicelib/parser/inpgmod.c`

Setting an OSDI parameter and its **aliasparam** to different values was accepted in silence and one spelling lost. An OSDI parameter and each of its alias names are registered by `osdiinit.c` as SEPARATE `IFparm` entries that all carry the SAME `.id`, so `N1 a 0 md w=1 width=4` writes ONE slot twice; setting the same name twice does likewise. Both are now reported, on the instance line (`INPdevParse`) and on the model card (`create_model`), by tracking the ids written while parsing that one line or card. SCOPED to OSDI devices -- `aliasparam` is a Verilog-A construct, so no built-in device's parsing changes -- and deliberately placed in the DECK parser, because `alter` legitimately re-sets a parameter after the deck is read and must stay silent; an instance line overriding a model-card default is likewise a different thing and is not reported. THE MODEL-CARD MESSAGE DELIBERATELY DOES NOT NAME A WINNER, because the two channels on a card disagree: a MODEL parameter is written straight through so the LAST one on the card wins,

while an INSTANCE-parameter default is pushed onto INPmodfast->defaults with wl_cons and therefore replayed in REVERSE, so the FIRST one wins -- stating a rule would be wrong half the time; the instance-line message, which has no such split, does say the last value is used. The rest of E-395 is compiler-side and is recorded in the openvaf report. Verify (examples/langguard_examples): 11 checks on this item alone -- 5 doubly-set forms reported, 6 legitimate forms silent AND giving the same number.

E-396 *src/osdi/osdiregistry.c, src/osdi/osdiinit.c*

\\$limit with an unresolvable limiting function SEGFAULTED the simulator with ZERO output. `osdi_create_registry_entry` resolves the model's OSDI_LIM_TABLE against a fixed set -- `pnjlim` (2 extra args), `fetlim` (1), `limitlog` (1), `limvds` (0). On an unknown NAME or a mismatched ARITY it printf'd "ignoring..." and left `func_ptr` NULL; the compiled model then CALLED that pointer, so the next evaluation jumped to address zero. The warning never reached the user either: it went to STDOUT and the crash destroyed the buffer before it was flushed -- which is exactly why the symptom was a silent death rather than a warning followed by a crash. **fetlim** takes 1 extra argument here and 2 in the LRM (**vto**, **vgst**), so a model following the standard was one of the ways to crash. There is no safe continuation -- the **\\$limit** call site is already compiled to an indirect call -- so an unresolvable entry is now a HARD LOAD FAILURE (INVALID_OBJECT) with a message naming the function, the arity given, the arity expected and the supported set, written to STDERR where a later abort cannot swallow it. A collision warning fired on almost the whole industry corpus. E-335's check compared KEYWORDS only, but a model that declares one of the loader's own names has two IFparm entries under that keyword BY DESIGN: `osdi_create_registry_entry` routes the built-in to the model's parameter (`dtemp -> dt = param_id, m -> has_m, temp` suppresses the loader entry), so both address the SAME id and nothing is unreachable. `dtemp` is a conventional CMC instance parameter, so 69 warnings fired across the 124-file corpus (PSP, MEXTRAM, VBIC, BSIM, HiSIM, L-UTSOI, EKV, MVSG, ASM-HEMT, JUNCAP200, r2/r3_cmc), none of them a real problem, and each worded "declared more than once differing only in case" when the spellings are IDENTICAL and there is no second declaration to find. `osdi_warn_case_collisions` now skips a pair whose `.ids MATCH`; the corpus emits 5, all genuine different-id clashes. A finding was WITHDRAWN here: this began as "a model parameter named `m` silently defeats E-394's subcircuit multiplier" because such a model gave `1x` under `X1 ... m=3` -- but the probe declared `m` and never USED it. A real CMC model declares `m` and scales by it, which is what `has_m` exists for, so the appended `m={m}` lands there and the multiplier is DELIVERED through it (verified `1x/3x/5x`). The warning added for it was removed. The compiler-side half of this release is recorded in the openvaf report. Verify (examples/limguard_examples): the crash check asserts a POSITIVE return code and a message naming the supported set, because the defect's signature was `rc = -11` with an EMPTY output stream -- a naive "did it fail?" test scores that as a pass. E-394's multiplier is re-pinned end to end (`1x/3x/5x`) THROUGH a model-declared `m`. Load sweep over the corpus: 107 loaded, 0 failures, 0 crashes, warnings 69 -> 5.

E-397 *src/osdi/osdiinit.c, src/osdi/osdiregistry.c, src/osdi/osdiparam.c, src/osdi/osdisetup.c, examples/instknobs_examples/**

temp, **dtemp** and **dt** could be WRITTEN and never READ on an OSDI device. `osdiinit.c` registered the loader's own three entries IF_REAL + IF_SET with no IF_ASK, while a model's own parameters get both -- nothing in the source defends the asymmetry. So `print @n1[temp]` answered "no such parameter" where every built-in reports one (`@r1[temp] -> 27, @r1[dtemp] -> 0`), show `n1` listed none of the three, and -- worst -- a **sweep** over any of them ended with a spurious error AFTER completing correctly: the swept data was right and printed, then ngspice's end-of-run vector bookkeeping failed to resolve `@n1[temp]`. A diagnostic that fires on SUCCESS is worse than none; it cost this project a wrong answer (see below). It also made the three look as though they existed only when the Verilog-A declared them -- declaring one turns it into a MODEL parameter, which is askable -- when in fact ngspice supplies all three to every OSDI device by default.

WHY A FLAG WAS NOT ENOUGH: the ids COLLIDED. `osdiregistry.c` gave the synthesized knobs `num_params + num_opvars + 1`, which is exactly the base `osdiinit.c` gives E-394's synthesized TERMINAL CURRENTS -- `dt`'s id WAS terminal 0's id and `temp`'s was terminal 1's, survivable only because the two groups were disjoint by DIRECTION (temperature knobs set-only, terminal currents ask-only). Adding IF_ASK on top would have made `@n1[temp]` return a terminal current: a wrong number where there had been an honest error. FIX: the synthesized ids move ABOVE the terminal range (`+ num_terminals`), making the three id spaces disjoint; OSDI-param dispatches on `param == entry->dt` explicitly so the write side follows unchanged; OSDIask serves the two new ids, guarded on `>= cur_base` so a model that DECLARES `dtemp/temperature` (whose routed id is BELOW that base) still falls through to the ordinary readable-parameter path. CONVENTION MEASURED, NOT ASSUMED: a built-in resistor reports `temp` as the BASE temperature in DEGREES CELSIUS, following `.temp/.option temp` when the instance does not set it and NEVER including `dtemp`; `dtemp` is the offset, 0 when unset. An OSDI device now answers identically in all seven cases tested. ONE BEHAVIOUR CHANGE, not merely reporting: `temp=` overrides `dtemp=` (E-394 established that and prints the same message `restemp.c` does) -- but `restemp.c` also FORCES `RESdtemp = 0`, which was invisible while `dtemp` could not be read. Leaving the written value would now report an offset that has no effect, so it is cleared at both `osdisetup.c` sites; the device temperature is unchanged (`temp=75 dtemp=10` is 348.15 K before and after). THE WRONG ANSWER THIS COST, RECORDED: asked whether sweep works with these knobs, this project answered no, on the strength of that trailing error -- the sweep had completed correctly and its five points were printed immediately above the message. The lesson is not "read more carefully" but that a diagnostic emitted after a successful operation will be read as a failure of it, by tools and people alike. Verify (examples/instknobs_examples): 127/127 fixed, 92/127 shipped -- 35 checks pin this. Every read-back case is checked twice, against the expected value AND against a built-in resistor in the same deck. The suite re-pins what the id move could have broken: E-394's terminal currents still resolve on a 3-terminal device and satisfy KCL to 1e-12 with `@n1[temp]` readable alongside; the 2-terminal bare `i` alias still resolves; a model-declared `dtemp` is still read from the model's own parameter; the physics is unmoved; and sweep over each of the four knobs yields the right point count with NO diagnostic. Corpus load sweep 107 loaded / 0 failures / 5 warnings -- identical to what E-396 left. Full regression 321/321.

E-399 *ngspice-46/src/spicelib/parser/inpgval.c, ngspice-46/src/spicelib/analysis/cktop.c, ngspice-46/src/osdi/osdiload.c*

Three ngspice-side defects from round 13: a value that depended on which door it came through, a diagnostic that stated the opposite of the truth, and three answerable \$simparam names. (1) AN INTEGER PARAMETER ROUNDED DIFFERENTLY FROM THE NETLIST THAN IN-MODEL: `INPgetValue` converted with `floor(0.5 + x)`, rounding .5 toward +infinity, while Verilog-A's own conversion inside a model rounds half AWAY FROM ZERO per the LRM. `.model mm dut(n=-2.5)` therefore gave -2 where `ii = -2.5` gave -3, and EVERY negative half-boundary disagreed (`-0.5 -> 0` vs `-1, -1.5, -2.5, -3.5`) while positives always agreed. Now round(), exactly the LRM rule. This is a GENERIC parser path shared by every device; the two forms differ only at exact negative half-integers, where no built-in uses a meaningful value and the old answer was the surprising one, and the full 322/322 regression confirms nothing depended on it. (2) A PLAIN CONVERGENCE FAILURE WAS ANNOUNCED AS A MODEL ABORT: CKTop's ordinary 'every convergence aid failed' exit FELL THROUGH into the `fatal:` label, so a model containing no \$fatal at all was reported with 'a Verilog-A device raised \$fatal ... This is not a convergence failure -- see the OSDI(fatal) message above', with no OSDI(fatal) line anywhere above to read. Both claims false, and the text denied the true cause -- a singular matrix that had just defeated `gmin` stepping, source stepping and `optran`, each of which printed its own correct diagnostic immediately above. E-378 added that block for a genuine device-raised abort and it landed in the fall-through path of the normal failure route; the path returns converged now, leaving E-378's own `goto fatal` jumps as the only way in. Verified both directions. (3) THREE MORE STANDARD \$simparam NAMES -- `iteration` (STATnumIter), `abstime` (CKTtime), `simulatorSubversion` (0) -- the ones ngspice

can answer truthfully. shrink, imax, imelt, rthresh stay out, keeping E-394's stated reasoning. What made the absence matter: an unknown name is not a soft miss, \\${simparam("iteration")} with no default aborts the ENTIRE run.

E-401 *src/include/ngspice/osdiitf.h, src/osdi/osdiregistry.c, src/osdi/osdisetup.c*

V(a,b) <+ 0 between two module TERMINALS connected nothing at all -- a silent OPEN CIRCUIT where the model wrote a short. The LRM's own page-155 parares, whose purpose is to degenerate to a wire, gave $i(v1) = 0.0$ at $r=1e-6$ against $-5.0e-4$ for a real short, while $r=1e-2$ conducted correctly: crossing the model's own threshold flipped it from wire to open. This project's diode_lim.va test model read 1000 V of self-heating rise at its DEFAULT $rth=0$, where it ties its thermal terminal to ground. TWO CAUSES, NEITHER WRONG ALONE: collapse_nodes skips any collapse pair whose endpoints are simulator-allocated (*"Terminals can never be collapsed in ngspice because they are allocated by ngspice instead of OSDI"*), and the compiler's DAE build emits no equation for a trivial potential contribution -- so the collapse was dropped and nothing replaced it. THE COMPILER-ONLY FIX IS WRONG AND THE REGRESSION PROVES IT: emitting the 0 V source and stopping there passes every targeted case and cargo test 210/0, then breaks FIVE regression examples, because a 0 V source between two nodes the NETLIST has already shorted is SINGULAR -- node collapsing is idempotent, an equation is not (HICUM with its thermal terminal grounded, the normal wiring: *"Warning: singular matrix: check node n1#xf1"*). Only the simulator knows the netlist, so the fix is two-sided: the compiler emits the source AND lists the branch in two new ADDITIVE exported symbols (OSDI_TERM_SHORT_COUNTS/_INFOS, the absdelay/last_crossing pattern -- no descriptor layout change, no ABI version bump), and ngspice drops the branch current when the terminals do not resolve to two distinct connected circuit nodes. Recorded only for branches whose current the model never reads, which is what makes dropping it safe. Terminal-to-INTERNAL collapse is untouched and still exact (BSIM4's $V(s,si) <+ 0$), as is the op-dependent switch branch (the LRM page-114 relay). VERIFIED: regression 322/322 (the five failures gone), cargo test --workspace 210/0 with six snapshots regenerated, corpus 107 compiled by both, 0 rc differences, 20 byte differences -- the FIRST release in this run to change emitted bytes, all 20 thermal models (HICUM L0/L2, HiSIM-HV/SOI, BSIM-BULK/CMG/IMG, ASM-HEMT, MVSG, MOSVAR) that ground a thermal terminal, correlating exactly with the new metadata (changed models export TERM_SHORT symbols, the 87 unchanged ones export none). Structure changes, answers do not: HICUM L2 on a normally-wired circuit gives $-1.86927e-02$ on both binaries, self-heating off and on.

E-402 *src/spicelib/parser/inp2n.c*

An OSDI instance line with FEWER nodes than the model has terminals was accepted in silence, while every built-in rejects the same mistake (diode *"could not find a valid modelname"*, resistor *"is not a valid resistor instance line, ignored!"*, MOSFET *"Simulation interrupted"*). A 4-terminal device given 3 nodes still simulates and answers differently -- $v(g)$ 1.333 fully connected against 2.000 mistyped, $rc=0$, empty diagnostic stream -- because the omitted pins DANGLE and every branch touching them goes dead. ROOT CAUSE: inp2n.c bounded the count on one side only ($numnodes > *dev \rightarrow terms$ errors; too few was never asked about), then bound the missing terminals to -1. REJECTING IT WOULD BE WRONG: that -1 is the same sentinel osdisetup.c reads as *"deliberately unconnected"*, i.e. the LRM optional-terminal feature 32 corpus models query via $\$port_connected$; and a dangling node is no substitute, since it reports $\$port_connected == 1$ where an omission reports 0. The lower bound therefore WARNS, naming each absent terminal from $dev \rightarrow termNames$ and spelling out that they are NOT grounded -- the natural assumption, and wrong by 67% on the measured node. Validated against BSIMSOI compiled with PORT_CONNECTED and instantiated with 4 of 7 terminals: the three it names (p, b, t) are exactly the optional ones its own logic is written for, and that case keeps working. Semantics unchanged; regression 322/322 with the warning firing 0 times across the suite, so every future firing is signal.

E-403 *src/spicelib/devices/res/resnoise.c, src/spicelib/devices/bjt/bjtnoise.c, src/spicelib/devices/dio/*

dionoise.c, src/spicelib/devices/vbic/vbicnoise.c, src/spicelib/devices/hicum2/hicum2noise.c, openvaf/hir_ty/src/inference.rs

Writing `temp=` equal to the ambient inflated an instance's thermal noise power by 9%. `rd x 0 1k` and `rd x 0 1k temp=27` are the same circuit at the same temperature, yet read 4.069337e-09 and 4.248250e-09. Five device noise routines computed the delta handed to `NevalSrcInstanceTemp` (which evaluates $4k(CKTtemp + param2)G$, so `param2` is a DIFFERENCE) as `inst->Xtemp - ckt->CKTtemp + (model->Xtnom - CONSTCtoK)` -- the third term adding the NOMINAL temperature in CELSIUS, 27 by default, to a difference. Instance temperatures are stored in Kelvin so the first two terms were the whole answer; the third is dropped in `res`, `bjt`, `dio`, `vbic` and `hicum2` alike. HOW IT SURFACED: a bug hunt found `sens` silently changing every later noise in the session (2.878894e-09 alone against 3.005592e-09 after `sens`, sticky, no analysis clearing it). The route to the cause matters because the symptom pointed away from it -- the shift was EXACTLY constant at 1.08996 in power across resistor values and topologies (so a global scale, not a perturbation); `onoise` and `inoise` scaled equally (so the sources changed, not the transfer function); $1.089955 = 327.15/300.15 = +27$ K, and repeating at 100 C and -50 C gave +27.00 K EVERY time (a constant, not a scaling); and an OSDI device's own `white_noise` was unaffected at ratio 1.00000 while the built-in resistor beside it shifted, which located the fault in the built-in device rather than in `ckt->CKTtemp`. `sens` restores parameter VALUES but leaves `...tempGiven` set, flipping the noise routine into that branch; correcting the arithmetic fixes both, since an instance at ambient then contributes zero either way. ALSO (compiler side): a type mismatch on `ac_stim` reported "*expected string literal, string literal or string literal*" -- `hir_ty` collected the requirement at the failing argument from every surviving CANDIDATE SIGNATURE and joined the list verbatim, and three of `AC_STIM`'s four signatures take a string there; the list is now deduplicated, leaving `\$limit`'s genuinely distinct "*string literal or function*" untouched. DELIBERATELY NOT SHIPPED: a save/restore for `CKTnumStates`, which `cktsens.c` overwrites with one device's state offset and never restores -- it was written, changed no measured result, and was reverted, because a fix must be no wider than its evidence. Regression 322/322, cargo test --workspace 210/0, corpus 107 compiled by both, 0 rc differences, 0 byte differences.

E-408 *src/frontend/inpcom.c, src/frontend/parse.c, src/frontend/vectors.c, src/frontend/device.c, src/spicelib/analysis/dctrcurv.c*

Three ways a bracketed name missed what it pointed at. (1) A LEADING ZERO split a node in two: `n[1]`, `n[01]` and `n[001]` were three distinct nodes -- `print all` listed all three -- while the vector lookup canonicalised the index first, so every probe returned the `n[1]` value (1.0 against the 2 V and 3 V actually driven). With one node it is worse: a netlist that built `m[02]` was unreachable by EVERY spelling including its own, since the lookup only ever asked for `m[2]`. E-221's own header settles it (the scalar names a bus produces use the same bracket form, so `a[0:1]` and `a[0]` denote one node), so the netlist now canonicalises the index, negatives included, as a byte-identical no-op when already canonical; a contradictory deck that used to answer 1.0 now correctly reports a singular matrix. (2) `@dev[param]` COULD NOT NAME ANY BRACKETED PARAMETER: show lists the bus terminal currents `i_a[0]..i_a[3]` (E-394) and array parameter elements `ap[0]..ap[2]` with correct values and the instance line can set them, but four separate splitters each cut the name at the INNER] -- *no such parameter i_a[0]*. -- so read failed, alter failed SILENTLY (the current stayed put) and a dc sweep was a FATAL ERROR, with no workaround: the name the simulator prints was not one it accepts back. All four had to change because each serves a different command (`PPlx` lexes the token at all, `vec_get` serves `print/let`, `com_alter` serves `alter`, `DCTfindInstParam` serves E-62's sweep). E-269's wildcard alias `@*[param]` works BECAUSE of the old first-] split, so a name beginning with] deliberately keeps it -- all four wildcard spellings measured unchanged and `wildparam` still passes. (3) A `[lo:hi]` BUS RANGE expanded on instance lines, R/C node lists, subcircuit calls and port lists but NOT on the cards that name nodes rather than connect them: `.save/.print/.plot` produced nothing and `.ic/.nodeset` warned and ignored. Those were skipped by design -- `inp_expand_buses` admits lines carrying BARE

node fields and an output card writes `v(a[0:3])` -- so a wrapper-aware form emits one copy of the whole token per index, carrying the wrapper and any `=value`; the spaced `.ic v(a[0:1]) = 0.25` absorbs its value into the token first. The four expanded `.print` columns check against the resistor dividers (0.5, 1/3, 0.2, 1/9), not merely against each other. DELIBERATELY UNCHANGED: a token naming two ranges has no unambiguous expansion and stays literal; and one unparseable probe voiding its whole `.print` card is general behaviour for any unresolvable probe, no longer reached by the bus case, and measured to stop at exactly one card. Compiler untouched. New example `busname_examples 30/30`; regression 325/325.

E-409 `src/frontend/com_sweep.c, src/frontend/spiceif.c, src/include/ngspice/ftext.h`

A wildcard **sweep** knob was never put back, and said so in a way that looked like a broken parameter name. `sweep @[wavelength_nm] ...` ran correctly and then printed *Error: no such device or model name / PError: syntax error in line segment @[wavelength_nm] near wavelength_nm]** -- the stray `]` because the expression lexer's `specials` set contains `*`, so `@*[p]` cannot lex as ONE token and the parser reports the leftover text. THE MESSAGE WAS THE HARMLESS PART: `sw_read_knob()` is how E-385 captures the nominal it restores at the end of a sweep, it parses the knob name as an EXPRESSION, and that call can never succeed for a wildcard -- so the read reported failure and E-385's deliberate ALL-OR-NOTHING rule skipped restoring EVERYTHING. Measured: `@*[wavelength]` left two model cards that started at 2.0 and 9.0 both sitting at 3.0; `@#*[scale]` and the E-269 alias `@*[scale]` the same; and with `-vs` it took a CONCRETE co-knob down with it (`@dev1[wavelength]` left at 3.0 although it restores perfectly alone), so any following analysis silently ran against the last swept point. ONE NUMBER COULD NOT FIX IT: a wildcard sets every matching model or instance to one value but the values it overwrites all differ, so undoing it needs one reading PER TARGET. New `if_saveparam_wildcard()/if_restoreparam_wildcard()` walk exactly the same targets in exactly the same order as the existing `if_setparam_wildcard{,_instance}` (same device-type loop, same model and instance chains, same `paramlookup(.., inout=set)` predicate), reading through the parameter's askable twin and the same `doask/doset` pair a plain alter/altermod uses; saving stays all-or-nothing (a set-only or vector-valued parameter refuses the whole capture) and the replay is pinned to the circuit the readings came from, so a `.param` co-knob that re-sources the deck cannot have them replayed onto a different one. `sw_read_knob()` now recognises the three wildcard spellings and returns without parsing, which removes the diagnostic. WHY IT SURVIVED THE AUDIT THAT EXISTS FOR THIS CLASS: E-385's own oracle `staterestore_examples/audit/audit_state.py` sweeps `@r1[resistance]`, a CONCRETE knob, and reports sweep clean on the DEFECTIVE binary too -- no wildcard form was ever exercised; the audit is unchanged and still reports the same four known offenders (`hb`, `pz`, `sens`, `sens_ac`) on both binaries. New example `wildrestore_examples 24/24`, and `14/24` on the pre-409 binary, so it detects the defect rather than describing it. Compiler untouched; regression 326/326.

E-410 `src/frontend/spiceif.c, src/frontend/gens.c, src/frontend/com_sweep.c, src/spicelib/analysis/dctrcurv.c, src/include/ngspice/ftext.h`

A device inside a subcircuit could not be named the way the node beside it is. `@r.x1.r1[resistance]` worked; `@x1.r1[resistance]` -- the obvious spelling, and the one that matches the node `x1.m` -- named nothing. THE TYPE LETTER IS NOT AN ACCESSOR CONVENTION: `r.x1.r1` is the device's real name, because ngspice flattens a subcircuit by rewriting its cards back into the deck and re-parsing them as ordinary element lines, and the parser takes the device type from the FIRST CHARACTER of the card (`inppas2.c: c = *(current->line); switch (c)`) -- the card `x1.r1 a m 1k` would be re-read as another SUBCIRCUIT CALL. Two details prove that is the reason rather than a plausible story: `translate_inst_name()` EXEMPTS `x` devices (`if (tolower_c(*name) != 'x')`) because a subcircuit instance already starts with the right letter, so `x2` in `x1` is plain `x1.x2`; and NODES, which have no type, keep plain hierarchical paths (`x1.m`, `x1.x2.q`) -- which is exactly the asymmetry. THE RECONSTRUCTION NEEDS NO SEARCH: the letter prepended is literally the leaf name's own first character (`bxx_`

putc(buffer, *name)) and a device name must already begin with its type letter, so x1.r1 can only mean r.x1.r1; two device types cannot share a leaf name. New if_find_instance_hier() is consulted ONLY after the exact lookups fail, so every name that resolves today resolves to exactly what it does today. Following E-408's lesson that @dev[param] is resolved in several independent places, every consumer is routed through it: finddev() (print/let/alter/altermod), finddev_special(), DCTfindInstParam() for E-62's .dc @inst[param] sweep, and sw_fp_bind() for the sweep knob -- where the old behaviour was NOT EVEN AN ERROR: the knob silently failed to bind and the three swept points came out IDENTICAL. show needed its own answer, since its grammar takes the query's first character as the device-TYPE selector (type = *word++) and uses :/# rather than . as the subcircuit delimiter, so x1.r1 parsed as type x; dgen_hier_match() is consulted ALONGSIDE that grammar as a whole-word alternative and can only add a match (a first attempt widened the strcmp inside the decomposition instead and was measured as DEAD CODE, then reverted). BACKWARD COMPATIBILITY DIFFED, NOT ASSUMED: across 17 show forms 14 are byte-identical and the only 3 that changed are the new spellings; legacy :r1/#dmod/showmod/-v unchanged; an x leaf gets no phantom prefix; bogus names and bogus parameters still reported; hierarchical NODES untouched. E-49's hiername_examples is a different subject (hierarchical names INSIDE Verilog-A) and is unaffected. New example hierdev_examples 32/32, and 15/32 on the pre-410 binary. Compiler untouched; regression 327/327.

E-411 src/frontend/inpcom.c

A descending netlist bus range binds nodes to a device's terminals in REVERSE order, and was silent. nd1 d[3:0] 0 mm against a model declaring inout [3:0] a compiles clean, simulates clean and is wired backwards -- and it is the spelling that LOOKS consistent, since both sides say [3:0]. TWO CONVENTIONS DISAGREE: the COMPILER declares a bus port's terminals in ASCENDING bit order whatever direction is written (hir_def item_tree lowering sorts the end-points and loops upward -- *declare from lsb to msb (ascending), matching natural bit order; direction of the original [msb:lsb] only affects range checks*), so [3:0] and [0:3] declarations give the SAME positional terminal list; while the NETLIST does honour direction, E-221 expanding d[3:0] to d[3] d[2] d[1] d[0]. The instance line's Nth node binds the model's Nth terminal, so a descending list reverses the mapping. MEASURED with one conductance per bit (a[k] -> (k+1) mS, so a node's current names the bit it landed on): wired d[0:3] the nodes read 1,2,3,4 mS, wired d[3:0] they read 4,3,2,1 -- and the [3:0] and [0:3] model rows are IDENTICAL, confirming the declaration contributes nothing. inp_expand_bus_token() now reports msb-first expansion and inp_expand_buses() warns on element instance lines only. SCOPE CAME FROM SCANNING THE SUITE, not taste: a .subckt descending PORT list is a deliberate interface choice E-221 documents and busnodes_examples tests; .save/.print/.ic bind nothing; and a range too wide to expand never binds (E-338's guard), so warning there would be false. Verilog-A part-selects (v[3:2], .i(v[1:0])) are compiler-side and untouched. THE OPT-OUT WAS CORRECTED AFTER TESTING: set nobusdirwarn does NOT work inside .control, because bus ranges expand while the netlist is read; it works from .spiceinit, verified both ways, and the message says which. New example busdir_examples 20/20 (13/20 on the pre-411 binary), pinning the binding itself as well as the diagnostic. Regression 327/327 with the warning firing 0 times across the suite, so every future firing is signal. Compiler untouched; no binding changed, only a diagnostic added.

E-412 src/osdi/osdiload.c

An OSDI operating-point variable read after .ac/.noise returned the SMALL-SIGNAL solution, and CHANGED WITH FREQUENCY. An opvar defined as nothing but vbias = V(a,c) read 0.4939733 after op -- the true bias, from solving the circuit's KCL cubic by hand -- but 0.4823410 / 0.4819028 / 0.0473587 after .ac at 1 k / 10 k / 1 meg, a multi-point sweep leaving the LAST point's. An operating point cannot depend on frequency, which is what separates this from a tolerance argument: identical at reltol 1e-3, 1e-6 and 1e-10. CAUSE: E-53 fires @(final_step) by issuing one dedicated evaluation per instance when an analysis completes, at CKTrhs0ld -- which

after a sweep holds the small-signal solution, not a bias. The model is genuinely RE-EVALUATED rather than mis-read, confirmed with opvars chosen to distinguish the two ($3V+1$ and V^2 matched the ac solution to nine digits while a constant opvar stayed 42). Built-ins are unaffected, having no such post-analysis evaluation -- the recurring *works for a built-in, silently not for OSDI* vein. NOT WRONG, established BEFORE fixing anything: the analyses themselves. A bias-dependent `white_noise(kf*V^2)` source integrated to `onoise_spectrum = 1.342573180366e-01`, matching the dc-bias prediction EXACTLY and not the ac-bias one ($7.21e-02$) -- so this is a READBACK defect, not a numerical one, and it is not sticky (any following op/dc/tran restores the correct value). THE EVALUATION CANNOT SIMPLY BE SKIPPED: it is the only thing that fires `@(final_step)`, and `@(final_step("ac"))` plus the noise variant are supported and tested by `finalstep_examples`. `OSDIfinalStep` therefore snapshots the instance data around the call and restores it, for `MODEAC` and `MODEACNOISE` only -- event bodies still run and their `$strobe` output stands, while what the evaluation wrote is discarded, which is correct because those results are by design never loaded into the matrix or RHS. DC, DC-sweep and transient are deliberately NOT snapshotted, their final solution being a real operating point (measured unchanged). WHY IT SURVIVED: with a purely resistive device the dc and ac solutions coincide, so the natural small test cannot see it; the example is reactive on purpose. New example `opvarac_examples` 19/19 (12/19 on the pre-412 binary); `finalstep` and `simctrl` pass unchanged; regression 329/329. Compiler untouched.

E-413 `src/frontend/breakp2.c, src/osdi/osdiparam.c, src/include/ngspice/osdiitf.h`

.options savecurrents registered an EMPTY vector for a multi-terminal OSDI device. It produced `@nd1[i] : current, real, 0 long` -- present in the listing, empty in use, with no diagnostic -- while a built-in BJT in the same deck produced four filled waveforms (`@q1[ic]/[ie]/[ib]/[is]`). CAUSE: `inp_savecurrents()` is a TEXTUAL pre-pass over the deck, running long before any `.osdi` is loaded, so it cannot know a compact model's terminal names and emits the same bare `@dev[i]` that R/C/L use -- E-394's own comment says exactly that. E-394 defines the bare `i` alias only for TWO-terminal devices, where it is unambiguous; for anything wider the save named a parameter that does not exist. THE VALUES WERE NEVER WRONG, only uncapturable: read as scalars after op the terminal currents were exact ($1.000e-3 / 4.000e-3 / 1.200e-2 / -1.700e-2$, KCL sum 0) and an explicit `.save @nd1[i_d]` always worked -- what was missing was the WAVEFORM, which is what `savecurrents` exists to provide. Since the names cannot be known during a textual pre-pass, the expansion moves to where they can: `ft_getSaves()` runs at analysis start with the circuit set up, so a bare `@dev[i]` belonging to an OSDI instance that does not define it is expanded there into one entry per terminal, through a new `OSDIterminalNames()`. The two-terminal case is deliberately untouched (`@dev[i]` is real there, byte-identical to the pre-413 binary) and built-ins never enter the path. Also explains why `meas tran ... MAX @nd1[i_a]` had quietly returned a scalar snapshot instead of a maximum over time. New example `savecur_examples` 15/15 (7/15 on the pre-413 binary); regression 330/330. Compiler untouched.

E-415 `src/osdi/osdinoise.c, src/spicelib/analysis/cktsgen.c, examples/evtnoise_examples/ (new)`

A per-device noise total that was the wrong quantity, and one knob counted twice. (1) `nVar(i,j)` indexed `noise_vals` with a stride of `num_noise_src`, but each of the `NSTATVARS` state variables needs one slot per source PLUS one for the whole-device total accumulated at index `num_noise_src` -- which is exactly what `osdiregistry.c` allocates (`NSTATVARS * (num_noise_src + 1)`). With `n` sources `nVar(OUTNOIZ, n) = noise_vals[1*n+n]` and `nVar(INNOIZ, 0) = noise_vals[2*n+0]` are THE SAME ADDRESS, so `onoise_total<dev>` came out bit-identical to `inoise_total<dev>_<first source>`: an input-referred, single-source number reported as the device's output-referred total (2.24x out in the measured case), while every built-in device's total matched its own components exactly. The spectra and the grand totals were never affected -- `onoise_spectrum` matched two real 1k resistors to $9e-08$ relative and the grand total already equalled the quadrature of the per-SOURCE entries -- so the regression pins the spectrum bit-identically against the pre-415 binary rather than against a derived value. (2) `m` is registered as

an alias of `\$mfactor` with the SAME parameter id, so anything walking the parameter table saw the multiplier twice: `sens` listed both `<inst>_m` and `<inst>__mfactor` with identical non-zero sensitivities and summing the table double-counted it. `IF_REDUNDANT` -- the marking built-in devices give their alias keywords -- was tried and REJECTED by the regression: that flag also hides the keyword from listings, and `show n1` is required to list `m` (E-397 went 126/127). Marking the other spelling would invert the flag's meaning, since `spiceif.c` walks BACKWARDS from a redundant entry to find its principal. The de-duplication therefore happens where the repeat mattered: the sensitivity generator skips a parameter whose id an earlier keyword in the same table already claimed. `sens` lists it once, `show` still lists `m`, and `m` remains settable from an instance line, a `.model` card and `alter` (all three verified identical).

E-439 *src/maths/KLU/klusmp.c, examples/klusingular_examples/ (new)*

A SUCCESSFUL **klu_refactor** IS NOT NECESSARILY A USABLE ONE. KLU FAILED an operating point that SPARSE solves -- the midpoint of two series capacitors, a node with no DC path: SPARSE gave $v(nb)=0.5$ in 289 iterations, KLU produced NaN and burned 33,911 before giving up with 'timestep too small', naming neither the node nor the cause. Instrumentation settled it in one run after four wrong hypotheses. TWO facts, both necessary: (1) `LoadGmin_CSC` skips diagonal pointers that are NULL ('Not all the elements on the diagonal are present') and a node with no DC path has NO diagonal entry -- so the one row needing regularisation is the one row that cannot receive `Gmin`; (2) `klu_refactor` REUSES the pivot ordering from the last full `klu_factor` and only refills values -- no pivoting, no singularity test -- so once `gmin` stepping ramped to $1e-12$ it filled the LU with zero/NaN pivots and returned `ret=1, status=KLU_OK`. NaN then neither converges nor trips a singularity check, so the Newton loop AND every rung of CKTop's ladder (dynamic `gmin`, static `gmin`, source stepping, pseudo-transient, `optran`) ran to its full iteration budget: 33,835 further non-finite solves with refactor reporting OK 33,191 times, and a count INVARIANT at 33,911 for any circuit size -- the signature of a fixed budget, not real work. SPARSE solves it because its refactor path DOES detect the zero pivot and forces a reorder. Fix: `klu_rcond` walks `diag(U)` and sets `rcond=0/status=KLU_SINGULAR` on the first zero or NaN pivot; calling it after a successful refactor and returning `E_SINGULAR` routes the caller into the reorder-and-factor-again path it already had -- the same recovery SPARSE performs, no new plumbing. Cost measured: +8-11% on the factor step, nothing measurable on total analysis time (0.105->0.105s, 0.418->0.416s, 1.489->1.454s at 500/2000/6000 nodes), iteration counts on healthy circuits identical (2032/2032, 2035/2035, 2037/2037). NOT fixed, deliberately: `Gmin` still cannot reach a row with no diagonal entry -- the CSC pattern is fixed at symbolic-analysis time; allocating a diagonal for every node would address the root at the cost of a denser pattern for every circuit, and has its own performance case to make. FOUR WRONG HYPOTHESES recorded in the write-up so they are not re-walked, the first of which (short-circuit CKTop on `E_SINGULAR`) would have BROKEN working decks, since `gmin` stepping legitimately rescues a floating node. Verified `klusingular` 16/16 (split closed, no NaN, rescue survives with each aid disabled, plus healthy-circuit and transient controls), op campaign 105/105 with 0 splits (was 3), regression 350/350 both solvers.

E-440 *src/spicelib/analysis/cktsens.c, src/spicelib/analysis/dcpss.c, src/spicelib/analysis/cktsopt.c, src/spicelib/parser/ptfuncs.c, src/spicelib/devices/asrc/asrload.c, src/spicelib/devices/asrc/asrdefs.h, src/frontend/options.c, src/frontend/parse.c, src/frontend/com_measure2.c, src/frontend/spiceif.c, examples/_setup.py, examples/sensrestore_examples/ (new), examples/argguard_examples/ (new)*

sens LEFT THE CIRCUIT ALTERED, AND FIVE INPUTS WENT UNCHECKED. Perturbing a parameter marks it GIVEN and nothing un-marks it, so a BJT fell into `bjttemp.c`'s `BJTBESatCurGiven` && `BJTBCsatCurGiven` branch with `ibe/ibc` both 0, lost its junction saturation currents, and every later analysis in the session solved a dead transistor -- 12.8% wrong on one stage, 101% on a differential pair, silent, sticky, cured only by `reset`. Not a tolerance artifact (identical at rel_tol $1e-3$ and $1e-12$), not a stale guess (`.nodeset` did not move it; `sens` run FIRST corrupted too), and 1 of 20 commands -- `alter`, `altermod`, `sweep`, `montecarlo`, `optimize`, `tf`, `pz`, `disto`, `noise`, `hb`, `pss`, `sp` all restored perfectly. Bisectioning the loop showed the damage came from `sens_setp`

setting a parameter to its own current value, and needed `ibe` AND `ibc` together -- which is the &&. Fix: snapshot every model struct before the loop, restore after (and on error paths), then `CKT-temp()`; generic via `DEVmodSize`, because the hazard is any given-flag branch, not the BJT. ALSO: `pss` validated none of its numeric arguments where `tran/ac/dc/hb/sp/noise` all do -- a negative `stbtime` sets `CKTfinalTime` behind the current time and `pss 1k -1m 0 1024 5 50 1u` ran past 100 s with no output (E-348 had guarded `fguess/points/harmonics`, not these three); `.meas` took an inverted window and, because `m_to == 0.0` doubles as 'no upper limit', `FROM=1m TO=0` measured `[1m, end]` and returned a confident number (the parser now records whether each bound was written; `dc` still normalises a downward window but says so); `pow(0, -1)` and `0**--1` returned a raw infinity where every other singular case is clamped to HUGE and reported (`x/0`, `sqrt(-x)`, `log(0)`, and `pwr(0, -1)` since E-256) -- an overflow like `1e300*1e300` is now reported but deliberately NOT clamped, since there is no defensible finite substitute for arbitrary user arithmetic; the temperature guard was one-sided, `.temp 1e6` silently giving a diode divider `-2.7e-15 V`; set `curplotname=x` before the first analysis called `FREE()` on a string LITERAL (`plot_cur` is the static constantplot until an analysis runs) and killed ngspice with a heap abort, while unset ignored its `isset` flag and performed the assignment too; and `PP_mkfnode()` copied an unbounded function name into `char buf[B_SIZE_SP]` with `strcpy+sprintf`, so `fourier 1k <600-char-name>(v(a))` smashed the stack. Plus E-438's own `warn_physics` reported from `dosim()`, once per analysis RUN, repeating one unchanged warning 20 times for `montecarlo 20` and 6 for a 6-point sweep; now once per circuit, keyed on the VALUE so a sweep through a bad range still reports each distinct one. And `examples/_setup.py` pointed `SPICE_LIB_DIR` at the committed `bin/` bundle while preferring the LOCAL ngspice, so code-model edits were never exercised -- and with no `SPICE_LIB_DIR` at all ngspice loads a THIRD PARTY's models from its compiled-in prefix, which is how four 'crashes' (`d_source`, `s_xfer`, `table2D`, `table3D`) were attributed to this tree when they belong to an unrelated Feb-2025 /usr/local install and do not reproduce here at all. Verified `sensrestore 16/16` (including that `sens` is still NUMERICALLY right: both signs of a divider's analytic `2.5e-4 V/ohm`) and `argguard 44/44`, both solvers; regression 352/352.

E-441 *src/frontend/inpcom.c, src/frontend/parse.c, src/frontend/vectors.c, src/frontend/device.c, src/include/ngspice/ftext.h, examples/arrayinst_examples/ (new), examples/xspicemodel_examples/*

ARRAY INSTANCES. A range on the instance NAME makes an array of devices: `R[0:3] a b r=1k` is four resistors in parallel, `N[0:3] a[0:3] b model` gives each element its own bit, `N[0:3] a[0:3] a[1:4] model` is a chain. The name selects the reading — E-221 already made a range in a NODE field mean one device with a WIDE PORT (`X1 bus[0:3] sub`), decks depend on it, and the two cannot both apply to a line; a scalar name keeps E-221's in-place expansion, a ranged name switches to one card per element with node ranges indexed IN STEP. Nothing that parses today changes meaning (a name carrying `[lo:hi]` was not a usable device name). New pass runs BEFORE `inp_expand_buses` and both read ranges through one shared parser, so they cannot drift on what a range is (an XSPICE port group `[d1 d2]` has no base and is invisible to both). REFUSED: a width mismatch is named and the deck is rejected — an earlier revision printed the error and let the line through, so ngspice built ONE resistor named `r[0:3]` and completed the run; and an A-device array, since XSPICE already uses brackets for port groups. E-221 also no longer expands the FIRST token of an element line (a name, never a node list). ADDRESSING: the lexer ended the token at the first `]` and the splits took everything before the first `[` as the device name, so `@r[2][resistance]` sought a device `r` with parameter 2; `ft_accessor_param_start()` fixes it with a narrow rule — an integer-only group immediately followed by another `[` belongs to the name — leaving `@nd1[i_a[0]]` (E-408) and `@*[p]` (E-269) untouched. The split lives in FIVE places: verifying against subcircuit hierarchy and the whole `.control` surface found two more beyond the lexer/`vectors.c/device.c` set — `dctrurv.c`, where `.dc` on an array element failed FATALLY ("not in the circuit", card and command alike), and `outitf.c`'s `parseSpecial`, where `save @r[1][i]` matched nothing and a save list that matches nothing loses the WHOLE plot, so the run ended with "no data saved ... analysis not run" instead of dropping one vector. ALSO: `inp_rem_levels()` reads `p->line->level`, so the scope tree must be freed BEFORE the deck; the

vdmos rejection site had them reversed and was a latent use-after-free — copying it crashed the new path with EXC_BAD_ACCESS at the level member. Verified arrayinst 36/36 both solvers (every structural check paired with a hand-computed electrical one; the Verilog-A array is bit-identical to the resistor array; a subckt array instantiated twice gives 8 uniquely-named devices, arrays NEST as `r.x1.x[0].r[0]`, and the `.dc/save` paths are checked analytically), regression 353/353.

[E-442](#) *src/frontend/inp.c, src/frontend/commands.c, examples/listtree_examples/ (new)*

listing tree — THE HIERARCHY, DRAWN. `listing e` gives one line per device with the structure only implicit in the dotted names; this draws the subcircuit tree, each X labelled with what it instantiates and placed where it is USED, plus a summary of instances/devices/depth. `listing t` too. Walks `ci_origdeck` (the expanded deck no longer knows which subcircuit an instance came from), which also means array instances ([E-441](#)) already show as the instances they are. The subcircuit name is NOT 'the last token before the parameters' — `numparam` rewrites `X1` in `0 sub PARAMS: rv=2k` into `x1 in 0 sub 2k`, so counting tokens picks `2k`; resolve against the collected definitions instead. A **.control** block's commands begin with a letter like element lines, so unskipped they were drawn AND counted as devices and broke last-child detection (final entry kept a tee where it needed a corner). No 'never instantiated' report — `ngspice` comments those definitions out first, so it could never fire. Verified `listtree` 23/23 both solvers (shape checks are specific: last-child corner, blank-not-bar subtree indent, 8-level alignment, hand-computed summary counts), regression 353/353.

[E-443](#) *src/frontend/inpcom.c, examples/idxlist_examples/ (new)*

BRACKETED INDEX LISTS. `a[1,3,5]`, `R[1,3,5]` and the mixed `a[0:1,7]` join the ranges of [E-221/E-441](#) — only the SOURCE of the indices is new, so every established reading carries over (node list = consecutive terminals; instance list = one card per element; node list on an array instance = taken IN STEP). Indices in written order, not sorted or deduplicated, since the order binds nodes to terminals — so a list has no direction to mistake and [E-411](#)'s reversed-binding warning still fires for `a[1:0]` but not `a[3,1]`. A lone **a[2]** is NOT a list — it is a scalar bus bit and stays a node name ([E-221](#)'s contract); that one condition is what makes the change additive, since the only new spellings are ones that previously meant nothing. Parse split into `inp_bus_indices()` (contents) + `inp_bus_index_parse()` (base), shared by FOUR consumers including `inp_bus_rewrite_wrapped` so `.save v(a[1,3,5])` expands — its old range-specific parse REMOVED rather than left beside the shared one. New diagnostic: a malformed list is NOT harmless — `R2 a[1,] 0 1k` re-tokenises the line and `ngspice` built a resistor with NO VALUE, warning only 'too small, set to 1e-12'; a group that looks like an index list and is not one now says so, on a test tight enough that `mem[addr]` and `a[2]` stay silent. Verified `idxlist` 26/26 both solvers (structural checks paired with analytic electrical ones; the CONTROLS are the point — scalar bit, non-numeric name, [E-221](#) range, [E-441](#) range instance, [E-411](#) warning scope), regression 354/354.

[E-451](#) *src/frontend/inp.c, src/frontend/spiceif.c, examples/optname_examples/ (new)*

AN OPTION MATCHED BY SUBSTRING, AND THREE CALLED UNKNOWN. Both from asking what [E-450](#) had just answered for `savecurrents`: which other options are decided by searching the line for a word? (1) `eval_opt()` found `seed=` and `cshunt=` with a bare `strstr`, so any option whose NAME MERELY ENDED in the watched text was taken as that option -- `.options myseed=7, noseed=7 and xseed=7` all set the RNG seed identically to `seed=7`, and for `cshunt` a node reading `1.0000000000e+00` reads `6.9209970972e-07`, six orders of magnitude, under `nocshunt=/mycshunt=/xcshunt=`. Each ALSO printed "unknown option" -- told not recognised, changes the answer anyway; for Monte Carlo a mistyped option name silently changes the random sequence. Whole-token matched now on [E-450](#)'s boundary rule, `seedinfo` included so `noseed=info` no longer switches it on. NOT a defect though it reads like one: the loop has no `.options` test, but the function receives the option deck already sorted (its own comment says so) -- a `.param seed=7`, a model parameter named `seed` and a comment containing `seed=` were each measured

and leave the sequence at baseline. (2) Three options took effect while being reported unknown: `scale=2` doubles `@m1[w]`, `rseries=100` moves a transient, and **autostop** truncates one from 567 rows to 2 -- the case E-447 fixed for `savecurrents/seed/numdgt`, with names it did not cover; `scale` and `rseries` are read from the deck's own option cards so they never reach the parameter table the check consults. **scaln** is flagged too and deliberately NOT registered, having no demonstrable effect -- adding a name on the strength of it appearing in the source would be the same mistake in the other direction. Verified `optname 24/24` both solvers and `14/24` on the previous binary, the ten failures there being exactly the defect-specific checks while every control passes on both; regression 364/364.

E-450 *src/frontend/inp.c, examples/savecuroff_examples/ (new)*

savecurrents COULD BE REQUESTED BUT NEVER DECLINED. Whether `.options savecurrents` was in force was decided by `strstr(options->line, "savecurrents")` -- a bare substring search -- so EVERY option line merely CONTAINING the word switched it on, whatever the line actually said: `savecurrents=0 ON`, `savecurrents=false ON`, **nosavecurrents** ON and even `mysavecurrentsxyz ON`. The no spelling is the one that matters, because `noacct`, `noinit`, `nomod` and `nopage` are all real ngspice options, so a no prefix is exactly what a user reaches for -- and it did the OPPOSITE. Once on there was no way back. Silent throughout, because a deck that quietly saves every terminal current still simulates correctly; it just carries vectors nobody asked for -- on a large deck the difference between a small rawfile and a very large one, and on an OSDI device one vector per TERMINAL (`i_<term>`, E-394), which is how it surfaced. Reachable ONLY from an option CARD (deck, `.include`, `.lib`) and never from `spinit/.spiceinit/set/.control`, since `inp_savecurrents()` is a netlist pre-pass -- the same "too late" ordering E-411 recorded and E-449 had to work around -- which narrows the search but makes the wrong answer stickier. Now read as TOKENS: exactly `savecurrents` or a declared `savecurrents_<variant>`, a no prefix or a `0/false/no/off` value turns it off, later card wins. WHICH card is retained is deliberately unchanged, because the caller re-searches that same line for `_bsim3/_bsim4/_mos1` to choose the MOS current set -- those four sets stay BYTE-IDENTICAL (4/6/3/11 params). Before/after over 15 option spellings: 8/15 -> 15/15, every changed case moving the intended way and no control moving. Verified `savecuroff 20/20` both solvers; regression 363/363.

E-449 *src/frontend/subckt.c, src/frontend/subckt.h, src/frontend/inp.c, examples/subbus_examples/ (new)*

AUTOBUS ACROSS A SUBCIRCUIT BOUNDARY. E-444 expands a bus port in `INP2N`, where the model is known; inside a `.subckt` that is too late. A definition declaring `a[0:4]` has the formals `a[0]..a[4]`, so a device line writing the bare `a` matched NONE of them, became the local node `x1.a`, and `INP2N` expanded THAT into five FLOATING `x1.a[0]..x1.a[4]` with the device wired to those rather than the ports -- all five bits reading one identical voltage, the device contributing nothing. Nothing was reported, because every terminal DID receive a node, so E-402's under-connected warning had nothing to say: turning the option ON removed the diagnostic the same deck gets with it OFF -- the shape E-445 recorded when autobus indexed ground. Fixed at the SUBSTITUTION: a token that is not a formal but for which formals `token[i]` exist expands to those formals' actuals, needing only the `.subckt` line. The option is not reachable from there the obvious way -- `cp_getvar` is false during flattening (`inp_subcktexpand` runs before `inp_dodeck` publishes options) AND the `.option` cards have already been split out of the deck, so a deck scan finds nothing; `inp.c` reads them from the option lists, as it already does for `scale` just above. Emitted by ascending BIT INDEX, because a compiled model orders bus terminals that way whatever direction the Verilog-A declared (`inout [4:0] a` still yields `a[0]..a[4]`, measured) and the line is positional -- so a DESCENDING `.subckt` declaration binds in reverse, E-411's rule one level up. OSDI lines only, so a bare `a` on an R line stays a local node. The other form was already correct: `.subckt mysub a b` with SCALAR formals passes the bus base to the caller's scope and expands there -- verified for one and two instances, unchanged. Out of scope: one bit where the bus is expected (`N1 a[0] b`) still floats silently and is BYTE-IDENTICAL before and

after, because after flattening nothing distinguishes it from a legitimate top-level line; `Xi a b` on a subcircuit instance fails loudly with "too few nodes". Verified subbus 16/16 both solvers, every check a differential against the written-out form on a ladder where all five bits differ; regression 361/361.

[E-448](#) *src/frontend/evaluate.c, src/frontend/evaluate.h, src/frontend/parse.c, src/frontend/vectors.c, examples/constname_examples/ (new)*

A NODE NAMED AFTER A CONSTANT. ngspice's permanent `const` plot holds twelve vectors (`c`, `e`, `i`, `pi`, `kelvin`, `boltz`, `echarge`, `planck`, `TRUE`, `FALSE`, `yes`, `no`) and `vec_get()` falls back to it — which is how a bare `pi` resolves, and stays. (1) `v(X)` answered with the speed of light: a node context resolved the name before considering the context, so `v(c)` on a renamed or mistyped node gave `2.9979245800e+08` where `v(nosuch)` correctly failed — and the other eleven are more dangerous than the speed of light because they look like results, `v(no)` a flat 0.0 and `v(i)/v(yes)/v(TRUE)` a flat 1.0. It defeated an existing guard: [E-431](#) refuses an `-output` that never resolves, and the constant walked through it BECAUSE the name did resolve, drawing a flat curve where `v(nosuch)` was refused. Only `v` has a null handler in the function table, which is why `v(c,0)`, `vm/vp/vr/vi/vdb(c)`, `i(c)` and `mag(v(c))` were already refusing. (2) A bus bit `c[0]` was UNREACHABLE — exactly what `.option autobus` ([E-444](#)) builds for a bus called `c`: `display` listed it and `print all` printed `5.0000000000e-01` while `print c[0]` said "indexing a scalar (c)" and `print v(c[0])` said "bad v() syntax"; [E-444](#)'s own five-bit ladder with the bus renamed `a->c` printed NOTHING. [E-224](#) preferred the literal name only when the base was UNRESOLVED, and a vector called `c` always exists. (3) The silent one needs no constant: the same gate returned the wrong vector's element whenever the base was LONGER, so a circuit with both a node `q` and a bus bit `q[0]` answered `print q[0]` with node `q`'s value at `t=0` as a SCALAR instead of the bit's waveform — and errored in `op` instead, so the failure mode flipped with the analysis type. The gate is now the literal vector's OWN existence, shared as `e448_literal_index()` rather than re-derived in the four places that had it ([E-408](#) recorded what drift costs); the three sites that render the index as text round through [E-274](#)'s `idx_floor()` rather than a bare `(int)` cast -- UB for `1e308/inf/NaN`, and the validating and the resolving site must round identically -- with 16 index expressions bit-identical before and after). A BARE name stays ambiguous and still resolves to the constant, now warning only when the name also exists elsewhere. Deliberate change of meaning: with both `x` and a literal `x[0]` present, `x[0]` is the literal one — the only working expression whose meaning changes, replacing the silent-wrong-answer reading; ordinary indexing and [E-433](#)'s quoting are untouched. One change was made and WITHDRAWN: a `checkvalid()` hook to report `v(c)` once per command rested on a probe artifact — the two paths use different wordings and the filter matched one; counted properly both emit 23 lines on a 21-point sweep, which is pre-existing sweep behaviour. Baseline captured on the PRE-FIX binary so the controls are measured: six defects reproduced, thirteen controls already held. Verified `constname` 57/57 both solvers, the decisive check being [E-444](#)'s ladder run with the bus named `a` and named `c` reading bit-identical; regression 360/360.

[E-447](#) *src/spicelib/analysis/cktsopt.c, src/spicelib/devices/res/restemp.c, src/spicelib/devices/dio/diosetup.c, src/spicelib/devices/vsrc/vsrcask.c, src/spicelib/devices/vsrc/vsrcpar.c, src/spicelib/devices/isrc/isrcask.c, src/spicelib/devices/isrc/isrcpar.c, src/spicelib/devices/isrc/isrc.c, src/spicelib/devices/isrc/isrcdefs.h, src/spicelib/parser/inpdpar.c, src/frontend/spiceif.c, src/frontend/commands.c, examples/guardspell_examples/ (new)*

THE GUARD THAT COVERED ONE SPELLING OF A BAD VALUE — eight of them. In every case a working guard already sat a few lines away covering a DIFFERENT spelling. (1) A negative `gmin` wrecked the operating point in silence — a diode 0.63 V drop became -0.001 V, `rc=0`, the error growing SMOOTHLY with magnitude so there is no threshold at which it announces itself — while `reitol/abstol/vntol/chgtol/trtol/maxord/temp` were all already guarded by [E-426](#); `gmin+gshunt` were the holes, zero stays legal. (2) `scale=0` SHORTED a resistor while a resistance written as `0` warned AND was clamped; the scale route was silent and unclamped (exactly

0.0). (3) **trrandom** TYPE outside 1..4 = a silently DEAD source (0/5/9/-1/100 gave rms 0.0). (4) Diode **level=99/-1** accepted silently while **level=2** was Fatal — the check tested one value, not a range. (5) **cshunt** from **.control** silently ignored: it works by adding node capacitors at setup from **eval_opt()**'s **cshunt_value**, so the control path stores a value nothing reads — the only card-only option of nineteen tested, and the card form is BIT-IDENTICAL to writing the capacitor by hand; the notice is gated on that variable being absent so correct decks stay quiet. (6) **show** claimed a source had ALL EIGHT transient waveforms — one shared coefficient array answered every query, so a sin-only source reported 0/2/3000 eight times; a dc-only source correctly showed –, so it separated "none" from "some" but not WHICH. Returning EMPTY rather than an error is what makes **show** render – instead of a column of **<<NAN, error>>** — pick the return the generic printer already renders well. (7) **savecurrents**, **seed**, **numdgt** flagged unknown while working (without **savecurrents @r1[i]** is a scalar, with it a waveform) — E-446 downgraded this class for shell variables, which these three are not. (8) **snload**'s help named one file while the command required two, and **pwl r=** on a current source failed as "unknown parameter"; now refused by name — a precise diagnostic, NOT feature parity, since porting repeat/delay means replacing ISRC's **pwl** evaluator. THREE reported defects were NOT: **m=0** is E-426's documented "disable this instance" idiom and **@r[resistance]** is deliberately NOMINAL with **1/@r1[conductance]** as the documented effective route — both settled by E-426, both caught by its suite failing in regression, both left unchanged and now pinned as controls; and **option method=bogus** DOES print "unsupported integration method" (the pre-E-447 binary too) — my probe's filter missed it. Verified guardspell 54/54 both solvers; regression 359/359.

E-446 *src/spicelib/devices/vsrc/vsrcload.c, src/spicelib/devices/vsrc/vsrcpar.c, src/spicelib/devices/isrc/isrcload.c, src/spicelib/devices/isrc/isrcpar.c, src/spicelib/devices/cap/capask.c, src/spicelib/parser/inp2dot.c, src/spicelib/parser/ptfuncs.c, examples/argdiscard_examples/(new)*

WHAT THE NETLIST WROTE, QUIETLY DISCARDED — six of them. ZERO USED AS THE "NOT SUPPLIED" SENTINEL accounts for three: the source loaders tested **coeffs[n] != 0.0**, so a legitimately written 0 was indistinguishable from an omitted argument. (1) An explicit **TD1=0** on an EXP source became **ckt->CKTstep** — the waveform started one step late, every printed row carrying the exact analytic value of the PREVIOUS timepoint to 10 decimals, and because the substitute IS the timestep the answer depended on the timestep: err -3.7e-03 at 10 us, -4.0e-02 (~4 %) at 100 us, with **1e-30** instead of 0 making it exact. Both source types; SIN/PULSE/PWL/SFFM exact throughout, which pinned it to EXP. (2) PULSE **PW=0/PER=0** meant different things to a V and an I source — **PW=0** gave no pulse on one and an endless one on the other, **PER=0** disagreed the other way; a zero WIDTH is now the degenerate pulse asked for and a zero PERIOD means "do not repeat", on both, while a zero rise/fall still falls back to the timestep. (3) A PWL list with an ODD token count had a value invented — **pwl(0 0 1m 1 2m)** on a V source was BYTE-IDENTICAL to **pwl(... 2m 0)**, ramping the stimulus to zero at a time never assigned one, while the I source ate the stray token AS a value; two guesses at one input, so it is refused. Then (4) a third **.dc** sweep source was neither run nor refused — 9 rows not 27 with V3 pinned, proven by making V3 dominant (**R3=1 ohm** vs **1k**) and watching nothing move — plus surplus **.ac** arguments dropped in silence while **.tran/.dc** refuse them. (5) **pow(fabs(x),y)** DROPPED THE SIGN of a negative base: **(-2)**3 = +8**, **(-2)**1 = +2**, so odd exponents wrong and even ones coincidentally right — while **pwr(-2,3)**, a Verilog-A model's own **pow(-2,3)**, and both the LTspice and HSPICE branches all give -8 (**E-399**'s rule: an expression must not mean different things depending on whether the netlist or the model computed it). An integer exponent now carries its sign; a non-integer one keeps the historical magnitude rather than a NaN that would poison the Jacobian (**E-256/E-440**). (6) **@c[capacitance]** folded in **m=** while **@r[resistance]** reported the written value; the capacitor now reports what it was given and the simulation does not move, since **capload.c** applies **m** to the stamps itself. One reported defect was NOT one: an AM source with a 5th argument emits a constant because ngspice's AM is **AM(V0 VM0 VMA FM FC TD PHASEM PHASEC)** — the 5th argument is the CARRIER FREQUENCY, so **am(1 0 100 1k 0)** sets **FC=0**; against the right order AM matches analytic to 1e-12, unchanged. Verified argdiscard

37/37 both solvers, every fix paired with a control that must not move (omitted arguments, positive bases, complete PWL lists, one- and two-source .dc); regression 358/358.

[E-445](#) *src/frontend/fourier.c, src/frontend/inpcom.c, src/frontend/subckt.c, src/frontend/com_sweep.c, src/frontend/spiceif.c, src/spicelib/analysis/cktsort.c, src/spicelib/parser/inp2n.c, src/spicelib/parser/inp2r.c, src/spicelib/parser/inp2c.c, src/spicelib/parser/inp2l.c, examples/guardgaps_examples/ (new)*

NINE SILENT FAILURES MADE LOUD. (1) A bare **.four** card SEGFAULTED — dotcards.c warns "no nodes given" but does not DROP the card, so `fourier()` read the fundamental off a NULL wordlist; the only one of 25 zero-arg dot-cards that died — **.print/.plot** print the identical message from the same function and were always safe. (2) **.four 1e400** overflows to +INF and produced a full report, THD 201.971 %, harmonic frequencies printed inf — while 0, -1000, abc, literal inf/nan and even 1e-400 (underflows to 0) were already refused; overflow was the one hole. (3) **R1 in nb 1,5k** built a 5 kOhm resistor: , is a token separator, so an unlabeled trailing number OVERWROTE the value already read (that overwrite is how **R1 a b rmod 2k** works, but there nothing had been read) — decisive because every other separator obeys the documented ignore-trailing-text rule (1;5, 1:5, 1_000 → 1). R, C and L. (4) An array instance past 8192 collapsed to ONE device named `r[0:8192]` — 9x wrong at the boundary, 21x at `R[0:19999]`, silent; the same range in a NODE field already got [E-443](#)'s warning, only the instance-NAME slot was quiet because [E-441](#) made `inp_expand_buses` skip the first token. (5) **.option autobus** indexed ground into five FLOATING `0[0]..0[4]` nodes (and a `0[0]` into a `0[0][0]`), device contributing nothing: correct -1.9375e-3 vs 5.4e-20 — and the same line with the option OFF is reported by [E-402](#), so switching the feature on removed an existing diagnostic. (6) A failed **sweep** point republished the PREVIOUS solution — three priors gave three different "results" (0.5/0.8/0.2), a flat plausible shelf that `wrdata` wrote unmarked; now NaN, which keeps the scale alignment [E-438](#) preserved the points for. **montecarlo** was the guarded sibling all along. (7) A legal 20-deep hierarchy was refused as "infinite subckt recursion" — a DEPTH cap (MAXNEST 21), not name length; now 256, with runaway recursion still bounded by the million-instantiation cap. (8) `itl1/itl2/itl4` below NI-iter's floor of 100 were stored, echoed by option, and never applied (13 iterations under `itl1=3` with both fallbacks off); [E-426](#) judged that floor deliberate and upstream and that judgement STANDS — it is now announced instead of silent, and only for an explicit setting since the shipped `itl2=50/itl4=10` are themselves below it. (9) `nfreqs/nperiods/polydegree/fourgridsize` were reported "unknown option" while demonstrably taking effect (`fourgridsize=10` moves the grid 200→10); registered — scope deliberately those four, since 176 of the 179 `cp_getvar` names are flagged and most are set variables. One reported defect was NOT one: an OSDI integer param taking `n=0.5` as 1 is [E-399](#)'s LRM round-half-away-from-zero conversion, and 1 is in range — unchanged. Verified guardgaps 59/59 both solvers, every fix paired with the sibling that must not move; regression 357/357.

[E-444](#) *src/spicelib/parser/inp2n.c, src/frontend/spiceif.c, examples/autobus_examples/ (new)*

.option autobus — CONNECT A VERILOG-A BUS PORT BY ONE NAME. `inout [0:4] a` compiles to five OSDI terminals `a[0]..a[4]`; with the opt-in option **N1 a b busdev** expands to the full six-token form, and `a[2]` elsewhere binds to the same node. The information was already in hand: INP2N resolves the model BEFORE binding nodes and `dev->termNames[]` holds those names — [E-402](#) was already reading that table just to REPORT unconnected terminals. Terminals group into ports (consecutive `base[i] = one bus port`); a line with one token per PORT expands. Indices copied from the model's own terminal name, so a `[4:1]` port gives `a[1]..a[4]`, not an assumed `0..n-1`. OPT-IN because a short line already means `\$port_connected` (BSIMSOI, BSIMCMG/IMG/BULK, BSIM6, PSP-HV) — and even with the option on, an all-scalar model has port count == terminal count so it can never be mistaken for the shorthand. Option name registered with [E-438](#)'s checker. Expansion is at DEVICE-PARSE time (it needs the model), so unlike [E-441](#) it does not appear in listing `e`. Verified autobus 12/12 both solvers — every check a differential: the short form BIT-IDENTICAL to the explicit one on a ladder where all five bits differ, a

[4:1] bus matching explicit, two buses on one device, \[extract_itex]port_connected identical on/off, and the explicit form unaffected by the option — regression 356/356.

E-438 *src/frontend/com_sweep.c, src/frontend/com_optimize.c, src/frontend/com_measure2.c, src/frontend/spiceif.c, src/frontend/inp.c, src/frontend/inpcom.c, src/frontend/runcoms.c, src/frontend/com_hb.c, src/spicelib/devices/ind/muttemp.c, src/include/ngspice/ftext.h, src/include/ngspice/cktdefs.h, examples/failacct_examples/ (new), examples/warnphysics_examples/ (new)*

A FAILED SIMULATION MUST NOT BECOME DATA. Every command that drives an analysis in a loop read its metric back out of whatever plot was left behind -- and ngspice leaves the PREVIOUS point's plot in place when a run fails, so the read-back succeeded and returned stale or zero data. **montecarlo** COUNTED FAILED SAMPLES AS PASSING: 6 'DC solution failed' lines immediately above 'yield: 100.000% (20/20 pass)', monotonic in the failure count (0/3/4/6/7 all reported 100%), and with a built-in diode at sigma=10 14 of 20 samples failed and yield still read 100%. Reproduced by a second mechanism (illegal tstep from a sampled param): 37 errors, 100% (12/12). NOT OSDI-specific. The machinery was otherwise correct -- with converging samples yield reads 85/70/65%, an impossible spec gives 0%, LHS and plain bracket the analytic 49.8% -- only the FAILURE path was wrong, which is why the statistical/yield audit passed it. The signal already existed: runcoms.c publishes `sim_status` per analysis. Samples that never solved now leave the yield population and are reported; sweep names how many points did not converge (their outputs are 0.0 and persist into `wrdata`); `optimize` scores a failed evaluation as worst-case instead of scoring the previous point's plot, and reports the count rather than claiming it CONVERGED. NEW . **option warn_physics** -- opt-in, because every value it flags is one a simulator has reason to accept (negative resistance is a small-signal equivalent): `abs(k) > 1` makes the inductance matrix indefinite and the pair GENERATES energy (measured `abs(v(sec)) = 1.178 > abs(v(pri)) = 0.9986` on a 1:1 transformer), `ron=-1` makes a switch a -1 ohm resistor (divider reads -1/999 V), `l=-1u` pushes a node ABOVE the supply (1.0306 V from 1 V). THE CONTROLS SHAPED IT: flagging ZERO warned six times on a healthy deck (`1/w` sit at 0 on every device not using them) so only strictly-negative is flagged; and `is` is dropped entirely because it is the saturation current on a diode model but the SOURCE CURRENT on a MOSFET instance. The coupling check lives in `MUTtemp` -- the askable `k` returns the mutual inductance `M`, not the coefficient, so only the device layer has both. Also: a bogus CO-KNOB no longer cancels the whole sweep restore (E-385's all-or-nothing premise fails when the failing knob is a different knob from the one that moved -- the circuit is NOT untouched; the next op read 0.3333 instead of 0.5) and capture is now order-independent; . **meas** ... WHEN respects its own MEASUREMENT_FAILURE instead of printing a fabricated 0 formatted like a real measurement (`meas ac ... WHEN mag(v(nb))=0.707` recorded 0 Hz as the -3 dB point while its sibling FIND failed loudly); an unknown .options NAME is reported, on the card path only (`if_option` is called by `cp_usrset` for EVERY shell variable, so warning there makes ngspice complain about its own `rndseed`); an unmatched .if now sets the exit status like its .subckt/.ends sibling; and a successful hb exits 0 (main.c reads `sim_status`, which hb never set, so every good HB run in batch mode exited 1 with 'incomplete or empty netlist'). DEFERRED with reasons: `pow(v, 0<e<1)` aborting (the solver is right to object to an unbounded derivative), numparam's 1-ULP decimal scanner, XSPICE pwl array-length mismatch, PWL odd token count, and subckt parameter typos (positional dollar-placeholder substitution in numparam). Verified failacct 9/9, warnphysics 18/18 (every positive paired with the option OFF, plus five clean decks), regression 347/347 both solvers.

E-437 *src/frontend/com_sweep.c, src/frontend/spiceif.c, src/frontend/inp.c, src/include/ngspice/ftext.h, examples/modelwild_examples/, examples/sweepguard_examples/*

A SWEPT MODEL WILDCARD IS PUT BACK, AND A .temp WITH NO VALUE. Two silent wrong answers found by a hunt run against the binary shipped the same day, both cases where the guard that should have caught them already existed and simply was not reached. (1) E-436 gave `@*: rmod[param]` a SET path but never the capture-and-replay path E-409 built for `@*[param]`, so sweep `@*: rmod[res] 1k 3k 1k` ran correctly and then left all three matched models at 3000

-- the next analysis silently used the wrong circuit. That is exactly the defect E-409 exists to fix (*a wildcard knob was never put back*), reappearing for a newer spelling. ONE OMISSION, TWO SYMPTOMS: `sw_wildcard_knob()` is consulted in exactly two places -- to NOT read a wildcard as a scalar (* is in the lexer's specials, so PParse emits a stray-] pair; E-409 added that guard and its comment quotes the very lines `@*:rmod[res]` printed) and to capture per target. Matching neither, the new form fell through to the scalar reader (spurious no such device or model name + PError) with nothing captured (no restore); teaching that one function fixes both. The new `if_{save,restore}param_wildcard_model_named()` reuse E-409's `wild_ask_scalar/wild_set_scalar` and its all-or-nothing rule, differing only by the `eq(model_leaf(nm), leaf)` filter -- they MUST walk exactly the targets `if_setparam_wildcard_model_named()` walks, in the same order, or index `i` names a different model on the way out and a decoy is restored to a value that was never its own (pinned by moving an unrelated `omod` to 7000 first and requiring it to still read 7000). (2) A .temp card with NO VALUE ran the circuit at 0 C, `rc=0`, nothing on stderr but the routine `TEMP = 0.000000` line: `strtod("")` returns 0.0 AND leaves the end pointer at the start, so the trailing-garbage test that correctly rejects .temp abc saw a clean parse. A divider with `tc1=0.01` answered 0.5780 instead of 0.5000 -- 15.6% wrong, silently, no Verilog-A involved. Its own sibling .options temp= already refused the identical mistake; probing all 14 value-taking dot cards showed .temp is a SINGLETON (.tnom/.include/.lib abort, .four/.print/.plot warn, .ic/.nodeset/.save/.param/.global are silent but INERT) -- the only card both undiagnosed and able to change the answer, so it is fixed alone. DELIBERATELY NOT CHANGED: .temp 1e9 (E-426 drew its line at the physically IMPOSSIBLE, not the implausible); @(initial_step) firing once per sens re-solve (each re-solve IS an initial step); tran uic omitting t=0 (long-standing, general ngspice, and emitting it would change every uic transient's row count). One finding WITHDRAWN: a malformed optimize IS diagnosed (needs hi > lo) -- stdout/stderr ordering hid it. Verified modelwild 21/21 (up from 16, with the swept curve pinned so the fix cannot hide an answer change) and sweepguard 43/43 (up from 35, including a positive control that .temp 125 still SETS 125 C); both suites run against the UNFIXED binary as negative controls, where exactly the defect-specific checks fail. Full regression 347/347 both solvers, including wildrestore.

E-436 *src/frontend/spiceif.c, src/frontend/device.c, src/frontend/com_sweep.c, src/include/ngspice/fteext.h, examples/modelwild_examples/ (new)*

ONE MODEL NAME, EVERY INSTANCE PATH. Subcircuit expansion renames a .model rmod inside x1 to x1:rmod, so a deck declaring rmod at top level AND inside a subcircuit ends up with several distinct models -- and only two ways to reach them, one too narrow and one too wide: `@rmod[param]` hits the TOP-LEVEL CARD ONLY (every subcircuit copy silently keeps its old value -- easy to write while meaning *the model*), and `@*[param]` hits every model that has param, sweeping up unrelated ones. Neither expresses the ordinary want: one definition, several instantiations, plus a top-level copy. New **@*:rmod[param]**: * stands for the INSTANCE PATH and matches any path including none, so the top-level card is covered alongside every <path>:rmod. Matching is on the LEAF NAME (after the last ':', or the whole name), which is depth-independent -- `rmod, x1:rmod, x1.x2:rmod` all match -- with no pattern syntax. `@*.rmod[param]` accepted too (E-433/435 taught the dotted spelling elsewhere). Deliberately NOT a glob: `@x*:r*` stays an error, since E-269's wildcards are a fixed token set and a mistyped pattern would match nothing silently. **@rmod** was deliberately NOT broadened: that would change what existing decks DO with no diagnostic, and every hierarchical fix here (E-410/428/433/435) was safe precisely because it could only turn an ERROR into a hit, never one working behaviour into a different one; targeting a single card also has to stay reachable for mismatch work. What was missing was information, so it now reports how many copies it left untouched and names `@*:rmod[...]`. Two distinct failure messages replace a shared not-found: an unknown model name explains the flattening, a known model with an unknown parameter says so. A GAP THAT TURNED OUT NOT TO EXIST: the instance wildcards `@#*[param]/@*[param]` looked like they missed subcircuit instances -- they do not. The probe measured a divider in which the wildcard changed BOTH resistors equally so

the ratio stayed 0.5; reading the parameter back shows `r.x1.rx` and `r.x2.rx` going 1000 -> 3000 like the top-level one, and `if_setparam_wildcard_instance()` walks every instance in the circuit, where flattened ones are ordinary members. Fourth non-discriminating probe of the session. Verify (examples/modelwild_examples, 11/11, new): the deck makes every level of reach distinguishable (`v(a)/v(b)` subcircuit copies, `v(e)` top-level card, `v(c)` an unrelated omod that must not move); both spellings checked, both existing forms checked UNCHANGED, sweep checked on a copy and the top-level card in one sweep, both diagnostics pinned. Full regression 347/347.

E-435 *src/frontend/com_sweep.c, examples/hierdev_examples/ (extended)*

The FOURTH consumer of a hierarchical model name, and the one E-433 missed. A `.model` inside a subcircuit is renamed `<instance-path>:<model>`, and E-433 taught the name funnels to accept the dotted `@x1.rmod[res]` users actually write -- reaching `altermod`, `optimize -mparam` and `@-readback` because all three resolve through `finddev/finddev_special`. **sweep** does not: `sw_kind()` calls `ft_sim->findModel()` itself, right where it classifies the knob. The failure was worse than a refusal: an unrecognised `@name[param]` falls through to the `INSTANCE` branch, `alter` reports a no-such-parameter error for a `MODEL` parameter, and the sweep runs on with a knob that never moved -- a full set of points, `rc=0`, and a plottable FLAT curve behind a diagnostic that scrolled past mid-run. Same shape as the zero column E-431 removed, except it is the KNOB not the output, and a constant curve is easier to mistake for a result than a zero one. Fix is one condition mirroring the two funnels; application already worked, since `sw_set_inplace` issues `altermod` which E-433 taught the dotted form. SCOPE CHECKED, NOT ASSUMED: `sw_kind()` has exactly ONE caller -- `montecarlo/highsigma/wcd perturb Gaussian .param` values rather than named knobs, and `optimize` classifies its own via `-param/-mparam/-dparam` whose model path already worked. Verify (examples/hierdev_examples, 38 -> 43 checks): all four spellings (`@x1.rmod[res]`, `@x1:rmod[res]`, and both nested forms) measured against the analytic divider `v(out)=1k/(res+1k) -> 0.5/0.3333/0.25`, plus a top-level `@rmod[res]` the new fallback must not disturb. POSITIVE-CONTROLLED: reverted and rebuilt, exactly the two dotted checks fail with the flat 0.5/0.5/0.5 that is the whole point; the colon forms and the top-level model pass either way. Full regression 346/346. KNOWN NOT FIXED: a knob that fails to apply for ANY reason still yields a flat curve at `rc=0` (`sweep @x1.nosuch[res]`); catching that means verifying after the first application that the knob took (E-385's `sw_read_knob` can read one back) and refusing the sweep otherwise -- a behaviour change with its own regression surface, so recorded rather than bundled.

E-434 *src/frontend/com_sweep.c, src/osdi/osdiload.c, src/spicelib/analysis/tfanal.c, examples/silentloss_examples/ (new)*

Three silent wrong answers, a silent truncation, and a swallowed request, from the round-35 hunt. (1) FOUND, NOT FIXED -- and the attempt is the finding. A save list that resolves to NOTHING aborts the analysis (no data saved ...; analysis not run) and every result is lost; save is alone in this (`.probe/.print/print/wrdata` diagnose and carry on) and disagrees even with itself, since a bad `@dev[param]` merely warns (E-418). The obvious fix -- fall back to recording everything, as when no save list is given -- was written and withdrawn because it creates a CRASH: the regression's `nameovf` suite went SIGTRAP. The abort is LOAD-BEARING, stopping execution before a latent long-name stack overflow on the `.four/gettoks` path that E-237 hardened but did not fully reach (`.four 1k i(<600 chars>): rc=1 -> rc=133`). E-399's lesson running backwards -- there, REJECTING a wrong answer created a crash; here ACCEPTING one does. Removing the abort must follow a `dotcards.c` fix, not ride along with this one. (2) `\$simparam("temp")` was not supplied though `tnom` was, despite `ckt->CKTtemp` being right there: with a default it returned the caller's value (silently the wrong temperature), without one it killed the run with OSDI(fatal). Returned in CELSIUS like `tnom`. `temperature/timestep/maxstep/freq` deliberately stay OUT -- no simulator answers those names and `\$temperature` already works, so supplying them would invent an interface rather than complete one. (3) `\$simparam("abstime")` returned the DC SWEEP VALUE (`dc V1 0 5 1 -> 5.0; -3 -1 -> -1.0`; a different swept source -> that source's

value) because `.dc` reuses `CKTtime` as its abscissa -- an aging model read a `VOLTAGE` as a time. Guarded on `MODETRAN`, exactly as `OSDIfinalStep` in the same file already did for E-412; the pollution never reached the solver (`ddt` exactly 0 in DC, verified to 10 digits). (4) `montecarlo/highsigma/wcd` silently truncated `-analysis` at a 512-byte `strncat` buffer and ran a `DIFFERENT` command past 509 chars; they now refuse. DISCLOSURE: for `wcd` that exposure was created by E-433 three commits earlier, which gave it the same multi-token collector as its siblings; the other two always had it. (5) `.tf` was the one analysis that reports a result while swallowing a model's `\$finish` -- `dcop/dctrcurv/acan/noisean/dctran` each have E-426's notice, `tfanal` had none; it now reports and, as in `dcop`, does NOT act (a single computed point, nothing left to truncate). WITHDRAWN WHILE FIXING: the `x1:rmod` model-name collision is real (1000 -> 7777 ohm, solution moves) but `ngspice` already warns model `"x1:rmod"` is already defined; the hunt missed it by checking only values on that probe, and a second warning would duplicate a diagnostic -- the very thing E-430 removed from `.probe` -- so NOTHING was changed and the existing warning is pinned instead. Also re-confirmed: `\$finish` at an operating point is reported but not obeyed, E-426's deliberate decision (the hunt's 'ignored' came from a head -4 eaten by three `\$strobe` lines). KNOWN NOT FIXED: a refused `alter` still reads back the `REFUSED` value; `opvars` named `temp/dt/m/dtemp` are shadowed by E-397's instance knobs (`dtemp` a fourth case not previously recorded). Verify (examples/silentloss_examples, 23/23, new): every fix checked in BOTH directions -- the failure is gone AND the neighbouring good case is unchanged. Full regression 346/346 (the new suite included).

E-433 `src/frontend/{com_sweep,com_optimize,spiceif}.c, src/include/ngspice/ftext.h, examples/{sweepguard,hierdev}_examples/ (extended)`

Two spellings that worked everywhere except in one place. (1) QUOTING: `ngspice`'s lexer treats the two quote characters differently and says so in its own comments (`parser/lexical.c`) -- `'...'` forms one word WITHOUT the quotes, `"..."` forms one word INCLUDING them, leaving each command to strip the survivors with `cp_unquote()`, which ~17 frontend files do (`com_echo.c` among them, hence `echo "a b"` looking fine). `com_sweep.c` and `com_optimize.c` called it ZERO times, so `-analysis "tran 1n 20n"` reached the command lookup with its quotes attached (unknown command `"tran 1n 20n"`) while the single-quoted spelling worked; for `-output` the residue was worse than an error -- `"v(d)"` recorded a vector literally NAMED with quotes and `"gain=v(d)"` split at the wrong = so nothing resolved. Tokens are unquoted INDIVIDUALLY, since stripping one outer pair from a joined `"tran" "1n" "20n"` would eat the wrong two characters. The same flag was also collected THREE ways by the five commands that document it identically: `sweep/optimize` used `collect_until_flag()`, `montecarlo/highsigma` stopped at ANY leading - (so a legitimately negative analysis argument, `disto lin 3 1e5 1e6 -0.5`, ended the list and came back as an unexpected token), and `wcd strncpyd` ONE token so it alone REQUIRED quoting (`-analysis tran 1n 20n` kept `tran` and rejected `1n`). All five now use `is_flag()` -- followed by a LETTER -- the same distinction E-432 drew for the `-output` terminator, and for the same reason: `-v(d)` and `-0.5` are values, not flags. (2) HIERARCHICAL MODEL: `modtranslate()` renames a subcircuit-local `.model` as `fprintf("%s:%s", scname, model_name)`, so a model in `x1` is `x1:rmod` and one in `x1/x2` is `x1.x2:rmod` -- levels joined with `.`, the model attached with `:`. Nothing else in the hierarchy is spelled that way (`devices @x1.rx[p]` per E-410, nodes `v(x1.mid)`), so the natural `@x1.rmod[res]` was refused and the colon had to be found by reading the expansion code. `if_find_model_hier()` turns the LAST `.` into `:` -- exactly the instance-path/model boundary at any depth -- wired into BOTH funnels (`finddev`, `finddev_special`) AFTER the exact lookups and after E-410's instance fallback, so the rule can only turn a former error into a hit; a name already containing `:` is skipped. One fallback reaches `altermod`, `optimize -mparam` and `@readback` together, since all three resolve through those funnels. Verify (`sweepguard` 23 -> 35, `hierdev` 32 -> 38): five commands x three quote styles + the negative argument; the dotted model resolves to the same model as the colon one, `altermod` drives it (`v(out) 0.5 -> 0.25`), `optimize -mparam` finds `res=3000`. Both sets POSITIVE-CONTROLLED -- reverted and rebuilt, 9 of 12 quoting checks and 3 of 6 model checks fail; the rest are deliberate no-regression

controls, incl. a hierarchical DEVICE that must keep resolving as a device. Full regression 345/345. FOUND BY two questions, not a hunt. Three method notes, all one mistake: a pass/fail filter keyed on error WORDING scored two real failures as ok; a probe using a bad analysis NAME could not discriminate (sw_run_cmd reports only the FIRST word, so a collected nosuch lin 3 1e5 and a truncated nosuch print identically); and **optimize** prints 'converged' with a nonzero residual when the knob never moved the metric (0.0625 = the square of the untouched starting error).

E-432 *src/frontend/com_sweep.c, examples/sweepguard_examples/ (extended)*

sweep -output read only its FIRST token, though its usage line has always advertised `-output <expr> ...`; every token after the first fell through to the unrecognized token branch and the sweep ran on with a silently shorter output list. The obvious termination rule is wrong here: stopping at the next token beginning with `-` would break `-v(d)`, a legitimate NEGATED output expression that the old single-token form accepted because it took the next token unconditionally. So the list ends at the flags sweep actually knows (`-vs/-family`, `-analysis/-a`, `-overlay/-ov`, `-output/-o`) and at nothing else, which leaves a MISTYPED flag to be read as an expression -- the right outcome, since Enhancement-431 then names it in full (`sweep -output -ouptut` never resolved). One case improves rather than merely surviving: a flag directly after `-output` is now a missing expression rather than an output NAMED `-vs`, so the outer knob that follows is honoured instead of consumed. Each element takes the same path as before, so `name=expr` and E-267 bus expansion base[lo:hi] work from any position, not only the first. SECOND HALF -- an Enhancement-431 regression, and the worse of the two. E-431 gives each output a tally of the points at which it failed to resolve and zeroes it with `for (k = 0; k < nout; k++)`. With no `-output` at all the outputs are auto-collected from the first analysis's plot INSIDE the point loop, below that line, so `nout` is still 0 there and every auto-collected output inherited stack garbage -- which E-431 read as a resolve failure. On a 25-node deck, a plain `sweep @r1[resistance] 1k 2k 1k -analysis op DELETED` six of the twenty-five node curves with never resolved and falsely warned about others. That is the default invocation of the command. Fixed by zeroing the whole array; the heap arrays sized per output (data, ovy) were checked and are fine, being allocated AFTER the auto-collect. Verify (examples/sweepguard_examples, 9 -> 23 checks): the variadic list agrees value-for-value with the one-flag-each form it replaces, each terminator ends it, a negated expression survives, an empty `-output` is diagnosed without swallowing the flag after it, a bus range expands from a non-first position, and both auto-collect checks were POSITIVE-CONTROLLED -- with the fix reverted they fail, naming the six nodes that disappear. Full regression 345/345.

E-431 *src/frontend/com_sweep.c, examples/sweepguard_examples/ (extended)*

A **sweep -output** naming a vector that never resolves was recorded as a full column of zeros -- a clean, plottable, entirely fictional flat line, behind nothing louder than a `checkvalid` warning. The cause was already written down in this file: `sw_eval_expr()` returns 0.0 on failure, and Enhancement-385's comment says exactly why that is a problem -- "indistinguishable from a knob that is legitimately zero" -- having solved it for the knob-restore path by adding an `*ok` out-param to `sw_read_knob`. The `-output` path has the same defect and a worse symptom, because the zero is not used once and discarded but recorded, named and drawn. Same fix shape: `sw_eval_expr_ok()` reports whether the expression RESOLVED, independently of its value; the five other callers (`optimize` and the metric evaluators) keep the old signature through a thin wrapper and are untouched. Reported PER OUTPUT, from a per-output tally of failing points: failed EVERYWHERE -> named in an error and the curve is not emitted; failed at SOME points -> a warning with the count, curve still emitted because the points that resolved are real; resolved to ZERO -> recorded, unremarked, which is the distinction the change exists for and is pinned with `v(d)-v(d)`. A bad output does not cost the good ones. Verify (examples/sweepguard_examples, +9 checks): the refusal, the surviving siblings, and both directions of the distinction. Full regression 345/345. NOTED NOT FIXED: `-output` consumes only its FIRST token, so `-output v(a) v(`

b) silently records `v(a)` alone (one `-output` per expression is the working syntax) -- pre-existing, identical before and after, and the usage line's `[-output <expr> ...]` implies otherwise, so which spelling is right wants its own decision.

E-430 *src/frontend/{inpc_probe,inpcom,spiceif,inp}.c, src/spicelib/analysis/optran.c, examples/probeshort_examples/ (extended)*

A diagnostic-only change: what **.probe** says when it refuses a token. `.probe @r1[i]` printed "Warning: Strange parameter in line *probe @r1[i], ignored" -- TWICE -- naming neither the accepted forms nor the right tool, echoing the card in its internal `*probe` form, and misspelling "ignored". The refusal itself is CORRECT and is unchanged: this was raised as a capability gap and is not one. Per the manual 11.7.1 `.probe` MEASURES a current by placing a voltage source in series with the device's node -- hence its `<inst>#branch` results -- while `@device[param]` is read out of the device and needs no extra node, which 11.7.3 describes as the `.options savecurrents` route "because the data are retrieved [internally]", generating `.save @r1[i]` lines. Three distinct documented mechanisms; teaching `.probe` the fourth spelling would conflate a circuit-MODIFYING command with a vector-SELECTION one, and the `vipVIP` gate is principled because `.probe` needs something it can put a source IN SERIES WITH. The message now names the accepted forms (`v(...)`, `i(...)`, `p(...)`, `alli`), points an `@` token at `.save/.options savecurrents` (and only an `@` token -- a plain `.probe` nonsense gets no irrelevant advice), fires ONCE (the type test warned and advanced past the token, then the `!nextnode` test warned again on the wreckage; a rejected flag suppresses the second), and echoes the card as the user wrote it. `ignored -> ignored` also in five further places that have nothing to do with `.probe`: `inpcom.c` (`x2`, `.hdl` and `.biaschk`), `spiceif.c` (`snsave`), `inp.c` (a comment) and `optran.c`. Verify (`examples/probeshort_examples`, 36 -> 48 checks): the message content, the single firing, the card echo, the absent typo, and in the other direction that `.probe i(r1)`, `.probe alli` and `.probe v(b)` stay silent and still produce `r1#branch`. Full regression 345/345.

E-429 *src/include/ngspice/{cktdefs,inpdefs}.h, src/spicelib/parser/{inpsymt,inp2dot}.c, src/spicelib/analysis/{tfanal,noisean}.c, examples/inputguard_examples/ (extended)*

A **.tf** CARD answered **transfer_function = 0.000000e+00** for a node that does not exist, with no diagnostic -- a plain typo or a device-internal node alike, while the `.control` command form of the same thing refuses it properly. E-426's bounds guard is not wrong, it simply cannot fire here: a node named by a COMMAND is created after `CKTsetup()` sized the matrix so its equation number lands past the end (the out-of-bounds read E-426 documented), while a node named by a CARD is created BEFORE setup and is counted when the matrix is sized -- the phantom sits inside it and the analysis merely reads an all-zero row. E-426's own write-up recorded this ("created early ... it lands inside the array and the read is merely wrong") and did not act on it. Refusing at parse time is not available: E-349 deliberately lets an analysis card name a node the deck has not reached yet, and that case is pinned in `examples/`. So the question is not "does this node exist" -- the card made it -- but "did anything OTHER than the card ever refer to it". One bit on `CKTnode`, `devRef`, records it: every REAL creation path sets it (`INPtermInsert` for a device card, whether it finds or creates -- finding is what makes a forward-referenced node real; `INPmkTerm` for the simulator's own internal nodes; `INPgndInsert` for ground), and exactly one path clears it -- `inp_analysis_node`, the only inventor. `CKTnodePhantom()` is the test; ground is never phantom. The default is REAL, not phantom, deliberately: a creation path added later and not taught about the flag behaves exactly as today rather than being silently rejected. Also solver-agnostic, unlike the obvious alternative of asking the matrix whether the node has any entries -- `SMPfindElt` asserts Sparse and would need a KLU special case, and the regression runs both. `noisean.c` has the same shape and gets the same guard; `sens` needs none, its card path fails earlier. Verify (`examples/inputguard_examples`, 88 -> 93 checks): the refusals plus both directions of the boundary -- a real node still answers 0.75 and E-349's card-before-its-devices case still answers 0.75. Full regression 345/345.

E-428 *src/frontend/vectors.c, src/frontend/outitf.c, src/include/ngspice/ftext.h, examples/hiernode_*

examples/ (new)

The last place the device-type letter leaked out was a NODE name. Enhancement-410 made `@x1.r1[resistance]` work; it left a device's own INTERNAL nodes behind. `v(n.x1.n1#mid)` resolved and `v(x1.n1#mid)` did not -- and a node is the one place a user never expects a device letter, since the plain node beside it is `x1.m`. The letter is INHERITED, not chosen: subcircuit expansion re-parses the emitted card and dispatches on its FIRST CHARACTER, so the flattened refdes must keep a type letter (`n1` inside `x1` becomes `n.x1.n1`), and an internal node is named `<instance>#<node>`, so it comes along for the ride. The reconstruction needs no search and cannot be ambiguous: the letter is literally the leaf instance name's own first character and ngspice requires a device name to begin with its type letter, so `x1.n1` can only ever mean `n.x1.n1` -- one-to-one at any depth (`x1.x2.n1#mid`), with `x` instances exempt exactly as `translate_inst_name` exempts them. STRICTLY a fallback, consulted only after the exact lookups fail, so every name that resolves today is unchanged. TWO independent resolution paths, and fixing one was not enough (the E-408 lesson, which nearly bit again): `findvec` (`vectors.c`) serves `print/let/meas/wrdata/write` and the `.print/.plot` cards, while `.save` matches on its own comparator `name_eq` (`outtf.c`). With only the first fixed, `save v(x1.n1#mid)` silently matched nothing -- and that failure is destructive, not local: the run ends with "no data saved for Transient analysis; analysis not run", losing the WHOLE plot rather than one vector. Both paths now share one helper, `cp_hier_devname()`. Verify (`examples/hiernode_examples`): 17 checks on a 1k/3k divider, so the internal node sits at 0.75 -- a value nothing else in the circuit takes, which makes a wrong resolution visible rather than plausible; both spellings in every consumer, at one and two nesting levels, case-insensitively, plus the things that must not move (the device-letter form, an ordinary node, an ordinary HIERARCHICAL node, and a genuinely bad name). Full regression 345/345.

E-427 *src/spicelib/analysis/dctrcurv.c, src/include/ngspice/trcvdefs.h, examples/sweepparam_examples/ (new)*

A swept parameter value the device REFUSED was applied anyway, and the sweep set one value past its own stop. (1) `.dc @inst[param]` was the ONE parameter-supply path that applied a value a Verilog-A `from` range forbids, published the results and exited `rc=0`: `dc @n1[r] -2000 -1000 500 on r ... from (0:inf)` gave three rows at `R = -2000, -1500, -1000` with "Parameter `r` is out of bounds!" printed four times. The instance line, `alter + a run`, and the sweep command all refuse the same value. The range is not checked where you would look: `OSDIparam` has no range check at all (which is why `alter @n1[r]=-5` stores -5 and `print` shows it), it is enforced when the device is set up again inside `DEVtemperature -> OSDItemp -> setup_instance`; `DCTsetInstParam` was void and discarded BOTH returns, so a fix keyed on `DEVparam` would have changed nothing. Keyed on the device refusing, never on the value: a negative resistance is legitimate for a built-in resistor (`resparam.c` has an explicit branch, per E-426) and that sweep is unchanged. The abort exits through the `RESTORE` path, not where it stands -- leaving the instance holding the rejected value is the E-381/E-382/E-385 class. (2) The sweep handed the device one value PAST **stop** -- it advances then tests the stop criterion -- so a sweep that legitimately ENDS AT a range edge stepped outside it: `k ... from [0:1]` with `dc @n1[k] 0 1 0.25` printed a spurious "out of bounds" while producing five correct rows, and that predates this enhancement. The first version of the fix turned that valid sweep into a hard error (`rc=1` where it was 0) and an adversarial review of the FIX caught it; the past-stop value is no longer applied, which removes the spurious message too. The `TEMP_CODE` arm has always declined its own overshoot for the same reason. (3) An `INTEGER` instance parameter could not be swept and the refusal lied: "Voltage source, current source, or resistor named `@n1[n]` is not in the circuit" -- every clause false, from an `IF_REAL` test folded into the keyword match so a wrong-type hit was indistinguishable from a miss, sharing a catch-all with ten distinct causes. Integer sweeps now work; a `FRACTIONAL` one is refused, because the accumulator must stay real (a rounded one with a 0.25 step never advances -- the E-362/E-426 non-advancing-loop class in this same function) and rounding only at the device boundary would publish duplicate points under an abscissa that disagrees with the value applied. (4) NOT FIXED,

documented instead: `.ic/.nodeset` on a device-INTERNAL node. INPpas3 resolves those names before `CKTsetup()`, which is when every device -- built-in and OSDI alike -- creates its internal nodes, and inppas3.c's own header comment says so; a built-in diode's `d1#internal` is rejected identically. Verify (examples/sweepparam_examples): 32 checks; full regression 344/344.

E-426 `src/spicelib/parser/{inp2dot,inpeval,inpdpdr,inpgmod,inpmkmod}.c, src/frontend/{spiceif,com_measure2}.c, src/include/ngspice/inpdefs.h, src/spicelib/analysis/{cktsopt,dctrcurv,noisean,span,acan,dcop,tfanal}.c, src/spicelib/devices/res/res.c, src/osdi/osdisetup.c, examples/inputguard_examples/(new)`

Inputs that were taken at face value, and the confident wrong answers that followed. (1) A mistyped analysis OUTPUT node was invented. E-349 already refuses one, but gated the check on `CKTisSetup -- false` for the SESSION'S FIRST analysis, since nothing has run so `CKTsetup()` has not either. `tf v(nosuch) v1` returned `transfer_function = 0.0`, `tf v(a,nosuch) v1` returned exactly 1.0, `sens v(nosuch)` printed every sensitivity as `-0.0` and `noise v(nosuch)` reported `onoise_total = 0.0`, all silently; the same typo after any op was diagnosed correctly, which is what hid it. A card `if_run()` synthesises from a `.control` command is by construction not deck parsing, so that path now says so instead of inferring it. The phantom node is also an out-of-bounds heap access -- `CKTrhs/CKTrhsOld` are sized from `SMPmatSize()` but indexed by `node->number` from `CKTmaxEqNum`; ASAN reports heap-buffer-overflow reads at `tfanal.c:121` and `noisean.c:465` and `tfanal.c:153` WRITES through the same index, the shipped binary printing what it read as `transfer_function = 3.999110e+252`. A DECK card may legitimately precede its devices, so a bounds check at the analysis entry is the last line of defence. It immediately surfaced a live bug in this repo: `rpfss_examples` ran `.pss` on node 1, which that circuit does not have, and the PSS self-check had been printing `osc-node swing [0, 0]` ever since. (2) Sweep arguments: `.tran` validates, almost nothing else did. `.ac` SUBSTITUTED defaults and ran a different sweep -- `ac dec 10 100k 1k` became 31 points over `1e5..1e8`, `ac dec 10 -1k 100k` became 51 points from 1 Hz -- under one generic warning; noise with an inverted range published `onoise_total = 0.0` manufactured by a loop that never executed; `sp` had no validation at all; `.dc` with a step pointing away from stop produced a plot with no vector. The boundary is the whole difficulty: re-measurement WITHDREW most of the `.dc` finding -- `dc v1 1 1 1, 0 1 1.5` and `0 1 2` correctly give ONE point, and 13 decks here depend on that while 2 more sweep descending -- so the predicate is exactly `(stop-start)*step < 0`, false for equal endpoints and false for a genuine descending sweep. Equal endpoints are pinned in all four (19 `.ac`, 4 `.noise`, 9 `.sp`, 13 `.dc` cards). (3) `meas` consumed a one-point vector as a waveform. `@dev[param]` is a documented scalar snapshot unless `.saved` and `save all` deliberately does not cover it, but every `meas` loop runs `i < d->v_length` against the full-length scale, so MAX/MIN/PP/INTEG reported that one sample as the extremum -- the `at= 0.00000e+00` was the tell. Six entry points; XSPICE event vectors carry their own `d->v_scale` and are exempt. The same function also had a heap write overflow: `measure_rms_integral` sized three buffers from the scale and filled them from the data vector. (4) A temperature below absolute zero was accepted on all six supply paths and became a NEGATIVE thermal voltage -- a model read `\$vt = -0.0195 V` in silence. Three paths share the `CKTsetOpt` funnel; `.dc temp` writes `CKTtemp` directly and instance `dtemp` is composed per device. -25 C is ordinary and one deck uses it: the line is absolute zero, not freezing. `dtemp=-300` is WITHDRAWN -- 0.15 K is physical. (5) Non-positive tolerances and iteration limits. A tolerance `<= 0` made the convergence test unsatisfiable and the run blamed the CIRCUIT. `itl2=0` is the one with teeth: `CKTdCTrcvMaxIter` is an UNFLOORED /4 and 3*/4 continuation threshold in four `ckttop.c` heuristics, so zero collapses the `gmin` ramp -- 736,920 Newton iterations against 55. `itl1/itl4` are floored at 100 by `NIiter` (left alone, upstream and deliberate) so those two are reporting hygiene. Keyed on the `OPT_enum`, never the name: `itl6` is a SYNONYM for `srcsteps` where 0 is documented and four decks rely on it. (6) A negative multiplier inverts a device -- `m=-1` stamps -2000 ohm on a 2k resistor and makes OSDI noise NaN via `sqrt(\$mfactor)`. Identified by parameter id, so both `m` and `_mfactor` are covered; `m=0` is deliberately untouched, being the ordinary disable idiom. Duplicate same-

name .model cards (first wins) and m= written on a .model card (discarded, while _mfactor= works) are now announced rather than silent. (7) Numbers. 1e400 became inf and a resistor an open circuit; 0e400 became NaN and the OP failed five levels away. The exponent accumulated into a plain int, so 1e2147483648 was signed overflow yielding pow(10, INT_MIN); it now saturates. An exponent marker with no digits was SWALLOWED, letting the next letter be read as a scale factor -- 10Emitter = 0.01, 1em = 1e-3, both against src/ngspice.txt:499. Documented forms are pinned unchanged (1k2, 2meg5, 1e5x, 1kk, 0x10, 5kohms), and 1..2 is deliberately NOT touched: version=4.8.2 is the universal BSIM4 spelling and reads as 4.8 by exactly that rule. (8) A SIGSEGV. A failing OSDI setup_model had its error OVERWRITTEN by the next model's success, so OSDIsetup returned OK while that model's instances were never set up and ngspice dereferenced a NULL jacobian: two .model cards gave rc=139 and ZERO bytes, while putting the bad one first gave a clean diagnostic -- deck order decided crash versus message. First error now latched. (9) The effective resistance became readable via new READ-ONLY conductance/acconduct entries whose handlers already existed in resask.c as dead code (1/@r1[conductance] = 1200.0 where @r1[resistance] correctly still reads the nominal 1000). @r1[resistance] is deliberately NOT changed: the capacitor scales its readback and that CORRUPTS the round trip (1n -> 1.2n -> 1.44n permanently under a sweep at 227 C) -- a separate defect, flagged not fixed. (10) \\${finish} reached three analyses out of eleven. The reported headline is WITHDRAWN -- ngspice does print Note: \\${finish} requested ...; the hunt harness filtered every line containing "Note:". The real gap: under .ac the request was dropped in silence and the whole sweep ran. .ac now aborts per the E-56 precedent, .op reports and keeps its result. Verify (examples/inputguard_examples): 88 checks, every boundary pinned from both sides; full regression 343/343.

E-419 *src/maths/ni/nicomcof.c, src/maths/ni/niinteg.c, src/spicelib/analysis/dctran.c, src/spicelib/analysis/cktterr.c, src/spicelib/analysis/cktopt.c, src/spicelib/analysis/cktdojob.c, src/spicelib/analysis/cktntask.c, src/include/ngspice/{cktdefs,tskdefs,optdefs}.h, examples/integmethod_examples/(new)*

Three new integration methods, all OPT-IN, plus the measurements that say when to use them. ngspice shipped trapezoidal (A-stable but NOT L-stable, so it RINGS at a sharp transition) and Gear 1-6; the only cures for the ringing were xmu<0.5, which costs an order, or method=gear, which costs accuracy everywhere. (1) **method=trbdf2** -- a one-step COMPOSITE: trapezoidal over [t,t+gh] then BDF2 across t, t+gh, t+h. With $g = 2\text{-}\sqrt{2}$ (the root of g^2-4g+2) both sub-steps share the leading coefficient $2/(gh) == (2-g)/((1-g)h) == 3.4142/h$, so the Jacobian scaling does not change between stages. Order 2 and L-STABLE. Neither sub-step needed a new formula -- stage 1 is the existing trapezoidal order-2 rule at $\Delta t=gh$, stage 2 the Gear order-2 shape with unequal-step BDF2 coefficients -- and the two earlier charges stage 2 needs are supplied by ROTATING the state slots once mid-step (the rotation dctran already does on acceptance), the interior slot being spliced back out so the accepted history never contains the stage value. Self-starting, so it never needs the order-1 starter E-181 showed imposes an $O(h^2)$ floor. (2) **method=sdirk** -- Alexander's 3-stage order-3 L-stable SDIRK, restricted to STIFFLY ACCURATE tableaux ($a[s][j]==b[j]$, $c[s]==1$) so the last stage IS the answer; without that a step ends in a weighted COMBINATION of stage values and every value a device holds has to come out of a solve, not an assignment. Coefficients derived from gamma in code, since a mistyped Butcher digit does not crash -- it quietly costs an order. Fully implicit RK is out of scope: coupled stages need one $s_n \times s_n$ system. (3) **method=adams** -- Adams-Moulton at order maxord, weights derived PER STEP from the actual spacing (the textbook (5.8,-1)/12 are fixed-step and ngspice never takes fixed steps); normalising to $\tau=(t-t_{n+1})/h$ keeps the Vandermonde entries $O(1)$ where raw seconds would underflow. NOT stiffly stable above order 2. Error constants are derived, not borrowed: for an RK method $R(z)=1+\sum z^{k+1} b^T A^k e$, so $C = b^T A^p e - 1/(p+1)!$ -- a formula that reproduces the two constants ALREADY in cktterr.c exactly (1/12 trapezoidal, 1/2 backward Euler), which is what makes it trustworthy on new tableaux. TR-BDF2 0.04044 (trapezoidal's 1/12 overstated it 2.06x) and SDIRK 0.025897 (Gear-3's 3/22 overstated it 5.27x); the derived TR-BDF2 constant predicts a 2.06x per-step advantage and the fixed-step test

independently MEASURES 2.02x. What measuring showed: per STEP the new methods win (SDIRK observed order 2.76 vs 1.80 for the second-order trio, 1.26e-07 vs trap 1.66e-05 at $h=2.5e-8$), but per NEWTON ITERATION -- the honest cost unit, since TR-BDF2 solves twice per step and SDIRK three times -- trapezoidal or Gear wins three of four circuits, and Adams-3 wins the oscillatory non-stiff case by ~100x. Recalibrating the constants barely moved that, which is the finding: the (cost,error) curve belongs to the method, so a wrong constant costs predictability of `trtol`, not efficiency. So TR-BDF2's case is L-stability alone -- a 1ns edge arriving at a step pinned to 20tau leaves trapezoidal ringing 25 times and still 2.7e-02 off, while TR-BDF2 settles in 2 steps; trapezoidal's measured ratio is -0.8182, exactly the theoretical $(1-h/2\tau)/(1+h/2\tau) = -9/11$. None becomes the default. Two earlier claims corrected in the write-up: trap/gear do NOT hit an accuracy floor (that was a `trtol` sweep stopping at 0.5), and `verify_syntaxhl.py` was wrongly named as the orphan source. Explicit methods (forward Euler, Adams-Bashforth, explicit RK) are NOT included and cannot be: with `ag[0]=0` a reactive element becomes a fixed current source and a series RC returns $v_{n+1}=v_n$ -- at Newton convergence the element's current IS state0[ccap] whatever `ag[0]` is, so charge, being the dependent quantity $q(v)$, cannot be prescribed by a conductance+current stamp. OSDI is verified explicitly, since it loads its Jacobian with CKTag[0] ALONE while TR-BDF2 stage 2 and Adams order ≥ 3 need `ag[1]/ag[2]` -- it routes its reactive residual through NIntegrate, and all five methods agree to 0.0e+00 between SPARSE and KLU on an OSDI device. Verify (examples/integmethod_examples): 13 checks, both solvers; full regression 336/336.

E-418 `src/osdi/osdisetup.c, src/osdi/osdipzld.c, src/osdi/osdidefs.h, src/frontend/outitf.c, src/frontend/com_measure2.c, examples/saveguard_examples/ (new)`

Four defects that share a shape: something was accepted, counted, or concluded, and never verified -- the fourth found by fixing the first, and older than it. (1) `absdelay()` and `last_crossing()` read 0.0 whenever their result was only OBSERVED rather than contributed -- the E-415 open item, now CLOSED. They are the only operators whose output row the compiler deliberately leaves empty (ngspice fills it from its history buffer via elements OSDIsetup allocates later), so the node appeared in no descriptor Jacobian entry and E-116's decoupled-node scan tied it to ground; `eval()` then read CKTrhs0ld[0]. A contribution -- even at weight 1e-30 -- is what put the node into a Jacobian entry, which is why only it rescued the value, and why `transition()` (same lowering) worked: its output feeds the slew residual. Confirmed rather than inferred: before the fix `n1#implicit_equation_1` does not exist as a circuit node. Five lines, idempotent for contributed models, no OpenVAF change and no ABI change -- E-415 had looked in the compiler and reverted when the descriptor came out byte-identical, correctly. The rule is now written at the site: a node coupled only by ngspice-side stamping must be marked there. (2) **.save** never validated a device name. `addSpecialDesc` only interns the string and `getSpecial's` caller discards INPName's `E_NODEV/E_BADPARM`, so `@nosuchdev[i]`, `@r1[nosuchparam]` and `@*[i_p]` each produced a registered vector that stayed 0 long, silently, while `print/meas/wrdata` all report the same name loudly. The check now calls `INPName` -- the very routine the per-point read uses -- and a HIERARCHICAL spelling is rewritten rather than rejected, so E-410's `@x1.n1[i_p]` yields a full waveform under the user's own spelling. Unresolvable names are warned about and still added (dropping them would change the plot; a bracket-less `@name` is a statistic on another path). Wildcards are diagnosed, not expanded: every wildcard mechanism in the tree is setter-side and none yields instance NAMES. Also corrects a round-25 note -- a bad NODE name is silently dropped too, so **.save** validated nothing. (3) **meas . . . when** invented a time: $-\text{inf}$, 1.15292e+05 seconds in a 3 us run, negative times, silently. The initialisation counted a crossing in the FIRST interval and the evaluation -- which deliberately starts one sample in -- applied that leftover count to a later, crossing-free interval, dividing by a difference that was exactly zero or ONE ULP (hence every bad value being a power of two times a power of ten). The first interval spans the operating point to the first timepoint, so it straddles thresholds for reasons unrelated to the waveform: an earlier attempt to evaluate it properly broke `defaulttransition` (expected 5.0e-07, got 1.0002e-10), which is what identified starting one sample in as deliberate rather than as the bug. So the count is no longer taken, every interpolation is gated on the interval actually BRACKETING the target (

mirroring `measure_at`, which has always refused to interpolate outside a bracketing pair), and a `.dc` restart resets to sample 0 so the previous sweep's last point is never an endpoint. Clamping to `[tstart, tstop]` was rejected -- it would turn $1.15292e+05$ into `tstop`, a plausible wrong answer in a delay measure. `cross=1` improves outright: garbage becomes the genuine later crossing at $1.00025e-06$. (4) `pz` filled the simulator-stamped row nowhere. Those rows are the simulator's to fill, so every load path has to fill them -- `osdi_load.c` does for `dc/tran` and `osdi_acld.c` for `ac`, but `osdi_pzld.c` did for neither: the row was identically zero, the matrix singular at EVERY trial `s`, every trial looked like a root, and `pz` blamed the netlist ("the input signal is shorted on the way to the output", naming neither cause nor device). This predates (1) for a model that CONTRIBUTES the value -- its row was already live -- and (1) would have widened it to every observed-only model. The AC stamp cannot be reused: it is exact and bounded there ('

E-417 *src/osdi/osdisetup.c, src/osdi/osdiparam.c, src/osdi/osdiinit.c, src/osdi/osdiregistry.c, src/osdi/osdidefs.h, src/include/ngspice/osdiitf.h, src/spicelib/analysis/cktsens.c, src/frontend/breakp2.c, examples/collapsestate_examples/ (new)*

A node collapse is decided once in `setup_instance`, and `OSDItemp` re-decides it on every temperature update without being able to rebuild anything derived from it. Three consumers reported over the mismatch. (1) `sens` printed `ROUNDOff` as a derivative: with the collapse taken, four Jacobian entries map onto ONE matrix element, so the perturbed device's $+g/-g$ ($g = 1/\delta = 1e6$) cancel on that diagonal and what survives is $\text{eps} * E / (Y * \delta^2)$ -- $2.5289e-02$ where the true derivative is $2.2676e-04$ (112x; the hunt's original case was 14000x and sign-flipped), with a magnitude tracking $1/\text{rd}$ and a `SIGN` that moves with unrelated parameters. It cannot be computed here -- `DEVsetup` runs only at the base value and re-running it would allocate nodes and trip `cktsens`'s own `CKTlastNode` guard -- so it is reported as 0 with a warning naming instance and parameter, leaving every other row of the table untouched. (2) A `.dc temp` sweep ran the whole range on ONE topology, in both directions: sweeping INTO a collapse leaves the terminal floating (exactly 0 A), sweeping OUT of one keeps the collapsed topology so the series element never appears (1 mA where the truth is 500 uA -- a plausible 2x error, the more dangerous of the two). `.temp` and `set temp` were always right because each re-does the setup; the sweep now warns once per instance, naming instance, model type and temperature. Rebuilding mid-sweep is not a fix but a re-architecture (nodes would be allocated and deleted, `SMPmakeEl`t called on an ordered and factored matrix -- a hard no under KLU -- and every other device's cached element pointers invalidated), and a hard `ERROR` would mean propagating five bare `CKTtemp(ckt)`; returns in `dctrcurv.c`, changing error handling for every device model in the tree. (3) `.options savecurrents` expanded `@dev[i_<term>]` at three or more terminals but bailed out at exactly two, where `@dev[i_p]` stayed a length-1 `SCALAR` and `meas/plot` silently used one point -- the shape E-413 existed to remove, surviving at a different arity. The expansion is now unconditional and the bare `@dev[i]` E-394 defines is `KEPT` beside it, with synthesized names de-duplicated against explicit `.save` entries. The detection has one trap that is the whole fix: `OpenVAF`'s collapse callback only ever `STORES TRUE`, so `OSDItemp` must `CLEAR collapsed[]` before re-deciding or a collapse that stopped happening is invisible -- precisely the direction the defect takes -- and must restore the snapshot before the error check, because the matrix implements the snapshot. E-416's own "deliberately not changed" note is corrected in place: its $\sim 2e-3$ terminal-vs-source gap was defect (3) measuring a scalar against a waveform (the plain `R`

E-416 *src/osdi/osdiparam.c, src/osdi/osdisetup.c, src/osdi/osdiregistry.c, src/osdi/osdiinit.c, src/osdi/osdidefs.h, examples/collapsecur_examples/ (new)*

A terminal current read exactly 0.0 whenever that terminal was `COLLAPSED` onto an internal node. `if (rd == 0) V(d,di) <+ 0; else I(d,di) <+ V(d,di)/rd;` is the standard compact-model idiom for an optional series resistance and `rd = 0` is the shipped default (BSIM `rdsmo=0`, `HICUM re=0`), so the broken path is the `COMMON` one. The model writes its current into `di`'s residual and terminal `d`'s own slot stays zero, while `OSDIask` read only `descr->nodes[t].resist_residual_off` -- 0.0 for a terminal carrying the device's full 2.001 mA,

through `@dev[i_<term>]`, `show` and `.options savecurrents` alike. With `rd` AND `rs` both zero EVERY terminal read 0.0 and the reported currents summed to zero, so a Kirchhoff check passed on `0 = 0`. The solution was never wrong: `osdload.c` stamps a collapse group into one matrix row, so the collapsed and uncollapsed forms give identical node voltages and source currents -- which is why nothing a user plots ever moved. `OSDIask` now sums the terminal's whole collapse group (a group holds at most one terminal, since `collapse_nodes` refuses to merge two simulator-allocated terminals), resistive residual plus the integrated reactive residual in a transient, advancing the state cursor past non-members exactly as `osdload.c` does -- the reactive half is load-bearing, not a refinement: 1 nF at 1 V/us makes the terminal current ~1 mA of pure displacement current against a few uA of conduction. The grouping cannot be recovered after setup, because `write_node_mapping` replaces the local mapping with GLOBAL node numbers and those cannot express a group -- two terminals on one net share a global number without being collapsed, and ground additionally collects grounded terminals, nodes collapsed to ground, E-116's decoupled internal nodes and E-401's dropped term-short flow nodes (grouping by global index makes a grounded terminal read 1.1 mA where 1.0 mA is correct). It is therefore snapshotted the moment `collapse_nodes` returns, into one `uint32_t` per descriptor node trailing `OsdiExtraInstData`, holding `t + 1` so the calloced zero still means "setup has not run". An uncollapsed terminal owns only itself, so every uncollapsed number is bit-identical to before; an unconnected terminal (E-402) is unowned and reports 0.0. A differential over the 40 real compact models in `integration_tests/` shows how common the broken path was: 23 bit-identical, 14 differing in terminal-current lines only (all previously all-zero: HICUM/L2, PSP103/_nqs/103t, ASMHMT, BSIM3, BSIMBULK, BSIMSOI, HiSIM2, HiSIM-HV n4/n5, DIODE, DIODE_CMC, BSIMCMG), and nothing else moved anywhere -- the models that were already right (BSIM4 `rdsmode=1`, MEXTRAM 505, VBIC 1.3, BSIM6, PSP102, JUNCAP200, EKV, MVSG) are bit-identical, as an uncollapsed terminal owning only itself requires. One cosmetic consequence: the sum accumulates into `0.0`, so a residual of exactly negative zero now prints `0` rather than `-0` in `print/show` (numerically identical; nothing in the suite depends on it). Verify (examples/collapsecur_examples): 31 checks over four `rd/rs` combinations, a three-node collapse chain, a ground-collapsed internal node beside a grounded terminal, a capacitive transient, `show`, `savecurrents` and E-397's instance knobs; 13/31 on the pre-416 binary. The grown instance block was checked rather than assumed: size-vs-offset recomputed against all 589 descriptors in the repo, and the suites re-run under guard `malloc` (faults on any access past the allocation) -- clean. Both solvers; full regression 333/333.

E-454 *src/frontend/inp.c, src/frontend/subckt.c, src/spicelib/parser/inp2n.c, examples/autobusopt_examples/(new)*

ONE OPTION, TWO READERS, TWO ANSWERS. `.option autobus` is decided in two places and they disagreed, so the same card switched the feature ON inside a subcircuit and off at the top level, or the reverse, depending only on the spelling. (1) The SUBCIRCUIT reader (E-449) must read the option cards directly, because the variable is not published until after expansion. Its `strstr` plus token-boundary test correctly ignored `noautobus`, `myautobus` and `autobusx` -- but `=` is neither alphanumeric nor `_`, so it passed as a clean terminator and the value was never read: `.option autobus=0, =false` and `=no` each switched the feature ON, silently. E-450 had already fixed this exact shape for `savecurrents`, which gets all four spellings right; `autobus` was its unguarded sibling. (2) The TOP-LEVEL reader (E-444, in `INP2N`) asks `cp_getvar(..., CP_BOOL, ...)`, but the spelling decides the published TYPE -- bare is a `BOOL`, `autobus=1` a `NUMBER`, `autobus=true` a `STRING` -- so only the bare form was ever seen and `.option autobus=1`, an ordinary way to write a boolean option and NOT reported as unknown, left a top-level bus port unbound. Both readers now answer by the same words: `e454_value_is_off` is the single list of off-words and `savecurrents` was moved onto it so the two cannot drift again. Precedence is deliberately matched to the OPTIONS MACHINERY, not to `savecurrents` -- within one card the later token wins, but across cards `ngspice` keeps the FIRST, and since the top-level reader can only follow what is published, making the card reader last-card-wins (as `savecurrents` is) would

put the two paths back into disagreement; both orders are pinned. A deck card now also beats a set autobus from .spiceinit, which used to be OR'd so an init file could turn the feature on and no deck could turn it off. (3) A bus could be bound by name at the level that DECLARED it but not passed down: Xi a b inner failed with "too few nodes" because the expansion was allowed on an OSDI device line only. It now applies to a subcircuit call too, so a bus travels by name through any depth. The subtlety is the node budget -- numnodes for an OSDI line counts the TOKENS written, so a token costs one however many nodes it expands to, while for an X line it is the CALLEE'S FORMAL COUNT, so a token standing for five nodes must spend five; e449_expand_bus_port therefore returns the number of actuals emitted instead of a flag. A short X call at the top level still fails cleanly (no enclosing formal list to take bits from), and with the option off X lines are untouched. The audit also confirmed much of the feature CORRECT and left it alone: ascending and descending .subckt bus declarations both match the explicit spelling, a bus port that is not first, a one-bit bus, array instances (E-441), the >1024-bit guard, and the sparse/over-wide formal lists which warn and error. Verified autobusopt 27/27 both solvers vs 15/27 on the previous binary, the twelve failures being exactly the defect checks while every control passes on both; autobus, subbus, busportsub, busdir, busoverflow, busname, busnodes, sweepbus, savecur, savecuroff and optname all still pass; regression 368/368.

E-478 *src/frontend/com_sweep.c, src/frontend/spec.c, src/frontend/evaluate.c, src/frontend/fourier.c, src/frontend/outitf.c, examples/guardmatch_examples/ (new)*

FIVE DEFECTS OF ONE SHAPE from bug-hunt round 46: a check was performed and then something ELSE was used -- a different parser, a different end of the range, a different lookup, or a different loop's state. (1) A COUNT WAS VALIDATED WITH A FLOAT PARSER AND CONSUMED WITH **atoi()**, which stops at the e. sw_isfinitenum accepts 2e2 as 200 and 1e6 as 1000000; atoi returned 2 and 1, and the float the validator had already computed was thrown away. So sweep lin 2e2 ran TWO points, montecarlo 2e2 drew two samples and printed a yield computed from them, highsigma 2e2 the same, and sweep lin 1e6 ran ONE point while the identical number written 1000000 was correctly refused as too many. Five sites read a count this way (sweep, montecarlo, highsigma, wcd -maxiter, wcd -is); they now share sw_count_arg(), which reads the value the validator produced. The same block also SILENTLY REWROTE a count below 1 -- for lin a one-point run, for dec/oct a silent change of the SPACING rather than the count, and on a -vs knob the collapse of a whole sweep dimension -- while every sibling names it (ac number of points is invalid, dc, tran, montecarlo, highsigma, wcd, .for). E-270's too many points wording is kept verbatim because sweepbounds_examples pins it. (2) **spec** CHECKED ITS STEP AGAINST THE SPAN BUT NEVER AGAINST ZERO (stepf > stopf - startf, upper end only): a negative step makes fpts = (stopf-startf)/stepf + 1 NEGATIVE, that count reaches the allocator (can't allocate -784 bytes), and the returned NULL is taken into the fill loop -- a DETERMINISTIC SIGSEGV (EXC_BAD_ACCESS at 0x0 in **com_spec**), present in the shipped binary; threshold at about -1e4. The guard now also requires finite and positive. (3) INDEXING A **@dev[param]** WAVEFORM RETURNED THE DEVICE'S LIVE VALUE, NOT THE ELEMENT: e448_literal_index() builds the name <base>[<k>] and asks vec_get() for it -- a clean miss for an ordinary base, which is what makes E-448's probe safe -- but **vec_get** answers a name beginning with @ from the DEVICE rather than the plot, so @c1[i][50] was read as *parameter i of device c1*, the index discarded, and every element of a saved waveform read back as the value after the run (@c1[i][50] and @c1[i][218] were EQUAL). length(), maximum() and wrdata were right all along -- they take the vector whole and never build that name, which is why this survived; E-441's @r[2][resistance] is unaffected because the lexer keeps it in ONE token. (4) **fourier** GUARDED A FUNDAMENTAL BELOW THE TIME SPAN AND NOT ONE ABOVE THE SAMPLE RATE, printing a full report whose every normalised magnitude was nan, summarised as *THD: nan %*, in silence. It now WARNS rather than refuses, because E-445's suite pins that a large but FINITE fundamental still runs -- the run stands and the nan is explained. (5) A LOOP COMMAND RUN AS ANOTHER'S **-analysis** OVERWROTE THE SHARED PROGRESS STATE (E-477): the inner begin() took the outer's label, total and index and the inner end() cleared outp_

loop_active for both, so the outer bar vanished and the line was left reading *montecarlo: sample 6/6 100%* while the outer sweep still had points to go. A nested begin is now counted and ignored so the OUTER loop keeps the line -- the same reasoning that made this feature not reuse the per-analysis bar. Display only; the numbers were byte-identical throughout. NOT CHANGED, pinned by checks: `.four 1e30` still runs, and `psd 0` still clamps to 1 -- because it ANNOUNCES it (*Number of averaged data points: 1*); round 46 reported that as a sixth, silent clamp and was WRONG, because the probe filtered for error/warning keywords while this is a plain informational line. Suite guardmatch_examples 21/21 (4/21 against the shipped pre-fix binary, the four being exactly the pinned decisions).

E-477 *src/frontend/outitf.c, src/frontend/com_sweep.c, src/include/ngspice/ftext.h, examples/loopbar_examples/ (new)*

A PROGRESS LINE FOR THE COMMANDS THAT RUN N ANALYSES IN A LOOP -- sweep, montecarlo, highsigma, wcd. All four set `ft_optimizing` to silence per-point chatter (Enhancement-130) and `outp_print_reference()` returns early on that flag, so all four printed a banner and then nothing at all: a forty-point sweep of a slow transient looked hung for minutes. LETTING THE INNER ANALYSIS BAR THROUGH IS THE OBVIOUS FIX AND THE WRONG ONE -- it runs 0 to 100% for EVERY point, so it resets N times and never answers the question being asked (how far is the SWEEP), and it redraws the same terminal line with a carriage return, so an outer bar and the inner bar would overwrite each other. But the inner number is not worthless either: with three points and a one-minute transient each, an outer-only bar sits at 1/3 and reads as frozen. So ONE LINE CARRIES BOTH -- sweep: point 7/40 [=====] 17% (tran 63%) -- and the inner fraction also advances the OUTER bar within the point, so it moves smoothly instead of stepping once per analysis and the percentage is continuous across a point boundary (32% at the end of point 1 of 3, 34% at the start of point 2). TWO DRIVERS ARE REQUIRED AND NEITHER COVERS BOTH REGIMES: while a point runs the loop command is blocked inside the analysis, so only `outitf.c` -- on the analysis's own data path -- can refresh; but the DEFAULT analysis is **op**, which produces no swept data points and never reaches that path, so the command also draws at each point boundary. The suite pins one driver each: a check that the percentage advances WITHIN point 1, and a plain `op` sweep where only the boundary driver can fire. Both draws use the same 0.25 s throttle as the analysis bar, since a sweep may run to `SW_MAXPTS` -- a 4000-point `op` sweep finishing in 69 ms draws twice, which is correct. **wcd** IS INDETERMINATE: it iterates a Hasofer-Lind search to convergence, usually well before `maxiter`, so it gets a counter rather than a bar that would sit low and jump to done; its line is released right after the loop, because the `ft_optimizing` restore it first used runs at the END of that function and put the last frame UNDERNEATH the worst-case distance it was meant to precede. THE FIRST POINT'S FRAME IS NOT DRAWN: a frame ends with a carriage return, leaving the cursor at column 0 of a line filled to a constant width, so anything printed before the next redraw overwrites only its LEADING columns and the rest survives as a tail -- and the first analysis prints the solver announcement, which is shorter than a frame (Using SPARSE 1.3 as Direct Linear Solver] 0%). Every LATER point is safe because its frame follows the preceding analysis's output; only the first has nothing in front of it, so it waits for the first intra-point refresh or the second point's boundary. The check that pins this had to be written twice: the residue does NOT exist in the byte stream (the text after the last carriage return is just the banner, so the line must be COMPOSITED the way a terminal does), and **subprocess(text=True)** applies universal-newline translation that turns every carriage return into a newline before the check can see it -- both earlier versions passed against the build that had the bug. Auto-enabled only when `stdout` is a terminal (the line is redrawn in place; a redirected run would collect one enormous line of frames -- the existing analysis bar does NOT make that distinction, which is why capturing a plain `tran` to a file yields a screenful of frames); `loopbar/no-loopbar` force it either way and all eight spellings are tested, because Enhancements 450, 451, 454, 466 and 467 each shipped an option here where a spelling meaning OFF turned the feature ON. Two checks allocate a real `pty`, since auto-on-a-terminal is the default path and is

otherwise indistinguishable from forced-off. NOT CHANGED: a standalone tran/ac/dc/noise keeps its own bar (progressbar_examples still 22/22), optimize stays silent, and a sweep's numbers are byte-identical with the bar on and off. outp_bar_shown is deliberately left alone by the new branch so outp_finish_reference() cannot stamp a stray 100% "Reference value" line over the loop line at the end of every point. Suite loopbar_examples 18/18 (5/18 against the shipped pre-fix binary).

E-476 *src/osdi/osdinoise.c, src/osdi/osdiparam.c, src/osdi/osdiload.c, src/osdi/osdidefs.h, src/frontend/spiceif.c, examples/reportguard_examples/ (new)*

FOUR DEFECTS OF ONE SHAPE from bug-hunt round 45: the simulator's account of itself did not match what it does. None raised an error. (1) AN OSDI OPERATING-POINT VARIABLE ANSWERED WITH A NUMBER WHEN NOTHING HAD COMPUTED ONE -- the opvar storage lives in the calloc'd instance block, so a read after *op simulation(s) aborted*, with no analysis run at all, or after a \$fatal killed an evaluation part-way, returned a clean 0.0 -- and 0.0 is a perfectly ordinary current, voltage or conductance, so nothing distinguished it from a result. **i(v1)** in the SAME **print** correctly reported *vector ... is not available*, and the equivalent built-in read produced nothing -- OSDI was the only channel manufacturing a plausible answer. ngspice already had the rule and applies it elsewhere: **param_forall()** in **device.c** will not even ask for an ask-only parameter unless **ckt->CKTrhsOld** exists, which is why **show** never displayed a fabricated opvar -- the direct @dev[opvar] read was the one path without that guard. A per-instance bit is now set in eval(), the SINGLE FUNNEL every evaluation passes through, gated on \$fatal because an evaluation that raised one was abandoned part-way; OSDIask returns E_NOTFOUND (the parameter exists, it has no value) and doask() lets the handler's own message stand instead of adding a vaguer one. Parameters are deliberately NOT gated -- they are inputs. (2) EVERY OSDI DEVICE'S INTEGRATED NOISE TOTAL WAS ADVERTISED AND UNREACHABLE -- onoise_total_<dev> and inoise_total_<dev> were built with a " " suffix instead of "", so tprintf stored the name with a TRAILING SPACE: display pads the name column so the blank is invisible and the vector looks present and *1 long*, while every read matches literally and misses. Its own N_DENS sibling eleven lines above already passed "", and every built-in passes a names array whose element 0 is "" -- which is why **onoise_total_r9** read back and only OSDI's did not. The per-source children and the grand total were unaffected, so a noise run looked complete and only the per-device attribution -- the number a contributor ranking is built from -- was missing. These were the ONLY two occurrences of " " in the whole src/osdi/ tree. (3) AN INSTANCE-SCOPE WRITE TO A ROUTED MODEL PARAMETER WAS DISCARDED IN SILENCE -- E-397 routes ngspice's own dt/dtemp/temp knobs onto the model's parameter when the model declares one (deliberate: PSP, MEXTRAM, VBIC, HiSIM and BSIM all declare dtemp), and that also places the name in the instance table, so alter @n1[dtemp]=20 found it, stored the value in inst->dt where nothing reads it once the model owns the name, and said nothing while @n1[dtemp] went on reporting the old value. Every other model-scope parameter reaching that setter already returned **E_BADPARAM** -- only the two routed names were quiet. param < num_params separates a routed id from the ids the loader synthesizes above the parameter/opvar/terminal ranges; it is now refused with E-467's wording, issued from OSDIparam because doset() discards the setter's return code at the alter call site. An INSTANCE-scope dtemp -- the corpus spelling -- never reaches that branch and is unaffected, altermod still works, and the READ still routes to the model's parameter by design. NOT CHANGED, pinned by checks: the routed read, case-sensitive \$simplparam matching, and an opvar staying readable after a LATER analysis fails once one has succeeded. LEFT OPEN: @n1[mul] reads 0.0 before any analysis rather than its declared default, because OSDI parameter defaults are applied during setup -- same family, predates this work, needs its own evidence. Suite reportguard_examples 31/31 both solvers (17/31 against the shipped pre-fix binaries); full regression 390/390.

E-475 *src/spicelib/devices/vsrc/vsrcload.c, src/spicelib/devices/isrc/isrcload.c, src/spicelib/analysis/transetp.c, src/spicelib/analysis/cktsopt.c, src/frontend/inpcom.c, src/frontend/measure.c, exam-*

ples/explicitvalue_examples/ (new)

SEVEN DEFECTS OF ONE SHAPE from bug-hunt round 44: a value the deck STATED was discarded and something else quietly put in its place, or a refusal named a fault other than the one it found. None raised an error. (1) **sin** WITH AN EXPLICIT **freq=0** TOOK ITS FREQUENCY FROM TSTOP -- `functionOrder > 2` already asks the only question that matters (was a frequency SUPPLIED?), but an `&& coeffs[2] != 0.0` made an explicitly-written zero fall through to `1/CKTfinalTime`, so `sin(0 1 0)` became ONE CYCLE PER SIMULATION and lengthening the run changed the stimulus (first 0.9 crossing 1.78/3.57/7.13/17.8 us for TSTOP 10/20/40/100 us, exactly linear; an explicit 50 kHz stays at 3.56 us). **TD** and **THETA** on the next two lines test only **functionOrder** -- **FREQ** was the odd one out, **PULSE** directly above **DOCUMENTS** its own zero-substitutions in a comment and **SINE** had none, and unlike a zero rise time a zero frequency is meaningful: it is DC. Omitting the frequency still defaults to `1/TSTOP`. The identical line in `is-rcload.c` is fixed with it. (2) AN UNKNOWN PARAMETER ON A SUBCIRCUIT CALL WAS SILENTLY DROPPED -- `X1 a 0 div rr=5k` against `.subckt div p n r=1k` built a 1k divider, the default, exactly as if nothing had been passed; one transposed character and the circuit is not the one written. The two sibling constructs both speak: an unknown param on a **.model** WARNS and on a device instance is a hard ERROR -- only the `subckt` call was quiet. Now a warning (not an error: the name is unusable in the body either way, so nothing that works stops working); `m exempt`. (3) A FAILED **.measure** LEFT THE PREVIOUS ANSWER UNDER ITS NAME -- and in a loop that is the last iteration's number standing in for a measurement that never happened, undetectable because **sim_status** is not set by **meas**. The name is now dropped, so a read after a failure says *not available* -- which is already what happens when the FIRST measurement fails; only that case was honest before. Same family as E-438. (4) **tran** TMAX HAD NO VALIDATION AT ALL: a negative one reached the integrator and came back as *singular matrix: check node b*, blaming a trivial RC that solves fine with any other TMAX, while TSTEP/TSTOP/TSTART all say *is invalid, must be...* (5) SIX OPTIONS ACCEPTED NONSENSE IN SILENCE -- `pivtol/pivrel` were the only tolerance-family members with no check (`reltol/abstol/vntol/trtol/chgtol` refuse `<=0`, `gmin/gshunt` refuse negative) and `minbreak/srcsteps/gminsteps/ramptime` had none while sibling counts `itl1/itl2/itl4` refuse a negative; all six now use the same `E426_BAD_OPT` path, `pivrel` additionally refusing `>1` because Sparse silently clamps anything above 1 back to its own default. Zero stays legal wherever zero is its meaning. (6)+(7) TWO E-474 REFUSALS NAMED THE WRONG FAULT (defects in an enhancement shipped hours earlier, contradicting its own one-fault-one-message rule): every unevaluable `{{ }}` reported *outside any .for loop* -- 13 ways of being wrong, all INSIDE a loop -- now split by asking whether anything remains that a later binding could fill in (`{{1/0}}`, `{{1+}}`, `{{}}`) named where they stand; a surviving NAME left for an inner loop and, if still there at the end, reported as *never resolved*; only a deck with no `.for` at all keeps the original message); and a nested `.for` reusing the enclosing index produced duplicate lines and died with *device already exists*, naming the symptom -- now refused during the walk that already looks for the matching `.endfor`. THREE LOOKALIKES FROM THE SAME ROUND ARE DELIBERATELY NOT CHANGED, each re-confirmed by READING THE CODE rather than the behaviour: negative R/C/L stay unflagged (E-438: *negative passives are the very idiom this project's own examples use*), `pow(-4,0.5)=2` and `1/0=1e32` stay finite (E-256/446: *a NaN here poisons the Newton Jacobian*), and `pulse tr=0` still takes the timestep (documented in a comment above the code) -- checks [20]-[22] pin them so a later round does not fix them. LEFT OPEN: no `pivtol` value changes a result on either solver, including `1e30` which should reject every pivot, though the plumbing into `spOrderAndFactor's` (Rel,Abs) order is correct and `AbsThreshold` IS consulted in the pivot search -- validating an input is a separate question from whether it then does anything. Suite `examples/explicitvalue_examples/ 41/41`, both solvers; full regression 389/389.

E-474 *src/frontend/inpcom.c, examples/forloop_examples/ (new)*

NEW **.for** / **.endfor** NETLIST CONSTRUCT. A deck often needs a run of near-identical instance lines differing only by an index -- a ladder, a stack of periodic sections, a bank of taps -- and writ-

ten out by hand they are long and wrong in ways that are hard to see: one node name out of step in the middle of forty lines still parses, still simulates and still gives an answer. `.for i in range(1,4) + XP{{i}} P{{i}} P{{i+1}} hl_periodic n1={nL} + .endfor` now expands to exactly the four instance lines. Forms: `range(first,last)`, `range(first,last,step)` and an explicit list `[7,2,9]`. **range** INCLUDES BOTH BOUNDS -- **range(1,4)** is FOUR iterations, not Python's three -- deliberate, because an index range in a netlist reads as *1 through 4* the way a bus range does, and it has its own check because it is the one thing a Python-literate reader assumes wrongly. `range(3,1)` counts down; a step contradicting the bounds is refused rather than silently yielding nothing. `{{i}}` is the index and `{{expression}}` is integer arithmetic over it (+ - * / %, parens, unary minus), substituted as TEXT so it builds node names, instance names, `.model` names and parameter values alike. Loops NEST, and an inner bound may be an expression over an outer index (`range(1,{{i}})`) -- which falls out of the design: one binding is applied at a time, so `{{i+j}}` becomes `{{2+j}}` and waits for the inner loop. **WHY {{ }} AND NOT { }**: numparam owns single braces and the body carries BOTH kinds; doubling the brace needs no ordering rule from the user, and the pass removes every `{{ }}` it introduces so numparam never sees one. **WHY IT RUNS FIRST**: at the top of `inp_readall`, before the syntax check, the scope tree, numparam, bus and subcircuit expansion -- it emits ordinary netlist lines and nothing else, so every later stage sees what the user would have typed and none of them needs to know the construct exists (`.include/.lib` are already pulled in, so it works in an included file and inside a `.subckt`). The cost of that placement is that the BOUNDS CANNOT BE **.param** VALUES, and that is refused by name rather than misread. `.control` is skipped entirely -- it has its own `foreach`. ONE FORWARD PASS SUFFICES: copies are appended AFTER the closing `.endfor`, never into the body still being read, which is the use-after-modify E-441 records for the array expander; the walk then continues into the copies so a nested `.for` is reached later in the same pass. 14 refusals, each producing exactly ONE message -- a header that fails to parse also retires its own `.endfor`, or the walk reports a second error about a line that is perfectly correct. THE SUITE'S ORACLE IS THE HAND-WRITTEN DECK: every check runs the `.for` version and the lines it stands for and requires them identical -- expansion line for line, solved result character for character -- because *expands to something plausible* is the failure that matters and an analytic comparison would not catch a ladder wired one node out of step (a 2000-section ladder is built both ways). Suite `examples/forloop_examples/ 31/31`, both solvers; full regression 388/388.

E-473 `src/frontend/com_sweep.c, examples/mcarming_examples/ (new), examples/reuseloops_examples/verify_reuseloops.py`

THE MONTE CARLO FAST PATH ARMED ON A DRAW IT COULD NOT PUSH -- A SILENTLY WRONG AND UNSTABLE YIELD. E-346's fast path re-draws the random values and pushes them into the live circuit instead of re-sourcing per sample, which is sound only if EVERY use is pushed. Only a BRACED expression is ever captured, and NUMPARAM DECIDES THE BRACES: a quoted `rd='rv'` reaches the capture pass as `rd={ (agauss(...)) }` and is captured, but a BARE `v=rv` in a B-source reaches it as `v= ( agauss(...))` with NO BRACES -- `sw_fp_scan_valueline()` walked past it, found no swept token and reported the line ELIGIBLE. A B-source's value is substituted TEXTUALLY AT PARSE TIME, so nothing short of a re-source moves it and every sample after the first saw the FIRST draw. A 40-sample run whose spec depended only on that value reported 100% or 0%, DIFFERING BETWEEN RUNS OF THE SAME DECK AND SEED (the frozen value depended on surviving state), where the correct sampled answer is ~45% -- `rv ~ N(100, 60)` against a `0 <= v <= 100` spec is $P = 0.452$. Same family as E-438, where a Monte Carlo counted failed samples as PASSES. REACH: the deck must ARM at all, so it needs a capturable random value ALONGSIDE the uncapturable one (`.model rd='rv' plus B1 bs 0 v=rv`); a bare `.param` name is not legal as an ordinary device value (ngspice rejects `R1 a b rv`), so the hole is confined to expression-valued elements. FIX: a random draw OUTSIDE ANY BRACES is ineligible -- the identifier walk already rejected a SWEPT token found outside a brace, it now rejects a random call there too, so the deck disarms and takes the reset path. Decks that were always fine are untouched (a quoted `.model` value still arms, a braced device value still arms) and **v=rv** /

`v='rv' / v={rv}` -- the same circuit written three ways -- now AGREE. THE **-warm** GATE, AND WHY E-188's SUITE CAUGHT IT: not tearing the circuit down also leaves the previous sample's SOLUTION in place, which WARM-STARTS the next one -- and E-188 made that OPT-IN deliberately (it keeps its guess in a buffer OUTSIDE the CKTcircuit precisely because a reset destroys the solution, and DCop asks MODEINITJCT rather than MODEINITFLOAT when it is off). An unconditional reuse cut E-188's COLD path from 20606 to 1416 iterations per sample -- the homotopy that option exists to avoid, avoided WITHOUT BEING ASKED. Every yield still matched exactly (215/215, 214/214, LHS 258/258, KLU 215/215), so nothing was wrong -- but a starting point decides which operating point a sample finds on a circuit that has more than one, and MC samples are meant to be independent. Silently turning an opt-in on is the fault E-450/451/454/466 each shipped once already, so the reuse is offered exactly where the user already asked for state to carry between samples. WHAT IT UNBLOCKED: E-472 gave optimize the E-471 setup reuse but withheld it from montecarlo precisely because the per-sample teardown was all that limited this defect's damage. With arming honest, arming MEANS every varying value is pushed -- the guarantee sweep already relies on -- so montecarlo keeps the circuit standing too (1.29x on a 1200-instance ladder, yield unchanged) -- BUT ONLY UNDER **-warm**, with every E-471 guard unchanged: *setup reused at 8 of 20 samples, 11 rebuilt after a node collapse moved*, yield 50% with the reuse on and off alike on a deck whose spec passes only a COLLAPSED sample. Suite examples/mcarming_examples/ 20/20, both solvers; E-472's two montecarlo checks flipped from *never reuses* to *does*; full regression 387/387.

E-472 `src/frontend/com_optimize.c, src/frontend/com_sweep.c, src/frontend/com_sweep.h, examples/reuseloops_examples/ (new)`

E-471 GAVE THE SETUP REUSE TO **sweep** AND NOTHING ELSE. optimize never re-sources the deck for `-param/-mparam` knobs -- nor for `-dparam` once E-322's fast path arms -- yet still tore down and rebuilt a circuit it had not touched on EVERY evaluation, hundreds of times in a fit. It now asks for the same reuse on the same terms: only when a circuit is standing that this evaluation did not re-source, never after an analysis that failed, never under `-center` (its inner Monte Carlo resets per sample). Every guard is E-471's, unchanged. THE WORK WAS PROVING IT HOLDS WHEN A SEARCH STEP MOVES THE COLLAPSE, NOT A SWEEP POINT -- E-417's `cs_gate` collapses at exactly `rd == 0`, which a sweep can step onto but an optimizer never lands on, so a new `cs_thresh` collapses below a THRESHOLD and a search range straddling it moves the topology on its own: *setup reused at 15 of 17 analyses, 1 rebuilt after a node collapse moved*, with the identical residual, evaluation count and parameter as the reuse-off run. THE TRAJECTORY QUESTION IS REAL BUT BOUNDED: on a well-posed fit `lm` and `nm` both return the identical parameters to 13 digits in the identical evaluation count (the FD Jacobian step is $1e-3$ normalised against a $5e-12$ perturbation, so the derivative error is $\sim 5e-9$), but a DEGENERATE fit -- unreachable target, pinned at a bound -- took 8 evaluations with the reuse and 17 without, same residual to ten digits, because LM's accept/reject test was comparing costs that agree to $1e-12$ and its damping diverged on noise. **montecarlo** WAS BUILT, MEASURED AT 1.29x, VERIFIED COLLAPSE-SAFE (11 rebuilds in 20 samples, the collapse-sensitive yield tracking draw for draw at 50%) AND THEN TAKEN BACK OUT: its fast path can ARM WHILE A RANDOM `.param` STILL HAS A USE IT CANNOT PUSH (a B-source value is substituted textually at parse time), and the per-sample rebuild is only what accidentally limits the damage. THAT DEFECT IS ALREADY LIVE WITH NO REUSE INVOLVED -- on the SHIPPED binary such a deck reports a 100%/0% yield that DIFFERS BETWEEN RUNS OF THE SAME DECK AND SEED, where re-sourcing every sample gives the correct 45%. Seeding the fast path's closure from the random `.param` names -- what makes the equivalent sweep disarm -- did NOT disarm it, so the fix lies further in, in what pass 2 accepts as a capture; it needs its own change and its own evidence. montecarlo is left exactly as it was and the suite asserts it, so it cannot be switched on by accident. highsigma/wcd re-source every sample by design and never have a setup to keep. Suite examples/reuseloops_examples/ 17/17, both solvers; full regression 386/386.

E-471 *src/spicelib/analysis/cktdojob.c, src/frontend/com_sweep.c, src/osdi/osdiparam.c, src/osdi/osdisetup.c, src/include/ngspice/cktdefs.h, src/include/ngspice/osdiitf.h, examples/reusesetup_examples/ (new)*

A REPEATED ANALYSIS TORE THE WHOLE CIRCUIT DOWN AND BUILT IT AGAIN FOR EVERY POINT even though only a parameter VALUE had changed. .dc has never done that -- including the parameter sweeps of E-427 it sets up ONCE and walks its points inside the analysis. After E-470 removed the quadratic teardown, what remained of that per-point rebuild was still most of the run: a 1001-point sweep over a 2448-unknown stack, 7.24 s -> 0.57 s (12.8x) under SPARSE with BYTE-IDENTICAL results (all 5005) and 6.35 s -> 1.38 s (4.6x) under KLU, where reusing the matrix ordering can put a solve on a slightly different Newton path -- exactly as it can under .dc, which has always reused both -- and moved ONE number of 5005 by 4.7e-09 relative, five orders of magnitude inside `reltol`. THE TWO SOLVERS SWAPPED PLACES: KLU was the faster of them on this deck before the change (6.35 s vs 7.24 s), but reuse lets SPARSE skip the `spOrderAndFactor` E-470's profile left dominating at 51% and KLU gains less from the same trick, so SPARSE is now 2.4x the faster. With E-470 the deck went 29.78 s -> 0.57 s. WHY IT COULD NOT SIMPLY BE DONE: NODE COLLAPSE. A device may decide, at setup and from its parameters, to merge two of its nodes, and the matrix is built for that topology; reuse the setup and the topology FREEZES at whatever the first point decided. A FIRST VERSION DID EXACTLY THAT -- sweeping `cs_gate's rd` across the value that collapses its internal node returned `0.5 0.5 0.5 0.5 0.5` instead of `0.5 0.3333 0.25 0.2 0.1667`, no error and no warning, a plausible flat line. It was measured at 3.5x, confirmed numerically sound on the deck that motivated the work (which simply never moved the collapse) and REVERTED. THE FIX IS SAFE BY CONSTRUCTION, NOT BY ARGUMENT: on a reused point `CKTdoJob` skips the `unsetup/setup` pair but STILL RUNS `CKTtemp`, which for an OSDI device is what re-decides the collapse and compares it against the snapshot the matrix was built from (E-417) -- so collapse is ALWAYS re-decided, never assumed -- and if any instance reports a change it throws the reused matrix away and does a genuine `unsetup/setup/temp`. THIS IS THE THING E-417's OWN WARNING SAID WAS IMPOSSIBLE (*the matrix was built for the collapse decided at setup and cannot be rebuilt here*); the warning is suppressed while the request is live, with its once-per-instance flag left unburned. A BUILT-IN device decides its collapse in `DEVsetup` and nowhere else, so there is nothing to re-check and no honest way to detect a change: reuse is offered ONLY to circuits built entirely from types whose topology is known fixed (the linear elements, the four controlled sources, ASRC, the independent sources) plus OSDI, and anything else keeps the old behaviour exactly. A FAILED POINT HANDS NOTHING ON -- THE REGRESSION FOUND THIS, NOT THE DESIGN: a point whose analysis fails leaves a matrix factored to NaN and states half-written, and reusing it carried the wreckage into every later point -- E-445's `guardgaps` suite requires the FORBIDDEN points of a range-violating sweep to be NaN and the LEGAL ones to keep real values, and with the first cut all five were NaN (KLU failed the whole sweep). Reuse is now declined after any failed point; `sw_run_failed()` from E-438 already knew, so the fix was to carry its answer one iteration forward. Sweeps driving `tran`, `ac`, `noise` and `dc` are each checked identical on and off. THE PIVOT-ORDER REUSE IS NOT E-439's HAZARD: `NIiter` already refactors with the existing ordering between Newton iterations, and BOTH solvers force a full reorder on a singular refactor. Default on; `reusesetup=0`, `noreusesetup` and the set form turn it off -- and because E-450/451/454/466 each shipped an option here whose off-spellings silently meant ON, the value is read as a NUMBER and a STRING before `CP_B00L` and every spelling is a check. **set ngdebug** now reports the decision (*setup reused at 3 of 5 points, 1 rebuilt after a node collapse moved*), which is what lets the suite assert the MECHANISM rather than infer it from a clock the circuits are far too small to move. FOLLOW-UP FIX: the option WORKED off a `.options` card but was also reported as an *unknown option ... ignored* -- `if_is_option()` did not list it. That is exactly the failure E-445's note on that list describes (a warning that fires on a setting the run then honours teaches the user to ignore the check), and the deck timing is what exposed it: 0.55 s with the option vs 6.81 s, i.e. demonstrably not ignored. `reusesetup/noreusesetup` registered, and the suite now asserts no `.options` spelling is called unknown. Also hardened the REGRESSION

RUNNER: several suites need numpy/matplotlib, and an interpreter without them (most easily by putting /usr/bin ahead of the real python3 on PATH) did not say so -- they exited rc=1 in 0.0s and the sweep ended with ~22 unrelated-looking FAILURES (every pyplot suite plus a few that import numpy directly), which has cost a full 13-minute run more than once. run_regression.py now preflights those imports and exits 2 with one clear line naming the interpreter. Suite examples/reusesetup_examples/ 37/37, both solvers; full regression 385/385 at release, 387/387 after E-472/E-473.

E-470 *src/spicelib/analysis/ckdltln.c, src/osdi/osdisetup.c, src/include/ngspice/cktdefs.h, examples/teardown_examples/ (new)*

TEARING AN OSDI CIRCUIT DOWN WAS QUADRATIC IN ITS NODE COUNT. CKTdltNNum() finds the node it is asked to delete by scanning the circuit's node list from the HEAD, and OSDIunsetup() calls it once per internal node -- $O(k*N)$, and EVERY repeated analysis pays it: a sweep, an optimize, a montecarlo, anything that runs an analysis more than once. A profile of a 1001-point parameter sweep over a 2448-unknown circuit found 77% of the ENTIRE RUN inside it -- not in the solve, not in setup, but in the teardown BETWEEN points (com_sweep -> sw_run_cmd -> dosim -> if_run -> CKTdoJob -> CKTunsetup -> OSDIunsetup -> CKTdltNNum). THE SCOPE FOR THIS WORK WAS WRONG, AND INSTRUMENTING IS WHAT SHOWED IT: it was planned (and shipped as a scope note beside E-469) as *move sweep's loop into the analysis kernel the way .dc does*, on the theory that the ~20 ms/point gap between a sweep point (~30 ms) and an op analysis (9.4 ms) was per-point SETUP. That note put instrumentation first and said in as many words that the whole case rested on it -- the profile overturned the hypothesis, and the kernel-loop rewrite would have been a large, risky change aimed at the wrong 20 ms. The fix: OSDIunsetup() knows every node number it wants gone before it deletes any of them, so it marks them and ONE walk of the list removes them all -- $O(N)$ for the whole unsetup, in a new CKTdltNodeSet(); CKTdltNNum() is unchanged for every other caller. A SECOND PROFILE WAS NEEDED: the first version sized its mark array from ckt->CKTmaxEqNum, which SHRINKS as nodes are deleted, and OSDIunsetup() runs once per model TYPE -- so the second type's numbers sat above the now-smaller bound, failed the range test and fell back to the per-node path, giving 23% with CKTdltNNum still at 64%. The bound is now the highest node number actually present, taken by walking the list once. Measured per sweep point: 5 periods 1.7 -> 1.2 ms (1.4x), 10 periods 4.0 -> 2.3 ms (1.8x), 25 periods / 2448 unknowns 32.9 -> 7.6 ms (4.3x) -- the speedup GROWING with circuit size is the signature of removing a quadratic -- and the full 1001-point deck 29.78 s -> 7.12 s with byte-identical results. 7.6 ms/point is close to the 9.4 ms a single op reports, the honest ceiling here. Teardown fell from 80% to 9% of the run, what remains of it is IFdelUid rather than list scanning, and spOrderAndFactor now dominates at 51%. Reusing the matrix ORDERING between points is the next opportunity and is deliberately NOT taken: that is what klu_refactor does, and E-439 recorded it reusing the old pivot order with NO pivoting and NO singularity test, producing a NaN solve on a circuit SPARSE handled -- that hazard has to be answered in its own change. Verified teardown 11/11 both solvers, nine of the eleven checks about the numbers and the node bookkeeping rather than the clock (a teardown that frees the wrong node, frees one twice or leaves one behind corrupts the NEXT analysis instead of failing loudly), the two timing checks asserting SCALING rather than milliseconds; regression 384/384.

E-469 *src/frontend/dotcards.c, src/frontend/inp.c, src/frontend/spiceif.c, src/include/ngspice/ftext.h, examples/saveused_examples/ (new)*

.option saveused -- SAVE ONLY THE VECTORS THE CONTROL BLOCK ACTUALLY READS. An analysis stores every node at every point unless a save says otherwise. On a small circuit that costs nothing; on a large one it IS the run -- a 2448-unknown dielectric stack over a 201-point parameter sweep costs 104.73 s (521 ms/point) with everything stored and 7.22 s (35.9 ms/point) with a hand-written save of the four vectors it writes, a factor of 14.5 from one line the author has to remember and then keep in step with the wrdata beside it; nothing warns when the two drift apart. With .option saveused and no save line at all the same deck runs in 7.08 s (35.2 ms/

point) and its written results are byte-identical to the hand-saved run. WHAT IS COLLECTED IS DELIBERATELY WIDER THAN THE LETTER OF THE REQUEST: every `v(...)`, `i(...)` and `@dev[param]` reference ANYWHERE in the block, not merely the arguments of the output command, plus the plain node names given to one. The reason is correctness -- let `r = v(out) - v(mid)` followed by `wrdata f r` names only `r`, and `r` is not a node, so a narrow scan would leave the two vectors that build it unsaved and a deck that worked before would fail. Under-saving turns a performance option into a wrong answer, so the scan errs the other way; over-saving costs a little memory. It STANDS ASIDE ENTIRELY -- the run is exactly what it would have been -- for an explicit `save/.save` (the author has already said what they want, and quietly adding to it would make their line mean something other than what it says), for `all` as an argument, and for a block with no output command; the explicit-save case is asserted IDENTICAL with the option on and off. Runs in `ft_saveused()`, called immediately after `ft_dotsaves()` and immediately before the controls execute -- that ordering is the design: the explicit `.save` is already installed so the stand-aside test can see it, and no analysis has run so the list is complete before it is needed. The option cards are read in `inp.c` beside `autobus`, because the option variable is not published until `inp_dodeck()` runs -- reading it any other way is what E-454, E-462 and E-464 each had to repair here. NOT CALLED **autosave**, THE NAME ASKED FOR, BECAUSE E-192 ALREADY OWNS IT for `set autosave=<file>`, a checkpoint written when a transient is interrupted -- and a checkpoint FILENAME is a string that is not an off-word, so reading `autosave` as a boolean here would have switched vector filtering on for every deck using E-192's checkpointing, silently discarding nodes it never asked to lose. The first implementation did exactly that and its suite file overwrote E-192's; the tell was that the regression suite count did not rise when a suite was added. E-192's suite is restored and passing, and the two options are pinned independent. All ten spellings pinned on and off, per the four repairs (E-450/451/454/466) this class has already needed. TWO CORRECTIONS SINCE IT SHIPPED. (1) A WILDCARD ACCESSOR IS A KNOB, NOT A VECTOR: the scan takes `@dev[param]` from every line, and the commonest one in a control block is the sweep knob itself -- `sweep @*[wavelength_nm] ...` -- which `save` cannot expand, so a real deck that dropped its `save` line (exactly what the option is for) got "a wildcard device name is not expanded here, so this vector will stay empty" once per sweep point. Results were right; the noise was this feature's own invention. Wildcards are no longer collected, a NAMED accessor still is. (2) THE 14.5x NO LONGER HOLDS FOR SWEEPS, BECAUSE E-470 REMOVED ITS CAUSE: that factor was never the cost of STORING vectors, it was the quadratic node teardown between sweep points. On the same deck at 1001 points today: hand-written `save` 6.67 s / 88.4 MB, `saveused` 6.71 s / 83.6 MB, no `save` at all 6.72 s / 86.6 MB -- identical, and identical again at 5001 points. The surviving benefit is TRANSIENTS, where every timepoint accumulates into one plot instead of being torn down: a 601-node chain over 20001 timepoints peaks at 56.6 MB with no `save`, 9.2 MB with a hand-written one and 9.3 MB with **saveused** -- six times less, scaling with nodes x timepoints. Verified `saveused` 26/26 both solvers, comparing vector NAMES rather than counts so a check cannot pass on the right number of the wrong vectors; regression 384/384.

E-468 `src/frontend/com_fft.c`, `src/frontend/com_measure2.c`, `src/frontend/numparam/xpressn.c`, `src/spicelib/analysis/cktsens.c`, `src/spicelib/parser/inpdpar.c`, `src/spicelib/parser/inpgmod.c`, `src/spicelib/parser/inppas3.c`, `src/xspice/icm/analog/limit/cfunc.mod`, `examples/mathguard_examples/(new)`

SEVEN NUMBERS THAT WERE WRONG -- TWO OF THEM IN CODE WITH NO TEST COVERAGE AT ALL. **psd** reported a total power set by the WINDOW, not the signal: a constant 1 V has total power exactly 1 V², and **psd** said 1.499908 under the default hanning window, 1.362744 hamming, 1.726652 blackman, 1.215261 gaussian -- and 1.000000 under **specwindow=none**, which is what identified the cause. `fft_windows` scales every window for unit COHERENT gain, which is right for the amplitude spectra `fft` and `spec` produce; a PSD sums SQUARED bins and needs the POWER gain, `sum(w^2)/length`. Zero padding inflated it again by `N/length` -- E-241 corrected this normalisation from `N^2` to `length^2`, and the padding factor it removed belongs on ONE of the two lengths, not neither -- so the same signal reported 1.5 V² at one stop time and 3.0 V² at

another, a total power that depended on where the transient happened to end. Parseval over the padded sequence gives sum

E-467 *src/frontend/variable.c, src/frontend/subckt.c, src/frontend/inpcom.c, src/frontend/spiceif.c, src/frontend/com_measure2.c, src/frontend/numparam/xpressn.c, src/frontend/numparam/numpaif.h, src/spicelib/analysis/cktsopt.c, src/spicelib/parser/inpdpar.c, src/spicelib/parser/inp2n.c, examples/silentaccept_examples/ (new)*

ELEVEN SILENT ACCEPTANCES -- THE DECK SAID ONE THING AND NGSPICE QUIETLY DID ANOTHER. The root of a class four enhancements had patched one site at a time: an option's SPELLING decides its published TYPE (bare a BOOL, =1 a NUMBER, =true a STRING) and cp_getvar's coercion table had no CP_BOOL case at all, so all ~110 cp_getvar(..., CP_BOOL, ...) readers saw only the bare word. set sqrnoise=1 reported V/sqrt(Hz) where the bare word gives V^2/Hz; set interp=1 left 165 transient rows where the bare word interpolates to 101; set autostop=1 ran to the end. Answered once in cp_getvar, so e454_autobus_var and autoadapt_mode stop being the only readers that get it right; a number is on when non-zero, so =0 still means off. The fix found its own hazard: a cascade that leads with CP_BOOL to mean "was the bare flag given?" worked only while cp_getvar refused to coerce, so autoadapt_mode returned "on, quiet" before it could see =debug -- it now reads the STRING first, and two existing suites caught this, which is what they are for. One word, one unreachable feature: case OPT_DEFAS assigned TSKdefaultMosAD, so the source-area default could NEVER be set and asking for it silently overwrote the DRAIN area (defas=7e-10 gave @m1[ad]=7e-10, @m1[as]=0). An instance temperature below absolute zero was accepted although .option temp= has warned since E-426 and osdisetup.c makes the same check for OSDI: R1 ... temp=-300 answered -0.998 V from a +1 V source, temp=-1e6 the same and dtemp=-400 by the other road -- the guard sits in INPdevParse, the one place every built-in applies its name=value pairs, and -25 C stays an ordinary temperature. The KiCad bit spelling never reached the SUBCIRCUIT formals: E-462 made a bit a_0_ rather than a[0], INP2N learned it and subckt.c did not, so formals written the way KiCad actually emits them matched nothing and the device floated at 1.0 against 0.5238095 -- with no not-connected warning, because every terminal WAS connected, to nodes nothing drives; two readers of one rule for the third time in this feature (E-454, E-464, here), and the bracket spelling is still always accepted. .func sqrt(x) silently replaced the built-in for the whole deck (2236.07 became 100000, ln 621.46 became 150000, abs +1000 became -998); the user's definition still wins, checked against fmathS exported rather than copied. .adapt validated the adapter MODEL but not its NODE list, so one typo switched the feature off and the deck answered the unadapted 0.7560976; reported per member, so .adapt b, nosuchnode still adapts b, and naming a model that is also a device model is reported too while our own injected n_adapt<N>_ lines stay silent for idempotency. meas lost every WINDOW function over a .dc of a device parameter -- max/min/avg/rms/integ all said "out of interval" at any range and with an exact from/to, while find-when and when worked on the SAME plot -- because the dc scale was a fixed list of four names and a parameter sweep is called param-sweep, the plot's [default scale] all along; the list is now a fast path with the plot's own scale behind it, written once where it had been duplicated. And alter @dm[is] blamed the parameter ("no such parameter is") when the cause was that dm is a model. Four candidates withdrawn rather than shipped: the instance wildcard does NOT drop resistance writes -- the probe measured a voltage-divider ratio, invariant when @#* moves every resistor together, and the current shows it working (-5e-4 to -2.5e-4); out-of-range vector indexing does warn; and a NEGATIVE-GEOMETRY GUARD WAS WRITTEN, TESTED AND THEN WITHDRAWN -- .option defw=-1e-5 and an instance w=-1e-5 each flip i(vd) from -1.8e-04 to +1.8e-04, but E-438's .option warn_physics ALREADY reports both routes and deliberately KEEPS the value, and the new guard's ignore-the-value contract silenced it; E-438's own suite caught that, which is what it is for. Three of the four were rejected by re-verifying the measurement, the fourth by the regression suite -- both cheaper than shipping the fix. Verified silentaccept 42/42 both solvers, roughly half the checks pinning what must NOT change, including E-438's contract; regression 381/381.

E-466 *src/spicelib/parser/inp2n.c, examples/adaptquiet_examples/ (new), examples/autoadapt_examples/*

.option autoadapt IS QUIET BY DEFAULT, AND ITS OFF-WORDS FINALLY WORK. E-463 reported per NODE -- a line for every adapter injected and another for every node that did not qualify -- which on a deck with many shared bus nodes buried the run's own output. The reporting is now opt-in: **.option autoadapt=debug**. **ERRORS ARE NEVER SILENCED**: a missing or wrong-shaped adapter model, a name collision, the same node on both ports of one device, **autoadapt** without **autobus** -- each means the option cannot do what the deck asked for, and a deck that asked for an adapter and did not get one must not run on in silence. Only the two INFORMATIONAL classes moved behind **=debug**: the per-split note and the per-node did-not-qualify notices. Found while making the value mean something: E-463 never looked at the value, only at its PRESENCE, so every spelling that means off turned the feature ON -- **autoadapt=0**, **=false**, **=no** and **=off** each injected an adapter and moved $v(a[0])$ from 0.7560976 to 0.7590361, a deck silently gaining an adapter its author had just switched off; only **noautoadapt** worked. This is the FOURTH appearance of that defect in a sibling option: E-450 found **savecurrents=0** and **nosavecurrents** both turning it on, E-451 found **nocshunt=** moving a node by six orders of magnitude while printing "unknown option", E-454 found **autobus=0** meaning ON at the top level and off inside a subcircuit. The word list is the one they share, **e454_value_is_off** -- a new boolean-ish option in this code base should be assumed to have this bug until its off-words are tested. An unrecognised value (**autoadapt=bogus**) is reported once and then proceeds quietly, the same shape E-462 gave an unknown **autobus** style. Verified **adaptquiet 22/22** both solvers, every check testing the VALUE as well as the text -- because "quiet" has to mean the messages stopped and not that the adapters did; E-463's own suite now asks for **=debug** where it inspects what the feature did, 26/26 otherwise unchanged; regression 380/380.

E-465 *src/frontend/com_sweep.c, examples/sweepsubfast_examples/ (new)*

sweep STOPPED TEARING THE CIRCUIT DOWN FOR SUBCIRCUIT AND DERIVED PARAMS. **sweep** has two ways to move a **.param** knob: E-320's FAST path writes each point straight into the live circuit, while the fallback stages **alterparam** and issues a **reset** -- re-sourcing the WHOLE DECK once per point. Measured 1.58s vs 0.50s on a 3000-element deck at 201 points and 0.54s vs 0.07s on 800 subcircuit instances, the gap widening with both deck size and point count because the rebuild is $O(\text{deck})$ per point. Almost anything involving a subcircuit or a derived parameter fell off the fast path, and silently -- there is a banner when it arms and nothing at all when it does not, so one unused **.param b={rv*2}** anywhere in a deck turned a 0.50s sweep into 1.58s with no indication why. Now on the fast path: derived params and chains, a param passed on the X line, two instances carrying DIFFERENT expressions, nested X calls, a swept param in a **.subckt** header default, a derived **.param** inside a subcircuit, and a local shadow beside a real dependence. The obstacle was that a subckt param is bound PER INSTANCE: **r.x1.r1** and **r.x2.r1** share ONE template -- keyed by source line -- but need different values, so a single global **nupa_eval_expr("r")** cannot serve both. Nothing is evaluated per instance at run time: each captured expression is rewritten ONCE at build time into global names by walking the instance's scope chain outward exactly as **numparam** would -- the subcircuit's own **.params** first (they may reference its formals), then the formals bound to that call's actuals or to the header default when the call omits them, then the enclosing frame -- so **{r}** becomes **{(rv)}** and the point loop is untouched. A local **.param** shadowing the swept name resolves to its own constant, which is what makes the shadow correct rather than merely refused. **numparam** has already rewritten the call by the time the original deck is readable -- **X1 a 0 sub r={rv}** arrives as the positional **x1 a 0 sub {rv}** -- so the **.subckt** header supplies the parameter ORDER; E-442 met the same rewrite from the other side. Derived TOP-LEVEL params needed one more thing: matching used the swept names only, so a device carrying **{rd}** was never captured; it now uses the transitive CLOSURE of derived names, the values having been refreshed by **nupa_recompute_params** since E-320. SECOND HALF -- the reset path never

put the knob back. E-385 closed that hole for the fast path and left it open here: `alterparam` rewrites the DECK TEXT, and nothing re-sourced it, so the devices kept the LAST POINT's values and every later analysis was quietly wrong -- after sweep `rv 900 1100` a fresh `op read v(a) = 0.99099099` with `@rtop = 1100` where nominal `rv=1000` gives `0.9900990099`. Fixed with a reset after the nominal `alterparam`, which the next comment in that same function already assumed this path would do. Still falling back deliberately: structural dot-cards (`.if/.temp/.tran`), which change the deck's shape, and an ISOLATED local shadow, pinned by E-321. Two name collisions were hit while building this, both from the matching set growing beyond the swept names: a DEVICE called `Rd` collided with `.param rd` so a line matched on its NAME and the scan disarmed on a line carrying no value (matching now skips the leading instance/model token), and a deck TITLE beginning with `X` was read as a subcircuit call whose callee could not be found, disarming the whole sweep -- the near-identical deck beside it passed only because its title began with "t". Verified sweeps `subfast 28/28` both solvers, every case against an ANALYTIC value rather than the other code path; regression 379/379.

E-464 *src/frontend/subckt.c, src/frontend/subckt.h, src/frontend/inp.c, src/spicelib/parser/inp2n.c, src/include/ngspice/inpdefs.h, examples/busmix_examples/ (new)*

A BUS FORMAL AND A LOCAL BUS ON THE SAME OSDI INSTANCE LINE GAVE A WRONG ANSWER. Inside `.subckt s a[0] a[1] a[2] a[3]`, the line `N1 a b mymodel1` had E-449 expand `a` into the caller's four actuals -- the formals exist -- while `b`, having none, stayed ONE token. The line then carried five node tokens where autobus needs two (one per port) or eight (one per terminal); being neither, INP2N expanded nothing and the tokens bound positionally: `a[0..3]` correctly, `x1.b` onto `b[0]`, and the top three bits dangling. Measured 1.0 against 0.5238095 for the same circuit flattened by hand. It warned -- 3 of the 8 terminals ... are not connected -- but the warning named the missing terminals rather than the cause, and the deck still ran. The fix: once ANY port on the line has been expanded from formals the line cannot still be in shorthand, so every remaining bus port is expanded at the same point, to `x1.b[k]` -- exactly the node INP2N would have produced, and the same one `a b[0]` written elsewhere in the subcircuit translates to, so references still unify. The widths are reachable at flattening time even though the model TABLE is not: `pre_osdi` has already registered the module and the deck's own `.model` cards map a model name onto it, so `.model mymodel1 chan` plus `INPtypelook("chan")` gives the IFdevice, and the port grouping is `INPbusPorts` -- INP2N's own function, exported rather than copied, so the two cannot drift. A SECOND DEFECT SURFACED WHILE FIXING THE FIRST: the initial version spelled the local bus's bits with brackets even under `.option autobus=kicad`, so `x1.b[k]` never met the `b_0_` written beside it and the bits floated -- the very failure being fixed, in a new place. The cause is the one E-454 already repaired in this same option: the **autobus** variable is not published at flattening time, which is the entire reason `inp_set_autobus()` exists, so asking `INPbusKicadStyle()` there silently returned "brackets" for every deck. The style now travels with the flag, same precedence -- a deck card wins, otherwise a `set autobus=kicad` from `.spiceinit`. Two readers of one option disagreeing, for the THIRD time in this feature's history. Only mixed lines change: all-formal, all-local, bus-BASE-formal (`.subckt s2 a`, which always worked) and scalar-beside-bus forms are each pinned unchanged. Note for E-463: a mixed line now leaves flattening fully explicit, and `.option autoadapt` only considers shorthand lines, so a local bus shared across mixed lines is not auto-adapted -- a narrowing of reach, not of correctness, recorded rather than fixed. Verified `busmix 13/13` both solvers, every check a differential against the flat circuit on a ladder where all four bits differ; `subbus 16/16`, `autobus 12/12`, `autobuskicad 27/27`, `autoadapt 26/26` and `busportsub 9/9` all still pass on the changed path; regression 378/378.

E-463 *src/frontend/spiceif.c, src/frontend/inpcom.c, src/spicelib/parser/inp2n.c, src/include/ngspice/inpdefs.h, examples/autoadapt_examples/ (new)*

.option autoadapt adapter=<model> -- INJECT A TWO-BUS-PORT ADAPTER BETWEEN TWO OSDI DEVICES THAT SHARE A BUS NODE. `N1 a b m1 / N2 b c m2` becomes `N1 a b_f m1 / N2 b_r c m2` plus `n_adapt1_ b_f b_r <adapter>`, instead of the deck renaming the shared

node on both instances and keeping the halves consistent by hand. It runs between INPpas1 and INPpas2 -- the only seam where all three requirements hold at once: pas1 has built the model table, so a line's PORT STRUCTURE (and therefore "is this token a bus node, and how wide?") is knowable, which a textual pass in inpcom.c cannot answer; subcircuits are already flattened, so "inside a subcircuit" needs no second implementation, the bill E-449 paid to give autobus one; and INP2N has NOT run, so the rewrite is at TOKEN level and `.option autobus` then expands all three lines -- the bus handling is a consequence of the seam rather than extra work. The forward side comes from the PORT INDEX, not the deck order: a SPICE deck is order-independent, and making a reordering change the circuit would be a worse bug than the one being fixed, so the suite reverses the two instance lines and requires the same answer. Every non-qualifying case is REFUSED AND REPORTED rather than guessed at -- a node also touched by a resistor (three occurrences), three OSDI ports, both ports of one device, mismatched widths, a missing or wrong-shaped adapter, autoadapt without autobus, an existing `b_f/b_r`. Three defects the prototype surfaced. The adapter's `.model` card was DELETED by `inp_rem_unused_models` before the pass could use it -- nothing references the adapter until the instance is injected -- and the symptom blamed the injected line (Unable to find definition of model) rather than the card silently removed thousands of lines earlier; the cull now protects the model named by `adapter=`. The pass was NOT IDEMPOTENT: a deck already carrying adapters had them adapted in turn, `b_f` becoming `b_f_f/b_f_r` with a second adapter between and the answer moving from 0.7590 to 0.7647. And a reference to a bus BIT is a use of the bus -- the occurrence count looked only for the bare token, so `Rb0 b[0] 0 1k` was invisible and the resistor was left on an orphan node, silently; both `b[0]` and E-462's `b_0_` now count. Optional `.adapt n1, n2` restricts the node set by WHOLE TOKEN (a substring test would make `.adapt bb select b`), accepting a flattened node's trailing component so one card covers every instance of a subcircuit. Known limitation, tracked separately: an instance line mixing an expanded bus FORMAL with a local bus node is not adapted -- correct for an interface port, and it falls out of the exactly-twice rule -- but that configuration is ALREADY broken without this feature (control: adapter hand-written, autoadapt off, 3 of the 8 terminals not connected, 1.0 where the flat equivalent gives 0.5238095), because E-449 expands the formal into four actuals while the local bus base stays one token. Verified autoadapt 26/26 both solvers, every value check a differential against the hand-written deck on a ladder where every bit differs; regression 377/377.

[E-462](#) *src/spicelib/parser/inp2n.c, examples/autobuskicad_examples/ (new)*

.option autobus=kicad -- THE BIT SPELLING A SCHEMATIC CAN ACTUALLY WRITE. E-444 names an expanded bus port's terminals `a[0]..a[4]`, copying the bracket text from the model's own terminal names -- right everywhere except the one place a bus port is most useful: a schematic. KiCad's SPICE exporter rewrites every `[` and `]` in a net name to `_`, so a sheet labelling a wire `AA[0]` puts `/AA_0_` in the netlist; measured, with multi-digit indices intact (`ZA[10]` -> `/ZA_10_`), while KiCad's INTERNAL net name keeps the brackets -- the `kicadxml` export shows them -- so the rewrite belongs to the SPICE exporter alone and nothing on the schematic side can prevent it. `/AA_0_` never unifies with autobus's `/AA[0]`, so under KiCad a bus port's bits could not be labelled (the label made a DIFFERENT node and the bit floated), wired to ordinary parts (a resistor drawn to `AA[0]` landed on `/AA_0_`), or plotted -- the simulator's signal list is built from schematic nets, so it offered `/AA`, which after expansion has no device on it at all, and answered vector `V(/AA)` not or not yet available. The only workable sheet was one where every net was a whole bus. The option changes the generated SPELLING and nothing else: indices still come from the model's terminal names, so a port declared `[4:1]` still expands `1..4` and never invents an `a_0_`. Only one reader synthesises a name -- `inp2n.c` concatenates token and index; the subcircuit path (`e449_expand_bus_port`, E-449) maps a bus base onto the FORMALS the `.subckt` line already declares, so it has no spelling to choose, and the suite pins both modes giving bit-identical answers there rather than leaving it to reasoning -- two readers disagreeing about one option is exactly what E-454 had to repair here. An unknown style is REPORTED: `autobus=kicad2` would otherwise fall back to the default spelling and keep solving, with the bus bound to fresh `a[k]` nodes and every

a_k_ dangling at the source voltage -- a plausible-looking number as the only symptom, the silent-degenerate shape E-447/451/455 each had to go back and fix. Verified autobuskicad 27/27 both solvers, every check a differential (each spelling must produce ONLY its own node names, on a ladder where all five bits differ); measured end to end through KiCad's own kicad-cli netlist and through libngspice, ERC clean; regression 376/376.

E-461 *src/frontend/inpcom.c, src/spicelib/parser/inpgstr.c, examples/strparam_examples/ (new)*

A QUOTED STRING PARAMETER VALUE WAS CORRUPTED BEFORE THE MODEL SAW IT. `.model mm dut (ty="PMOS")` reached the Verilog-A model as **pmos**, and `ty="with space"` as **with** -- the remainder then reported as "unrecognized parameter (space)". So a string parameter could never match a mixed-case value, and a `from '{...}'` set written in upper case rejected every one of its own members (the compiler half of that report is the companion row in [openvaf_changes_full-report.md](#)). Two causes. The netlist reader lower-cases whole lines, with case retention wired only to Cider `.model` cards, `ic.file`, and a FIXED LIST of XSPICE code models -- an OSDI model card matches none of them -- and the existing helper only preserves a line carrying EXACTLY ONE pair of quotes, which a model card with two string parameters already exceeds. Separately `inp_casefix` turns quotes into SPACES unless the line is `.param`, `.subckt` or an X line, which is what cut the value at its first space. A quoted value on a model card or a device instance line is DATA -- a selector compared with `==`, a file name, a `from` set member -- not an identifier SPICE may fold, so both paths now keep it verbatim, on the card, on its + continuations and on instance lines, gated on the line actually containing `=\"` so nothing else changes. A trap inside the fix: adding `.model` to `keepquotes` made the case WORSE before it made it better -- the quoted text came back lower-cased even though the quotes now survived, because with `keepquotes` set the code never stepped PAST the opening quote, so the skip-to-closing-quote loop exited immediately and the tail of the same loop lower-cased the value one character at a time, keeping the quotes while losing the case they exist to protect. One else `string++`; Verified `strparam` 33/33 both solvers (value checked verbatim on model cards, continuation lines and instance lines); regression 375/375.

E-480 *src/spicelib/parser/inpgmod.c, src/frontend/com_measure2.c, src/mathscmaths/cmath2.c, src/frontend/control.c, src/frontend/inp.c, src/frontend/inpcom.c, src/frontend/spiceif.c, src/include/ngspice/cpextern.h, src/spicelib/analysis/dctrcurv.c, src/spicelib/devices/tra/trasetup.c, src/xspice/icm/analog/limit/cfunc.mod, src/xspice/icm/analog/pwl/cfunc.mod, examples/gatecheck_examples/ (new)*

A CHECK THAT COULD NOT FIRE WHERE IT MATTERED -- most of round 49 is not a MISSING check but a written one made unreachable by something upstream. THE DUPLICATE-PARAMETER TEST ON A `.model` CARD FAILED TWICE OVER: it was gated on the tracking list not being FULL, so once every distinct parameter had been seen once a repeat was never even LOOKED UP and a device with a single model parameter could never report one (`.model mm dut (r=1k r=4k)` took 4k in silence); and it counted the model TYPE token as a parameter, because a built-in card leaves that token in the line for the parse while an OSDI card has it removed -- harmless where the type is not also a parameter name (d is not a diode parameter, which is why `d(is=..)` was always clean) and A FALSE ACCUSATION ON THE MOST ORDINARY CARD THERE IS for r, c and l: `.model rmod r(r=1k)` was told *parameter 'r' is set more than once ... remove one*. `.measure`'s EDGE COUNT COLLIDED WITH A SENTINEL -- the field's *not given* and *LAST* markers are -1 and -2, so `fall=-1` was read as *no fall given* and answered with A RISING EDGE (0.5 ms where `fall=1` correctly gives 1.5 ms), `cross=-2` was read as *LAST*, and `cross=-3` simply failed; the guard tests the TOKEN THE USER WROTE, since by the time the value is a number *LAST* has already become -2. % DISAGREED WITH ITSELF: `.param` and a B-source call `fmod`, while the control-language operator took `floor(fabs())` of BOTH operands and did an integer % -- `(0.5) % 3` was 0, the value gone entirely, and `(-5) % 3` was 2 where C and `.param` give -2; the manual lists % as *modulo* and gives a SEPARATE \ for *integer divide*, and E-273's RANGE CHECK IS KEPT so `1e30 % 5` still errors as `mathcast_examples pins`. AN UNTERMINATED CONTROL BLOCK WAS CHECKED PER LINE, NEVER AT THE END OF THE SECTION, so an

if with no end swallowed every command after it and exited 0 in silence while a stray end has always been reported; the state is now compared BEFORE AND AFTER feeding the section, because reset inside a running dowlile re-enters that path with a block legitimately open -- a first version tore down the live structures there and hung lhs_examples. Guards that were genuinely absent: a transmission line's nl/f (f=0 returned A TABLE OF nan PRINTED AS AN ORDINARY AC RESULT, every row, rc 0), a switch's vh/ih (the only members of that model missing from the physics table), the code-model PWL's breakpoint ORDER (it checked lengths, and the SOURCE pwl warns about exactly this), a duplicate subcircuit call parameter (an UNKNOWN one already warned), and a .dc step larger than its span. And the limiter's reversed-limits message carried CLIMIT's TIME != 0 guard, which is never true in op or dc -- silent in those analyses and 214 messages in a transient; both its faults are model-card properties and now fire once at INIT. NOT CHANGED, each reported and withdrawn on reading the code: vector(-4) == vector(4) (cx_vector documents the MAGNITUDE), a negative pulse TR/PW (vsrload documents negative = use the default), .dc with start == stop (E-426: 13 decks rely on it) and ac lin 1 (span.c: nine cards use it). AND ONE FIX BUILT, MEASURED AND REVERTED: the constant plot IS writable -- let pi = 3 redefines pi for the session while destroy const is refused -- but shadowing the write into the current plot leaves READS still resolving to the constant first, and run is itself a built-in constant that lhs_examples writes and loops on, so its dowlile never advanced; making that safe means changing name resolution across the interpreter, far wider than the evidence, and check [13] pins the behaviour as it stands. Suite gatecheck_examples 47/47 (24/47 against the shipped pre-fix binary); full regression 394/394.

E-481 *src/spicelib/parser/inp2n.c, src/frontend/spiceif.c, examples/silentports_examples/ (new)*

.option silentports -- AN OPT-OUT FOR A NETLIST NOBODY TYPED. E-402 made an omitted OSDI terminal audible and that stays the DEFAULT, for good reason: an omitted terminal looks exactly like a typo, and it DANGLES rather than grounding, so the natural assumption is the wrong one. What it did not account for is a netlist written by a TOOL -- a schematic front end emits the short form for every instance of a model that declares an optional thermal port, and the author cannot add the pin from the schematic, so a five-device sheet collects TWENTY-FIVE LINES about a choice nobody made. KiCad's SPICE exporter is the case that prompted it. The option suppresses that warning and nothing else: the terminal is still absent, \$port_connected() still reports 0, the branch still carries no current, and the circuit solved is the same one. Opt-IN, so every existing deck is untouched, and a front end that cannot edit netlists can ship set silentports in .spiceinit instead. IT SILENCES A WARNING; IT DOES NOT MAKE AN ILL-POSED CIRCUIT WELL POSED -- measured on misc/bsimbulk_thermal_repro, the reproducer E-402 was decided on: with t written as 0 the deck is clean either way, and with t OMITTED the option removes the five warning lines and leaves all six **singular matrix: check node nl#t**, because BSIM-BULK pins its thermal node with a POTENTIAL tie-off so an absent terminal carries no current. That is E-402's decided territory and the answer is unchanged -- write 0 for the pin -- and checks [7]-[9] pin the distinction so a later reader does not take silentports for a fix. NO openvaf-r CHANGE: the warning is raised entirely inside ngspice, in INP2N, from numnodes < *dev->terms against the terminal count already carried in the .osdi descriptor; compiling a model with an unused optional port the compiler prints nothing at all. REGISTERED IN BOTH PLACES -- if_is_option()'s name list AND the type dispatch beside it: the first build had the option working while .options silentports still printed *unknown option ... ignored*, which is E-451's defect and is how the second registration point was found. Every OFF spelling is tested, because cp_getvar(..., CP_B00L, ...) reports a merely PRESENT variable as true and =0, =false, =no, =off would each have turned the feature ON -- the defect E-450, 451, 454, 466 and 467 each shipped exactly once. The suite compiles BOTH model shapes from source (thermal network contributed unconditionally, which stays well posed with the port absent; and one gated on \$port_connected, whose node floats) so it exercises the real compile-and-simulate path rather than a canned .osdi. Suite silentports_examples 24/24 (11/24 against the shipped pre-fix binary); full regression 395/395.

E-482 *src/spicelib/parser/inp2n.c, src/frontend/spiceif.c, examples/groundports_examples/ (new)*

.option silentports=ground -- GROUND THE TERMINALS A NETLIST LEAVES OUT. E-481 turned the absent-terminal warning off and that job is unchanged here -- the bare card still means exactly what it shipped and `silentports_examples` still scores 24/24 -- but what E-481 could not do was make the deck RUN. An omitted OSDI terminal is NOT GROUNDED: INP2N binds every terminal the line did not reach to -1 and `osdisetup.c` builds a private node `<inst>#<term>` for each, UPSTREAM behaviour that E-402 only started saying out loud. Ten of the twelve corpus models with an optional pin tie it off with a POTENTIAL contribution (`Temp(t) <+ 0.0`), which puts nothing into a node `ngspice` allocated itself, so the operating point dies on `singular matrix: check node n1#t`. E-402's answer was write 0 for the pin, and a schematic front end that hides the pin cannot write anything -- on the E-402 reproducer, silencing removed five warning lines and left all six singular-matrix reports, quiet and still broken, which is worse because the warning had been the only clue. THE OPTION WRITES THE 0: every omitted terminal is bound to ground in the parser, so `$port_connected()` reports 1, the skipped branches are built with the node held at 0, and no private node is created. `four.cir` now returns `i(vd) = -6.58515e-07`, EXACTLY the hand-grounded `gnd.cir` value, and `ngspice`'s own three broken BSIM-BULK decks go from 36, 12 and 12 singular reports with every sweep aborting to zero of both. THREE STATES, AND THE BARE CARD KEEPS E-481's MEANING: bare -- equally =dangle or =quiet -- silences and leaves the terminal DANGLING, held to the warned default VALUE FOR VALUE and on the gated shape just as singular; =ground alone binds the pin, asked for BY NAME because it changes the circuit; 1/true/yes/on mean the bare card, 0/false/no/off turn it off. THE READ ORDER IS LOAD-BEARING: `silentports_mode()` asks `cp_getvar` for the CP_STRING FIRST, CP_REAL next, CP_BOOL LAST where it now means only the bare card -- because E-467 gave `cp_getvar` a CP_BOOL COERCION answering TRUE for any value that is not an off-word, right for a two-state option and fatal for a three-state one since it swallows every value word. Measured with the BOOL query still first: =quiet GROUNDED the terminal, and so did =bananna. An unrecognised word is named once per distinct spelling and falls back to the DEFAULT, not to either ON state, following `ngspice`'s own `.options method=banana` handling. WHY THE PARSER: `osdisetup.c` reads the -1 sentinel to count connected terminals and passes it to `setup_instance`, which is what the model reads back through `$port_connected` -- bind in INP2N and the count, `$port_connected`, the absent private node and the untouched collapse machinery all follow; INP2N inserts ground before any device card so "0" never creates a node, checked with the OSDI card FIRST in the deck and from inside a SUBCIRCUIT (global ground -- where a front end actually puts the instance). EVERY omitted terminal is grounded, not just the first: a two-terminal model reads the current back to prove the LAST one was too. Grounding is NOT stock `ngspice` behaviour and never was -- the pre-E-402 `inp2n.c` binds -1 and `osdisetup.c`'s private-node loop arrived with the vanilla import untouched by this fork. NO `openvaf-r` CHANGE. Suite `groundports_examples` 59/59 (41/59 against the E-481 binary; eighteen checks discriminate, every one about the terminal's STATE rather than the message), `silentports_examples` 24/24 unchanged; full regression 396/396.

E-483 *src/spicelib/analysis/dcpss.c, src/frontend/com_qpss.c, examples/hbconv_examples/ (new)*

set qpss_tol / set qpss_maxiter, AND HARMONIC-BALANCE STALL DETECTION. QPSShb was called with its convergence bound and iteration cap COMPILED IN (0, 0, 60, 1e-10), and `tol` is an ABSOLUTE bound on the residual norm, so what is reachable depends on the circuit -- the diode two-tone deck settles at 2.3e-15 while an FET amplifier carrying tens of milliamps floors near 1e-8 and can never satisfy 1e-10. THE WAY IT FAILED WAS THE DEFECT: the Newton iteration reached a residual of 9.482e-09 at iteration 4, NINE ORDERS down and quadratic, then sat FLAT TO SEVEN DIGITS for the remaining 55; having failed the level the E-135 ladder halved its step and walked back to `lambda=0` -- 1022 Newton iterations -- and returned a bare E_ITERLIM. A good answer found in five iterations was discarded after minutes of work, six to seven of them at hb 4 4, which is indistinguishable from a hang and was

reported as one. Both numbers are now readable from a deck via `qpss_knob()`, asking BOTH published types (E-454: the spelling decides the type) and parsing with `strtod` not `atoi` so `qpss_maxiter=2e2` is 200 and not 2 (E-478's trap); a present-but-unusable value is REPORTED and the default kept. The stall test: `QP_STALL_RUN` (4) consecutive iterates improving the residual by less than `QP_STALL_RATIO` (0.1%) is a stall, accepted as the answer only if it sits **QP_STALL_ACCEPT** (1e6) below the residual the level opened with -- a stall that never came down is still a failure, reaching that verdict in a few iterations instead of `maxiter`. The design is not give-up-early; it is stop pretending a converged answer has not converged, and the closing message names which test carried it. ACCEPTANCE DOES NOT CHANGE THE ANSWER: against the same circuit under a loosened bound the fundamentals are BIT-IDENTICAL and the third-order products agree to 1.1e-05, about 0.0001 dB on OIP3; the counterweight is that `hb 3 3` still fails at a residual of 5.5e-05 at `lambda=0` and is refused, a check asserting the PROPERTY rather than the digits. On the ATF-34143 two-tone IP3 deck that prompted it, `hb 2 2` goes from `E_ITERLIM` after 14 s to converging in 10 iterations under a second, giving OIP3 = +34 dBm with both sidebands agreeing to 0.1 dB. DELIBERATELY NOT FIXED: `hb 3 3` and above still fail and the bound does not rescue them (1e-8, 1e-6, 1e-4 all die at the same residual before the continuation takes its first step) -- a separate fault; what changes is the COST OF FINDING OUT, 48 s instead of 6-7 minutes. NO `openvaf-r` CHANGE. Suite `hbconv_examples` 23/23 (6/23 pre-fix; seventeen checks discriminate) while `qpssleak`, `distoexact` both solvers, `plotorder` 25/25 and `pzklu` 4/4 are unchanged; full regression 397/397.

E-484 *src/spicelib/analysis/dcpss.c, src/frontend/com_qpss.c, examples/hbtrust_examples/ (new)*

A CONVERGED FLAG IS NOT A CORRECT ANSWER. **E-483** made the harmonic-balance bound reachable and taught the Newton loop to spot a stalled residual, and left a hole: the STALL path reported what it settled for, the ORDINARY `tol` path reported nothing, so a deck that loosened the bound far enough had its solution accepted with a clean *converged in 3 iterations*. On a two-tone FET amplifier set `qpss_tol=1e-1` at `K=4` accepts a residual that came down by only ~100x and the third-order products land 6 dB out -- 27.818 dBm against the trustworthy 33.976 dBm of the `K=2` run -- silently. WORSE THAN THE FAILURE **E-483** FIXED: a refusal is honest, a wrong number is not, and **E-483** had handed users a knob whose obvious misuse was silent corruption. The reduction behind an accepted solution is now reported on the ordinary path too, naming the residual, the reduction, the bound that permitted the early stop and both remedies; the run is not aborted and the spectrum still prints, since the user may know exactly what they are doing. THE THRESHOLD IS CALIBRATED, NOT CHOSEN: `QP_LOWRED_WARN` is deliberately a different constant from `QP_STALL_ACCEPT` -- one decides whether to accept, the other whether to believe -- and its value comes from measurement, 101x putting OIP3 6.16 dB out and warned against 55345x agreeing to 0.033 dB and silent, with two suite checks holding the bar from both sides because a warning that fires on a good answer is one people learn to ignore (**E-445's** note). IT DOES NOT FIX **hb 4 4**, whose Newton step degrades as harmonics are added (5.3e8x at `K=2`, 7.8e4x at `K=3`, 1.5e2x at `K=4`) with drive amplitude, beta, is, alpha, cgs/cgd, the passive network, the topology and the DFT oversampling each ruled out by measurement, `pss_solve` pivoting and `tmalloc` being `calloc` -- leaving `qp_build_matrix's` harmonic-domain Jacobian as the open question. What it guarantees is narrower and worth having: when the solver cannot give a correct answer it does not quietly give a wrong one. Also folds a `const char *qualifier fix` in `com_qpss.c` that **E-483** introduced, two discards-qualifiers warnings in a tree kept at zero, surfaced by a `build-shared` rebuild whose output had been filtered to errors only. NO `openvaf-r` CHANGE. Suite `hbtrust_examples` 16/16 (10/16 against the **E-483** binary; six checks discriminate) with `hbconv` 23/23, `qpssleak`, `distoexact` both solvers, `plotorder` 25/25 and `pzklu` 4/4 unchanged; full regression 398/398.

E-485 *src/xspice/cm/cmutil.c, src/xspice/icm/analog/{limit,int,d_dt,pwl,hyst,slew}/cfunc.mod, src/spicelib/parser/inp2dot.c, src/frontend/inpcom.c, src/frontend/inp.c, src/frontend/com_measure2.c, examples/guardsweep_examples/ (new)*

EIGHT GUARDS THAT DETECTED A FAULT AND THEN USED THE BAD VALUE. An hour-long hunt found the frontend and OSDI paths clean almost everywhere and the defects concentrated in the XSPICE code models, all of one shape. THE HEADLINE: `cm_climit_fcn` had its bail-out COMMENTED OUT -- it detected the crossed linear range, printed, and used the bad value regardless. It could NOT be uncommented: those lines assign the LOCALS and return, while the out-parameters are written at the end of the function, so restoring them verbatim would have left `*out_final` uninitialised -- very likely why they were disabled rather than repaired. Fixing the INPUT instead (clamping the smoothing half-width to half the limit span, leaving hard limiting at the declared bounds) takes `ilimit` from 24.48 to 0.437 against rails of +/-1, its messages from 26 per op to one, and its text from naming CLIMIT in a deck with none. SAME SHAPE FOUR MORE TIMES: `limit`, `int`, `d_dt` never computed `linear_range`, so an oversized `limit_range` reached 24.5057, 95.04 and 24.25 against limits of +/-1 and 249999.75 at 1e6 -- unbounded and silent, with those limits MANDATORY parameters; E-468's own comment says it added its checks *as the CLIMIT sibling already does* and ported two, not CLIMIT's own. `pwl`'s monotonicity guard ended in break, leaving only the CHECKING loop, so the table was built from rejected data and `x=[0 2 1]` answered 5.5 for an input of 0.5; it cannot return in place (STATIC_VAR-owned arrays), so the test moved beside the `size_error` check. `hyst` and `slew` had NO checks: an inverted pair and an oversized `hyst` left the block dead at 0.0, a negative `rise_slope` drove the output to -2.0. THREE IN THE FRONTEND: `dot_sens` validated nothing while `dot_ac` in the same file names the same faults -- a reversed range swept a FABRICATED decade 1e6 to 1e7 ascending; `dot_disto` reported a DEVICE-parameter fault it does not have; `.include/source` of a DIRECTORY succeeded in SILENCE (`fopen()` on a directory succeeds on macOS/BSD/glibc) so the deck solved a different circuit, `v(out)` 1.0 not 0.5, while `.lib` was already guarded; `meas` clamped a negative FROM and then reported the window the user asked for. THE REPAIR IS THE CODEBASE'S OWN PATTERN -- `sine/square/triangle` already detect, report, substitute a safe value -- which is what makes these defects and not design. THREE WITHDRAWN, one at fix time: sweep's negative step is deliberate (`com_sweep.c:2119`), XSPICE Limits: ARE enforced, and `.nodeset` on an OSDI internal node really is ignored (the counter-evidence was a diagnostic ECHOING the card). Suite `guardsweep_examples 37/37` (24/37 pre-fix; thirteen checks discriminate), every fix paired with the control that must not move; full regression 399/399.

E-486 `src/spicelib/analysis/cktsens.c, src/xspice/enh/enhtrans.c, src/xspice/mif/mifmpara.c, src/xspice/icm/table/{table2D,table3D}/cfunc.mod, src/xspice/icm/digital/d_state/cfunc.mod, src/xspice/icm/xtrdev/{capacitor,inductor,core}/cfunc.mod, src/xspice/icm/tlines/{m1in,cpline,cpmlin}/cfunc.mod, src/xspice/icm/analog/{file_source,xfer}/cfunc.mod, five ifspec.ifs Limits, examples/guardpair_examples/(new)`

TWELVE GUARDS THAT ONE SIBLING HAD AND THE OTHER DID NOT. An hour-long hunt found OSDI itself clean -- state restoration across all nine analyses including E-483/484 hb, array instantiation, instance-parameter ranges, round-34 still fixed -- and one shape repeated: the check EXISTS in the tree and is absent next door. THE HEADLINE: `table2D`'s data-row loop is driven by the FILE and wrote `table_data[lLineCount-1]` unbounded, while the x row, the y row, each row's WIDTH and even a premature EOF were all checked. 3 declared y values and 5 supplied rows indexed past the allocation and SEGFAULTED (EXC_BAD_ACCESS at 0, rc 139, no diagnostic) from a twelve-line file; too FEW rows left `calloc`'s zeros in place and a probe there returned 0.0, a plausible no-current. E-247 had already fixed the OOB READ and the too-small case *in this same file* and left the WRITE; the sibling `table3D` refuses the truncated file, so the two disagreed and only one crashed. THE SECOND: `count_steps()` returns a POINT COUNT, and E-362's guard signalled failure with `return(E_PARMVAL)` -- `E_PARMVAL` is 11, the caller never tested it, so an impossible sweep ran eleven points with `*stepsize` never written. Two silent repairs sat below it, rewriting a stated 0 Hz start to 1e-3 and a stated stop to a decade above the start -- in the LOCAL copy only, so the count used the repaired bounds while the sweep ran from `job->start_freq: .sens ac dec 5 0 1meg` printed a table of all 0 Hz rows, `.sens ac dec 5 1k 1k` swept a full decade past the stop (6 rows where `.ac` gives 1). The new rules are `.ac`'s own. TEN

MORE: `d_state` ran ROW 0 for a state the file never defines; `file_source` lacked the monotonicity check its sibling `pwl` has had since E-480; `xfer`'s file path validated nothing while its table path ten lines above refuses the same data; `hyst`'s unbounded `input_domain` drove the output to 250000 against declared limits of 0 and 1; core indexed B with H's size; `m1in/cpline/cpmlin` accepted impossible geometry (a NEGATIVE length returning a plausible 0.5011 against 0.4992); `poly()` reported each fault with the OTHER's message; and four models were the only ones of twenty-plus with unbounded delays. DELIBERATELY UNCHANGED, with the built-ins as oracle: NEGATIVE capacitance and inductance are legitimate -- built-in C and L behave identically -- so only ZERO is fixed; `m1in`'s `rho=0` looked like a perfect-conductor idealisation until it measured as NaN; `xfer` duplicates stay legal because the table path allows them. Both headlines trace by `git -S` to 4f29ffad, the vanilla UPSTREAM import. Verified `guardpair` 65/65 (24/65 pre-fix, 41 discriminating). No `openvaf-r` change.

E-487 `src/spicelib/analysis/dcpss.c`, `src/include/ngspice/cktdefs.h`, `src/frontend/com_hbosc.c`, `src/frontend/com_hb.c`, `src/frontend/com_hb.h`, `src/frontend/com_loadpull.c`, `src/frontend/typesdef.c`, `examples/rfplots_examples/` (new)

EVERY RF ANALYSIS LEAVES ITS RESULTS IN A NUTMEG PLOT. An RF result that is only PRINTED cannot be plotted, printed again, `wrdata'd`, diffed or read by a script. Six of the eight RF entry points published; **hbosc** and **phasenoise** printed a table and stored NOTHING, leaving the session with `hbosc`'s own STARTUP TRANSIENT as the current plot. THE CAUSE: E-209 added `struct hbspectrum *out` to `HBanalyze` so `com_hb` could publish; `HBOSCanalyze` computes the SAME (2K+1)*N spectrum in the SAME layout and never got the parameter -- one signature extended, its sibling not, the shape E-486 spent its length on. Both now hand their results back, and rather than let `com_hbosc` grow a second copy of `hb_publish()` the two share ONE parameterised publisher. `hbosc` also stores **oscfreq**, the frequency an autonomous circuit SOLVES for. THE SUBTLE PART: `ft_plotabbrev()` matches by SUBSTRING, so `hbosc` hit the `hb` pattern and came out named `hb1` -- indistinguishable from a driven run -- while `phasenoise` hit `noise`. Both get entries AHEAD of the shadowing pattern, per E-383's rule; `plotorder_examples` asserts that rule for every name in the tree but its NAMES table is HAND-MAINTAINED and was passing over these two vacuously. TYPES: `loadpull`'s `pout_dbm/gain_db` are dB and now carry `SV_DB`, while `gamma_re/gamma_im`, `pae`, `eff` and `stb`'s `loopgain` stay untyped -- dimensionless ratios with no enum member, where inventing one would be worse. NOT CHANGED: `loadpull` leaves 61 intermediate transient plots at `-n 9` and 333 at `-n 21`; discarding plots a user may inspect is a behaviour change, so it is recorded. Verified `rfplots` 17/17 (7/17 pre-fix, 10 discriminating) with checks [1]-[6] and [8] passing BEFORE and AFTER as the control that the six working analyses did not move; `hbosc` 13/13 (6/13 pre-fix). No `openvaf-r` change.

E-488 `src/frontend/com_sweep.c`, `examples/sweeptemp_examples/` (new)

sweep temp SWEPT NOTHING. `sweep` resolves a knob as a `.model` param, an instance/device param, or a deck `.param`; the GLOBAL circuit temperature is none of those, so a bare `temp` fell through to instance/device, `alter temp=` found no such device, and the sweep ran to completion over a knob that never moved -- a full set of points, `rc=0`, and a plottable FLAT curve reading as *temperature has no effect on this circuit*. E-431 removed that shape for an unresolved -output and E-435's own comment here describes it for subckt-local model names; `temp` was a third instance, and the one users meet, because every OTHER route worked (`.option temp=`, `set temp=`, `alter @dev[temp]=`, `sweep @#*[temp]`). THE OBVIOUS FIX DOES NOT WORK: writing `ckt->CKTtemp` (what `.dc temp` does) left the curve flat, because `CKTdoJob` opens with `ckt->CKTtemp = task->TSKtemp` -- `.dc` survives only because its whole sweep runs INSIDE one `CKTdoJob`, while sweep runs a fresh analysis per point, so the write was discarded before the next point solved. The TASK must move, so it issues `option temp=`, which also carries the value through the guarded `OPT_TEMP` funnel and inherits E-426's absolute-zero refusal and E-440's sanity check. DELIBERATELY NOT IDENTICAL TO **dc temp**: over a temperature-driven node collapse `dc temp` returns 0.0 where a static `op` returns 0.5 -- it holds one setup for the whole sweep and never rebuilds (

round-24, still open, NOT fixed here) -- while sweep rebuilds via E-471 and is right; the suite pins all three against the static op and asserts dc temp DISAGREES, so "fixing" sweep into agreement FAILS. STRICTLY ADDITIVE: a deck .param temp is tested first and still wins. Also matched: an unphysical range is refused UP FRONT (left to the funnel, bad points are ignored one at a time and the axis still carries them, so a row LABELLED -600 C held the 27 C answer), and the axis carries SV_TEMP. Verified sweeptemp 20/20 (10/20 pre-fix, 10 discriminating). No openvaf-r change.

E-490 *src/spicelib/parser/inp2n.c, src/frontend/subckt.c, src/include/ngspice/inpdefs.h, examples/busmixed_examples/ (new)*

MIXING SHORTHAND AND WRITTEN-OUT BITS ON A BUS PORT SILENTLY MISWIRED THE DEVICE. E-444's .option autobus fires on a token count equal to the PORT count; positional binding covers a count equal to the TERMINAL count. A line that leaves one port in shorthand while writing another port's bits out is NEITHER, so N1 a b[0:2] against inout [0:4] a; inout [0:2] b fell through both and bound positionally against the flat terminal list -- a onto a[0], b[0] onto a[1], b[1] onto a[2]. Measured on a model whose eight bits each carry a different conductance, every node sat one or more terminals off and every current was wrong. The only thing said was E-402's warning about the terminals left over at the TAIL -- the symptom, not the cause -- so a user who supplies the two nodes it asks for still has a circuit wired entirely wrong, now with no warning at all. E-445's diagnostic (*already carries an index, so it cannot be expanded as the bus port*) exists for exactly this mistake but sits INSIDE the port-count check: the guard and the failure were one if apart. NOTHING HERE IS AMBIGUOUS, so the fix does not guess -- it walks the ports left to right and lets each token declare its form: a bare name on a bus port is shorthand for that port's bits, and a token already carrying an index (or ground, which E-445 established can never be indexed) means that port was written out, so take one token per bit. N1 a 0 0 0 reads correctly under the same rule. The rewrite is accepted only when the walk consumes EXACTLY the tokens the line has, and the walk is bounded by that count because gettok_instance cannot tell where the node tokens end -- without the bound a port claiming more bits than the line has left would swallow the MODEL NAME and report against it. WHERE NO READING EXISTS IT REFUSES: if the counts do not reconcile, the bits written do not match the width the model declares and nothing can repair that, so it is refused where the port and both counts are still in hand to say so -- naming the port, its width, the tokens written and the two counts that would have worked. That replaces a wrong answer, not a working deck; warning and binding anyway is the *detect, announce, then use the bad value regardless* shape E-485 had to undo eight times in one round. ONE RULE, ONE READER: whether a token already carries a bit index is spelling-dependent (a[0] always, a_0_ only under autobus=kicad), and subckt.c already answered it for .subckt formals -- a second copy in inp2n.c would have been free to disagree, the two-readers-of-one-rule shape E-454 had to repair in this same option -- so the rule moved to INPbusTokenIndexed beside INPbusBitSuffix and subckt.c now delegates. TWO SMALLER FIXES IN THE SAME OPTION: .option autobus=1 reported a style that does not exist, the on-word list holding true/yes/on but not 1, the mirror of the 0 in the off-word list -- the comment on the early return says *bare flag, or =1: a NUMBER, not a string*, true of set autobus=1 but not of a deck .option card, which publishes a STRING; and E-445's own message named a TERMINAL where a PORT was meant, reporting a [0:2] port as *the bus port 'b[0]'*, an index the user never wrote. UNCHANGED BY DESIGN: with autobus off nothing moves, all-shorthand and all-explicit lines take the paths they always did, a short line with no shorthand token keeps E-402's warning and the \$port_connected idiom it protects, and E-445's two refusals still fire on a full-width line. Eighteen further shapes measured and needing no fix: unequal port widths, descending [4:0] and non-zero-based [1:5] ranges, a 16-bit port, three buses of sizes 2/3/4, bus/scalar/bus interleaving, one token feeding two ports (which superposes exactly), uppercase tokens, the option placed AFTER the instance, two instances in parallel, m= multipliers, all four subcircuit shapes including reversed formals, autobus=kicad on unequal widths, and too-many-nodes as a hard error; a [0:0] port connects correctly in both spellings, its node merely named a rather than a[0], which is inherent. Verified busmixed 46/46 (20/46 pre-fix, 26 discriminating). No openvaf-

r change.

E-491 *src/misc/printnum.c, src/misc/printnum.h, src/include/ngspice/cpstd.h, src/xspice/evt/evtprint.c, src/spicelib/parser/ptfuncs.c, src/spicelib/parser/inpptree.c, src/spicelib/analysis/cktsens.c, src/frontend/measure.c, src/frontend/device.c, src/frontend/inpcom.c, src/frontend/{diff,dotcards,postcoms}.c (18 call sites), src/xspice/icm/analog/s_xfer/cfunc.mod, examples/numguard_examples/(new)*

AN UNBOUNDED NUMBER USED AS A LENGTH, AND FOUR WRONG ONES. Ten defects of one shape from a bug hunt: a number the deck supplies, used without being measured against what it was about to control. THE CRASHES: set numdgt is the user's print precision and nothing bounded it -- printnum() sprintf'd `%.*e` into caller buffers of BSIZE_SP(512) and evtprint into a char[100], so numdgt=510 with print ABORTED and numdgt=94 with eprint TRAPPED, both from a plain batch deck with no interactivity and both exactly where n+9 bytes says they must. printnum()'s OWN COMMENT recorded the hazard -- *it can cause buffer overruns, the size of buf is unknown* -- without bounding it, while the DSTRING sibling printnum_ds() directly below could not overflow, which is precisely why fourier, wrdata, write, display and diff were unaffected and only the print family crashed. It now takes the buffer size and CLAMPS rather than truncates -- a truncated number is not the number computed, and past ~17 digits the rest is zero padding -- saying so once; evtprint uses the same clamp so the two printing paths cannot disagree. Both trace by git -S to 4f29ffad, the vanilla UPSTREAM import. THE WRONG NUMBERS: PTdivide added PTfudge_factor to EVERY divisor rather than a zero one, and that factor is gmin*1e-20 -- not even a fixed perturbation but one scaling with an unrelated convergence option -- so 1/boltz came out 42% low under .option gmin=1e-3 and 88% low at gmin=1e-2, while 1/0 returned 1e26/1e32/1e50 by gmin alone; the nudge now applies only to an EXACT zero with a fixed epsilon, so a non-zero divisor is used as written and 1/0 is the same number in every deck. And B-source trig reduced with x-(int)(x/2pi)*2pi, undefined above 2^31*2pi, so sin(1e20) returned +0.9993 where the answer is -0.6453 -- while numparam and a Verilog-A model each already matched libm exactly, making the B-source the SOLE OUTLIER, the divergence E-399 forbids and ptfuncs.c's own comment quotes. THE SILENT REFUSALS: a sens filter matching no parameter returned OK with no plot and no message, leaving the PREVIOUS analysis current for a following print (and the trailing words on a sens line ARE filters, so sens v(b) r8 legitimately works -- which made a typo indistinguishable from a filter doing its job); the interactive meas enforced its analysis keyword for INTERVAL measurements (E-468) and ignored it for POINT ones, so meas tran ... FIND AT= read a DC sweep as a transient; s_xfer blamed the SOLVER for a model error, an all-zero denominator going NaN and being reported as *Dynamic gmin stepping failed*, and num>den announced once per evaluation (1238x over 300 steps) with rc=0 and the device contributing nothing; printf("oops ") reached users on stdout from two default: arms in inpptree.c, one of them the very printer the *internal check of parse tree failed* diagnostic uses to SHOW the offending expression; show printed ?????????? where print/alter/altermod/wrdata/sweep all name the parameter; and a duplicate user .func was silently last-wins while shadowing a builtin warned (E-467). ONE COMMENT CORRECTED RATHER THAN OBEYED: E-440's claim that the singular routes *clamp to HUGE so Nlitter can reach for gmin* is not what the code does -- ifeval treats a result equal to HUGE as an ERROR FLAG and aborts, the opposite of continuing, and the routes were never uniform; recorded rather than unified, since making log(0) abort or pow(0,-1) not abort would each move an answer a working deck gets today. THE FIX'S OWN TRAP: the new meas gate first refused BEFORE the standard failure path, so it skipped both the failed! line a script tests on and E-475's vec_remove -- mathguard caught it, and the gate now refuses the way every other refused measurement does. UNCHANGED: ordinary precisions, 1/0 still finite so a solve continues, interval measurements, an unfiltered or matching sens, a valid s_xfer, and .func last-wins. Verified numguard 68/68 (18/68 pre-fix, 50 discriminating), mathguard 57/57, regression 405/405. No openvaf-r change.

E-492 *src/spicelib/parser/inp2dot.c, src/spicelib/parser/inp2{e,g,s}.c, src/spicelib/parser/inppas3.c, src/*

include/ngspice/inpdefs.h, src/include/ngspice/cktdefs.h, src/spicelib/analysis/cktop.c, src/osdi/osdiloadd.c, src/math/KLU/klusmp.c, examples/ctrlnode_examples/ (new)

A NODE NAMED ONLY IN A CONTROL POSITION WAS SILENTLY CREATED. E, G and S take a controlling node PAIR, bound exactly the way the output pair is -- INPtermInsert, which CREATES the node -- so a typo simply invented a node and the run continued against it. For E/G the invented node has no path to ground, the matrix goes singular, and the user is told *singular matrix: check node nosuch* -- a node they never wrote, reported as a fault in their circuit. S IS WORSE: a switch only READS its control voltage to decide open or closed and stamps nothing for it, so the matrix stays non-singular, the solve succeeds, and the answer is silently wrong -- S1 a b c t l 0 sw gives 0.999001 while S1 a b nosuch 0 sw gives 9.99999e-07, a factor of a million from one mistyped character, rc=0, with no diagnostic at all. A phantom reference is dangerous exactly where it is READ but not STAMPED. EVERY OTHER ROUTE ALREADY ANSWERED THIS: .ic/.nodeset report *IC on non-existent node*; F, H, W and a B-source i() report *unknown controlling source*; all thirteen output constructs name a missing vector; and warn_physics even validates the switch's own ron/roff/vh -- only the controlling-node pair skipped it, under any option. THE MECHANISM IS E-429's, UNCHANGED: devRef records whether a node was ever named by a device or made by the simulator, and a control reference is the same kind -- it names without making real -- so it no longer sets devRef and CKTnodePhantom() answers as it stands. Marking is left to INPtermInsert for every other position, so a control node that IS connected somewhere, including one that is the source's own output (E1 out 0 out 0 2), stays marked and is never reported; the check runs in PASS 3 for the same reason .ic's does, because only once every device card has been read is *did anything connect to this?* answerable. It REFUSES rather than warns -- E/G already fail on their own, while a switch would carry on and answer from a node that is not in the circuit, the detect-announce-then-use-it-anyway shape E-485 had to undo eight times in one round. TWO DIAGNOSTICS THAT NAMED THE WRONG THING: CKTop blamed Verilog-A for faults that were not its own -- E-378 added the message and E-399 narrowed its entry to E_PANIC checks, arguing the test exact because E_PANIC(1) and E_ITERLIM(103) differ, which is true, but the invariant the MESSAGE relies on is **E_PANIC** = a Verilog-A fatal, and **E_PANIC** has ~10 producers (cktsetup x2, osdiparam x2, osdiloadd, osdisetup, dcpss x2, ckt-pzstr, ifeval, CIDER's twosolve), so .option klu with a current-source-only matrix and no Verilog-A device anywhere claimed one had raised it and pointed at an OSDI(fatal) line that does not exist, through op, dc, ac and tran alike since each calls CKTop; CKTvFatalRaised is now set only where one is actually detected and cleared on entry so a previous analysis cannot be inherited. And KLU printed NINE LINES FOR ONE CONDITION -- its own PreOrder comment says an empty matrix is legitimate (*XSPICE pure digital circuits produce empty KLU matrix*) and returns success for one, yet all three sites reported it per call and Solve added two NULL-object errors that are its consequences, re-entered per Newton iteration; it now says once, in words, what the state is. UNCHANGED: a control node connected anywhere else -- the source's own output, one defined later, ground, or one through a subcircuit port; F/H/W and B-source i(); a real Verilog-A fatal, still named and still pointing at its OSDI(fatal) line; and either solver on an ordinary circuit or a mixed-signal KLU run. Verified ctrlnode 37/37 (19/37 pre-fix, 18 discriminating), regression 406/406. No openvaf-r change.

E-493 *src/frontend/device.c, src/frontend/outitf.c, src/spicelib/parser/inp2r.c, examples/namelookup_examples/ (new)*

THREE NAMES THE SIMULATOR WOULD NOT LOOK UP. One shape, three times: a name the user wrote was not looked up where it was written, and what came back described something else. **showmod <MODEL>** COULD NOT FIND THE MODEL: the device generator reads a bare word as an INSTANCE name and only #name as a MODEL, so with .model dm d used by D1, both showmod d1 and showmod #dm printed it while **showmod dm** answered *No matching instances or models* -- of a model that plainly exists. The command's own help calls its argument *models* and its write sibling takes the model name directly (altermod dm is=1e-12 works), so the one command dedicated

to models was the one that could not be handed one; OSDI models were affected identically, the defect being in the name grammar rather than in any device. **RETRY, NOT REINTERPRET:** the bare-name-is-an-instance reading runs first and unchanged, so `showmod d1 --` and any name that is both a device and a model -- behaves exactly as before, and only when **NOTHING** matched is each bare word retried with the `#` the grammar wants. A **SAVED NAME THAT MATCHED NOTHING WAS DROPPED IN SILENCE:** Pass 1 marks the `.save/.probe` items it places and left the rest unmarked, so `.save v(n) v(nosuch)` recorded `v(n)`, discarded the typo and said nothing -- the analysis succeeded and the vector the user asked for was merely absent. `.probe v(nosuch)` shares that path, which is why a mistyped **NODE** there was silent while a mistyped **SOURCE** in the same card is reported by the measure-source pass, and why E-418 already spoke for the `@dev [param]` spelling -- the plain node spelling was the one route left quiet. It **WARNS** rather than refuses (an absent vector is not a wrong answer, and a deck saving a node it does not always build is a real idiom) and honours `savesused[]` first, so the `all/allv` keywords and items belonging to another analysis are never reported, falling back to matching the name itself under `save all --` which `.probe` turns on and where Pass 1 never runs. A **RESISTOR MODEL NAMED r WAS UNREACHABLE:** **INP2R** excluded the token `r` outright, which `R1 a b r=1k` needs since there `r` is the keyword that writes the resistance, but that also locked out a model actually **CALLED r** -- `.model r r rsh=1k` with `R1 a 0 r l=1u w=1u` bound neither model nor value, so the device fell through to the default and came out 1 mOhm with *resistance too low or not given*: the symptom, when the cause is that the model named on the line was never looked up. Every other name works, including every other resistor keyword (`rsh`, `l`, `w`, `tc1`, `tc2`, `temp`, `dtemp`, `m`, `scale`, `ac`, `dc`, `noisy`, `short`, `narrow`); only this one letter was unreachable. The two spellings differ by what **FOLLOWS** the token -- the keyword form is `r=`, a model name is not -- so asking that keeps `r=1k` meaning what it always did, and a deck with no such model is unaffected because **INPlookMod** fails and the existing else branch restores the line exactly as for any other unrecognised token. **UNCHANGED:** `showmod` by device name, by `#model` and bare, and `show`, including a name that is both; `altermod <model>`; `.save all/allv` and `.probe alli`, and items belonging to another analysis; `@dev [param]` saves; and `r=1k`, `r = 1k`, `R=2k`, a plain value, `r=1k tc1=0`, a value with `m=`, and **`r=4k`** winning when a model named `r` also exists. Verified `namelookup 39/39` (29/39 pre-fix, 10 discriminating), regression 407/407. No `openvaf-r` change.

E-494 `src/spicelib/parser/ptfuncs.c, src/frontend/inpcom.c, examples/bexprvalue_examples/ (new)`

TWO WAYS A B SOURCE EXPRESSION DISAGREED WITH EVERY OTHER VALUE PATH. Round 54 compared the B source path against the paths that read the *same number on the same deck*. **x/0 LOST ITS SIGN:** E-491 replaced a `gmin`-derived fudge factor in `PTdivide` with a fixed epsilon and wrote the nudge as `arg2 = (arg1 >= 0.0) ? PTDIV_EPS : -PTDIV_EPS;`, meaning to keep the sign of the numerator -- but a negative numerator was then divided by a **NEGATIVE** epsilon, so the quotient came out **POSITIVE** either way. `v(p)/0` with `v(p) = -3` returned `+3e+32` instead of `-3e+32`, and `-1/0`, `(0-2)/0`, `-1.0/0.0`, `-1/(1-1)` and `E0 b0 0 vol='-1/0'` were all positive too. Every case E-491 measured had a positive numerator, which is why its own suite did not see it. A divisor of exactly zero carries no sign to recover, so the side zero is approached from is now chosen **ONCE** rather than per operand: the epsilon is unconditionally positive, and the quotient keeps the numerator's sign -- the limit of `x/eps` as `eps` tends to zero from above, and what the comment E-491 left in place already described. **NUMERIC LITERALS WERE ROUNDED TO 11 SIGNIFICANT DIGITS:** every B line is rewritten before parsing by `inp_modify_exp()`, which re-emitted each literal with `"%18.10e"`, so `v=1.2345678901234567` reached the parser as `1.2345678901e+00` -- a relative error of `1.9e-11` -- while the same literal on an R, C or V card, in a `.param`, or as an OSDI `.model` parameter kept all seventeen. The B source was the only path in the simulator that could not carry a double, and it reached the arithmetic rather than just the `printout`: `tan(1.5707963267948966)` returned `-1.96e11` against `libm`'s `1.633123935319537e+16`, so the tangent was right and its argument was not. **MORE DIGITS WOULD NOT HAVE FIXED IT:** `INPevaluate()` reads the text back and accumulates the mantissa as `mantis = 10 * mantis + digit`, losing a bit past `2^53`, so a fixed `"%.17e"` turns the sixteen digits of `0.7853981633974483`

into eighteen and brings it back 1 ulp out when the user's own text would have survived -- the new `inp_num_text()` emits the SHORTEST text that round-trips, trying 15 then 16 then 17 digits. DELIBERATELY NOT WIDENED: `INPevaluate`'s own scaling still costs 1-2 ulp, but a `V` source shows it identically on the same literals, so the suite pins parity and a 2 ulp bound rather than bit-exactness, against an 11-digit rounding that had put those values 1.0e5-1.6e5 ulp out; and `mkb()` in `inpptree.c` folds a constant / with a raw division and no zero guard, bypassing `PTdivide`, but probed from B, E and G in constant, parenthesised, nested and mixed forms it never produced an infinity, so it appears unreachable from user input and was left alone rather than fixed past the evidence. UNCHANGED: E-491's magnitude and `gmin` independence; ordinary divisors; the SPICE suffixes, leading-dot and exponent forms; `pwl` B sources, which `inp_bsource_compat()` excludes from the rewrite; current B sources, node references mixed with literals, and the value surviving a transient; and E/G expressions, which never passed through `inp_modify_exp()` at all since only `b` lines do. Verified `bexprvalue` 63/63 (37/63 pre-fix, 26 discriminating), regression 408/408. No `openvaf-r` change.

E-495 *src/spicelib/parser/inpgmod.c, src/spicelib/analysis/dctrcurv.c, src/frontend/numparam/xpressn.c, src/frontend/rawfile.c, examples/binstale_examples/(new)*

SEVEN WAYS A DECISION MADE ONCE WAS NEVER REVISITED. Round 55 probed MOSFET model binning, untouched by any earlier round, and the shape it exposed led into the DC sweep. THE TOLERANCE WAS ABSOLUTE: `is_equal` tested `fabs(a-b) < 1e-9` on values in metres -- a fixed one-nanometre slop -- so a device up to 1 nm outside *every* declared bin was silently placed in one (`l=31n` bound to a bin reaching `30n`; `31.1n` refused, so the edge really was the constant). The magnitude does not scale, so as a *fraction* it grows without bound: 0.03% of a 3 um width but 5% of a 20 nm channel. Now relative. ADJACENT BINS OVERLAPPED: `in_range`'s own comment states `min <= value < max` while the code also accepted `is_equal(value, max)`, closing the interval, so a device on a shared boundary matched both neighbours and `.model` declaration order decided -- reversing two cards moved `l=1.999u`, unambiguously inside the LOWER bin, into the upper one and changed `i(V1)` by 2.95x. Selection now runs the strict rule first and the closed one only where it matched nothing, which keeps the top bin's `lmax` binding and means nothing that works today moves. AN OSDI MODEL COULD NOT BE BINNED: the set was eleven hardcoded built-ins, so a Verilog-A model written as a PDK writes one died with *Unable to find definition of model nv* for a model defined twice; OSDI is now asked through E-323's predicate and the four bin limits consumed like `level`. A DEGENERATE **agauss/gauss** WAS SILENTLY FLAT: a zero or negative variation or sigma returns the nominal for every draw, so a montecarlo over a parameter that never moves reported 100% yield where the real spec gave 12% -- one character apart, silent under `warn_physics`. Now audible; behaviour unchanged; `unif/aunif/limit` deliberately untouched. **.dc** NEVER REVISITED WHAT SETUP DECIDED: E-471's own comment says *.dc sets the circuit up once and walks its points inside the analysis* and that a frozen topology makes *the sweep quietly draw a flat line* -- it gave sweep the machinery to rebuild and `.dc` never got it, so a swept `l/w` leaving its bin kept the parse-time model (2.9x out) and a swept parameter changing an OSDI node collapse kept the old matrix (a flat line: -1.000e-03 at `rs = 0, 500, 1000` where the answers are -1e-3, -6.67e-4, -5e-4). `alter` and sweep both get these right, so `.dc` now REFUSES the point it cannot compute and names sweep (E-485's rule). Detection is free when nothing moves: the collapse is read from a flag `OSDItemp` already sets and `CKTdoJob` already consumes (E-417) -- `DCTsetInstParam` calls `DEVtemperature` and never asked -- and the bin from the four limits on the bound model. The refusal deliberately does NOT borrow E-427's *the device refused ... also refused by alter*, false on both counts here. THE ASCII RAWFILE WAS ONE DIGIT SHORT: `DEFPREC 15` emits sixteen significant digits where a double needs seventeen, so `write+load` changed the last ulp while the BINARY format was exact (`%.15e` fails to round-trip 51390 of 200000 random doubles, `%.16e` none). UNCHANGED: bin selection that works today in either card order incl. the top `lmax`; `alter`; sweep; `unif/aunif/limit`; a genuine device refusal; source and `m` sweeps; an unbinned model's sweep; the binary rawfile and an explicit `rawfileprec`. Verified `binstale` 64/64 (40/64 pre-fix, 24 discriminating), regression 409/409. No `openvaf-r` change.

E-496 *src/frontend/dotcards.c, src/frontend/breakp2.c, src/frontend/outitf.c, src/include/ngspice/ftedebug.h, src/include/ngspice/ftedefs.h, src/include/ngspice/ftext.h, examples/savekw_examples/ (new)*

A PLOT KEYWORD IS NOT A SIGNAL NAME. Reported from use: with `.option saveused, pyplot v(a[0]) xlabel 'something'` printed *save 'xlabel': nothing of that name is in this analysis*. E-469's bare-word scan took EVERY argument of an output command that was not a number, a redirection or an expression, with no knowledge of those commands' own keywords -- all 22 plot keywords, hardcopy likewise, and meas worst (meas tran m1 FIND v(b) AT 50u offered tran, m1, find, at). THE ANSWER WAS NEVER AFFECTED: E-469 over-collects deliberately (its own comment says under-saving would cost the answer) and a save matching nothing produces nothing -- the vector asked for returns bit-identical either way. The names were gathered in SILENCE until E-493 added the unmatched-save warning, which exposed the flaw rather than causing it. FIXED IN TWO PARTS, THE OBVIOUS ONE SECOND. (1) An INFERRED save is never reported: E-493's warning exists to catch a name the AUTHOR wrote and got wrong, so registration records which is which (db_auto -> save_info.autosaved) and the warning skips what saveused guessed; the flag is set only around ft_saveused's own com_save, leaving .save, save, .probe and savecurrents untouched. This covers every keyword of every command, present and future. (2) The plot family now knows its own grammar and meas contributes no bare words (its vectors arrive as `v(...)/i(...)`, already taken by the reference scan) -- but alone that would be the E-487 trap, a hand-maintained list that rots invisibly (it mirrors CT_PLOTKEYWORDS, unusable here because it is a tab-completion table built only in an interactive session), which is why (1) answers the report. UNCHANGED: everything about WHAT IS SAVED, pinned against the same run without the option (plain plot, plot with keywords, `plot ... vs ...`, meas then plot, let then plot); all still means everything; wrdata's file name is still not a vector; an explicit .save still makes saveused stand aside; a node genuinely named title or vs keeps its value; .save/.probe of a name matching nothing still warns. Verified savekw 46/46 (15/46 pre-fix, 31 discriminating), regression 410/410. No openvaf-r change.

E-497 *src/spicelib/analysis/dsetparm.c, src/spicelib/analysis/nsetparm.c, src/math/misc/randnumb.c, src/frontend/inpcom.c, src/xspice/icm/analog/{s_xfer,sine,square,triangle,oneshot}/cfunc.mod, examples/argcheck_examples/ (new)*

CONSTRAINTS THE DOCUMENTATION STATES AND THE CODE DID NOT CHECK. Round 56 mined the manual for stated numeric constraints; three of five findings came from one grep. **disto's** f2overf1 was unvalidated -- the manual says *between (and not equal to) 0.0 and 1.0*, yet 0, 1, 1.5, 2 and -0.5 were accepted in silence and move the answer (2F1-F2 read 1.695 / 1.630 / 1.477 / 1.580 vs 1.711 for a legal 0.5); at ratio 1 the plot labelled *IM: f1-f2* holds a product at DC, at a negative ratio the second tone is at a negative frequency. Both neighbouring cases in the SAME switch already return E_PARMVAL. Refused, not clamped -- the ratio IS the experiment. **setseed** truncated a fractional seed: %d stops at the first unusable character so 2.5 became 2, while 0/-3/abc are each named and sibling repeat names a fractional count. **s_xfer** indexed **int_ic** by the WRONG array's size -- `PARAM(int_ic[den_size - 2 - i])` with nothing consulting `PARAM_SIZE(int_ic)`: too short and the unreachable ICs read zero, `v(out)` at 10 us 7.00005 -> 4.99995e-05. The oscillator family went SILENTLY DEAD -- sine/square/triangle/oneshot announced an array mismatch and returned without setting an output, every evaluation: rc=0, source at zero, 2025 message blocks per 2 ms (~1e6 for 1 s). Exactly the shape E-491 fixed in **s_xfer** and named there; **d_osc/d_pwm** check inside INIT and are untouched. A duplicate **.param** was the only one of its family to pass in silence -- **.func** (E-491), **.model** and **.subckt** all warn -- and it takes the LAST where they keep the FIRST, so two includes that each set vdd resolve by include ORDER and a deck's own **.param** is displaced by a later library. Which value wins is unchanged; only made audible, scoped to top-level cards so subcircuit parameters are not mistaken for duplicates. Also: **D_STOP/N_STOP** reset the start field on a refused stop frequency (no observable consequence). UNCHANGED: legal ratios and single-tone disto; setseed with an integer or blanks; **s_xfer** cor-

rect/absent int_ic and E-491's num>den guard; matching oscillator arrays; a single .param, two names on one card, subcircuit parameters, and the .func/.model/.subckt messages. Verified argcheck 52/52 (33/52 pre-fix, 19 discriminating), regression 411/411. No openvaf-r change.

E-498 *src/spicelib/devices/vsrc/vsrc-temp.c, src/osdi/osdisetup.c, examples/reusestate_examples/ (new)*

PER-RUN STATE LEFT STALE BY THE SETUP-REUSE FAST PATH. E-471 keeps a circuit standing between points -- CKTdoJob skips CKTunsetup/CKTsetup and re-runs only CKTtemp. That reasoning was about topology and for topology it is right; what was never asked is which state DEVsetup initialises that is not topology at all. A VOLTAGE SOURCE STOPPED SCHEDULING BREAKPOINTS ENTIRELY -- VSRcbreak_time is the source's breakpoint cursor, armed only when CKTtime >= VSRcbreak_time and seeded to -1.0 in vsrctest.c; reused, the instance carried the PREVIOUS run's break time, at or past that run's TSTOP, so the test was false at time zero and stayed false, and a PULSE or PWL source armed no breakpoints at all (0 of 5 PWL corners landed on, against 5 of 5; 81 native timepoints against 103). Against a standalone run of the identical circuit maximum(v(n)) -- grid-INDEPENDENT, so a different ANSWER not a different sampling -- was +16%, +44%, -10% across five points and 106% at other spacings, in silence; a 0.3 ppm change of the swept value sufficed, since the schedule is either armed or it is not. Only the FIRST point of a sweep kept a correct schedule, so the same resistance gave two different answers by direction of travel (R=1000: 0.40217511 up, 0.33963131 down). **optimize** RETURNED A WRONG FIT AND CALLED IT CONVERGED -- fitting R for maximum(v(n))=0.20 gave 2501.777336 with reuse off (peak 0.2000000003) against 2156.690117 with it on (peak 0.2264, 13% out), in 106 evaluations rather than 67. Contradicts E-471's own comment that reuse *changes nothing a user can see except the time ... a few ulp*, whose stated cause is refuted too: KLU and Sparse give bit-identical deviations. OSDI's **last_crossing** CACHE -- per-run state whose contract osdiaccept.c states outright, *starts at 0.0 before any crossing has been observed*; only OSDIsetup seeded it, so a reused point held the previous point's crossing and a 100 kHz sine over 40 us read 3e-05 at time zero of points 2 to 5 instead of 0. Sibling absdelay was already right, re-seeding on is_init_tran. Both re-armed in **DEVtemperature**, which CKTtemp runs once per job on BOTH paths and which does NOT run on a resume. Suites stayed green because reusesetup_examples and reuseloops_examples contain no PULSE or PWL source and the two shipped sweep+tran+PULSE examples use -overlay, whose resampling damps the error below their tolerances -- both moved when this was fixed. UNCHANGED: the fast path itself (still reused at 4 of 5 points, 0 rebuilt), sources registering no breakpoints (SIN, EXP, dc-only), a PULSE current source (ISRC recomputes from CKTtime and keeps no cursor), and op/ac/tf/noise. REPORTED, NOT FIXED: under KLU, sweep -analysis ac with reuse returns 0.0 for reused points and crashes in about one run in ten, non-deterministically -- a different defect in a different layer, reproducing on the shipped binary. Verified reusestate 34/34 both solvers (17/34 pre-fix, 17 discriminating), regression 412/412. No openvaf-r change.

E-499 *src/maths/KLU/klusmp.c, src/include/ngspice/klu.h, src/frontend/com_optimize.c, src/frontend/com_sweep.c, examples/loopguard_examples/ (new)*

LOOP-COMMAND ARGUMENTS, VERDICTS, AND A KLU REFACTOR KIND. Round 58 re-probed the fast path after E-498; the reuse arithmetic was clean, everything around it was not. A COMPLEX REFACTOR HANDED A REAL FACTORISATION -- klu_z_refactor refills in place, walking the L/U index arrays klu_z_factor built; given a REAL Numeric it walks half-sized arrays with complex strides and a later klu_free_numeric frees what it scribbled (malloc names object 0x3ff0000000000000, the bit pattern of the double 1.0 -- a matrix VALUE freed as a pointer). Reachable because every AC/SP/NOISE run follows a real operating point and E-471's reuse keeps the matrix standing: under .option klu, sweep -analysis ac/noise gave 0.0 at every reused point, optimize -analysis ac fitted 10x wrong saying *converged*, sp SIGSEGV'd 9 of 10 runs. SPARSE and every real analysis were always fine; both directions now check the kind. FOUR ARGUMENTS PARSED BY A WEAKER PARSER THAN THE ONE IN THE SAME FILE -- optnum() served bounds/-target/-spec while -maxiter/-samples/-swarmsize used bare atoi and -

tol bare atof, so **1k** meant 1000 in a bound and 1 in **-maxiter** on one line; **-samples 2e2** ran 2 MC samples yet design centering printed a yield and a Wilson interval. **sweep lin 2e2** and **montecarlo 2e2** were already right, and **sweep** grew a strict count parser in E-478. **-seed** had the matching hole: E-497 guarded the COMMAND, the OPTION truncated 3.7 silently and 0/-3 left the run unseeded and NOT reproducible; all four sites now apply the rule. CONVERGED WHEN NOTHING WAS OPTIMISED -- a parameter the objective cannot depend on, or one that does not resolve, hands the STARTING value back after 3 evaluations; **-target x 0.4 0** printed *residual = 0* for a fit that never happened; a bound-pinned answer was not distinguished from a minimum. A KNOB THAT NAMES NOTHING fell through **sw_kind's SW_ALTER** fallback to a full set of points, **rc=0** and a plottable FLAT curve named after the typo (the shape E-435/E-488/E-431 each removed elsewhere); **list 1k inf 2k 3k** ran ONE point calling the valid 2k/3k unrecognized. **-overlay** DISCARDED THE SOLVER'S TIMEPOINTS -- a UNIFORM grid of the same count reported a peak 39% low (0.244 vs 0.402) from the same 121 points; now the union of the runs' own points, bounded, and it says which grid. Its %g names collided (three vectors all **vn_1000**), now shortest-round-trip per E-494. TWO WARNINGS CONTRADICTED EACH OTHER: *recorded as NaN* (true) and *those entries are zero* (false). UNCHANGED: a real fit is not annotated; every legitimate knob spelling, a good **list** and a two-knob sweep still parse; a transient sweep and the reuse tally are untouched; a legal **-seed** still pins the run. Verified loopguard 49/49 both solvers (18/49 pre-fix sparse, 13/49 klu -- 31 and 36 discriminating), regression 413/413. No **openvaf-r** change.

E-500 *src/frontend/com_dl.c, src/frontend/inp.c, src/frontend/com_presnp.c, src/frontend/commands.c, src/osdi/osdiregistry.c, src/include/ngspice/osdiitf.h, examples/vacompile_examples/ (new)*

pre_osdi -va COMPILES VERILOG-A AND LOADS IT IN ONE STEP. A deck shipping its own models needed a separate build first. **pre_osdi -va rmod.va gmod.va** now invokes **openvaf-r** and loads what it produces. Objects go in an **osdi/** directory BESIDE THE NETLIST -- a model directory stays clean, and an input/output collision becomes structurally impossible (E-452: **openvaf-r m.va -o m.va** destroyed the source, exit 0). The **.va** resolves against the netlist too, as **load_osdi** already does -- the first version got this wrong and worked from one directory but not another. Compiler lookup is **pre_snp's**, exported not reimplemented: **openvaf** variable, **\$OPENVAF**, **\$SPICE_LIB_DIR/openvaf-r**, **PATH** -- a bare **PATH** search would miss the compiler in the shipped bundle. RECOMPILING IS THE DEFAULT, **.option osdicache** opts out: a **.va** timestamp says whether the SOURCE changed, nothing about the COMPILER, so while **openvaf-r** is under development a cached object may come from a compiler that no longer exists (E-453's shape). Read from the options CARDS, not **cp_getvar** (E-464: nothing is published at **pre_time**), matched as a whole word not by **strstr** (E-451), every off-spelling off. **-f** FORCES THE REBUILD -- E-229 added it for the edit/recompile/re-source loop and the compile is now part of that loop, so forcing only the load reloaded the object being replaced; suite edits a **.va** and checks 0.5 -> 0.3333 only with **-f**. Staleness is strictly newer, never **>=**: **st_mtime** is 1 s coarse on POSIX, 2 s on FAT, so a tie rebuilds -- a tie costs a compile, the other way costs a wrong answer. Portable: **stat/st_mtime** as **inpcom.c** has used for years, **mkdir** guarded on **_WIN32 -> _mkdir**. Mixes with plain **pre_osdi** in every arrangement, since **-va** only changes how a **.va** ARGUMENT becomes a path. Refusals name the cause; a read-only directory falls back beside the netlist. UNCHANGED: plain **pre_osdi file.osdi**; **-va** with no **.va** argument is a no-op. Verified **vacompile** 33/33 both solvers (pre-fix the suite cannot complete: **-va** was taken as a filename), regression 414/414. No **openvaf-r** change.

E-501 *src/frontend/com_aging.c, src/frontend/com_aging.h, src/frontend/com_sweep.c, src/frontend/com_optimize.c, src/frontend/runcoms2.c, examples/dcenter_examples/dcenter_demo.cir, examples/agestate_examples/ (new)*

THE AGED DEVICE CAME BACK FRESH. **aging** (E-157) writes an accumulated dose into each device's age with **alter** and leaves the circuit standing. That dose lives ONLY in the running circuit -- it was never in the deck -- so every command that re-resources between evaluations threw

it away on the first one and ran the rest of the loop on a FRESH circuit, silently: wcd, highsigma, optimize -center. A 95-year dose moves the worst-case distance 8.66 sigma -> 1.81 sigma; wcd printed 8.66 for both. montecarlo escaped by accident -- E-346's fast path arms whenever a random value binds and does *no per-sample reset*. Doses are now recorded as written and replayed after an INTERNAL reset; a reset the USER types drops the record, because that is what reset means. **aging** ACCEPTED TARGETS IT COULD NOT AGE TO: aging nan wrote NaN doses and reported them aged (every comparison with NaN is false, so $t \leq 0$ let it through), aging abc aged to 0 s, dynamic verbose ate verbose as the stop time then blamed agerate -- E-497's shape. A NaN SPEC BOUND IS NEVER VIOLATED, so it is identical to omitting it: montecarlo -max nan reported 100% yield with a Wilson interval. -max/-min/-tol/-step/-scale now parsed across all four statistical commands -- with ft_numparse, NOT strtod: the first guard refused dynamic 20u, E-157's own documented spelling, and the existing aging suite caught it. A NEGATIVE bound stays legal, which is why these are not E-499's -tol reader. A YIELD FROM SAMPLES THAT NEVER VARIED is one point measured N times; E-495 warns per agauss call, this is the whole-run version. **optimize** PUBLISHED THE SCORE BUT NOT THE ANSWER -- dcenter_yield/dcenter_cpk but never the centred knob, so the shipped dcenter demo printed *checkvalid: vector xc is not available* on every run since it shipped; now optimize_<name>. **aging 1e8 param w** wrote 4.095e8 into a MOSFET WIDTH (409,500 km), reported 1 device aged, rc=0 -- warns and proceeds when the device also has an age. Verified agestate 30/30 both solvers, 19 fail pre-fix, regression 415/415. No openvaf-r change.

E-502 *src/frontend/parser/numparse.c, src/include/ngspice/ftext.h, src/frontend/com_envelope.c, src/frontend/com_emir.c, src/frontend/com_eye.c, src/frontend/com_qpss.c, src/frontend/com_reduce.c, src/frontend/com_hbosc.c, src/frontend/com_compose.c, src/frontend/com_rfstab.c, examples/nanguard_examples/ (new)*

THE GUARD THAT REFUSES THE WRONG VALUE AND ADMITS NaN. Six report-producing commands wrote `if (x <= 0.0) refuse;;` every comparison with NaN is false, so that refuses 0, refuses a negative number, and lets NaN straight through. **envelope <node> nan <tstop>** SIGSEGVd 5/5 -- $T = 1/f_c$ NaN, internal tran nan nan 0 nan REFUSED, refusal never checked, EFanalysis derefs a matrix never built (SMPmatSize, 0x10). Both halves fixed: the argument, and the unchecked internal run that any refused transient could trigger (E-438's shape). A LIMIT NEVER EXCEEDED: emir jmax nan = 0 segments over Jmax, all ok, on a grid with 2 genuine violations; reduce nan = 26 nodes -> 26 nodes (1.0x) and a reduced.sp that reduced nothing; qpss v(o) nan 1.1e6 = a 25-row all-nan spectrum; eye -ui nan = eye height 0, a fully closed link -- E-501's NaN spec bound one command over. TWO RETURNED A DIFFERENT ANSWER, NOT A BLANK: eye -tstart nan never skipped (the skip test is a comparison too) so the startup transient was folded in for 660x the RMS jitter; qpss maxorder nan silently became order <= 1, dropping every intermodulation product. Both clamped through an undefined double->int conversion, as did compose lin=1e12 -- INT_MAX, 17 GB really allocated. **emir** COULD NOT TELL A GIVEN WIDTH FROM THE DEFAULT: @r[w] answers 1e-5 for an undimensioned wire, so the likely oversight was analysed as a 10 um conductor and reported ok, while w=0/w=-0.5u were skipped and mislabelled *no width given*; now RESwidthGiven, two distinct messages. The idiom was already in the tree (hbosc K >= 1, compose's per-parameter NaN refusal, stb delegating to ac); guards now share ft_argpos/ft_argfinite/ft_argcount in numparse.c, parsing with ft_numparse so thick 0.5u and 1meg keep working. Verified nanguard 34/34 both solvers, 23 fail pre-fix, regression 416/416. No openvaf-r change.

E-503 *src/spicelib/analysis/cktdojob.c, src/include/ngspice/cktdefs.h, src/frontend/com_sweep.c, src/frontend/com_sweep.h, examples/reusedev_examples/ (new)*

ONE TRANSISTOR NO LONGER DISABLES SETUP REUSE FOR THE WHOLE CIRCUIT. E-471 offered reuse only to circuits built entirely from types whose topology is fixed UNCONDITIONALLY (linear elements, sources) plus OSDI, so a single BJT in an otherwise linear deck cost 3.46x at 300 sections, 10.09x at 1200. A per-TYPE gate for a per-PARAMETER hazard: a built-

in semiconductor decides its collapse in DEVsetup from a knowable set and nothing else -- BJT rc/rb/re/**rco**, diode rs/rsw/vp/tt, JFET rd/rs, Mos1/2/3/6/9 rd/rs/rsh/nrd/nrs -- each read off the device's own setup by auditing EVERY CKTmkVolt and its guard, not from the parameter docs, which do not say which parameters build nodes (rco gates a FOURTH BJT node via its Given flag; missed on the first pass). The sweep now DECLARES what it varies: 2.28x / 3.25x / 6.69x recovered. THE ABSENCE OF A DECLARATION KEEPS IT SAFE -- sw_request_reuse() CLEARS it, so montecarlo and any .param knob inherit nothing and keep the strict gate; whole-token match, so rsh is not rs. The declaration is a file-static, NOT a **CKTcircuit** field: in the struct it grew CKTcircuit 256 bytes and made argguard's warn_physics count random -- PADDING ALONE reproduces that, so a stale-layout read exists elsewhere and is recorded as an OPEN finding rather than made reachable here. Verified reusedev 16/16 both solvers, 8 fail pre-fix, every reuse check BIT-IDENTICAL against reusedev=0 after asserting the curve moves. Regression 417/417. No openvaf-r change.

E-504 *src/osdi/osditrunc.c, src/osdi/osdidefs.h, examples/rtdomain_examples/ (new)*

A MODEL MAY NOT DEMAND AN UNBOUNDED STEP COUNT. Two ways to hang the transient >150 s with no output, no error, no *timestep too small*. \$bound_step(1e-18) is a perfectly LEGAL positive request taken literally -- that check compares against CKTdelmin (~5e-20 for a 12 ns analysis), far BELOW what was asked, so nothing fired: the step was not too small for the solver, it was too small to FINISH. No clamp makes 1.2e10 steps work, so the rule is the only one that bounds the run -- a model may force at most 10⁶ steps across the analysis window, then it is clamped and the instance NAMED ONCE. \$discontinuity(0) outside any conditional pinned the step to the last accepted delta so it could never grow (E-55 already made the RETRY edge-triggered for this reason; the CAP needed the same floor); \$discontinuity(-1) means *no discontinuity*, is fine, stays silent. The compiler-side halves are in the openvaf-r report. The noise sign is deliberately NOT fixed here: by osdinoise.c the contribution factor is folded in as 'fac'

E-505 *src/osdi/osdiinit.c, examples/dropguard_examples/ (new)*

AN OPVAR NAME LOST TO THE SIMULATOR. ngspice writes its own instance parameters -- m, temp, dtemp, dt -- into the lookup table BEFORE anything the model declared, so a model operating-point variable with one of those names is computed on every evaluation and can never be read back: @n1[temp] returned the ambient temperature, @n1[m] the multiplier. Legal Verilog-A, and nothing said a word. Named at load time, not refused -- the rest of the model works. OP-VARS ONLY: a model may legitimately declare its own m/dtemp PARAMETER (E-394's has_m; a CMC-style model scales by it and the loader routes the deck value in), and the first, too-broad version warned for those too until limguard caught it. The three compiler-side halves are in the openvaf-r report. Verified dropguard 22/22 both solvers, 11 fail pre-fix, regression 419/419.

E-507 *src/spicelib/parser/{inpgval.c,inpgmod.c,inpptree.c}, src/include/ngspice/inpdefs.h, src/frontend/oututf.c, examples/scanfmt_examples/ (new)*

A VALUE NOTHING CHECKED HAD BEEN READ. numparam substitutes the TEXT of a { ... } expression, and {1/0} substitutes inf, which INPevaluate refuses. INPgetValue's IF_REALVEC path honours that refusal; its IF_REAL and IF_INTEGER paths took the same error and THREW IT AWAY, so the value landed as ZERO. .model nm nmos level=1 vto=0.5 kp={1/0} therefore built a transistor conducting 1e-12 against 1.25e-4 -- eight orders of magnitude, exit code 0, no diagnostic. Writing inf/nan/1e400 DIRECTLY on the same card is refused, and so is {1/0} on an instance line, as a built-in device's value, in an analysis argument (TSTEP/TSTOP/AC points) and through altermod: the model card was one path out of six, for OSDI and built-in devices alike, and the zeroing happened BEFORE the range check so a from clause could not catch it either. INPgetValue cannot simply return NULL for these types the way its vector path does -- none of its call sites test for NULL, which would trade a wrong number for a null dereference -- so the failure is RECORDED and the model-card path asks for it, that being the one caller with no other way to find out; it reports and keeps the parameter's DEFAULT, as the surrounding .model con-

ventions already do for an unknown or duplicated parameter. AN ORDINARY LEAF WAS NOT IN THE RELEASE SWITCH: PT_PLACEHOLDER is node type 0 and a normal leaf (for *i(something)*), so freeing one fell to default and printed *Internal error: unhandled parse-tree node type 0* -- on an ERROR path, where a half-built tree is released, ahead of the real and correct diagnostic. AND **save** CALLED AN OPERATING-POINT VARIABLE A MISSING PARAMETER: INPaName both FINDS a name and ASKS the device for its value, and any failure of the second half arrived as the first half's message, so a name the device does have -- and that print and meas both resolve -- was reported as *device has no parameter*, while the netlist . save form reported the same case correctly. Only E_BADPARM produces that message now. The compiler-side halves are in the openvaf-r report. Verified scanfmt 31/31 both solvers (19 fail pre-fix), regression 421/421.

E-509 *src/spicelib/parser/inpgval.c, src/spicelib/parser/inpgmod.c, src/include/ngspice/inpdefs.h*

AN INTEGER MODEL PARAMETER THAT DOES NOT FIT WAS SATURATED BEFORE ITS RANGE CHECK. (int) round(tmp) is undefined behaviour once the rounded value leaves the integer range; on this target it saturates, so a card saying *n=1e300* silently applied 2147483647 -- 291 orders of magnitude from what was written. The saturation happening BEFORE the parameter's own range check is what made it dangerous rather than merely surprising: a parameter declared from [0:2147483647] -- the idiomatic *any non-negative integer* -- accepted 1e300 because the clamped value landed exactly on its upper bound and passed. *sp=200* against [0:100] was refused correctly the whole time; only the values that OVERFLOWED slipped through, and those are the absurd ones. That is E-507's shape, a value mangled before the check that would have caught it. Flagged now the same way E-507 flags a {...} that evaluates non-finite: the model-card path reports it and keeps the parameter's DEFAULT rather than applying a fabricated number, via a new INPlastRangeError. Exactly INT_MAX, ordinary values and in-range values that ROUND are all still accepted. Verified in domainrt (28/28 both solvers), regression 423/423.

E-510 *src/spicelib/parser/inpgmod.c, src/frontend/com_dl.c*

EVERY RESISTOR .model CARD REPORTED A PARAMETER THAT IS NOT ON IT. .model mm r, .model mm r rsh=1 and .model mm res tc1=0.01 all produced *Error on .model mm : parameter (r) is not a number this parser accepts* -- and for the res card the message names the model TYPE token itself. The first token of a .model card IS the model type, and ngspice's resistor model has a parameter named r, so the type token matched a real parameter and the value check fired on whatever followed. The simulation was right and the message was not, on every resistor model card in every deck, on the shipped binary. **first_tok** already existed for exactly this collision -- it gates the duplicate-parameter warning a few lines above -- and the value checks added by E-507 and E-509 simply never consulted it; they are guarded by it now, and a genuinely unparseable value on a model card is still refused (E-507's contract is asserted by check [21]). Also: pre_osdi -va printed system()'s WAIT STATUS as the compiler's exit code, so 101 came out as 25856 (101 « 8) and 2 as 512 -- the comment at that site quotes *exit 512* as if it were an exit code, which is the same encoding gone unnoticed. Decoded with WEXITSTATUS now, and a signal death is named as such rather than printed as a status word. Verified in lrmntrin (21/21 both solvers), regression 424/424.

E-511 *src/frontend/spiceif.c, examples/optknown_examples/(new)*

AN OPTION THE RUN HONOURS, REPORTED AS UNKNOWN. User-reported: .option osdicache printed *Warning: unknown option 'osdicache' on a .options card; ignored.* and then cached the compile anyway -- the run said it was ignoring a setting it had already acted on. Two enhancements colliding. E-500 reads osdicache straight off the OPTION CARDS in inp.c, deliberately NOT through cp_getvar, because pre_osdi -va runs before any option has been published and cp_getvar would answer for the previous deck (the trap E-464 recorded for autobus). E-438 later made an unrecognised name on a .options card a warning, and rightly so -- a misspelling used to be silently inert, so .options reltoll=1e-12 left reltol at its default while the user believed the tolerance had been tightened. osdicache never reaches the simulator option table at all, so

E-438's check called it unknown. This is precisely the failure the allow-list in **spiceif.c** exists to prevent, and its own comment states it: *a warning that fires on a setting the run then honours is worse than no warning: it teaches the user to ignore the check E-438 added.* `autobus`, `saveused`, `klu`, `reusesetup` and `noinit` are all listed there; `osdicache` was simply never added. **seedinfo** had the IDENTICAL defect, found by checking whether `osdicache` was alone: read from the deck the same way and acted on by `setseedinfo()`, which makes `randnumb.c` print the seed it chose. The check must still bite, or the fix would undo E-438 -- a misspelt `osdicaches` and E-438's own `reltoll=1e-12` are both still reported (checks [4] and [5]). NOT A DEFECT, worth knowing: a `.va` and its `.osdi` written within the SAME SECOND skip the cache and the model rebuilds -- `st_mtime` has one-second granularity so the staleness test is STRICTLY newer; a tie costs one needless recompile, the other way loads the object built from the PREVIOUS text. Also observed and left alone: `seedinfo` and `seed=` must share one option card. Verified `optknown` 13/13 both solvers (6 fail pre-fix, measured 7/13 before), regression 425/425.

E-513 `src/spicelib/devices/vsrc/vsrcdefs.h`, `src/spicelib/devices/vsrc/vsrcrtemp.c`, `src/spicelib/devices/isrc/isrcdefs.h`, `src/spicelib/devices/isrc/isrcrtemp.c`, `examples/srcnote_examples/` (new)

A NOTE ABOUT THE DECK, REPEATED ONCE PER ANALYSIS. User-reported from a 100-sample Monte Carlo: *Note: v1: dc value used for op instead of transient time=0 value.* printed 100 times. The note describes a STATIC property of the deck -- the source carries both a DC value and a transient function whose `t=0` value differs, so the operating point uses one and the transient starts from the other -- but it is emitted from `VSRCtemp`, which `CKTdoJob` runs once per ANALYSIS. Any loop command repeats it per point, and three plain `tran` commands in one `.control` block printed it three times for the same reason. The worse half is the collision: E-477's loop bar redraws with `\r`, so every copy landed on that line and mangled both -- `...time=0 value.ran 47%` is the note's tail and the bar's `ran 47%` sharing a row. Two features that are individually correct, producing corrupt output together. FIX: latch the note per instance (`VSRCdcNoteDone`, and the same for `ISRC`), following the bitfield idiom already in those structs. Device instances are rebuilt when the deck is re-sourced, so *once per instance* is exactly *once per deck load*, and again after a reset, with no extra bookkeeping and no global state. Three `trans` -> 1 note, 100 MC samples -> 1, across a reset -> 2 (said again, correctly), and a source whose `dc` EQUALS its `t=0` value or has no `dc` at all stays silent, as before. DELIBERATE TRADE: an `alter` that later creates or removes the mismatch is not re-announced -- an informational note about a deck property, where *once per load* is the normal treatment, but a trade rather than a free win. NOT A DEFECT: a deck can avoid it entirely -- `V1 in 0 dc 1 PULSE(0 1 ...)` asks for 1 V at the operating point and 0 V at `t=0`, and dropping the `dc 1` removes the disagreement. Verified `srcnote` 12/12 both solvers (3 fail pre-fix, measured 9/12), with checks [7]/[8] asserting no progress frame carries the note's text and the bar still reaches 100%; regression 427/427.

E-514 `src/osdi/osdiload.c`, `src/osdi/osdisetup.c`, `src/osdi/osdidefs.h`, `examples/lrmops_examples/` (new)

Analog-operator state seeded with ZERO instead of the converged operating point. LRM 4.5.7 states `Output(t) = Input(max(t-td, 0))`, so before one delay has elapsed `absdelay` must return the input at `t=0`; `absdelay_stamp_tran` stored a literal `0.0`, so a delay around any non-zero bias read 0 for the whole first `td` and then stepped to the bias. Seeded from `CKTrhs0ld` at `MODEINITTRAN`, where the operating point has converged. LRM 4.5.10 requires `last_crossing` to return a negative value until the expression has crossed -- the only way a model can distinguish *not yet* from *crossed at t=0* -- and it returned `0.0`; the sentinel is now `OSDI_LAST_CROSSING_NONE`, and the crossing history is seeded from the operating point too, because for an expression already above zero the zero-filled history made the first accepted point look like a rising edge and overwrote the sentinel before any model could read it. Three existing suites were updated to match: `reusestate` expected the literal 0.0 sentinel, and `limguard/defaulttransition` are covered in the `openvaf` report.

E-516 `src/osdi/{osdicallbacks,osdiload,osdiaccept,osdiregistry}.c`, `src/osdi/osdidefs.h`, `src/include/`

ngspice/osdiitf.h, src/spicelib/analysis/dctrcurv.c

THE SIMULATOR HALF OF LRM 9.4.6/9.5.9 DEFERRAL. Every Verilog-A display funnels through `osdi_log`, so the accepted-iteration buffer lives THERE: DISPLAY/MONITOR-class messages without the compiler's immediate tag are held, OSDIload drops the superseded iteration's output when a new Newton iteration starts, and OSDIaccept / OSDIfinalStep / a new per-point flush in the .dc sweep print what the accepted point produced. Detecting 'new iteration' is the subtle half: STATnumIter is bulk-updated only when NIiter RETURNS, so it cannot separate the iterations of one solve -- but NIiter swaps CKTrhs/CKTrhsOld once per iteration, so the composite (circuit, RHS pointer, time, mode, iteration total) differs between adjacent iterations. \$monitor change detection compares the k-th monitor text of the flush with the k-th of the previous flush. The file-side hooks (`osdi_io_iter_begin/osdi_io_flush`) are resolved per loaded object with dlsym and are OPTIONAL both ways: an old .osdi lacks them and writes through; an old ngspice never calls them and the runtime stays unmanaged. Display output produced BEFORE the first Newton iteration (instance setup evaluating hoisted constant statements) prints through immediately, preserving the documented init-time behavior.

E-517 *src/spicelib/parser/{inpgmod,inpdpar,inpgval}.c, src/include/ngspice/inpdefs.h*

THE NETLIST HALF OF LRM 3.4.7 AND 3.4.1. Setting a parameter by its original name AND an aliasparam (or two aliases) on one .model card or instance line is an ERROR now -- the LRM says 'it shall be an error ... regardless of how the override is done', and E-395's warning let one value silently win (last-wins for model params, first-wins for instance defaults, deliberately unspecified). Setting the SAME spelling twice stays the E-395 warning. And the silent lossy round is audible: a non-integral value applied to an integer parameter is still rounded per the LRM rule (E-399's round-half-away-from-zero), but the E-507/E-509 flag pattern in INPgetVaLue now records that rounding CHANGED the value, and both the .model-card and instance-line applicers print a warning naming the parameter -- the footgun being an UNTYPED Verilog-A parameter whose type froze as integer from its default, where untyped=2.5 simulated as 3 without a word. Integral values stay silent, verified.

E-520 *src/osdi/{osdiregistry,osdicallbacks}.c, src/osdi/osdidefs.h*

\$LIMIT FALLBACK PER LRM 9.17.3, AND THE vdsLIM ALIAS. 'If the string refers to an unknown or unsupported function, the simulator is responsible for determining the appropriate limiting algorithm, just as if no string had been supplied.' E-396 made an unresolved \$limit name a hard LOAD FAILURE -- justified then, because the alternative was a NULL function pointer the compiled model still calls (SIGSEGV, no output). The mandated fallback exists now: the slot binds `osdi_limit_unknown`, a pass-through returning the access-function value -- one adapter serves every arity, because extra C arguments are simply never read -- and a load-time WARNING names the function, the arity, and the supported set. The model loads and runs; only its convergence aid is absent, exactly the no-string behavior, and the SIGSEGV stays fixed because every slot always gets a real function. Annex E Table E.2's preferred name `vdsLim` -- which this tree spells `limvds`, so a model written against the LRM's own table was REFUSED -- now binds to the same DEVlimvds implementation, in both the compiler's L020 lint set and this loader.

E-522 *src/osdi/osdiloadd.c*

THE SIMULATOR HALF OF LRM TABLE 5-1: STEP EVENTS FIRED FOR THE WRONG ANALYSES. `@(initial_step('dc'))` fired at the operating point of an .ac and a .noise analysis -- the phase runs with MODEDCOP set, so it carried ANALYSIS_DC in addition to the owning analysis's flag from E-53's job consultation -- and `@(final_step('ac'))` fired at the end of a .noise run, whose CK-Tmode carries MODEAC

E-523 *src/osdi/osdiloadd.c, src/osdi/osdidefs.h, src/osdi/osdiregistry.c*

THE SIMULATOR HALF OF LRM 4.5.7: ABSDELAY'S FROZEN td. The compiler's absdelay

info structs grew a flags word (absdelay_info_size 12 -> 16, gated on the descriptor's declared size so older .osdi files load unchanged) with OSDI_ABSDELAY_TD_FROZEN marking the two-argument form; the instance data lazily allocates a latched-td array, captures the live td value at the first transient evaluation (MODEINITTRAN, negative values clamped to 0), and every later evaluation of a frozen entry reads the latch while maxdelay entries keep reading the live value -- so `absdelay(V(in), td_expr)` holds the delay 4.5.7 fixes at initialization (the audit's deck reads the input 4.99ms back where tracking gave 3.99), and the three-argument form keeps its bounded tracking semantics. Verified under `lrmcoreops` 18/18 both solvers.

E-527 *src/osdi/osdiload.c*

`$SIMPARAM$STR SERVES` analysis_type AND cwd (LRM Table 9-28). The string channel served only analysis_name and the non-LRM 'simulator'; requesting any other Table 9-28 name was a compile warning and then FATAL at run time. analysis_type answers with the same derivation as analysis_name (ngspice gives analyses no user-chosen names distinct from their type -- the honest answer) and cwd answers the working directory, refreshed per query so a .control cd is followed. module/instance/path stay OUT, documented: the simparam channel is filled per circuit with no instance identity in reach, and inventing one would be worse than the honest warn-then-fatal. The compiler's served-name list moved in lockstep so L025 stays truthful. Verified under `lrmkernel` 43/43 both solvers.

E-528 *src/osdi/{osdiload,osdiacld,osdinoise}.c*

ANALYSIS PHASES AND NOISE, AUDITED AGAINST LRM 4.6 (simulator half). Table 4-22's TRAN OP column holds: analysis('ic')/analysis('static') are 1 at the t=0 operating point that precedes a transient and 0 at the timepoints -- the bits rode MODEINITTRAN, ngspice's FIRST ACCEPTED TIMESTEP, so the LRM's own if(analysis('ic')) initial-condition idiom fired mid-transient; they map to MODETRANOP now. analysis('nodeset') has a writer: the ANALYSIS_NODESET flag existed in the shared header, the compiler tested it, and nothing ever set it -- it is raised exactly while ngspice enforces .nodeset values (MODEINITJCT/MODEINITFIX with CKThadNodeset), latch-verified 1 with a card and 0 without. ac_stim matches the RUNNING small-signal analysis (4.6.3): .noise reuses the AC loader, and the activation compared the literal 'ac' -- an ac_stim('ac') source was injected into the noise gain solve, poisoning the input-referred noise 1000x (measured 5.75e-12 against a 5.76e-9 control), while ac_stim('noise') could never activate even though this simulator names the analysis 'noise'; the two decks swap roles exactly now. And the noise bridge correlates by CALL SITE: the compiler exports each source's call id after a \x1f separator in the name; grouping compares the full string (one call's multi-branch entries keep summing coherently), the per-label summary vectors strip the suffix, and a reporting fold combines a label's powers onto one entry per 4.6.4.1 -- two separate same-labelled calls no longer cancel as signed amplitudes where the LRM requires uncorrelated power addition. Verified `lrmnoise` 17/17 both solvers.

E-529 *src/osdi/{osdiregistry,osdiinit,osdiload,osdiparam}.c, src/osdi/osdi.h, README_OSDI.md*

THE OSDI LAYER, AUDITED -- the conformance audit's last area. Original-OpenVAF v0.3 objects are REJECTED with a recompile message: the 'must be version 0.3' path accepted them and read them through the EXTENDED in-repo layout -- the Osdinode stride grew 48 -> 56 bytes (E-45), so every node record past the first was misread (a spec-conformant fake object listed its second terminal's current as i_V, the units field); osdiacld.c/osditrnoise.c read num_ac_stim_src and friends past the 0.3 descriptor's end; and 0.3's five-argument load_noise was called with four -- subtly wrong DC metadata and a transient SIGSEGV with zero diagnostics. osdi.h drops its stale 'matches the OSDI specification' claim and the v0.3 CURR constants for an explicit divergence note. The phantom IFparm row is gone: the synthesized m-alias slot was counted whenever the model lacked its own m but only WRITTEN when a \$mfactor parameter exists, leaving the table's last row zeroed -- keyword NULL, a '(null)' devhelp row one strcmp from a crash for foreign objects; the count now mirrors the writer's condition. The OpenMP eval branch gets the serial

branch's initial-step gating (a task-local `OsdiSimInfo`, one task per instance): an `--enable-openmp` build never fired `@(initial_step)` in any OSDI device. A NEGATIVE multiplicity is warned and ignored on every route `--alter` included, which bypassed E-447's parser warning and APPLIED the value: a 1k resistor model sourcing +4mA, and `.noise` printing 'onoise_spectrum = nan' through the compiled `sqrt(m)` factor; `m=0` stays E-426's silent disable-this-instance idiom (the `inputguard/instknobs` suites caught the first over-broad cut of this guard immediately). `README_OSDI` gained a version-support and deliberate-bounds section, the `$bound_step` (`tstop-tstart`)/1e6 per-model floor (E-504) now disclosed there and in the compliance doc. Verified `lrmosdi` 10/10 both solvers, including a fake v0.3 object the suite compiles itself.

E-530 `src/osdi/{osdisetup,osdiregistry,osdiparam}.c, src/osdi/{osdi.h,osdidefs.h}, src/include/ngspice/osdiitf.h, src/frontend/spiceif.c, README_OSDI.md`

.OPTION OSDIMC -- THE SIMULATOR RUNS THE WHOLE MONTE-CARLO LOOP (simulator half). With the option set (alias `automc`), every run-class command starts a fresh trial: each parameter the Verilog-A declared with `(* std= *)` statistics is written nominal+draw through the ORDINARY parameter setter -- the `alter` path -- so there is no reset, no netlist re-expansion, and no `gauss()`/`agauss()` expressions in the deck; the existing setup/temperature machinery derives everything downstream, node-collapse rebuilds (E-471) included. The FIRST run after sourcing is the nominal BASELINE -- defaults of parameters the deck never set resolve during setup (E-476), so nominals are only knowable after one pass; draws begin with run two, uniformly. A MODEL parameter draws once per card per trial (process: instances sharing the card move in lockstep, distinct cards independently); a `(* type="instance" *)` parameter draws per instance (mismatch). Draws are PURE functions of (mcseed, trial, owner name, param id) -- `splitmix64` into Box-Muller, no stored RNG state, the compiler's (seed, salt) \$random philosophy -- so a fresh process reproduces every value bit-for-bit, `.option mcseed=` swaps whole ensembles, and resume never redraws. `alter/altermod RECENTER` a parameter's nominal (a hook in `OSDIparam/OSDIParam` -- draws are nominal+delta, never a random walk); dropping the option restores every drawn parameter exactly; a re-source invalidates the table; `.option osdimc_verbose` prints every draw; all four option names joined if `is_option's` list so no honoured card is called unknown (the E-471/473 lesson). The suite MEASURES the machinery: over 300 trials `gauss` sigma=25 reads 25.1, uniform fills exactly [nominal-std, nominal+std], `std_rel` 5% of 2.0 reads 0.0998, mismatch sigma 9.65/10 with per-instance independence beside per-card lockstep; an attribute-free model is provably untouched and the draw path adds no measurable overhead beside a PSP103. DELIBERATE BOUND, documented in `README_OSDI` and handbook 3.6: a draw violating the parameter's Verilog-A range fails that run with the device's own located error, exactly as the same `alter` would (the descriptor exports no ranges) -- size sigmas accordingly; correlated parameters are future work. Verified `osdimc` 24/24 both solvers; full sweep 444/444 ALL OK.

E-531 `src/frontend/{spiceif,device,dotcards,inp,rawfile}.c, src/include/ngspice/{fiteext,osdiitf}.h, src/osdi/{osdisetup,osdiparam,osditrnoise}.c, examples/huntfix_examples/(new), examples/osdimc_examples/(24 -> 29 checks)`

THE BUG-HUNT ROUND (simulator half): the E-530 machinery hardened and the `alter/batch/rawfile` surface made honest. MACHINE WRITES NO LONGER RECENTER MONTE-CARLO NOMINALS: the recenter hook lived in the raw parameter setter, so a `.dc` sweep's save/restore -- and the sweep command's, and sensitivity's perturbations -- permanently shifted a statistical parameter's nominal to whatever value was in flight (a random walk seeded by one sweep). Recentering moved into a `doset_user` wrapper on the frontend's user `alter/altermod` sites (wildcards included); the pinned check shows the post-sweep draw landing on the exact nominal-0 delta. `reset/re-source` RESTARTS the MC cleanly (`inp_dodeck` drops the nominal table; `reset` reuses the CKTcircuit pointer, so the old identity check missed the reallocation -- entries leaked and a heap-address reuse could have matched a stale nominal to a fresh device). A FAILED TRIAL SAYS SO IN-BAND: the range error names the model and value ("Parameter rd of 'mm' is out of bounds (value -0.489)!") and a once-per-trial notice states the previous run's vectors remain cur-

rent -- a 1000-trial script can no longer silently duplicate its last good sample. Draws are isfinite-checked (an overflowing sigma wrote -inf into parameter storage; only a declared range caught it). THE SURFACE: altermod reaches STRING parameters (@mm[mode] = "quad" died as no such vector "quad"; a new if_setparam_string routes it with altermod's CKTtemp propagation); whole-array altermod gets per-element guidance instead of "has no parameter cf" two lines above a showmod that prints it; alter REFUSES non-representable values (1e400 stored +inf: a built-in resistor silently became an OPEN -- i=0, rc=0, a perfectly plausible wrong answer -- and an OSDI device died in convergence noise pointing nowhere near the cause); repeated run of an .op+.ac+.tran deck keeps ALL its jobs (the batch epilogue's .op save-all was analysis-restricted, so the automatic run's other analyses failed with "no data saved ... analysis not run", plots silently lost); rawfile roundtrips keep qualified cross-plot names (dc2.i(v1) came back as the unaddressable i(dc2.i(v1))); same for v(...)); alter of a model parameter through a device name points at altermod instead of denying the parameter exists; the transient-noise noise_table warning prints the user's label instead of the raw 0x1F-suffixed correlation name (od-verified: the control byte reached the terminal). Three hunt findings were RETRACTED as artifacts of the hunt harness itself and are recorded as such; the handbook's stale ".disto not supported" row was corrected - E-352's OSDIdisto matches an analytic ground truth (pointwise periodic solve + FFT) to <1%. Verified huntfix 12/12 and osdimc 29/29 both solvers; full sweep 445/445 ALL OK.

E-532 *src/osdi/{osdisetup,osdiloadd,osdiacld,osdipzld}.c, src/include -- via src/osdi/osdidefs.h, examples/huntfix2_examples/(new)*

THE OPENVAF-R HUNT ROUND (simulator half): a chained terminal short was silently an OPEN circuit. $V(a,m) <+ 0.0$; $V(m,b) <+ 0.0$; with m internal is physically the SAME circuit as $V(a,b) <+ 0.0$ -- and the direct spelling shorts correctly (E-401 emits a real 0 V source for a terminal-terminal 0 V contribution). The chained spelling conducted NOTHING: 0 A and the full source voltage across the device, zero diagnostics from either tool. The first collapse merged m onto one terminal; the second then faced a terminal-terminal merge, which node collapsing cannot perform (ngspice allocates terminal nodes), and the old loop silently continued -- the $V = 0$ EQUATION ITSELF vanished from the system. Chains of any depth failed identically; a chain landing on an internal node was fine, which is exactly why the hole survived (BSIM4-class gate ladders chain collapses like this, saved only by their conditions rarely all firing). THE FIX STAMPS WHAT THE HINT EXPRESSED: when collapse_nodes refuses a terminal-terminal or terminal-ground merge it records the pair, and OSDIsetup builds a SYNTHETIC IDEAL 0 V SOURCE between the two global nodes -- a branch equation $V(n1)-V(n2)=0$ plus the +-1 current entries, the vsrc stamp at dc 0 -- following the absdelay extra-entry pattern end to end: SMPmakeElt at setup, KLU COO->CSC rebind and real/complex pointer switching on every DC<->AC transition, stamps in the DC/tran, AC and PZ loads (OpenMP branch included), branch-equation REUSE across sens's double-setup (which forbids node allocation), deletion in OSDIunsetup, netlist-tied endpoints and duplicate pairs dropped as redundant. REWRITING THE LOOP AS PROPER GROUP MERGES FIXED TWO LATENT CORRUPTIONS, both reachable through hint chains: a group already collapsed to GROUND, merged again, had its members re-mapped onto the other group -- quietly UN-grounding them -- while the node count decremented for a row that never disappeared; and a redundant hint between nodes ALREADY IN ONE GROUP (a collapse triangle) renumbered every node above the representative and corrupted the count the same way. Verified huntfix2 21/21 both solvers -- direct/one-internal/two-internal spellings identical in op, AC and transient, across reset, through sens, plus the ground-chain, un-ground and triangle shapes against closed-form currents; full sweep 446/446 ALL OK.

E-533 *src/frontend/com_sweep.c, examples/sweepdc_examples/(new), examples/{loopbar,reusedev}_examples/(re-pointed)*

sweep HANDS ELIGIBLE OP-POINT SWEEPS TO ONE DC ANALYSIS. With the default -- analysis op, a single dc-sweepable knob and evenly spaced points, the point loop -- npt COLD operating points, a full op job with the whole gmin/source-stepping fallback chain and one

retained plot per point -- becomes ONE dc analysis: a warm NIiter continuation from point to point (E-258), the sweep's outputs served from the dc plot. Measured on the motivating deck, a 1000-device OSDI Monte-Carlo ladder swept over 9900 resistance points: 21.2 s -> 2.16 s, bit-identical to a direct `.dc`, within Newton tolerance of the per-point loop ($\sim 2e-4$ relative on a clamped diode divider); `-perpoint` forces the old loop. SAFE BECAUSE THE TWO ENGINES WERE ALREADY COMPLEMENTS, each pointing at the other: `.dc` REFUSES a point that moves an OSDI node collapse or leaves a model bin (E-495 -- its own message recommends sweep), aborts when a device rejects a value (E-427), and stops on non-convergence -- every such outcome reaches the handover as a failed run and FALLS BACK to the per-point loop unchanged, which re-decides topology per point and is the correct instrument for exactly those sweeps (pinned: an instance-parameter sweep across a collapse boundary falls back and lands on closed-form series-resistance values to $3e-13$). Both engines set the swept value through the machine-write path and restore it afterwards, so neither recenters `osdimc` nominals (E-531) and frozen `agauss()` draws stay frozen. ELIGIBLE: bare source/resistor names, non-wildcard `@inst[param]` (E-62's `.dc` spelling), spacing judged from the VALUES (positional, `lin`, and a uniform list qualify; `dec/oct/uneven` lists cannot be represented as `start + kstep`); `@r2[resistance]` and `@v1[dc]` are rewritten to the bare spelling so `.dc`'s classic resistor/source arms step the value directly instead of routing every point through `CKTsetInstParam` + a full `CKTtemp` pass over every device. STAYING ON THE LOOP BY DESIGN: `model-parameter` and `.param` knobs (no dc arm), `-vs` families, `-overlay`, live `@dev[param]` outputs (only the loop can read the circuit at each point -- prescreened BEFORE any dc is run, so none is wasted), and sweep `temp` whenever the deck contains an OSDI device: `dc temp` holds one setup for the whole sweep and cannot follow a temperature-moved collapse -- the known-open finding the sweep`temp` suite pins, whose own check names the per-point path as the correct instrument; built-in-only decks keep the `temp` speedup. FLOAT CARE, twice: `.dc` accumulates `value += step` against an ABSOLUTE overshoot test, so a `span/(N-1)` step can lose its endpoint to accumulated rounding -- the issued stop is padded by `0.49step` (the endpoint always lands, the next point always exceeds the pad and is dropped) and the dc plot's own scale values are ADOPTED into the sweep scale rather than assumed, with a count mismatch falling back. STATED HONESTLY: dc points run `MODEDCTRANCURVE` -- a Verilog-A analysis ("dc") is true where the loop's op had `analysis("static")` -- and a warm continuation TRACKS a solution branch where independent cold ops re-decide it at every point; both are `.dc`'s own readings, restored by `-perpoint`. The announce line names the engine, so a log always says which path computed the numbers. Two suites were re-pointed at the machinery they actually pin: `loopbar`'s op-sweep vehicles carry `-perpoint` (the point-loop bar needs a point loop; a new check pins that the eligible default hands over and draws none) and `reusedev`'s eligible knobs likewise (the E-471/E-503 reuse decision lives in the loop). Verified sweepdc 15/15 both solvers; full sweep 447/447 ALL OK. No `openvaf-r` change.

E-534 `src/spicelib/analysis/{dctrcurv,dctsetp}.c`, `src/spicelib/parser/inp2dot.c`, `src/include/ngspice/trcvdefs.h`, `src/frontend/com_sweep.c`, `examples/dcxswEEP_examples/ (new)`, `examples/sweepdc_examples/ (15 -> 17 checks)`

.dc LEARNS THE REST OF THE PARAMETER SURFACE: model knobs, wildcards, subcircuits, and `lin/dec/oct` scales. The sweep variables the sweep/`altermod` family established now have a NATIVE dc arm, card and command alike: MODEL parameters (`@mod[p]`, with the dotted subcircuit `@x1.rmod[p]` resolved through the same E-433 hierarchy funnel the rest of the tooling uses) and the wildcard families `@*[p]` (every model with p), `@#*[p] / @*[ [p] ]` (every instance with p), `@*: leaf[p]` (every model named leaf, wherever expansion put it). Every knob kind, old and new, takes `'lin`

E-535 `src/osdi/{osdisetup,osdiparam,osdicallbacks}.c`, `src/osdi/osdidefs.h`, `src/frontend/{com_sweep,com_optimize,com_loadpull,evaluate}.c`, `src/include/ngspice/{osdiitf,trcvdefs}.h`, `src/spicelib/analysis/dctrcurv.c`, `examples/mcpolicy_examples/ (new)`

THE OSDIMC TRIAL POLICY FOR LOOP COMMANDS — AND THE HUNT ROUND THAT

MADE IT HONEST. E-530's per-run draw was right for a lone op and wrong at both ends of the loop-command spectrum: a swept "curve" was N DIFFERENT samples stitched together, optimize fit a STOCHASTIC objective (measured "converged" 2.4% off a closed-form optimum, reported with full confidence), and montecarlo — the one command that WANTS a fresh draw per sample — got none, its per-sample internal reset restarting the counter so every sample re-computed the nominal. THREE LEVERS: HOLD (sweep's per-point path, optimize, wcd, load-pull bracket their inner analyses — the whole loop consumes exactly ONE trial; draws are pure functions of (mcseed, trial, owner, id), so the held sample survives even an internal re-source), PRESERVE (a command's own internal reset keeps the sequence running where a USER reset still restarts at the baseline), SCALE (highsigma -scale inflates attribute-declared sigmas along with the netlist ones). A ONE-HOUR ADVERSARIAL HUNT over the first cut produced seventeen findings; the five correctness fixes ship with the policy: (1) DRAWS NOW PRECEDE THE INIT-RESIDENT CODE — the first cut applied a preserved trial at the END of OSDItemp, after every setup_instance had cached its intermediates from the NOMINALS, so a parameter feeding only a hoisted assignment (geff=1/rr) was drawn in storage and nominal in the physics: every montecarlo sample computed the exact nominal while osdimc_verbose logged distinct draws (5/5 threshold violations where the drawn samples give 3 — the log said the samples differed, the answers said one circuit). The apply moved to the end of OSDIsetup: defaults are resolved there and CKTtemp/OSDItemp comes after, so the hoisted code evaluates the drawn values; a draw that moves node collapse is caught by the E-417/E-495 machinery like any between-setups change. (2) MACHINE WRITES PIN THEIR PARAMETER — the late-apply also ran on every OSDItemp (per .dc point, per sens perturbation) and the held re-apply per sweep point, overwriting machine writes with nominal+delta: dc @n1[dr] 0 1000 500 returned ONE POINT THREE TIMES (flat 0.49596 where the curve runs 0.50→0.33), sweep @n1[dr] likewise, sens reported ~1e-14 where -2.5e-4 is closed-form. E-531 said machine writes never RECENTER; the missing complement: they must WIN — every successful OSDIparam/OSDIparam store pins its osdimc entry, the applier skips pinned entries, pins clear at bracket edges and at every un-held run (so alter recentering still lands next run, exactly as osdimc check 19 pins). (3) THE BASELINE IS EXPLICIT — trial-1-never-draws was an empty-table accident that an optimize flow or an unset/set osdimc toggle defeated (both then DREW on the run every contract calls the baseline); a trial<2 guard replaces it. (4) SETUP DISPLAY PRINTS PER SETUP — display_managed latched at the first Newton iteration forever, so a re-setup's hoisted \$strobe was deferred-then-dropped: a .dc temp sweep printed the FIRST point's strobe, dropped points 2..N, then flushed the RESTORE's strobe where it read like a sweep point; OSDIsetup and OSDItemp now re-enter setup-phase display and every point prints at its own temperature. (5) \$MONITOR HISTORY RESETS PER ANALYSIS — a second identical op printed nothing (its first accepted point matched the PREVIOUS run's last flush); reset per analysis, deliberately NOT per temperature point. ALSO: a nested .dc REFUSES two levels that move the same knob in any spelling (v1 vs @v1[dc], @r2[resistance] vs r2, wildcard-list overlaps — the first point was computed with the outer start and labeled with the inner value, and the restore parked the knob on the wrong "nominal"), with a full reverse-order restore on refusal; dec/oct point counts keep the PARSED N apart from the derived total (a re-run of a still-loaded card no longer refines its own grid 5→11→23); evaluate.c warns once when two genuine multi-element vectors of different lengths meet (the silent flat-extension of a truncated import now says so). KNOWN-OPEN, recorded with repro decks: interrupt longjmps leak the hold/sigma-scale/preserve flags (an interrupted highsigma measured continuing its sequence through a user re-source); optimize's internal runner lacks the preserve call (-center sees no osdimc variation); highsigma -scale carries no likelihood-ratio term for the inflated OSDI draws (P(fail) 0.42 reported where 0.354 is true — use netlist .param statistics with -scale until the weight channel exists); the hold is a flag, not a depth counter (a loop nested as another's -analysis releases the outer bracket); loop commands never poll ft_intrpt; a failed later-level .dc resolution leaves earlier levels applied. Verified mcpolicy 12/12 (new; closed-form expectations from the run's own readbacks, both solvers) and osdimc 29/29, optimize 43/43, sweepdc 17/17, dcxsweep 20/20, montecarlo/highsigma/wcd green; full sweep ALL OK. No openvaf-r change.

E-536 *src/frontend/{signal_handler,com_sweep,com_optimize,com_loadpull,outitf}.c, src/include/ngspice/{fteext,osdiitf}.h, src/osdi/osdisetup.c, src/spicelib/analysis/dctrcurv.c, examples/mcpolicy_examples/ (12 -> 18 checks)*

THE OSDIMC LEDGER CLOSED: interrupt safety, nested holds, weighted high-sigma, and the `.dc` restore. E-535 shipped its loop-command policy with an honest known-open ledger of nine hunt findings; this closes all nine, each repro now a pinned check. THE THROUGH-LINE IS STATE THAT OUTLIVED ITS COMMAND. AN INTERRUPT NO LONGER POISONS THE SESSION: `ft_sigintr` LONGJMPs to the prompt, and every loop command's cleanup sat AFTER that point of no return -- an interrupted `highsigma -scale 5` left the INFLATION ARMED (every later `osdimc` draw in the session used 5x sigmas, silently), a dangling hold froze all later runs on one Monte-Carlo sample, a dangling preserve carried the interrupted run's trial sequence through a USER re-source (measured: a fresh source continuing at trial 91828), `mc_wcd_off/mc_sss_off` never ran so the netlist gaussians stayed pinned to a chosen u-vector, `ft_optimizing` left TRUE muted every later banner, `aging_internal_reset` left elevated made later USER resets keep age records they must drop, and a half-drawn progress bar kept the terminal line; `ft_sigintr_cleanup()` -- which exists for exactly this -- now resets all of it. THE HOLD IS A DEPTH, NOT A FLAG: a loop command nested as another's -analysis (`optimize -analysis sweep . . .`, a swept-curve objective) had the inner sweep's release un-hold the OUTER optimizer, so every evaluation took a fresh trial -- measured trials 2..15 across 14 evaluations, the stochastic-objective bias E-535 exists to eliminate, reintroduced through nesting; only the outermost edges now reset/release, and every caller releases exactly what it took (an error path that never took the bracket cannot pop someone else's). OPTIMIZE JOINS THE PRESERVE PROTOCOL: `opt_run_cmd` -- the optimizer's own copy of the runner E-535 taught -- never got the call, so a `-dparam` fit's per-evaluation re-source zeroed the counter (the objective silently changed mid-search) and `-center` saw ZERO `osdimc` variation (every sample of every candidate computed the nominal, so its yield ignored the model's declared statistics entirely); `-center` now runs WITHOUT the hold its samples must not have and instead REWINDS THE TRIAL COUNTER to a pre-search checkpoint, replaying the same trial window per candidate -- variation sampled, determinism kept, exact because draws are pure functions of (seed, trial, owner, id). HIGHSIGMA NOW WEIGHTS THE DRAWS IT INFLATES: `-scale` inflated the attribute-declared sigmas but `mc_sample_weight()` carries likelihood ratios only for netlist `.param` draws, so a metric depending on an OSDI statistical parameter had every sample at weight 1 and $P(\text{fail})$ was just the lambda-inflated failure fraction (measured 0.42 reported where the true value is 0.354, with a confident ± 0.03 bar under it); a new `OSDI mcSampleLogLR()` recomputes each gauss dimension's standard-normal deviate and sums $\log(\lambda) - n^2(\lambda^2 - 1)/2$, multiplied into the weight beside the netlist term -- recomputed rather than accumulated, so the E-471 double-setup is harmless by construction, and uniforms are now explicitly NOT inflated (matching the netlist SSS policy: an inflated uniform can land where the true density is zero, which no finite weight represents). Validated against quadrature: 0.29670536 closed-form, the seeded suite run reporting 0.341 against a raw inflated fraction of 0.454. CTRL-C STOPS A LOOP COMMAND: none of them polled `ft_intrpt`, and with `ft_setflag` an interrupt aborts only the inner analysis it lands in while the next `dosim()` clears the flag -- so a Ctrl-C mid-sweep still announced "100000 points into plot 'sweep1'", every remaining point an insta-aborted analysis stitched in as data; each loop now polls at its iteration boundary (sweep marks un-run points nan per E-445 and says how far it got, montecarlo/highsigma report over completed samples, `wcd` refuses a half-converged MPFP, `loadpull` stops the grid, the optimizer methods break out and `opt_eval` short-circuits so a population method owes no further analyses), with the documented limitation that an interrupt arriving INSIDE a long inner analysis is consumed by that analysis's own `OUTstopnow()` -- ngspice's contract, shared by every multi-analysis command, left alone. THE `.dc` RESTORE COMPLETED: resolution APPLIES each nest level's start as it walks the levels, and a failure at a LATER level returned without putting the earlier ones back -- `dc v1 0.5 1.5 0.5 @n1 [nonesuch] 1 2 0.5` left `v1` parked at 0.5 V, a temp outer level left every temper-baked expression at the sweep's start temperature; the reverse-order restore

E-535 wrote for its clash refusal is now a shared helper every resolution-failure path calls. Plus: the clash guard catches a WILDCARD instance level covering a source/resistor level's principal parameter (dc v1 0.5 1.5 0.5 @#*[dc] 0 2 1 moved v1's dc from both levels and left it corrupted), and DCTresolveXParam's integer-refusal paths drop their collected target list instead of leaving dangling owner pointers on the job. Verified mcpolicy 18/18 (12 -> 18, both solvers, every new check a former ledger entry) and osdimc 29/29, optimize 43/43, sweepdc 17/17, dcxsweep 20/20, sweeptemp 20/20, montecarlo/highsigma/wcd/nested sweep green; full sweep ALL OK. No openvaf-r change.

E-537 *src/frontend/{com_sweep,com_optimize,com_aging,com_emir,com_loadpull,evaluate,spiceif}.c, src/osdi/osdisetup.c, src/include/ngspice/osdiitf.h, examples/mcpolicy_examples/ (18 -> 28 checks)*

THE SECOND HUNT ROUND: WHAT THE STATISTICAL COMMANDS WERE NOT SAYING. A second one-hour adversarial hunt over the shipped E-535/536 work produced sixteen findings (fourteen with repro decks); this resolves all sixteen, two of them regressions E-536 itself introduced. THE HEADLINE IS A MIS-ESTIMATE E-536 SHIPPED: its highsigma -scale importance weight is a PRODUCT over every inflated statistical parameter in the circuit -- including ones the metric cannot depend on -- so its variance grows exponentially with their number. Measured with bystander devices on a DISCONNECTED subcircuit (identical failures observed every run, proving the physics untouched) against a true P(fail) of 0.2967: 0 bystanders 0.282 correct, 2 -> 0.072, 5 -> 0.018, 10 -> 0.0015, 20 -> 2.5e-11, ten orders of magnitude low and printed with the same confidence as a good answer. E-536 had traded a BOUNDED bias (0.42 against a true 0.354) for an UNBOUNDED variance -- better in one dimension, far worse in the decks people have, where per-instance mismatch supplies dimensions automatically. Unbiased-but-useless is what importance sampling DOES in high dimension, so the command now measures its own reliability: it computes the EFFECTIVE SAMPLE SIZE $(\sum w)^2 / \sum w^2$ and, when that falls under a tenth of the run, states that the weights collapsed, gives the number (measured 1.2 out of 400), explains the cause and says P(fail) is not trustworthy -- a guard that also covers the pre-existing netlist SSS path, which degenerates identically (20 extra gaussians -> $2.4e-04$; that weakness predates E-536, what E-536 changed is how easily an ordinary deck reaches it). Two more honesty repairs in the same summary: a weighted mean is NOT automatically a probability (it printed 1.0445 where all 200 samples failed and the true answer is exactly 1) so it is clamped into [0,1] and says when; and the equivalent-sigma guard substituted 0.000 at the boundary, which READS AS $P = 0.5$ -- the opposite of certain failure -- and now reports n/a. FIVE COMMANDS WERE READING ANOTHER RUN'S NUMBERS: ngspice keeps the previous plot when an analysis fails, and the E-438/E-445 guard was applied in sweep and optimize but MISSING in highsigma, wcd, aging, emir and loadpull. highsigma counted failed samples using the previous sample's metric -- on one deck montecarlo reported "10 of 60 samples failed to simulate and are EXCLUDED" while highsigma said nothing and folded all 60 in, and the bias has a DIRECTION: with -scale the failures cluster in the TAIL, exactly the region being measured, so the rare-failure probability came out systematically LOW; it now excludes, averages over what solved, and states that direction. wcd differenced its gradient against a stale value, making that component exactly zero -- with one dimension the whole gradient went to zero and the command announced "the metric does not respond to any statistical parameter", blaming the model for a point the SEARCH had chosen out of range (proven false on the same deck: at -max 0.6 it finds $\beta = 0.3704$, and only when the search steps into the failing region does it accuse the model); it now names what happened, separately at the nominal point, at an iteration and at a gradient probe. aging computed a dose from a foreign bias point and then WROTE AND PERSISTED it; emir reprinted a complete reliability sign-off ("0 segments over Jmax") on an operating point that no longer existed; loadpull entered a failed grid point as a duplicate of its neighbour, where the reported optimum could sit on it. A MISTYPED MODEL PARAMETER NO LONGER KILLS THE SESSION: altermod with a value an OSDI Verilog-A range refuses called controlled_exit(1) -- harmless with no analysis yet and after op (the block is guarded by CKTtime > 0) and FATAL after any dc or tran, the state a real session is almost always in, while the same refusal on an INSTANCE parameter and on every built-

in model was only a warning; it now refuses and says the update did not take effect, as E-531 established for `alter` and as the sibling string-parameter path already did (E-534's comment shows the hazard was known -- it declined to reuse these setters for `.dc` points precisely because they "controlled_exit() on a CKTtemp error"). WHAT THE SAMPLING COMMANDS WERE CLAIMING: `montecarlo` drew N-1 samples on a freshly sourced deck (sample 1 was `osdimc`'s nominal baseline, a deterministic point folded into the yield and its Wilson interval, with the effective count depending on session history) and now steps past it; `-seed` did not vary the draws at all (`-seed 1` and `-seed 999` were byte-identical, so every "independent" replication returned the same points and an estimate looked perfectly stable when nothing had been re-sampled) and is now mixed into the draw key when given; `-lhs` silently did nothing for model-declared variability (it stratifies the netlist draw functions, which `osdimc`'s pure-hash draws never enter -- byte-identical with and without the flag on E-530's headline deck) and now says so. `highsigma` also gained E-501's degenerate-metric check: a metric that never varies is a mistyped node far more often than a rare failure, and the old "increase -scale or N" sent the user to spend a bigger run chasing a quantity that does not exist. THE TWO E-536 REGRESSIONS: its sweep interrupt poll filled the results array at point 0, which is not allocated until the first point has run, so an interrupt arriving before that wrote NaN through a NULL pointer (now skips to the restore); and an interrupted `optimize` printed "converged" -- E-499 had already qualified that word for three other situations, and interruption is the fourth, now "INTERRUPTED -- best point so far". ALSO: aging's dose recentred an `osdimc` nominal because it writes through the `alter` USER channel while being a machine writer, so a statistically-declared aging parameter had the draw's randomness written permanently into its nominal and random-walked across repeated runs (0 -> 0.495253 -> 0.49919 -> 0.499013, non-monotonic, proving noise not accumulation) -- aging's writes and its E-501 replays are now marked machine writes, restoring E-531's rule; result variables were never invalidated, so after a `wcd` reporting "cannot locate an MPFP" the previous run's `$wcd_beta = 0.37037` was still readable by the scripts they exist for, and each command now clears its namespace on entry; and the E-535 vector-length warning repeated per operation despite promising "once" (50 identical lines from a 50-iteration loop) and now latches. The handbook records the one finding that is a contract rather than a defect: under `.option osdimc` the two sweep engines are separate run-class commands, so a default run and a `-perpoint` run compare DIFFERENT Monte-Carlo samples (~2% apart), each correct for its own. Verified `mcpolicy` 28/28 (18 -> 28, both solvers, every new check a hunt finding with its own repro); full sweep ALL OK. No `openvaf-r` change.

E-538 *src/osdi/osdisetup.c, src/include/ngspice/osdiitf.h, src/frontend/{com_sweep,commands}.c, examples/mcpolicy_examples/ (28 -> 33 checks)*

highsigma -scale BECOMES SCOPEABLE, AND SO BECOMES USABLE. E-537 gave the command an effective-sample-size guard that says when its importance weights have collapsed -- half an answer, because nothing could name a subset of the variability to inflate, so the diagnosis came with no way out. WHY IT COULD NOT BE RESCUED BY ARITHMETIC: scaled-sigma importance sampling corrects an inflated density with a likelihood ratio per inflated dimension, and the weight is their PRODUCT, so its variance grows exponentially with their number -- while `-scale` inflated EVERY gauss statistical parameter in the circuit, including ones the metric cannot depend on. Under `.option osdimc` that count is not a modelling choice: a `(* type="instance" *)` mismatch parameter contributes one dimension PER INSTANCE, so an ordinary deck arrives in the degenerate regime without anyone asking for it (E-537 measured a true $P(\text{fail})$ of 0.297 reported as $2.5e-11$ once twenty statistically-declared bystander devices joined the product). The estimator's algebra is not wrong -- it stays unbiased in expectation -- so there is nothing to correct; the only fix is to stop putting irrelevant dimensions into the product, which needs a way to say which ones matter. THE SPELLING: `-inflate <param>`, repeatable, taking either a bare parameter name (`rr`, meaning that parameter wherever it occurs) or the project's ordinary accessor `@owner [param]` with `*` allowed as the owner (`@mm [rr]`, `@*[rr]`), matched case-insensitively like every other name in this surface. With no `-inflate` the behaviour is byte-for-byte what it was, so no existing deck changes; the announce line says when a run IS scoped ("scale (sigma inflation) = 3 on the -inflate

parameters only”), so a log never leaves it ambiguous which mode produced the numbers. THE WEIGHT FOLLOWS THE SCOPE EXACTLY: a parameter left out of scope is drawn at its TRUE sigma and therefore carries no likelihood ratio at all -- including one anyway would bias the estimate and omitting an inflated one would too -- so the draw applier and the weight walker consult the SAME predicate on the same (owner, parameter) pair and cannot disagree about which value was drawn or how it is weighted. That equivalence is pinned: the same circuit, once with twenty bystander devices and `-inflate rr` and once without the bystanders at all, produces IDENTICAL numbers (0.29665 both ways) -- the bystanders now cost exactly nothing, which is the whole point. SAYING WHEN THE SCOPING DID NOT WORK: a spec matching NO statistical parameter inflates nothing, so the run samples the nominal spread while looking like a scoped result, and now reports that nothing was inflated; a MALFORMED spec (`@bad I`) is refused before the run rather than silently ignored, so a typo cannot quietly widen the scope back to everything; and E-537’s collapse note now distinguishes the two cases it must advise differently -- an unscoped run is told to name the parameters that matter, a run already scoped and still degenerate is told to narrow further. MEASURED on the deck E-537 used to demonstrate the collapse (metric depends on one parameter, twenty statistically-declared bystanders on a DISCONNECTED subcircuit, true $P(\text{fail}) = 0.29670536$ by quadrature): unscoped at 2000 samples $3.35\text{e-}05$ with the weights flagged as collapsed; `-inflate rr` at the same 2000 samples 0.2967; at 8000 samples 0.28944 ± 0.00715 ; and the no-bystanders deck 0.29665, identical to the scoped run. Deliberately unchanged: netlist `.param` gaussians are inflated wholesale as always, since `-inflate` names OSDI statistical parameters -- a deck mixing both with many netlist gaussians can still degenerate, which the E-537 guard reports, and that side has its own established idiom; the `osdimc` side is where dimensions appear without being asked for, which is why it is the side that needed naming. Verified `mcpolicy 33/33` (28 -> 33, both solvers); full sweep ALL OK. No `openvaf-r` change.

E-539 *OpenVAF: `openvaf/osdi/stdlib.c`, `openvaf/hir_lower/src/{fmt,callbacks,expr}.rs`, `openvaf/osdi/src/{lib,compilation_unit}.rs`; ngspice: `src/osdi/{osdi.h,osdidefs.h,osdicallbacks,osdiregistry,osdisetup}.c`, `src/include/ngspice/{osdiitf,cktdefs}.h`, `src/math/si/niconv.c`, `src/spicelib/analysis/{cktdojob,ckntask}.c`, `examples/lrmio_examples/` (new, 17 checks)*

THE SECOND LRM AUDIT WORKED: FIVE DIVERGENCES FIXED, ONE FINDING WITHDRAWN. The 2026-09-02 re-audit raised six findings; five were real and are fixed here, and the sixth is withdrawn because it was never a defect. THE WITHDRAWAL FIRST, because the audit was wrong in a way worth recording: it reported that `$random`’s seed never advances and listed it among defects producing wrong numbers “with NO DIAGNOSTIC ON ANY CHANNEL”. Both halves are false. The behaviour is DELIBERATE -- `hir_lower/src/expr.rs` calls it a “DELIBERATE DEVIATION from LRM 9.13.1’s inout seed” because an advancing seed returns a different value on every Newton iteration and DESTROYS CONVERGENCE, which Enhancement-10 and `examples/rng_examples/README.md` also record -- and it is already reported by default as lint L019, naming the cause, the reason and the remedy. The audit tested the simulator and never read the compiler’s own output. Three further findings (`%m`, multichannel descriptors, `$fscanf`) were likewise already on record in E-417, E-516 and E-11, so the audit’s framing as “findings the first pass did not reach” was inaccurate for four of six; the audit page is corrected rather than quietly amended. The DIVERGENCES were real. `%m` NAMES THE INSTANCE (LRM 9.4.4): one module instantiated four ways printed the same string every time because `%m` expanded at compile time to the MODULE name, defeating the clause’s entire purpose. E-417 had declined to fix it for a real reason -- a handle-to-name callback would have a model built by a new `openvaf-r` call a NULL slot on an older ngspice, the E-396 hazard -- but that argues against a NAIVE fix, not against the fix: the null check now lives INSIDE the model (`osdi_inst_name` in `stdlib.c`), so an old simulator gets exactly today’s module-name behaviour and the crash cannot occur (verified both directions against the shipped binaries). Measured: `na`, `nb`, `n.x1.nsub`, `n.x2.nsub`, hierarchical where the hierarchy is real, against `fmt` four times before. MULTICHANNEL DESCRIPTORS ARE ONE-HOT BITS (LRM 9.5.1): both descriptor kinds came from one namespace of consecutive small integers, so `$fopen("a")=1` and `$fopen(`

"b")=2 made an OR-ed `$fdisplay` compute 3 -- the descriptor of an unrelated THIRD file, which silently received the text meant for both -- and the first user file was handed descriptor 1, the value the LRM reserves for stdout. E-516 recorded this as "documented-missing"; a write landing in the wrong file is worse than missing, which is why it is implemented rather than re-deferred. **\$fscanf** LEAVES WHAT IT DID NOT CONSUME: lowered as "read a line, then scan it like `sscanf`", it took the WHOLE line however little the format matched, so the scan -- the -- numbers -- take -- the -- trailing -- label idiom lost everything after the last conversion and a following \f the NEXT line. A distinct `FgetsScancallback` now records where the line began and each field scanner leaves the descriptor at the first unconsumed byte -- distinct precisely because `fgets` proper IS specified to consume the whole line, so a plain `fgets+sscanf` must `*fclose+fopen`REWINDS**`: the same -- named edup returned the live handle with its read position then -- open read line 2; there open now rewinds and cancel the deferred close, but only when the stream mode stream for reading is a mode change needing a real `fopen` (the first version missed that and bro then -- reopen -- to -- read round trip in `stringio_examples`). **\*\*ANATURE'S ABSTOL REACHES** `*:openvaf-r` has always written each declared abstol into the .osdi nature tables and NOTHING IN wide abstol/vntol whatever its nature declared. No ABI change was needed because the data was already there. The loader now reads `OSDI_NATURES/OSDI_DISCIPLINES/OSDI_ATTRIBUTES`, resolves 06 on a my voltage potential and 1e-12 on a my current flow, and it works for models compiled BEFORE `*ANEXPLICIT`option`STILLWINS, AND THAT IS NOT A DETAIL * * :`disciplines.vams` declares abstol on the STANDARD nature too (`Voltage` = 1u, `Current` = 1p), so this path runs for every OSDI node in every deck, not just exotic ones. Those -- but they stop being harmless the moment a user LOOSENs a tolerance, and a first version of this cho 1e-4` set to get a stubborn circuit to converge, from a declaration the user never wrote. The nature's sval ngspice already records which tolerance option the user set (`TSKtolGiven`, E -- 110), but the analysis task inherits every tolerance VALUE from the default task and then CLEAR given vntol reached the solver with the right value and no memory of who chose it; the flags now travel w 110's preset rule is unaffected -- a preset and the explicit options land on the same task, so an explicit. *COMPATIBILITY**: the descriptor encoding changes, visible only to a model that PRINTS a de returns 0 on failure and every success is non-zero, and no fd > 0 idiom exists in the tree. Both ngspice-side additions are optional -- symbol lookups, so an .osdi from an older compiler is handled exactly as before. **TWO EXISTING CHECKS ASSERTED THE BEHAVIOUR BEING CORRECTED** and were updated with the reason rather than worked around: display's "%m prints the module path" becomes the instance path, and lrmfuncs' FTELL 19 becomes 18, since $fscanf leaves the trailing newline exactly as C's own fscanf does. Verified lrmio' 17/17 (new suite, both solvers), scoring 6/17 against the previously shipped binaries so every check discriminates; full sweep 450/450 ALL OK. openvaf-r changed (first compiler change since E-534).`

E-542 ngspice: `src/spicelib/devices/dev.c, src/spicelib/parser/{inp2n,inpdomod,inpmod,inpkmmod}.c, src/frontend/inpcom.c, src/include/ngspice/{inpdefs,osdiitf}.h, examples/multimod_examples/` (16 -> 24 checks); `OpenVAF: openvaf/hir_def/src/nameres/{collect,diagnostics}.rs, openvaf/test_data/ui/reserved_module_name.{va,log}` (+6 regenerated logs)

A VERILOG-A MODULE NAMED LIKE AN NGSPICE BUILT-IN IS USABLE, NEVER SILENTLY REPLACED, AND CASE-TWINS ARE CAUGHT. The 2026-09-04 hunt's two registry findings. A module whose name is one of ngspice's own -- a built-in device's name (diode) or a .model type keyword (res, d, sw, nmos ...) -- could not be used from a netlist: on an n line the deck aborted with "incorrect model type" and no cause; on the built-in's own device letter ngspice's built-in simulated in the module's place, taking the card's parameters where the names coincided (the junction diode took is), and finished with a plausible, wrong answer. Worse, `pre_osdi -f` on such a module replaced the built-in for the session: after a forced reload of a module named diode, every plain .model ... d card ran the Verilog-A model. The keyword family (c cpl csw d l ltra nhfet njf nmf nmos npn nsoi phfet pjf pmf pmos pnp psoi r res sw txl urc vdmos ...) was not even refused -- registered and then never

selectable, because INPdomodel matches those keywords before it consults the device table. THE FIX DECIDES THE AMBIGUITY WHERE IT IS NOT AMBIGUOUS: an n-line instance can mean nothing but an OSDI device, and ngspice materialises a model lazily, at the first instance line that uses it. So the loader registers a colliding module under its own slot as SHADOWED (a new INPbuiltinModelTypeKeyword reads the same keyword table as INPdomodel), saying what the name costs; INPdomodel keeps the card's type token; the N-line parser re-binds a card whose type resolved to a built-in to the shadowed module BEFORE INPgetMod materialises it (Note(osdi): ... this n-line instance can only mean the Verilog-A module "res" ... binding the card to it)-- .model mm res r=2000 on an n line now reads -5.00000e-04; a card materialised as a built-in says so at that moment (create_model), the one place it is certain a built-in device letter is using it; one card cannot serve both letters (n-then-d fails on the d line, d-then-n fails on the n line with the explanation); the deck-reading pre-pass no longer pre-judges n lines (it runs before any pre_osdi has loaded a library, so it cannot know, and INP2N reports every genuine mismatch with the cause); and -f swaps the shadowed module's own slot, never the built-in's. F2: two modules in one .osdi whose names differ only in case (Foo/foo, legal in Verilog-A) collapsed into one with no diagnostic -- the second unreachable, every card asking for it getting the first (1 mA where the module implements 7 mA). The hunt blamed a case-sensitive comparison; the cause was that the duplicate scan stopped at DEVNUM, advanced only after the whole file, so a module never saw the ones registered earlier in the same load. The scan now covers them and reports both spellings. THE COMPILER extends L018 (reserved_module_name) to the keyword family with its own wording, both matching the simulator (.model <name> res resolves to the built-in; ngspice re-binds only for an n-line instance), with the _va convention in the help. Verified multimod 24/24 (was 16; the vcvs E-29 fixture and res now RUN from an n line, the d-line card warns at materialisation, -f keeps the built-in, both mixed orderings fail loudly), cargo 209 fast / 235 slow, full sweep 453/453 ALL OK.

E-541 *ngspice: src/osdi/{osdicallbacks,osdiload,osdiparam}.c, src/osdi/{osdi.h,osdidefs.h}, src/include/ngspice/cktddefs.h, src/spicelib/analysis/cktop.c, examples/lrmvoice_examples/ (new, 28 checks), examples/simparamdiag_examples/ (assertion updated); OpenVAF: openvaf/hir_lower/src/{ctx,lib,fmt,callbacks,expr,state}.rs, openvaf/hir_ty/src/validation{,/body}.rs, openvaf/hir_def/src/builtin.rs, openvaf/hir/src/elaborate.rs, openvaf/osdi/src/compilation_unit.rs, openvaf/osdi/src/metadata/osdi_0_4.rs, openvaf/osdi/header/osdi_0_4.h, openvaf/osdi/stdlib.c, openvaf/openvaf/tests/load/osdi_0_4.rs, examples/lrmvoice_examples/ (new, 28 checks)*

THE THIRD LRM AUDIT WORKED: NINE FINDINGS, ALL ABOUT A MODEL'S VOICE. The 2026-09-02 round-3 audit went at what the first two swept past -- the TIMING and ADDRESSING of everything a model says to the outside world -- and five of the nine reduce to one sentence: a guard exists, is correct, and does not cover every route into the thing it guards. (1) AN **analog initial** BLOCK SAID NOTHING AT ALL. LRM 5.2.1 forbids access functions, analog operators, contributions and event controls inside it, so initialising variables and REPORTING are the only two things it can do -- and the reporting half produced zero output in every analysis, with the file it opened created and left at 0 bytes. \$debug and \$info printed once per analysis throughout, which is what proves the block RAN and only its output was thrown away. Cause: hir_lower/src/lib.rs gates the block on IsInitialStep inside eval (E-456, and right for what it was for), which puts its statements on the instance's FIRST Newton iteration -- exactly the one the 9.4.6/9.5.9 deferral treats as superseded. The compiler already had the concept: stmt.rs sets in_event_ctx for event-controlled statements and the comment above it states the hazard verbatim for the neighbouring construct. A new in_analog_initial joins it. THIRD APPEARANCE OF ONE HAZARD -- E-516 handled @(initial_step), E-535 handled re-entered setup, this route was still open. (2) THE SEVERITY TASKS RAN ON EVERY NEWTON ITERATION. LRM 9.7.3 states the rule in its OWN words rather than by reference to 9.4.6 -- "non-fatal system severity tasks called during a rejected iteration shall have no effect" -- which is very likely why it was missed. One diode .op, \$strobe and \$warning on adjacent lines: 1 line versus 21, walking the unconverged sequence 0 -> 1.000000 -> 0.974135 -> ... -> 0.629443. That is verbatim the August

audit's headline surviving on the one family its remediation did not sweep. Fixed in `osdicall-backs.c` by adding INFO/WARN/ERR to the deferral gate; DEBUG and FATAL deliberately stay out, each for its own clause, and the suite pins `$debug`'s exemption as a check because a fix that deferred everything would also stop the 21 lines. Prior art is PARTIAL and is quoted rather than paraphrased: E-516 carries the parenthesis "(`$fdebug/severity` tasks keep their immediate paths)" with no clause cited, and states the exemption 34 lines earlier as "only `$fdebug` exempt" -- so the behaviour was chosen, by grouping the severity tasks with the one task 9.4.6 exempts and 9.7.3 does not cover. (3) **\$error** IN AN **analog initial** BLOCK LET THE RUN PROCEED. 9.7.3: "the message is issued and the initialization continues. However, the simulation shall not proceed past initialization." It printed and handed the deck `v(1) = 1`. The mechanism was already wired up and correct -- the same model with `$fatal` aborts exactly as it should -- and `$fatal` lowers to a display PLUS `SetRetFlag(RetFlag::Abort)` while `$error` emitted the display and nothing else. New `RetFlag::InitErr / EVAL_RET_FLAG_INITERR` carries the weaker requirement, and the distinction is real: `$fatal` abandons the evaluation, `$error` must let initialization FINISH and only then stop the analysis. CKTop reports it under its own flag so the message names the task that actually ran. (4) NONE OF THE FOUR REPORTED WHEN IT WAS CALLED. Same clause: "shall also report the simulation run time", or the swept value during a `.dc`, or "that the call was made during initialization". Messages carried the instance name and nothing else. Now a trailing parenthetical, so the `OSDI(warn) <inst>`: head every reader matches on is unchanged -- and RESOLVED AT OUTPUT TIME, not at format time, because during a `.dc` CKTtime only takes the point's swept value AFTER the solve that produced the message; the first implementation reported the previous point's value and (at sweep value 0.5) on a message raised at 0.75 is how it was caught. (5) A NEGATIVE **\$mfactor** FROM VERILOG-A SIGN-INVERTED THE DEVICE. LRM 9.18 Table 9-29 says `$mfactor > 0; #(. $mfactor(-3))` made a plain resistor model SOURCE 3 mA with no diagnostic anywhere, while the identical `m=-3` on the netlist line has been refused all along by a message that spells out the consequence. A second partial guard existed too -- with a `white_noise` contribution the constant-folded `sqrt(-3)` trips the domain check -- so a model without noise got no luck at all. Literal instance overrides are now judged against Table 9-29 where they are collected; only a literal, which is the same boundary the constant-only `sqrt` check draws. `$hflip/$vflip` get the same on both routes. `m=0` on a netlist line is UNTOUCHED (E-426's disable-this-instance idiom, a SPICE convention this does not relitigate). (6) **\$angle** WAS NEVER REDUCED MODULO 360. Table 9-29's resolution rule is a sum "modulo 360 degrees" into `0 <= angle < 360`; two levels of 200 degrees read back as 400. Harmless to a model that uses the angle trigonometrically, wrong for one that COMPARES it. Normalised wherever a value is materialised, so every route lands in range, with `x - 360*floor(x/360)` rather than a `%` -- the value is a real and this form is correct for negative angles too. (7) THE 31ST MULTICHANNEL DESCRIPTOR WAS THE RESERVED BIT. 9.5.1 reserves bit 31 and says so precisely so an implementation can tell the two descriptor kinds apart -- and it is the bit E-539 took for file descriptors, with `0x8000_0000` being STDIN, so the 31st `$fopen(name)` aliased a read-only stream and every write to it was discarded. The allocator now stops at bit 30 and the 31st open returns the clause's own failure value, 0. (8) A REOPEN IN A DIFFERENT MODE WAS IGNORED. Table 9-24 gives "w" as "truncate to zero length or create for writing"; read a file, `$fclose`, reopen "w" and the same-name dedup handed back the READ-mode stream, so nothing was truncated and every write vanished without an error. E-539's own comment names the general rule and implemented it for write -> read; read -> write was the other half, and it failed silently where the case E-539 fixed at least produced visibly wrong data. The descriptor's queued deferred writes are performed BEFORE the reopen -- the model has closed the file, so a reopen for reading must see them. ONE VERSION OF THAT FIX WAS WRONG AND THE SUITE ESTATE CAUGHT IT: it also applied 9.5.1.1's append-on-rewrite rule there, which looks like consistency and is not -- that clause is about content written from following ANALYSES, while ngspice runs instance initialization twice per analysis (setup and temperature), so a write-read-write model appended its second run's output onto its first's and broke E-516's byte-for-byte re-run guarantee. No check output moved; `stringio_examples`' committed round-trip artefacts did, one line becoming two, which is the whole

reason those files are in the repository. (9) **\$write** COULD NOT COMPOSE A LINE. 9.4.1 gives it "the same capabilities as \$strobe, but with no newline", and assembling one line from several calls is the only thing that distinguishes it -- which the per-call OSDI <inst>: head defeated. The head is now written only at the start of a line, tracked per stream; every other task ends its text with a newline and is unchanged, which the suite pins with a \$strobe immediately after three \$writes. PLUS LRM 9.20's OTHER TWO ERROR RULES (a port as the aliased net, and a target that is another call's aliased net), the second judged only where it does not need hierarchy resolution because a false error on a legal alias would be worse than the missing one. COMPATIBILITY: two ADDITIVE ABI bits on E-55's precedent (EVAL_RET_FLAG_INITERR, LOG_FLAG_INIT), so an older simulator ignores an unknown return bit and an older .osdi never sets the log flag; nothing crashes and nothing changes meaning. Two behaviours change for existing decks, both deliberately: a per-iteration \$warning becomes one per accepted point (the fix), and the three non-fatal severity messages gain a trailing (at ...) -- with the head unchanged, so matches on OSDI (warn) <inst>: <text> still hold. One existing check was updated with its reason: simparam-diag's "newline-terminated" assertion tested the property by requiring the line to end at a closing quote, and now admits that one parenthetical. Verified lrmvoice 28/28 (new suite, both solvers), scoring 11/28 against the audited binaries where the eleven passes are the compile checks plus four deliberate controls; full sweep ALL OK.

E-543 *src/frontend/{com_sweep,spiceif}.c, src/frontend/numparam/xpressn.c, src/maths/misc/randnumb.c, src/maths/KLU/klusmp.c, src/osdi/{osdiload,osdiregistry,osdisetup,osdicallbacks}.c, src/osdi/osdidefs.h, src/spicelib/analysis/cktacct.c, src/include/ngspice/{osdiitf,randnumb,klu}.h, examples/osdilimit_examples/ (new, 12 checks), examples/{montecarlo,highsigma,wcd,mcpolicy,yield,display,klu_tuning}_examples/ (+34 checks), examples/benchmark_examples/large_bench.py (new)*

THE MONTE CARLO COMMANDS ANSWER HONESTLY, AND A COMPILED MOSFET CONVERGES LIKE A BUILT-IN ONE. Two 2026-09-04 hunts folded, nine findings. The Monte Carlo commands: a quoted or unresolvable -spec/-metric was scored as 0 and reported as a confident 0% or 100% yield (the evaluator's validity flag was discarded) -- now unquoted like -analysis and refused with the spec, sample and plot named; every banner states its seed and an un-seeded run says a rerun repeats the netlist draws (montecarlo_seed/highsigma_seed/wcd_seed for scripts); wcd searches over model-declared statistics through a new osdimc WALK mode (OSDimcWalk: every Gaussian statistical parameter takes nominal + sigma*z[k]; uniforms held) -- the osdimc-only deck reports beta = 4.0000 where it was refused, and a 1-ohm netlist dimension beside those statistics 3.9601 where it read 106.7; mvnorm(3) against a 2x2 matrix is a .param error and mcorr reports an index the deck already used beyond it; -max below -min is refused. Large circuits (to 40 000 BSIM4 MOSFETs, 67 000 OSDI instances, both solvers agreeing to 1e-15): an OSDI MOSFET or BJT whose model calls no \$limit now gets the built-ins' Newton step limiting -- recognized by terminal names (d,g,s[,b] / c,b,e[,s]) with polarity from type and threshold from vth0/vto, the cold-start guess and DEVfetlim/DEVlimvds/DEVpnjlim applied in the type-normalized frame across the model's live internal DI/SI/GP/BI nodes; models with a thermal terminal, another live internal node (MEXTRAM) or their own \$limit are left alone. A 100-stage BSIM4 or PSP103 chain converges in 8 iterations (was 333/387 with gmin stepping; built-in 9), a 40x40 grid in 8 and 0.56 s (was 167, 5.2 s), every operating point identical to 1e-16; .option noosdilim / set osdilim verbose. KLU's refactor reused a pivot order the matrix had outgrown without any test (137 iterations with gmin stepping where Sparse took 8): a collapse of its rcond estimate by more than 1e-6 relative to the last full factorization now reorders, as Sparse's small-pivot test does. rusage no longer reports a negative KLU fill-in (the diagonal was counted twice, the off-block entries never) nor a zero Sparse-mode total in a KLU build; a model line repeated by thousands of instances is shown five times and then counted.

E-544 *src/frontend/{com_aging.c,com_aging.h,control.c,device.c,spiceif.c,com_sweep.c,com_optimize.c,runcoms2.c}, src/include/ngspice/ftext.h, examples/mcpolicy_examples/ (34 -> 41 checks)*

THE USER'S **alter/altermod** WRITES SURVIVE THE LOOP COMMANDS' INTERNAL RESETS. Finding F3 of the 2026-09-04 ngspice + OSDI hunt. The sampling commands redraw a deck's random .param values with an internal reset -- a full re-source -- on highsigma and wcd at every evaluation and on montecarlo whenever E-346's fast path cannot arm (every deck whose only variability is model-declared), and the re-source discarded every alter/altermod the user had typed, statistical or not: wcd reported $\beta = 4.00$ for a nominal the user had just moved 4 sigma, highsigma $P(\text{fail}) = 2e-5$ where the recentred nominal made it 0.35, and the op after montecarlo read the un-altered circuit. The user's writes are now JOURNALED at the command dispatcher (only commands the user typed; the optimizer's, sweep's, temper's and aging's own writes stay out), one entry per target with the value as evaluated, and REPLAYED after every internal reset beside E-501's aging replay, under ft_optimizing, with a circuit-title check; a user reset forgets them; optimize journals its final optimum. Seven checks in mcpolicy_examples (34 -> 41), twenty related suites green on both solvers. The hunt also records F14 -- set temp dropped by every reset -- as pre-existing and unfixed.

E-547 *src/frontend/plotting/pyplot.c, examples/pyplot_examples/ (17 -> 24 checks), docs/internals/ngspice_internals/{ngspice_pyplot,ngspice_commands}.md*

pyplot QUOTES ITS LAUNCH AND JUDGES ITS EXIT STATUS. Defects 1 and 2 of a review of the pyplot command (six defects, five improvements; E-548 to E-551 work through the rest). The command line that ran the generated matplotlib script was built unquoted through system(), so with the deck-folder output of E-183 a folder named My Circuits handed Python the file My and an apostrophe left the shell waiting for a closing quote -- the .py and .data were written, no image was, and ngspice went on with exit status 0. And only a -1 from system() was ever looked at, so a missing interpreter or a missing matplotlib printed Python's own complaint and nothing else: a batch deck finished green with no figure. One launch path (pyplot_run) now serves the five renderers: the script path is shell-quoted on both platforms (pyplot_python is still spliced verbatim so it can carry options; a value naming an executable path with a space is one word), a hardcopy's non-zero exit is named together with the image that was not written, every pyplot publishes **pyplot_status** as shell publishes shellstatus (so if \$pyplot_status ne 0 . . . quit 1), and the script's own 'wrote' line quotes the file name as a Python literal. pyplot_examples 17 -> 24, both solvers; the seven other pyplot suites, eye and rplots pass.

E-548 *src/frontend/plotting/pyplot.c, examples/pyplot_examples/ (24 -> 29 checks), docs/internals/ngspice_internals/ngspice_pyplot.md*

pyplot WRITES EXACT DATA, HONOURS ylimit UNDER ylog, GIVES MIXED UNITS TWO SCALES, AND ITS SCRIPT RUNS FROM ANYWHERE. Defects 3 to 6 of the pyplot review. The data table was written with %e, six significant digits, in the file the reference calls the export: a time axis offset to 1 s with 1 ns steps collapsed to ONE distinct x value in 59 rows, a microvolt ripple on 1 V to eight levels -- every table now carries 17 significant digits. ylimit lo hi under ylog set the log scale and silently no limits; it is applied now (a non-positive bound never reaches the backend: plotit refuses it first). A voltage and a current on one plot shared one axis with no label at all, the milliamps flat along the bottom of the volt scale; within a panel the first trace's type now owns the left axis and any other type is drawn on a twinx() axis on the right, each labelled with its type, the legends combined, every trace given an explicit colour (a twin axis restarts the colour cycle). And the generated script looked for its data relative to the directory ngspice ran in, so the reference's own advice -- edit it and run it again -- failed from anywhere else; it now resolves its data table and its image against its own location. pyplot_examples 24 -> 29, both solvers; the twelve suites that exercise pyplot pass; full sweep 455 of 455.

E-549 *src/frontend/{com_pyplot.c,commands.c}, src/frontend/plotting/{plotit.c,pyplot.c,pyplot.h}, examples/pyplot_examples/ (29 -> 35 checks), examples/{pyplotcontour,pyplotsmith,pyplotmore,pyplohist}_examples/ (read either format), .gitignore, docs/internals/ngspice_internals/{ngspice_pyplot,ngspice_commands}.md*

****pyplot WRITES ITS DATA TABLE AS .npz, 'set pyplot_export=bin**

E-550 *src/frontend/plotting/pyplot.c, src/frontend/commands.c, examples/pyplot_examples/ (35 -> 40 checks), docs/internals/ngspice_internals/ngspice_pyplot.md*

pyplot DRAWS A LONG TRACE AS ITS PER-PIXEL MIN/MAX ENVELOPE, RE-DECIMATED ON ZOOM. Improvement 2 of the pyplot review. A pixel column can only show its extremes, so a trace with more samples than twice the axis width in pixels is drawn as its min/max envelope per column: the same picture, drawn in milliseconds -- matplotlib took 4.8 s of a 5.2 s run to draw four million points onto a 640-pixel canvas; the million-point, four-trace hardcopy now takes 2.34 s end to end against 6.81 s without. The envelope bins by x (adaptive time steps are handled), keeps each column's actual extreme samples in time order, skips NaN padding, and leaves point plots, step plots and a vs plot whose x runs backwards alone. An interactive window keeps the full data beside each line and re-decimates on every zoom, pan and resize, so zooming into a slice redraws it in full detail; a hardcopy decimates once at its dpi; the script says what it did. 'set pyplot_decimate=auto

E-551 *src/frontend/plotting/pyplot.c, src/frontend/commands.c, examples/pyplot_examples/ (40 -> 47 checks), docs/internals/ngspice_internals/ngspice_pyplot.md*

pyplot TICKS IN ENGINEERING UNITS, AXES LABELLED BY TYPE, THE DECK'S * OFF THE TITLE. Improvement 3 of the pyplot review. The default figure showed 0.0005, 0.0010 on an axis labelled s, V on the other, and * rc lowpass for a title. Each axis is now formatted by matplotlib's EngFormatter in the unit of the vector type it carries -- 500 us, 1 ms, -500 mV, 10 kHz, 1 kOhm -- with prefixes only on SI units (s, Hz, V, A, W, F, C, Ohm, S); dB, rad, Celsius and the noise densities stay plain; the formatter is installed after the axis scale, since set_xscale('log') replaces whatever was there. The default labels read time [s], voltage [V], frequency [Hz], decibel [dB], a mixed plot voltage [V] left and current [A] right; a label or title the user gives is kept verbatim; the default title drops the deck's leading * . set pyplot_eng=off keeps plain numbers. The Bode frequency axis, the eye's time axis, a contour's axes and colour bar and a histogram's value axis get the same ticks. pyplot_examples 40 -> 47, both solvers, reading the rendered tick texts; the twelve suites that exercise pyplot pass; full sweep 455 of 455.

E-552 *src/frontend/{com_sweep.c,commands.c,typesdef.c}, examples/mcrecord_examples/ (new, 13 checks), docs/internals/ngspice_internals/{ngspice_statistics,ngspice_commands}.md, docs/handbook/03-ngspice-workflows.md*

montecarlo -expr RECORDS A VALUE PER SAMPLE INTO A **montecarlo<n>** PLOT, AND A YIELD IS JUDGED ONLY WHERE A SPEC HAS A LIMIT. The packaged Monte Carlo command answered one question, the yield: a -spec was mandatory, a limit on it was mandatory, and nothing per sample survived the run (the analyses run under the loop commands' plot recycling), so 'run the sweep N times and keep a value from each' had to be hand-written around reset. -expr [name=]<expression> is evaluated after every sample and recorded, unjudged, into a plot of its own -- montecarlo1, montecarlo2, ... one per invocation, named in \$montecarlo_plot -- with sample (1..N) as its scale. A scalar per sample is an N-long vector; a waveform per sample (a dc/ac/tran output of L points) is an N x L two-dimensional vector with the analysis scale copied beside it, which plot draws as a family and vo[k] indexes as sample k. A complex value is its magnitude, a failed sample is nan, a waveform whose point count differs between samples is refused with the reason (the scalars beside it still recorded), an expression that never varies is noted, and one that resolves to nothing is refused on sample 1. A -spec without -max/-min is refused with a pointer to -expr; with no -spec there is no yield; nothing to judge and nothing to record is refused naming both; -spec and -expr combine; a limited -spec alone is unchanged and creates no plot. montecarlo is registered as a plot type (the plots came out as unknown1 without it). mcrecord_examples (new, 13 checks, both solvers); the 22 suites that use montecarlo pass unchanged; full sweep 456 of 456.

E-553 *src/frontend/{control.c, inpcom.c, quote.c}, src/frontend/parser/{lexical.c, glob.c}, src/include/ngspice/cpextern.h, examples/rawfstring_examples/ (new, 14 checks), docs/handbook/{03-ngspice-workflows, 04-limitations-and-gotchas}.md, docs/internals/ngspice_internals/{ngspice_commands, ngspice_pyplot}.md*

RAW STRINGS `r"..."` KEEP THEIR CASE THROUGH THE DECK READER; F-STRINGS `f"..."` `{expr: .3f}...` EVALUATE IN A CONTROL SCRIPT. A deck is folded to lower case as it is read, and the fold reached the text a control script hands to a command: `set t="ABC"` stored `abc`, `pyplot ... title "RC Low-Pass"` printed `rc low-pass`; the fold was command-specific and uneven (echo lines and the title tokens of `plot` and `gnuplot` were exempt, `pyplot`'s were not), and the same reading pass dropped the spaces around an `=` inside a quoted string. After Python: `r"..."` / `r'...'` is a RAW string, copied through the deck reader as written, case and spaces kept; `f"..."` is an F-STRING, every `{expression}` evaluated with the control-language evaluator when the command runs and replaced by its text, with an optional format after a colon (`.3f`, `.4g`, `e`, `d`; a scalar prints with `%g`, a vector as its elements, a complex value as `re`, `im`; `\{` and `\}` are literal braces -- `{ }` stays with the netlist's `.for`); `rf"..."` is both. The prefix counts only at a token start (a resistor `r1`, a variable `f`, `set r=...` are untouched). An expression that resolves to nothing, an unbalanced brace or a bad format is an error naming the string, and the command is not run. `$variables` are substituted before the braces are evaluated; interactive input, never folded, simply accepts the prefixes; a plain string folds as before. `pyplot` keeps the case of its `title/xlabel/ylabel` tokens the way `plot` and `gnuplot` always did. `prefixed_span_end` in `inpcom.c` (the control-line fold and the whitespace pass skip a prefixed literal), `cp_string_prefix_len` in the lexer (a prefixed `'...'` becomes the double-quoted form, backslashes kept), `cp_unquote` strips the prefix with the quotes, `cp_doglob` leaves a prefixed word alone, `cp_fstringsubst` in `control.c` evaluates after globbing, also in `foreach` lists. `rawfstring_examples` (new, 14 checks, both solvers); full sweep 457 of 457.

E-554 *src/osdi/{osdi.h, osdiregistry.c, osdisetup.c}, src/include/ngspice/osdiitf.h, src/frontend/com_sweep.c, examples/osdidist_examples/ (new, 18 checks), docs/handbook/03-ngspice-workflows.md, README_OSDI.md, docs/internals/ngspice_internals/ngspice_statistics.md*

THE LOGNORMAL AND TRUNCATED DRAWS, THEIR IMPORTANCE WEIGHT, AND A `wcd` THAT NAMES A CLAMPED WALK. The registry merges the INFOS records and the optional TRUNCS array into in-memory `OsdiStatParam` records. A truncated draw is deterministic rejection: the first attempt is the plain key's deviate, so declaring a truncation changes only the draws that would have exceeded it (282 of 300 identical, 18 redrawn inside the window); a rejected attempt is redrawn from a sub-key of the same trial, 64 times, then clamped. A lognormal is nominal $\exp(s z)$. Both inflate under `highsigma` -scale with the matching weight: with a truncation both densities are confined to the same value window and their normalisers $\text{erf}(t/\sqrt{2})$ and $\text{erf}(t/(sc \sqrt{2}))$ join the likelihood ratio -- 0.0346 measured against 0.0336 exact for a 2-sigma window, 0.024 without them; the untruncated control 0.0568 against 0.0548. Both take a `wcd` walk coordinate, clamped at the truncation, and `wcd` now says *the walk is held at the trunc truncation of N model-declared parameters ... the spec boundary lies beyond the declared window instead of zero gradient*. Measured over 300 draws: lognormal `std_rel=0.2` gives $\ln(v/\text{nom})$ sd 0.203, every draw positive; `std_rel=1.0` on $(0:\text{inf})$ loses no trial where the gauss control loses 47; gauss `trunc=1` stays within ± 1 sigma with sd 0.54 sigma and no clamp. `osdimc_verbose` names the shape. The 13 suites that share the draw machinery pass; full sweep 458 of 458.

E-555 *src/osdi/{osdi.h, osdiregistry.c, osdisetup.c}, src/include/ngspice/{osdiitf, trcvdefs}.h, src/spicelib/analysis/dctrcurv.c, src/frontend/com_sweep.c, examples/paramgiven_examples/ (new, 16 checks), examples/elabguard_examples/ (adjusted), docs/handbook/03-ngspice-workflows.md, README_OSDI.md, docs/internals/ngspice_internals/ngspice_statistics.md*

A GATED PARAMETER THE DECK NEVER GAVE IS NOT DRAWN, AND EVERY SWEEP RE-STORE PUTS THE GIVEN FLAG BACK. The simulator half of F1. `osdimc` draws a parameter the

model tests with `$param_given` only when the deck gave it, and says so once -- `mos_va:toxp` is not given by the deck and the model tests `$param_given(toxp)`: a draw would switch the model to its "given" branch instead of varying it -- not drawn. Give it on the card, or altermod it, to vary it -- and such a parameter is no dimension of the wcd walk and no factor of the highsigma weight. The registry loads the object's given-flag entry point; `OSDIparamGiven / OSDIparamGivenByName` let the .dc restores (instance and model targets, `dctrcurv.c`), the sweep command's restore and un-set `osdimc` clear the flag of a parameter the deck never gave -- where before a dc `@mos_va[toxp]` with the option off left BSIM4 at -76.30 uA for good, and the built-in BSIM4 under the same sweep read -108.96 uA before and after. BSIM4 with only `toxe` on the card now reads -112.39 uA on every trial, after the sweep, and in a recorded ensemble; with `toxp` on the card it is drawn as before. F2's runtime effect: `altermod mm lmin=1.5` with `l` defaulted, dc `@mm[lmin] 0.5 2.0 0.5` and the draws of `lmin` past `l` are refused with the model's own range error; `elabguard's freeze(aa=5)` check had set exactly that shape and now sets `aa=3`, with a new check pinning the refusal. The statistics, sweep, range and CMC runtime suites pass; full sweep 459 of 459.

E-556 `src/frontend/parser/{lexical,glob}.c, src/frontend/{control,cpitf,com_pyplot}.c, src/include/ngspice/cpextern.h, examples/rawfstring_examples/ (14 -> 18 checks), examples/pyplot_examples/ (47 -> 49 checks), docs/handbook/03-ngspice-workflows.md, docs/internals/ngspice_internals/ngspice_pyplot.md`

AN F-STRING AFTER `name=` IS EVALUATED, ITS RESULT IS PLAIN TEXT, AND `pyplot` UNQUOTES ITS NAME. F3, F4 and F5 of the 2026-09-05 hunt, on E-553's string forms. The lexer accepted a string prefix at a word's start or after `=`, `(`, `,`; the glob skip and the f-string pass looked at the word's start only, and the deck reader collapses `let z = f"{7}"` into `let z=f"{7}"` and `if f"{1+1}" = 2` into `if f"{1+1}"=2` -- so in a deck an f-string after `name=` had its braces globbed away and was never evaluated, and no set variable could take one at all. `cp_string_prefix_at` finds a prefixed string wherever the lexer does, with a tail after the closing quote kept and more than one per word; `cp_doglob` skips a word carrying one, `cp_fstringsubst` rebuilds head + text + tail, and `cp_istrue` -- the if/while condition path, which never saw the pass -- runs it. The substituted text was a quoted word that `let`, `alter`, `setplot`, file names and numeric options refused; it is plain text now, quoted only when it has whitespace so a command that re-reads its arguments (`set u=rf"{x:.3f} V"`) still sees one word (`foreach x f"{3}"` yields `x=3`, not `x="3"`). And `pyplot` used a quoted name verbatim, quotes and all, so `pyplot -export "my dir/sig"` -- the one spelling that can carry a space -- wrote `"my dir/sig".npv` nowhere; the name is unquoted, the eye path included. `rawfstring_examples 14 -> 18, pyplot_examples 47 -> 49`; full sweep 459 of 459.

E-557 `src/frontend/{com_pyplot,com_sweep}.c, src/frontend/plotting/pyplot.c, examples/pyplot_examples/ (49 -> 53 checks), examples/mcrecord_examples/ (13 -> 14 checks), examples/paramgiven_examples/ (the pre-E-555 compiler from git), docs/internals/ngspice_internals/{ngspice_pyplot,ngspice_statistics}.md`

`pyplot_status` ON EVERY `pyplot`, A COLLIDING `-expr` NAME REFUSED, A BIN COUNT BELOW 2 SAID. F6, F7 and F8 of the 2026-09-05 hunt. E-547 promised that every `pyplot` publishes `pyplot_status`, and five paths did not -- a refusal before the backend ran (a usage error, plotit's own *X values must be > 0 for log scale*), a table or script that could not be opened (a read-only file, a missing directory), and a successful `-export` -- so the deck's `if $pyplot_status ne 0` died with *no such variable* on exactly the failures it was written to catch; `com_pyplot` assumes a failure at entry (status 1) and every path that succeeds says so, the launch path with Python's exit status, the export path with 0. `montecarlo` refused two `-expr` of one name and let one take the record plot's own -- `sample`, its scale, or a result such as `montecarlo_n` -- which then shadowed the record; those names are refused with the reason. `'set pyplot_decimate=1`

E-558 `src/osdi/{osdiregistry.c,osdisetup.c}, src/include/ngspice/osdiitf.h, src/frontend/{inp.c,outitf.c, postcoms.c,com_gnuplot.c}, examples/paramgiven_examples/ (16 -> 18 checks), examples/`

rawfstring_examples/ (18 -> 19 checks), docs/handbook/03-ngspice-workflows.md, README_OSDI.md

THE RANGE ERROR SAYS THE DECLARED RANGE AND WHAT MOVED IT, QUOTED FILE NAMES ARE UNQUOTED, **.save** TAKES A MODEL CARD'S PARAMETER. F9, F10 and F11 of the 2026-09-05 hunt. The registry slices the compiler's new range texts per descriptor and the out-of-bounds message appends the text and the current value of every parameter it names, read where the failing parameter lives: Parameter 1 of 'mm' is out of bounds (value 1.2; range from [lmin:inf), lmin = 1.5)!, (value 2; range from (0:w], w = 1) for a per-instance range, (value -5; range from (0:inf)) for a static one. wrdata "o3.txt", write "w2.raw" and source "sub1.cir" used the quoted name verbatim, quotes and all -- the spelling for a path with a space, and what an f-string with whitespace yields -- and source reported the errno the last call had left (*Inappropriate ioctl for device* for a file that does not exist); each unquotes its name, and source probes the file once more and says *No such file or directory*. **.save @mm[s]** of a model card's parameter was refused as *no such device* while **print @mm[s]** read it; a name that resolves to a model card is accepted at save time, checked against the card's parameters (an unknown one warns), and read per point through the resolver **print** uses. paramgiven_examples 16 -> 18, rawfstring_examples 18 -> 19; full sweep 459 of 459.

E-559 *src/osdi/osdisetup.c, src/include/ngspice/osdiitf.h, src/frontend/{com_sweep,control}.c, src/frontend/parser/lexical.c, examples/mcpolicy_examples/ (41 -> 46 checks), examples/rawfstring_examples/ (19 -> 23 checks), docs/handbook/03-ngspice-workflows.md, README_OSDI.md*

A LOOP COMMAND'S **-seed** PINS THE MODEL-DECLARED DRAWS, AND THE F-STRING EDGES. F13 and F12 of the 2026-09-05 hunt. E-537 mixed a loop command's **-seed** into the **osdimc** draw key but left the session-wide trial counter in it, so **montecarlo 3 -seed 1** run twice reproduced the netlist **agauss** values and not the model-declared draws (1020.17 / 815.62 / 719.15, then 1213.59 / 1198.65 / 1133.03), **highsigma 1000 -scale 2 -seed 3** gave 0.0594 and 0.0481 for one seed, and the published **montecarlo_seed** could not regenerate an ensemble -- what the draws depended on was how many run-class commands had happened before. Inside **montecarlo**, **highsigma** and **wcd** the key now carries (**-seed**, sample number counted from the command's **start**); the seed defaults to 1 exactly as the netlist half's always has, so an unseeded run repeats itself whole and equals **-seed 1**, and the NOTE says so. The trial counter still advances, so the plain run after the command is a fresh trial as before; **osdimc_verbose** shows [sample 2 of -seed 1]. A never-run deck on **montecarlo**'s fast path (no per-sample reset) spent sample 0 on the nominal baseline because the first run's new circuit pointer restarted the count; the baseline skip blesses that pointer. F12: (long) of a value beyond a long's range is undefined and {1e20:d} printed 9223372036854775807; :d/:i print such a whole number exactly, :x/:X/:u refuse a negative or too-large value with the value named, {{1+1}} is told braces do not nest (the message named a vector {1}, { } is an empty {}, a colon tail with a format's shape but no conversion letter ({x:.3}, {x:.3q}) gets a hint, a ternary's colon does not, and an unterminated **r'abc** passes through as typed instead of re-quoted as **r"abc**. mcpolicy_examples 41 -> 46 (one one-sided limit widened to a two-sided band), rawfstring_examples 19 -> 23; full sweep 459 of 459.

E-560 *src/frontend/plotting/pyplot.c, src/frontend/{spiceif,device,inp}.c, src/spicelib/analysis/dctrcurv.c, src/spicelib/devices/dev.c, src/osdi/osdisetup.c, src/include/ngspice/{ftedefs,osdiitf}.h, examples/pyplot_examples/ (53 -> 54 checks), examples/instdep_examples/ (20 -> 23 checks), examples/osdireload_examples/ (5 -> 6 checks), docs/handbook/03-ngspice-workflows.md, docs/internals/ngspice_internals/{ngspice_pyplot,ngspice_commands}.md*

pyplot LABELS AN UNTYPED AXIS BY NAME, A PROMOTED PARAMETER IS NAMED INSTANCE-LEVEL THROUGH ITS MODEL, AND A CIRCUIT BUILT AGAINST A RELOADED OBJECT IS REFUSED UNTIL REBUILT. F14, F15 and F16 of the 2026-09-05 hunt. **pyplot mcv rr** on a **montecarlo** plot (scale sample, value **rr**, both untyped) wrote a script with neither **set_**

`xlabel` nor `set_ylabel` -- E-551 labels by vector TYPE and `plotit` hands a NULL abbreviation for notype; the scale's name and a single signal's name are the labels now, `-hist` of an untyped value labels its x-axis by name, a user's label still wins. `altermod mm l=10` on a parameter E-546 resolves per instance (`l = 2.0*w`, lint L028) was refused as *model 'mm' has no parameter l* after a stray MOS width probe (any name starting with `m` and a `w/l`), while the card and `alter n1 l=10` worked; `altermod, print @mm[l]` and `dc @mm[l]` now say the parameter is INSTANCE-level and point at `alter @<instance>[l]=...` and at `l=...` on the card, where it is the instances' default, and the bin probe runs for instance writes only. The hunt's *unlabelled run after pre_osdi -f* was the first run's nominal baseline, miscounted -- but the reload itself was unguarded: `osdi -f` swaps the device type in place, so a circuit loaded earlier would run the NEW object's code on the OLD layout's data (E-229's documented caveat); the reload now names every such circuit and its next run is refused until `reset` or a re-source rebuilds it, while the deck whose control block carries the hoisted `pre_osdi -f` is built against the new object as before. `pyplot_examples 53 -> 54`, `instdep_examples 20 -> 23`, `osdireload_examples 5 -> 6`; full sweep 459 of 459.